

Assignment1: CancerMortality

Joan Acero, Anna Monsó, Walter Troiani

2024-10-21

Contents

Introduction	2
First Steps	3
Explanatory Data Analysis	6
Univariate Descriptive Analysis	6
Definition of factors	152
Data Quality Report	157
Target variable: Profiling and correlation analysis.	177
Modeling	188
Model Exploration	188
Bias-variance tradeoff	296
Influential data study	302
Model evaluation	303
Conclusion	304

Introduction

To complement this report, an ANNEX document is included, containing all the results, visualizations, and plots necessary to interpret and validate the findings. This additional resource provides a thorough understanding of the analysis and facilitates the reproducibility of the study.

This report documents the process and results of the Modeling Assignment for the Statistical Modeling and Inference course. The primary aim of this project is to analyze the Cancer Mortality dataset and build an accurate and interpretable linear model to predict `target_deathrate`, which represents the average cancer mortality rate per 100,000 individuals.

The project begins with data preparation and exploratory analysis, focusing on cleaning and enhancing the dataset to ensure its quality. This involves generating a Data Quality Report, identifying and handling missing values through imputation, and analyzing variable distributions and relationships. Univariate descriptive analysis is performed to examine each variable's characteristics.

The second stage involves modeling using the training dataset, where several key analyses are conducted to optimize the model:

- Initial model exploration addresses potential issues such as multicollinearity and the contribution of each predictor.
- Model transformations are applied to improve variable relationships and address non-linearity.
- An influential data study identifies and evaluates the impact of unusual observations on the model.
- The model is further refined through validation and diagnostics, ensuring it adheres to key assumptions like normality, homoscedasticity, and independence.

Finally, the evaluation process assesses the model's generalization performance on the test dataset, providing insights into its predictive capabilities. A detailed interpretation of the model results is presented, highlighting how the predictors influence the target variable and their practical implications for understanding cancer mortality trends.

This report follows a systematic and methodical approach to ensure the development of a reliable model.

First Steps

To begin, we import the necessary data, load the required libraries, and create functions to streamline repetitive processes.

```
calculate_outliers <- function(data) {
  # Check for regular outliers
  Boxplot(variable, main = "Boxplot")
  outliers <- boxplot.stats(data)$out

  varsumm <- summary(data)
  Q1 <- varsumm[2]
  Q3 <- varsumm[5]
  IQR <- Q3 - Q1

  lower_severe_bound <- Q1 - 3 * IQR
  upper_severe_bound <- Q3 + 3 * IQR
  lower_mild_bound <- Q1 - 1.5 * IQR
  upper_mild_bound <- Q3 + 1.5 * IQR

  # Identify outliers
  mild_outliers <- which((data < lower_mild_bound) |
                        (data > upper_mild_bound))
  severe_outliers <- which((data < lower_severe_bound) |
                          (data > upper_severe_bound))
  num_mild_outliers <- length(mild_outliers)
  num_severe_outliers <- length(severe_outliers)

  # Return the results as a list
  return(list(
    num_outliers = num_mild_outliers,
    num_severe_outliers = num_severe_outliers,
    outliers_indexes = mild_outliers
  ))
}

summary_and_normality_check <- function(data) {
  #Basic summary
  print("Summary: ")
  print(summary(data))

  # Plot normality
  print("Mean")
  mm <- mean(data, na.rm = TRUE); print(mm)
  print("Standard deviation")
  ss <- sd(data, na.rm = TRUE); print(ss)
  hist(data, freq = F, 100, main = "Histogram")
  curve(dnorm(x, mean = mm, sd = ss), col="red", lwd = 2, add=T)

  #Compute skewness
  print("Skewness")
  print(skewness(data, na.rm = TRUE))
}
```

```

#Number of outliers
out.res <- calculate_outliers(data)
print("Num Outliers:")
print(out.res$num_outliers)
print("Num. severe outliers:")
print(out.res$num_severe_outliers)

#Shapiro test
print("Shapiro test: ")
print(shapiro.test(data))

# Most repeated value:
freq_table <- table(variable)
most_repeated_value <- names(freq_table)[which.max(freq_table)]
most_repeated_value
most_repeated_count <- max(freq_table); most_repeated_count
print("Most repeated value ad times it occurs: ")
print(most_repeated_value)
print(most_repeated_count)

return(out.res$outliers_indexes)
}

evaluate_model <- function(model, data, plot_results = TRUE, print_metrics = T){
  # Generate predictions
  predictions<-predict.lm(
    model,
    newdata = data,
    interval="prediction",
    level=0.95,
  )

  inverse_transform <- function(y){
    (lambda*(y)+1)^(1/lambda)
  }

  predictions <- inverse_transform(predictions)
  predicted <- (predictions[, "fit"])
  target_test <- test$target_deathrate

  # Calculate metrics
  mse <- mean((predicted-target_test)^2); mse
  rmse <- sqrt(mean((predicted-target_test)^2)); rmse
  ssres <- sum((target_test-predicted)^2);ssres
  sstot <- sum((target_test-mean(target_test))^2);sstot
  rsquared <- cor(predicted, target_test)^2;rsquared

  # Optionally Print metrics
  if(print_metrics) {
    cat("Mean Squared Error (MSE):", mse, "\n")
    cat("Root Mean Squared Error (RMSE):", rmse, "\n")
    cat("SStotal is: ", sstot, "\n")
    cat("SSresidual is: ",ssres, "\n")
  }
}

```

```

cat("R-squared ( $R^2$ ):", rsquared, "\n")

}

# Optionally plot predictions vs actual values
if (plot_results) {
  # Plot the actual vs predicted values
  plot(target_test, predicted,
       col = "grey", pch = 19, xlab = "Actual Values",
       ylab = "Predicted Values", main = "Predicted vs. Actual Values")
  abline(a = 0, b = 1, col = "red", lwd = 2.5)
  mean_actual <- mean(target_test, na.rm=T)
  abline(h=mean_actual, col = "cyan", lwd=2.5)
  # Display SSresidual, SStotal, RMSE, MSE, and  $R^2$  on the plot
  text(x = max(target_test) * 0.4, y = max(predicted) * 0.82,
       labels = paste(
         "SS Residual:", round(ssres, 2), "\n",
         "SS Total:", round(sstot, 2), "\n",
         "RMSE:", round(rmse, 2), "\n",
         "MSE:", round(mse, 2), "\n",
         " $R^2$ :", round(rsquared, 2), "\n",
         "Mean Actual:", round(mean_actual, 2))
       , pos = 4, cex = 0.8, col = "green"
  )
  legend("topright",
       legend = c("Predicted by the model", "Mean of Actual Values"),
       col = c("red", "cyan"), pch = 19, cex = 0.8, box.lwd = 0.5
  )
}

}

outliers_df <- df
outliers_df[,] <- 0

```

Explanatory Data Analysis

In this section, we undertake a crucial step in any data science project, particularly for modeling purposes. A detailed descriptive analysis is performed on each variable, highlighting their most relevant characteristics. We follow a column-by-column examination of the dataset, and then generate a data quality report and prepare the data for the subsequent modeling stage by addressing missing values through imputation methods, and deleting non-informative variables and creating new ones.

Note that the explanatory data analysis is assessed using exclusively the train data, even though new variables and variable removals also impact the test dataset.

Univariate Descriptive Analysis

In this initial phase, we analyze all variables in the dataset, highlighting their most significant characteristics. We also convert numeric variables that represent qualitative concepts into factors and consider creating new variables based on existing numeric ones.

Subsequently, we present a data quality report that assesses the overall quality of the data and discusses how it may affect the modeling process.

Variable *avganncount*

avganncount is a numerical variable representing the “Mean number of reported cancer cases diagnosed annually (2010-2015).” First, note that it contains 126 observations with the value “1962.667684”, which we consider inconsistent and that represent NAs. After setting this individuals as NAs, we start the variable analysis.

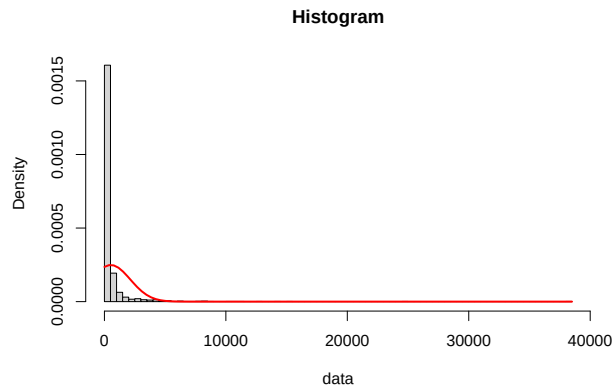
The difference between the mean (524.24) and the median (155) indicates that the distribution is not symmetric and deviates from normality. Additionally, the first and third quartiles (74 and 383, respectively) are much lower than the mean and the maximum value of 38150, further suggesting skewness. As expected, the Shapiro-Wilk normality test gets rejected. The boxplot confirms this skewness, with an extended upper whisker and numerous outliers, indicating a heavy right tail. It has 207 mild outliers and 131 severe outliers.

```
variable <- df$avganncount
l11 <- which(variable == 1962.667684 ); length(l11)
```

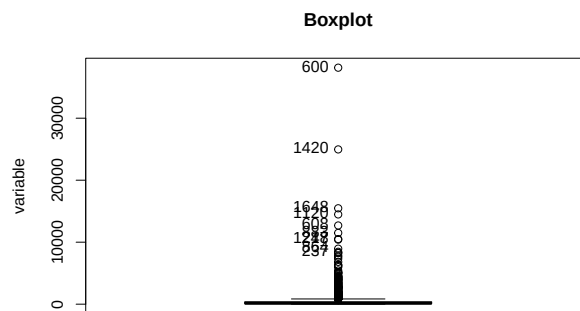
```
## [1] 126
```

```
variable[l11] <- NA
df$avganncount[l11] <- NA
res.var <- summary_and_normality_check(variable)
```

```
## [1] "Summary: "
##      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.    NA's
##       7.0   74.0   155.0   524.2   383.0 38150.0     126
## [1] "Mean"
## [1] 524.2393
## [1] "Standard deviation"
## [1] 1601.779
```



```
## [1] "Skewness"
## [1] 12.16498
```



```
## [1] "Num Outliers:"
## [1] 207
## [1] "Num. severe outliers:"
## [1] 131
## [1] "Shapiro test: "
##
## Shapiro-Wilk normality test
##
## data: data
## W = 0.2677, p-value < 2.2e-16
##
## [1] "Most repeated value ad times it occurs: "
## [1] "81"
## [1] 12
```

```
outliers_df[res.var,"avganncount"] <- 1
outliers <- which(outliers_df$avganncount == 1); length(outliers)
```

```
## [1] 207
```

Variable *avgdeathsperyear*

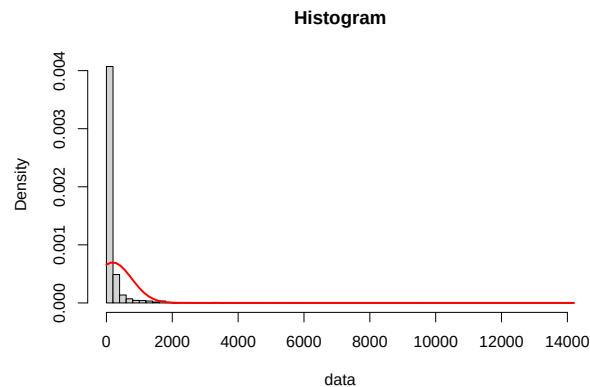
avgdeathsperyear is a numerical variable representing the “Average number of reported cancer-related mortalities per year.”

This variable is complete, with no missing values. Descriptive statistics reveal a mean of 191.6 and a median of 62, indicating a strongly non-symmetric distribution. This asymmetry is further supported by the first quartile (29) and third quartile (140.5), both significantly lower than the mean, suggesting a right-skewed distribution. Visual exploration using a boxplot confirms this skewness, displaying a pronounced right tail.

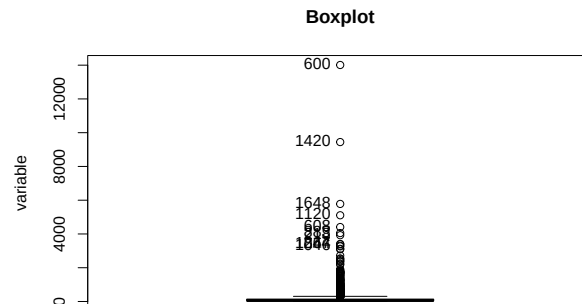
Additionally, a histogram with an overlaid normal curve underscores the substantial deviation from normality, a conclusion reinforced by the results of the Shapiro-Wilk test. The distribution exhibits a significant number of outliers, with 225 classified as mild and 138 as severe, all of which are above the third quartile. These outliers contribute to the variable's extended right tail and may warrant further investigation to understand their context and potential implications.

```
df$avgdeathsperyear <- as.numeric(df$avgdeathsperyear)
variable <- df$avgdeathsperyear
res.var <- summary_and_normality_check(variable)
```

```
## [1] "Summary: "
##      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      3.0   29.0   62.0   191.6   140.5 14010.0
## [1] "Mean"
## [1] 191.6237
## [1] "Standard deviation"
## [1] 571.8763
```



```
## [1] "Skewness"
## [1] 12.4999
```



```
## [1] "Num Outliers:"
## [1] 225
```



```
## [1] "Num. severe outliers:"
## [1] 138
## [1] "Shapiro test: "
##
## Shapiro-Wilk normality test
##
## data: data
## W = 0.26769, p-value < 2.2e-16
##
## [1] "Most repeated value ad times it occurs: "
## [1] "13"
## [1] 26

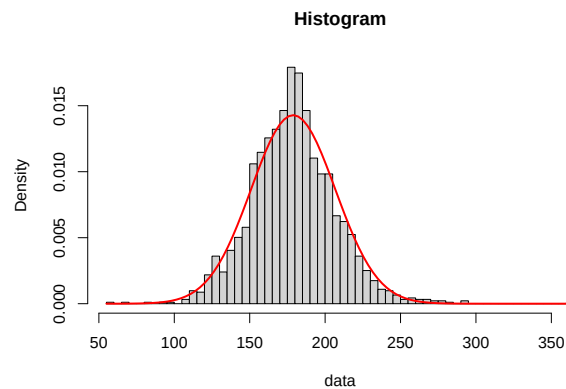
outliers_df[res.var,"avgdeathsperyear"] <- 1
```

Variable *target_deathrate* :

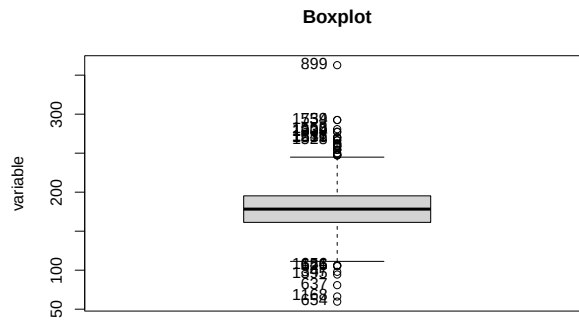
target_deathrate is the numeric target variable of the dataset. We analyze it carefully in the subsequent section **Target variable: Profiling and serial correlation**.

```
variable <- df$target_deathrate;
res.var <- summary_and_normality_check(variable)

## [1] "Summary: "
##      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      59.7  161.3   178.3   178.8   195.3   362.8
## [1] "Mean"
## [1] 178.8197
## [1] "Standard deviation"
## [1] 27.93042
```



```
## [1] "Skewness"
## [1] 0.3243216
```



```
## [1] "Num Outliers:"
## [1] 35
## [1] "Num. severe outliers:"
## [1] 1
## [1] "Shapiro test: "
##
## Shapiro-Wilk normality test
##
## data: data
## W = 0.98647, p-value = 4.149e-12
##
## [1] "Most repeated value ad times it occurs: "
## [1] "173.6"
## [1] 8
```

```
outliers_df[res.var,"target_deathrate"] <- 1
```

Variable *incidencerate* :

incidencerate is a numerical variable representing the “Mean per capita cancer diagnosis rate per 100,000 individuals.”

Anomalies were identified in the data: the same 126 individuals who had a value of “1962.667684” in the *avganncount* variable also showed a value of “453.5494221” in the *incidencerate* variable. These values are considered non-coherent, and thus, we set them as missing (NA). After setting these values to NA, we proceed with the analysis of the column’s distribution.

The mean of the variable is 448.7, the median is 448.7, and the standard deviation is 57.85, indicating moderate variability in the data. The skewness of 1.40 suggests a positively skewed distribution, with the tail extending toward higher values. This is further confirmed by the presence of 36 outliers, including 4 severe outliers, which are all above the upper threshold. The Shapiro-Wilk strongly rejects the null hypothesis of normality.

These results suggest that while the variable is mostly symmetric within its central range, the skewness and presence of outliers may have implications for modeling and analysis.

```
variable <- df$incidencerate;
l11 <- which(variable == 453.5494221 ); length(l11)
```

```
## [1] 126
```

```
variable[l11] <- NA
df$incidencerate[l11] <- NA
```



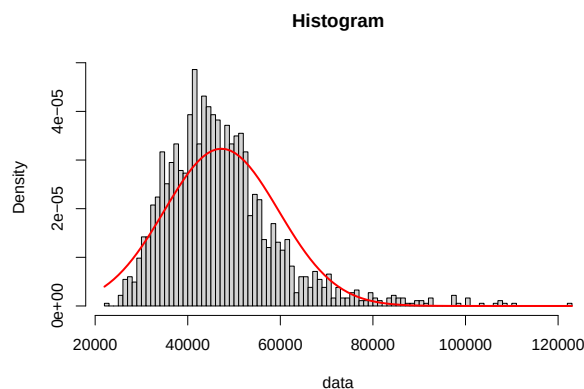
```
## [1] 7
outliers_df[res.var,"incidencerate"] <- 1
```

Variable *medincome* :

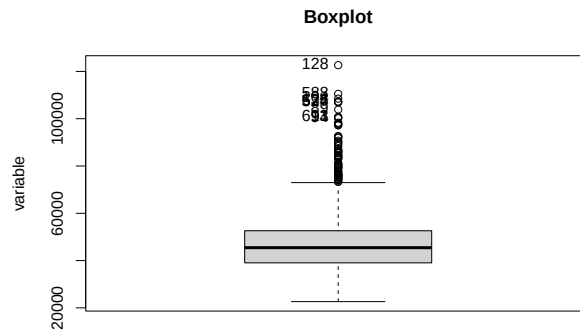
medincome is a numerical variable representing the “Median income per county.” The distribution is neither normal nor symmetric, as evidenced by the histogram, Shapiro-Wilk test, and skewness value (1.47), all indicating a right-skewed distribution with a long tail. Descriptive statistics show a minimum of 22,640, a maximum of 122,641 (Loudoun County, Virginia), and a mean of 47,278 with a standard deviation of 12,342. The median income is 45,454, with the first and third quartiles at 39,031 and 52,612, respectively. There are no missing values, but the variable contains 69 outliers, 15 of which are severe, all above the upper quartile. These findings suggest substantial variability and a pronounced concentration of counties with lower incomes compared to a smaller group with exceptionally high incomes.

```
df$medincome <- as.numeric(df$medincome)
variable <- df$medincome;
res.var <- summary_and_normality_check(variable)
```

```
## [1] "Summary: "
##      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 22640  39031   45454   47278   52612  122641
## [1] "Mean"
## [1] 47277.94
## [1] "Standard deviation"
## [1] 12341.69
```



```
## [1] "Skewness"
## [1] 1.466811
```



```
## [1] "Num Outliers:"
## [1] 69
## [1] "Num. severe outliers:"
## [1] 15
## [1] "Shapiro test: "
##
## Shapiro-Wilk normality test
##
## data: data
## W = 0.9105, p-value < 2.2e-16
##
## [1] "Most repeated value ad times it occurs: "
## [1] "34116"
## [1] 3
```

```
outliers_df[res.var,"medincome"] <- 1
```

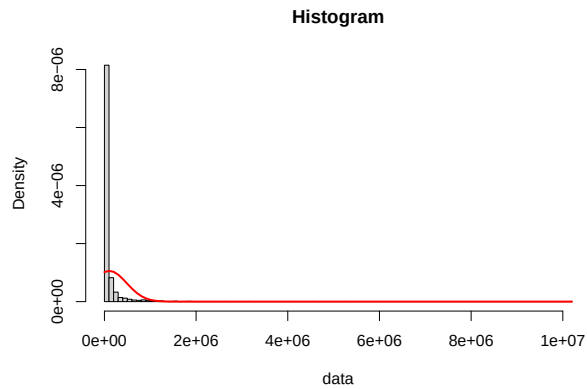
Variable *popest2015* :

popest2015 is a numerical variable representing the population of each county in 2015.

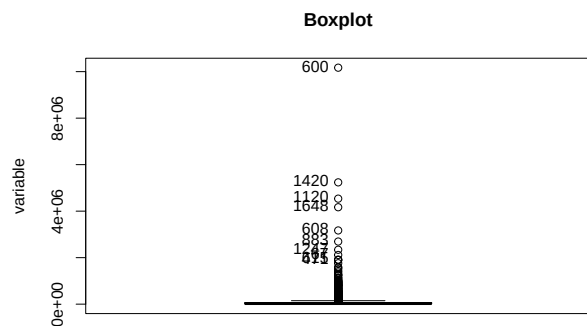
It contains no missing values but has an extensive range and an extremely skewed distribution, as highlighted by a skewness value of 14.71 and a Shapiro-Wilk test confirming strong deviation from normality. The distribution is heavily right-skewed, with a mean of 106,841 significantly exceeding the median of 27,158, and a wide spread between the 1st quartile (12,191) and 3rd quartile (66,880). The variable includes 252 mild outliers and 162 severe outliers, all exceeding the upper whisker of the boxplot. The highest value, 10,170,292, corresponds to Los Angeles County, California. The standard deviation of 376,494 reflects the substantial variability, driven by a small number of extremely populous counties. This skewed distribution highlights the vast disparity in county populations, with most counties having smaller populations and a few extreme outliers significantly influencing the overall statistics.

```
df$popest2015 <- as.numeric(df$popest2015)
variable <- df$popest2015;
res.var <- summary_and_normality_check(variable)
```

```
## [1] "Summary: "
##      Min.   1st Qu.   Median     Mean   3rd Qu.    Max.
##      829   12191    27158   106841   66880 10170292
## [1] "Mean"
## [1] 106841.2
## [1] "Standard deviation"
## [1] 376494
```



```
## [1] "Skewness"
## [1] 14.70818
```



```
## [1] "Num Outliers:"
## [1] 252
## [1] "Num. severe outliers:"
## [1] 162
## [1] "Shapiro test: "
##
## Shapiro-Wilk normality test
##
## data: data
## W = 0.22666, p-value < 2.2e-16
##
## [1] "Most repeated value ad times it occurs: "
## [1] "1426"
## [1] 2
```

```
outliers_df[res.var,"popest2015"] <- 1
```

Variable *povertypersent* :

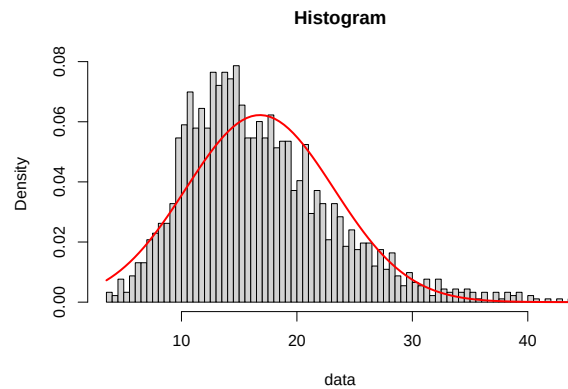
povertypersent is a numerical variable representing the percentage of the population in poverty. It has no missing values and a moderate positive skew, as indicated by the skewness value of 0.89 and the Shapiro-Wilk test, confirming the variable does not follow a normal distribution. The mean is 16.79, slightly higher than the median of 15.7, reflecting the slight skew. The 1st quartile (12.15) and 3rd quartile (20.4) show a moderate

spread, with values ranging from a minimum of 3.7 to a maximum of 44. The standard deviation is 6.41, indicating variability in poverty levels across counties.

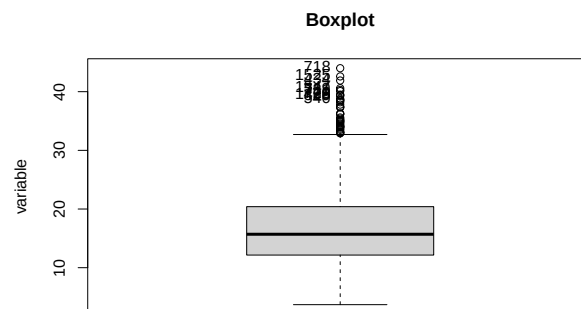
The variable has 42 mild outliers, but no severe ones. The most frequent value is 14.7, occurring 22 times. While this variable shares a non-normal distribution with other numerical variables like medincome and incidencerate, its skewness and outlier characteristics are less pronounced, indicating a more balanced distribution overall.

```
variable <- df$povertypercent;
res.var <- summary_and_normality_check(variable)
```

```
## [1] "Summary: "
##      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      3.70  12.15   15.70   16.79   20.40   44.00
## [1] "Mean"
## [1] 16.79175
## [1] "Standard deviation"
## [1] 6.411161
```



```
## [1] "Skewness"
## [1] 0.8905168
```



```
## [1] "Num Outliers:"
## [1] 42
## [1] "Num. severe outliers:"
## [1] 0
## [1] "Shapiro test: "
##
```

```
## Shapiro-Wilk normality test
##
## data: data
## W = 0.95557, p-value < 2.2e-16
##
## [1] "Most repeated value ad times it occurs: "
## [1] "14.7"
## [1] 22
outliers_df[res.var,"povertypercent"] <- 1
```

Variable *studypercap* :

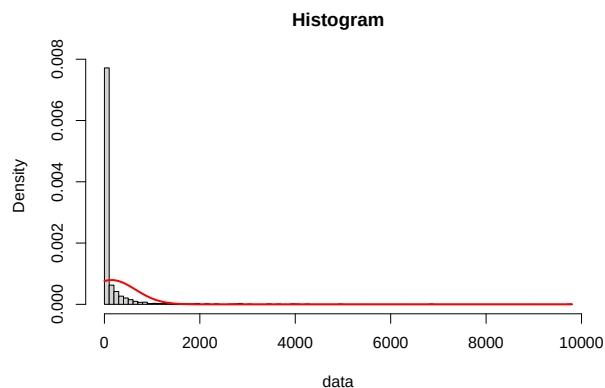
studypercap is a numerical variable representing the “Per capita number of cancer-related clinical trials per county.”

A significant feature of this variable is that 1,162 out of 1,831 counties (63.5%) have a value of 0, making this the most frequent value and dominating the distribution. Nearly all non-zero values are flagged as outliers due to their extreme deviation from the majority, with 307 identified as mild outliers, and 222 severe ones.

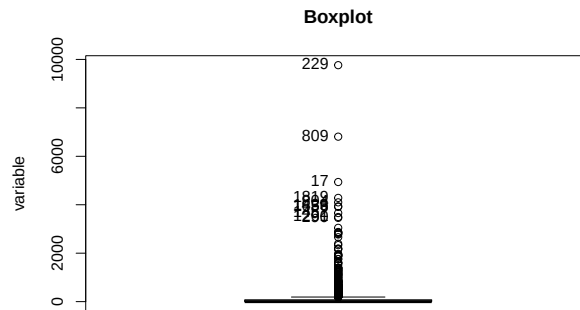
Given the predominance of zero values and the highly skewed nature, this variable may contribute more noise than predictive value to the model. Removing it from both the training and test datasets is a reasonable decision to simplify the model and enhance its performance.

```
variable <- df$studypercap;
res.var <- summary_and_normality_check(variable)
```

```
## [1] "Summary: "
##      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      0.0      0.0      0.0  148.2   76.0  9762.3
## [1] "Mean"
## [1] 148.2024
## [1] "Standard deviation"
## [1] 501.867
```



```
## [1] "Skewness"
## [1] 8.477802
```

```
## [1] "Num Outliers:"
## [1] 307
## [1] "Num. severe outliers:"
## [1] 222
## [1] "Shapiro test: "
##
## Shapiro-Wilk normality test
##
## data: data
## W = 0.30754, p-value < 2.2e-16
##
## [1] "Most repeated value ad times it occurs: "
## [1] "0"
## [1] 1162

# Delete from train, test
lltrain <- which(names(df) == "studypercap");lltrain

## [1] 8

lltest <- which(names(test) == "studypercap");lltest

## [1] 8

df <- df[,-lltrain]
test <- test[,-lltest]
```

Variable *binnedi* :

The variable *binnedi* represents median income per capita binned into deciles, categorizing counties into ten groups based on their income distribution. The frequency table shows a relatively balanced distribution across these bins, with counts ranging from 164 to 195 observations per bin. For instance, the smallest bin (37413.8, 40362.7] contains 164 observations, while the largest bin (48021.6, 51046.4] contains 195. The proportion table confirms this balance, with percentages varying narrowly from 8.96% to 10.65%. It contains no missing values.

Given its nature, *binnedi* is likely related to *medincome*, the median income per county, as both variables represent different perspectives of income distribution. While *medincome* is a continuous variable, *binnedi* simplifies analysis by discretizing income into clear categories. This potential relationship highlights the need to evaluate *binnedi* alongside *medincome* to assess redundancy or complementarity in modeling efforts.

```
variable <- df$binnedinc;
summary <- summary(variable); summary
```

```
##      Length      Class      Mode
##      1831 character character

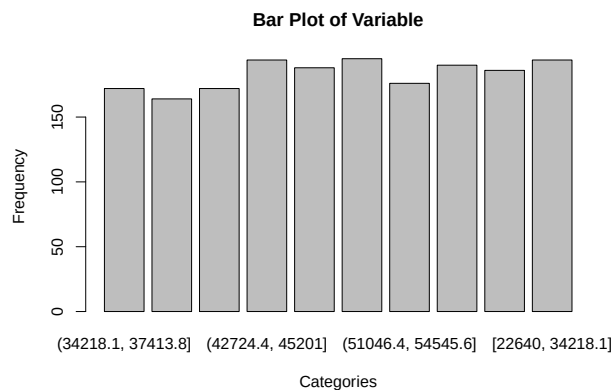
table(variable)

## variable
## (34218.1, 37413.8] (37413.8, 40362.7] (40362.7, 42724.4] (42724.4, 45201]
##              172              164              172              194
## (45201, 48021.6] (48021.6, 51046.4] (51046.4, 54545.6] (54545.6, 61494.5]
##              188              195              176              190
## (61494.5, 125635] [22640, 34218.1]
##              186              194

prop.table(table(variable))

## variable
## (34218.1, 37413.8] (37413.8, 40362.7] (40362.7, 42724.4] (42724.4, 45201]
##      0.09393774      0.08956854      0.09393774      0.10595303
## (45201, 48021.6] (48021.6, 51046.4] (51046.4, 54545.6] (54545.6, 61494.5]
##      0.10267613      0.10649918      0.09612234      0.10376843
## (61494.5, 125635] [22640, 34218.1]
##      0.10158383      0.10595303

barplot(table(variable), main="Bar Plot of Variable",
        xlab="Categories", ylab="Frequency")
```



Variable *medianage* :

medianage is a numerical variable that expresses the median age of county residents.

Note that initially, the variable contained 18 nonsensical values, ranging from 349.2 to 624, which were considered incoherent. These values were replaced with NAs, leaving the variable with no other missing values. The cleaned data shows a median age ranging from 23.3 to 59.0 years, with a mean of 40.85 and a standard deviation of 5.14. The skewness is near zero (-0.024), indicating a nearly symmetric distribution. However, the Boxplot and Shapiro-Wilk test for normality suggests the variable is not normally distributed despite its symmetry. The data contains 52 mild outliers (both above the upper bound and below the lower bound), but none are classified as severe. The most frequent value is 38.8, which occurs 29 times.

```
variable <- df$medianage
table(variable)

## variable
## 23.3 24.2 24.4 24.6 25 25.9 26.1 26.3 26.4 26.5 26.6 26.8 27.2
```

```
##      1      1      1      1      1      1      2      1      1      1      1      1      1
## 27.5 27.8 27.9      28 28.1 28.3 28.4 28.5 28.6 28.8 29.1 29.4 29.8
##      1      1      2      1      1      2      1      1      1      2      2      3      3
## 29.9      30 30.1 30.2 30.3 30.4 30.5 30.6 30.7 30.8 30.9      31 31.1
##      1      3      1      5      4      4      2      2      5      3      2      1      4
## 31.2 31.4 31.5 31.6 31.7 31.8 31.9      32 32.1 32.2 32.3 32.4 32.5
##      1      2      4      2      4      2      4      2      2      4      1      4      3
## 32.6 32.7 32.8 32.9      33 33.1 33.2 33.3 33.4 33.5 33.6 33.7 33.8
##      1      6      5      4      4      1      2      4      1      6      2      2      5
## 33.9      34 34.1 34.2 34.3 34.4 34.5 34.6 34.7 34.8 34.9      35 35.1
##      3      7      7      3      2      3      3      7      7      12      6      5      4
## 35.2 35.3 35.4 35.5 35.6 35.7 35.8 35.9      36 36.1 36.2 36.3 36.4
##      6      7      5      10      11      7      7      5      10      14      6      7      7
## 36.5 36.6 36.7 36.8 36.9      37 37.1 37.2 37.3 37.4 37.5 37.6 37.7
##      10      9      10      11      9      10      7      8      8      11      10      13      6
## 37.8 37.9      38 38.1 38.2 38.3 38.4 38.5 38.6 38.7 38.8 38.9      39
##      10      12      13      10      9      9      15      17      10      7      29      13      21
## 39.1 39.2 39.3 39.4 39.5 39.6 39.7 39.8 39.9      40 40.1 40.2 40.3
##      17      14      16      15      14      10      18      17      18      14      20      11      16
## 40.4 40.5 40.6 40.7 40.8 40.9      41 41.1 41.2 41.3 41.4 41.5 41.6
##      12      14      17      22      22      13      24      17      17      19      16      19      15
## 41.7 41.8 41.9      42 42.1 42.2 42.3 42.4 42.5 42.6 42.7 42.8 42.9
##      11      17      17      15      16      19      18      14      14      22      9      15      16
##      43 43.1 43.2 43.3 43.4 43.5 43.6 43.7 43.8 43.9      44 44.1 44.2
##      14      10      12      10      9      17      13      16      11      10      5      14      14
## 44.3 44.4 44.5 44.6 44.7 44.8 44.9      45 45.1 45.2 45.3 45.4 45.5
##      10      10      11      8      8      5      10      8      8      7      5      11      6
## 45.6 45.7 45.8 45.9      46 46.1 46.2 46.3 46.4 46.5 46.6 46.7 46.8
##      8      8      8      7      10      8      6      5      8      6      7      6      6
## 46.9      47 47.1 47.2 47.3 47.4 47.5 47.6 47.7 47.8 47.9      48 48.1
##      5      4      8      6      4      4      6      7      4      5      5      4      3
## 48.2 48.3 48.4 48.5 48.6 48.7 48.8 48.9      49 49.1 49.2 49.3 49.4
##      4      7      4      3      4      2      1      2      4      8      5      2      5
## 49.5 49.6 49.7 49.8 49.9      50 50.1 50.2 50.3 50.4 50.5 50.6 50.7
##      4      4      1      6      3      3      2      3      1      2      2      2      4
## 50.8 50.9      51 51.1 51.2 51.3 51.4 51.5 51.7 51.8 51.9      52 52.1
##      3      4      1      2      2      5      1      1      1      2      2      1      2
## 52.3 52.4 52.5 52.6 52.8 52.9 53.1 53.2 53.3 53.4 53.5 54.1 54.4
##      2      1      1      1      1      1      1      1      1      1      2      1      2
## 54.5 54.6 54.7 54.8 55.4 55.6 56.3 56.4 56.5 56.6      59 349.2 406.8
##      1      2      1      1      1      3      1      1      1      1      1      1      1
## 424.8 429.6 430.8 481.2 496.8      498 500.4 502.8 508.8 523.2 525.6 536.4      546
##      1      1      1      1      2      1      1      1      2      1      1      1      1
##      624
##      1
```

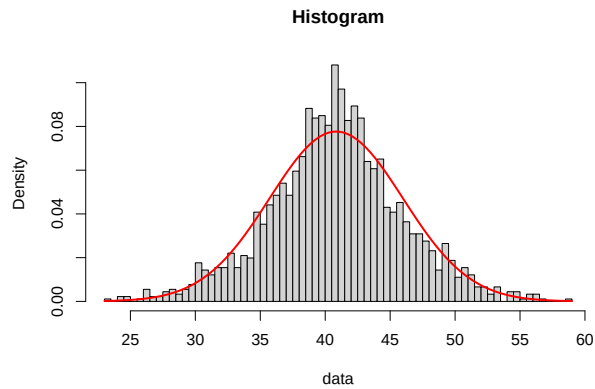
```
df[which(df$medianage > 60), c("medianage")] <- NA
summary(df$medianage)
```

```
##      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.    NA's
##    23.30   37.90   40.90   40.85   43.80   59.00      18
```

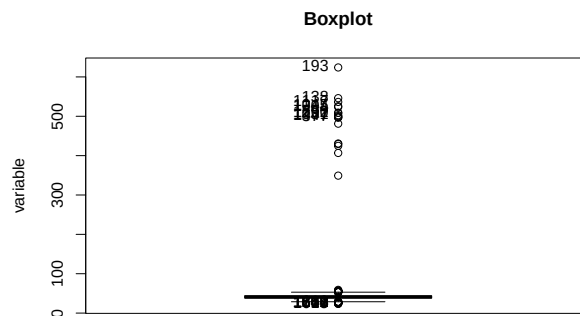
```
res.var <- summary_and_normality_check(df$medianage)
```

```
## [1] "Summary: "
```

```
##      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.     NA's
##    23.30  37.90   40.90   40.85  43.80   59.00        18
## [1] "Mean"
## [1] 40.85063
## [1] "Standard deviation"
## [1] 5.13671
```



```
## [1] "Skewness"
## [1] -0.02432015
```



```
## [1] "Num Outliers:"
## [1] 52
## [1] "Num. severe outliers:"
## [1] 0
## [1] "Shapiro test: "
##
##  Shapiro-Wilk normality test
##
## data:  data
## W = 0.9951, p-value = 1.161e-05
##
## [1] "Most repeated value ad times it occurs: "
## [1] "38.8"
## [1] 29
```

```
outliers_df[res.var,"medianage"] <- 1
```

medianagemale represents the median age of male county residents.

Comparing *medianagemale* with *medianage*, the two variables exhibit similar distributions and characteristics, which is expected as *medianage* includes both genders.

```
## [1] "Summary: "
```

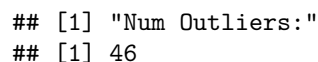
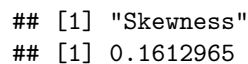
##	Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
##	23.00	36.40	39.50	39.59	42.60	60.20

```
## [1] "Mean"
```

```
## [1] 39.58564
```

```
## [1] "Standard deviation"
```

```
## [1] 5.188667
```



```
## [1] "Num. severe outliers:"
## [1] 0
## [1] "Shapiro test: "
##
## Shapiro-Wilk normality test
##
## data: data
## W = 0.99404, p-value = 9.877e-07
##
## [1] "Most repeated value ad times it occurs: "
## [1] "39.2"
## [1] 23
```

```
table(variable)
```

```
## variable
## 23 24 24.3 24.4 24.6 24.8 25 25.5 25.6 26.1 26.2 26.3 26.4 26.5 26.6 26.8
## 1 1 1 1 1 1 1 1 1 1 2 1 1 1 2 2
## 27.4 27.5 27.6 27.8 28 28.1 28.4 28.5 28.8 28.9 29 29.1 29.2 29.3 29.4 29.5
## 1 2 2 3 2 1 3 2 2 1 1 5 1 4 1 6
## 29.6 29.7 29.8 29.9 30.2 30.3 30.4 30.5 30.6 30.7 30.8 30.9 31 31.1 31.2 31.3
## 3 2 1 4 1 1 3 9 3 5 2 2 6 5 1 2
## 31.4 31.5 31.6 31.7 31.8 31.9 32 32.1 32.2 32.3 32.4 32.5 32.6 32.7 32.8 32.9
## 2 3 3 2 5 4 4 1 4 4 6 5 6 2 3 3
## 33 33.1 33.2 33.3 33.4 33.5 33.6 33.7 33.8 33.9 34 34.1 34.2 34.3 34.4 34.5
## 1 7 9 6 7 10 2 3 12 5 6 8 9 7 9 7
## 34.6 34.7 34.8 34.9 35 35.1 35.2 35.3 35.4 35.5 35.6 35.7 35.8 35.9 36 36.1
## 4 12 5 10 9 7 12 17 6 14 19 11 8 15 6 14
## 36.2 36.3 36.4 36.5 36.6 36.7 36.8 36.9 37 37.1 37.2 37.3 37.4 37.5 37.6 37.7
## 7 9 9 14 16 11 11 11 9 16 14 12 19 12 15 10
## 37.8 37.9 38 38.1 38.2 38.3 38.4 38.5 38.6 38.7 38.8 38.9 39 39.1 39.2 39.3
## 14 14 15 16 17 14 12 22 13 14 20 22 15 22 23 15
## 39.4 39.5 39.6 39.7 39.8 39.9 40 40.1 40.2 40.3 40.4 40.5 40.6 40.7 40.8 40.9
## 12 21 15 11 8 16 14 15 11 22 20 19 20 15 19 18
## 41 41.1 41.2 41.3 41.4 41.5 41.6 41.7 41.8 41.9 42 42.1 42.2 42.3 42.4 42.5
## 10 14 16 16 16 15 14 8 13 10 11 23 9 14 18 6
## 42.6 42.7 42.8 42.9 43 43.1 43.2 43.3 43.4 43.5 43.6 43.7 43.8 43.9 44 44.1
## 10 11 13 9 11 16 9 9 8 10 6 9 3 1 7 7
## 44.2 44.3 44.4 44.5 44.6 44.7 44.8 44.9 45 45.1 45.2 45.3 45.4 45.5 45.6 45.7
## 14 7 8 9 7 1 7 10 5 8 2 9 10 9 9 4
## 45.8 45.9 46 46.1 46.2 46.3 46.4 46.5 46.6 46.7 46.8 46.9 47 47.1 47.2 47.3
## 6 3 4 8 7 4 4 3 3 7 7 5 4 4 1 2
## 47.4 47.5 47.6 47.7 47.8 47.9 48 48.1 48.2 48.3 48.4 48.5 48.6 48.7 48.8 48.9
## 1 2 3 4 6 4 4 2 5 1 1 4 2 5 2 3
## 49 49.1 49.2 49.3 49.4 49.6 49.7 49.9 50 50.1 50.2 50.3 50.4 50.5 50.6 50.7
## 2 2 2 1 3 3 1 2 3 2 1 2 1 3 4 4
## 50.8 50.9 51.1 51.2 51.4 51.5 51.6 51.7 51.8 51.9 52 52.1 52.2 52.3 52.4 52.8
## 3 1 1 5 1 1 1 1 1 2 2 1 1 1 2 1
## 53.2 53.4 53.5 53.8 53.9 54 54.3 54.4 54.7 54.8 55.5 55.8 56.5 58.5 58.6 60.2
## 2 1 1 1 1 2 2 1 1 1 1 1 1 1 1 1
```

```
outliers_df[res.var,"medianagemale"] <- 1
```

Variable *medianagefemale* :

medianagefemale represents the median age of female county residents.

Its values range from 23.6 to 58.2 years. All entries are valid, showing no nonsensical or missing values. The average median age is 42.17 years, slightly higher than that of males, with a median of 42.40 years and a standard deviation of 5.20 years. The data is slightly skewed to the left, as indicated by a skewness of -0.206, and the Shapiro-Wilk test confirms a significant deviation from normality. There are 55 mild outliers.

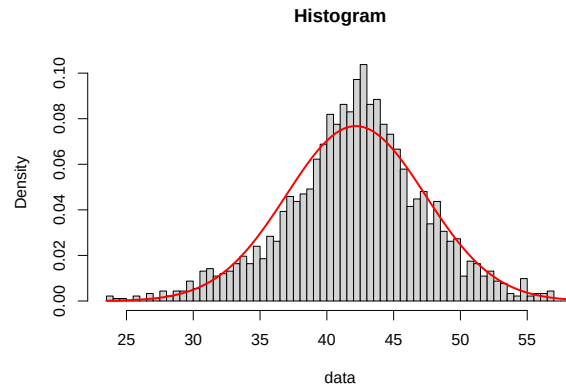
```
variable <- df$medianagefemale
table(variable)
```

```
## variable
## 23.6 23.9 24.5 24.6 25.8 26 26.6 26.7 27 27.3 27.7 27.9 28 28.2 28.3 28.6
## 1 1 1 1 1 1 1 1 1 1 1 2 1 1 1 1
## 28.8 29 29.3 29.6 29.7 29.9 30 30.1 30.3 30.4 30.6 30.7 30.8 30.9 31 31.1
## 2 1 4 3 3 1 1 2 1 2 1 2 2 3 4 3
## 31.2 31.3 31.4 31.5 31.6 31.7 31.8 31.9 32 32.1 32.2 32.3 32.4 32.5 32.6 32.7
## 3 2 4 1 1 3 4 1 1 2 1 3 2 3 2 2
## 32.8 32.9 33 33.2 33.3 33.4 33.5 33.6 33.8 33.9 34 34.1 34.2 34.3 34.4 34.5
## 2 4 2 7 1 5 2 6 3 3 6 3 4 5 2 1
## 34.6 34.7 34.8 34.9 35 35.1 35.2 35.3 35.4 35.5 35.6 35.7 35.8 35.9 36 36.1
## 3 6 3 2 8 3 4 2 4 4 6 8 3 5 4 8
## 36.2 36.3 36.4 36.5 36.6 36.7 36.8 36.9 37 37.1 37.2 37.3 37.4 37.5 37.6 37.7
## 4 2 4 6 9 4 9 6 8 7 10 11 6 8 8 8
## 37.8 37.9 38 38.1 38.2 38.3 38.4 38.5 38.6 38.7 38.8 38.9 39 39.1 39.2 39.3
## 8 11 5 9 5 12 9 8 13 11 6 6 9 13 9 19
## 39.4 39.5 39.6 39.7 39.8 39.9 40 40.1 40.2 40.3 40.4 40.5 40.6 40.7 40.8 40.9
## 10 6 17 17 10 3 16 8 15 15 21 16 14 15 16 10
## 41 41.1 41.2 41.3 41.4 41.5 41.6 41.7 41.8 41.9 42 42.1 42.2 42.3 42.4 42.5
## 16 18 15 11 19 16 12 13 17 21 13 13 21 17 21 17
## 42.6 42.7 42.8 42.9 43 43.1 43.2 43.3 43.4 43.5 43.6 43.7 43.8 43.9 44 44.1
## 14 19 17 31 14 11 12 19 19 18 14 19 18 9 21 12
## 44.2 44.3 44.4 44.5 44.6 44.7 44.8 44.9 45 45.1 45.2 45.3 45.4 45.5 45.6 45.7
## 15 10 10 24 12 13 14 7 21 11 16 11 8 15 15 8
## 45.8 45.9 46 46.1 46.2 46.3 46.4 46.5 46.6 46.7 46.8 46.9 47 47.1 47.2 47.3
## 6 12 12 11 12 3 7 5 8 10 8 11 4 11 7 11
## 47.4 47.5 47.6 47.7 47.8 47.9 48 48.1 48.2 48.3 48.4 48.5 48.6 48.7 48.8 48.9
## 7 8 8 3 9 6 5 4 6 10 11 9 10 5 3 3
## 49 49.2 49.3 49.4 49.5 49.6 49.7 49.8 49.9 50 50.1 50.2 50.4 50.5 50.6 50.8
## 7 9 2 9 4 6 5 7 5 2 1 3 2 4 4 2
## 50.9 51 51.1 51.2 51.3 51.4 51.5 51.6 51.7 51.9 52 52.1 52.2 52.4 52.5 52.6
## 7 3 2 1 8 3 1 4 1 2 3 3 3 3 3 3
## 52.7 52.8 53.1 53.2 53.4 53.5 53.8 54 54.4 54.5 54.6 54.8 54.9 55 55.3 55.4
## 2 3 2 2 1 2 1 2 1 1 3 1 3 2 1 1
## 55.6 56.1 56.4 56.5 56.7 56.8 58.2
## 3 1 1 1 2 2 1
```

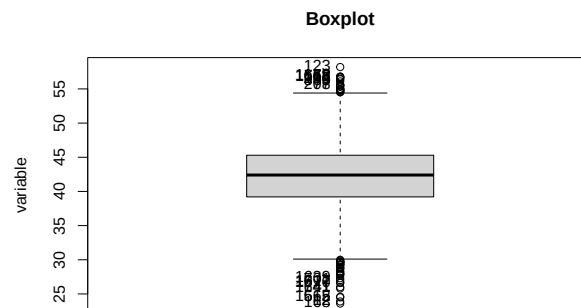
```
res.var <- summary_and_normality_check(variable)
```

```
## [1] "Summary: "
## Min. 1st Qu. Median Mean 3rd Qu. Max.
## 23.60 39.20 42.40 42.17 45.30 58.20
## [1] "Mean"
## [1] 42.16931
## [1] "Standard deviation"
```

```
## [1] 5.196007
```



```
## [1] "Skewness"
## [1] -0.2057034
```



```
## [1] "Num Outliers:"
## [1] 55
## [1] "Num. severe outliers:"
## [1] 0
## [1] "Shapiro test: "
##
## Shapiro-Wilk normality test
##
## data: data
## W = 0.99321, p-value = 1.817e-07
##
## [1] "Most repeated value ad times it occurs: "
## [1] "42.9"
## [1] 31
```

```
outliers_df[res.var,"medianagefemale"] <- 1
```

Variable *geography* :

The variable *geography* is a categorical variable that specifies the name of the county for each observation. It contains no missing values and serves as a unique identifier for each row, as all values are distinct. Additionally, it includes the name of the state to which each county belongs.


```
variable <- df$geography;
table(variable)
```

```
## variable
##          Acadia Parish, Louisiana          Ada County, Idaho
##                      1                      1
##          Adair County, Kentucky          Adair County, Missouri
##                      1                      1
##          Adams County, Idaho          Adams County, Illinois
##                      1                      1
##          Adams County, Iowa          Adams County, Mississippi
##                      1                      1
##          Adams County, Nebraska          Adams County, North Dakota
##                      1                      1
##          Adams County, Ohio          Aiken County, South Carolina
##                      1                      1
##          Alachua County, Florida          Albany County, Wyoming
##                      1                      1
##          Albemarle County, Virginia          Alcona County, Michigan
##                      1                      1
##          Alcorn County, Mississippi          Aleutians West Census Area, Alaska
##                      1                      1
##          Alexander County, Illinois          Alfalfa County, Oklahoma
##                      1                      1
##          Alger County, Michigan          Allegan County, Michigan
##                      1                      1
##          Allegany County, New York          Alleghany County, Virginia
##                      1                      1
##          Allen County, Kansas          Allen Parish, Louisiana
##                      1                      1
##          Alpena County, Michigan          Amador County, California
##                      1                      1
##          Amelia County, Virginia          Amite County, Mississippi
##                      1                      1
##          Anchorage Municipality, Alaska          Anderson County, Kentucky
##                      1                      1
##          Anderson County, South Carolina          Andrew County, Missouri
##                      1                      1
##          Androscoggin County, Maine          Anne Arundel County, Maryland
##                      1                      1
##          Anoka County, Minnesota          Anson County, North Carolina
##                      1                      1
##          Antelope County, Nebraska          Antrim County, Michigan
##                      1                      1
##          Appanoose County, Iowa          Appling County, Georgia
##                      1                      1
##          Arapahoe County, Colorado          Archer County, Texas
##                      1                      1
##          Archuleta County, Colorado          Arenac County, Michigan
##                      1                      1
##          Arlington County, Virginia          Armstrong County, Pennsylvania
##                      1                      1
##          Armstrong County, Texas          Aroostook County, Maine
##                      1                      1
```

##	Ascension Parish, Louisiana	Ashe County, North Carolina
##	1	1
##	Ashland County, Ohio	Ashtabula County, Ohio
##	1	1
##	Assumption Parish, Louisiana	Atchison County, Kansas
##	1	1
##	Atchison County, Missouri	Athens County, Ohio
##	1	1
##	Atlantic County, New Jersey	Atoka County, Oklahoma
##	1	1
##	Attala County, Mississippi	Audubon County, Iowa
##	1	1
##	Augusta County, Virginia	Aurora County, South Dakota
##	1	1
##	Austin County, Texas	Avoyelles Parish, Louisiana
##	1	1
##	Baca County, Colorado	Bacon County, Georgia
##	1	1
##	Baker County, Florida	Baker County, Georgia
##	1	1
##	Baker County, Oregon	Baltimore city, Maryland
##	1	1
##	Baltimore County, Maryland	Bamberg County, South Carolina
##	1	1
##	Banks County, Georgia	Bannock County, Idaho
##	1	1
##	Barbour County, Alabama	Barbour County, West Virginia
##	1	1
##	Barnes County, North Dakota	Barren County, Kentucky
##	1	1
##	Barron County, Wisconsin	Barrow County, Georgia
##	1	1
##	Barry County, Michigan	Barton County, Missouri
##	1	1
##	Bastrop County, Texas	Bates County, Missouri
##	1	1
##	Bath County, Virginia	Bayfield County, Wisconsin
##	1	1
##	Beadle County, South Dakota	Bear Lake County, Idaho
##	1	1
##	Beaufort County, North Carolina	Beaufort County, South Carolina
##	1	1
##	Beaver County, Oklahoma	Beaver County, Utah
##	1	1
##	Beaverhead County, Montana	Bedford County, Tennessee
##	1	1
##	Bee County, Texas	Belknap County, New Hampshire
##	1	1
##	Bell County, Texas	Beltrami County, Minnesota
##	1	1
##	Benewah County, Idaho	Bennington County, Vermont
##	1	1
##	Benson County, North Dakota	Benton County, Indiana
##	1	1

##	Benton County, Minnesota	Benton County, Mississippi
##	1	1
##	Benton County, Missouri	Benton County, Oregon
##	1	1
##	Benton County, Tennessee	Benton County, Washington
##	1	1
##	Berkeley County, South Carolina	Berkeley County, West Virginia
##	1	1
##	Berkshire County, Massachusetts	Bernalillo County, New Mexico
##	1	1
##	Berrien County, Georgia	Berrien County, Michigan
##	1	1
##	Bertie County, North Carolina	Bexar County, Texas
##	1	1
##	Bibb County, Georgia	Bienville Parish, Louisiana
##	1	1
##	Big Horn County, Montana	Big Horn County, Wyoming
##	1	1
##	Black Hawk County, Iowa	Blaine County, Idaho
##	1	1
##	Blaine County, Oklahoma	Blair County, Pennsylvania
##	1	1
##	Blanco County, Texas	Blue Earth County, Minnesota
##	1	1
##	Bolivar County, Mississippi	Bon Homme County, South Dakota
##	1	1
##	Bonner County, Idaho	Boone County, Arkansas
##	1	1
##	Boone County, Illinois	Boone County, Indiana
##	1	1
##	Bosque County, Texas	Botetourt County, Virginia
##	1	1
##	Bottineau County, North Dakota	Bourbon County, Kansas
##	1	1
##	Bowman County, North Dakota	Box Elder County, Utah
##	1	1
##	Boyd County, Kentucky	Boyle County, Kentucky
##	1	1
##	Bradford County, Pennsylvania	Bradley County, Arkansas
##	1	1
##	Bradley County, Tennessee	Brantley County, Georgia
##	1	1
##	Braxton County, West Virginia	Breathitt County, Kentucky
##	1	1
##	Breckinridge County, Kentucky	Bremer County, Iowa
##	1	1
##	Brevard County, Florida	Bristol County, Rhode Island
##	1	1
##	Broadwater County, Montana	Brookings County, South Dakota
##	1	1
##	Brooks County, Georgia	Brooks County, Texas
##	1	1
##	Broome County, New York	Broomfield County, Colorado
##	1	1

##	Broward County, Florida	1	Brown County, Kansas	1
##		1		1
##	Brown County, Ohio	1	Brown County, Texas	1
##		1		1
##	Brule County, South Dakota	1	Brunswick County, North Carolina	1
##		1		1
##	Brunswick County, Virginia	1	Bryan County, Oklahoma	1
##		1		1
##	Buchanan County, Iowa	1	Buchanan County, Missouri	1
##		1		1
##	Buchanan County, Virginia	1	Buckingham County, Virginia	1
##		1		1
##	Buena Vista County, Iowa	1	Buffalo County, Wisconsin	1
##		1		1
##	Bullitt County, Kentucky	1	Bulloch County, Georgia	1
##		1		1
##	Bullock County, Alabama	1	Buncombe County, North Carolina	1
##		1		1
##	Burke County, Georgia	1	Burke County, North Dakota	1
##		1		1
##	Burleson County, Texas	1	Burlington County, New Jersey	1
##		1		1
##	Burnet County, Texas	1	Burnett County, Wisconsin	1
##		1		1
##	Burt County, Nebraska	1	Butler County, Ohio	1
##		1		1
##	Butte County, South Dakota	1	Cabarrus County, North Carolina	1
##		1		1
##	Caddo County, Oklahoma	1	Calaveras County, California	1
##		1		1
##	Calcasieu Parish, Louisiana	1	Caldwell County, Kentucky	1
##		1		1
##	Caldwell County, Missouri	1	Caldwell County, North Carolina	1
##		1		1
##	Caldwell County, Texas	1	Caldwell Parish, Louisiana	1
##		1		1
##	Calhoun County, Arkansas	1	Calhoun County, Georgia	1
##		1		1
##	Calhoun County, Iowa	1	Calhoun County, Michigan	1
##		1		1
##	Calhoun County, South Carolina	1	Calhoun County, West Virginia	1
##		1		1
##	Callahan County, Texas	1	Calumet County, Wisconsin	1
##		1		1
##	Camden County, Missouri	1	Camden County, North Carolina	1
##		1		1
##	Cameron County, Pennsylvania	1	Cameron Parish, Louisiana	1
##		1		1
##	Campbell County, Kentucky	1	Campbell County, Virginia	1
##		1		1
##	Campbell County, Wyoming	1	Canadian County, Oklahoma	1
##		1		1
##	Cannon County, Tennessee	1	Cape Girardeau County, Missouri	1
##		1		1

##	Cape May County, New Jersey	Carbon County, Montana
##	1	1
##	Carbon County, Utah	Carbon County, Wyoming
##	1	1
##	Carlisle County, Kentucky	Carlton County, Minnesota
##	1	1
##	Caroline County, Virginia	Carroll County, Georgia
##	1	1
##	Carroll County, Illinois	Carroll County, Indiana
##	1	1
##	Carroll County, Iowa	Carroll County, Kentucky
##	1	1
##	Carroll County, Maryland	Carroll County, New Hampshire
##	1	1
##	Carroll County, Ohio	Carroll County, Tennessee
##	1	1
##	Carroll County, Virginia	Carson City, Nevada
##	1	1
##	Carson County, Texas	Carter County, Kentucky
##	1	1
##	Carter County, Tennessee	Carver County, Minnesota
##	1	1
##	Casey County, Kentucky	Cass County, Iowa
##	1	1
##	Cass County, Michigan	Cass County, Missouri
##	1	1
##	Cass County, Nebraska	Cass County, North Dakota
##	1	1
##	Cass County, Texas	Cassia County, Idaho
##	1	1
##	Castro County, Texas	Caswell County, North Carolina
##	1	1
##	Catahoula Parish, Louisiana	Catawba County, North Carolina
##	1	1
##	Catoosa County, Georgia	Cattaraugus County, New York
##	1	1
##	Cavalier County, North Dakota	Cayuga County, New York
##	1	1
##	Cedar County, Iowa	Cedar County, Missouri
##	1	1
##	Cedar County, Nebraska	Centre County, Pennsylvania
##	1	1
##	Cerro Gordo County, Iowa	Chaffee County, Colorado
##	1	1
##	Chambers County, Alabama	Chambers County, Texas
##	1	1
##	Champaign County, Ohio	Chariton County, Missouri
##	1	1
##	Charles Mix County, South Dakota	Charleston County, South Carolina
##	1	1
##	Charlotte County, Virginia	Charlton County, Georgia
##	1	1
##	Chase County, Kansas	Chatham County, Georgia
##	1	1

##	Chattahoochee County, Georgia	Chattooga County, Georgia
##	1	1
##	Cheatham County, Tennessee	Cheboygan County, Michigan
##	1	1
##	Chelan County, Washington	Chemung County, New York
##	1	1
##	Chenango County, New York	Cherokee County, Iowa
##	1	1
##	Cherokee County, North Carolina	Cherokee County, Oklahoma
##	1	1
##	Cherokee County, South Carolina	Chester County, Pennsylvania
##	1	1
##	Chester County, South Carolina	Chester County, Tennessee
##	1	1
##	Chesterfield County, South Carolina	Cheyenne County, Colorado
##	1	1
##	Chickasaw County, Iowa	Chickasaw County, Mississippi
##	1	1
##	Childress County, Texas	Chilton County, Alabama
##	1	1
##	Chippewa County, Michigan	Chippewa County, Minnesota
##	1	1
##	Chisago County, Minnesota	Chittenden County, Vermont
##	1	1
##	Choctaw County, Oklahoma	Chouteau County, Montana
##	1	1
##	Churchill County, Nevada	Cibola County, New Mexico
##	1	1
##	Cimarron County, Oklahoma	Citrus County, Florida
##	1	1
##	Claiborne County, Mississippi	Claiborne County, Tennessee
##	1	1
##	Clallam County, Washington	Clare County, Michigan
##	1	1
##	Clarendon County, South Carolina	Clark County, Missouri
##	1	1
##	Clark County, Nevada	Clark County, South Dakota
##	1	1
##	Clark County, Washington	Clark County, Wisconsin
##	1	1
##	Clarke County, Georgia	Clarke County, Mississippi
##	1	1
##	Clay County, Florida	Clay County, Georgia
##	1	1
##	Clay County, Illinois	Clay County, Iowa
##	1	1
##	Clay County, Kansas	Clay County, Kentucky
##	1	1
##	Clay County, Minnesota	Clay County, Mississippi
##	1	1
##	Clay County, Nebraska	Clay County, North Carolina
##	1	1
##	Clay County, Tennessee	Clay County, Texas
##	1	1

##	Clay County, West Virginia	1	Clayton County, Georgia	1
##	Clayton County, Iowa	1	Clearfield County, Pennsylvania	1
##	Cleburne County, Alabama	1	Cleburne County, Arkansas	1
##	Cleveland County, North Carolina	1	Clinch County, Georgia	1
##	Clinton County, Iowa	1	Clinton County, Michigan	1
##	Clinton County, New York	1	Clinton County, Pennsylvania	1
##	Cloud County, Kansas	1	Coahoma County, Mississippi	1
##	Cochise County, Arizona	1	Cochran County, Texas	1
##	Cocke County, Tennessee	1	Coconino County, Arizona	1
##	Coffee County, Alabama	1	Coffee County, Georgia	1
##	Coffee County, Tennessee	1	Coffey County, Kansas	1
##	Colbert County, Alabama	1	Colfax County, Nebraska	1
##	Colfax County, New Mexico	1	Collier County, Florida	1
##	Colonial Heights city, Virginia	1	Colquitt County, Georgia	1
##	Columbia County, Georgia	1	Columbia County, New York	1
##	Columbia County, Oregon	1	Columbia County, Washington	1
##	Columbia County, Wisconsin	1	Columbiana County, Ohio	1
##	Columbus County, North Carolina	1	Colusa County, California	1
##	Comal County, Texas	1	Concho County, Texas	1
##	Concordia Parish, Louisiana	1	Contra Costa County, California	1
##	Conway County, Arkansas	1	Cook County, Georgia	1
##	Cook County, Illinois	1	Cook County, Minnesota	1
##	Cooke County, Texas	1	Cooper County, Missouri	1
##	Coos County, New Hampshire	1	Coosa County, Alabama	1
##	Corson County, South Dakota	1	Cortland County, New York	1
##	Coryell County, Texas	1	Cottle County, Texas	1
##	Covington County, Mississippi	1	Cowley County, Kansas	1

##	Cowlitz County, Washington	Crane County, Texas
##	1	1
##	Craven County, North Carolina	Crawford County, Arkansas
##	1	1
##	Crawford County, Georgia	Crawford County, Indiana
##	1	1
##	Crawford County, Iowa	Crawford County, Michigan
##	1	1
##	Crawford County, Ohio	Crawford County, Wisconsin
##	1	1
##	Crenshaw County, Alabama	Crisp County, Georgia
##	1	1
##	Crittenden County, Arkansas	Crittenden County, Kentucky
##	1	1
##	Crockett County, Tennessee	Crockett County, Texas
##	1	1
##	Cross County, Arkansas	Crow Wing County, Minnesota
##	1	1
##	Cullman County, Alabama	Culpeper County, Virginia
##	1	1
##	Cumberland County, Kentucky	Cumberland County, Maine
##	1	1
##	Cumberland County, Tennessee	Cuming County, Nebraska
##	1	1
##	Curry County, New Mexico	Curry County, Oregon
##	1	1
##	Custer County, Colorado	Custer County, Oklahoma
##	1	1
##	Cuyahoga County, Ohio	Dade County, Georgia
##	1	1
##	Dade County, Missouri	Dakota County, Minnesota
##	1	1
##	Dale County, Alabama	Dallam County, Texas
##	1	1
##	Dallas County, Alabama	Dallas County, Iowa
##	1	1
##	Dallas County, Missouri	Dane County, Wisconsin
##	1	1
##	Danville city, Virginia	Dare County, North Carolina
##	1	1
##	Darlington County, South Carolina	Davidson County, North Carolina
##	1	1
##	Daviess County, Indiana	Daviess County, Kentucky
##	1	1
##	Daviess County, Missouri	Davis County, Utah
##	1	1
##	Davison County, South Dakota	Dawes County, Nebraska
##	1	1
##	Dawson County, Montana	Dawson County, Nebraska
##	1	1
##	Dawson County, Texas	Day County, South Dakota
##	1	1
##	De Baca County, New Mexico	De Soto Parish, Louisiana
##	1	1

##	Deaf Smith County, Texas	Dearborn County, Indiana
##	1	1
##	Decatur County, Indiana	Decatur County, Iowa
##	1	1
##	Decatur County, Kansas	Decatur County, Tennessee
##	1	1
##	Defiance County, Ohio	DeKalb County, Alabama
##	1	1
##	DeKalb County, Georgia	DeKalb County, Illinois
##	1	1
##	DeKalb County, Indiana	DeKalb County, Missouri
##	1	1
##	Del Norte County, California	Delaware County, Iowa
##	1	1
##	Delaware County, Ohio	Delaware County, Oklahoma
##	1	1
##	Delaware County, Pennsylvania	Delta County, Michigan
##	1	1
##	Delta County, Texas	Dent County, Missouri
##	1	1
##	Denver County, Colorado	Des Moines County, Iowa
##	1	1
##	Deschutes County, Oregon	Desha County, Arkansas
##	1	1
##	Deuel County, South Dakota	Dewey County, Oklahoma
##	1	1
##	Dewey County, South Dakota	Dickenson County, Virginia
##	1	1
##	Dickey County, North Dakota	Dickinson County, Michigan
##	1	1
##	Dickson County, Tennessee	Dillon County, South Carolina
##	1	1
##	Dinwiddie County, Virginia	Dixie County, Florida
##	1	1
##	Dixon County, Nebraska	Doddridge County, West Virginia
##	1	1
##	Dodge County, Minnesota	Dodge County, Nebraska
##	1	1
##	Dodge County, Wisconsin	Dolores County, Colorado
##	1	1
##	Doña Ana County, New Mexico	Doniphan County, Kansas
##	1	1
##	Donley County, Texas	Dooly County, Georgia
##	1	1
##	Dorchester County, South Carolina	Dougherty County, Georgia
##	1	1
##	Douglas County, Colorado	Douglas County, Illinois
##	1	1
##	Douglas County, Kansas	Douglas County, Minnesota
##	1	1
##	Douglas County, Missouri	Douglas County, Nebraska
##	1	1
##	Douglas County, Nevada	Douglas County, Washington
##	1	1

##	Drew County, Arkansas	Dubois County, Indiana
##	1	1
##	Dubuque County, Iowa	Duchesne County, Utah
##	1	1
##	Dukes County, Massachusetts	Dundy County, Nebraska
##	1	1
##	Dunklin County, Missouri	Dunn County, North Dakota
##	1	1
##	Dunn County, Wisconsin	Duplin County, North Carolina
##	1	1
##	Dutchess County, New York	Duval County, Florida
##	1	1
##	Duval County, Texas	Dyer County, Tennessee
##	1	1
##	Eagle County, Colorado	Early County, Georgia
##	1	1
##	East Baton Rouge Parish, Louisiana	East Feliciana Parish, Louisiana
##	1	1
##	Eastland County, Texas	Echols County, Georgia
##	1	1
##	Ector County, Texas	Eddy County, North Dakota
##	1	1
##	Edwards County, Kansas	Effingham County, Georgia
##	1	1
##	El Dorado County, California	El Paso County, Texas
##	1	1
##	Elbert County, Georgia	Elk County, Pennsylvania
##	1	1
##	Elkhart County, Indiana	Elko County, Nevada
##	1	1
##	Elliott County, Kentucky	Elmore County, Alabama
##	1	1
##	Elmore County, Idaho	Emanuel County, Georgia
##	1	1
##	Emery County, Utah	Emmet County, Michigan
##	1	1
##	Emmons County, North Dakota	Erath County, Texas
##	1	1
##	Erie County, New York	Erie County, Pennsylvania
##	1	1
##	Escambia County, Florida	Esmeralda County, Nevada
##	1	1
##	Essex County, Massachusetts	Essex County, New York
##	1	1
##	Etowah County, Alabama	Eureka County, Nevada
##	1	1
##	Fairbanks North Star Borough, Alaska	Fairfax County, Virginia
##	1	1
##	Fairfield County, Connecticut	Fairfield County, Ohio
##	1	1
##	Fairfield County, South Carolina	Fallon County, Montana
##	1	1
##	Fannin County, Georgia	Fannin County, Texas
##	1	1

##	Faribault County, Minnesota	Faulk County, South Dakota
##	1	1
##	Faulkner County, Arkansas	Fauquier County, Virginia
##	1	1
##	Fayette County, Indiana	Fayette County, Kentucky
##	1	1
##	Fayette County, Ohio	Fayette County, Pennsylvania
##	1	1
##	Fayette County, Tennessee	Fentress County, Tennessee
##	1	1
##	Finney County, Kansas	Fisher County, Texas
##	1	1
##	Flagler County, Florida	Fleming County, Kentucky
##	1	1
##	Florence County, Wisconsin	Floyd County, Indiana
##	1	1
##	Floyd County, Kentucky	Floyd County, Virginia
##	1	1
##	Fluvanna County, Virginia	Ford County, Illinois
##	1	1
##	Ford County, Kansas	Forsyth County, Georgia
##	1	1
##	Foster County, North Dakota	Franklin city, Virginia
##	1	1
##	Franklin County, Florida	Franklin County, Georgia
##	1	1
##	Franklin County, Illinois	Franklin County, Iowa
##	1	1
##	Franklin County, Kentucky	Franklin County, Missouri
##	1	1
##	Franklin County, Nebraska	Franklin County, North Carolina
##	1	1
##	Franklin County, Vermont	Franklin County, Virginia
##	1	1
##	Franklin County, Washington	Frederick County, Maryland
##	1	1
##	Frederick County, Virginia	Fredericksburg city, Virginia
##	1	1
##	Freeborn County, Minnesota	Freestone County, Texas
##	1	1
##	Fremont County, Idaho	Fremont County, Wyoming
##	1	1
##	Fresno County, California	Frontier County, Nebraska
##	1	1
##	Fulton County, Arkansas	Fulton County, New York
##	1	1
##	Fulton County, Pennsylvania	Furnas County, Nebraska
##	1	1
##	Gage County, Nebraska	Gallatin County, Illinois
##	1	1
##	Gallatin County, Montana	Gallia County, Ohio
##	1	1
##	Galveston County, Texas	Garden County, Nebraska
##	1	1

##	Garfield County, Colorado	Garfield County, Oklahoma
##	1	1
##	Garrard County, Kentucky	Garrett County, Maryland
##	1	1
##	Gasconade County, Missouri	Gem County, Idaho
##	1	1
##	Genesee County, Michigan	Geneva County, Alabama
##	1	1
##	George County, Mississippi	Gibson County, Indiana
##	1	1
##	Giles County, Tennessee	Gillespie County, Texas
##	1	1
##	Gilliam County, Oregon	Gilmer County, West Virginia
##	1	1
##	Gladwin County, Michigan	Glascock County, Georgia
##	1	1
##	Glenn County, California	Gloucester County, Virginia
##	1	1
##	Glynn County, Georgia	Gogebic County, Michigan
##	1	1
##	Goliad County, Texas	Gonzales County, Texas
##	1	1
##	Goodhue County, Minnesota	Gordon County, Georgia
##	1	1
##	Goshen County, Wyoming	Gosper County, Nebraska
##	1	1
##	Grady County, Georgia	Grady County, Oklahoma
##	1	1
##	Graham County, North Carolina	Grand County, Colorado
##	1	1
##	Grand County, Utah	Grand Isle County, Vermont
##	1	1
##	Grant County, Arkansas	Grant County, Kansas
##	1	1
##	Grant County, Kentucky	Grant County, Minnesota
##	1	1
##	Grant County, Oklahoma	Grant County, Oregon
##	1	1
##	Grant County, West Virginia	Grant Parish, Louisiana
##	1	1
##	Granville County, North Carolina	Graves County, Kentucky
##	1	1
##	Gray County, Kansas	Grays Harbor County, Washington
##	1	1
##	Grayson County, Texas	Grayson County, Virginia
##	1	1
##	Greeley County, Kansas	Greeley County, Nebraska
##	1	1
##	Green County, Kentucky	Green County, Wisconsin
##	1	1
##	Green Lake County, Wisconsin	Greene County, Illinois
##	1	1
##	Greene County, Indiana	Greene County, Iowa
##	1	1

##	Greene County, Missouri	Greene County, North Carolina
##	1	1
##	Greene County, Pennsylvania	Greene County, Tennessee
##	1	1
##	Greene County, Virginia	Greenlee County, Arizona
##	1	1
##	Greenup County, Kentucky	Greenville County, South Carolina
##	1	1
##	Greenwood County, South Carolina	Gregg County, Texas
##	1	1
##	Grenada County, Mississippi	Griggs County, North Dakota
##	1	1
##	Grundy County, Illinois	Grundy County, Iowa
##	1	1
##	Grundy County, Missouri	Guadalupe County, New Mexico
##	1	1
##	Gulf County, Florida	Guthrie County, Iowa
##	1	1
##	Haakon County, South Dakota	Habersham County, Georgia
##	1	1
##	Haines Borough, Alaska	Hale County, Alabama
##	1	1
##	Hale County, Texas	Halifax County, North Carolina
##	1	1
##	Halifax County, Virginia	Hall County, Georgia
##	1	1
##	Hall County, Nebraska	Hall County, Texas
##	1	1
##	Hamblen County, Tennessee	Hamilton County, Illinois
##	1	1
##	Hamilton County, Indiana	Hamilton County, Iowa
##	1	1
##	Hamilton County, Kansas	Hamilton County, Nebraska
##	1	1
##	Hamilton County, New York	Hamilton County, Ohio
##	1	1
##	Hamilton County, Texas	Hamlin County, South Dakota
##	1	1
##	Hampshire County, West Virginia	Hampton County, South Carolina
##	1	1
##	Hancock County, Indiana	Hancock County, Iowa
##	1	1
##	Hancock County, Kentucky	Hancock County, Maine
##	1	1
##	Hancock County, Mississippi	Hancock County, Tennessee
##	1	1
##	Hancock County, West Virginia	Hansford County, Texas
##	1	1
##	Haralson County, Georgia	Hardin County, Iowa
##	1	1
##	Hardin County, Kentucky	Hardin County, Ohio
##	1	1
##	Hardy County, West Virginia	Harford County, Maryland
##	1	1

##	Harlan County, Kentucky	Harlan County, Nebraska
##	1	1
##	Harmon County, Oklahoma	Harper County, Oklahoma
##	1	1
##	Harris County, Texas	Harrison County, Iowa
##	1	1
##	Harrison County, Mississippi	Harrison County, Texas
##	1	1
##	Harrison County, West Virginia	Hart County, Kentucky
##	1	1
##	Hartford County, Connecticut	Hartley County, Texas
##	1	1
##	Harvey County, Kansas	Haskell County, Texas
##	1	1
##	Hawkins County, Tennessee	Haywood County, North Carolina
##	1	1
##	Heard County, Georgia	Hempstead County, Arkansas
##	1	1
##	Henderson County, Illinois	Henderson County, North Carolina
##	1	1
##	Henderson County, Tennessee	Hendricks County, Indiana
##	1	1
##	Hennepin County, Minnesota	Henrico County, Virginia
##	1	1
##	Henry County, Georgia	Henry County, Illinois
##	1	1
##	Henry County, Indiana	Henry County, Iowa
##	1	1
##	Henry County, Missouri	Henry County, Tennessee
##	1	1
##	Herkimer County, New York	Hertford County, North Carolina
##	1	1
##	Hettinger County, North Dakota	Hickman County, Kentucky
##	1	1
##	Hickman County, Tennessee	Hickory County, Missouri
##	1	1
##	Hidalgo County, Texas	Highland County, Virginia
##	1	1
##	Highlands County, Florida	Hill County, Texas
##	1	1
##	Hillsborough County, New Hampshire	Hillsdale County, Michigan
##	1	1
##	Hockley County, Texas	Hoke County, North Carolina
##	1	1
##	Holmes County, Mississippi	Hood County, Texas
##	1	1
##	Hood River County, Oregon	Hopewell city, Virginia
##	1	1
##	Hopkins County, Texas	Horry County, South Carolina
##	1	1
##	Hot Spring County, Arkansas	Houghton County, Michigan
##	1	1
##	Houston County, Alabama	Houston County, Georgia
##	1	1

##	Houston County, Minnesota	Houston County, Texas
##	1	1
##	Howard County, Arkansas	Howard County, Iowa
##	1	1
##	Howard County, Maryland	Howard County, Missouri
##	1	1
##	Howard County, Nebraska	Howard County, Texas
##	1	1
##	Howell County, Missouri	Huerfano County, Colorado
##	1	1
##	Hughes County, Oklahoma	Hughes County, South Dakota
##	1	1
##	Humboldt County, Nevada	Humphreys County, Mississippi
##	1	1
##	Humphreys County, Tennessee	Hunt County, Texas
##	1	1
##	Hunterdon County, New Jersey	Huntingdon County, Pennsylvania
##	1	1
##	Huron County, Michigan	Hutchinson County, South Dakota
##	1	1
##	Hutchinson County, Texas	Hyde County, North Carolina
##	1	1
##	Iberia Parish, Louisiana	Ida County, Iowa
##	1	1
##	Imperial County, California	Independence County, Arkansas
##	1	1
##	Indian River County, Florida	Indiana County, Pennsylvania
##	1	1
##	Ingham County, Michigan	Inyo County, California
##	1	1
##	Iosco County, Michigan	Iowa County, Iowa
##	1	1
##	Iowa County, Wisconsin	Iredell County, North Carolina
##	1	1
##	Iron County, Missouri	Iron County, Utah
##	1	1
##	Iroquois County, Illinois	Irwin County, Georgia
##	1	1
##	Isanti County, Minnesota	Issaquena County, Mississippi
##	1	1
##	Itasca County, Minnesota	Izard County, Arkansas
##	1	1
##	Jack County, Texas	Jackson County, Arkansas
##	1	1
##	Jackson County, Georgia	Jackson County, Indiana
##	1	1
##	Jackson County, Iowa	Jackson County, Kansas
##	1	1
##	Jackson County, Kentucky	Jackson County, Mississippi
##	1	1
##	Jackson County, Missouri	Jackson County, North Carolina
##	1	1
##	Jackson County, Oregon	Jackson County, Texas
##	1	1

##	Jackson County, West Virginia	James City County, Virginia
##	1	1
##	Jasper County, Illinois	Jasper County, Indiana
##	1	1
##	Jasper County, Mississippi	Jasper County, Missouri
##	1	1
##	Jay County, Indiana	Jeff Davis County, Texas
##	1	1
##	Jefferson County, Arkansas	Jefferson County, Colorado
##	1	1
##	Jefferson County, Georgia	Jefferson County, Idaho
##	1	1
##	Jefferson County, Illinois	Jefferson County, Iowa
##	1	1
##	Jefferson County, Kansas	Jefferson County, Mississippi
##	1	1
##	Jefferson County, Montana	Jefferson County, Nebraska
##	1	1
##	Jefferson County, New York	Jefferson County, Oklahoma
##	1	1
##	Jefferson County, Pennsylvania	Jefferson County, Texas
##	1	1
##	Jefferson County, Washington	Jefferson County, West Virginia
##	1	1
##	Jefferson County, Wisconsin	Jefferson Davis County, Mississippi
##	1	1
##	Jefferson Davis Parish, Louisiana	Jefferson Parish, Louisiana
##	1	1
##	Jenkins County, Georgia	Jennings County, Indiana
##	1	1
##	Jerauld County, South Dakota	Jerome County, Idaho
##	1	1
##	Jersey County, Illinois	Jessamine County, Kentucky
##	1	1
##	Jewell County, Kansas	Jim Hogg County, Texas
##	1	1
##	Jo Daviess County, Illinois	Johnson County, Arkansas
##	1	1
##	Johnson County, Georgia	Johnson County, Iowa
##	1	1
##	Johnson County, Kansas	Johnson County, Nebraska
##	1	1
##	Johnson County, Tennessee	Johnston County, North Carolina
##	1	1
##	Jones County, Georgia	Jones County, Iowa
##	1	1
##	Jones County, Mississippi	Jones County, Texas
##	1	1
##	Josephine County, Oregon	Juab County, Utah
##	1	1
##	Judith Basin County, Montana	Juneau County, Wisconsin
##	1	1
##	Kalamazoo County, Michigan	Kanabec County, Minnesota
##	1	1

##	Kane County, Utah	Karnes County, Texas
##	1	1
##	Kauai County, Hawaii	Kay County, Oklahoma
##	1	1
##	Kearney County, Nebraska	Kearny County, Kansas
##	1	1
##	Keith County, Nebraska	Kemper County, Mississippi
##	1	1
##	Kendall County, Illinois	Kennebec County, Maine
##	1	1
##	Kent County, Delaware	Kent County, Maryland
##	1	1
##	Kent County, Rhode Island	Keokuk County, Iowa
##	1	1
##	Kern County, California	Kerr County, Texas
##	1	1
##	Kershaw County, South Carolina	Keweenaw County, Michigan
##	1	1
##	Kimball County, Nebraska	Kimble County, Texas
##	1	1
##	King and Queen County, Virginia	King William County, Virginia
##	1	1
##	Kinney County, Texas	Kiowa County, Colorado
##	1	1
##	Kiowa County, Kansas	Kit Carson County, Colorado
##	1	1
##	Kittson County, Minnesota	Knott County, Kentucky
##	1	1
##	Knox County, Illinois	Knox County, Indiana
##	1	1
##	Knox County, Kentucky	Knox County, Maine
##	1	1
##	Knox County, Missouri	Knox County, Ohio
##	1	1
##	Kodiak Island Borough, Alaska	Koochiching County, Minnesota
##	1	1
##	Kosciusko County, Indiana	La Paz County, Arizona
##	1	1
##	La Salle County, Texas	Lac qui Parle County, Minnesota
##	1	1
##	Lackawanna County, Pennsylvania	Lafayette County, Arkansas
##	1	1
##	Lafayette County, Mississippi	Lafayette County, Missouri
##	1	1
##	Lafayette County, Wisconsin	Lafourche Parish, Louisiana
##	1	1
##	Lake County, Florida	Lake County, Illinois
##	1	1
##	Lake County, Indiana	Lake County, Ohio
##	1	1
##	Lake County, South Dakota	Lake County, Tennessee
##	1	1
##	Lamar County, Alabama	Lamar County, Georgia
##	1	1

##	Lamar County, Mississippi	Lamoille County, Vermont
##	1	1
##	LaMoure County, North Dakota	Lampasas County, Texas
##	1	1
##	Lancaster County, Nebraska	Lancaster County, Pennsylvania
##	1	1
##	Lander County, Nevada	Larue County, Kentucky
##	1	1
##	Lassen County, California	Latimer County, Oklahoma
##	1	1
##	Lauderdale County, Alabama	Lauderdale County, Mississippi
##	1	1
##	Laurel County, Kentucky	Lawrence County, Alabama
##	1	1
##	Lawrence County, Arkansas	Lawrence County, Illinois
##	1	1
##	Lawrence County, Indiana	Lawrence County, Kentucky
##	1	1
##	Lawrence County, Ohio	Lawrence County, Pennsylvania
##	1	1
##	Lawrence County, South Dakota	Lawrence County, Tennessee
##	1	1
##	Le Sueur County, Minnesota	Leake County, Mississippi
##	1	1
##	Lebanon County, Pennsylvania	Lee County, Arkansas
##	1	1
##	Lee County, Georgia	Lee County, Iowa
##	1	1
##	Lee County, South Carolina	Lee County, Texas
##	1	1
##	Lee County, Virginia	Leflore County, Mississippi
##	1	1
##	Lehigh County, Pennsylvania	Lemhi County, Idaho
##	1	1
##	Lenawee County, Michigan	Leon County, Florida
##	1	1
##	Leslie County, Kentucky	Letcher County, Kentucky
##	1	1
##	Lewis County, Idaho	Lewis County, Kentucky
##	1	1
##	Lewis County, New York	Lewis County, Tennessee
##	1	1
##	Lexington County, South Carolina	Liberty County, Florida
##	1	1
##	Liberty County, Montana	Licking County, Ohio
##	1	1
##	Limestone County, Alabama	Lincoln County, Arkansas
##	1	1
##	Lincoln County, Colorado	Lincoln County, Georgia
##	1	1
##	Lincoln County, Idaho	Lincoln County, Kentucky
##	1	1
##	Lincoln County, Maine	Lincoln County, Minnesota
##	1	1

##	Lincoln County, Mississippi	Lincoln County, Montana
##	1	1
##	Lincoln County, Nebraska	Lincoln County, Nevada
##	1	1
##	Lincoln County, Oklahoma	Lincoln County, Oregon
##	1	1
##	Lincoln County, South Dakota	Lincoln County, Tennessee
##	1	1
##	Lincoln County, Washington	Lincoln County, Wisconsin
##	1	1
##	Lincoln County, Wyoming	Linn County, Kansas
##	1	1
##	Linn County, Oregon	Lipscomb County, Texas
##	1	1
##	Litchfield County, Connecticut	Live Oak County, Texas
##	1	1
##	Livingston County, Illinois	Livingston County, Kentucky
##	1	1
##	Livingston County, Michigan	Livingston County, Missouri
##	1	1
##	Livingston County, New York	Llano County, Texas
##	1	1
##	Logan County, Colorado	Logan County, Illinois
##	1	1
##	Lorain County, Ohio	Los Alamos County, New Mexico
##	1	1
##	Los Angeles County, California	Loudoun County, Virginia
##	1	1
##	Louisa County, Virginia	Lowndes County, Alabama
##	1	1
##	Lowndes County, Georgia	Lowndes County, Mississippi
##	1	1
##	Luna County, New Mexico	Lunenburg County, Virginia
##	1	1
##	Lycoming County, Pennsylvania	Lyman County, South Dakota
##	1	1
##	Lynn County, Texas	Lyon County, Kansas
##	1	1
##	Lyon County, Minnesota	Lyon County, Nevada
##	1	1
##	Macomb County, Michigan	Macon County, Georgia
##	1	1
##	Macon County, Illinois	Macon County, Tennessee
##	1	1
##	Macoupin County, Illinois	Madison County, Alabama
##	1	1
##	Madison County, Florida	Madison County, Indiana
##	1	1
##	Madison County, Kentucky	Madison County, Mississippi
##	1	1
##	Madison County, Missouri	Madison County, New York
##	1	1
##	Madison County, North Carolina	Madison County, Ohio
##	1	1

##	Madison County, Tennessee	1	Mahaska County, Iowa	1
##	Mahnomen County, Minnesota	1	Malheur County, Oregon	1
##	Manassas city, Virginia	1	Manitowoc County, Wisconsin	1
##	Marathon County, Wisconsin	1	Maricopa County, Arizona	1
##	Maries County, Missouri	1	Marinette County, Wisconsin	1
##	Marion County, Arkansas	1	Marion County, Georgia	1
##	Marion County, Illinois	1	Marion County, Indiana	1
##	Marion County, Iowa	1	Marion County, Kansas	1
##	Marion County, Kentucky	1	Marion County, Mississippi	1
##	Marion County, Missouri	1	Marion County, Texas	1
##	Marion County, West Virginia	1	Mariposa County, California	1
##	Marlboro County, South Carolina	1	Marquette County, Wisconsin	1
##	Marshall County, Alabama	1	Marshall County, Indiana	1
##	Marshall County, Iowa	1	Marshall County, Kentucky	1
##	Marshall County, Minnesota	1	Marshall County, Mississippi	1
##	Marshall County, South Dakota	1	Marshall County, Tennessee	1
##	Marshall County, West Virginia	1	Martin County, Florida	1
##	Martin County, Indiana	1	Martin County, Kentucky	1
##	Martin County, Minnesota	1	Martin County, North Carolina	1
##	Martin County, Texas	1	Mason County, Kentucky	1
##	Mason County, Michigan	1	Mason County, Texas	1
##	Mason County, Washington	1	Mason County, West Virginia	1
##	Massac County, Illinois	1	Matanuska-Susitna Borough, Alaska	1
##	Maui County, Hawaii	1	Mauzy County, Tennessee	1
##	Maverick County, Texas	1	Mayes County, Oklahoma	1
##	McClain County, Oklahoma	1	McCormick County, South Carolina	1
##	McCracken County, Kentucky	1	McCurtain County, Oklahoma	1

##	McDonald County, Missouri	1	McDowell County, North Carolina	1
##	McHenry County, North Dakota	1	McIntosh County, Georgia	1
##	McIntosh County, North Dakota	1	McIntosh County, Oklahoma	1
##	McKean County, Pennsylvania	1	McKenzie County, North Dakota	1
##	McLean County, Illinois	1	McLean County, Kentucky	1
##	McLean County, North Dakota	1	McLennan County, Texas	1
##	McMinn County, Tennessee	1	McPherson County, Kansas	1
##	Meade County, Kansas	1	Meade County, Kentucky	1
##	Meade County, South Dakota	1	Meagher County, Montana	1
##	Mecklenburg County, North Carolina	1	Mecosta County, Michigan	1
##	Meeker County, Minnesota	1	Meigs County, Tennessee	1
##	Mellette County, South Dakota	1	Menard County, Illinois	1
##	Menard County, Texas	1	Mendocino County, California	1
##	Menifee County, Kentucky	1	Menominee County, Michigan	1
##	Merced County, California	1	Mercer County, Kentucky	1
##	Mercer County, West Virginia	1	Meriwether County, Georgia	1
##	Merrick County, Nebraska	1	Merrimack County, New Hampshire	1
##	Mesa County, Colorado	1	Metcalf County, Kentucky	1
##	Miami County, Kansas	1	Miami County, Ohio	1
##	Miami-Dade County, Florida	1	Middlesex County, Connecticut	1
##	Middlesex County, Massachusetts	1	Mifflin County, Pennsylvania	1
##	Milam County, Texas	1	Millard County, Utah	1
##	Mille Lacs County, Minnesota	1	Miller County, Arkansas	1
##	Miller County, Georgia	1	Mills County, Iowa	1
##	Miner County, South Dakota	1	Mineral County, Nevada	1
##	Mineral County, West Virginia	1	Mingo County, West Virginia	1
##	Minidoka County, Idaho	1	Minnehaha County, South Dakota	1

##	Mississippi County, Arkansas	Mississippi County, Missouri
##	1	1
##	Mitchell County, Kansas	Mitchell County, North Carolina
##	1	1
##	Mitchell County, Texas	Moniteau County, Missouri
##	1	1
##	Monmouth County, New Jersey	Monona County, Iowa
##	1	1
##	Monroe County, Alabama	Monroe County, Arkansas
##	1	1
##	Monroe County, Florida	Monroe County, Georgia
##	1	1
##	Monroe County, Indiana	Monroe County, Kentucky
##	1	1
##	Monroe County, Mississippi	Monroe County, New York
##	1	1
##	Monroe County, Ohio	Monroe County, Pennsylvania
##	1	1
##	Monroe County, Tennessee	Monroe County, West Virginia
##	1	1
##	Monroe County, Wisconsin	Montague County, Texas
##	1	1
##	Montcalm County, Michigan	Monterey County, California
##	1	1
##	Montgomery County, Alabama	Montgomery County, Georgia
##	1	1
##	Montgomery County, Illinois	Montgomery County, Kansas
##	1	1
##	Montgomery County, Kentucky	Montgomery County, Maryland
##	1	1
##	Montgomery County, Mississippi	Montgomery County, Missouri
##	1	1
##	Montgomery County, New York	Montgomery County, North Carolina
##	1	1
##	Montgomery County, Pennsylvania	Montgomery County, Tennessee
##	1	1
##	Montgomery County, Texas	Montgomery County, Virginia
##	1	1
##	Montmorency County, Michigan	Montrose County, Colorado
##	1	1
##	Moore County, Tennessee	Moore County, Texas
##	1	1
##	Mora County, New Mexico	Morgan County, Alabama
##	1	1
##	Morgan County, Colorado	Morgan County, Georgia
##	1	1
##	Morgan County, Illinois	Morgan County, Indiana
##	1	1
##	Morgan County, Missouri	Morgan County, West Virginia
##	1	1
##	Morrill County, Nebraska	Morris County, New Jersey
##	1	1
##	Morrow County, Ohio	Morton County, Kansas
##	1	1

##	Moultrie County, Illinois	Mountrail County, North Dakota
##	1	1
##	Multnomah County, Oregon	Muscogee County, Georgia
##	1	1
##	Muskogee County, Oklahoma	Nacogdoches County, Texas
##	1	1
##	Nash County, North Carolina	Nassau County, Florida
##	1	1
##	Nassau County, New York	Natchitoches Parish, Louisiana
##	1	1
##	Natrona County, Wyoming	Nelson County, Virginia
##	1	1
##	Nemaha County, Nebraska	Neosho County, Kansas
##	1	1
##	Neshoba County, Mississippi	Ness County, Kansas
##	1	1
##	Nevada County, Arkansas	Nevada County, California
##	1	1
##	New Haven County, Connecticut	New Kent County, Virginia
##	1	1
##	New London County, Connecticut	New Madrid County, Missouri
##	1	1
##	Newaygo County, Michigan	Newport County, Rhode Island
##	1	1
##	Newton County, Georgia	Newton County, Missouri
##	1	1
##	Newton County, Texas	Nez Perce County, Idaho
##	1	1
##	Niagara County, New York	Nicollet County, Minnesota
##	1	1
##	Niobrara County, Wyoming	Noble County, Indiana
##	1	1
##	Noble County, Ohio	Noble County, Oklahoma
##	1	1
##	Nobles County, Minnesota	Nodaway County, Missouri
##	1	1
##	Nolan County, Texas	Nome Census Area, Alaska
##	1	1
##	Norfolk County, Massachusetts	Northumberland County, Pennsylvania
##	1	1
##	Northwest Arctic Borough, Alaska	Norton city, Virginia
##	1	1
##	Nottoway County, Virginia	Noxubee County, Mississippi
##	1	1
##	Nuckolls County, Nebraska	Nueces County, Texas
##	1	1
##	Oakland County, Michigan	Obion County, Tennessee
##	1	1
##	Ocean County, New Jersey	Oceana County, Michigan
##	1	1
##	Oconee County, South Carolina	Oconto County, Wisconsin
##	1	1
##	Ogemaw County, Michigan	Oglethorpe County, Georgia
##	1	1

##	Ohio County, Indiana	Ohio County, Kentucky
##	1	1
##	Ohio County, West Virginia	Okaloosa County, Florida
##	1	1
##	Okeechobee County, Florida	Okfuskee County, Oklahoma
##	1	1
##	Oktibbeha County, Mississippi	Oldham County, Kentucky
##	1	1
##	Oneida County, Idaho	Oneida County, New York
##	1	1
##	Onondaga County, New York	Onslow County, North Carolina
##	1	1
##	Ontario County, New York	Ontonagon County, Michigan
##	1	1
##	Orange County, California	Orange County, Florida
##	1	1
##	Orange County, New York	Orange County, North Carolina
##	1	1
##	Orange County, Vermont	Orange County, Virginia
##	1	1
##	Osage County, Oklahoma	Osborne County, Kansas
##	1	1
##	Osceola County, Florida	Oscoda County, Michigan
##	1	1
##	Otero County, Colorado	Otero County, New Mexico
##	1	1
##	Otoe County, Nebraska	Otsego County, Michigan
##	1	1
##	Ottawa County, Kansas	Ottawa County, Michigan
##	1	1
##	Ottawa County, Ohio	Ottawa County, Oklahoma
##	1	1
##	Otter Tail County, Minnesota	Ouachita County, Arkansas
##	1	1
##	Ouray County, Colorado	Overton County, Tennessee
##	1	1
##	Owen County, Indiana	Ozaukee County, Wisconsin
##	1	1
##	Pacific County, Washington	Page County, Iowa
##	1	1
##	Page County, Virginia	Palm Beach County, Florida
##	1	1
##	Pamlico County, North Carolina	Panola County, Mississippi
##	1	1
##	Park County, Colorado	Park County, Montana
##	1	1
##	Park County, Wyoming	Parke County, Indiana
##	1	1
##	Parker County, Texas	Pasco County, Florida
##	1	1
##	Pasquotank County, North Carolina	Patrick County, Virginia
##	1	1
##	Pawnee County, Nebraska	Pawnee County, Oklahoma
##	1	1

##	Payette County, Idaho	Payne County, Oklahoma
##	1	1
##	Peach County, Georgia	Pearl River County, Mississippi
##	1	1
##	Pecos County, Texas	Pembina County, North Dakota
##	1	1
##	Pender County, North Carolina	Pendleton County, Kentucky
##	1	1
##	Pennington County, Minnesota	Penobscot County, Maine
##	1	1
##	Peoria County, Illinois	Pepin County, Wisconsin
##	1	1
##	Perkins County, Nebraska	Perkins County, South Dakota
##	1	1
##	Perquimans County, North Carolina	Perry County, Illinois
##	1	1
##	Perry County, Indiana	Perry County, Kentucky
##	1	1
##	Perry County, Mississippi	Perry County, Missouri
##	1	1
##	Perry County, Ohio	Person County, North Carolina
##	1	1
##	Pettis County, Missouri	Phelps County, Missouri
##	1	1
##	Philadelphia County, Pennsylvania	Phillips County, Colorado
##	1	1
##	Pickaway County, Ohio	Pickens County, Alabama
##	1	1
##	Pickens County, Georgia	Pickett County, Tennessee
##	1	1
##	Pierce County, Georgia	Pierce County, Washington
##	1	1
##	Pike County, Georgia	Pike County, Illinois
##	1	1
##	Pike County, Indiana	Pike County, Mississippi
##	1	1
##	Pike County, Ohio	Pike County, Pennsylvania
##	1	1
##	Pima County, Arizona	Pine County, Minnesota
##	1	1
##	Pinellas County, Florida	Pitkin County, Colorado
##	1	1
##	Pitt County, North Carolina	Pittsylvania County, Virginia
##	1	1
##	Plaquemines Parish, Louisiana	Platte County, Missouri
##	1	1
##	Platte County, Nebraska	Platte County, Wyoming
##	1	1
##	Pleasants County, West Virginia	Plymouth County, Massachusetts
##	1	1
##	Pocahontas County, Iowa	Poinsett County, Arkansas
##	1	1
##	Pointe Coupee Parish, Louisiana	Polk County, Florida
##	1	1

##	Polk County, Iowa	Polk County, Missouri
##	1	1
##	Polk County, Nebraska	Polk County, North Carolina
##	1	1
##	Polk County, Texas	Pondera County, Montana
##	1	1
##	Pope County, Illinois	Portage County, Ohio
##	1	1
##	Portage County, Wisconsin	Posey County, Indiana
##	1	1
##	Pottawatomie County, Kansas	Pottawatomie County, Oklahoma
##	1	1
##	Pottawattamie County, Iowa	Potter County, Pennsylvania
##	1	1
##	Powell County, Kentucky	Power County, Idaho
##	1	1
##	Poweshiek County, Iowa	Powhatan County, Virginia
##	1	1
##	Prairie County, Arkansas	Pratt County, Kansas
##	1	1
##	Preble County, Ohio	Prentiss County, Mississippi
##	1	1
##	Presidio County, Texas	Presque Isle County, Michigan
##	1	1
##	Preston County, West Virginia	Prince George County, Virginia
##	1	1
##	Prince William County, Virginia	Prowers County, Colorado
##	1	1
##	Pueblo County, Colorado	Pulaski County, Arkansas
##	1	1
##	Pulaski County, Georgia	Pulaski County, Indiana
##	1	1
##	Pulaski County, Kentucky	Pulaski County, Missouri
##	1	1
##	Putnam County, Georgia	Putnam County, Illinois
##	1	1
##	Putnam County, Indiana	Putnam County, Tennessee
##	1	1
##	Quay County, New Mexico	Queen Anne's County, Maryland
##	1	1
##	Queens County, New York	Quitman County, Georgia
##	1	1
##	Quitman County, Mississippi	Racine County, Wisconsin
##	1	1
##	Rains County, Texas	Randall County, Texas
##	1	1
##	Randolph County, Alabama	Randolph County, Arkansas
##	1	1
##	Randolph County, Georgia	Randolph County, North Carolina
##	1	1
##	Randolph County, West Virginia	Rankin County, Mississippi
##	1	1
##	Ransom County, North Dakota	Rapides Parish, Louisiana
##	1	1

##	Rappahannock County, Virginia		Ravalli County, Montana
##	1		1
##	Rawlins County, Kansas		Ray County, Missouri
##	1		1
##	Real County, Texas		Red Willow County, Nebraska
##	1		1
##	Redwood County, Minnesota		Reeves County, Texas
##	1		1
##	Reno County, Kansas		Rensselaer County, New York
##	1		1
##	Renville County, North Dakota		Republic County, Kansas
##	1		1
##	Rhea County, Tennessee		Rice County, Kansas
##	1		1
##	Richardson County, Nebraska		Richland County, Illinois
##	1		1
##	Richland County, Ohio		Richland County, South Carolina
##	1		1
##	Richland Parish, Louisiana		Richmond County, New York
##	1		1
##	Richmond County, North Carolina		Rio Arriba County, New Mexico
##	1		1
##	Ripley County, Indiana		Roane County, Tennessee
##	1		1
##	Roane County, West Virginia		Roanoke County, Virginia
##	1		1
##	Roberts County, South Dakota		Robeson County, North Carolina
##	1		1
##	Rock County, Minnesota		Rock County, Wisconsin
##	1		1
##	Rock Island County, Illinois		Rockbridge County, Virginia
##	1		1
##	Rockingham County, North Carolina		Rockingham County, Virginia
##	1		1
##	Rockland County, New York		Rogers County, Oklahoma
##	1		1
##	Rooks County, Kansas		Roosevelt County, Montana
##	1		1
##	Roscommon County, Michigan		Roseau County, Minnesota
##	1		1
##	Ross County, Ohio		Rowan County, Kentucky
##	1		1
##	Runnels County, Texas		Rush County, Kansas
##	1		1
##	Rusk County, Texas		Rusk County, Wisconsin
##	1		1
##	Russell County, Alabama		Russell County, Kentucky
##	1		1
##	Russell County, Virginia		Rutherford County, North Carolina
##	1		1
##	Rutherford County, Tennessee		Sabine County, Texas
##	1		1
##	Sabine Parish, Louisiana		Sacramento County, California
##	1		1

##	Sagadahoc County, Maine	Saginaw County, Michigan
##	1	1
##	Salem city, Virginia	Salem County, New Jersey
##	1	1
##	Saline County, Illinois	Saline County, Nebraska
##	1	1
##	Salt Lake County, Utah	Saluda County, South Carolina
##	1	1
##	Sampson County, North Carolina	San Augustine County, Texas
##	1	1
##	San Benito County, California	San Jacinto County, Texas
##	1	1
##	San Juan County, Utah	San Juan County, Washington
##	1	1
##	San Luis Obispo County, California	San Mateo County, California
##	1	1
##	San Miguel County, Colorado	San Miguel County, New Mexico
##	1	1
##	San Patricio County, Texas	San Saba County, Texas
##	1	1
##	Sanborn County, South Dakota	Sanders County, Montana
##	1	1
##	Sandusky County, Ohio	Sanpete County, Utah
##	1	1
##	Santa Barbara County, California	Santa Clara County, California
##	1	1
##	Santa Cruz County, Arizona	Santa Cruz County, California
##	1	1
##	Santa Fe County, New Mexico	Santa Rosa County, Florida
##	1	1
##	Sauk County, Wisconsin	Saunders County, Nebraska
##	1	1
##	Sawyer County, Wisconsin	Schenectady County, New York
##	1	1
##	Schleicher County, Texas	Schley County, Georgia
##	1	1
##	Schoharie County, New York	Schuyler County, New York
##	1	1
##	Schuylkill County, Pennsylvania	Scioto County, Ohio
##	1	1
##	Scotland County, Missouri	Scott County, Arkansas
##	1	1
##	Scott County, Illinois	Scott County, Iowa
##	1	1
##	Scott County, Kansas	Scott County, Virginia
##	1	1
##	Scurry County, Texas	Searcy County, Arkansas
##	1	1
##	Sebastian County, Arkansas	Sedgwick County, Colorado
##	1	1
##	Seminole County, Georgia	Seminole County, Oklahoma
##	1	1
##	Seneca County, New York	Seneca County, Ohio
##	1	1

## Sequatchie County, Tennessee	1	Sequoyah County, Oklahoma	1
## Seward County, Kansas	1	Sharkey County, Mississippi	1
## Shawano County, Wisconsin	1	Shawnee County, Kansas	1
## Shelby County, Alabama	1	Shelby County, Illinois	1
## Shelby County, Kentucky	1	Shelby County, Missouri	1
## Shelby County, Tennessee	1	Shenandoah County, Virginia	1
## Sheridan County, Kansas	1	Sheridan County, Montana	1
## Sheridan County, North Dakota	1	Sherman County, Nebraska	1
## Sherman County, Oregon	1	Shiawassee County, Michigan	1
## Sierra County, New Mexico	1	Simpson County, Mississippi	1
## Siskiyou County, California	1	Sitka City and Borough, Alaska	1
## Skagit County, Washington	1	Skamania County, Washington	1
## Smith County, Kansas	1	Smith County, Mississippi	1
## Smith County, Tennessee	1	Smith County, Texas	1
## Smyth County, Virginia	1	Snohomish County, Washington	1
## Snyder County, Pennsylvania	1	Solano County, California	1
## Somerset County, Maine	1	Somerset County, New Jersey	1
## Somerset County, Pennsylvania	1	Sonoma County, California	1
## Southampton County, Virginia	1	Spalding County, Georgia	1
## Spartanburg County, South Carolina	1	Spencer County, Indiana	1
## Spencer County, Kentucky	1	Spink County, South Dakota	1
## St. Charles Parish, Louisiana	1	St. Clair County, Alabama	1
## St. Clair County, Michigan	1	St. Francois County, Missouri	1
## St. Helena Parish, Louisiana	1	St. James Parish, Louisiana	1
## St. John the Baptist Parish, Louisiana	1	St. Johns County, Florida	1
## St. Joseph County, Indiana	1	St. Landry Parish, Louisiana	1
## St. Lawrence County, New York	1	St. Lucie County, Florida	1

##	St. Martin Parish, Louisiana	St. Mary Parish, Louisiana
##	1	1
##	St. Mary's County, Maryland	St. Tammany Parish, Louisiana
##	1	1
##	Stafford County, Kansas	Stafford County, Virginia
##	1	1
##	Stanley County, South Dakota	Stanly County, North Carolina
##	1	1
##	Stanton County, Nebraska	Starke County, Indiana
##	1	1
##	Starr County, Texas	Staunton city, Virginia
##	1	1
##	Ste. Genevieve County, Missouri	Stearns County, Minnesota
##	1	1
##	Steele County, Minnesota	Steele County, North Dakota
##	1	1
##	Stephens County, Georgia	Stephens County, Oklahoma
##	1	1
##	Stephens County, Texas	Steuben County, Indiana
##	1	1
##	Stevens County, Kansas	Stevens County, Washington
##	1	1
##	Stewart County, Georgia	Stillwater County, Montana
##	1	1
##	Stoddard County, Missouri	Stone County, Missouri
##	1	1
##	Stonewall County, Texas	Storey County, Nevada
##	1	1
##	Story County, Iowa	Strafford County, New Hampshire
##	1	1
##	Stutsman County, North Dakota	Suffolk city, Virginia
##	1	1
##	Suffolk County, Massachusetts	Sullivan County, Missouri
##	1	1
##	Sullivan County, New York	Sullivan County, Pennsylvania
##	1	1
##	Sullivan County, Tennessee	Sully County, South Dakota
##	1	1
##	Summers County, West Virginia	Summit County, Utah
##	1	1
##	Sumner County, Kansas	Sumner County, Tennessee
##	1	1
##	Sumter County, Georgia	Sunflower County, Mississippi
##	1	1
##	Surry County, Virginia	Susquehanna County, Pennsylvania
##	1	1
##	Suwannee County, Florida	Sweetwater County, Wyoming
##	1	1
##	Swift County, Minnesota	Swisher County, Texas
##	1	1
##	Switzerland County, Indiana	Talbot County, Maryland
##	1	1
##	Taliaferro County, Georgia	Tallahatchie County, Mississippi
##	1	1

##	Tama County, Iowa	Taos County, New Mexico
##	1	1
##	Tate County, Mississippi	Taylor County, Florida
##	1	1
##	Taylor County, Iowa	Taylor County, Wisconsin
##	1	1
##	Terrebonne Parish, Louisiana	Terry County, Texas
##	1	1
##	Teton County, Idaho	Texas County, Oklahoma
##	1	1
##	Thayer County, Nebraska	Thomas County, Georgia
##	1	1
##	Throckmorton County, Texas	Thurston County, Nebraska
##	1	1
##	Tift County, Georgia	Tioga County, Pennsylvania
##	1	1
##	Tippah County, Mississippi	Tippecanoe County, Indiana
##	1	1
##	Tipton County, Indiana	Tishomingo County, Mississippi
##	1	1
##	Todd County, Minnesota	Tolland County, Connecticut
##	1	1
##	Tompkins County, New York	Tooele County, Utah
##	1	1
##	Toombs County, Georgia	Torrance County, New Mexico
##	1	1
##	Towns County, Georgia	Traill County, North Dakota
##	1	1
##	Traverse County, Minnesota	Trego County, Kansas
##	1	1
##	Trigg County, Kentucky	Tripp County, South Dakota
##	1	1
##	Trousdale County, Tennessee	Trumbull County, Ohio
##	1	1
##	Tucker County, West Virginia	Tulsa County, Oklahoma
##	1	1
##	Tuolumne County, California	Turner County, Georgia
##	1	1
##	Turner County, South Dakota	Tuscaloosa County, Alabama
##	1	1
##	Tuscarawas County, Ohio	Twiggs County, Georgia
##	1	1
##	Twin Falls County, Idaho	Tyler County, Texas
##	1	1
##	Tyrrell County, North Carolina	Umatilla County, Oregon
##	1	1
##	Unicoi County, Tennessee	Union County, Florida
##	1	1
##	Union County, Georgia	Union County, Iowa
##	1	1
##	Union County, Kentucky	Union County, New Jersey
##	1	1
##	Union County, New Mexico	Union County, North Carolina
##	1	1

##	Union County, Ohio	Union County, Oregon
##	1	1
##	Union County, Pennsylvania	Union County, South Carolina
##	1	1
##	Union County, South Dakota	Upson County, Georgia
##	1	1
##	Upton County, Texas	Utah County, Utah
##	1	1
##	Uvalde County, Texas	Val Verde County, Texas
##	1	1
##	Valencia County, New Mexico	Valley County, Idaho
##	1	1
##	Van Buren County, Michigan	Van Buren County, Tennessee
##	1	1
##	Van Wert County, Ohio	Van Zandt County, Texas
##	1	1
##	Vermilion County, Illinois	Vermilion Parish, Louisiana
##	1	1
##	Vermillion County, Indiana	Vernon County, Missouri
##	1	1
##	Vernon County, Wisconsin	Vernon Parish, Louisiana
##	1	1
##	Vigo County, Indiana	Vilas County, Wisconsin
##	1	1
##	Virginia Beach city, Virginia	Wabash County, Illinois
##	1	1
##	Wabaunsee County, Kansas	Wadena County, Minnesota
##	1	1
##	Wagoner County, Oklahoma	Wake County, North Carolina
##	1	1
##	Wakulla County, Florida	Walla Walla County, Washington
##	1	1
##	Walsh County, North Dakota	Walthall County, Mississippi
##	1	1
##	Walton County, Georgia	Walworth County, South Dakota
##	1	1
##	Wapello County, Iowa	Ward County, North Dakota
##	1	1
##	Ware County, Georgia	Warren County, Illinois
##	1	1
##	Warren County, Indiana	Warren County, Iowa
##	1	1
##	Warren County, Kentucky	Warren County, Mississippi
##	1	1
##	Warren County, New Jersey	Warren County, North Carolina
##	1	1
##	Warren County, Ohio	Warren County, Pennsylvania
##	1	1
##	Warrick County, Indiana	Wasatch County, Utah
##	1	1
##	Washakie County, Wyoming	Washburn County, Wisconsin
##	1	1
##	Washington County, Alabama	Washington County, Arkansas
##	1	1

##	Washington County, Colorado	Washington County, Idaho
##	1	1
##	Washington County, Illinois	Washington County, Indiana
##	1	1
##	Washington County, Kansas	Washington County, Kentucky
##	1	1
##	Washington County, Maryland	Washington County, Minnesota
##	1	1
##	Washington County, Missouri	Washington County, Nebraska
##	1	1
##	Washington County, North Carolina	Washington County, Ohio
##	1	1
##	Washington County, Texas	Washington County, Virginia
##	1	1
##	Washington County, Wisconsin	Washington Parish, Louisiana
##	1	1
##	Washita County, Oklahoma	Waushara County, Wisconsin
##	1	1
##	Wayne County, Georgia	Wayne County, Illinois
##	1	1
##	Wayne County, Iowa	Wayne County, Michigan
##	1	1
##	Wayne County, Mississippi	Wayne County, Missouri
##	1	1
##	Wayne County, New York	Wayne County, Ohio
##	1	1
##	Wayne County, Pennsylvania	Wayne County, Tennessee
##	1	1
##	Wayne County, West Virginia	Waynesboro city, Virginia
##	1	1
##	Weakley County, Tennessee	Webb County, Texas
##	1	1
##	Webster County, Missouri	Webster County, Nebraska
##	1	1
##	Webster County, West Virginia	Webster Parish, Louisiana
##	1	1
##	Weld County, Colorado	Wells County, Indiana
##	1	1
##	Wells County, North Dakota	West Carroll Parish, Louisiana
##	1	1
##	Westchester County, New York	Wetzel County, West Virginia
##	1	1
##	Wharton County, Texas	Whatcom County, Washington
##	1	1
##	Wheatland County, Montana	Wheeler County, Oregon
##	1	1
##	White County, Illinois	White Pine County, Nevada
##	1	1
##	Whiteside County, Illinois	Whitfield County, Georgia
##	1	1
##	Whitley County, Indiana	Whitley County, Kentucky
##	1	1
##	Whitman County, Washington	Wichita County, Kansas
##	1	1

##	Wilbarger County, Texas	1	Wilkes County, Georgia	1
##	Wilkes County, North Carolina	1	Wilkin County, Minnesota	1
##	Wilkinson County, Georgia	1	Wilkinson County, Mississippi	1
##	Will County, Illinois	1	Willacy County, Texas	1
##	Williams County, North Dakota	1	Williams County, Ohio	1
##	Williamsburg city, Virginia	1	Williamsburg County, South Carolina	1
##	Williamson County, Illinois	1	Williamson County, Tennessee	1
##	Wilson County, Kansas	1	Wilson County, Texas	1
##	Windham County, Vermont	1	Winkler County, Texas	1
##	Winn Parish, Louisiana	1	Winnebago County, Illinois	1
##	Wise County, Texas	1	Wise County, Virginia	1
##	Wolfe County, Kentucky	1	Wood County, Ohio	1
##	Wood County, West Virginia	1	Wood County, Wisconsin	1
##	Woodford County, Kentucky	1	Woodruff County, Arkansas	1
##	Woods County, Oklahoma	1	Woodward County, Oklahoma	1
##	Worcester County, Massachusetts	1	Worth County, Missouri	1
##	Wright County, Minnesota	1	Wyandot County, Ohio	1
##	Wyandotte County, Kansas	1	Wyoming County, Pennsylvania	1
##	Yamhill County, Oregon	1	Yancey County, North Carolina	1
##	Yankton County, South Dakota	1	Yates County, New York	1
##	Yell County, Arkansas	1	Yellowstone County, Montana	1
##	Yoakum County, Texas	1	Yolo County, California	1
##	York County, Nebraska	1	York County, Pennsylvania	1
##	York County, South Carolina	1	York County, Virginia	1
##	Yuba County, California	1	Yukon-Koyukuk Census Area, Alaska	1
##	Yuma County, Arizona	1	Yuma County, Colorado	1
##	Zavala County, Texas	1		

Variable *percentmarried*

The variable *percentmarried* represents the percentage of county residents who are married.

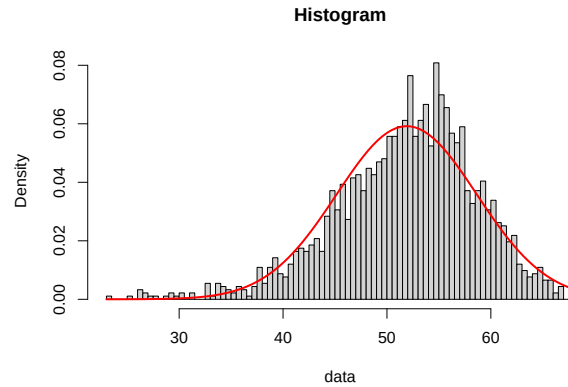
It contains no errors, as all values fall within the valid range of 0 to 100. The variable has a minimum of 23.1% and a maximum of 68.0%, with a mean of 51.9% and a median of 52.5%. The standard deviation is 6.74%, indicating moderate variability in marriage rates across counties. The distribution is slightly left-skewed, with a skewness of -0.65, suggesting a higher concentration of values near the upper end of the range. Although the Shapiro-Wilk test confirms that the distribution is not normal, this skewness does not indicate extreme asymmetry. There are 34 outliers, but none are classified as severe. The most frequent value is 52.2%, which appears 22 times.

```
variable <- df$percentmarried
table(variable)
```

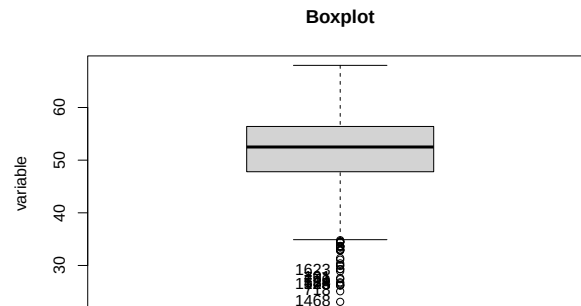
```
## variable
## 23.1 25.1 26.2 26.3 26.6 26.8 27.4 27.6 28.9 29.3 29.9 30.1 30.4 31.1 31.4 32.8
##    1    1    2    1    1    1    1    1    1    2    1    1    1    1    1    1
## 32.9 33 33.4 33.6 33.7 34.4 34.5 34.6 34.8 34.9 35.1 35.3 35.8 35.9 36 36.1
##    2    2    1    3    2    2    2    1    1    1    1    1    1    1    2    1
## 36.3 36.4 36.7 37.2 37.3 37.6 37.7 37.8 38 38.1 38.2 38.4 38.6 38.7 38.8 39
##    1    1    1    3    1    2    3    4    1    3    1    1    2    5    1    2
## 39.1 39.2 39.3 39.4 39.5 39.6 39.8 39.9 40 40.1 40.2 40.3 40.4 40.5 40.6 40.7
##    3    3    2    2    3    1    2    2    3    1    2    2    1    1    4    2
## 40.8 40.9 41 41.1 41.2 41.3 41.4 41.5 41.6 41.7 41.8 41.9 42 42.1 42.2 42.3
##    3    1    1    2    4    2    3    4    5    4    3    3    1    3    2    3
## 42.4 42.5 42.6 42.7 42.8 42.9 43 43.1 43.2 43.3 43.4 43.5 43.6 43.7 43.8 43.9
##    3    4    1    7    3    3    3    6    2    3    4    4    3    1    2    6
## 44 44.1 44.2 44.3 44.4 44.5 44.6 44.7 44.8 44.9 45 45.1 45.2 45.3 45.4 45.5
##    3    3    10    8    2    3    4    7    11    5    7    7    4    9    4    4
## 45.6 45.7 45.8 45.9 46 46.1 46.2 46.3 46.4 46.5 46.6 46.7 46.8 46.9 47 47.1
##    5    4    12    6    9    2    4    7    4    8    8    5    11    9    5    5
## 47.2 47.3 47.4 47.5 47.6 47.7 47.8 47.9 48 48.1 48.2 48.3 48.4 48.5 48.6 48.7
##   10    7    8    9    9    8    7    5    5    4    15    9    7    6    5    14
## 48.8 48.9 49 49.1 49.2 49.3 49.4 49.5 49.6 49.7 49.8 49.9 50 50.1 50.2 50.3
##    7    5    8    12    8    6    6    11    7    11    6    8    12    16    11    4
## 50.4 50.5 50.6 50.7 50.8 50.9 51 51.1 51.2 51.3 51.4 51.5 51.6 51.7 51.8 51.9
##   12    8    10    9    10    11    11    19    4    10    11    10    12    15    7    12
## 52 52.1 52.2 52.3 52.4 52.5 52.6 52.7 52.8 52.9 53 53.1 53.2 53.3 53.4 53.5
##   10   14   22   10   15    9    8   17   10    9    7    7   17   12    6   14
## 53.6 53.7 53.8 53.9 54 54.1 54.2 54.3 54.4 54.5 54.6 54.7 54.8 54.9 55 55.1
##   15    8    8   17   13    7   12    5   17    7   14   16   11   13   20   14
## 55.2 55.3 55.4 55.5 55.6 55.7 55.8 55.9 56 56.1 56.2 56.3 56.4 56.5 56.6 56.7
##   19   10    6   15   12   16   10   12   10    8    9   13   11   11   10   14
## 56.8 56.9 57 57.1 57.2 57.3 57.4 57.5 57.6 57.7 57.8 57.9 58 58.1 58.2 58.3
##   10    5   10    9   12    7   16   10    7    8    9    5    5    7    4    4
## 58.4 58.5 58.6 58.7 58.8 58.9 59 59.1 59.2 59.3 59.4 59.5 59.6 59.7 59.8 59.9
##    9    6   11    7    3    6    7    4    5    5   11   12    6    6   11    4
## 60 60.1 60.2 60.3 60.4 60.5 60.6 60.7 60.8 60.9 61 61.1 61.2 61.3 61.4 61.5
##    1    6    7    5    4    9    6    5    3    5    5    5    7    2    7    2
## 61.6 61.7 61.8 61.9 62 62.1 62.2 62.3 62.4 62.5 62.6 62.7 62.8 62.9 63 63.1
##    2    4    5    2    5    1    6    8    1    4    1    5    2    2    1    2
## 63.2 63.3 63.5 63.7 63.8 63.9 64 64.1 64.2 64.4 64.5 64.6 64.7 64.8 64.9 65
##    3    2    2    1    1    2    3    2    2    2    2    2    1    3    3    1
## 65.1 65.2 65.3 65.5 65.6 65.9 66 66.2 66.3 66.6 66.7 67 68
##    1    1    2    2    1    1    4    1    1    1    2    1    1
```

```
res.var <- summary_and_normality_check(variable)
```

```
## [1] "Summary: "
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##   23.1   47.8   52.5   51.9   56.4   68.0
## [1] "Mean"
## [1] 51.90087
## [1] "Standard deviation"
## [1] 6.740129
```



```
## [1] "Skewness"
## [1] -0.6474867
```



```
## [1] "Num Outliers:"
## [1] 34
## [1] "Num. severe outliers:"
## [1] 0
## [1] "Shapiro test: "
##
## Shapiro-Wilk normality test
##
## data: data
## W = 0.97753, p-value = 2.346e-16
##
## [1] "Most repeated value ad times it occurs: "
## [1] "52.2"
```

```
## [1] 22
```

```
outliers_df[res.var,"percentmarried"] <- 1
```

Variable *pctnohs18_24*

The numerical variable *pctnohs18_24* represents the percentage of county residents aged 18-24 whose highest level of education is less than high school. The values range from 0.5% to 59.1%, with a mean of 18.29% and a median of 17.2%. The distribution of the variable is positively skewed, with a skewness of 0.89, suggesting that a higher proportion of counties have lower values (fewer residents with less than high school education), while some counties exhibit higher percentages. The Shapiro-Wilk test for normality confirms a significant deviation from normality. There are 35 outliers, with 5 classified as severe.

```
variable <- df$pctnohs18_24
```

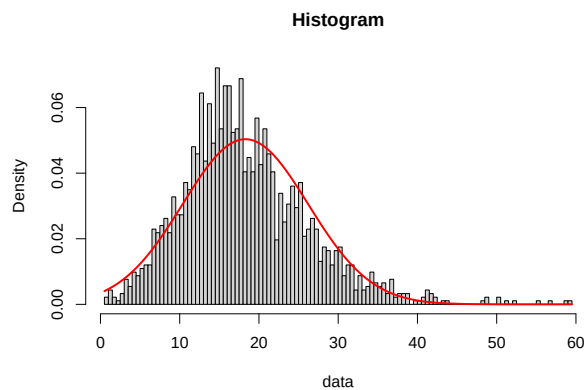
```
table(df$pctnohs18_24)
```

```
##
## 0.5 0.8 1.2 1.4 1.5 1.6 1.7 2.3 2.6 2.7 2.8 3.1 3.3 3.4 3.5 3.7
## 1 1 2 1 1 1 1 1 1 1 1 2 1 2 2 1
## 3.8 3.9 4.1 4.2 4.3 4.4 4.5 4.7 4.8 4.9 5 5.1 5.2 5.3 5.4 5.5
## 1 3 2 1 2 1 3 1 4 2 1 2 3 1 2 2
## 5.6 5.7 5.8 5.9 6 6.1 6.3 6.5 6.6 6.7 6.8 6.9 7 7.1 7.2 7.3
## 1 5 3 1 1 2 4 5 2 4 2 5 8 7 3 6
## 7.4 7.5 7.6 7.7 7.8 7.9 8 8.1 8.2 8.3 8.4 8.5 8.7 8.8 8.9 9
## 1 3 3 7 6 2 4 5 7 3 5 4 7 3 4 6
## 9.1 9.2 9.3 9.4 9.5 9.6 9.7 9.8 9.9 10 10.1 10.2 10.3 10.4 10.5 10.6
## 4 7 9 6 4 5 6 3 5 6 2 4 2 6 11 5
## 10.7 10.8 10.9 11 11.1 11.2 11.3 11.4 11.5 11.6 11.7 11.8 11.9 12 12.1 12.2
## 10 8 8 3 8 5 6 4 9 2 9 8 15 10 10 6
## 12.3 12.4 12.5 12.6 12.7 12.8 12.9 13 13.1 13.2 13.3 13.4 13.5 13.6 13.7 13.8
## 7 8 11 14 11 15 7 12 9 9 6 8 8 12 10 6
## 13.9 14 14.1 14.2 14.3 14.4 14.5 14.6 14.7 14.8 14.9 15 15.1 15.2 15.3 15.4
## 11 17 8 10 6 12 9 10 16 16 11 13 6 13 11 11
## 15.5 15.6 15.7 15.8 15.9 16 16.1 16.2 16.3 16.4 16.5 16.6 16.7 16.8 16.9 17
## 8 14 13 12 13 9 13 10 16 11 11 8 11 9 9 11
## 17.1 17.2 17.3 17.4 17.5 17.6 17.7 17.8 17.9 18 18.1 18.2 18.3 18.4 18.5 18.6
## 15 8 12 7 7 14 17 8 15 9 2 6 9 7 13 10
## 18.7 18.8 18.9 19 19.1 19.2 19.3 19.4 19.5 19.6 19.7 19.8 19.9 20 20.1 20.2
## 11 6 9 5 7 7 6 9 8 8 7 14 11 12 6 5
## 20.3 20.4 20.5 20.6 20.7 20.8 20.9 21 21.1 21.2 21.3 21.4 21.5 21.6 21.7 21.8
## 13 8 7 9 6 12 15 7 11 6 8 9 8 5 10 5
## 21.9 22 22.2 22.3 22.4 22.5 22.6 22.7 22.8 22.9 23 23.1 23.2 23.3 23.4 23.5
## 9 8 4 8 3 3 6 8 6 3 8 6 3 3 6 5
## 23.6 23.7 23.8 23.9 24 24.1 24.2 24.3 24.4 24.5 24.6 24.7 24.8 24.9 25 25.1
## 5 5 5 7 6 9 2 5 8 9 4 8 4 5 6 13
## 25.2 25.3 25.4 25.5 25.6 25.7 25.8 25.9 26 26.1 26.2 26.3 26.4 26.5 26.6 26.7
## 10 5 2 4 3 7 5 3 1 4 3 6 4 4 3 9
## 26.8 26.9 27 27.1 27.2 27.3 27.4 27.5 27.6 27.7 27.8 27.9 28 28.1 28.2 28.3
## 3 6 3 6 3 7 4 1 4 2 2 1 3 4 1 4
## 28.4 28.5 28.6 28.7 28.8 28.9 29 29.1 29.2 29.3 29.4 29.5 29.6 29.8 29.9 30
## 3 4 4 1 3 4 3 3 2 2 3 1 5 3 3 4
## 30.1 30.2 30.3 30.4 30.5 30.6 30.9 31 31.1 31.2 31.3 31.4 31.5 31.6 31.7 31.9
## 3 4 2 4 3 1 2 5 2 2 2 1 4 3 2 1
## 32 32.2 32.3 32.4 32.6 32.7 32.8 32.9 33 33.1 33.2 33.3 33.5 33.6 33.8 33.9
## 5 1 1 2 1 1 1 1 1 4 1 1 1 1 1 1
```

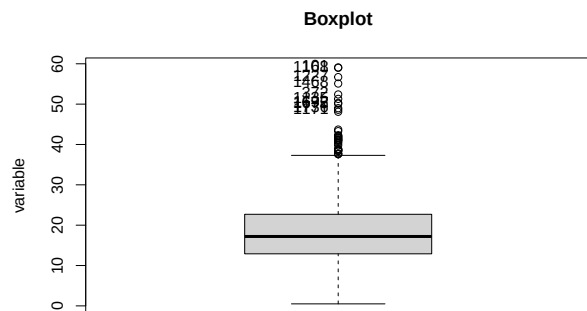
```
## 34 34.1 34.2 34.3 34.5 34.6 34.7 34.9 35 35.1 35.2 35.3 35.6 35.7 35.8 35.9
## 2 2 4 1 2 1 3 1 1 3 1 1 1 1 2 1
## 36 36.1 36.4 36.5 36.6 36.7 36.8 36.9 37 37.1 37.3 37.6 37.8 38.4 38.5 38.7
## 1 1 1 1 1 1 2 2 1 1 1 2 1 2 1 1
## 38.8 38.9 39.8 40.3 40.7 40.9 41.1 41.2 41.5 41.7 42 42.1 42.3 43.3 43.7 48.1
## 1 1 1 1 1 1 1 2 1 2 1 1 1 1 1 1
## 48.6 49 50.1 50.3 51.3 52.4 55.1 56.7 59 59.1
## 1 1 1 1 1 1 1 1 1 1
```

```
res.var <- summary_and_normality_check(variable)
```

```
## [1] "Summary: "
## Min. 1st Qu. Median Mean 3rd Qu. Max.
## 0.50 12.90 17.20 18.29 22.70 59.10
## [1] "Mean"
## [1] 18.28777
## [1] "Standard deviation"
## [1] 7.92004
```



```
## [1] "Skewness"
## [1] 0.8925103
```



```
## [1] "Num Outliers:"
## [1] 35
## [1] "Num. severe outliers:"
## [1] 5
## [1] "Shapiro test: "
```

```
##
## Shapiro-Wilk normality test
##
## data: data
## W = 0.96205, p-value < 2.2e-16
##
## [1] "Most repeated value ad times it occurs: "
## [1] "14"
## [1] 17
```

```
outliers_df[res.var,"pctnohs18_24"] <- 1
```

Variable *ppcths18_24*

The variable *pcths18_24* represents the percentage of county residents aged 18-24 whose highest level of education is a high school diploma.

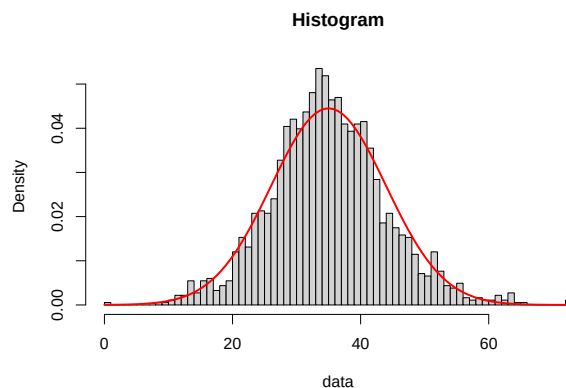
Note that there is a county with a 0% (Greeley County, Kansas), which potentially is an error.

The values range from 0% to 72.5%, with a mean of 35% and a median of 34.7%. The standard deviation is 8.96%, indicating substantial variability in high school graduation rates across counties. The skewness is 0.22, suggesting a slight positive skew, meaning that most counties have lower percentages of residents with only a high school diploma, while a few counties exhibit higher values. The Shapiro-Wilk test confirms that the distribution is not normal. There are 33 outliers, none of which are severe.

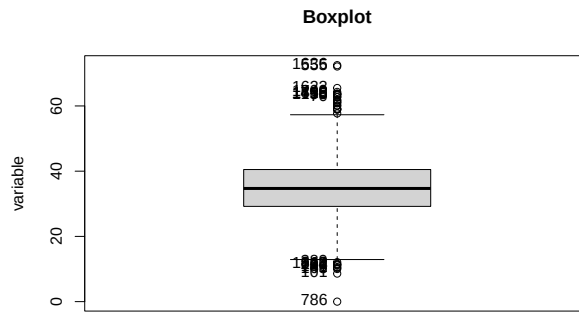
This variable is related to *pctnohs18_24* (the percentage of residents aged 18-24 without a high school diploma), and we can consider creating a factor variable for characterizing this relation.

```
variable <- df$pcths18_24
ll<- which(df$pcths18_24 == 0);
res.var <- summary_and_normality_check(df$pcths18_24)
```

```
## [1] "Summary: "
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##   0.0   29.2   34.7   35.0   40.5   72.5
## [1] "Mean"
## [1] 34.99978
## [1] "Standard deviation"
## [1] 8.964462
```



```
## [1] "Skewness"
## [1] 0.2205398
```



```
## [1] "Num Outliers:"
## [1] 33
## [1] "Num. severe outliers:"
## [1] 0
## [1] "Shapiro test: "
##
## Shapiro-Wilk normality test
##
## data: data
## W = 0.99323, p-value = 1.922e-07
##
## [1] "Most repeated value ad times it occurs: "
## [1] "33.7"
## [1] 17
```

```
outliers_df[res.var,"pcths18_24"] <- 1
table(df$pcths18_24)
```

```
##
##      0  8.6   10 10.3 10.6 11.1 11.3 11.4 11.6 12.1 12.9   13 13.1 13.3 13.4 13.6
##      1   1    1   1   1   1   1   1   1   1   2   1   3   1   2   1
## 13.7   14 14.2 14.7 14.8   15 15.1 15.4 15.5 15.7 15.9   16 16.1 16.2 16.4 16.5
##      2   1   1   1   2   1   3   2   1   2   1   1   2   1   2   1
## 16.6 16.7 16.9 17.2 17.3 17.5 17.8   18 18.1 18.2 18.3 18.5 18.7 18.9 19.2 19.5
##      3   1   1   2   1   1   1   1   1   1   1   1   2   2   3   1
## 19.6 19.9   20 20.1 20.2 20.3 20.4 20.5 20.6 20.7 20.8 20.9   21 21.1 21.2 21.3
##      3   1   2   1   3   4   3   1   2   3   1   3   1   4   1   4
## 21.4 21.5 21.6 21.7 21.9   22 22.1 22.2 22.4 22.5 22.6 22.7 22.8 22.9   23 23.1
##      1   1   5   3   5   4   1   3   5   2   4   2   4   2   1   3
## 23.2 23.3 23.4 23.5 23.6 23.7 23.8 23.9   24 24.1 24.2 24.3 24.4 24.5 24.6 24.7
##      3   3   1   9   5   4   4   4   2   3   2   4   2   4   4   4
## 24.8 24.9   25 25.1 25.2 25.3 25.4 25.5 25.6 25.7 25.8 25.9   26 26.1 26.2 26.3
##      3   5   8   5   6   2   2   3   5   2   4   5   4   3   4   6
## 26.4 26.5 26.6 26.7 26.8 26.9   27 27.1 27.2 27.3 27.4 27.5 27.6 27.7 27.8 27.9
##      3   2   7   3   7   3   6   4   3   5   7   9   4   8   4   10
##   28 28.1 28.2 28.3 28.4 28.5 28.6 28.7 28.8 28.9   29 29.1 29.2 29.3 29.4 29.5
##      6   6   5   8   6   10   6   7   12   3   11   12   9   8   7   6
## 29.6 29.7 29.8 29.9   30 30.1 30.2 30.3 30.4 30.5 30.6 30.7 30.8 30.9   31 31.1
##      4   10   5   12   4   11   8   5   7   9   5   5   6   9   8   12
## 31.2 31.3 31.4 31.5 31.6 31.7 31.8 31.9   32 32.1 32.2 32.3 32.4 32.5 32.6 32.7
```



```
##      6      8      11      9      5      5      10      5      9      7      5      11      12      8      8      12
## 32.8 32.9      33 33.1 33.2 33.3 33.4 33.5 33.6 33.7 33.8 33.9      34 34.1 34.2 34.3
##      8      8      9      5      9      12      5      4      11      17      10      11      14      7      5      6
## 34.4 34.5 34.6 34.7 34.8 34.9      35 35.1 35.2 35.3 35.4 35.5 35.6 35.7 35.8 35.9
##      15      13      12      10      10      5      12      12      12      9      5      12      9      3      7      11
##      36 36.1 36.2 36.3 36.4 36.5 36.6 36.7 36.8 36.9      37 37.1 37.2 37.3 37.4 37.5
##      5      9      10      10      11      13      9      4      1      10      9      7      8      6      4      7
## 37.6 37.7 37.8 37.9      38 38.1 38.2 38.3 38.4 38.5 38.6 38.7 38.8 38.9      39 39.1
##      12      11      7      8      5      9      5      7      6      8      9      7      5      11      5      9
## 39.2 39.3 39.4 39.5 39.6 39.7 39.8 39.9      40 40.1 40.2 40.3 40.4 40.5 40.6 40.7
##      9      8      10      7      5      7      5      5      10      10      7      2      9      8      7      14
## 40.8 40.9      41 41.1 41.2 41.3 41.4 41.5 41.6 41.7 41.8 41.9      42 42.1 42.2 42.3
##      6      6      7      7      6      12      9      5      5      1      9      5      6      9      4      8
## 42.4 42.5 42.6 42.7 42.8 42.9      43 43.1 43.2 43.3 43.4 43.5 43.6 43.7 43.8 43.9
##      3      7      2      2      4      7      6      2      1      5      5      5      4      2      1      7
##      44 44.1 44.2 44.3 44.4 44.5 44.6 44.7 44.8 44.9      45 45.1 45.2 45.3 45.4 45.5
##      2      5      2      5      7      8      2      2      2      4      1      2      4      4      2      5
## 45.6 45.7 45.8 45.9      46 46.1 46.2 46.3 46.4 46.6 46.7 46.8 46.9      47 47.1 47.2
##      6      2      4      2      1      3      6      2      3      3      4      4      3      1      4      3
## 47.3 47.4 47.5 47.6 47.7 47.8 47.9      48 48.2 48.3 48.4 48.5 48.7 48.8 48.9      49
##      3      3      1      6      2      3      2      1      1      1      4      5      3      3      2      2
## 49.3 49.4 49.5 49.6 49.7 49.9      50 50.1 50.2 50.3 50.4 50.5 50.6 50.8 50.9      51
##      3      3      1      2      1      1      2      2      2      1      1      1      1      1      1      2
## 51.1 51.2 51.3 51.4 51.5 51.6 51.7 51.8      52 52.1 52.2 52.3 52.4 52.5 52.8      53
##      4      1      2      2      1      4      2      3      3      3      1      3      1      2      3      1
## 53.1 53.3 53.5 53.6 53.8 53.9 54.2 54.3 54.4 54.6 54.7 55.1 55.2 55.3 55.5 55.6
##      1      1      2      1      1      2      3      1      1      1      1      1      1      3      1      1
## 55.8 55.9 56.4 56.8 57.3 57.7 58.9      59 59.2      60 60.2 60.8 61.1 61.3 61.7 61.8
##      1      1      2      1      1      1      2      1      1      1      1      1      1      1      1      1
## 62.7 63.2 63.5 63.6 63.7 63.8 64.3 65.5 72.1 72.5
##      2      1      1      1      1      1      1      1      1      1      1
```

Variable *pctsomecol18_24*

The variable *pctsomecol18_24* represents the percentage of county residents aged 18-24 whose highest level of education attained is some college. However, this variable contains 1,376 missing values, which accounts for approximately 75% of the total observations. Given the high proportion of missing data, further steps will be taken to address the missing values, either through imputation methods to obtain valid results or by deleting the variable if it cannot be reasonably imputed.

```
summary(df$pctsomecol18_24)
```

```
##      Min. 1st Qu.  Median      Mean 3rd Qu.     Max.    NA's
##      9.60   33.25   40.10   40.48   46.10   78.30   1376
```

Variable *pctbachdeg18_24*

The variable *pctbachdeg18_24* represents the percentage of county residents aged 18-24 who have attained a bachelor's degree. It ranges from 0% to 51.8%, with a mean of 6.22% and a median of 5.4%. The standard deviation is 4.52%, indicating notable variability across counties. A key feature of this variable is the presence of 73 zero values, meaning that in many counties, no residents aged 18-24 have a bachelor's degree. The skewness is 1.98, reflecting a strongly positive skew, where most counties have relatively low percentages, while a few counties exhibit exceptionally high values. The Shapiro-Wilk test indicates that the normality assumption is rejected. The variable contains 56 mild outliers, and 10 of them classified as severe.

It may also be particularly insightful when combined with related variables, such as *peths18_24* and

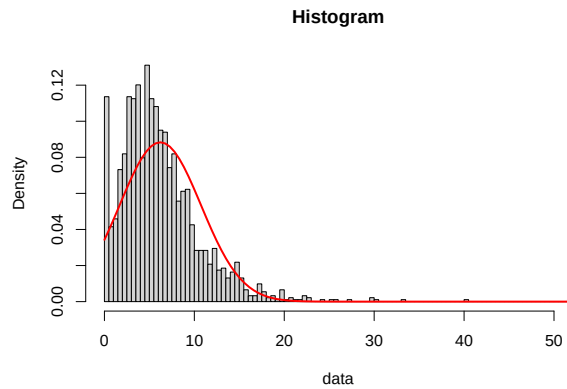
pctsomecol18_24, to explore the full spectrum of educational attainment.

```
variable <- df$pctbachdeg18_24
table(df$pctbachdeg18_24)
```

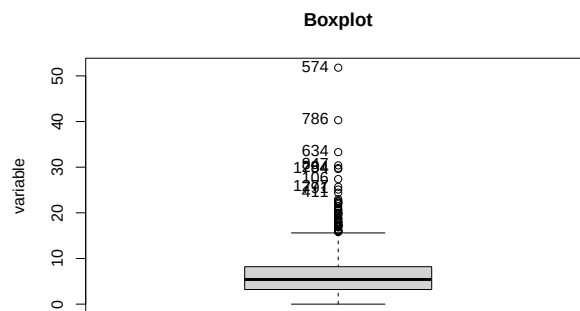
```
##
##      0 0.1 0.2 0.3 0.4 0.5 0.6 0.7 0.8 0.9  1  1.1  1.2  1.3  1.4  1.5
##    73   4   8   2   4  13   6   7   7   7  11  13   7   8   9   5
##   1.6 1.7 1.8 1.9   2 2.1 2.2 2.3 2.4 2.5 2.6 2.7 2.8 2.9   3 3.1
##   14   8  25   9  11  11  14  17  18  15  20  29  24  17  14  24
##   3.2 3.3 3.4 3.5 3.6 3.7 3.8 3.9   4 4.1 4.2 4.3 4.4 4.5 4.6 4.7
##   19  21  15  24  22  22  19  23  24  12  17  14  10  20  21  28
##   4.8 4.9   5 5.1 5.2 5.3 5.4 5.5 5.6 5.7 5.8 5.9   6 6.1 6.2 6.3
##   25  22  24  26  19  26  11  21  15  26  19  19  20  25   9  16
##   6.4 6.5 6.6 6.7 6.8 6.9   7 7.1 7.2 7.3 7.4 7.5 7.6 7.7 7.8 7.9
##   19  18   9  21  20  16  20  10  16  11  14  17  19  17  14  12
##    8 8.1 8.2 8.3 8.4 8.5 8.6 8.7 8.8 8.9   9 9.1 9.2 9.3 9.4 9.5
##   13  16  10  10  11   4  10  17   8  11  10  10  19  12   7   9
##   9.6 9.7 9.8 9.9  10 10.1 10.2 10.3 10.4 10.5 10.6 10.7 10.8 10.9  11 11.1
##    6   6   2   9  16   6   3   5   5   7   6   6   5   6   3   3
##  11.2 11.3 11.4 11.5 11.6 11.7 11.8 11.9  12 12.1 12.2 12.3 12.4 12.5 12.6 12.7
##    4   8   2   9   6   6   1   3   3   8   1   7   5   6   4   4
##  12.8 12.9  13 13.1 13.2 13.3 13.4 13.5 13.6 13.7 13.8 13.9  14 14.1 14.2 14.3
##    1   2   5   2   4   1   6   4   1   4   4   1   2   2   4   2
##  14.4 14.5 14.6 14.7 14.8 14.9  15 15.1 15.2 15.3 15.5 15.6 15.8 15.9 16.1 16.2
##    4   3   4   4   4   4   4   3   3   2   4   2   1   3   1   1
##  16.3 16.8 16.9  17 17.1 17.2 17.3 17.4 17.7 17.8 17.9  18 18.1 18.5 18.7 18.8
##    1   1   1   1   3   2   2   2   1   1   1   2   1   1   1   1
##  18.9 19.4 19.6 19.7 19.9  20 20.4 20.6 20.8 21.4 21.9 22.2 22.5 22.6  23 24.4
##    1   1   1   1   3   1   1   1   1   1   1   2   1   1   1   1
##  25.1 25.7 27.4 29.6 29.8 30.4 33.3 40.3 51.8
##    1   1   1   1   1   1   1   1   1
```

```
res.var <- summary_and_normality_check(variable)
```

```
## [1] "Summary: "
##      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##    0.000  3.200   5.400   6.216   8.200  51.800
## [1] "Mean"
## [1] 6.216221
## [1] "Standard deviation"
## [1] 4.517556
```



```
## [1] "Skewness"
## [1] 1.980584
```



```
## [1] "Num Outliers:"
## [1] 56
## [1] "Num. severe outliers:"
## [1] 10
## [1] "Shapiro test: "
##
## Shapiro-Wilk normality test
##
## data: data
## W = 0.87959, p-value < 2.2e-16
##
## [1] "Most repeated value ad times it occurs: "
## [1] "0"
## [1] 73
```

```
outliers_df[res.var,"pctbachdeg18_24"] <- 1
```

Variable *pcths25_over*

The variable *pcths25_over* represents the percentage of county residents aged 25 and over whose highest level of education is a high school diploma.

It ranges from 8.3% to 52.7%, with a mean of 34.73% and a median of 35.3%. The standard deviation is 7.07%. The skewness of -0.37 reflects a slight left skew, suggesting a concentration of values toward the higher

end of the range. The Shapiro-Wilk test indicates that the normality assumption is rejected. The variable has 18 outliers, none of which are severe. The most frequent value is 35.6%, occurring 19 times.

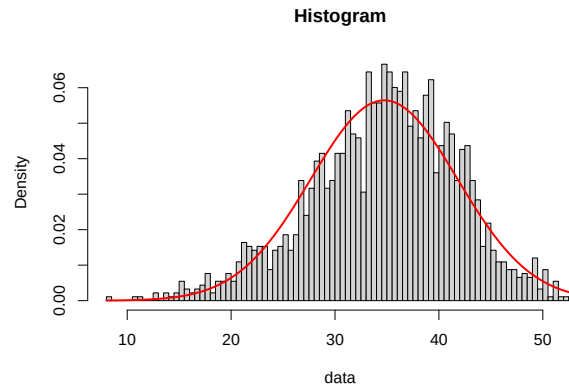
```
variable <- df$pcths25_over
table(variable)
```

```
## variable
## 8.3 10.8 11.5 12.9 13 13.8 14 14.2 14.7 14.9 15.2 15.5 15.6 15.9 16 16.4
## 1 1 1 1 1 1 1 1 1 1 2 3 1 1 1 1
## 16.5 16.8 17 17.1 17.2 17.3 17.4 17.7 17.8 17.9 18 18.2 18.4 18.7 18.8 19
## 1 2 1 1 1 1 1 3 1 1 2 1 1 2 1 2
## 19.1 19.3 19.4 19.6 19.7 19.8 19.9 20 20.1 20.2 20.3 20.4 20.6 20.7 20.8 20.9
## 1 2 2 1 2 1 1 2 2 1 1 1 4 2 1 1
## 21 21.1 21.2 21.3 21.4 21.5 21.6 21.7 21.8 21.9 22 22.1 22.2 22.3 22.4 22.5
## 2 4 3 4 3 1 3 2 4 3 2 3 1 2 4 3
## 22.6 22.7 22.8 22.9 23 23.1 23.2 23.3 23.4 23.6 23.7 23.8 23.9 24 24.1 24.2
## 3 1 4 1 5 4 5 1 4 2 1 3 1 1 3 1
## 24.3 24.4 24.5 24.6 24.7 24.8 24.9 25 25.1 25.2 25.3 25.4 25.5 25.6 25.7 25.8
## 3 4 2 1 3 3 2 5 1 7 3 1 5 6 1 3
## 26 26.1 26.2 26.3 26.4 26.5 26.6 26.7 26.8 26.9 27 27.1 27.2 27.3 27.4 27.5
## 3 6 5 2 2 2 5 5 3 7 11 4 6 6 3 3
## 27.6 27.7 27.8 27.9 28 28.1 28.2 28.3 28.4 28.5 28.6 28.7 28.8 28.9 29 29.1
## 5 4 4 11 5 8 6 6 10 6 12 4 6 12 4 9
## 29.2 29.3 29.5 29.6 29.7 29.8 29.9 30 30.1 30.2 30.3 30.4 30.5 30.6 30.7 30.8
## 5 8 7 6 7 5 6 7 8 7 7 8 8 8 6 10 7
## 30.9 31 31.1 31.2 31.3 31.4 31.5 31.6 31.7 31.8 31.9 32 32.1 32.2 32.3 32.4
## 7 8 10 10 11 11 7 8 7 13 6 9 16 8 7 2
## 32.5 32.6 32.7 32.8 32.9 33 33.1 33.2 33.3 33.4 33.5 33.6 33.7 33.8 33.9 34
## 9 6 4 3 10 5 12 11 9 17 10 9 13 11 9 9
## 34.1 34.2 34.3 34.4 34.5 34.6 34.7 34.8 34.9 35 35.1 35.2 35.3 35.4 35.5 35.6
## 6 6 13 13 13 9 15 12 12 13 12 6 13 15 13 19
## 35.7 35.8 35.9 36 36.1 36.2 36.3 36.4 36.5 36.6 36.7 36.8 36.9 37 37.1 37.2
## 7 11 8 10 9 7 12 14 12 16 10 13 11 9 11 10
## 37.3 37.4 37.5 37.6 37.7 37.8 37.9 38 38.1 38.2 38.3 38.4 38.5 38.6 38.7 38.8
## 9 6 9 14 11 9 7 8 9 6 13 5 9 12 7 13
## 38.9 39 39.1 39.2 39.3 39.4 39.5 39.6 39.7 39.8 39.9 40 40.1 40.2 40.3 40.4
## 12 9 13 13 10 8 13 4 6 9 7 7 6 10 5 3
## 40.5 40.6 40.7 40.8 40.9 41 41.1 41.2 41.3 41.4 41.5 41.6 41.7 41.8 41.9 42
## 16 11 9 6 8 12 10 12 12 5 4 6 3 6 7 9
## 42.1 42.2 42.3 42.4 42.5 42.6 42.7 42.8 42.9 43 43.1 43.2 43.3 43.4 43.5 43.6
## 9 12 14 3 1 6 7 15 7 5 4 8 8 6 5 5
## 43.7 43.8 43.9 44 44.1 44.2 44.3 44.4 44.5 44.6 44.7 44.8 44.9 45 45.1 45.2
## 4 3 6 8 1 4 4 1 4 4 2 8 4 2 3 2
## 45.3 45.4 45.5 45.6 45.7 45.8 45.9 46 46.1 46.2 46.3 46.4 46.5 46.6 46.7 46.8
## 5 1 2 2 2 1 1 4 2 2 2 2 2 2 3 1
## 46.9 47.1 47.2 47.3 47.4 47.5 47.7 47.9 48 48.1 48.2 48.6 48.7 48.9 49.1 49.2
## 2 1 1 3 1 2 2 1 3 2 5 1 3 2 3 2
## 49.3 49.4 49.7 49.9 50 50.1 50.2 50.3 50.4 50.9 51.1 51.2 51.7 52.1 52.7
## 2 4 1 1 1 3 3 1 1 1 1 4 1 1 1
```

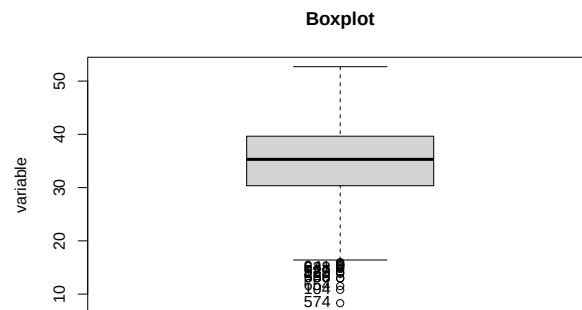
```
res.var <- summary_and_normality_check(variable)
```

```
## [1] "Summary: "
## Min. 1st Qu. Median Mean 3rd Qu. Max.
## 8.30 30.35 35.30 34.73 39.65 52.70
```

```
## [1] "Mean"
## [1] 34.73255
## [1] "Standard deviation"
## [1] 7.06599
```



```
## [1] "Skewness"
## [1] -0.3660496
```



```
## [1] "Num Outliers:"
## [1] 18
## [1] "Num. severe outliers:"
## [1] 0
## [1] "Shapiro test: "
##
## Shapiro-Wilk normality test
##
## data: data
## W = 0.99107, p-value = 3.741e-09
##
## [1] "Most repeated value ad times it occurs: "
## [1] "35.6"
## [1] 19
```

```
outliers_df[res.var,"pcths25_over"] <- 1
```

Variable *pctbachdeg25_over*

The variable *pctbachdeg25_over* captures the percentage of county residents aged 25 and over with a bachelor's degree as their highest level of education.

The distribution is positively skewed, with most counties at the lower end of the range (2.5% to 42.2%), while a few have much higher values. The mean is 13.3%, close to the median of 12.3%, with a standard deviation of 5.47%. Outliers are identified in 59 counties, including 3 severe ones. The variable does not follow a normal distribution.

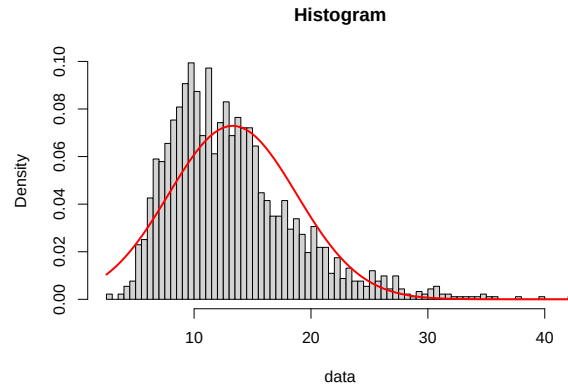
This variable is essential for understanding higher education attainment in adults and its variability across counties. When analyzed alongside related variables like *pcths25_over*, it can help reveal regional educational trends.

```
variable <- df$pctbachdeg25_over
table(variable)
```

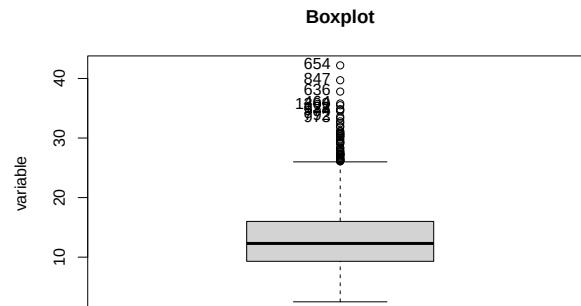
```
## variable
##  2.5  2.7  3.9  4.3  4.4  4.6  4.7  4.8  4.9   5  5.1  5.2  5.3  5.4  5.5  5.6
##    1    1    2    3    2    2    2    1    1    1    6    2    5    4    4    5
##  5.7  5.8  5.9    6  6.1  6.2  6.3  6.4  6.5  6.6  6.7  6.8  6.9    7  7.1  7.2
##    5    2    4    7   12    8    6    5    8   11   13   11   11    8    8   10
##  7.3  7.4  7.5  7.6  7.7  7.8  7.9    8  8.1  8.2  8.3  8.4  8.5  8.6  8.7  8.8
##   12   11   12   12   17    7   11   13   16   19    4   16   14    6   16   11
##  8.9    9  9.1  9.2  9.3  9.4  9.5  9.6  9.7  9.8  9.9   10 10.1 10.2 10.3 10.4
##   15   26   16   15   19   20   13   11   18   18   21   23   22   11   19   11
## 10.5 10.6 10.7 10.8 10.9   11 11.1 11.2 11.3 11.4 11.5 11.6 11.7 11.8 11.9   12
##   17   14   17   12    8   12   18   16   17   16   22   13   12   12    9   10
## 12.1 12.2 12.3 12.4 12.5 12.6 12.7 12.8 12.9   13 13.1 13.2 13.3 13.4 13.5 13.6
##   10   17   19   11   11   15   17   14   14   16   15   11    9   13   15   19
## 13.7 13.8 13.9   14 14.1 14.2 14.3 14.4 14.5 14.6 14.7 14.8 14.9   15 15.1 15.2
##   14   11   12   14   14   19    7   12   14   15   17    7   16   11   13   17
## 15.3 15.4 15.5 15.6 15.7 15.8 15.9   16 16.1 16.2 16.3 16.4 16.5 16.6 16.7 16.8
##   10   10    9   12    7    6    8    8    5    9    6   12    6    6    6    8
## 16.9   17 17.1 17.2 17.3 17.4 17.5 17.6 17.7 17.8 17.9   18 18.1 18.2 18.3 18.4
##    8    4    7    5    7    4    9    9    9    7    7    6    5    6    5    6
## 18.5 18.6 18.7 18.8 18.9   19 19.1 19.2 19.3 19.4 19.5 19.6 19.7 19.8 19.9   20
##    5    5   11    5    5    5    5    5    6    3    6    2    5    7    1    3
## 20.1 20.2 20.3 20.4 20.5 20.6 20.7 20.8 20.9   21 21.1 21.2 21.3 21.4 21.5 21.6
##    2    6    4   12    4    2    4    7    5    2    5    3    3    3    6    5
## 21.7 21.9 22.1 22.2 22.3 22.5 22.6 22.7 22.8 22.9   23 23.1 23.2 23.3 23.5 23.6
##    4    1    2    7    4    3    2    2    2    1    1    2    5    2    3    1
## 23.7 23.9   24 24.1 24.2 24.3 24.4 24.7 24.8 24.9   25 25.1 25.2 25.3 25.4 25.5
##    2    2    2    3    2    1    1    1    1    1    2    4    2    1    3    1
## 25.6 25.7 25.8 25.9   26 26.1 26.2 26.3 26.4 26.5 26.6 26.7 26.9 27.1 27.2 27.3
##    1    2    2    1    1    2    2    2    2    1    1    1    2    2    1    4
## 27.4 27.6 27.7 27.8   28 28.3 28.4   29 29.2 29.3 29.6 29.8 30.1 30.3 30.6 30.7
##    2    1    1    1    1    1    1    1    1    2    1    1    1    3    2    1
## 30.8 30.9 31.1 31.2 31.8   32 32.5 32.8 33.3 33.6 34.5 34.8 35.5 35.8 37.8 39.7
##    1    1    1    1    1    1    1    1    1    1    1    2    1    1    1    1
## 42.2
##    1
```

```
res.var <- summary_and_normality_check(variable)
```

```
## [1] "Summary: "
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##    2.5    9.3   12.3   13.3   16.0   42.2
## [1] "Mean"
## [1] 13.29754
## [1] "Standard deviation"
## [1] 5.470341
```



```
## [1] "Skewness"
## [1] 1.159947
```



```
## [1] "Num Outliers:"
## [1] 59
## [1] "Num. severe outliers:"
## [1] 3
## [1] "Shapiro test: "
##
## Shapiro-Wilk normality test
##
## data: data
## W = 0.92998, p-value < 2.2e-16
##
## [1] "Most repeated value ad times it occurs: "
## [1] "9"
```

```
## [1] 26
```

```
outliers_df[res.var,"pctbachdeg25_over"] <- 1
```

Variable *pctemployed16_over*

pctemployed16_over represents the percentage of county residents aged 16 and over who are employed. It ranges from 23.9% to 80.1%, with a mean of 54.21% and a median of 54.5%. The data has a standard deviation of 8.22%. Its skewness of -0.34 suggests a slight left skew, meaning there are slightly more counties with higher employment percentages. The Shapiro-Wilk test rejects the normality assumption. There are 11 outliers, none of which are severe.

Additionally, 82 missing values (NA) are present, which require further handling (imputation section).

```
variable<-df$pctemployed16_over  
table(variable)
```

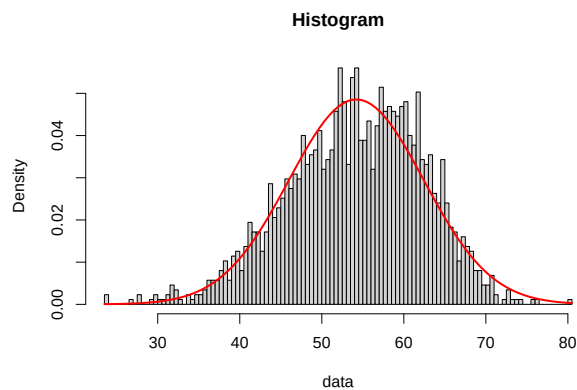
```
## variable  
## 23.9 24 27 27.8 27.9 29.3 29.6 29.7 30.3 30.9 31.2 31.6 31.7 31.9 32.2 32.3  
## 1 1 1 1 1 1 1 1 1 1 2 1 2 1 1 1  
## 32.4 32.8 33.9 34.4 34.9 35 35.2 35.3 35.6 35.8 35.9 36.1 36.2 36.5 36.6 36.7  
## 1 1 2 1 1 1 1 1 1 1 1 3 1 1 1 1  
## 36.8 36.9 37 37.1 37.3 37.5 37.6 37.8 37.9 38 38.1 38.2 38.3 38.4 38.6 38.7  
## 1 1 1 2 2 1 1 2 2 2 3 2 3 1 1 1  
## 38.9 39 39.2 39.3 39.4 39.5 39.6 39.7 39.8 39.9 40 40.1 40.2 40.4 40.5 40.6  
## 1 2 1 4 2 3 2 3 1 4 1 1 2 3 1 1  
## 40.8 40.9 41 41.1 41.2 41.3 41.4 41.5 41.6 41.7 41.8 41.9 42 42.1 42.2 42.3  
## 3 5 3 2 3 6 3 3 3 3 5 2 2 1 5 4  
## 42.4 42.5 42.6 42.7 42.8 42.9 43 43.2 43.3 43.5 43.6 43.7 43.8 43.9 44 44.1  
## 3 2 1 1 2 3 4 9 5 1 5 4 4 9 3 5  
## 44.2 44.3 44.4 44.5 44.6 44.7 44.8 44.9 45 45.1 45.2 45.3 45.4 45.5 45.6 45.7  
## 3 3 2 5 4 1 5 5 5 5 3 5 4 5 5 2  
## 45.8 45.9 46 46.1 46.2 46.3 46.4 46.5 46.6 46.7 46.8 46.9 47 47.1 47.2 47.3  
## 5 4 10 2 8 5 3 6 5 6 3 4 9 3 3 4  
## 47.4 47.5 47.6 47.7 47.8 47.9 48 48.1 48.2 48.3 48.4 48.5 48.6 48.7 48.8 48.9  
## 3 13 13 7 6 1 8 7 7 8 3 4 11 3 5 9  
## 49 49.1 49.2 49.3 49.4 49.5 49.6 49.7 49.8 49.9 50 50.1 50.2 50.3 50.4 50.5  
## 3 3 8 8 5 8 10 9 8 4 5 4 8 6 7 3  
## 50.6 50.7 50.8 50.9 51 51.1 51.3 51.4 51.5 51.6 51.7 51.8 51.9 52 52.1 52.2  
## 7 7 6 6 4 10 7 10 5 10 4 9 10 7 11 7  
## 52.3 52.4 52.5 52.6 52.7 52.8 52.9 53 53.1 53.2 53.3 53.4 53.5 53.6 53.7 53.8  
## 7 13 11 5 7 9 12 9 2 11 5 6 5 6 10 13  
## 53.9 54 54.1 54.2 54.3 54.4 54.5 54.6 54.7 54.8 54.9 55 55.1 55.2 55.3 55.4  
## 6 12 11 16 9 11 2 10 6 3 7 8 4 6 11 6  
## 55.5 55.6 55.7 55.8 55.9 56 56.1 56.2 56.3 56.4 56.5 56.6 56.7 56.8 56.9 57  
## 7 11 8 10 1 8 5 12 4 3 4 8 4 6 10 9  
## 57.1 57.2 57.3 57.4 57.5 57.6 57.7 57.8 57.9 58 58.1 58.2 58.3 58.4 58.5 58.6  
## 9 8 10 12 6 5 9 11 10 5 7 11 4 6 13 7  
## 58.7 58.8 58.9 59 59.1 59.2 59.3 59.4 59.5 59.6 59.7 59.8 59.9 60 60.1 60.2  
## 13 5 8 7 6 7 9 6 11 7 12 6 8 8 11 7  
## 60.3 60.4 60.5 60.6 60.7 60.8 60.9 61 61.1 61.2 61.3 61.4 61.5 61.6 61.7 61.8  
## 9 7 8 4 5 7 14 5 6 7 8 3 9 4 9 10  
## 61.9 62 62.1 62.2 62.3 62.4 62.5 62.6 62.7 62.8 62.9 63 63.1 63.2 63.3 63.4  
## 13 8 9 4 4 7 6 5 3 9 5 7 5 10 3 5  
## 63.5 63.6 63.7 63.8 63.9 64 64.1 64.2 64.3 64.4 64.5 64.6 64.7 64.8 64.9 65  
## 8 3 7 5 4 4 4 6 1 2 8 4 8 8 5 5
```



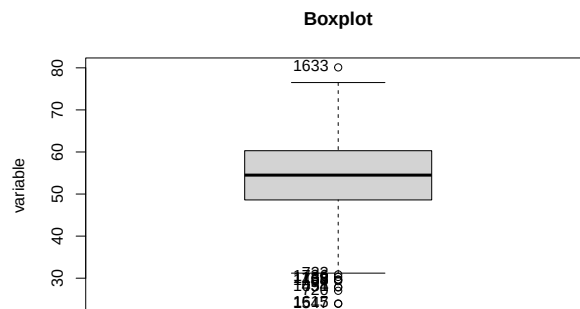
```
## 65.1 65.2 65.3 65.4 65.5 65.6 65.7 65.9 66 66.1 66.2 66.3 66.4 66.5 66.6 66.7
## 2 3 7 3 6 3 5 3 5 2 5 2 2 4 2 1
## 66.8 66.9 67 67.1 67.2 67.3 67.4 67.5 67.6 67.7 67.8 67.9 68 68.1 68.2 68.3
## 1 2 3 3 2 1 5 3 4 2 1 2 3 2 1 2
## 68.4 68.5 68.7 68.8 68.9 69 69.1 69.2 69.4 69.5 69.6 69.8 70 70.1 70.3 70.4
## 2 4 1 2 1 3 1 1 1 4 1 2 1 2 1 1
## 70.6 70.8 70.9 71.2 71.4 72.4 72.6 72.7 73.2 73.6 74.4 75.9 76.5 80.1
## 2 3 1 1 1 1 2 1 1 1 1 1 1 1
```

```
res.var <- summary_and_normality_check(variable)
```

```
## [1] "Summary: "
## Min. 1st Qu. Median Mean 3rd Qu. Max. NA's
## 23.90 48.60 54.50 54.21 60.30 80.10 82
## [1] "Mean"
## [1] 54.21172
## [1] "Standard deviation"
## [1] 8.216153
```



```
## [1] "Skewness"
## [1] -0.3362687
```



```
## [1] "Num Outliers:"
## [1] 11
## [1] "Num. severe outliers:"
## [1] 0
## [1] "Shapiro test: "
```

```
##
## Shapiro-Wilk normality test
##
## data: data
## W = 0.99196, p-value = 3.371e-08
##
## [1] "Most repeated value ad times it occurs: "
## [1] "54.2"
## [1] 16
outliers_df[res.var,"pctemployed16_over"] <- 1
```

Variable *pctunemployed16_over*

The variable *pctunemployed16_over* represents the percentage of county residents aged 16 and over who are unemployed. It spans from a minimum of 0.7% to a maximum of 29.4%, with a mean of 7.86% and a median of 7.5%. The standard deviation is 3.46%, indicating moderate variability across counties. The distribution is positively skewed, with a skewness of 0.92, suggesting that while most counties have lower unemployment rates, a few exhibit significantly higher percentages. The Shapiro-Wilk test confirms that the normality assumption is rejected. There are 42 outliers, including 4 severe ones.

This variable, in conjunction with *pctemployed16_over*, provides a comprehensive view of the labor market across counties.

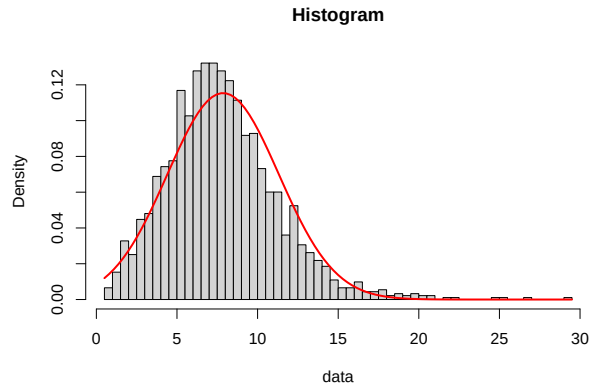
```
variable<-df$pctunemployed16_over
table(variable)
```

```
## variable
## 0.7 0.8 0.9 1 1.1 1.2 1.3 1.4 1.5 1.6 1.7 1.8 1.9 2 2.1 2.2
## 2 1 2 1 4 4 2 2 2 5 6 4 7 8 5 2
## 2.3 2.4 2.5 2.6 2.7 2.8 2.9 3 3.1 3.2 3.3 3.4 3.5 3.6 3.7 3.8
## 5 3 8 11 5 8 7 10 10 6 10 7 11 9 10 16
## 3.9 4 4.1 4.2 4.3 4.4 4.5 4.6 4.7 4.8 4.9 5 5.1 5.2 5.3 5.4
## 12 16 11 10 18 12 17 7 13 16 21 14 24 18 14 25
## 5.5 5.6 5.7 5.8 5.9 6 6.1 6.2 6.3 6.4 6.5 6.6 6.7 6.8 6.9 7
## 26 17 20 16 18 23 31 16 20 29 21 21 21 25 30 24
## 7.1 7.2 7.3 7.4 7.5 7.6 7.7 7.8 7.9 8 8.1 8.2 8.3 8.4 8.5 8.6
## 21 31 18 22 29 23 26 24 28 16 19 18 27 28 20 16
## 8.7 8.8 8.9 9 9.1 9.2 9.3 9.4 9.5 9.6 9.7 9.8 9.9 10 10.1 10.2
## 20 20 19 27 11 17 17 22 17 15 23 16 11 20 17 11
## 10.3 10.4 10.5 10.6 10.7 10.8 10.9 11 11.1 11.2 11.3 11.4 11.5 11.6 11.7 11.8
## 11 11 17 15 10 9 13 8 12 13 10 7 13 5 7 6
## 11.9 12 12.1 12.2 12.3 12.4 12.5 12.6 12.7 12.8 12.9 13 13.1 13.2 13.3 13.4
## 6 9 12 16 7 9 4 7 8 4 6 3 7 10 3 2
## 13.5 13.6 13.7 13.8 13.9 14 14.1 14.2 14.3 14.4 14.5 14.6 14.7 14.8 14.9 15
## 2 2 5 7 4 2 4 3 5 4 1 1 2 2 2 3
## 15.3 15.4 15.5 15.6 15.7 15.9 16 16.2 16.3 16.4 16.5 17 17.1 17.3 17.4 17.5
## 2 1 3 1 1 3 1 3 2 1 3 4 1 1 1 1
## 17.6 17.7 18 18.1 18.2 18.7 19 19.2 19.8 19.9 20 20.1 20.3 20.6 20.7 21.9
## 2 1 2 1 1 1 2 2 1 1 1 1 1 1 1 1
## 22.1 24.6 25.3 26.8 29.4
## 1 1 1 1 1
```

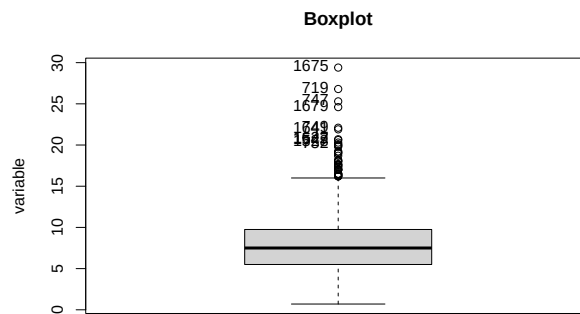
```
res.var <- summary_and_normality_check(variable)
```

```
## [1] "Summary: "
## Min. 1st Qu. Median Mean 3rd Qu. Max.
```

```
##      0.700      5.500      7.500      7.861      9.750      29.400
## [1] "Mean"
## [1] 7.86095
## [1] "Standard deviation"
## [1] 3.457111
```



```
## [1] "Skewness"
## [1] 0.9189968
```



```
## [1] "Num Outliers:"
## [1] 42
## [1] "Num. severe outliers:"
## [1] 4
## [1] "Shapiro test: "
##
## Shapiro-Wilk normality test
##
## data: data
## W = 0.9612, p-value < 2.2e-16
##
## [1] "Most repeated value ad times it occurs: "
## [1] "6.1"
## [1] 31
```

```
outliers_df[res.var,"pctunemployed16_over"] <- 1
```

Variable *pctprivatecoverage*

pctprivatecoverage is a numerical variable that reflects the percentage of county residents with private health coverage. The values span from 23.4% to 89.6%, with a median of 65.2% and a mean of 64.47%. The interquartile range is relatively narrow, from 57.5% (1st quartile) to 72.1% (3rd quartile), indicating that most counties have moderate to high levels of private health coverage. The standard deviation is 10.64%, showing some variability, while the skewness of -0.38 points to a slight left skew, with more counties concentrated at higher percentages. The distribution rejects the normality assumption, as confirmed by the Shapiro-Wilk test.

The dataset identifies 17 outliers, but none are severe. These represent counties with exceptionally low private health coverage rates, offering insights into regional disparities.

This variable holds significant importance due to its potential connection with the target variable, cancer target deathrate. Private health coverage likely plays a crucial role in access to healthcare services, early detection, and treatment of cancer, making it a key factor in understanding regional variations in cancer-related mortality.

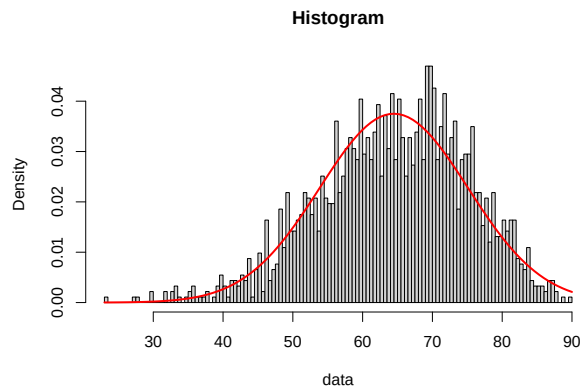
```
variable<- df$pctprivatecoverage  
table(variable)
```

```
## variable  
## 23.4 27.2 27.8 29.6 29.9 32 32.6 32.9 33.4 33.5 33.8 34.9 35.2 35.6 35.8 36  
## 1 1 1 1 1 2 1 1 1 2 1 1 2 1 1 1  
## 36.4 36.6 37.5 37.7 37.9 38.9 39.2 39.3 39.5 39.6 39.7 39.9 40 40.2 40.3 40.4  
## 1 1 1 1 1 1 1 1 1 1 1 2 1 1 1 1  
## 40.9 41.2 41.3 41.5 41.7 41.8 42 42.3 42.4 42.6 42.7 42.8 43 43.3 43.4 43.5  
## 1 2 1 1 1 1 2 2 1 1 1 1 2 1 2 1  
## 43.6 43.7 43.8 43.9 44 44.2 44.6 44.7 44.8 44.9 45 45.1 45.2 45.4 45.5 45.9  
## 2 1 1 3 1 1 1 1 1 2 1 2 4 2 1 1  
## 46 46.1 46.2 46.3 46.4 46.5 46.7 46.8 47 47.1 47.3 47.4 47.6 47.7 47.9 48  
## 1 1 2 3 5 4 2 1 1 2 3 1 2 1 2 2  
## 48.1 48.2 48.3 48.4 48.5 48.6 48.7 48.8 48.9 49 49.1 49.2 49.3 49.4 49.5 49.6  
## 3 2 6 3 3 2 2 2 3 1 2 3 5 5 5 7  
## 49.7 49.8 50 50.1 50.2 50.3 50.4 50.5 50.6 50.7 50.8 50.9 51 51.1 51.2 51.3  
## 1 3 2 1 2 2 4 4 1 5 5 2 2 4 4 2  
## 51.4 51.5 51.6 51.7 51.8 51.9 52 52.1 52.2 52.3 52.4 52.5 52.6 52.7 52.8 52.9  
## 4 2 8 3 5 2 2 5 4 1 7 2 3 6 3 4  
## 53.1 53.2 53.3 53.4 53.5 53.6 53.7 53.8 53.9 54 54.1 54.2 54.3 54.4 54.5 54.6  
## 6 3 4 5 1 4 4 1 1 3 2 9 6 2 4 3  
## 54.7 54.8 54.9 55 55.1 55.2 55.3 55.4 55.5 55.6 55.7 55.8 55.9 56 56.1 56.2  
## 8 2 3 3 2 7 5 1 3 3 7 2 4 2 9 5  
## 56.3 56.4 56.5 56.6 56.7 56.8 56.9 57 57.1 57.2 57.3 57.4 57.5 57.6 57.7 57.8  
## 5 8 6 3 3 4 5 5 5 5 3 4 6 3 8 5  
## 57.9 58 58.1 58.2 58.3 58.4 58.5 58.6 58.7 58.8 58.9 59 59.1 59.2 59.3 59.4  
## 7 5 6 3 6 10 5 9 4 4 5 6 2 9 3 6  
## 59.5 59.6 59.7 59.8 59.9 60 60.1 60.2 60.3 60.4 60.5 60.6 60.7 60.8 60.9 61  
## 6 9 7 8 9 4 6 7 6 4 4 8 6 3 5 8  
## 61.1 61.2 61.3 61.4 61.5 61.6 61.7 61.8 61.9 62 62.1 62.2 62.3 62.4 62.5 62.6  
## 4 4 7 5 6 8 7 3 8 5 7 8 11 4 6 4  
## 62.7 62.8 62.9 63 63.1 63.2 63.3 63.4 63.5 63.6 63.7 63.8 63.9 64 64.1 64.2  
## 5 1 7 6 7 8 3 5 11 6 5 8 4 5 5 6  
## 64.3 64.4 64.5 64.6 64.7 64.8 64.9 65 65.1 65.2 65.3 65.4 65.5 65.6 65.7 65.8  
## 9 11 7 6 5 4 5 6 7 9 8 5 8 5 6 5  
## 65.9 66 66.1 66.2 66.3 66.4 66.5 66.6 66.7 66.8 66.9 67 67.1 67.2 67.3 67.4  
## 6 8 3 9 3 3 5 8 5 6 6 5 2 4 6 11  
## 67.5 67.6 67.7 67.8 67.9 68 68.1 68.2 68.3 68.4 68.5 68.6 68.7 68.8 68.9 69
```

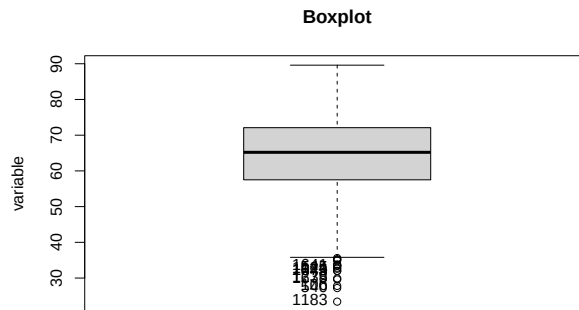
```
##      2      9      3      6      7      6      12      4      6      9      6      4      5      4      7      6
## 69.1 69.2 69.3 69.4 69.5 69.6 69.7 69.8 69.9 70 70.1 70.2 70.3 70.4 70.5 70.6
##      12      7      8      7      9      9      10      12      6      6      9      5      12      5      8      7
## 70.7 70.8 70.9 71 71.1 71.2 71.3 71.4 71.5 71.6 71.7 71.8 71.9 72 72.1 72.2
##      6      6      4      3      3      8      5      8      8      12      5      9      8      4      7      4
## 72.3 72.4 72.5 72.6 72.7 72.8 72.9 73 73.1 73.2 73.3 73.4 73.5 73.6 73.7 73.8
##      7      4      5      1      10      5      7      7      11      3      6      8      5      4      3      4
## 73.9 74 74.1 74.2 74.3 74.4 74.5 74.6 74.7 74.8 74.9 75 75.1 75.2 75.3 75.4
##      3      3      7      7      6      5      1      7      6      5      4      5      7      11      1      3
## 75.5 75.6 75.7 75.8 75.9 76 76.1 76.2 76.3 76.4 76.5 76.6 76.7 76.8 76.9 77
##      5      9      7      4      3      9      7      2      3      3      5      2      6      2      4      6
## 77.1 77.2 77.3 77.4 77.5 77.6 77.7 77.8 77.9 78 78.1 78.2 78.3 78.4 78.5 78.6
##      3      3      2      2      4      4      7      2      4      2      2      4      3      1      1      6
## 78.7 78.8 78.9 79 79.1 79.2 79.3 79.4 79.5 79.6 79.7 79.8 79.9 80 80.1 80.2
##      2      7      2      3      1      2      5      1      3      4      2      1      2      3      4      4
## 80.3 80.4 80.5 80.6 80.7 80.8 80.9 81 81.1 81.2 81.3 81.4 81.5 81.6 81.8 81.9
##      2      3      2      4      6      1      1      1      2      5      2      4      2      3      5      5
## 82 82.2 82.3 82.4 82.5 82.6 82.7 82.8 82.9 83 83.1 83.2 83.3 83.4 83.5 83.6
##      2      2      1      3      2      1      1      2      2      1      1      1      2      1      1      2
## 83.7 83.8 83.9 84 84.1 84.5 84.7 84.8 85.1 85.2 85.7 86 86.2 86.4 86.6 86.7
##      2      2      1      3      3      1      1      2      1      2      2      1      1      1      2      1
## 86.9 87.2 87.4 87.5 87.9 88 88.9 89.6
##      1      1      1      2      1      1      1      1
```

```
res.var <- summary_and_normality_check(variable);
```

```
## [1] "Summary: "
##      Min. 1st Qu.  Median      Mean 3rd Qu.      Max.
##    23.40  57.50   65.20   64.47   72.10   89.60
## [1] "Mean"
## [1] 64.4746
## [1] "Standard deviation"
## [1] 10.63614
```



```
## [1] "Skewness"
## [1] -0.3796678
```



```
## [1] "Num Outliers:"
## [1] 17
## [1] "Num. severe outliers:"
## [1] 0
## [1] "Shapiro test: "
##
## Shapiro-Wilk normality test
##
## data: data
## W = 0.98964, p-value = 3.725e-10
##
## [1] "Most repeated value ad times it occurs: "
## [1] "68.1"
## [1] 12
```

```
outliers_df[res.var,"pctprivatecoverage"] <- 1
```

Variable *pctprivatecoveragealone*

The variable *pctprivatecoveragealone* represents the percentage of county residents with private health coverage without any public assistance. It ranges from 16.8% to 78.9%, with a mean of 48.65% and a median of 49.0%, indicating that approximately half of the counties have coverage levels close to the central tendency. The interquartile range spans from 41.5% to 55.5%, with a standard deviation of 10.07%. The skewness of 0.02 indicates an almost symmetrical distribution, and the Shapiro-Wilk test suggests the data does not deviate significantly from normality. However, the variable contains 356 missing values (approximately 20% of the dataset), which may require imputation or exclusion strategies. There are 4 outliers, none of which are severe.

This variable is closely related to *pctprivatecoverage*, which captures overall private health coverage, including public assistance.

```
variable<-df$pctprivatecoveragealone
table(variable)
```

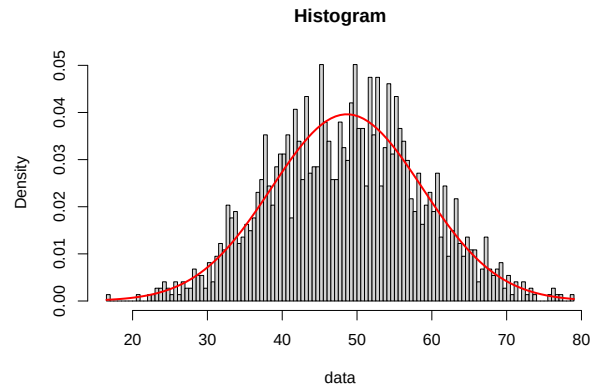
```
## variable
## 16.8 20.7 22.2 23 23.2 23.4 23.7 24 24.1 24.2 24.4 24.6 24.7 25.1 25.7 25.8
## 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1
## 25.9 26.3 26.6 26.7 27.3 27.4 27.6 27.8 28.1 28.2 28.3 28.5 28.7 28.8 28.9 29
## 1 1 1 2 1 1 1 1 1 1 2 1 1 1 1 1
## 29.3 29.4 29.5 29.8 30 30.1 30.2 30.4 30.5 30.7 30.9 31 31.2 31.3 31.4 31.5
## 1 2 1 1 1 2 1 1 2 1 1 1 3 1 2 1
## 31.6 31.7 31.8 31.9 32 32.1 32.2 32.3 32.4 32.5 32.6 32.7 32.8 32.9 33 33.1
## 2 1 2 1 3 2 2 1 1 2 1 4 3 3 4 3
```

```
## 33.2 33.3 33.4 33.5 33.6 33.7 33.8 33.9 34 34.1 34.2 34.4 34.5 34.6 34.7 34.9
## 1 3 4 2 5 5 1 2 1 2 2 1 4 4 1 2
## 35 35.1 35.3 35.4 35.5 35.6 35.7 35.8 35.9 36 36.1 36.2 36.4 36.5 36.6 36.7
## 3 4 3 3 2 4 2 2 1 2 5 1 5 2 1 3
## 36.8 36.9 37 37.1 37.2 37.3 37.4 37.5 37.6 37.7 37.8 37.9 38 38.1 38.2 38.3
## 6 2 5 2 5 7 2 3 5 7 2 7 5 6 5 2
## 38.4 38.5 38.6 38.7 38.8 38.9 39 39.1 39.2 39.3 39.4 39.5 39.6 39.7 39.8 39.9
## 2 3 2 6 4 2 1 4 3 5 3 6 5 5 6 2
## 40 40.1 40.2 40.3 40.4 40.5 40.6 40.7 40.8 40.9 41 41.1 41.2 41.3 41.4 41.5
## 5 7 4 3 5 4 5 6 2 4 9 4 2 3 1 3
## 41.6 41.7 41.8 41.9 42 42.1 42.2 42.3 42.4 42.5 42.6 42.7 42.8 42.9 43 43.1
## 6 4 4 11 5 4 9 7 1 4 3 5 6 4 1 7
## 43.2 43.3 43.4 43.5 43.6 43.7 43.8 43.9 44 44.1 44.2 44.3 44.4 44.5 44.6 44.7
## 6 9 5 5 4 6 2 6 2 2 6 7 3 3 4 1
## 44.8 44.9 45 45.1 45.2 45.3 45.4 45.5 45.6 45.7 45.8 45.9 46 46.1 46.2 46.3
## 3 5 8 10 4 8 7 8 5 6 7 2 8 3 5 4
## 46.4 46.5 46.6 46.7 46.8 46.9 47 47.1 47.2 47.3 47.4 47.5 47.6 47.7 47.8 47.9
## 8 5 7 2 4 4 2 3 3 7 4 2 9 3 5 5
## 48 48.1 48.2 48.3 48.4 48.5 48.6 48.7 48.8 48.9 49 49.1 49.2 49.3 49.4 49.5
## 6 3 7 9 3 2 3 5 4 4 6 5 3 10 7 6
## 49.6 49.7 49.8 49.9 50 50.1 50.2 50.3 50.4 50.5 50.6 50.7 50.8 50.9 51 51.1
## 11 9 8 6 3 5 3 4 10 5 7 4 5 10 1 3
## 51.2 51.3 51.4 51.5 51.6 51.7 51.8 51.9 52 52.1 52.2 52.3 52.4 52.5 52.6 52.7
## 5 2 6 2 9 10 9 5 2 4 6 7 6 3 3 7
## 52.8 52.9 53 53.1 53.2 53.3 53.4 53.5 53.6 53.7 53.8 53.9 54 54.1 54.2 54.3
## 10 6 9 4 3 5 4 2 3 5 8 3 5 3 5 8
## 54.4 54.5 54.6 54.7 54.8 54.9 55 55.1 55.2 55.3 55.4 55.5 55.6 55.7 55.8 55.9
## 10 8 3 2 5 9 4 4 10 4 5 9 5 5 5 7
## 56 56.1 56.2 56.3 56.4 56.5 56.6 56.7 56.8 56.9 57 57.1 57.2 57.3 57.4 57.5
## 5 9 5 4 3 4 2 5 7 4 4 2 2 2 5 5
## 57.6 57.8 57.9 58 58.1 58.2 58.3 58.4 58.5 58.6 58.7 58.8 58.9 59 59.1 59.2
## 5 2 3 4 3 8 3 2 4 3 2 2 3 2 2 4
## 59.3 59.4 59.5 59.6 59.7 59.8 59.9 60 60.1 60.3 60.4 60.5 60.6 60.7 60.8 60.9
## 4 3 2 4 2 3 2 6 3 2 6 3 6 3 3 6
## 61 61.1 61.2 61.3 61.4 61.5 61.6 61.8 61.9 62 62.1 62.2 62.4 62.5 62.6 62.7
## 2 1 1 4 1 3 9 2 4 3 1 1 3 2 3 4
## 62.9 63 63.1 63.2 63.3 63.4 63.5 63.7 63.8 63.9 64.1 64.2 64.4 64.6 64.7 64.8
## 3 1 4 3 2 3 4 3 2 4 1 3 3 2 2 1
## 64.9 65 65.2 65.3 65.4 65.5 65.6 65.7 65.8 66 66.1 66.2 66.6 66.8 66.9 67
## 3 2 3 2 1 2 3 2 2 1 2 1 1 1 2 1
## 67.1 67.3 67.4 67.5 67.6 67.7 67.9 68.2 68.4 68.7 68.9 69 69.1 69.2 69.4 69.5
## 6 1 2 1 3 1 1 2 2 1 2 2 2 1 1 2
## 69.8 69.9 70.1 70.2 70.3 70.9 71.1 71.3 71.4 72.2 72.3 72.5 72.8 73.3 74 75.7
## 1 1 2 1 1 1 1 1 1 1 1 1 1 2 1 1
## 76.3 76.5 76.6 77.1 78.9
## 1 1 1 1 1
```

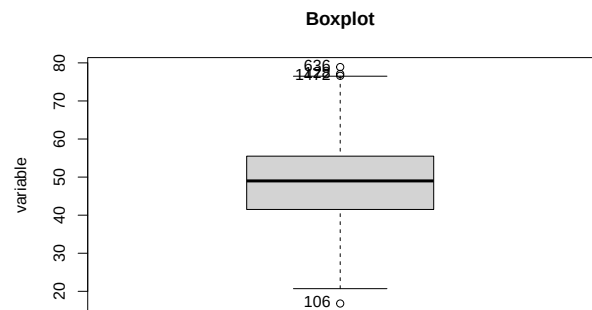
```
res.var <- summary_and_normality_check(variable)
```

```
## [1] "Summary: "
##      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.   NA's
## 16.80  41.50   49.00  48.65  55.50  78.90   356
## [1] "Mean"
## [1] 48.64807
## [1] "Standard deviation"
```

```
## [1] 10.06977
```



```
## [1] "Skewness"  
## [1] 0.01553065
```



```
## [1] "Num Outliers:"  
## [1] 4  
## [1] "Num. severe outliers:"  
## [1] 0  
## [1] "Shapiro test: "  
##  
## Shapiro-Wilk normality test  
##  
## data: data  
## W = 0.99831, p-value = 0.1453  
##  
## [1] "Most repeated value ad times it occurs: "  
## [1] "41.9"  
## [1] 11
```

```
outliers_df[res.var,"pctprivatecoveragealone"] <- 1
```

Variable *pctempprivcoverage*

pctempprivcoverage represents the percentage of county residents with employee-provided private health coverage.

Its values range from 14.3% to 70.2%, with a mean of 41.29% and a median of 41.1%, indicating a balanced

central tendency. The interquartile range spans from 34.6% to 47.7%, with a standard deviation of 9.53%. The skewness is close to zero (0.11), suggesting a nearly symmetrical distribution, while the Shapiro-Wilk test indicates a slight deviation from normality for a confidence level of 99%, but is not rejected in a confidence of 95% (p-value = 0.02861). The variable contains 7 outliers, none of which are severe.

This variable is integral to understanding employment-related health benefits, especially when analyzed alongside *pctprivatecoverage* and *pctprivatecoveragealone*.

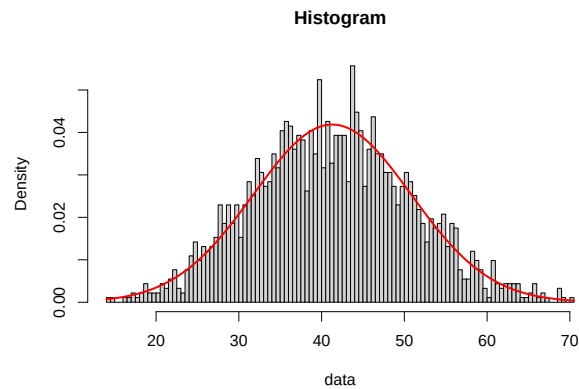
```
variable<-df$pctempprivcoverage
table(variable)
```

```
## variable
## 14.3 15 16.3 16.8 17.2 17.4 17.7 18.3 18.4 18.6 18.8 19 19.3 19.5 19.6 19.8
## 1 1 1 1 1 1 1 1 1 2 1 1 1 1 1 1
## 20.2 20.4 20.6 20.9 21 21.1 21.2 21.4 21.6 21.7 21.8 21.9 22.1 22.2 22.4 22.5
## 1 1 1 2 1 1 1 1 2 1 1 1 3 2 1 1
## 22.7 22.8 22.9 23.4 23.5 23.6 23.8 23.9 24 24.1 24.2 24.3 24.4 24.5 24.6 24.7
## 1 1 1 1 1 2 3 1 1 2 3 3 1 1 2 5
## 24.8 25 25.1 25.2 25.3 25.4 25.5 25.6 25.9 26 26.1 26.2 26.3 26.4 26.5 26.6
## 3 3 1 3 2 2 1 3 4 5 2 1 1 6 1 2
## 26.7 26.8 27 27.3 27.4 27.5 27.6 27.7 27.8 27.9 28 28.1 28.2 28.3 28.4 28.5
## 2 2 6 1 9 4 3 5 3 4 6 2 4 2 3 6
## 28.6 28.7 28.8 28.9 29 29.1 29.2 29.3 29.4 29.5 29.6 29.7 29.8 29.9 30 30.1
## 4 6 2 6 3 4 5 4 1 3 5 5 3 3 5 1
## 30.2 30.3 30.4 30.5 30.6 30.7 30.8 30.9 31 31.1 31.2 31.3 31.4 31.5 31.6 31.7
## 4 3 1 5 2 5 4 8 2 2 8 7 5 4 7 2
## 31.8 31.9 32 32.1 32.2 32.3 32.4 32.5 32.6 32.7 32.8 32.9 33 33.1 33.2 33.3
## 2 4 8 5 5 5 8 8 13 1 5 8 1 3 4 7
## 33.4 33.5 33.6 33.7 33.8 33.9 34 34.1 34.2 34.3 34.4 34.5 34.6 34.7 34.8 34.9
## 4 7 5 3 7 6 5 9 4 6 6 7 11 5 3 5
## 35 35.1 35.2 35.3 35.4 35.5 35.6 35.7 35.8 35.9 36 36.1 36.2 36.3 36.4 36.5
## 5 7 5 11 5 9 8 9 6 7 9 5 7 8 10 8
## 36.6 36.7 36.8 36.9 37 37.1 37.2 37.3 37.4 37.5 37.6 37.7 37.8 37.9 38 38.1
## 5 7 7 6 8 11 8 5 7 5 6 4 5 9 11 7
## 38.2 38.3 38.4 38.5 38.6 38.7 38.8 38.9 39 39.1 39.2 39.3 39.4 39.5 39.6 39.7
## 2 5 6 4 12 1 8 11 5 4 5 5 12 6 4 14
## 39.8 39.9 40 40.1 40.2 40.3 40.4 40.5 40.6 40.7 40.8 40.9 41 41.1 41.2 41.3
## 10 6 14 5 7 4 5 8 6 10 8 9 6 9 4 7
## 41.4 41.5 41.6 41.7 41.8 41.9 42 42.1 42.2 42.3 42.4 42.5 42.6 42.7 42.8 42.9
## 5 5 9 9 3 5 10 9 6 5 9 7 6 6 8 8
## 43 43.1 43.2 43.3 43.4 43.5 43.6 43.7 43.8 43.9 44 44.1 44.2 44.3 44.4 44.5
## 8 5 4 6 5 6 11 16 8 5 11 8 3 6 12 12
## 44.6 44.7 44.8 44.9 45 45.1 45.2 45.3 45.4 45.5 45.6 45.7 45.8 45.9 46 46.1
## 10 5 6 8 8 10 2 3 9 1 6 6 6 10 5 11
## 46.2 46.3 46.4 46.5 46.6 46.7 46.8 46.9 47 47.1 47.2 47.3 47.4 47.5 47.6 47.7
## 7 7 9 6 4 10 5 5 8 7 7 7 5 6 3 7
## 47.8 47.9 48 48.1 48.2 48.3 48.4 48.5 48.6 48.7 48.8 48.9 49 49.1 49.2 49.3
## 9 5 4 6 2 6 6 8 4 7 4 4 6 7 6 5
## 49.4 49.5 49.6 49.7 49.8 49.9 50 50.1 50.2 50.3 50.4 50.5 50.6 50.7 50.8 50.9
## 2 1 4 4 5 9 3 6 9 5 5 3 5 4 5 5
## 51 51.1 51.2 51.3 51.4 51.5 51.6 51.7 51.8 51.9 52 52.1 52.2 52.3 52.5 52.6
## 7 4 5 3 7 4 3 3 5 4 5 7 4 1 5 4
## 52.7 52.9 53 53.1 53.2 53.3 53.4 53.5 53.6 53.7 53.8 53.9 54 54.1 54.2 54.3
## 5 2 2 5 5 3 3 2 2 4 3 3 4 2 2 3
## 54.4 54.5 54.6 54.7 54.8 54.9 55 55.1 55.2 55.3 55.4 55.5 55.6 55.7 55.8 55.9
```

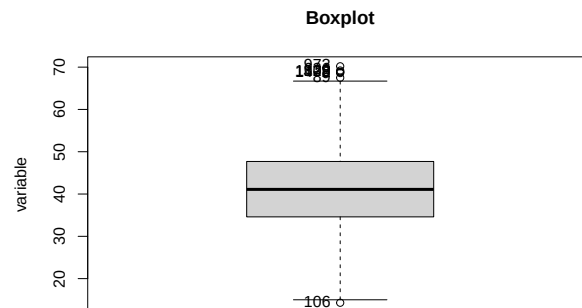
```
##      4      6      3      5      3      4      4      2      1      2      4      3      4      2      2      5
##    56 56.1 56.2 56.3 56.4 56.5 56.6 56.7 56.8 56.9 57.3 57.4 57.5 57.7 57.8 58
##      4      4      2      4      2      4      1      1      3      2      2      1      2      1      3      1
##    58.1 58.2 58.3 58.5 58.6 58.9 59 59.1 59.2 59.3 59.5 60 60.1 60.6 60.7 60.8
##      3      5      2      1      3      4      2      1      3      1      2      3      1      4      1      3
##    60.9 61.4 61.5 61.6 61.8 62 62.1 62.4 62.5 62.8 62.9 63 63.1 63.2 63.4 63.7
##      1      2      2      1      1      1      1      2      1      2      1      1      2      1      1      2
##    63.8 64 64.4 64.9 65.1 65.6 65.8 66.3 66.6 66.7 67.5 68.6 68.8 68.9 69.2 70.2
##      1      1      1      1      2      1      3      1      1      1      1      1      1      1      1      1
```

```
res.var <- summary_and_normality_check(variable)
```

```
## [1] "Summary: "
##      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##    14.30  34.60   41.10   41.29   47.70   70.20
## [1] "Mean"
## [1] 41.29388
## [1] "Standard deviation"
## [1] 9.528869
```



```
## [1] "Skewness"
## [1] 0.1082829
```



```
## [1] "Num Outliers:"
## [1] 7
## [1] "Num. severe outliers:"
## [1] 0
```

```
## [1] "Shapiro test: "
##
## Shapiro-Wilk normality test
##
## data: data
## W = 0.99807, p-value = 0.02861
##
## [1] "Most repeated value ad times it occurs: "
## [1] "43.7"
## [1] 16
```

```
outliers_df[res.var,"pctempprivcoverage"] <- 1
```

Variable *pctpubliccoverage*

The numeric variable *pctpubliccoverage* represents the percentage of county residents with government-provided health coverage, which is not exclusive and may overlap with other types of coverage. It ranges from 11.2% to 62.7%, with a mean of 36.15% and a median of 36.3%, suggesting a balanced distribution around the central tendency. The interquartile range extends from 30.9% to 41.4%, and the standard deviation of 7.84% . The skewness of -0.03 reflects a nearly symmetrical distribution. The Shapiro-Wilk test suggests that the data does not significantly deviate from normality. There are 13 outliers, though none are classified as severe.

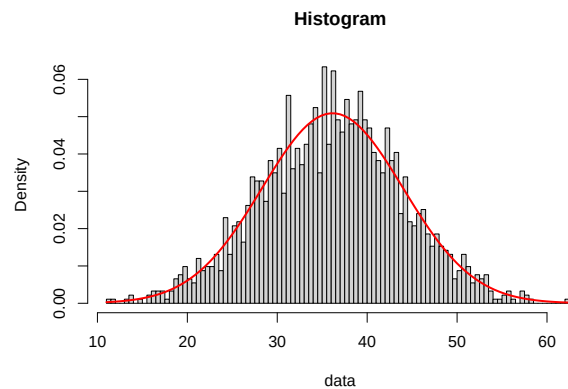
```
variable<- df$pctpubliccoverage
table(variable)
```

```
## variable
## 11.2 11.8 13.5 13.8 14 14.8 15.7 16 16.1 16.5 16.8 17 17.1 17.9 18.1 18.4
## 1 1 1 1 1 1 1 1 2 1 1 2 3 1 1 1
## 18.5 18.6 18.7 18.8 18.9 19 19.1 19.2 19.3 19.4 19.6 19.7 19.8 19.9 20 20.1
## 1 1 1 1 1 2 1 1 4 1 1 2 2 3 1 1
## 20.2 20.4 20.5 20.6 20.8 20.9 21.1 21.2 21.3 21.4 21.5 21.6 21.7 21.9 22 22.1
## 2 2 1 1 1 3 1 3 3 2 2 1 2 3 2 2
## 22.2 22.3 22.4 22.5 22.6 22.7 22.9 23 23.1 23.2 23.3 23.4 23.5 23.6 23.7 23.8
## 2 1 3 1 2 2 3 2 1 2 4 2 3 2 2 2
## 23.9 24.1 24.2 24.3 24.4 24.5 24.6 24.7 24.8 24.9 25 25.1 25.2 25.3 25.4 25.5
## 2 2 5 2 8 4 2 3 2 2 3 3 7 4 3 2
## 25.6 25.7 25.8 25.9 26 26.1 26.2 26.3 26.4 26.5 26.6 26.7 26.8 26.9 27 27.1
## 3 8 2 1 6 1 1 8 2 3 4 2 7 6 5 4
## 27.2 27.3 27.4 27.5 27.6 27.7 27.8 27.9 28 28.1 28.2 28.3 28.4 28.5 28.6 28.7
## 8 6 5 8 6 4 6 4 10 3 9 6 4 8 3 2
## 28.8 28.9 29 29.1 29.2 29.3 29.4 29.5 29.6 29.7 29.8 29.9 30 30.1 30.2 30.3
## 8 4 8 4 5 12 7 7 6 9 6 8 3 7 6 9
## 30.4 30.5 30.6 30.7 30.8 30.9 31 31.1 31.2 31.3 31.4 31.5 31.6 31.7 31.8 31.9
## 7 9 6 2 6 6 7 11 5 14 10 11 4 6 6 11
## 32 32.1 32.2 32.3 32.4 32.5 32.6 32.7 32.8 32.9 33 33.1 33.2 33.3 33.4 33.5
## 6 11 9 5 4 9 5 8 8 10 3 6 10 4 14 5
## 33.6 33.7 33.8 33.9 34 34.1 34.2 34.3 34.4 34.5 34.6 34.7 34.8 34.9 35 35.1
## 5 12 7 14 6 7 12 12 7 10 3 7 7 4 11 15
## 35.2 35.3 35.4 35.5 35.6 35.7 35.8 35.9 36 36.1 36.2 36.3 36.4 36.5 36.6 36.7
## 12 11 10 10 2 6 12 13 6 12 12 12 7 14 8 9
## 36.8 36.9 37 37.1 37.2 37.3 37.4 37.5 37.6 37.7 37.8 37.9 38 38.1 38.2 38.3
## 8 6 14 9 9 5 9 10 12 11 6 11 10 10 8 8
## 38.4 38.5 38.6 38.7 38.8 38.9 39 39.1 39.2 39.3 39.4 39.5 39.6 39.7 39.8 39.9
## 11 7 10 10 12 6 7 11 13 11 11 6 10 7 14 7
## 40 40.1 40.2 40.3 40.4 40.5 40.6 40.7 40.8 40.9 41 41.1 41.2 41.3 41.4 41.5
```

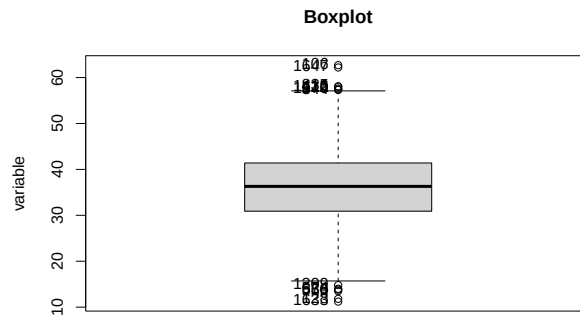
```
##      7      9      5      8      10      11      10      12      4      5      6      5      6      11      8      5
## 41.6 41.7 41.8 41.9      42 42.1 42.2 42.3 42.4 42.5 42.6 42.7 42.8 42.9      43 43.1
##      7      7      8      5      5      7      7      9      13      7      7      9      7      9      3      5
## 43.2 43.3 43.4 43.5 43.6 43.7 43.8 43.9      44 44.1 44.2 44.3 44.4 44.5 44.6 44.7
##      9      6      9      8      5      1      2      8      6      9      6      4      7      5      5      8
## 44.8 44.9      45 45.1 45.2 45.3 45.5 45.6 45.7 45.8 45.9      46 46.1 46.2 46.3 46.4
##      2      2      3      3      5      9      2      7      7      3      2      3      5      6      3      6
## 46.5 46.6 46.7 46.8 46.9      47 47.1 47.2 47.3 47.4 47.5 47.6 47.7 47.8 47.9      48
##      3      3      3      5      3      3      5      3      3      1      2      6      5      3      2      1
## 48.1 48.2 48.3 48.4 48.6 48.7 48.8 48.9      49 49.1 49.2 49.3 49.4 49.5 49.6 49.7
##      4      5      2      3      2      1      4      1      5      2      3      4      2      1      2      1
## 49.8 49.9      50 50.3 50.4 50.5 50.6 50.8 50.9      51 51.1 51.2 51.3 51.4 51.5 51.8
##      1      1      1      3      4      1      4      1      3      4      2      2      1      2      2      1
## 51.9      52 52.1 52.2 52.3 52.5 52.6 52.8 52.9      53 53.1 53.2 53.3 53.4 53.6 53.7
##      2      2      2      2      1      2      2      2      1      1      1      3      1      2      1      1
##      54 54.3 54.6 55.1 55.5 55.6      56 56.5 57.1 57.3 57.4 57.6      58 58.1 62.2 62.7
##      1      1      1      1      1      1      2      1      1      1      1      1      1      1      1      1
```

```
res.var <- summary_and_normality_check(variable)
```

```
## [1] "Summary: "
##      Min. 1st Qu.  Median      Mean 3rd Qu.      Max.
##      11.20  30.90   36.30   36.15   41.40   62.70
## [1] "Mean"
## [1] 36.15309
## [1] "Standard deviation"
## [1] 7.835075
```



```
## [1] "Skewness"
## [1] -0.02812824
```



```
## [1] "Num Outliers:"
## [1] 13
## [1] "Num. severe outliers:"
## [1] 0
## [1] "Shapiro test: "
##
## Shapiro-Wilk normality test
##
## data: data
## W = 0.99947, p-value = 0.9186
##
## [1] "Most repeated value ad times it occurs: "
## [1] "35.1"
## [1] 15
```

```
outliers_df[res.var,"pctpubliccoverage"] <- 1
```

Variable *pctpubliccoveragealone*

The variable *pctpubliccoveragealone* represents the percentage of county residents with government-provided health coverage exclusively. The data ranges from 2.6% to 46.6%, with a mean of 19.15% and a median of 18.7%. Most counties fall within an interquartile range of 14.9% to 23%, while the standard deviation of 6.06% indicates moderate variability. The skewness of 0.47 suggests a slight positive skew, with more counties having lower values. The Shapiro-Wilk test indicates that the normality assumption is rejected, and there are 21 outliers, though none are severe.

This variable should complement *pctprivatecoveragealone*, which measures private coverage without public assistance. Together, they should provide a complete view of the reliance on public versus private health coverage. Counties with higher *pctpubliccoveragealone* may reflect areas with greater socioeconomic challenges or fewer employer-provided insurance options.

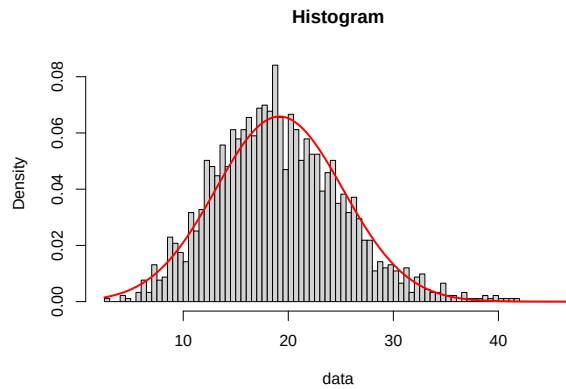
```
variable<-df$pctpubliccoveragealone
table(variable)
```

```
## variable
## 2.6 4.2 4.6 5.6 5.7 5.9 6.1 6.2 6.3 6.6 6.8 7 7.1 7.2 7.3 7.4
## 1 2 1 1 1 1 1 4 2 1 1 1 1 1 2 5
## 7.5 7.7 7.8 7.9 8 8.1 8.2 8.3 8.4 8.5 8.6 8.7 8.8 8.9 9 9.1
## 3 1 1 2 3 2 2 1 2 1 3 4 4 6 4 1
## 9.2 9.3 9.4 9.5 9.6 9.7 9.8 9.9 10 10.1 10.2 10.3 10.4 10.5 10.6 10.7
## 5 7 2 4 4 3 2 4 3 1 1 3 6 2 10 5
## 10.8 10.9 11 11.1 11.2 11.3 11.4 11.5 11.6 11.7 11.8 11.9 12 12.1 12.2 12.3
```

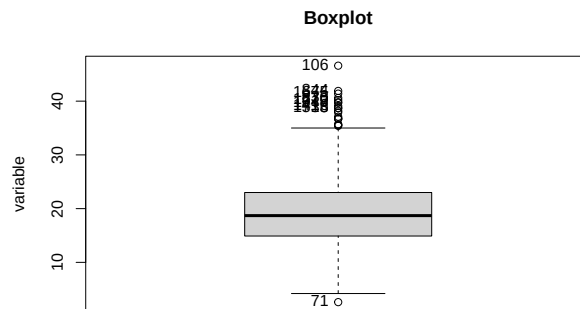
```
##      6      2      6      5      1      10      2      5      6      4      4      10      6      9      8      4
## 12.4 12.5 12.6 12.7 12.8 12.9      13 13.1 13.2 13.3 13.4 13.5 13.6 13.7 13.8 13.9
##      14      11      4      9      6      12      13      6      8      7      11      9      9      11      12      11
##      14 14.1 14.2 14.3 14.4 14.5 14.6 14.7 14.8 14.9      15 15.1 15.2 15.3 15.4 15.5
##      8      10      7      7      9      11      13      7      12      10      14      5      15      13      10      10
## 15.6 15.7 15.8 15.9      16 16.1 16.2 16.3 16.4 16.5 16.6 16.7 16.8 16.9      17 17.1
##      10      11      14      6      15      9      13      9      17      12      9      13      14      9      9      15
## 17.2 17.3 17.4 17.5 17.6 17.7 17.8 17.9      18 18.1 18.2 18.3 18.4 18.5 18.6 18.7
##      13      14      11      10      11      14      14      18      7      14      13      15      10      10      14      23
## 18.8 18.9      19 19.1 19.2 19.3 19.4 19.5 19.6 19.7 19.8 19.9      20 20.1 20.2 20.3
##      11      16      13      18      7      11      16      8      12      9      9      8      5      15      17      10
## 20.4 20.5 20.6 20.7 20.8 20.9      21 21.1 21.2 21.3 21.4 21.5 21.6 21.7 21.8 21.9
##      4      15      14      14      8      11      9      10      8      11      13      4      10      7      15      9
##      22 22.1 22.2 22.3 22.4 22.5 22.6 22.7 22.8 22.9      23 23.1 23.2 23.3 23.4 23.5
##      12      7      7      10      6      18      10      14      4      11      9      9      7      8      5      7
## 23.6 23.7 23.8 23.9      24 24.1 24.2 24.3 24.4 24.5 24.6 24.7 24.8 24.9      25 25.1
##      6      10      8      9      9      15      8      8      9      6      7      7      10      6      2      4
## 25.2 25.3 25.4 25.5 25.6 25.7 25.8 25.9      26 26.1 26.2 26.3 26.4 26.5 26.6 26.7
##      6      10      11      4      6      6      4      7      6      8      8      8      4      6      4      10
## 26.8 26.9      27 27.1 27.3 27.4 27.5 27.6 27.7 27.8 27.9      28 28.1 28.2 28.3 28.4
##      7      2      4      8      4      3      5      5      4      4      5      2      2      1      4      2
## 28.5 28.6 28.7 28.8 28.9      29 29.1 29.2 29.3 29.4 29.5 29.6 29.7 29.8 29.9      30
##      1      2      3      2      4      2      1      2      2      2      4      2      2      3      4      1
## 30.1 30.2 30.3 30.4 30.5 30.8 30.9      31 31.1 31.2 31.3 31.4 31.5 31.8      32 32.1
##      1      1      3      3      2      3      2      1      3      1      3      1      3      1      2      2
## 32.2 32.3 32.4 32.5 32.6 32.7 32.8 32.9      33 33.2 33.4 33.5 33.7      34 34.1 34.2
##      2      2      1      1      1      3      1      2      2      1      2      1      1      2      1      1
## 34.3 34.6 34.7      35 35.4 35.5 35.6 35.7 36.7 36.8 37.1 37.9 38.2 38.7 39.1 39.7
##      1      3      2      1      1      1      1      1      2      1      1      1      1      2      1      1
## 39.8 40.2 40.6 41.4 41.9 46.6
##      1      1      1      1      1      1
```

```
res.var <- summary_and_normality_check(variable)
```

```
## [1] "Summary: "
##      Min. 1st Qu.  Median      Mean 3rd Qu.      Max.
##      2.60  14.90   18.70   19.15   23.00   46.60
## [1] "Mean"
## [1] 19.14992
## [1] "Standard deviation"
## [1] 6.056619
```



```
## [1] "Skewness"
## [1] 0.4695236
```



```
## [1] "Num Outliers:"
## [1] 21
## [1] "Num. severe outliers:"
## [1] 0
## [1] "Shapiro test: "
##
## Shapiro-Wilk normality test
##
## data: data
## W = 0.98784, p-value = 2.648e-11
##
## [1] "Most repeated value ad times it occurs: "
## [1] "18.7"
## [1] 23
```

```
outliers_df[res.var,"pctpubliccoveragealone"] <- 1
```

Variable *pctwhite*

The variable *pctwhite* represents the percentage of county residents who identify as White. The values range from 12.27% to 99.69%, with a mean of 83.85% and a median of 89.9%. The interquartile range extends from 77.31% to 95.57%, indicating that the majority of counties have a high percentage of White residents. However, the standard deviation of 16.04% highlights notable variability across counties. The skewness of -1.63 indicates a strong left skew, meaning there are a small number of counties with significantly lower

percentages of White residents. The Shapiro-Wilk test confirms that the data is not normally distributed. Outliers are present in 97 counties, of which 7 are severe.

This variable is significant for understanding the racial composition of counties and how it may intersect with socioeconomic and health-related outcomes. For instance, its relationship with variables like *pctpubliccoverage* may provide insights into racial disparities in healthcare access or disease prevalence.

```
variable<- df$pctwhite
table(variable)
```

```
## variable
## 12.27366787 13.6227224 13.93356182 16.54059237 17.45484067 21.72432336
##          1          1          1          1          1          1
## 21.9145019 22.60685663 22.61027956 23.24592488 25.39662394 25.49804506
##          1          1          1          1          1          1
## 25.78553259 26.64356621 27.76631069 27.92924037 28.66869099 28.87105404
##          1          1          1          1          1          1
## 28.88114886 28.93956964 29.77164281 30.07614684 30.31292272 30.99542058
##          1          1          1          1          1          1
## 31.56228008 31.79768041 32.01456165 32.93429392 33.12427891 33.2623088
##          1          1          1          1          1          1
## 33.28837296 33.94976113 34.3144849 34.76418319 34.89180234 35.14492602
##          1          1          1          1          1          1
## 35.58765594 35.8281493 36.20723267 36.81737273 36.83282734 36.96712428
##          1          1          1          1          1          1
## 38.40549368 38.85538607 38.92380927 39.11199585 39.27105726 39.35762224
##          1          1          1          1          1          1
## 39.96460409 40.03486345 40.07339862 40.1648627 40.20138216 40.50002466
##          1          1          1          1          1          1
## 41.14631928 41.29961062 41.41952203 41.5007886 41.67215409 41.9019492
##          1          1          1          1          1          1
## 42.05747152 42.10842971 42.51941582 42.57532852 42.71415955 42.7252659
##          1          1          1          1          1          1
## 42.94775734 43.3675339 43.37886203 44.12649549 44.56461453 45.5681704
##          1          1          1          1          1          1
## 45.61946351 45.81820905 46.20023557 46.26235833 46.48262359 46.71613514
##          1          1          1          1          1          1
## 46.89255314 47.36373088 47.38030857 47.53378129 47.7455336 47.82892163
##          1          1          1          1          1          1
## 48.12703271 48.30471231 48.44051729 48.64130068 48.66974596 48.76273654
##          1          1          1          1          1          1
## 48.81447648 49.20159406 49.57101208 49.61306965 49.63041183 49.70186963
##          1          1          1          1          1          1
## 49.88286614 50.5164034 52.22217423 52.36238745 52.39630661 52.64153689
##          1          1          1          1          1          1
## 52.70434041 52.71237566 53.19134731 53.19787809 53.22630441 53.25871046
##          1          1          1          1          1          1
## 53.33558093 53.34252592 53.5454893 53.67825558 53.7446487 53.79654601
##          1          1          1          1          1          1
## 53.81430078 53.8903663 53.90910419 54.02562084 54.20510763 54.39918921
##          1          1          1          1          1          1
## 54.46688045 54.66991748 54.74701246 54.75245817 54.97635332 55.43052617
##          1          1          1          1          1          1
## 55.53585398 55.53702932 55.53807455 55.59848322 55.6128568 55.62676166
##          1          1          1          1          1          1
```


##	55.85850276	55.91857902	56.15221987	56.17949234	56.35900656	56.36324495
##	1	1	1	1	1	1
##	56.41978981	56.58286023	56.63260799	56.67573474	56.73240055	56.82006238
##	1	1	1	1	1	1
##	56.84258229	57.15141351	57.56679345	57.59345794	57.66601133	57.72739119
##	1	1	1	1	1	1
##	57.79175464	57.86687417	57.88546256	57.88547117	58.04367558	58.08640831
##	1	1	1	1	1	1
##	58.13573666	58.32653727	58.59394601	58.6211205	58.6265536	58.67961657
##	1	1	1	1	1	1
##	58.77102643	59.1032873	59.15456929	59.36114777	59.37956518	59.56744087
##	1	1	1	1	1	1
##	59.58677393	59.623893	59.75922953	59.87126212	59.8719436	59.89884393
##	1	1	1	1	1	1
##	59.90531614	59.92440395	60.08416267	60.10226395	60.49400101	60.52598926
##	1	1	1	1	1	1
##	60.59253971	60.69434951	60.88033012	60.91797936	60.99173554	61.08838151
##	1	1	1	1	1	1
##	61.17475415	61.23401124	61.4488498	61.47665866	61.53215423	61.53351895
##	1	1	1	1	1	1
##	61.54792594	61.62411733	61.71100581	61.72878031	61.72986196	61.73178789
##	1	1	1	1	1	1
##	61.81716834	61.82519059	61.86231141	62.05440484	62.22626705	62.28867429
##	1	1	1	1	1	1
##	62.335	62.46330077	62.51470625	62.57913008	62.61746183	62.64290518
##	1	1	1	1	1	1
##	62.68663157	62.7090301	62.79081711	62.81157622	62.81592673	62.88179465
##	1	1	1	1	1	1
##	62.90432897	63.0489418	63.07203357	63.08526869	63.09661192	63.12172864
##	1	1	1	1	1	1
##	63.22067964	63.33377483	63.41120675	63.5620553	63.5929834	63.60774199
##	1	1	1	1	1	1
##	63.64806867	63.65120836	63.75048389	63.8563798	63.86387253	63.91384349
##	1	1	1	1	1	1
##	63.93200613	64.04462091	64.15782871	64.18530742	64.25086425	64.47102167
##	1	1	1	1	1	1
##	64.47883316	64.70700136	64.71187697	64.71609399	64.72685497	64.85519401
##	1	1	1	1	1	1
##	64.94465191	65.41905444	65.46317765	65.5462692	65.60753955	65.63511208
##	1	1	1	1	1	1
##	65.63971528	65.7627638	65.77030677	65.82313915	65.82386482	65.83781952
##	1	1	1	1	1	1
##	65.92020556	66.06017241	66.08576114	66.1082892	66.14851989	66.19170984
##	1	1	1	1	1	1
##	66.20782922	66.22561217	66.30233239	66.40205807	66.45688889	66.56640794
##	1	1	1	1	1	1
##	66.56947803	66.58389459	66.68856767	66.78247679	66.81680913	67.00235523
##	1	1	1	1	1	1
##	67.02505497	67.07372866	67.14492185	67.18136571	67.18981008	67.19187159
##	1	1	1	1	1	1
##	67.20882077	67.340003	67.35286116	67.36241231	67.37511237	67.52513103
##	1	1	1	1	1	1
##	67.64519953	67.66506824	67.69575926	67.74074818	67.77024968	68.01405213
##	1	1	1	1	1	1

##	68.06368596	68.09498399	68.09951624	68.23529412	68.24146108	68.32789264
##	1	1	1	1	1	1
##	68.40778724	68.44707822	68.56930272	68.66576819	68.66713189	68.73943021
##	1	1	1	1	1	1
##	68.75480295	68.76623707	68.81508584	68.92979408	68.94149762	68.96159986
##	1	1	1	1	1	1
##	69.05513273	69.06379987	69.07589216	69.13638232	69.24869698	69.41070373
##	1	1	1	1	1	1
##	69.61920956	69.66552468	69.67262123	69.70656982	69.82983006	69.85536302
##	1	1	1	1	1	1
##	69.8660205	69.92613732	70.06581022	70.08382643	70.0872857	70.13161253
##	1	1	1	1	1	1
##	70.2302209	70.3068456	70.46766744	70.52848253	70.57884838	70.61894788
##	1	1	1	1	1	1
##	70.80255453	70.82865589	70.87584032	70.88537021	70.94553333	70.94958707
##	1	1	1	1	1	1
##	71.04446245	71.23077469	71.41253551	71.43312962	71.50236763	71.52033417
##	1	1	1	1	1	1
##	71.56481625	71.70072051	71.83549124	71.84702564	71.99867107	72.06055058
##	1	1	1	1	1	1
##	72.19973009	72.28382951	72.45970568	72.49834327	72.58333913	72.63181718
##	1	1	1	1	1	1
##	72.79982954	72.92711203	72.99306414	73.02563343	73.06965199	73.1630591
##	1	1	1	1	1	1
##	73.30694308	73.33771055	73.34272521	73.3605796	73.64076537	73.64998029
##	1	1	1	1	1	1
##	73.70574631	73.7964527	73.80173858	73.85106122	73.8603352	73.91591125
##	1	1	1	1	1	1
##	73.92672732	73.95730899	73.96484761	74.08988863	74.18130582	74.20537897
##	1	1	1	1	1	1
##	74.24044345	74.27382081	74.28381324	74.33184764	74.35931572	74.58309373
##	1	1	1	1	1	1
##	74.65169011	74.66409979	74.67242527	74.69748284	74.72536378	74.72966791
##	1	1	1	1	1	1
##	74.74629196	74.77168565	74.80624858	74.81990158	74.84473039	74.86874585
##	1	1	1	1	1	1
##	74.87236267	74.87261693	74.88129882	74.88816592	74.98196821	74.98799686
##	1	1	1	1	1	1
##	75.01240977	75.03195944	75.03972915	75.09536901	75.13846844	75.17619421
##	1	1	1	1	1	1
##	75.19642582	75.27013065	75.33128316	75.42221085	75.46526055	75.47942792
##	1	1	1	1	1	1
##	75.50680621	75.51029252	75.516016	75.64577627	75.6954831	75.70624512
##	1	1	1	1	1	1
##	75.75059434	75.8002887	75.80933956	75.91359136	75.95609174	75.95762228
##	1	1	1	1	1	1
##	76.02764256	76.03874073	76.08080609	76.1058924	76.16090095	76.19911156
##	1	1	1	1	1	1
##	76.24483316	76.27705471	76.30367718	76.39656816	76.42257649	76.464661
##	1	1	1	1	1	1
##	76.52357582	76.64911489	76.67372536	76.68511436	76.77658989	76.95875202
##	1	1	1	1	1	1
##	76.97804711	77.04863775	77.11176675	77.14139329	77.20770046	77.21459367
##	1	1	1	1	1	1

##	77.25732107	77.27160086	77.3477044	77.46779355	77.52178408	77.54663465
##	1	1	1	1	1	1
##	77.71456838	77.88384406	77.9217492	77.93436819	77.93619676	77.94741844
##	1	1	1	1	1	1
##	77.97468354	78.08697329	78.09877234	78.16041594	78.20248875	78.21998423
##	1	1	1	1	1	1
##	78.31513261	78.36663771	78.38815983	78.42838316	78.44850781	78.50286938
##	1	1	1	1	1	1
##	78.62225601	78.6541433	78.68568419	78.69895062	78.71971318	78.76225088
##	1	1	1	1	1	1
##	78.76957152	78.77316752	78.80733945	78.83273756	78.97302254	78.9999163
##	1	1	1	1	1	1
##	79.05310401	79.06554999	79.09656603	79.12401701	79.20616905	79.21319939
##	1	1	1	1	1	1
##	79.2331147	79.30350155	79.33280274	79.52859912	79.5422369	79.55146573
##	1	1	1	1	1	1
##	79.58076654	79.58742968	79.58798977	79.61972599	79.65743648	79.74433568
##	1	1	1	1	1	1
##	79.74804352	79.75484885	79.79679363	79.79692321	79.8169586	79.82404747
##	1	1	1	1	1	1
##	79.93850491	80.00278125	80.00626062	80.02036299	80.04426307	80.04950495
##	1	1	1	1	1	1
##	80.07217007	80.09481517	80.16330503	80.25184275	80.28525745	80.38026581
##	1	1	1	1	1	1
##	80.39517183	80.43407133	80.46575342	80.47696821	80.50602677	80.54538432
##	1	1	1	1	1	1
##	80.546875	80.58301517	80.69633566	80.7158717	80.74235423	80.76214761
##	1	1	1	1	1	1
##	80.8318264	80.83532158	80.98399963	80.99152707	81.00160921	81.04412853
##	1	1	1	1	1	1
##	81.05213911	81.05399471	81.09964473	81.1536643	81.23411503	81.25056679
##	1	1	1	1	1	1
##	81.25204788	81.26041115	81.29129129	81.3079153	81.4076827	81.41253981
##	1	1	1	1	1	1
##	81.43951478	81.45469085	81.46216936	81.46860186	81.52519688	81.55751196
##	1	1	1	1	1	1
##	81.55946321	81.59594901	81.6313018	81.64110429	81.67343116	81.80033447
##	1	1	1	1	1	1
##	81.84067293	81.85877666	81.96473552	81.99174407	82.00864161	82.04656541
##	1	1	1	1	1	1
##	82.0494186	82.10702341	82.10760824	82.14676443	82.15511161	82.18380346
##	1	1	1	1	1	1
##	82.20704859	82.22687722	82.23233419	82.2868042	82.31597108	82.34111059
##	1	1	1	1	1	1
##	82.50172227	82.58469187	82.60289104	82.64219084	82.68996377	82.72845314
##	1	1	1	1	1	1
##	82.72943097	82.73009307	82.73862931	82.76404906	82.76407845	82.77600959
##	1	1	1	1	1	1
##	82.78916982	82.79565081	82.82266167	82.82413741	82.82920589	82.83177307
##	1	1	1	1	1	1
##	82.83653317	82.86884451	82.97482192	83.03941909	83.0508444	83.1942379
##	1	1	1	1	1	1
##	83.24428311	83.24503311	83.30470629	83.30638048	83.35592685	83.42018001
##	1	1	1	1	1	1

##	83.43598502	83.46784596	83.58260051	83.58814217	83.61168813	83.64624506
##	1	1	1	1	1	1
##	83.65384615	83.6628002	83.71072553	83.73185107	83.73394798	83.76027487
##	1	1	1	1	1	1
##	83.76766498	83.78819687	83.79022646	83.79984663	83.89419841	83.95981087
##	1	1	1	1	1	1
##	83.96028051	84.02649255	84.03981265	84.04454761	84.04573359	84.05531771
##	1	1	1	1	1	1
##	84.05855701	84.06049018	84.0702263	84.08399159	84.09626036	84.13195057
##	1	1	1	1	1	1
##	84.13691572	84.14684873	84.15819708	84.27265804	84.29288405	84.40749963
##	1	1	1	1	1	1
##	84.4213423	84.45108928	84.51989777	84.53185538	84.56202933	84.57622123
##	1	1	1	1	1	1
##	84.59823343	84.72754024	84.75732189	84.79787436	84.85195798	84.85390555
##	1	1	1	1	1	1
##	84.85793486	84.88263065	84.89254943	84.89373923	84.90389123	84.90513779
##	1	1	1	1	1	1
##	84.93705856	84.94101319	84.95441156	84.97536794	85.00654328	85.00670845
##	1	1	1	1	1	1
##	85.01071662	85.01149476	85.04271989	85.13685183	85.17656189	85.21917643
##	1	1	1	1	1	1
##	85.25812551	85.29690558	85.31033937	85.33468845	85.35108054	85.38405594
##	1	1	1	1	1	1
##	85.40975841	85.42860782	85.45906829	85.46395975	85.4703042	85.49665307
##	1	1	1	1	1	1
##	85.50305117	85.5051108	85.59027341	85.66004386	85.6646515	85.71305174
##	1	1	1	1	1	1
##	85.71383309	85.72697332	85.73337788	85.73668622	85.74292951	85.76898532
##	1	1	1	1	1	1
##	85.77846047	85.80561892	85.83386993	85.85898734	85.89548453	85.94828568
##	1	1	1	1	1	1
##	85.98849108	86.00202429	86.03229738	86.09969256	86.12235182	86.15384615
##	1	1	1	1	1	1
##	86.17375723	86.18892432	86.21969754	86.23882396	86.2503845	86.26361271
##	1	1	1	1	1	1
##	86.27619868	86.31195278	86.37873754	86.39309945	86.40590864	86.4172342
##	1	1	1	1	1	1
##	86.42420414	86.42653428	86.50009323	86.51439384	86.53313323	86.56506918
##	1	1	1	1	1	1
##	86.56988996	86.61109314	86.68923923	86.70494453	86.7100877	86.71483405
##	1	1	1	1	1	1
##	86.72084754	86.74607489	86.76610298	86.78174509	86.78477113	86.79003579
##	1	1	1	1	1	1
##	86.84657734	86.86268122	86.90110201	86.91210873	86.91885988	86.92646948
##	1	1	1	1	1	1
##	86.9474417	86.97920091	86.98498267	87.00855394	87.02253254	87.02602088
##	1	1	1	1	1	1
##	87.06275268	87.07928037	87.11028723	87.16773602	87.17320261	87.205294
##	1	1	1	1	1	1
##	87.26641651	87.27895291	87.31753874	87.33964037	87.36651881	87.37949167
##	1	1	1	1	1	1
##	87.38145926	87.38413642	87.39131044	87.39401845	87.41156153	87.43626799
##	1	1	1	1	1	1

##	87.470675	87.51616359	87.52183679	87.52474469	87.56534937	87.61940299
##	1	1	1	1	1	1
##	87.6218248	87.70890127	87.71861748	87.72465609	87.80061626	87.80122144
##	1	1	1	1	1	1
##	87.8192448	87.82364969	87.85019442	87.85836487	87.87217459	87.88763045
##	1	1	1	1	1	1
##	87.89630119	87.89981761	87.92244871	87.95050198	87.96162931	87.97370456
##	1	1	1	1	1	1
##	88.01957558	88.02697811	88.03759743	88.05636199	88.06493706	88.08337109
##	1	1	1	1	1	1
##	88.08504763	88.08928547	88.09306889	88.10515195	88.11047089	88.13192317
##	1	1	1	1	1	1
##	88.13661223	88.18659973	88.19864721	88.21465263	88.24152542	88.24211251
##	1	1	1	1	1	1
##	88.25069295	88.26242952	88.31275932	88.31837381	88.31850331	88.33127127
##	1	1	1	1	1	1
##	88.36185204	88.37058633	88.37558531	88.39750109	88.42885054	88.44436958
##	1	1	1	1	1	1
##	88.44601413	88.44650966	88.45548166	88.51752295	88.5499087	88.55258989
##	1	1	1	1	1	1
##	88.56807251	88.57476084	88.6091954	88.6121673	88.66663129	88.67778872
##	1	1	1	1	1	1
##	88.68356313	88.71332924	88.73123586	88.7742098	88.80986144	88.82818542
##	1	1	1	1	1	1
##	88.83325385	88.86076458	88.95823332	88.96690449	88.97717685	88.97918449
##	1	1	1	1	1	1
##	88.98530012	88.98798172	88.99017523	89.02461341	89.0265234	89.03818953
##	1	1	1	1	1	1
##	89.05792123	89.09704342	89.10027536	89.15127277	89.1518848	89.16273988
##	1	1	1	1	1	1
##	89.17745936	89.19202518	89.22265154	89.2326564	89.26378539	89.27891274
##	1	1	1	1	1	1
##	89.29579301	89.30604203	89.31518563	89.31961576	89.33153275	89.34737712
##	1	1	1	1	1	1
##	89.35844963	89.36052311	89.36743798	89.40663599	89.41366119	89.42117531
##	1	1	1	1	1	1
##	89.43900189	89.44015237	89.44221841	89.46253186	89.47312738	89.47517044
##	1	1	1	1	1	1
##	89.47641265	89.47942905	89.48482061	89.4875549	89.49456134	89.51295286
##	1	1	1	1	1	1
##	89.53510377	89.60478086	89.60699588	89.65284779	89.65381034	89.67242325
##	1	1	1	1	1	1
##	89.67332982	89.67664111	89.69862148	89.73762869	89.75338747	89.77218033
##	1	1	1	1	1	1
##	89.79512402	89.8024309	89.8877193	89.89629391	89.8993987	89.98906873
##	1	1	1	1	1	1
##	89.99770062	90.05977419	90.07563997	90.10192952	90.11553753	90.11837365
##	1	1	1	1	1	1
##	90.12553846	90.24904215	90.25568538	90.31879004	90.33413505	90.34852547
##	1	1	1	1	1	1
##	90.37347713	90.38019661	90.39008962	90.39324535	90.40500136	90.40530075
##	1	1	1	1	1	1
##	90.40843215	90.41344843	90.43205476	90.43706043	90.48232695	90.48964062
##	1	1	1	1	1	1

##	90.51255155	90.52279166	90.53846667	90.57295737	90.64304967	90.6525953
##	1	1	1	1	1	1
##	90.65876859	90.6911225	90.71312709	90.75258239	90.78351965	90.79853315
##	1	1	1	1	1	1
##	90.81785988	90.84310069	90.84703304	90.86021505	90.86640525	90.872137
##	1	1	1	1	1	1
##	90.88725986	90.88987266	90.93897778	90.95027649	90.95564592	90.98819714
##	1	1	1	1	1	1
##	90.98930935	91.01781299	91.02981969	91.03797236	91.04164619	91.04257087
##	1	1	1	1	1	1
##	91.06217822	91.07108763	91.08030896	91.14249245	91.16086235	91.17538216
##	1	1	1	1	1	1
##	91.18760757	91.19845713	91.2134134	91.21839954	91.23809524	91.25473137
##	1	1	1	1	1	1
##	91.26661242	91.31982475	91.33148975	91.3352791	91.39044339	91.3968591
##	1	1	1	1	1	1
##	91.4024158	91.40556625	91.40972416	91.42519801	91.43344225	91.46599743
##	1	1	1	1	1	1
##	91.52562022	91.5416721	91.54757418	91.547682	91.55763487	91.5779126
##	1	1	1	1	1	1
##	91.58384815	91.59412923	91.60869151	91.61133548	91.6213557	91.62256267
##	1	1	1	1	1	1
##	91.63235998	91.66175604	91.66838642	91.70700163	91.71247357	91.72160155
##	1	1	1	1	1	1
##	91.72922764	91.74946217	91.75231888	91.75826601	91.822907	91.84420259
##	1	1	1	1	1	1
##	91.8443972	91.85675342	91.86629526	91.89355958	91.92262256	91.94593057
##	1	1	1	1	1	1
##	91.94709615	91.95178376	91.98705179	91.9911339	91.99266142	92.0039308
##	1	1	1	1	1	1
##	92.01189296	92.01318942	92.05441508	92.06909789	92.07261552	92.09648203
##	1	1	1	1	1	1
##	92.13347628	92.13347996	92.14676934	92.1491744	92.15538847	92.16433254
##	1	1	1	1	1	1
##	92.19793413	92.20324135	92.21431815	92.21624756	92.21993641	92.23297282
##	1	1	1	1	1	1
##	92.24736204	92.25083453	92.2839857	92.29077508	92.29545856	92.31460482
##	1	1	1	1	1	1
##	92.33556299	92.34807973	92.36887491	92.37533393	92.37831389	92.38325932
##	1	1	1	1	1	1
##	92.41373316	92.42127395	92.43055694	92.44469574	92.44677749	92.46692056
##	1	1	1	1	1	1
##	92.47380217	92.48321792	92.48826291	92.49862189	92.5076629	92.54476897
##	1	1	1	1	1	1
##	92.55102423	92.55220387	92.57332665	92.57608654	92.58115889	92.58126107
##	1	1	1	1	1	1
##	92.58539351	92.5888522	92.60408051	92.60500337	92.60910371	92.65554343
##	1	1	1	1	1	1
##	92.66039623	92.66962744	92.69248226	92.69836362	92.73314866	92.73739636
##	1	1	1	1	1	1
##	92.7459942	92.75233583	92.75482094	92.76733474	92.79800317	92.82062207
##	1	1	1	1	1	1
##	92.84715098	92.84977978	92.90157137	92.90871087	92.91160442	92.91517152
##	1	1	1	1	1	1

##	92.92330053	92.95879342	92.9941143	92.99624984	93.00939159	93.01632757
##	1	1	1	1	1	1
##	93.04489073	93.04584704	93.04706681	93.05904661	93.07993519	93.09818643
##	1	1	1	1	1	1
##	93.10674118	93.11429466	93.11764023	93.12798427	93.12987769	93.13285866
##	1	1	1	1	1	1
##	93.13499044	93.13642416	93.13655843	93.14510012	93.14729051	93.15272244
##	1	1	1	1	1	1
##	93.1880109	93.19889639	93.19952881	93.20076092	93.20349762	93.24811807
##	1	1	1	1	1	1
##	93.25074754	93.25596079	93.26500732	93.27246906	93.2775984	93.28714063
##	1	1	1	1	1	1
##	93.2907489	93.29113924	93.29127701	93.29350079	93.33290746	93.35622574
##	1	1	1	1	1	1
##	93.36394999	93.36398271	93.37558685	93.39362832	93.39534621	93.40067714
##	1	1	1	1	1	1
##	93.40417735	93.41732112	93.41812683	93.46380246	93.51382686	93.52895149
##	1	1	1	1	1	1
##	93.53891336	93.57257409	93.60379689	93.60756816	93.61511367	93.61702128
##	1	1	1	1	1	1
##	93.62083707	93.62142045	93.62646404	93.6386991	93.65235013	93.66007786
##	1	1	1	1	1	1
##	93.68868289	93.69924701	93.71840049	93.75334165	93.75620128	93.77950569
##	1	1	1	1	1	1
##	93.77990431	93.78134403	93.82361909	93.82408331	93.84213759	93.85548091
##	1	1	1	1	1	1
##	93.8785585	93.89365655	93.90173005	93.91193215	93.92749942	93.93398397
##	1	1	1	1	1	1
##	93.95492164	93.95538137	93.97413678	93.98094353	94.00995325	94.03143478
##	1	1	1	1	1	1
##	94.03410346	94.03680145	94.04320314	94.08618785	94.08816494	94.10402393
##	1	1	1	1	1	1
##	94.10405614	94.11282671	94.11502452	94.14825724	94.16267161	94.16427578
##	1	1	1	1	1	1
##	94.16838239	94.18381648	94.19527371	94.21051691	94.21758569	94.23408035
##	1	1	1	1	1	1
##	94.24790023	94.25115866	94.25142857	94.25467635	94.25893572	94.26490288
##	1	1	1	1	1	1
##	94.27091895	94.27637469	94.28022465	94.28868544	94.29760603	94.29995957
##	1	1	1	1	1	1
##	94.30355829	94.30398661	94.31427042	94.32701744	94.32849365	94.34724092
##	1	1	1	1	1	1
##	94.35150207	94.35507646	94.36309243	94.36448633	94.37769693	94.37966711
##	1	1	1	1	1	1
##	94.40625677	94.42242804	94.42647004	94.42868303	94.42893232	94.43153934
##	1	1	1	1	1	1
##	94.44316877	94.44317518	94.44819929	94.44894317	94.45115864	94.46289619
##	1	1	1	1	1	1
##	94.46546789	94.48016963	94.49229194	94.51182643	94.51203028	94.51549967
##	1	1	1	1	1	1
##	94.52367722	94.52728899	94.53366584	94.53384419	94.53400891	94.5637224
##	1	1	1	1	1	1
##	94.56911862	94.57660464	94.57692308	94.59855855	94.60512696	94.61717839
##	1	1	1	1	1	1

##	94.62531729	94.64440645	94.67168155	94.67455621	94.67772455	94.69001148
##	1	1	1	1	1	1
##	94.69010674	94.71999449	94.72108614	94.74246064	94.74415847	94.76028356
##	1	1	1	1	1	1
##	94.77124183	94.78005865	94.78326268	94.7913826	94.81534859	94.81794954
##	1	1	1	1	1	1
##	94.81820784	94.82015118	94.83025929	94.83362522	94.83617988	94.84354341
##	1	1	1	1	1	1
##	94.85151263	94.8526791	94.85328829	94.85610053	94.89655172	94.90148231
##	1	1	1	1	1	1
##	94.90628848	94.90850559	94.92232451	94.92269852	94.92561614	94.92716748
##	1	1	1	1	1	1
##	94.94144032	94.96734126	94.9773429	94.9824123	94.99675728	94.99843266
##	1	1	1	1	1	1
##	95.00266951	95.03029058	95.03564597	95.03993274	95.04151238	95.04270211
##	1	1	1	1	1	1
##	95.04940903	95.05000773	95.05622746	95.08121711	95.08122957	95.09868421
##	1	1	1	1	1	1
##	95.10234811	95.12574036	95.13269905	95.15302511	95.17001962	95.1777687
##	1	1	1	1	1	1
##	95.19832985	95.20063609	95.2155464	95.2239283	95.23133117	95.23966555
##	1	1	1	1	1	1
##	95.24914775	95.24918491	95.251778	95.25210084	95.25602197	95.27499731
##	1	1	1	1	1	1
##	95.286051	95.29648609	95.30010499	95.33943392	95.34544639	95.3463349
##	1	1	1	1	1	1
##	95.35568275	95.35758553	95.36579204	95.39845052	95.4084507	95.4095657
##	1	1	1	1	1	1
##	95.40995667	95.46011814	95.46347562	95.46355825	95.46529968	95.46611711
##	1	1	1	1	1	1
##	95.46862737	95.4904724	95.49619057	95.51288633	95.51697268	95.5215301
##	1	1	1	1	1	1
##	95.5304457	95.53248008	95.53503185	95.53692183	95.54753687	95.54874607
##	1	1	1	1	1	1
##	95.55001313	95.55854643	95.56038497	95.56690837	95.56767815	95.57608788
##	1	1	1	1	1	1
##	95.5787037	95.58247675	95.58337212	95.58755225	95.60940326	95.61284047
##	1	1	1	1	1	1
##	95.61601029	95.62860047	95.63393868	95.6346111	95.63524679	95.64007917
##	1	1	1	1	1	1
##	95.64161119	95.65633451	95.66169067	95.66721829	95.67041654	95.67367993
##	1	1	1	1	1	1
##	95.67378765	95.67589438	95.68737433	95.69320077	95.71537366	95.73618167
##	1	1	1	1	1	1
##	95.74567868	95.74720069	95.75790327	95.76259095	95.76457071	95.76563474
##	1	1	1	1	1	1
##	95.76891782	95.77516532	95.77991146	95.78386606	95.78697121	95.78741122
##	1	1	1	1	1	1
##	95.78947368	95.7962963	95.80587178	95.80796156	95.82573179	95.84633385
##	1	1	1	1	1	1
##	95.87472638	95.87953003	95.87960728	95.8860955	95.88800871	95.8964372
##	1	1	1	1	1	1
##	95.90009425	95.90042754	95.90932777	95.9160105	95.9221475	95.93192459
##	1	1	1	1	1	1

##	95.96111082	95.96388741	95.9796948	95.98660053	96.00132188	96.0061761
##	1	1	1	1	1	1
##	96.00980226	96.0271106	96.02836879	96.03204884	96.0375391	96.03977583
##	1	1	1	1	1	1
##	96.06303692	96.07598745	96.08621668	96.08962246	96.09682553	96.11409891
##	1	1	1	1	1	1
##	96.12228149	96.1423255	96.14582103	96.15263423	96.15694977	96.16110472
##	1	1	1	1	1	1
##	96.16488229	96.16930716	96.17779481	96.18284913	96.18689581	96.18988205
##	1	1	1	1	1	1
##	96.19140625	96.19537275	96.20092866	96.21345989	96.21874724	96.23094512
##	1	1	1	1	1	1
##	96.23130035	96.23248195	96.24210378	96.24445203	96.24618996	96.2504498
##	1	1	1	1	1	1
##	96.25054013	96.25277866	96.258032	96.26046355	96.26149132	96.27059844
##	1	1	1	1	1	1
##	96.27578304	96.28820961	96.29150626	96.29171817	96.3036963	96.30562552
##	1	1	1	1	1	1
##	96.31372549	96.3188486	96.33098516	96.33487654	96.34408602	96.34462458
##	1	1	1	1	1	1
##	96.34463821	96.35136243	96.36188643	96.37006103	96.3724155	96.37956347
##	1	1	1	1	1	1
##	96.38798245	96.38901306	96.39508685	96.39845648	96.41161533	96.42578886
##	1	1	1	1	1	1
##	96.43596229	96.44406335	96.45119493	96.45357884	96.45565702	96.45720338
##	1	1	1	1	1	1
##	96.4590209	96.46416628	96.46907216	96.46914199	96.47066359	96.47125783
##	1	1	1	1	1	1
##	96.47679843	96.47709587	96.48323022	96.5029668	96.50624386	96.5143121
##	1	1	1	1	1	1
##	96.51584424	96.5232358	96.52575958	96.52811164	96.52919791	96.52962822
##	1	1	1	1	1	1
##	96.53630653	96.53854553	96.54014559	96.54704719	96.55017457	96.56526006
##	1	1	1	1	1	1
##	96.56902087	96.57754759	96.57917122	96.59491194	96.60148489	96.60156851
##	1	1	1	1	1	1
##	96.6047088	96.61953728	96.62005945	96.62034464	96.62637363	96.633514
##	1	1	1	1	1	1
##	96.6395892	96.64856056	96.6573969	96.65749821	96.66330553	96.67113767
##	1	1	1	1	1	1
##	96.67868177	96.684036	96.68428724	96.68466847	96.69964504	96.70548575
##	1	1	1	1	1	1
##	96.706247	96.71060444	96.71251258	96.71713147	96.72536149	96.73173316
##	1	1	1	1	1	1
##	96.73180129	96.73794853	96.74343867	96.74801784	96.77217173	96.78810223
##	1	1	1	1	1	1
##	96.79317107	96.79799748	96.80049006	96.8035688	96.80886243	96.81008902
##	1	1	1	1	1	1
##	96.81021681	96.81116716	96.81388013	96.81685196	96.81835712	96.82267067
##	1	1	1	1	1	1
##	96.82328393	96.82566063	96.82806781	96.84418146	96.84525314	96.85714286
##	1	1	1	1	1	1
##	96.85867521	96.86165274	96.880078	96.90447812	96.90452072	96.91019347
##	1	1	1	1	1	1

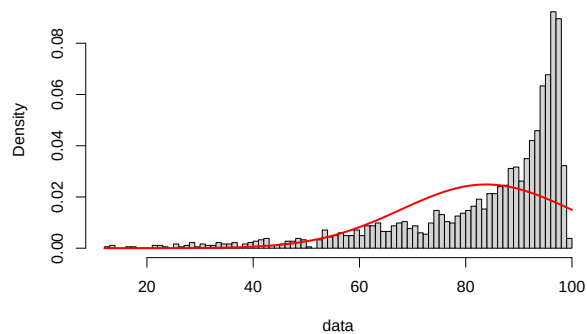
##	96.91401649	96.93098739	96.93733667	96.95109474	96.95286914	96.95962667
##	1	1	1	1	1	1
##	96.96182407	96.9634648	96.96498691	96.96587537	96.97084438	97.01238601
##	1	1	1	1	1	1
##	97.01417004	97.01455712	97.02440655	97.03071969	97.03912779	97.05146533
##	1	1	1	1	1	1
##	97.05350495	97.05368377	97.07208299	97.07397178	97.08737864	97.08799135
##	1	1	1	1	2	1
##	97.08815106	97.10536472	97.1147479	97.11707746	97.11749431	97.12034301
##	1	1	1	1	1	1
##	97.12941827	97.13627386	97.14510215	97.14582702	97.15827033	97.15981348
##	1	1	1	1	1	1
##	97.16864709	97.16917457	97.1747612	97.17705098	97.18000171	97.18796251
##	1	1	1	1	1	1
##	97.19118091	97.19192679	97.19792679	97.20172047	97.20353156	97.20481282
##	1	1	1	1	1	1
##	97.2107438	97.21254355	97.21720659	97.21907146	97.22124516	97.2359708
##	1	1	1	1	1	1
##	97.24643844	97.24843391	97.25051403	97.25480532	97.25524862	97.25714286
##	1	1	1	1	1	1
##	97.25807487	97.26791488	97.26931042	97.27172482	97.27255568	97.27435631
##	1	1	1	1	1	1
##	97.27995261	97.29185244	97.29312716	97.2987257	97.29959399	97.30101427
##	1	1	1	1	1	1
##	97.30226855	97.30869822	97.30936469	97.31413613	97.31792762	97.32594556
##	1	1	1	1	1	1
##	97.327852	97.32802975	97.32833423	97.3357536	97.35622318	97.35846717
##	1	1	1	1	1	1
##	97.36697751	97.36779324	97.36842105	97.37056763	97.37621667	97.39650201
##	1	1	1	1	1	1
##	97.40224228	97.40634006	97.40667976	97.40837129	97.41041855	97.41653819
##	1	1	1	1	1	1
##	97.42660384	97.4281767	97.45791824	97.45836293	97.47399703	97.48219263
##	1	1	1	1	1	1
##	97.48663102	97.49230423	97.50592029	97.52023547	97.53397081	97.53554502
##	1	1	1	1	1	1
##	97.53734356	97.56690998	97.57656562	97.57739558	97.59358289	97.61255116
##	1	1	1	1	1	1
##	97.625	97.62851463	97.63001236	97.63599917	97.63642935	97.65285851
##	1	1	1	1	1	1
##	97.65619224	97.65987351	97.66570114	97.6660401	97.66625616	97.66957969
##	1	1	1	1	1	1
##	97.67318879	97.67873303	97.68491269	97.6900149	97.69165513	97.69326856
##	1	1	1	1	1	1
##	97.69501857	97.70198005	97.71880819	97.71885647	97.72255598	97.72925764
##	1	1	1	1	1	1
##	97.73741567	97.74701811	97.75220765	97.75254555	97.75371806	97.75565403
##	1	1	1	1	1	1
##	97.76552114	97.76951673	97.7759009	97.77894737	97.79765667	97.80653543
##	1	1	1	1	1	1
##	97.81362355	97.82906764	97.85410869	97.86400434	97.86584244	97.87137681
##	1	1	1	1	1	1
##	97.87166455	97.88602941	97.90596602	97.90712884	97.90794979	97.90864982
##	1	1	1	1	1	1

```
## 97.91234619 97.92846739 97.9300706 97.9365947 97.94820047 97.95643661
##          1          1          1          1          1          1
## 97.95798873 97.95918367 97.97312685 97.98364512 97.98726638 97.99958754
##          1          1          1          1          1          1
## 98.02484583 98.02568982 98.02586794 98.04178974 98.04662502 98.06207657
##          1          1          1          1          1          1
## 98.06301468 98.06431274 98.08069258 98.08152415 98.08772559 98.10267857
##          1          1          1          1          1          1
## 98.11448725 98.1206504 98.12185066 98.1227652 98.12279464 98.13197322
##          1          1          1          1          1          1
## 98.13204509 98.14698492 98.15350696 98.15435181 98.16408491 98.17477876
##          1          1          1          1          1          1
## 98.18069689 98.22335025 98.25106879 98.25152728 98.25327511 98.25783972
##          1          1          1          1          1          1
## 98.27973074 98.29218107 98.29934714 98.32162373 98.33974061 98.34410339
##          1          1          1          1          1          1
## 98.35119538 98.3586789 98.36214741 98.37518464 98.37677036 98.37969814
##          1          1          1          1          1          1
## 98.44030451 98.44767151 98.45844504 98.47207587 98.47614741 98.51116625
##          1          1          1          1          1          1
## 98.51689764 98.57233791 98.63355143 98.6533282 98.67435159 98.76015391
##          1          1          1          1          1          1
## 98.80976811 98.81559014 98.8372093 98.90601691 98.92803878 99.00944483
##          1          1          1          1          1          1
## 99.08290536 99.31506849 99.37731536 99.50980392 99.646286 99.69304485
##          1          1          1          1          1          1
```

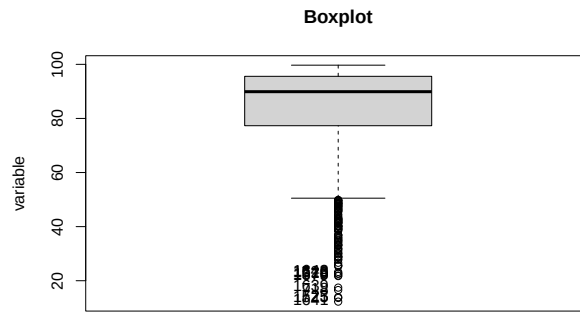
```
res.var <- summary_and_normality_check(variable)
```

```
## [1] "Summary: "
##      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##    12.27  77.31   89.90   83.85   95.57   99.69
## [1] "Mean"
## [1] 83.84699
## [1] "Standard deviation"
## [1] 16.03708
```

Histogram



```
## [1] "Skewness"
## [1] -1.625604
```



```
## [1] "Num Outliers:"
## [1] 97
## [1] "Num. severe outliers:"
## [1] 7
## [1] "Shapiro test: "
##
## Shapiro-Wilk normality test
##
## data: data
## W = 0.80758, p-value < 2.2e-16
##
## [1] "Most repeated value ad times it occurs: "
## [1] "97.08737864"
## [1] 2
```

```
outliers_df[res.var,"pctwhite"] <- 1
```

Variable *pctblack*

The variable *pctblack* represents the percentage of county residents who identify as Black. The values range from 0% to 85.95%, with a mean of 9.08% and a median of 2.32%, indicating that most counties have a relatively small Black population. The interquartile range extends from 0.65% to 10.87%, and the standard deviation of 14.45% highlights substantial variability across counties. The skewness of 2.27 indicates a strong positive skew, meaning a small number of counties have significantly higher percentages of Black residents. The Shapiro-Wilk test confirms that the data is not normally distributed. There are 224 outliers, with 99 classified as severe, and the most frequent value is 0%, occurring in 44 counties.

This variable provides critical insights into the racial diversity of counties and the distribution of Black populations across regions. Comparing *pctblack* with *pctwhite*, *pctpubliccoverage*, or *pctpubliccoveragealone*, can shed light on racial disparities in access to healthcare, insurance coverage, or socioeconomic opportunities. Analyzing its relationship with cancer *target_deathrate* may help identify potential inequities in health outcomes tied to racial demographics.

```
variable<-df$pctblack
table(variable)

## variable
##          0 0.009738995 0.012794268 0.013453518 0.013691128 0.020497404
##          44          1          1          1          1          1
## 0.022291574 0.024594196 0.033046927 0.034285714 0.035848718 0.036580905
##          1          1          1          1          1          1
## 0.0390625 0.041152263 0.043294729 0.044920267 0.045289855 0.045547711
```

##	1	1	1	1	1	1
##	0.051506567	0.05171433	0.054097917	0.054501853	0.055096419	0.055298265
##	1	1	1	1	1	1
##	0.057463925	0.057892706	0.058848256	0.059189109	0.060012753	0.065832785
##	1	1	1	1	1	1
##	0.068013991	0.068317677	0.077120823	0.078483458	0.078698846	0.07902015
##	1	1	1	1	1	1
##	0.079660117	0.080691643	0.081286127	0.083090985	0.083874101	0.084647127
##	1	1	1	1	1	1
##	0.085203067	0.086380651	0.089164203	0.090843023	0.091317251	0.09262935
##	1	1	1	1	1	1
##	0.093052109	0.095351609	0.099255583	0.10063737	0.100777426	0.101992032
##	1	1	1	1	1	1
##	0.103831006	0.103861769	0.105769231	0.110288325	0.113622017	0.11701772
##	1	1	1	1	1	1
##	0.117109732	0.117596088	0.118835413	0.118880363	0.119832235	0.121565767
##	1	1	1	1	1	1
##	0.122709424	0.122774708	0.123221687	0.123342584	0.124185036	0.125921928
##	1	1	1	1	1	1
##	0.128755365	0.128824477	0.129093867	0.129157249	0.134544231	0.136798906
##	1	1	1	1	1	1
##	0.143751925	0.145651269	0.147412099	0.147800383	0.14983254	0.151697111
##	1	1	1	1	1	1
##	0.152695068	0.154112888	0.154508332	0.154774803	0.156128025	0.16014642
##	1	1	1	1	1	1
##	0.16173108	0.164563906	0.16499472	0.165593376	0.167785235	0.171397965
##	1	1	1	1	1	1
##	0.171569864	0.171909919	0.175327522	0.178784267	0.178826896	0.180505415
##	1	1	1	1	1	1
##	0.183031869	0.185070944	0.188283624	0.189170017	0.190106037	0.191391819
##	1	1	1	1	1	1
##	0.191611667	0.19374481	0.194087787	0.194174757	0.196705188	0.199434934
##	1	1	1	1	1	1
##	0.200181984	0.201072386	0.20242915	0.206114737	0.209238693	0.212445175
##	1	1	1	1	1	1
##	0.213583938	0.222768036	0.22866943	0.229489386	0.23139221	0.233349538
##	1	1	1	1	1	1
##	0.234497134	0.235914516	0.243790949	0.245343026	0.245490447	0.246632518
##	1	1	1	1	1	1
##	0.253192426	0.253323164	0.253888581	0.255195042	0.255419885	0.259134491
##	1	1	1	1	1	1
##	0.260153201	0.261058738	0.264466307	0.265632247	0.267893021	0.26925746
##	1	1	1	1	1	1
##	0.269564485	0.270192029	0.270343336	0.27124774	0.274881517	0.275013096
##	1	1	1	1	1	1
##	0.2757184	0.276208412	0.277321863	0.278219396	0.286111338	0.287601193
##	1	1	1	1	1	1
##	0.288718331	0.28922632	0.289969685	0.290941368	0.292340674	0.294117647
##	1	1	1	1	1	1
##	0.295698925	0.296944174	0.297619048	0.2997003	0.301204819	0.301300737
##	1	1	1	1	1	1
##	0.302998086	0.304281678	0.304981362	0.305158634	0.309973714	0.310998021
##	1	1	1	1	1	1
##	0.313620072	0.31617349	0.316668913	0.317561131	0.321789764	0.322344952

##	1	1	1	1	1	1
##	0.323331609	0.323373432	0.323734878	0.325726584	0.326205147	0.326522014
##	1	1	1	1	1	1
##	0.326797386	0.327868852	0.329851567	0.333835882	0.334433728	0.334963489
##	1	1	1	1	1	1
##	0.337637281	0.338382211	0.338609339	0.339411161	0.339762166	0.3409177
##	1	1	1	1	1	1
##	0.341236689	0.341689746	0.342501297	0.346598203	0.348894349	0.350519406
##	1	1	1	1	1	1
##	0.350965154	0.352685838	0.353634578	0.354581352	0.355349556	0.3580774
##	1	1	1	1	1	1
##	0.358397751	0.359901501	0.360770166	0.361454493	0.361663653	0.361783816
##	1	1	1	1	1	1
##	0.363036304	0.370576695	0.373241459	0.373248352	0.376226307	0.376355988
##	1	1	1	1	1	1
##	0.378787879	0.381094404	0.381194409	0.382585285	0.38406145	0.386100386
##	1	1	1	1	1	1
##	0.387789364	0.389972145	0.391836735	0.391913894	0.39209001	0.392166721
##	1	1	1	1	1	1
##	0.393920631	0.396407557	0.396615547	0.397130413	0.398033247	0.398644608
##	1	1	1	1	1	1
##	0.400995575	0.402688452	0.403078051	0.405760658	0.409737286	0.409900678
##	1	1	1	1	1	1
##	0.411058274	0.412301738	0.413080895	0.415480495	0.416747432	0.419426049
##	1	1	1	1	1	1
##	0.419440583	0.419564977	0.421236687	0.424309392	0.426179604	0.426253552
##	1	1	1	1	1	1
##	0.426541804	0.427807487	0.428311382	0.431169156	0.431718062	0.432680071
##	1	1	1	1	1	1
##	0.434054691	0.434920972	0.435931308	0.436681223	0.437921847	0.438180957
##	1	1	1	1	1	1
##	0.44153064	0.442105263	0.448180429	0.451163063	0.451671183	0.452621296
##	1	1	1	1	1	1
##	0.453721597	0.454403008	0.455580866	0.456053068	0.4563065	0.45780764
##	1	1	1	1	1	1
##	0.458669136	0.459875834	0.46022858	0.460597337	0.464902186	0.465717018
##	1	1	1	1	1	1
##	0.469924812	0.471459842	0.472339512	0.472937467	0.473008484	0.473151408
##	1	1	1	1	1	1
##	0.475325038	0.475814453	0.47614249	0.476240872	0.479463624	0.479535608
##	1	1	1	1	1	1
##	0.480113206	0.481173845	0.482587714	0.483376273	0.48421265	0.484976866
##	1	1	1	1	1	1
##	0.485615481	0.485691783	0.486168506	0.486255187	0.486922414	0.487978087
##	1	1	1	1	1	1
##	0.488073142	0.488236199	0.488683128	0.489115459	0.489482736	0.49012934
##	1	1	1	1	1	1
##	0.491951133	0.493139983	0.495771362	0.497718789	0.501998084	0.50271339
##	1	1	1	1	1	1
##	0.504201681	0.504686373	0.506853682	0.507474626	0.507660437	0.509182448
##	1	1	1	1	1	1
##	0.51072523	0.513950073	0.515317509	0.51728832	0.51969172	0.519987
##	1	1	1	1	1	1
##	0.521920668	0.525394046	0.525424315	0.525525526	0.52567023	0.527303602

##	1	1	1	1	1	1
##	0.531220593	0.532177254	0.532346106	0.533305448	0.53498857	0.535031847
##	1	1	1	1	1	1
##	0.537210448	0.537804492	0.538342868	0.538461538	0.539172535	0.540229885
##	1	1	1	1	1	1
##	0.543055349	0.544838757	0.545078905	0.545209941	0.547206795	0.547401347
##	1	1	1	1	1	1
##	0.547491724	0.551548579	0.551661768	0.553967119	0.554083353	0.554391756
##	1	1	1	1	1	1
##	0.554400554	0.557103064	0.559118566	0.561730608	0.564046579	0.566180171
##	1	1	1	1	1	1
##	0.567634512	0.570012392	0.571861651	0.573443372	0.575473327	0.576187875
##	1	1	1	1	1	1
##	0.576970292	0.578243585	0.578687368	0.58189922	0.5822199	0.582306246
##	1	1	1	1	1	1
##	0.583018741	0.583941606	0.586832936	0.589838593	0.591908182	0.593507963
##	1	1	1	1	1	1
##	0.594553126	0.597839625	0.602678571	0.603763459	0.605628785	0.605980983
##	1	1	1	1	1	1
##	0.606173645	0.607837205	0.608720272	0.61482821	0.614898103	0.616142945
##	1	1	1	1	1	1
##	0.616197183	0.616890023	0.619160404	0.619515722	0.620104319	0.620552706
##	1	1	1	1	1	1
##	0.621371335	0.625959608	0.626512886	0.628990756	0.634501278	0.639195534
##	1	1	1	1	1	1
##	0.646821022	0.64911648	0.651792429	0.652370885	0.654656664	0.655094917
##	1	1	1	1	1	1
##	0.659750949	0.659996957	0.662035145	0.66311184	0.664451827	0.666176407
##	1	1	1	1	1	1
##	0.666797411	0.668235402	0.670100874	0.671641791	0.67341378	0.676309424
##	1	1	1	1	1	1
##	0.676714857	0.678571429	0.679706971	0.682004183	0.685238553	0.685253181
##	1	1	1	1	1	1
##	0.685992476	0.686591276	0.687512138	0.69074687	0.691085003	0.691621671
##	1	1	1	1	1	1
##	0.692695214	0.69373073	0.694773409	0.695009242	0.695111613	0.696848242
##	1	1	1	1	1	1
##	0.697281307	0.69978368	0.702782278	0.705739391	0.709186659	0.712332252
##	1	1	1	1	1	1
##	0.71355143	0.715717149	0.718759626	0.718948065	0.719871452	0.719956937
##	1	1	1	1	1	1
##	0.72262368	0.724877654	0.724914155	0.728319181	0.730667749	0.732360652
##	1	1	1	1	1	1
##	0.7352553	0.738459699	0.740125891	0.741953082	0.742804085	0.74553659
##	1	1	1	1	1	1
##	0.746783555	0.747571887	0.748076807	0.754189944	0.755754036	0.756243405
##	1	1	1	1	1	1
##	0.757642889	0.759660905	0.76317577	0.772532189	0.777439024	0.778572799
##	1	1	1	1	1	1
##	0.779016869	0.780096252	0.780326318	0.781058814	0.786731384	0.790177687
##	1	1	1	1	1	1
##	0.790575916	0.791452315	0.793091648	0.793871866	0.796038611	0.800164136
##	1	1	1	1	1	1
##	0.800832033	0.802479385	0.804677999	0.806079954	0.809954751	0.81366965

##	1	1	1	1	1	1
##	0.814214249	0.814546651	0.818115138	0.81875	0.818770084	0.819032761
##	1	1	1	1	1	1
##	0.819732123	0.820232399	0.825746072	0.828739107	0.833514997	0.835654596
##	1	1	1	1	1	1
##	0.835747716	0.836639155	0.836770068	0.838228096	0.843566471	0.844170659
##	1	1	1	1	1	1
##	0.84468753	0.844970204	0.845521024	0.845688405	0.849663969	0.85514382
##	1	1	1	1	1	1
##	0.858990148	0.860323887	0.860951368	0.863583873	0.864422202	0.866613062
##	1	1	1	1	1	1
##	0.867507886	0.87037037	0.872520849	0.874103066	0.875307125	0.879819643
##	1	1	1	1	1	1
##	0.880733945	0.880956029	0.881704629	0.889357853	0.89263051	0.894043736
##	1	1	1	1	1	1
##	0.896043137	0.89846732	0.89869281	0.898920003	0.899356103	0.902708124
##	1	1	1	1	1	1
##	0.902934537	0.903564975	0.904890258	0.905179639	0.905259124	0.910934595
##	1	1	1	1	1	1
##	0.917165076	0.918006194	0.922984224	0.92778671	0.930274978	0.932270021
##	1	1	1	1	1	1
##	0.933092224	0.933676755	0.935461723	0.935583048	0.936949494	0.937190232
##	1	1	1	1	1	1
##	0.941406279	0.94844971	0.952104077	0.954032958	0.954258675	0.955414013
##	1	1	1	1	1	1
##	0.955921402	0.956833411	0.957907929	0.961379082	0.963679155	0.965456481
##	1	1	1	1	1	1
##	0.965517241	0.966855757	0.966911229	0.972222222	0.976572469	0.97788476
##	1	1	1	1	1	1
##	0.980486707	0.980817494	0.982099324	0.98537754	0.986083499	0.98687829
##	1	1	1	1	1	1
##	0.99047619	0.996133886	0.996156735	0.997109526	0.997907362	0.9981463
##	1	1	1	1	1	1
##	1.001251564	1.001663948	1.002857885	1.011336962	1.013279872	1.015265503
##	1	1	1	1	1	1
##	1.016210985	1.023400218	1.032278556	1.034096712	1.03460721	1.035953687
##	1	1	1	1	1	1
##	1.036564529	1.037351744	1.046082319	1.052034774	1.054029244	1.059393015
##	1	1	1	1	1	1
##	1.066782779	1.067785299	1.069947001	1.071193144	1.07931868	1.080221778
##	1	1	1	1	1	1
##	1.08464721	1.085233508	1.085323467	1.088995264	1.090240915	1.09621419
##	1	1	1	1	1	1
##	1.109158131	1.11262048	1.113266287	1.114846709	1.118957089	1.125620894
##	1	1	1	1	1	1
##	1.125863402	1.129559504	1.13202047	1.135566142	1.143973214	1.156905278
##	1	1	1	1	1	1
##	1.160117616	1.162833897	1.169092127	1.173907782	1.175869121	1.181567546
##	1	1	1	1	1	1
##	1.18373649	1.187074441	1.187901583	1.188867874	1.19063793	1.191567369
##	1	1	1	1	1	1
##	1.19563473	1.197839961	1.197932502	1.201780415	1.202124685	1.211453744
##	1	1	1	1	1	1
##	1.215311064	1.223044738	1.225330731	1.225852446	1.226048506	1.226215645

##	1	1	1	1	1	1
##	1.226461843	1.231783484	1.234965485	1.236167288	1.23860847	1.239444068
##	1	1	1	1	1	1
##	1.241038319	1.243821806	1.245169601	1.251121179	1.25463359	1.259782401
##	1	1	1	1	1	1
##	1.26484464	1.267346671	1.26789882	1.272384543	1.280447225	1.281922884
##	1	1	1	1	1	1
##	1.28378497	1.287374079	1.289709476	1.290607735	1.291959857	1.295164316
##	1	1	1	1	1	1
##	1.300087689	1.308472359	1.314060447	1.324220419	1.325927954	1.327331487
##	1	1	1	1	1	1
##	1.336881718	1.336882036	1.338239302	1.351623966	1.354695845	1.356142223
##	1	1	1	1	1	1
##	1.361021468	1.364683235	1.365273473	1.376251848	1.379602201	1.382013201
##	1	1	1	1	1	1
##	1.384175746	1.397649969	1.399144025	1.408362804	1.408646173	1.410508927
##	1	1	1	1	1	1
##	1.410803787	1.41163763	1.41473972	1.415294435	1.416350703	1.420487354
##	1	1	1	1	1	1
##	1.422988701	1.425178147	1.426386369	1.429154757	1.433615064	1.436746621
##	1	1	1	1	1	1
##	1.445640277	1.445864156	1.453945144	1.461075451	1.482194418	1.489752879
##	1	1	1	1	1	1
##	1.490166	1.490584286	1.49478835	1.494872241	1.496856046	1.499469884
##	1	1	1	1	1	1
##	1.513947038	1.517412401	1.519005291	1.522129042	1.526980482	1.536515381
##	1	1	1	1	1	1
##	1.540507592	1.541976014	1.548653775	1.561572116	1.563265814	1.565008026
##	1	1	1	1	1	1
##	1.56816852	1.568714096	1.568981064	1.576749808	1.579240647	1.594331267
##	1	1	1	1	1	1
##	1.596107887	1.59912927	1.610212519	1.612867784	1.615849018	1.616713421
##	1	1	1	1	1	1
##	1.618384852	1.635082889	1.635127321	1.636285345	1.637795276	1.642509208
##	1	1	1	1	1	1
##	1.644088412	1.645244216	1.649198004	1.649850014	1.653205244	1.658398299
##	1	1	1	1	1	1
##	1.662864333	1.677822492	1.68499883	1.685052734	1.699674845	1.702812606
##	1	1	1	1	1	1
##	1.704762633	1.714737518	1.720068984	1.73020569	1.730872516	1.735174145
##	1	1	1	1	1	1
##	1.741749074	1.751012146	1.765797362	1.767910967	1.784174969	1.787049109
##	1	1	1	1	1	1
##	1.789722745	1.790178211	1.807469543	1.817357332	1.820342619	1.822198458
##	1	1	1	1	1	1
##	1.824259176	1.832504299	1.837917849	1.842234942	1.850163746	1.869714759
##	1	1	1	1	1	1
##	1.882115515	1.896275988	1.901671956	1.911381407	1.916342412	1.916698804
##	1	1	1	1	1	1
##	1.920100087	1.923364766	1.924059254	1.925683896	1.929332787	1.933627641
##	1	1	1	1	1	1
##	1.94148437	1.945118181	1.94532191	1.960236346	1.968536846	1.986597129
##	1	1	1	1	1	1
##	1.988569408	2.002271484	2.002294367	2.004478672	2.005617701	2.008141937

##	1	1	1	1	1	1
##	2.0256917	2.048959642	2.05238279	2.054016021	2.054846537	2.05800354
##	1	1	1	1	1	1
##	2.061446977	2.0657277	2.073383029	2.077549067	2.086865026	2.088167053
##	1	1	1	1	1	1
##	2.096061978	2.096261541	2.102845026	2.109602328	2.11366539	2.115738683
##	1	1	1	1	1	1
##	2.11836226	2.128872838	2.129412001	2.132247245	2.136509737	2.154398564
##	1	1	1	1	1	1
##	2.154609838	2.170313293	2.174522059	2.197657587	2.202843626	2.206311246
##	1	1	1	1	1	1
##	2.207404001	2.211658094	2.222103186	2.234460603	2.23567769	2.236653468
##	1	1	1	1	1	1
##	2.238336479	2.238745906	2.257525084	2.274509804	2.296482372	2.30499967
##	1	1	1	1	1	1
##	2.311123125	2.313187763	2.322915109	2.3229246	2.325284924	2.326771335
##	1	1	1	1	1	1
##	2.328623757	2.329856165	2.329998147	2.332148869	2.337591737	2.347701366
##	1	1	1	1	1	1
##	2.34913012	2.356118845	2.370938479	2.371513851	2.375713884	2.380478088
##	1	1	1	1	1	1
##	2.383054392	2.388531878	2.410242515	2.417703277	2.418338109	2.425118031
##	1	1	1	1	1	1
##	2.429850619	2.436075682	2.446601942	2.44903628	2.45380672	2.4604294
##	1	1	1	1	1	1
##	2.482394366	2.488115083	2.499597704	2.5050523	2.50659389	2.508675124
##	1	1	1	1	1	1
##	2.518843007	2.537206383	2.551816663	2.563864375	2.576177285	2.581386139
##	1	1	1	1	1	1
##	2.585982167	2.595851085	2.596259538	2.599928348	2.604285948	2.606036431
##	1	1	1	1	1	1
##	2.606084092	2.609010226	2.609400896	2.623783326	2.636604843	2.638269024
##	1	1	1	1	1	1
##	2.640406176	2.640984162	2.645042567	2.659033479	2.669675551	2.68185158
##	1	1	1	1	1	1
##	2.683246073	2.714823292	2.726690457	2.727068299	2.729562629	2.741925127
##	1	1	1	1	1	1
##	2.742837769	2.748204429	2.749601678	2.761000273	2.779349904	2.781105596
##	1	1	1	1	1	1
##	2.788481961	2.795409904	2.798403564	2.800763845	2.834754552	2.837189
##	1	1	1	1	1	1
##	2.847329148	2.855136906	2.856500225	2.859484171	2.86936375	2.877066368
##	1	1	1	1	1	1
##	2.886137901	2.902259183	2.905684992	2.926183898	2.9264536	2.928288531
##	1	1	1	1	1	1
##	2.94627383	2.948700687	2.966719737	2.967533421	2.974094898	2.977877572
##	1	1	1	1	1	1
##	2.980111774	2.992395492	3.019593429	3.040631447	3.049345833	3.068462171
##	1	1	1	1	1	1
##	3.082419636	3.086306972	3.086787951	3.092227776	3.09760905	3.107869482
##	1	1	1	1	1	1
##	3.12536908	3.126509766	3.127385685	3.137346038	3.142235687	3.145379211
##	1	1	1	1	1	1
##	3.168047895	3.179076923	3.186646434	3.206795774	3.21045057	3.211806505

##	1	1	1	1	1	1
##	3.217455621	3.229166667	3.250464478	3.253027878	3.263283786	3.313093981
##	1	1	1	1	1	1
##	3.31974939	3.320037541	3.335298577	3.344363196	3.359918547	3.377140723
##	1	1	1	1	1	1
##	3.391097745	3.406596711	3.418155636	3.419016721	3.423649584	3.425703688
##	1	1	1	1	1	1
##	3.456319686	3.456492233	3.462271693	3.488372093	3.491634884	3.498480458
##	1	1	1	1	1	1
##	3.50591716	3.50877193	3.51668824	3.520281279	3.524169592	3.525142561
##	1	1	1	1	1	1
##	3.532809125	3.543473157	3.556127389	3.56835842	3.596447234	3.598394532
##	1	1	1	1	1	1
##	3.613754297	3.617592538	3.62211657	3.634613667	3.671207573	3.675906872
##	1	1	1	1	1	1
##	3.680331049	3.6900369	3.702105589	3.708013105	3.725621042	3.746518562
##	1	1	1	1	1	1
##	3.746670104	3.767219567	3.784976372	3.798703484	3.799711975	3.824486233
##	1	1	1	1	1	1
##	3.840758909	3.841645388	3.848186183	3.857660211	3.863348235	3.874366549
##	1	1	1	1	1	1
##	3.883428414	3.898487438	3.903716561	3.907186685	3.912684313	3.915492958
##	1	1	1	1	1	1
##	3.930892376	3.942532576	3.946272964	3.972315064	3.981402722	3.983953521
##	1	1	1	1	1	1
##	3.999408526	4.000268384	4.031375892	4.045751016	4.04906106	4.050161904
##	1	1	1	1	1	1
##	4.052445277	4.091758953	4.094270426	4.101838755	4.11560629	4.124833997
##	1	1	1	1	1	1
##	4.125293843	4.152439444	4.154831255	4.169345665	4.170844195	4.176349512
##	1	1	1	1	1	1
##	4.185170233	4.185463659	4.198567133	4.209668026	4.2229167	4.247210094
##	1	1	1	1	1	1
##	4.289238945	4.290873477	4.299697561	4.305764809	4.348637358	4.352840881
##	1	1	1	1	1	1
##	4.355020019	4.355625718	4.374899727	4.39424391	4.445537251	4.451408467
##	1	1	1	1	1	1
##	4.4575357	4.471615467	4.484407011	4.488774095	4.516516761	4.525040706
##	1	1	1	1	1	1
##	4.548091426	4.552129222	4.5690091	4.630518704	4.641309106	4.661597906
##	1	1	1	1	1	1
##	4.684599281	4.706903096	4.72646663	4.732529617	4.732582688	4.733215276
##	1	1	1	1	1	1
##	4.784123618	4.826421777	4.834727747	4.882669415	4.901347251	4.90648941
##	1	1	1	1	1	1
##	4.908104323	4.919631094	4.95364368	4.99195479	5.005123355	5.018399019
##	1	1	1	1	1	1
##	5.06497342	5.069145617	5.070229173	5.071432337	5.07553053	5.086830324
##	1	1	1	1	1	1
##	5.116428739	5.130784708	5.156157926	5.199353976	5.20119225	5.216718569
##	1	1	1	1	1	1
##	5.221043651	5.225488773	5.22667908	5.284712238	5.290675962	5.399653198
##	1	1	1	1	1	1
##	5.418208734	5.420679774	5.433811392	5.447660575	5.456625547	5.460829913

##	1	1	1	1	1	1
##	5.525990567	5.595503543	5.614035088	5.616717635	5.61707268	5.624133148
##	1	1	1	1	1	1
##	5.657748049	5.662349723	5.714088779	5.729687432	5.765959527	5.785185739
##	1	1	1	1	1	1
##	5.789395953	5.810985443	5.817918027	5.824107282	5.841296219	5.844992636
##	1	1	1	1	1	1
##	5.854426677	5.857071303	5.861119956	5.871238876	5.901412759	5.91099916
##	1	1	1	1	1	1
##	5.913334131	5.965955367	5.986506875	5.993537964	6.006925946	6.021202442
##	1	1	1	1	1	1
##	6.081964418	6.092289504	6.112570356	6.172693378	6.242912313	6.262568687
##	1	1	1	1	1	1
##	6.285700816	6.350872329	6.365875794	6.395094174	6.398697989	6.425935688
##	1	1	1	1	1	1
##	6.432748538	6.52387617	6.530794156	6.54800332	6.567988485	6.58464463
##	1	1	1	1	1	1
##	6.69413777	6.702359707	6.710854162	6.803366125	6.805041628	6.830142103
##	1	1	1	1	1	1
##	6.834058073	6.914562269	6.919462573	6.921192323	6.932296728	6.953355963
##	1	1	1	1	1	1
##	6.96562466	7.042128209	7.05432088	7.066798162	7.078049152	7.083042568
##	1	1	1	1	1	1
##	7.122121153	7.195352868	7.219146858	7.224465252	7.241474527	7.243170016
##	1	1	1	1	1	1
##	7.286251262	7.333247489	7.388746206	7.407407407	7.409987697	7.432026406
##	1	1	1	1	1	1
##	7.461754605	7.479148201	7.479233436	7.498432267	7.546456928	7.589333524
##	1	1	1	1	1	1
##	7.646316006	7.714978176	7.735331363	7.785514498	7.792569044	7.797191378
##	1	1	1	1	1	1
##	7.821907679	7.838501689	7.873954092	7.887144565	7.929362881	7.938041932
##	1	1	1	1	1	1
##	7.950127015	8.002241298	8.267344126	8.269310202	8.276139555	8.321515856
##	1	1	1	1	1	1
##	8.347991259	8.366100174	8.429995099	8.445612882	8.447281152	8.450679215
##	1	1	1	1	1	1
##	8.455647592	8.499002149	8.543421657	8.580298816	8.594261758	8.624376336
##	1	1	1	1	1	1
##	8.69232242	8.720413975	8.725355342	8.753577757	8.779226049	8.783524904
##	1	1	1	1	1	1
##	8.792982456	8.799950283	8.838191511	8.869576983	8.88065339	8.901734104
##	1	1	1	1	1	1
##	9.005981585	9.007635485	9.007769711	9.01979626	9.095473739	9.103588552
##	1	1	1	1	1	1
##	9.110169492	9.135126919	9.143708769	9.144961688	9.154993358	9.246426722
##	1	1	1	1	1	1
##	9.327686477	9.363244448	9.366877449	9.374527112	9.449514477	9.473120283
##	1	1	1	1	1	1
##	9.559840534	9.60651289	9.633092506	9.706397559	9.724612737	9.76168933
##	1	1	1	1	1	1
##	9.7617388	9.763596853	9.830420935	9.957416675	9.962582357	10.03488894
##	1	1	1	1	1	1
##	10.05636894	10.08655057	10.13781119	10.18320497	10.18372568	10.24050238

##	1	1	1	1	1	1
##	10.27333021	10.28087523	10.2884207	10.32049101	10.33341701	10.37985398
##	1	1	1	1	1	1
##	10.39844188	10.46357616	10.49057817	10.53037053	10.54029093	10.54964824
##	1	1	1	1	1	1
##	10.72857652	10.76168053	10.80344358	10.83685568	10.89773253	10.93701313
##	1	1	1	1	1	1
##	10.97128747	10.97175264	11.01207902	11.063967	11.06955027	11.07337437
##	1	1	1	1	1	1
##	11.08452951	11.08706935	11.12664244	11.15097128	11.15370573	11.21220841
##	1	1	1	1	1	1
##	11.24307567	11.2732989	11.2767481	11.30187778	11.32083624	11.36069274
##	1	1	1	1	1	1
##	11.39219201	11.39551338	11.43067459	11.43518519	11.45715477	11.46783485
##	1	1	1	1	1	1
##	11.47136847	11.57862578	11.65506063	11.67522576	11.69079969	11.72329318
##	1	1	1	1	1	1
##	11.74761278	11.76058219	11.76239375	11.83148879	11.88655785	11.96093354
##	1	1	1	1	1	1
##	12.03694931	12.15687742	12.16571379	12.19750195	12.21014343	12.25351298
##	1	1	1	1	1	1
##	12.25460123	12.25936824	12.28607684	12.3184956	12.34899979	12.46258045
##	1	1	1	1	1	1
##	12.53344076	12.74135738	12.74231678	12.84109466	12.86639266	12.9214282
##	1	1	1	1	1	1
##	12.94094332	13.05737939	13.14383562	13.30947586	13.31016378	13.33836749
##	1	1	1	1	1	1
##	13.34419509	13.35227856	13.39795282	13.44057074	13.4846651	13.50965083
##	1	1	1	1	1	1
##	13.53595596	13.5426542	13.54933187	13.55714182	13.61440544	13.71272315
##	1	1	1	1	1	1
##	13.76832989	13.83759705	13.88414668	13.90759431	13.97876827	14.02487715
##	1	1	1	1	1	1
##	14.13973947	14.19621749	14.25365373	14.25766389	14.26747466	14.35032215
##	1	1	1	1	1	1
##	14.38542071	14.39048386	14.39970187	14.40762666	14.42136222	14.51262207
##	1	1	1	1	1	1
##	14.51561661	14.53426142	14.55108359	14.73354519	14.95277134	15.02458779
##	1	1	1	1	1	1
##	15.02612166	15.07748821	15.15211223	15.27721296	15.34744501	15.36690359
##	1	1	1	1	1	1
##	15.37920996	15.4434015	15.47481367	15.50722981	15.56381093	15.57196385
##	1	1	1	1	1	1
##	15.79654188	15.87127215	15.98473336	16.00163236	16.12030469	16.1588181
##	1	1	1	1	1	1
##	16.19689542	16.20545287	16.38626677	16.49794293	16.52802893	16.60964436
##	1	1	1	1	1	1
##	16.67451303	16.73651126	16.74777301	17.11924011	17.12931569	17.14569459
##	1	1	1	1	1	1
##	17.14703824	17.27446584	17.33818264	17.35123025	17.43480849	17.52249472
##	1	1	1	1	1	1
##	17.53067681	17.56756757	17.6079398	17.93686107	17.97243412	17.97275895
##	1	1	1	1	1	1
##	18.13623136	18.27973291	18.33635989	18.33918025	18.40400246	18.40628386

##	1	1	1	1	1	1
##	18.47910969	18.48121445	18.53435277	18.61159725	18.6422952	18.86174749
##	1	1	1	1	1	1
##	18.89015583	18.92983888	19.03404714	19.2257232	19.29994514	19.30826028
##	1	1	1	1	1	1
##	19.54401441	19.6486584	19.68915448	19.89630947	19.8991225	19.95761215
##	1	1	1	1	1	1
##	20.01715199	20.08321304	20.13855325	20.14483106	20.42012725	20.4657981
##	1	1	1	1	1	1
##	20.56405641	20.74262899	20.76687384	20.80414778	20.80926643	20.84834016
##	1	1	1	1	1	1
##	20.89795647	20.9050664	20.93309039	21.21887498	21.30720336	21.31779765
##	1	1	1	1	1	1
##	21.41765137	21.50319594	21.53256799	21.59173195	21.60403752	21.66463405
##	1	1	1	1	1	1
##	21.75586405	21.80007377	21.82603998	21.88376998	22.17349857	22.23116734
##	1	1	1	1	1	1
##	22.36997936	22.39316712	22.92154567	22.93643831	23.02627812	23.09881801
##	1	1	1	1	1	1
##	23.17877095	23.27327939	23.29987224	23.51126417	23.75103722	23.94560505
##	1	1	1	1	1	1
##	23.98259642	24.02235729	24.08425173	24.10673903	24.15381757	24.23282658
##	1	1	1	1	1	1
##	24.27026226	24.28018855	24.31198343	24.32495891	24.50248692	24.76471674
##	1	1	1	1	1	1
##	24.80232814	24.85144437	24.85313143	24.87484455	24.90262243	24.91752955
##	1	1	1	1	1	1
##	24.95248347	25.3541508	25.51007103	25.55666914	25.64530449	25.66213621
##	1	1	1	1	1	1
##	25.69276001	25.69346346	25.70812165	25.71195967	26.23	26.34358555
##	1	1	1	1	1	1
##	26.4161463	26.4251357	26.43762747	26.48635231	26.54095131	26.55804144
##	1	1	1	1	1	1
##	26.59198946	26.63819761	26.76500072	26.81149994	27.05619457	27.11581014
##	1	1	1	1	1	1
##	27.37761703	27.64981708	27.6991722	27.83747624	28.2468665	28.40395054
##	1	1	1	1	1	1
##	28.4350565	28.47576531	28.70685217	28.73667948	28.7886873	28.86958814
##	1	1	1	1	1	1
##	28.95770123	29.25823708	29.31896727	29.39720863	29.55648803	29.61957282
##	1	1	1	1	1	1
##	29.68819393	29.74249896	29.74722338	29.80502066	29.81733524	29.86077981
##	1	1	1	1	1	1
##	30.13145459	30.32707171	30.39499037	30.44557051	30.46579129	30.55095506
##	1	1	1	1	1	1
##	30.58346387	30.68831051	30.76310702	30.78303713	30.79534014	30.94009016
##	1	1	1	1	1	1
##	31.06469003	31.06954914	31.08296477	31.12709408	31.12897909	31.129657
##	1	1	1	1	1	1
##	31.30666667	31.42633017	31.47892471	31.53678911	31.69439256	32.02041974
##	1	1	1	1	1	1
##	32.070626	32.29537707	32.45828047	32.56476684	32.56815887	32.79105895
##	1	1	1	1	1	1
##	32.89652905	32.96335824	33.20055421	33.49298666	33.53508182	33.57204487

```

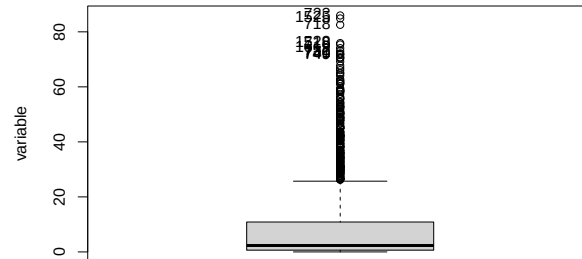
##          1          1          1          1          1          1
## 33.61147142 34.03408186 34.09000924 34.14711332 34.14782183 34.32552872
##          1          1          1          1          1          1
## 34.42422596 34.46801031 34.55176639 34.64470399 34.87541943 35.02628041
##          1          1          1          1          1          1
## 35.16019697 35.25382048 35.69942703 35.70146211 35.8495874 35.85845617
##          1          1          1          1          1          1
## 35.90096898 35.92069527 36.12648602 36.12801781 36.34002361 36.475995
##          1          1          1          1          1          1
## 36.92615333 36.9308811 37.0549173 37.11670201 37.81163435 37.99740404
##          1          1          1          1          1          1
## 38.08519233 38.19807428 38.1991512 38.23775973 38.31056017 38.40459516
##          1          1          1          1          1          1
## 38.71002189 38.97588018 39.24883695 39.41434617 39.43786029 39.55827127
##          1          1          1          1          1          1
## 40.18527392 40.31221573 40.4805492 40.62306764 40.68455854 40.86924282
##          1          1          1          1          1          1
## 40.98105061 41.22450976 41.35965102 41.73315797 41.79430057 41.8699187
##          1          1          1          1          1          1
## 41.94748515 42.02609002 42.18068536 42.2289238 42.35812231 42.3650282
##          1          1          1          1          1          1
## 42.40540179 42.45540635 42.7327249 42.75757007 43.16290641 43.40750963
##          1          1          1          1          1          1
## 43.56272915 43.6153953 43.83323004 44.76770602 44.91060186 45.04013761
##          1          1          1          1          1          1
## 45.06815184 45.13002206 45.1862732 45.53374261 45.70626892 45.85636053
##          1          1          1          1          1          1
## 46.64227772 46.76592901 47.59928542 47.66099635 47.94877135 48.05719363
##          1          1          1          1          1          1
## 48.1859525 48.65959953 48.68779823 48.81193801 48.91064099 48.94961671
##          1          1          1          1          1          1
## 49.22613929 49.81092711 50.17601024 50.17840901 50.31408776 50.36392854
##          1          1          1          1          1          1
## 50.7841246 50.8509997 50.88430721 51.66247733 52.18042579 52.47161572
##          1          1          1          1          1          1
## 52.68071732 52.71852839 52.95997106 53.09076581 53.72251096 53.84164509
##          1          1          1          1          1          1
## 54.34433097 54.50714547 55.52706464 55.58410834 55.78263438 56.08929683
##          1          1          1          1          1          1
## 56.41180012 57.43898217 58.23265447 58.24593603 58.34701248 58.60234126
##          1          1          1          1          1          1
## 58.86112725 59.01478797 59.49448202 61.05025773 61.23440248 61.2883341
##          1          1          1          1          1          1
## 61.88653452 61.97972512 62.30559876 62.83548664 64.04165311 64.81772385
##          1          1          1          1          1          1
## 65.14329977 66.06594861 66.81335459 67.73872039 69.18441904 70.24867929
##          1          1          1          1          1          1
## 70.65810594 70.70612474 71.55046826 71.93753926 72.17921056 73.0393035
##          1          1          1          1          1          1
## 74.05511078 75.40071238 75.89688762 82.55913062 84.86602358 85.94779858
##          1          1          1          1          1          1

```

```
res.var <- summary_and_normality_check(variable)
```

```
## [1] "Summary: "
```

```
outliers_df[res.var, "pctblack"] <- 1
```



```
outliers_df[res.var, "pctblack"] <- 1
```


Variable *pctasian*

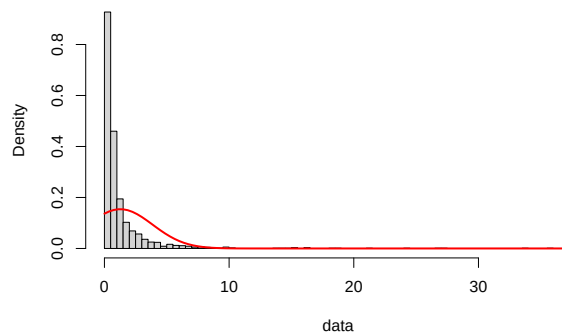
The variable *pctasian* represents the percentage of county residents who identify as Asian. The values range from 0% to 37.16%, with a mean of 1.27% and a median of 0.55%, indicating that the majority of counties have a very low percentage of Asian residents. The interquartile range extends from 0.26% to 1.25%, with a standard deviation of 2.59%. The skewness of 7.10 indicates a highly positive skew, where a few counties have significantly higher percentages of Asian residents. The Shapiro-Wilk test confirms the data deviates strongly from normality. There are 198 outliers, of which 107 are severe, and the most frequent value is 0%, occurring in 107 counties.

This variable highlights the uneven distribution of Asian populations across counties. The high skew and large number of outliers underscore the concentration of Asian residents in a few counties, likely in urban or metropolitan areas. Analyzing *pctasian* alongside other racial variables like *pctwhite* or *pctblack* and socioeconomic factors such as *pctprivatecoverage* or *pctpubliccoverage* can help explore racial and ethnic disparities in healthcare access, insurance coverage, and health outcomes.

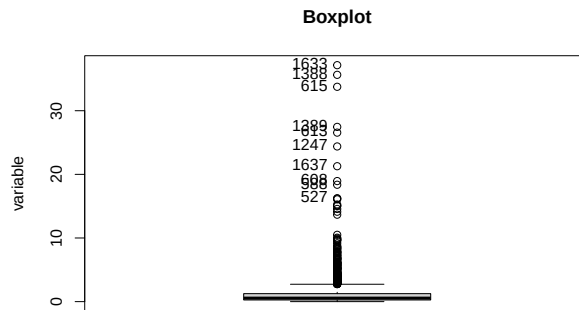
```
variable<-df$pctasian
res.var <- summary_and_normality_check(variable)
```

```
## [1] "Summary: "
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##  0.0000  0.2582  0.5495  1.2743  1.2515 37.1569
## [1] "Mean"
## [1] 1.274301
## [1] "Standard deviation"
## [1] 2.588845
```

Histogram

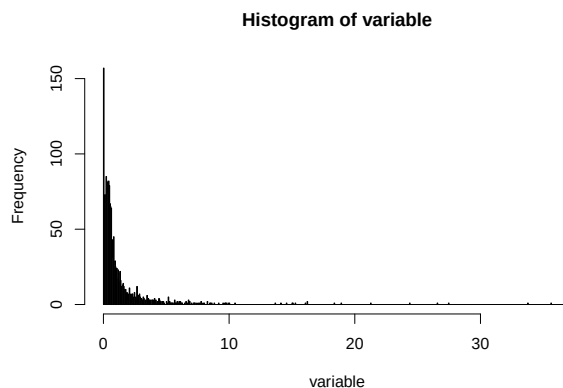


```
## [1] "Skewness"
## [1] 7.103922
```



```
## [1] "Num Outliers:"
## [1] 198
## [1] "Num. severe outliers:"
## [1] 107
## [1] "Shapiro test: "
##
## Shapiro-Wilk normality test
##
## data: data
## W = 0.41908, p-value < 2.2e-16
##
## [1] "Most repeated value ad times it occurs: "
## [1] "0"
## [1] 107
```

```
hist(variable , 1000 )
```



```
outliers_df[res.var,"pctasian"] <- 1
```

Variable *pctotherrace*

The variable *pctotherrace* represents the percentage of county residents identifying as a race other than White, Black, or Asian. It ranges from 0% to 41.93%, with a mean of 2.00% and a median of 0.78%. Most counties have very low values, as reflected in the interquartile range (0.29% to 2.11%) and the strong positive skew (4.99). Outliers are frequent (181 total, 102 severe), and the most common value is 0%, appearing in 79 counties. The Shapiro-Wilk test indicates a significant departure from normality.

```
variable<-df$pctotherrace
table(variable)
```

```
## variable
##      0 0.007952286 0.009243853 0.010645093 0.011117287 0.012302393
##      79      1      1      1      1      1
## 0.012604777 0.013071895 0.013615322 0.013827434 0.01445609 0.014638716
##      1      1      1      1      1      1
## 0.014715621 0.015320974 0.018382353 0.018570102 0.019961674 0.020420666
##      1      1      1      1      1      1
## 0.021059725 0.024803125 0.025824223 0.026465529 0.027360933 0.028293879
##      1      1      1      1      1      1
## 0.029884645 0.030646644 0.031578947 0.033496349 0.035212301 0.035234013
##      1      1      1      1      1      1
## 0.035277894 0.035848718 0.035991938 0.036543029 0.037924278 0.039635355
##      1      1      1      1      1      1
## 0.039984006 0.040371417 0.040387722 0.041265475 0.042979558 0.043758889
##      1      1      1      1      1      1
## 0.044385264 0.044776119 0.045010577 0.045495905 0.046023299 0.046253469
##      1      1      1      1      1      1
## 0.046532369 0.047442422 0.047850706 0.048323769 0.048459004 0.049160392
##      1      1      1      1      1      1
## 0.049739995 0.051093399 0.051440329 0.05252246 0.053399786 0.053475936
##      1      1      1      1      1      1
## 0.053649377 0.053814072 0.054854635 0.056306306 0.057197331 0.059029388
##      1      1      1      1      1      1
## 0.059174516 0.059417706 0.059548621 0.060192616 0.060208724 0.060222824
##      1      1      1      1      1      1
## 0.061525841 0.06281407 0.065866855 0.066021127 0.066615912 0.067228945
##      1      1      1      1      1      1
## 0.067393014 0.06875 0.070458622 0.073190368 0.073313783 0.075011892
##      1      1      1      1      1      1
## 0.075046904 0.075872534 0.077720207 0.077743705 0.077957513 0.080385852
##      1      1      1      1      1      1
## 0.080691643 0.082236842 0.082349712 0.08361204 0.083671635 0.084576349
##      1      1      1      1      1      1
## 0.085724083 0.086215155 0.086719524 0.086749078 0.08675535 0.08775351
##      1      1      1      1      1      1
## 0.087822014 0.087830585 0.087886334 0.087977532 0.08854078 0.089069856
##      1      1      1      1      1      1
## 0.089285714 0.089572292 0.091322597 0.091407678 0.091421606 0.092103865
##      1      1      1      1      1      1
## 0.092341119 0.09391876 0.097145254 0.097237569 0.09731413 0.097581018
##      1      1      1      1      1      1
## 0.097697732 0.099348714 0.100416729 0.101078167 0.101905184 0.102406554
##      1      1      1      1      1      1
## 0.102612677 0.102627621 0.10307286 0.104544616 0.10477956 0.104891244
##      1      1      1      1      1      1
## 0.105329682 0.105529759 0.105764146 0.105932203 0.106408134 0.106749478
##      1      1      1      1      1      1
## 0.10724819 0.107386669 0.107411386 0.108249629 0.108481262 0.109950522
##      1      1      1      1      1      1
## 0.110029623 0.111825552 0.114830538 0.115074799 0.115481134 0.116585704
##      1      1      1      1      1      1
```

##	0.116712688	0.117284108	0.117891817	0.118078429	0.120070779	0.120623955
##	1	1	1	1	1	1
##	0.121680471	0.121732904	0.121936349	0.122188927	0.122267785	0.122865217
##	1	1	1	1	1	1
##	0.122873957	0.123222749	0.123228589	0.123380629	0.125992188	0.128150363
##	1	1	1	1	1	1
##	0.128810648	0.129932002	0.13074744	0.130833861	0.130918612	0.132042254
##	1	1	1	1	1	1
##	0.132289131	0.132604011	0.132898393	0.133187528	0.133349497	0.13368984
##	1	1	1	1	1	1
##	0.134447887	0.134632137	0.135111111	0.135521759	0.136975877	0.137474224
##	1	1	1	1	1	1
##	0.138839804	0.139492902	0.140724425	0.141726704	0.142074285	0.142366636
##	1	1	1	1	1	1
##	0.142482617	0.142947903	0.143204926	0.143587903	0.144682903	0.145247936
##	1	1	1	1	1	1
##	0.145604562	0.14587892	0.147860251	0.148248811	0.14852822	0.15037594
##	1	1	1	1	1	1
##	0.15066212	0.151156346	0.154156921	0.154479918	0.155076959	0.155727493
##	1	1	1	1	1	1
##	0.157087915	0.15743073	0.157522539	0.158000169	0.1580403	0.158371188
##	1	1	1	1	1	1
##	0.159320234	0.159416989	0.160314423	0.160735813	0.160834389	0.161168471
##	1	1	1	1	1	1
##	0.16131328	0.16194332	0.162677202	0.163781115	0.164421343	0.166493236
##	1	1	1	1	1	1
##	0.16772895	0.168088333	0.168570249	0.169338858	0.169955623	0.170406135
##	1	1	1	1	1	1
##	0.170870627	0.171502072	0.172413793	0.174767886	0.175925926	0.177731819
##	1	1	3	1	1	1
##	0.178013833	0.178284767	0.179491074	0.179654166	0.179948586	0.180113496
##	1	1	1	1	1	1
##	0.180132538	0.181699251	0.182189724	0.182207931	0.182751719	0.18403171
##	1	1	1	1	1	1
##	0.184270039	0.184723774	0.185087255	0.186467768	0.187669701	0.187847109
##	1	1	1	1	1	1
##	0.1890657	0.19170579	0.191966923	0.192810908	0.193298969	0.194043702
##	1	1	1	1	1	1
##	0.19436346	0.195608587	0.19610105	0.19621422	0.196415419	0.196753566
##	1	1	1	1	1	1
##	0.197016606	0.198515107	0.201876262	0.203472599	0.203535083	0.203614151
##	1	1	1	1	1	1
##	0.204188482	0.204498978	0.204974037	0.211821666	0.212228398	0.212615167
##	1	1	1	1	1	1
##	0.213526509	0.214462999	0.214818989	0.215269805	0.217205091	0.21743182
##	1	1	1	1	1	1
##	0.218738607	0.218964403	0.219976924	0.221091526	0.22234827	0.222643297
##	1	1	1	1	1	1
##	0.222841226	0.222969096	0.223278182	0.223598464	0.223689817	0.224318262
##	2	1	1	1	1	1
##	0.224382947	0.224747961	0.224959829	0.228879165	0.232385661	0.234588107
##	1	1	1	1	1	1
##	0.235017626	0.23551258	0.23556582	0.236406619	0.236885624	0.236951649
##	1	1	1	1	1	1

##	0.236966825	0.237796705	0.239034301	0.239398085	0.241699867	0.244021474
##	1	1	1	1	1	1
##	0.244508741	0.24641139	0.24721436	0.247754723	0.248088498	0.248138958
##	1	1	1	1	1	1
##	0.248999918	0.249324745	0.250375563	0.250438853	0.251116071	0.251269714
##	1	1	1	1	1	1
##	0.251386513	0.251710568	0.251726288	0.251851338	0.251861586	0.252131108
##	1	1	1	1	1	1
##	0.25220681	0.252491694	0.253902813	0.254336768	0.255072612	0.255465129
##	1	1	1	1	1	1
##	0.255724747	0.257546101	0.258899676	0.259165017	0.260588853	0.260713704
##	1	1	1	1	1	1
##	0.261396238	0.261577214	0.261655566	0.261918387	0.262209859	0.262467192
##	1	1	1	1	1	1
##	0.262797075	0.264625541	0.265356265	0.266760944	0.268009421	0.26992755
##	1	1	1	1	1	1
##	0.270472464	0.271539028	0.272378358	0.272382757	0.272502768	0.273404771
##	1	1	1	1	1	1
##	0.27356379	0.275055871	0.275435502	0.27700831	0.279347816	0.279977659
##	1	1	1	1	1	1
##	0.280311361	0.281392895	0.282845425	0.28480064	0.285040441	0.288435374
##	1	1	1	1	1	1
##	0.288745789	0.289508754	0.289612841	0.28980788	0.289896062	0.290528762
##	1	1	1	1	1	1
##	0.290574061	0.29154519	0.291882031	0.292913059	0.295420975	0.295936289
##	1	1	1	1	1	1
##	0.296353563	0.296442688	0.297216968	0.297409408	0.297865299	0.298750679
##	1	1	1	1	1	1
##	0.298794152	0.299363057	0.2997003	0.300402601	0.300864987	0.301132257
##	1	1	1	1	1	1
##	0.302107673	0.305646584	0.306067039	0.306208763	0.306247448	0.308268193
##	1	1	1	1	1	1
##	0.309514197	0.311554715	0.311665693	0.312868414	0.313138189	0.313873195
##	1	1	1	1	1	1
##	0.315148757	0.315514521	0.316681181	0.318344608	0.320897533	0.32123159
##	1	1	1	1	1	1
##	0.321886504	0.322853027	0.325754554	0.327792989	0.327938221	0.327967422
##	1	1	1	1	1	1
##	0.328314946	0.329990013	0.330066694	0.330377326	0.333979746	0.334078877
##	1	1	1	1	1	1
##	0.334480586	0.335738939	0.336406678	0.336961174	0.338725878	0.339116719
##	1	1	1	1	1	1
##	0.340192962	0.34038932	0.34039575	0.342026507	0.342176014	0.342276692
##	1	1	1	1	1	1
##	0.344454286	0.345337938	0.346921075	0.349043744	0.350103734	0.350403446
##	1	1	1	1	1	1
##	0.350539556	0.351601956	0.353713997	0.354628422	0.355536965	0.355888551
##	1	1	1	1	1	1
##	0.35720965	0.359464627	0.359783144	0.359811791	0.360559235	0.361532899
##	1	1	1	1	1	1
##	0.362192659	0.362346192	0.363174782	0.364785992	0.3651953	0.365736285
##	1	1	1	1	1	1
##	0.365804092	0.36661112	0.36800644	0.368808625	0.369065593	0.370919881
##	1	1	1	1	1	1

##	0.371252899	0.371463949	0.372302385	0.374511986	0.376647834	0.376914016
##	1	1	1	1	1	1
##	0.379683804	0.380677343	0.3808617	0.381094404	0.381479933	0.383829321
##	1	1	1	1	1	1
##	0.384683672	0.385193038	0.388712928	0.389029372	0.390625	0.390932399
##	1	1	1	1	1	1
##	0.392650763	0.393513101	0.394823426	0.395935277	0.397135808	0.400562193
##	1	1	1	1	1	1
##	0.400593472	0.400686892	0.401955459	0.402073855	0.402665926	0.402813684
##	1	1	1	1	1	1
##	0.403191216	0.403228331	0.404438453	0.405616225	0.4056978	0.406235551
##	1	1	1	1	1	1
##	0.407398354	0.407437085	0.407744822	0.408444727	0.409735812	0.410322859
##	1	1	1	1	1	1
##	0.410770307	0.410783055	0.41113591	0.411710887	0.411783339	0.412229474
##	1	1	1	1	1	1
##	0.412371134	0.413080895	0.413359788	0.414259412	0.415186965	0.41716329
##	1	1	1	1	1	1
##	0.417342917	0.41807026	0.418167106	0.418410042	0.419064648	0.419727177
##	1	1	1	1	1	1
##	0.421718503	0.422068134	0.422123606	0.425864766	0.42840691	0.428477258
##	1	1	1	1	1	1
##	0.430292599	0.435876433	0.435883699	0.437578139	0.438095238	0.438105891
##	1	1	1	1	1	1
##	0.43921293	0.439674124	0.440615385	0.440852314	0.441956394	0.441984752
##	1	1	1	1	1	1
##	0.442747825	0.443786982	0.444498569	0.444750115	0.445611037	0.447672475
##	1	1	1	1	1	1
##	0.449755847	0.450617284	0.452898551	0.453645661	0.454861737	0.455796354
##	1	1	1	1	1	1
##	0.456238361	0.458063591	0.458546803	0.460891962	0.464543375	0.464583674
##	1	1	1	1	1	1
##	0.465170671	0.465928946	0.46870692	0.470840021	0.470879802	0.47151277
##	1	1	1	1	1	1
##	0.47167541	0.471957249	0.472498882	0.472600925	0.475554566	0.476144204
##	1	1	1	1	1	1
##	0.478314396	0.4784689	0.478607687	0.478675664	0.479422595	0.480570484
##	1	1	1	1	1	1
##	0.480786481	0.481410161	0.484079497	0.4857527	0.486168506	0.488624217
##	1	1	1	1	1	1
##	0.489064251	0.490109502	0.491355407	0.491671684	0.491832074	0.492135482
##	1	1	1	1	1	1
##	0.492300719	0.492800224	0.493370336	0.493513353	0.493745885	0.494929986
##	1	1	1	1	1	1
##	0.495161656	0.495237064	0.498589847	0.500136401	0.500374368	0.500560755
##	1	1	1	1	1	1
##	0.501296186	0.506871876	0.5078125	0.508800684	0.512618297	0.513930214
##	1	1	1	1	1	1
##	0.514949293	0.515092885	0.515489968	0.515507767	0.51716032	0.525799673
##	1	1	1	1	1	1
##	0.529083665	0.530047822	0.530610243	0.531223509	0.532780269	0.535173894
##	1	1	1	1	1	1
##	0.535552049	0.535586765	0.535659026	0.540511029	0.542215337	0.542543658
##	1	1	1	1	1	1

##	0.545502124	0.546163849	0.548262955	0.548439918	0.548825768	0.550050203
##	1	1	1	1	1	1
##	0.551068013	0.551466001	0.551952948	0.552147239	0.554002684	0.55531824
##	1	1	1	1	1	1
##	0.55589644	0.55919784	0.559642551	0.562246989	0.56335034	0.563544501
##	1	1	1	1	1	1
##	0.56424367	0.56424581	0.565939031	0.566591121	0.56688521	0.567056706
##	1	1	1	1	1	1
##	0.568594241	0.570121151	0.570190444	0.57039802	0.570658016	0.571114772
##	1	1	1	1	1	1
##	0.572012257	0.572663922	0.572787204	0.572921441	0.574116345	0.574175457
##	1	1	1	1	1	1
##	0.574181626	0.575277115	0.576553491	0.576776273	0.577769275	0.578344632
##	1	1	1	1	1	1
##	0.578854105	0.580734756	0.581996896	0.582954732	0.583176249	0.586195553
##	1	1	1	1	1	1
##	0.587669155	0.588485106	0.588584492	0.588857753	0.5892691	0.5907173
##	1	1	1	1	1	1
##	0.592540955	0.596065965	0.596284688	0.597971169	0.598591549	0.600109111
##	1	1	1	1	1	1
##	0.601656245	0.601876438	0.602190885	0.604845231	0.605060506	0.605811685
##	1	1	1	1	1	1
##	0.607010395	0.607130742	0.607242702	0.610164709	0.610950471	0.612776226
##	1	1	1	1	1	1
##	0.614765173	0.617708292	0.618492295	0.620642452	0.621148828	0.621539855
##	1	1	1	1	1	1
##	0.623639466	0.625760473	0.626476155	0.626657026	0.628424106	0.628601362
##	1	1	1	1	1	1
##	0.629059454	0.629731833	0.631604795	0.632225434	0.632414335	0.63463282
##	1	1	1	1	1	1
##	0.636350429	0.636472126	0.637332605	0.640011506	0.64074282	0.643192488
##	1	1	1	1	1	1
##	0.645145204	0.645185076	0.645796848	0.64652925	0.64690838	0.647630262
##	1	1	1	1	1	1
##	0.648916472	0.649218909	0.649637085	0.651080201	0.652049173	0.653178804
##	1	1	1	1	1	1
##	0.657998424	0.665228145	0.665796157	0.667050029	0.667127172	0.668093902
##	1	1	1	1	1	1
##	0.668825979	0.672260247	0.673203371	0.67703689	0.677436165	0.678204373
##	1	1	1	1	1	1
##	0.678528534	0.678665391	0.679221982	0.68006045	0.681826839	0.684251186
##	1	1	1	1	1	1
##	0.685082873	0.685449771	0.686341798	0.687172775	0.688239556	0.689786585
##	1	1	1	1	1	1
##	0.690896006	0.694444444	0.696485366	0.696685428	0.697503671	0.697723725
##	1	1	1	1	1	1
##	0.6991141	0.701055911	0.701627486	0.703722321	0.704982091	0.705044835
##	1	1	1	1	1	1
##	0.705716302	0.707236185	0.708513229	0.708999638	0.709158895	0.709188811
##	1	1	1	1	1	1
##	0.709606987	0.710196391	0.711494138	0.711623179	0.71222109	0.713320617
##	1	1	1	1	1	1
##	0.716644856	0.718711276	0.722104299	0.723411456	0.725806452	0.727738819
##	1	1	1	1	1	1

##	0.72894355	0.729927007	0.732857755	0.732953179	0.733272227	0.735021649
##	1	1	1	1	1	1
##	0.735294118	0.73664825	0.73674679	0.737355648	0.739023617	0.739750781
##	1	1	1	1	1	1
##	0.742513998	0.743508039	0.744945016	0.745770258	0.747471787	0.74784928
##	1	1	1	1	1	1
##	0.750040763	0.751219084	0.755166932	0.758097864	0.761294913	0.765880763
##	1	1	1	1	1	1
##	0.765972785	0.766832392	0.770807343	0.773770138	0.773814818	0.774305874
##	1	1	1	1	1	1
##	0.775636839	0.776206021	0.776869278	0.777882244	0.779559799	0.782171213
##	1	1	1	1	1	1
##	0.782594217	0.784898552	0.790115469	0.795824063	0.79616209	0.798597585
##	1	1	1	1	1	1
##	0.8	0.801455124	0.802026171	0.804871221	0.806700463	0.807819695
##	1	1	1	1	1	1
##	0.814205748	0.814446287	0.817843866	0.820170109	0.821857259	0.823883942
##	1	1	1	1	1	1
##	0.824603475	0.824731007	0.825916696	0.830184358	0.830328336	0.83245936
##	1	1	1	1	1	1
##	0.834326579	0.835654596	0.83908046	0.83968144	0.84279956	0.845461526
##	1	1	1	1	1	1
##	0.846190552	0.847264314	0.848412043	0.850666864	0.850864264	0.851154737
##	1	1	1	1	1	1
##	0.854360689	0.855553971	0.858126383	0.859835948	0.860463623	0.86424734
##	1	1	1	1	1	1
##	0.865711399	0.866968262	0.868131868	0.870927435	0.871024167	0.873657355
##	1	1	1	1	1	1
##	0.876221655	0.878984285	0.879130588	0.881428681	0.88808364	0.891630309
##	1	1	1	1	1	1
##	0.892579139	0.893136948	0.895368646	0.895948753	0.896026761	0.898705255
##	1	1	1	1	1	1
##	0.902843283	0.903609615	0.904488486	0.904890258	0.905209223	0.905583539
##	1	1	1	1	1	1
##	0.905963725	0.907175485	0.910155805	0.912741877	0.915401099	0.918290656
##	1	1	1	1	1	1
##	0.918448956	0.922398235	0.922971497	0.925925926	0.927191209	0.929626046
##	1	1	1	1	1	1
##	0.9307062	0.931190022	0.933267959	0.934223164	0.935863236	0.935916542
##	1	1	1	1	1	1
##	0.938928468	0.939583405	0.940757691	0.946663588	0.946825166	0.948487326
##	1	1	1	1	1	1
##	0.955045341	0.955525345	0.957679679	0.95788113	0.95810917	0.964589415
##	1	1	1	1	1	1
##	0.970767089	0.970873786	0.975080005	0.977054689	0.981219073	0.983154142
##	1	1	1	1	1	1
##	0.984147952	0.985032042	0.986150444	0.987681918	0.994803774	0.995492016
##	1	1	1	1	1	1
##	0.99718442	0.998843772	0.999868931	1.000564375	1.000595209	1.001508571
##	1	1	1	1	1	1
##	1.003251277	1.003689533	1.008787143	1.009421265	1.012545039	1.013060951
##	1	1	1	1	1	1
##	1.014353096	1.015449336	1.016604541	1.016826387	1.018719258	1.019115007
##	1	1	1	1	1	1

##	1.020633438	1.023846553	1.02873756	1.028985507	1.029387349	1.02965716
##	1	1	1	1	1	1
##	1.033885954	1.035048738	1.035642233	1.035673188	1.045725346	1.046735515
##	1	1	1	1	1	1
##	1.048317926	1.05126711	1.052732015	1.053217133	1.053679848	1.056159522
##	1	1	1	1	1	1
##	1.058175227	1.059354607	1.060908637	1.062536244	1.062753036	1.065449011
##	1	1	1	1	1	1
##	1.065891473	1.067384444	1.06931014	1.06955256	1.071723001	1.074537981
##	1	1	1	1	1	1
##	1.075943886	1.076931004	1.077869499	1.078515962	1.080155705	1.080989303
##	1	1	1	1	1	1
##	1.081220328	1.082044427	1.082372786	1.082431307	1.082989684	1.083880613
##	1	1	1	1	1	1
##	1.084507664	1.084525379	1.086471533	1.086593113	1.096350419	1.097490332
##	1	1	1	1	1	1
##	1.09827433	1.101477893	1.101810117	1.104893098	1.105311636	1.107499652
##	1	1	1	1	1	1
##	1.112781493	1.113284552	1.115075825	1.125502486	1.129964572	1.135545818
##	1	1	1	1	1	1
##	1.136363636	1.136624569	1.141621102	1.142528358	1.144961162	1.147028154
##	1	1	1	1	1	1
##	1.150835378	1.160212829	1.166861144	1.167565088	1.174022158	1.177221577
##	1	1	1	1	1	1
##	1.182490782	1.185730349	1.185861741	1.188375366	1.189187021	1.190239044
##	1	1	1	1	1	1
##	1.190680438	1.19269963	1.194413493	1.196799896	1.200260926	1.201004156
##	1	1	1	1	1	1
##	1.20240481	1.206500328	1.207261938	1.207488164	1.209366816	1.209409204
##	1	1	1	1	1	1
##	1.209729758	1.210585586	1.214860856	1.220919638	1.222077842	1.225845411
##	1	1	1	1	1	1
##	1.225934559	1.230845999	1.23228361	1.236447151	1.237276311	1.237736885
##	1	1	1	1	1	1
##	1.238774388	1.244461609	1.244495501	1.248693695	1.250900269	1.251443974
##	1	1	1	1	1	1
##	1.2538849	1.255199323	1.259294795	1.261076602	1.261847035	1.267217631
##	1	1	1	1	1	1
##	1.268910891	1.269440942	1.270529075	1.271470265	1.276374357	1.281776022
##	1	1	1	1	1	1
##	1.284308301	1.28687252	1.287258997	1.292572835	1.301547374	1.302017768
##	1	1	1	1	1	1
##	1.314367767	1.315327955	1.320132013	1.322373124	1.334364173	1.336908083
##	1	1	1	1	1	1
##	1.34218605	1.343502751	1.346076188	1.347131163	1.353540028	1.354343849
##	1	1	1	1	1	1
##	1.359770201	1.368889357	1.375328084	1.376518219	1.377052024	1.37861466
##	1	1	1	1	1	1
##	1.380775358	1.381045282	1.381561593	1.390057659	1.397558681	1.397968834
##	1	1	1	1	1	1
##	1.398998015	1.400012856	1.405283411	1.417162997	1.419628884	1.420203127
##	1	1	1	1	1	1
##	1.427006914	1.427913404	1.428571429	1.430412833	1.432207506	1.432274505
##	1	1	1	1	1	1

##	1.437986818	1.439760424	1.440439902	1.445793852	1.448258302	1.472423259
##	1	1	1	1	1	1
##	1.477693257	1.479163704	1.48230607	1.486348344	1.491956179	1.492844288
##	1	1	1	1	1	1
##	1.493007434	1.496753617	1.498954445	1.504236213	1.506292429	1.507215312
##	1	1	1	1	1	1
##	1.50940226	1.512163846	1.516871755	1.518390749	1.521328743	1.523702785
##	1	1	1	1	1	1
##	1.526369746	1.527461813	1.528786071	1.531603612	1.533289185	1.53380291
##	1	1	1	1	1	1
##	1.536575398	1.53733253	1.538325001	1.541372211	1.554209513	1.555414953
##	1	1	1	1	1	1
##	1.555682034	1.564065457	1.565155174	1.567545353	1.567851659	1.572701081
##	1	1	1	1	1	1
##	1.574074074	1.575525447	1.57658375	1.578403091	1.583723787	1.585981586
##	1	1	1	1	1	1
##	1.587393212	1.587621931	1.589843303	1.596009975	1.597349037	1.601138342
##	1	1	1	1	1	1
##	1.604920122	1.607504072	1.61160691	1.612552701	1.61542247	1.628048727
##	1	1	1	1	1	1
##	1.62964668	1.633826676	1.647406692	1.659551761	1.66005291	1.668810479
##	1	1	1	1	1	1
##	1.670324233	1.676605576	1.6788187	1.679780998	1.684088269	1.688730093
##	1	1	1	1	1	1
##	1.691331924	1.691361476	1.692453578	1.694099727	1.696768361	1.697973257
##	1	1	1	1	1	1
##	1.703180212	1.704580199	1.705387035	1.715238198	1.720209615	1.7216441
##	1	1	1	1	1	1
##	1.72224258	1.723958442	1.727982163	1.728810339	1.734180567	1.738290681
##	1	1	1	1	1	1
##	1.740752254	1.741364545	1.74320281	1.745301112	1.745893978	1.745975677
##	1	1	1	1	1	1
##	1.747200834	1.749058199	1.749071722	1.751616877	1.756535948	1.76490895
##	1	1	1	1	1	1
##	1.766784452	1.770997784	1.772190299	1.773748522	1.788546256	1.805655232
##	1	1	1	1	1	1
##	1.809315146	1.825293351	1.825345154	1.844553051	1.852779169	1.854909731
##	1	1	1	1	1	1
##	1.855263994	1.856877116	1.857923497	1.866100987	1.866288684	1.872218119
##	1	1	1	1	1	1
##	1.872504911	1.885542476	1.89306799	1.894897899	1.897835431	1.899961225
##	1	1	1	1	1	1
##	1.900722934	1.908738443	1.911381407	1.921033271	1.924023446	1.930084712
##	1	1	1	1	1	1
##	1.936797524	1.945356769	1.946619371	1.948564216	1.953190773	1.954015592
##	1	1	1	1	1	1
##	1.956745623	1.95886554	1.964735516	1.970299711	1.971663999	1.972010178
##	1	1	1	1	1	1
##	1.972745609	1.976059546	1.97799668	1.979937077	1.992724114	2.000171689
##	1	1	1	1	1	1
##	2.001253582	2.018769096	2.019049045	2.023789856	2.032411138	2.037074456
##	1	1	1	1	1	1
##	2.040210627	2.042677777	2.042926264	2.067723673	2.070349636	2.070626003
##	1	1	1	1	1	1

##	2.077779305	2.09012847	2.090182231	2.091241632	2.09473201	2.096762133
##	1	1	1	1	1	1
##	2.100171832	2.105934907	2.107185156	2.11556335	2.119182354	2.130325815
##	1	1	1	1	1	1
##	2.136773258	2.141142943	2.151166979	2.151621556	2.159065874	2.179926005
##	1	1	1	1	1	1
##	2.180667917	2.180944963	2.18227135	2.18229586	2.194505841	2.206070533
##	1	1	1	1	1	1
##	2.209795604	2.219548068	2.229739058	2.236771734	2.237402858	2.245862884
##	1	1	1	1	1	1
##	2.247923754	2.24855667	2.252719804	2.268083789	2.273607661	2.286267861
##	1	1	1	1	1	1
##	2.29570654	2.30180141	2.315724816	2.324551694	2.331428571	2.333790939
##	1	1	1	1	1	1
##	2.336103416	2.343647996	2.356942538	2.364643927	2.366571491	2.371342079
##	1	1	1	1	1	1
##	2.373603996	2.374048568	2.375354682	2.385636941	2.388125307	2.390561937
##	1	1	1	1	1	1
##	2.405498282	2.420147779	2.425501375	2.425881217	2.445890722	2.471323324
##	1	1	1	1	1	1
##	2.523013979	2.52822559	2.530105796	2.555838527	2.5574977	2.561964237
##	1	1	1	1	1	1
##	2.573003079	2.585037939	2.600278894	2.602118272	2.606228693	2.608227482
##	1	1	1	1	1	1
##	2.611473446	2.614018495	2.619371265	2.623794917	2.628624033	2.636460734
##	1	1	1	1	1	1
##	2.64237814	2.644067797	2.645465665	2.659527587	2.663357877	2.66523351
##	1	1	1	1	1	1
##	2.670723242	2.67075523	2.673076923	2.680100532	2.688366486	2.690149023
##	1	1	1	1	1	1
##	2.690596315	2.699184381	2.719624535	2.735377233	2.735606863	2.737053347
##	1	1	1	1	1	1
##	2.738089549	2.741052095	2.754062839	2.765861368	2.773386332	2.779238562
##	1	1	1	1	1	1
##	2.787501621	2.789142581	2.795688803	2.798253775	2.799638122	2.811896485
##	1	1	1	1	1	1
##	2.812546017	2.820616883	2.839067416	2.840577454	2.852946161	2.853785462
##	1	1	1	1	1	1
##	2.854285974	2.875753806	2.889043083	2.8970745	2.899575672	2.902334152
##	1	1	1	1	1	1
##	2.903811252	2.905096721	2.908389124	2.91314937	2.913566868	2.914549028
##	1	1	1	1	1	1
##	2.927200152	2.966386223	2.967144593	2.976028946	2.984531392	2.985809092
##	1	1	1	1	1	1
##	2.990537904	3.000491884	3.002631172	3.006985308	3.021712851	3.030546674
##	1	1	1	1	1	1
##	3.03094576	3.034047919	3.043494249	3.054017942	3.057199211	3.062697228
##	1	1	1	1	1	1
##	3.06695054	3.075764965	3.082227454	3.08437656	3.088425789	3.091182658
##	1	1	1	1	1	1
##	3.106726778	3.113118833	3.116360422	3.117670972	3.141664951	3.142370373
##	1	1	1	1	1	1
##	3.143651598	3.162650602	3.167675975	3.167866824	3.178079444	3.184025904
##	1	1	1	1	1	1

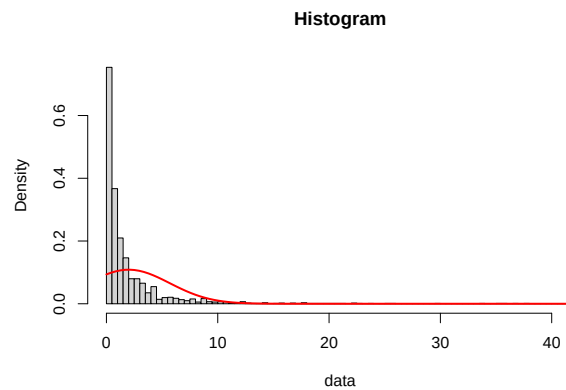
##	3.185494214	3.186119296	3.189415042	3.192453755	3.197610091	3.197637087
##	1	1	1	1	1	1
##	3.225806452	3.230015163	3.240580112	3.244085565	3.250970439	3.25165665
##	1	1	1	1	1	1
##	3.264814126	3.276673456	3.289376447	3.303015797	3.31463539	3.317915192
##	1	1	1	1	1	1
##	3.321315234	3.323075332	3.332678229	3.356737964	3.359237904	3.364888786
##	1	1	1	1	1	1
##	3.375139883	3.409384417	3.443664326	3.445715134	3.454960881	3.479995567
##	1	1	1	1	1	1
##	3.49986246	3.508424182	3.509933775	3.527312654	3.553736808	3.57060051
##	1	1	1	1	1	1
##	3.577150131	3.603840631	3.614272532	3.623105059	3.625732993	3.628950051
##	1	1	1	1	1	1
##	3.639536333	3.645060201	3.668552656	3.674856767	3.681615488	3.683078965
##	1	1	1	1	1	1
##	3.696361195	3.716577859	3.731749993	3.742593071	3.765966968	3.776911341
##	1	1	1	1	1	1
##	3.785820667	3.82387022	3.832739685	3.837187636	3.846876594	3.873869425
##	1	1	1	1	1	1
##	3.91593203	3.919904972	3.921646693	4.019367635	4.036176097	4.054323946
##	1	1	1	1	1	1
##	4.058108321	4.061643836	4.072416115	4.078914417	4.081511485	4.099713973
##	1	1	1	1	1	1
##	4.102341308	4.113436622	4.119217691	4.119475605	4.127602581	4.142857143
##	1	1	1	1	1	1
##	4.167266446	4.186885455	4.197354991	4.22401794	4.241662552	4.265348356
##	1	1	1	1	1	1
##	4.275384051	4.282339043	4.293886553	4.31230523	4.318265344	4.330778827
##	1	1	1	1	1	1
##	4.334716922	4.335701481	4.339654878	4.348445965	4.357038803	4.358300482
##	1	1	1	1	1	1
##	4.359344894	4.370686822	4.371008379	4.381618669	4.385672235	4.389623103
##	1	1	1	1	1	1
##	4.398211744	4.41748603	4.426145136	4.427970715	4.435058078	4.436119874
##	1	1	1	1	1	1
##	4.440026508	4.442734206	4.459111203	4.478707783	4.488439137	4.509538267
##	1	1	1	1	1	1
##	4.521158129	4.529639455	4.537163703	4.561949196	4.590753145	4.59188924
##	1	1	1	1	1	1
##	4.60125477	4.748105216	4.823739597	4.862290482	4.908792642	4.909264565
##	1	1	1	1	1	1
##	5.004828722	5.069127757	5.079431438	5.106835745	5.122170397	5.125232418
##	1	1	1	1	1	1
##	5.169186853	5.269810404	5.271426568	5.291257508	5.295566502	5.337210913
##	1	1	1	1	1	1
##	5.375760863	5.376813175	5.42617651	5.441834963	5.456673331	5.485656769
##	1	1	1	1	1	1
##	5.554589063	5.573246769	5.580515488	5.629666325	5.63573653	5.637089897
##	1	1	1	1	1	1
##	5.66783654	5.675706603	5.680705191	5.702146067	5.724673565	5.737324516
##	1	1	1	1	1	1
##	5.765975455	5.804689112	5.815538392	5.817264224	5.947442452	5.976719916
##	1	1	1	1	1	1

```
## 5.992571149 6.014032743 6.015766564 6.083386341 6.188306389 6.222178466
##      1      1      1      1      1      1
## 6.226590583 6.244146113 6.272401434 6.280455266 6.28585086 6.36784115
##      1      1      1      1      1      1
## 6.376594149 6.387093938 6.440072877 6.453117421 6.474408937 6.549672616
##      1      1      1      1      1      1
## 6.564629491 6.595977791 6.628260941 6.675081565 6.797513702 6.8865327
##      1      1      1      1      1      1
## 6.939208089 6.945460686 6.946614992 6.972730411 6.985117619 7.099306188
##      1      1      1      1      1      1
## 7.179426817 7.202471527 7.244043786 7.250100951 7.288251946 7.358393167
##      1      1      1      1      1      1
## 7.4039522 7.47108947 7.5024692 7.548874651 7.61357263 7.680608365
##      1      1      1      1      1      1
## 7.718034474 7.73395205 7.76119403 7.802113647 7.834766619 7.859929519
##      1      1      1      1      1      1
## 7.868383405 7.894277304 7.928721594 7.975460123 8.13681268 8.167921515
##      1      1      1      1      1      1
## 8.291741939 8.297444732 8.319848458 8.511018051 8.516900725 8.566632984
##      1      1      1      1      1      1
## 8.719844605 8.734237421 8.837825895 8.875437242 8.905228758 8.915218751
##      1      1      1      1      1      1
## 8.917418459 8.949758313 8.969153533 8.984509466 8.990337124 8.999224205
##      1      1      1      1      1      1
## 9.07549527 9.111029141 9.318656292 9.332922982 9.42051083 9.447451755
##      1      1      1      1      1      1
## 9.520113827 9.60371558 9.834279814 9.847732972 9.880649013 10.01947247
##      1      1      1      1      1      1
## 10.11 10.14589511 10.18211715 10.33491544 10.47189905 10.62252964
##      1      1      1      1      1      1
## 10.78188308 10.92088899 10.93019686 10.95647773 10.99870603 11.0474034
##      1      1      1      1      1      1
## 11.13148201 11.63048294 11.8168115 11.95125329 12.14105793 12.1518083
##      1      1      1      1      1      1
## 12.21413721 12.21700128 12.27433705 12.39527799 12.63084438 12.75344797
##      1      1      1      1      1      1
## 13.26786739 13.51850772 14.01244526 14.06276198 14.13028842 15.0990099
##      1      1      1      1      1      1
## 15.68327036 15.79281775 16.69528291 16.92585895 17.06498417 17.53578732
##      1      1      1      1      1      1
## 17.74362833 17.78612154 18.37112588 18.56503148 19.00423039 19.59152207
##      1      1      1      1      1      1
## 20.2889159 20.91767881 21.15709993 22.1468782 22.32569439 22.65191277
##      1      1      1      1      1      1
## 23.6662531 25.44525842 25.50491014 29.93039443 33.72992267 35.00734214
##      1      1      1      1      1      1
## 35.57939914 36.8285196 37.61099366 41.93025142
##      1      1      1      1
```

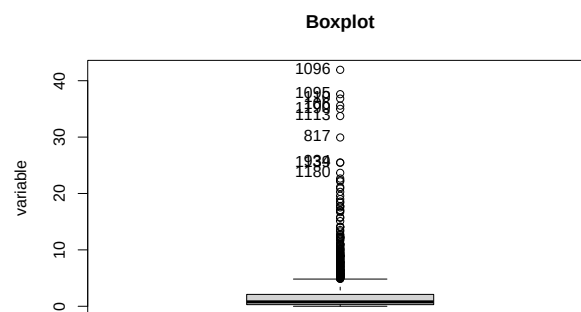
```
res.var <- summary_and_normality_check(variable)
```

```
## [1] "Summary: "
##      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.0000 0.2867 0.7826 2.0031 2.1066 41.9303
## [1] "Mean"
```

```
## [1] 2.003134
## [1] "Standard deviation"
## [1] 3.672809
```



```
## [1] "Skewness"
## [1] 4.987083
```



```
## [1] "Num Outliers:"
## [1] 181
## [1] "Num. severe outliers:"
## [1] 102
## [1] "Shapiro test: "
##
##  Shapiro-Wilk normality test
##
## data:  data
## W = 0.50981, p-value < 2.2e-16
##
## [1] "Most repeated value ad times it occurs: "
## [1] "0"
## [1] 79
```

```
outliers_df[res.var,"pctotherrace"] <- 1
```

Variable *pctmarriedhouseholds*

pctmarriedhouseholds represents the percentage of households that are married.

Values range from 22.99% to 71.40%, with a mean of 51.40% and a median of 51.73%, indicating a fairly balanced distribution. The interquartile range spans from 47.85% to 55.48%, with a standard deviation of 6.45%. The skewness of -0.50 suggests a slight left skew, and the Shapiro-Wilk test indicates non-normality. There are 57 outliers, including 2 severe ones, and the most common value is 50%, occurring in 2 counties.

```
variable<- df$pctmarriedhouseholds
table(variable)
```

```
## variable
## 22.99248989 24.02463388 25.50125313 25.54027505 26.46417445 26.98724239
##          1          1          1          1          1          1
## 27.45994321 28.06061913 28.93569845 29.76495727 30.50769584 30.84274544
##          1          1          1          1          1          1
## 30.98763176 31.49733484 31.61961436 31.79473256 31.9243604 32.24368918
##          1          1          1          1          1          1
## 32.64161526 32.83512377 33.11496888 33.19228961 33.29564726 33.34770192
##          1          1          1          1          1          1
## 33.68648224 33.8123415 33.83092514 34.21626984 34.42211055 34.62088698
##          1          1          1          1          1          1
## 35.20291838 35.62091503 35.63437926 35.70673712 35.79382487 35.92963409
##          1          1          1          1          1          1
## 36.05484575 36.08648401 36.27241993 36.27949323 36.33759365 36.37739657
##          1          1          1          1          1          1
## 36.76248109 36.82926829 36.89978083 36.90882667 36.95193176 37.15664456
##          1          1          1          1          1          1
## 37.34087694 37.60480715 37.6164403 37.7028181 37.74393475 37.81552964
##          1          1          1          1          1          1
## 37.82864464 37.93103448 38.21259647 38.22078907 38.23678433 38.32478632
##          1          1          1          1          1          1
## 38.43382929 38.45841411 38.52216749 38.6145629 38.76146789 38.79198202
##          1          1          1          1          1          1
## 38.80436161 38.83188635 38.91816921 38.96880138 38.97061322 39.01594283
##          1          1          1          1          1          1
## 39.05462999 39.1929128 39.24670876 39.24786107 39.36196935 39.41986227
##          1          1          1          1          1          1
## 39.58795563 39.61676485 39.62978544 39.63436072 39.73199772 39.83536149
##          1          1          1          1          1          1
## 39.93115318 40.13853119 40.16167871 40.26643207 40.29543112 40.42576242
##          1          1          1          1          1          1
## 40.43404785 40.46070836 40.5218061 40.52689699 40.55269457 40.55405483
##          1          1          1          1          1          1
## 40.56014847 40.57971014 40.59765561 40.60228453 40.63093964 40.63444109
##          1          1          1          1          1          1
## 40.6686994 40.67557054 40.73517512 40.78656398 40.80389769 40.84865449
##          1          1          1          1          1          1
## 40.87046446 41.00879123 41.0109432 41.05174525 41.0637149 41.07124285
##          1          1          1          1          1          1
## 41.13486198 41.16047144 41.18541033 41.2160804 41.29643117 41.35260379
##          1          1          1          1          1          1
## 41.36584844 41.38070246 41.44295302 41.49092693 41.50030248 41.56265602
##          1          1          1          1          1          1
## 41.56296102 41.62962963 41.69865643 41.74704724 41.77884615 41.78310495
##          1          1          1          1          1          1
## 41.79490974 41.8307372 41.84408516 41.89674177 41.92307692 41.92591941
##          1          1          1          1          1          1
```

##	41.97879859	41.98473282	42.01097206	42.01819686	42.03821656	42.10221794
##	1	1	1	1	1	1
##	42.16600744	42.20451527	42.24678817	42.25182366	42.25950308	42.29000242
##	1	1	1	1	1	1
##	42.32804233	42.3475685	42.40940255	42.4122807	42.47477961	42.50944076
##	1	1	1	1	1	1
##	42.52300614	42.65057979	42.65554216	42.68932243	42.69631298	42.70196877
##	1	1	1	1	1	1
##	42.70748576	42.72642886	42.7300221	42.80662505	42.83269148	42.8667293
##	1	1	1	1	1	1
##	42.88224956	42.92636963	42.92698827	42.93785311	42.97381269	42.98472784
##	1	1	1	1	1	1
##	43.04294896	43.06175402	43.12035159	43.16723441	43.17563049	43.18792376
##	1	1	1	1	1	1
##	43.203404	43.20718409	43.21167883	43.24014463	43.25842697	43.27147205
##	1	1	1	1	1	1
##	43.27694643	43.30106657	43.34945586	43.36469148	43.40527578	43.42926195
##	1	1	1	1	1	1
##	43.44441603	43.45582911	43.48336299	43.49984589	43.51071966	43.53518822
##	1	1	1	1	1	1
##	43.55086301	43.58974359	43.59130303	43.68165547	43.69406021	43.69722094
##	1	1	1	1	1	1
##	43.72557027	43.73804971	43.81306543	43.83862255	43.8495072	43.88561321
##	1	1	1	1	1	1
##	43.89416163	43.89529902	43.90160228	43.92135143	43.94933618	43.96226415
##	1	1	1	1	1	1
##	43.98362892	43.99577725	44.01303285	44.01690236	44.07408582	44.13291796
##	1	1	1	1	1	1
##	44.16932255	44.17130145	44.21204472	44.24762927	44.24802405	44.27786107
##	1	1	1	1	1	1
##	44.34058657	44.34453384	44.42383799	44.44522031	44.48737409	44.49216395
##	1	1	1	1	1	1
##	44.50887548	44.56321839	44.56630944	44.57106192	44.58164997	44.60242547
##	1	1	1	1	1	1
##	44.6376899	44.66265578	44.69463294	44.69791928	44.69897734	44.70367094
##	1	1	1	1	1	1
##	44.70658208	44.72000041	44.75893652	44.7662507	44.77432297	44.7804878
##	1	1	1	1	1	1
##	44.78928273	44.7998424	44.80543715	44.86582724	44.87314289	44.89449176
##	1	1	1	1	1	1
##	44.92905938	44.9580658	44.95813615	44.9714232	45.01341027	45.04306379
##	1	1	1	1	1	1
##	45.0853781	45.09297521	45.12060962	45.14153511	45.15519372	45.16008471
##	1	1	1	1	1	1
##	45.19096738	45.20152443	45.21959459	45.25418955	45.26843317	45.29652352
##	1	1	1	1	1	1
##	45.30500343	45.31301605	45.31508664	45.32906764	45.33935081	45.37901705
##	1	1	1	1	1	1
##	45.38266302	45.40726776	45.43318649	45.44265292	45.4432712	45.44421907
##	1	1	1	1	1	1
##	45.48274525	45.50658271	45.51036071	45.51689103	45.52995392	45.53971759
##	1	1	1	1	1	1
##	45.56004036	45.6067151	45.63246972	45.65002073	45.65139199	45.65665826
##	1	1	1	1	1	1

##	45.67653277	45.68305405	45.6838491	45.69355222	45.72944794	45.73721163
##	1	1	1	1	1	1
##	45.80894404	45.81880408	45.87134773	45.89769997	45.91882391	45.95862401
##	1	1	1	1	1	1
##	45.9592145	45.98629397	45.9899335	46.00270254	46.04255319	46.04377104
##	1	1	1	1	1	1
##	46.04447395	46.04772558	46.05161334	46.05451937	46.0656592	46.08439986
##	1	1	1	1	1	1
##	46.08887662	46.08938547	46.11948988	46.15248227	46.15710354	46.16229053
##	1	1	1	1	1	1
##	46.21993127	46.22883848	46.2430072	46.25023914	46.2518666	46.28352971
##	1	1	1	1	1	1
##	46.28481184	46.30368458	46.34128688	46.36348801	46.37281334	46.37607056
##	1	1	1	1	1	1
##	46.37692162	46.38007941	46.38557303	46.42487047	46.42554551	46.48961979
##	1	1	1	1	1	1
##	46.49507119	46.51047978	46.51854017	46.52714376	46.52861953	46.553108
##	1	1	1	1	1	1
##	46.5592921	46.56700481	46.59100754	46.61581243	46.64438503	46.64695662
##	1	1	1	1	1	1
##	46.67246629	46.68361787	46.70006452	46.70476871	46.71562217	46.76269132
##	1	1	1	1	1	1
##	46.76616915	46.77488253	46.81324986	46.81557501	46.83262121	46.83710651
##	1	1	1	1	1	1
##	46.83916084	46.8612555	46.86523551	46.86919273	46.87668573	46.89855697
##	1	1	1	1	1	1
##	46.90374585	46.91349545	46.93075295	46.93813131	46.95113161	46.97095568
##	1	1	1	1	1	1
##	47.01261604	47.01731879	47.03650145	47.05491585	47.0950042	47.10178856
##	1	1	1	1	1	1
##	47.107575	47.12271638	47.12882509	47.13228735	47.13990351	47.15063697
##	1	1	1	1	1	1
##	47.16324511	47.16886588	47.19623253	47.19879139	47.20175728	47.21497522
##	1	1	1	1	1	1
##	47.2318625	47.2359343	47.25165806	47.2534908	47.25688255	47.26089934
##	1	1	1	1	1	1
##	47.27745488	47.28648334	47.30477103	47.30857745	47.32220033	47.32708386
##	1	1	1	1	1	1
##	47.33045769	47.33299849	47.33818078	47.34207692	47.34389562	47.35119602
##	1	1	1	1	1	1
##	47.3540756	47.39858578	47.42739538	47.5137372	47.51381215	47.52860823
##	1	1	1	1	1	1
##	47.55445952	47.56717502	47.60257044	47.61023743	47.61904762	47.64763538
##	1	1	1	1	1	1
##	47.68200247	47.6859659	47.69147789	47.69150052	47.69427369	47.7128143
##	1	1	1	1	1	1
##	47.72313297	47.72838885	47.72943576	47.73130301	47.74235394	47.75248103
##	1	1	1	1	1	1
##	47.7758055	47.77908805	47.78911565	47.79240899	47.79304059	47.81021898
##	1	1	1	1	1	1
##	47.8139261	47.82182055	47.82259504	47.825159	47.83197832	47.84396444
##	1	1	1	1	1	1
##	47.84428631	47.85031847	47.85966011	47.86439665	47.87510841	47.88351017
##	1	1	1	1	1	1

##	47.95784899	47.95902368	47.96176046	47.97968254	47.98206278	47.99700086
##	1	1	1	1	1	1
##	48.0222068	48.02492212	48.02575315	48.03953338	48.04147465	48.05527124
##	1	1	1	1	1	1
##	48.06875024	48.07254169	48.07692308	48.10896588	48.11020036	48.1167057
##	1	1	1	1	1	1
##	48.13390165	48.14612868	48.15681763	48.16699694	48.18559034	48.18933417
##	1	1	1	1	1	1
##	48.20143885	48.20957407	48.21428571	48.23214537	48.23231138	48.25839578
##	1	1	1	1	1	1
##	48.27568997	48.28122973	48.31017873	48.36130259	48.3713803	48.37725381
##	1	1	1	1	1	1
##	48.38043678	48.38731814	48.39043119	48.41855381	48.42257744	48.43001597
##	1	1	1	1	1	1
##	48.43179478	48.44113541	48.44146848	48.44605751	48.48143982	48.48832759
##	1	1	1	1	1	1
##	48.49160546	48.51193606	48.51378199	48.5237484	48.53437795	48.53503185
##	1	1	1	1	1	1
##	48.54907179	48.57245681	48.58203096	48.58438317	48.58921443	48.61986525
##	1	1	1	1	1	1
##	48.6479635	48.67392304	48.6797142	48.6859826	48.69879518	48.70666718
##	1	1	1	1	1	1
##	48.71544318	48.71549802	48.7159544	48.72639165	48.72652431	48.74141876
##	1	1	1	1	1	1
##	48.7488741	48.75555434	48.77216166	48.78370003	48.78764346	48.79100208
##	1	1	1	1	1	1
##	48.81232613	48.81261304	48.83839413	48.84944116	48.87912088	48.88513921
##	1	1	1	1	1	1
##	48.88522854	48.88709025	48.88990826	48.91114983	48.91433083	48.94044517
##	1	1	1	1	1	1
##	48.94355093	48.96519285	48.96703297	48.96763592	48.97730666	48.98454077
##	1	1	1	1	1	1
##	49.00105152	49.030747	49.03846154	49.05771316	49.06037585	49.07775769
##	1	1	1	1	1	1
##	49.07904835	49.08438337	49.10109048	49.11068515	49.11764706	49.14080309
##	1	1	1	1	1	1
##	49.14308489	49.15220518	49.15232048	49.16363636	49.1981483	49.20323326
##	1	1	1	1	1	1
##	49.20913884	49.21375921	49.22178988	49.22182047	49.22668061	49.24208977
##	1	1	1	1	1	1
##	49.2531077	49.25567334	49.25638387	49.27138445	49.27268529	49.27293065
##	1	1	1	1	1	1
##	49.29082171	49.29176921	49.2949111	49.31100276	49.31470035	49.32366203
##	1	1	1	1	1	1
##	49.3270649	49.32749453	49.3301719	49.34500607	49.35606748	49.38118812
##	1	1	1	1	1	1
##	49.39434724	49.40247032	49.41006003	49.41654648	49.42212718	49.42512216
##	1	1	1	1	1	1
##	49.43957335	49.4429786	49.44348256	49.45798646	49.46399182	49.46891725
##	1	1	1	1	1	1
##	49.50424005	49.51241135	49.51324812	49.52362434	49.5254298	49.53981255
##	1	1	1	1	1	1
##	49.54318937	49.54671736	49.56634866	49.56736261	49.58100559	49.59331841
##	1	1	1	1	1	1

##	49.59542297	49.59658829	49.59953998	49.61094928	49.61990325	49.62191015
##	1	1	1	1	1	1
##	49.62527198	49.63051429	49.64669739	49.67055246	49.67188289	49.67565455
##	1	1	1	1	1	1
##	49.67789586	49.67867575	49.69012031	49.69144084	49.69820377	49.69862227
##	1	1	1	1	1	1
##	49.69920844	49.70554711	49.7342445	49.736708	49.75897994	49.76017406
##	1	1	1	1	1	1
##	49.77964744	49.78165939	49.7916778	49.82292109	49.82392595	49.83858598
##	1	1	1	1	1	1
##	49.8428187	49.84746797	49.85552712	49.86218664	49.86310746	49.86582478
##	1	1	1	1	1	1
##	49.8692911	49.87153482	49.87210143	49.88060494	49.88517745	49.91633035
##	1	1	1	1	1	1
##	49.9238965	49.94167427	49.95216455	49.96969945	49.97492434	50
##	1	1	1	1	1	2
##	50.00414559	50.00462036	50.00545197	50.00646078	50.01287664	50.01332445
##	1	1	1	1	1	1
##	50.01458151	50.02640264	50.03289474	50.03309067	50.0389206	50.03925967
##	1	1	1	1	1	1
##	50.04163197	50.04540982	50.04906771	50.06252605	50.06357342	50.06790403
##	1	1	1	1	1	1
##	50.08125909	50.08606949	50.09153318	50.10391083	50.1044481	50.13726303
##	1	1	1	1	1	1
##	50.14290988	50.1535907	50.15446127	50.1637513	50.17760329	50.18597731
##	1	1	1	1	1	1
##	50.20147284	50.22233202	50.22244192	50.23356548	50.25251435	50.25776603
##	1	1	1	1	1	1
##	50.26308502	50.26994601	50.2842472	50.28831243	50.28946353	50.28948094
##	1	1	1	1	1	1
##	50.28968714	50.29617663	50.30287152	50.30612245	50.31364349	50.31450694
##	1	1	1	1	1	1
##	50.31750197	50.32822757	50.33139185	50.35411163	50.36688036	50.38510482
##	1	1	1	1	1	1
##	50.39924354	50.40018121	50.40904224	50.41254125	50.4167353	50.42384468
##	1	1	1	1	1	1
##	50.43463668	50.44770774	50.44795086	50.4494786	50.45658567	50.45748116
##	1	1	1	1	1	1
##	50.45861171	50.46463114	50.47870034	50.47890956	50.49173	50.49348599
##	1	1	1	1	1	1
##	50.50714615	50.51013468	50.52141527	50.52309161	50.53822295	50.54616749
##	1	1	1	1	1	1
##	50.54618363	50.54828593	50.54919908	50.55243994	50.5574358	50.56079956
##	1	1	1	1	1	1
##	50.5624792	50.56657224	50.57139016	50.57606289	50.58787484	50.6080207
##	1	1	1	1	1	1
##	50.62934601	50.63562432	50.63985375	50.64194009	50.64813075	50.6545659
##	1	1	1	1	1	1
##	50.66310141	50.68079601	50.68270187	50.71046319	50.71982281	50.72282706
##	1	1	1	1	1	1
##	50.73056692	50.73243647	50.74327039	50.74358294	50.7538094	50.7687802
##	1	1	1	1	1	1
##	50.78380144	50.79339638	50.79365079	50.79961843	50.80108455	50.8032491
##	1	1	1	1	1	1

##	50.80739871	50.81604717	50.84037515	50.84943094	50.85190039	50.8534401
##	1	1	1	1	1	1
##	50.8656873	50.87762067	50.8948485	50.89713227	50.90702948	50.90715048
##	1	1	1	1	1	1
##	50.9203316	50.94274147	50.94954489	50.96534463	50.96921323	50.97956899
##	1	1	1	1	1	1
##	50.9848174	51.00742312	51.01043728	51.01314348	51.03572217	51.03967888
##	1	1	1	1	1	1
##	51.05653591	51.06417813	51.06536419	51.06597522	51.06761566	51.06802879
##	1	1	1	1	1	1
##	51.0731913	51.07628923	51.07688874	51.08041575	51.08732299	51.08794887
##	1	1	1	1	1	1
##	51.08834827	51.09841828	51.10481586	51.10805481	51.14075027	51.14469717
##	1	1	1	1	1	1
##	51.15022387	51.17860381	51.17885733	51.18215461	51.18649733	51.19894598
##	1	1	1	1	1	1
##	51.19917323	51.20654397	51.20807592	51.21833921	51.22035959	51.22478386
##	1	1	1	1	1	1
##	51.2302261	51.24645469	51.25894215	51.26568184	51.27585686	51.27798692
##	1	1	1	1	1	1
##	51.28205128	51.29502828	51.3107775	51.32292523	51.32490011	51.33849708
##	1	1	1	1	1	1
##	51.34890507	51.35794438	51.35914333	51.36070111	51.3855096	51.38875768
##	1	1	1	1	1	1
##	51.38981814	51.39359976	51.39821712	51.4017769	51.40329469	51.41445891
##	1	1	1	1	1	1
##	51.41582622	51.42490372	51.43165856	51.43251847	51.4377298	51.44144144
##	1	1	1	1	1	1
##	51.45213338	51.45951036	51.4608383	51.46743486	51.46959459	51.46987169
##	1	1	1	1	1	1
##	51.47665926	51.50548355	51.50644666	51.50873478	51.50922015	51.51280062
##	1	1	1	1	1	1
##	51.51515152	51.52294304	51.52350516	51.54017857	51.54628257	51.56424581
##	1	1	1	1	1	1
##	51.56720916	51.56748911	51.57885074	51.58581605	51.58614403	51.58730159
##	1	1	1	1	1	1
##	51.5913601	51.59235669	51.59717142	51.61691542	51.61931254	51.63615956
##	1	1	1	1	1	1
##	51.64319249	51.66994106	51.67364017	51.67824074	51.68165168	51.69230769
##	1	1	1	1	1	1
##	51.69798007	51.70068027	51.70842825	51.70872261	51.70881226	51.71176077
##	1	1	1	1	1	1
##	51.71564856	51.71667413	51.72605791	51.72964343	51.73501577	51.73547733
##	1	1	1	1	1	1
##	51.74340811	51.74558619	51.75634087	51.75792507	51.75879397	51.75930076
##	1	1	1	1	1	1
##	51.77545484	51.80709534	51.81047766	51.81544316	51.82028856	51.82077766
##	1	1	1	1	1	1
##	51.82132412	51.83637434	51.84020956	51.85447464	51.86110417	51.86159024
##	1	1	1	1	1	1
##	51.86251311	51.86264982	51.86838785	51.88659315	51.9092219	51.92620727
##	1	1	1	1	1	1
##	51.94532072	51.94562166	51.95195195	51.95520796	51.96295401	51.96451241
##	1	1	1	1	1	1

##	51.96681008	51.97306014	51.97446556	51.97613076	51.98325253	51.98366394
##	1	1	1	1	1	1
##	51.98386566	51.98701098	52.00381679	52.00769416	52.00793651	52.00899749
##	1	1	1	1	1	1
##	52.01918614	52.02079887	52.04494621	52.04708291	52.04755614	52.04957102
##	1	1	1	1	1	1
##	52.050581	52.05061305	52.0583875	52.07697724	52.09427184	52.09451499
##	1	1	1	1	1	1
##	52.10630899	52.10889571	52.11106885	52.12383009	52.13282937	52.14513361
##	1	1	1	1	1	1
##	52.14865363	52.15311005	52.19686649	52.19885742	52.22007099	52.23542728
##	1	1	1	1	1	1
##	52.24348917	52.2468299	52.26761016	52.27663799	52.28193357	52.29303157
##	1	1	1	1	1	1
##	52.29310276	52.2945571	52.30400334	52.3494794	52.354297	52.35531629
##	1	1	1	1	1	1
##	52.36294896	52.37213921	52.38235946	52.38662469	52.40220693	52.40904621
##	1	1	1	1	1	1
##	52.41587576	52.42167848	52.43500867	52.44819519	52.46457038	52.48109205
##	1	1	1	1	1	1
##	52.48119487	52.49303974	52.5142173	52.52586732	52.52962445	52.53632424
##	1	1	1	1	1	1
##	52.53893026	52.53935141	52.55536627	52.56518082	52.58746356	52.59030604
##	1	1	1	1	1	1
##	52.59370817	52.60510813	52.60735221	52.61361092	52.61639885	52.61752646
##	1	1	1	1	1	1
##	52.62411957	52.63250883	52.64644351	52.66229169	52.666999	52.66836086
##	1	1	1	1	1	1
##	52.67666271	52.6801744	52.68518519	52.6920083	52.69406393	52.6972873
##	1	1	1	1	1	1
##	52.70108043	52.71734183	52.72049671	52.72202545	52.7421835	52.74376066
##	1	1	1	1	1	1
##	52.75321855	52.76653171	52.77562023	52.7799604	52.79043982	52.79664691
##	1	1	1	1	1	1
##	52.80428595	52.80460624	52.80612245	52.81049885	52.81829642	52.822868
##	1	1	1	1	1	1
##	52.82673761	52.83877574	52.83939201	52.84421461	52.84685609	52.84922777
##	1	1	1	1	1	1
##	52.86757039	52.90453762	52.91262136	52.91387313	52.92240511	52.93732685
##	1	1	1	1	1	1
##	52.93762454	52.94398662	52.94771969	52.95223768	52.9590728	52.96595297
##	1	1	1	1	1	1
##	52.96638102	52.9829631	52.98314358	52.98358377	53.00896287	53.01573427
##	1	1	1	1	1	1
##	53.03021379	53.0572913	53.06008644	53.07888763	53.12265988	53.12759016
##	1	1	1	1	1	1
##	53.13706564	53.14527463	53.16042597	53.1743525	53.18127251	53.20351759
##	1	1	1	1	1	1
##	53.20602663	53.20651532	53.20852938	53.21361059	53.23821488	53.24203229
##	1	1	1	1	1	1
##	53.24756825	53.2505921	53.25584652	53.27269485	53.27640985	53.28205669
##	1	1	1	1	1	1
##	53.29996737	53.30250208	53.30729997	53.31825038	53.31902246	53.31920904
##	1	1	1	1	1	1

##	53.33333333	53.33401681	53.34713596	53.35134297	53.35436383	53.35894843
##	1	1	1	1	1	1
##	53.36016587	53.3734194	53.38283828	53.38491296	53.38800964	53.40436158
##	1	1	1	1	1	1
##	53.40548509	53.43425397	53.44377949	53.45121762	53.45761431	53.47605031
##	1	1	1	1	1	1
##	53.48140241	53.48532996	53.49090909	53.49462366	53.49988746	53.50906096
##	1	1	1	1	1	1
##	53.52258361	53.52986811	53.53820299	53.54345054	53.55691057	53.57704286
##	1	1	1	1	1	1
##	53.57809583	53.58111774	53.59477124	53.5982009	53.60330103	53.62200682
##	1	1	1	1	1	1
##	53.62409659	53.62776025	53.65028203	53.65382552	53.65425415	53.67536949
##	1	1	1	1	1	1
##	53.68635766	53.70241078	53.72316126	53.73980054	53.74561974	53.74565671
##	1	1	1	1	1	1
##	53.74742513	53.74910736	53.76058041	53.77251102	53.77866168	53.77892031
##	1	1	1	1	1	1
##	53.78479825	53.7874558	53.79832387	53.81660264	53.82644893	53.82685069
##	1	1	1	1	1	1
##	53.82925436	53.83079507	53.83226777	53.83759733	53.84242816	53.84371029
##	1	1	1	1	1	1
##	53.844404	53.85802469	53.8656602	53.87931034	53.89250928	53.89830508
##	1	1	1	1	1	1
##	53.90869293	53.92670157	53.9275695	53.92941176	53.94121576	53.94179533
##	1	1	1	1	1	1
##	53.94637462	53.95285545	53.95787945	53.96020289	53.97391536	53.97810219
##	1	1	1	1	1	1
##	53.98618958	53.99449786	54.00059341	54.00578752	54.01155144	54.01484866
##	1	1	1	1	1	1
##	54.0255049	54.02745995	54.02780201	54.02895028	54.03626428	54.04750742
##	1	1	1	1	1	1
##	54.065766	54.0721857	54.07513858	54.08074857	54.08412277	54.09965078
##	1	1	1	1	1	1
##	54.11020777	54.11392405	54.126895	54.12723449	54.12785095	54.12988651
##	1	1	1	1	1	1
##	54.13409439	54.14033382	54.15298854	54.15846326	54.15959253	54.17001877
##	1	1	1	1	1	1
##	54.17744518	54.17912625	54.17921687	54.18585648	54.22507008	54.23312883
##	1	1	1	1	1	1
##	54.25001996	54.25257732	54.25797503	54.26127527	54.27882684	54.2877907
##	1	1	1	1	1	1
##	54.28790199	54.29372197	54.30466505	54.30575593	54.32196669	54.32342007
##	1	1	1	1	1	1
##	54.32620153	54.33287483	54.37152392	54.37784929	54.38254081	54.38575454
##	1	1	1	1	1	1
##	54.4047619	54.41316794	54.43612713	54.47154472	54.47316103	54.49083187
##	1	1	1	1	1	1
##	54.49160425	54.49526814	54.50719046	54.50824606	54.51100686	54.51398136
##	1	1	1	1	1	1
##	54.52999083	54.53251946	54.53835338	54.57013575	54.58467264	54.58960661
##	1	1	1	1	1	1
##	54.60626758	54.61738163	54.62888711	54.64772054	54.65468838	54.66050864
##	1	1	1	1	1	1

##	54.66371797	54.6809039	54.68263699	54.69448584	54.70653872	54.73421158
##	1	1	1	1	1	1
##	54.73812424	54.74475777	54.77389453	54.77898249	54.78018144	54.8037466
##	1	1	1	1	1	1
##	54.80726628	54.80842619	54.81702095	54.81821391	54.8215151	54.84009752
##	1	1	1	1	1	1
##	54.84299238	54.8502573	54.85049834	54.85789339	54.86235142	54.87066593
##	1	1	1	1	1	1
##	54.87220447	54.88761108	54.88790688	54.88850772	54.91341991	54.91358025
##	1	1	1	1	1	1
##	54.94211932	54.94516044	54.94720247	54.9562195	54.9758293	55
##	1	1	1	1	1	1
##	55.00782632	55.01498074	55.02144979	55.02568893	55.03236361	55.04032258
##	1	1	1	1	1	1
##	55.04192647	55.04587156	55.07615872	55.08264463	55.08586526	55.08958456
##	1	1	1	1	1	1
##	55.09433962	55.10159652	55.11347716	55.11711088	55.1213419	55.1426102
##	1	1	1	1	1	1
##	55.15394913	55.17431337	55.17681595	55.18108863	55.19524618	55.21239889
##	1	1	1	1	1	1
##	55.2130898	55.21715624	55.22226238	55.22971652	55.25443787	55.2736352
##	1	1	1	1	1	1
##	55.28066751	55.28741411	55.28885946	55.30331224	55.30440345	55.31228394
##	1	1	1	1	1	1
##	55.3172673	55.31781651	55.32495512	55.33486122	55.33650578	55.35270379
##	1	1	1	1	1	1
##	55.37152408	55.38477444	55.38892222	55.39414973	55.39809327	55.4011976
##	1	1	1	1	1	1
##	55.41340462	55.41501976	55.41918203	55.42524041	55.42932052	55.42975288
##	1	1	1	1	1	1
##	55.44444444	55.44621649	55.44835414	55.45023697	55.45062062	55.45087483
##	1	1	1	1	1	1
##	55.46135117	55.46623794	55.46630361	55.46666667	55.48879131	55.49945907
##	1	1	1	1	1	1
##	55.50427582	55.52070942	55.52788845	55.5392636	55.55250891	55.57090946
##	1	1	1	1	1	1
##	55.57483731	55.57580175	55.58980682	55.59017891	55.60993976	55.62451513
##	1	1	1	1	1	1
##	55.63023079	55.63873845	55.65993424	55.67095741	55.67182922	55.68151859
##	1	1	1	1	1	1
##	55.69738191	55.69972196	55.70167996	55.70687419	55.70921552	55.71112749
##	1	1	1	1	1	1
##	55.71212884	55.73047859	55.73352377	55.74272588	55.74874687	55.75048733
##	1	1	1	1	1	1
##	55.77395577	55.77512375	55.78754661	55.78764143	55.79514825	55.80008519
##	1	1	1	1	1	1
##	55.84604212	55.8526976	55.86911357	55.87154667	55.88080861	55.89519651
##	1	1	1	1	1	1
##	55.90180473	55.91549296	55.92105263	55.92848181	55.93438405	55.93984962
##	1	1	1	1	1	1
##	55.94855305	55.95080801	55.95126523	55.9521805	55.96274794	55.96546311
##	1	1	1	1	1	1
##	55.9681182	56.00347308	56.00375235	56.03038936	56.06258149	56.06891428
##	1	1	1	1	1	1

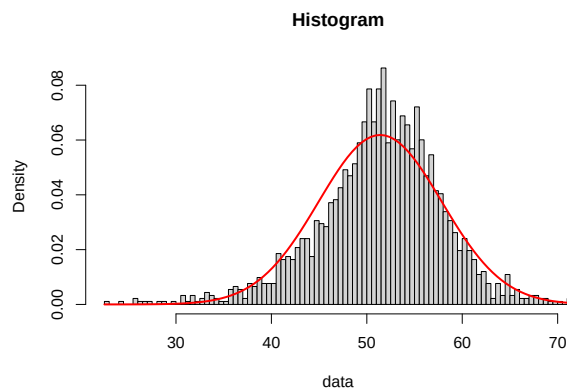
##	56.07021518	56.07321838	56.07787274	56.1044768	56.13529501	56.15238497
##	1	1	1	1	1	1
##	56.15648201	56.16692476	56.17542728	56.22047244	56.23599701	56.25722878
##	1	1	1	1	1	1
##	56.25967592	56.26874625	56.27056673	56.28948091	56.30311056	56.30872483
##	1	1	1	1	1	1
##	56.31128083	56.32238242	56.32384273	56.33534356	56.35323607	56.35803931
##	1	1	1	1	1	1
##	56.3676215	56.37254902	56.38106082	56.38540224	56.39295088	56.40196431
##	1	1	1	1	1	1
##	56.40479472	56.4098706	56.42070934	56.43371868	56.43678161	56.44622578
##	1	1	1	1	1	1
##	56.45325203	56.46618884	56.50806959	56.51105651	56.52616402	56.52954375
##	1	1	1	1	1	1
##	56.55389457	56.55539429	56.56596097	56.5676769	56.56848306	56.57916191
##	1	1	1	1	1	1
##	56.59293338	56.62865895	56.62913186	56.66280417	56.66372462	56.67977102
##	1	1	1	1	1	1
##	56.701726	56.71740569	56.73015194	56.73692428	56.7383997	56.74735811
##	1	1	1	1	1	1
##	56.75931775	56.75967986	56.78785857	56.79225717	56.79732758	56.80724824
##	1	1	1	1	1	1
##	56.81528662	56.81681682	56.83060109	56.86868687	56.86991246	56.87054533
##	1	1	1	1	1	1
##	56.87263556	56.87324321	56.87707641	56.88094652	56.88442211	56.89655172
##	1	1	1	1	1	1
##	56.89721545	56.90633869	56.91305397	56.9140625	56.91467003	56.94623656
##	1	1	1	1	1	1
##	56.95154185	56.98248001	56.98651056	56.99136868	57.01402806	57.01953465
##	1	1	1	1	1	1
##	57.03671953	57.03768405	57.07880669	57.09251101	57.09918902	57.14965878
##	1	1	1	1	1	1
##	57.15249094	57.15368581	57.17801166	57.18283582	57.1945047	57.19827586
##	1	1	1	1	1	1
##	57.20113892	57.21052632	57.22070845	57.24111578	57.26034123	57.2702332
##	1	1	1	1	1	1
##	57.2713258	57.275	57.3063573	57.31479005	57.33005733	57.34660926
##	1	1	1	1	1	1
##	57.34908136	57.35753743	57.37591114	57.3795509	57.38845144	57.39389024
##	1	1	1	1	1	1
##	57.40169622	57.41970774	57.43918818	57.44373479	57.49039693	57.49244324
##	1	1	1	1	1	1
##	57.50165909	57.5246773	57.52914198	57.53358925	57.53732709	57.54847645
##	1	1	1	1	1	1
##	57.56531726	57.56686236	57.58044327	57.61102603	57.62759264	57.63500931
##	1	1	1	1	1	1
##	57.63697582	57.66095994	57.66982409	57.68894714	57.69099432	57.69758915
##	1	1	1	1	1	1
##	57.7359535	57.76652195	57.82097649	57.86113328	57.86551511	57.87008756
##	1	1	1	1	1	1
##	57.8721885	57.88269551	57.88772281	57.89782736	57.92579144	57.92632259
##	1	1	1	1	1	1
##	57.93067227	57.94421328	57.96875716	57.97535001	57.9788197	57.98394291
##	1	1	1	1	1	1

##	57.98611111	58.04384486	58.05384979	58.05982551	58.08658795	58.09248555
##	1	1	1	1	1	1
##	58.1022409	58.11808118	58.12459859	58.13417794	58.13925734	58.14360771
##	1	1	1	1	1	1
##	58.16633267	58.19746835	58.22066261	58.24836321	58.25617107	58.25863279
##	1	1	1	1	1	1
##	58.26086957	58.26415094	58.30122355	58.32908813	58.33333333	58.33924769
##	1	1	1	1	1	1
##	58.35469775	58.37739174	58.40512737	58.43071786	58.44535359	58.48365534
##	1	1	1	1	1	1
##	58.48474785	58.48682908	58.52386683	58.53674286	58.5480978	58.55774444
##	1	1	1	1	1	1
##	58.5998978	58.61317052	58.630394	58.64154127	58.66180855	58.66550286
##	1	1	1	1	1	1
##	58.66885785	58.69187729	58.69722776	58.74613003	58.75343721	58.75781102
##	1	1	1	1	1	1
##	58.76956073	58.81427072	58.81656805	58.83239688	58.84967501	58.87762001
##	1	1	1	1	1	1
##	58.88459857	58.91069283	58.91095012	58.93703453	58.96067254	58.96535938
##	1	1	1	1	1	1
##	59.02578797	59.05396402	59.05511811	59.06443537	59.06553979	59.06907728
##	1	1	1	1	1	1
##	59.08585547	59.11931818	59.12515462	59.13853318	59.14859455	59.21402018
##	1	1	1	1	1	1
##	59.23236515	59.25061962	59.25306577	59.26801589	59.29628149	59.29662522
##	1	1	1	1	1	1
##	59.31461903	59.37019969	59.42297512	59.45375543	59.47831362	59.49197861
##	1	1	1	1	1	1
##	59.51840865	59.52341552	59.54720205	59.57669496	59.62751137	59.63230466
##	1	1	1	1	1	1
##	59.69746083	59.69795677	59.77522216	59.81993998	59.8323437	59.83515133
##	1	1	1	1	1	1
##	59.8441058	59.86962292	59.8932859	59.91237678	59.92968453	59.99343617
##	1	1	1	1	1	1
##	60.04896107	60.07342818	60.07486979	60.08018328	60.09506026	60.10540184
##	1	1	1	1	1	1
##	60.12052464	60.12813668	60.16096579	60.19490255	60.1961379	60.2118003
##	1	1	1	1	1	1
##	60.21316724	60.22216345	60.23950993	60.25241508	60.28214738	60.30391761
##	1	1	1	1	1	1
##	60.30637328	60.35044825	60.48309179	60.48707753	60.5823533	60.59837728
##	1	1	1	1	1	1
##	60.59983209	60.61479347	60.63639025	60.66267248	60.66311488	60.67132867
##	1	1	1	1	1	1
##	60.68268015	60.68418443	60.70604543	60.71467401	60.72740756	60.77909614
##	1	1	1	1	1	1
##	60.81118115	60.8137045	60.82474227	60.9652612	61.04842736	61.06304079
##	1	1	1	1	1	1
##	61.09799292	61.13161132	61.14462263	61.27588137	61.30384168	61.30713713
##	1	1	1	1	1	1
##	61.33424438	61.33661337	61.37873754	61.39920858	61.43309427	61.43432459
##	1	1	1	1	1	1
##	61.49898521	61.56462585	61.64104464	61.67647617	61.70039607	61.70681551
##	1	1	1	1	1	1

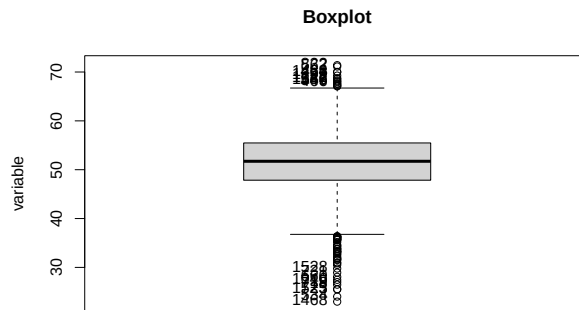
```
## 61.73052088 61.7563373 61.86285295 61.87173482      62 62.07381821
##           1           1           1           1           1           1
## 62.21363084 62.21920946 62.29758067 62.30679741 62.31263383 62.34448122
##           1           1           1           1           1           1
## 62.42960579 62.43695698 62.44671782 62.49372175 62.51145738 62.55361267
##           1           1           1           1           1           1
## 62.60986328 62.82316215 62.84304485 62.95743092 62.98229565 63.00594806
##           1           1           1           1           1           1
## 63.15789474 63.5566516 63.62573099 63.68381424 63.72496899 63.7754003
##           1           1           1           1           1           1
## 63.82715093 63.99600553 64.06140963 64.21699626 64.2777983 64.63878327
##           1           1           1           1           1           1
## 64.6571958 64.68172485 64.70421858 64.84977289 64.88372093 64.90147783
##           1           1           1           1           1           1
## 64.90258172 64.90514511 64.97461929 65.25373134 65.2892562 65.40865385
##           1           1           1           1           1           1
## 65.51325905 65.56748466 65.72456321 65.75840145 65.7719961 66.0330321
##           1           1           1           1           1           1
## 66.38405714 66.6645667 66.72032402 67.06315491 67.19123757 67.25663717
##           1           1           1           1           1           1
## 67.55122192 67.64327109 67.92717087 68.10336023 68.27869684 68.57473787
##           1           1           1           1           1           1
## 68.73940079 69.2651329 69.98906129 70.02327635 71.12700722 71.40009965
##           1           1           1           1           1           1
```

```
res.var <- summary_and_normality_check(variable)
```

```
## [1] "Summary: "
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 22.99 47.85   51.73   51.40  55.48   71.40
## [1] "Mean"
## [1] 51.40232
## [1] "Standard deviation"
## [1] 6.450696
```



```
## [1] "Skewness"
## [1] -0.5031018
```



```
## [1] "Num Outliers:"
## [1] 57
## [1] "Num. severe outliers:"
## [1] 2
## [1] "Shapiro test: "
##
## Shapiro-Wilk normality test
##
## data: data
## W = 0.9816, p-value = 1.341e-14
##
## [1] "Most repeated value ad times it occurs: "
## [1] "50"
## [1] 2
```

```
outliers_df[res.var, "pctmarriedhouseholds"] <- 1
```

Variable *birthrate*

The variable *birthrate* represents the number of births relative to the number of women in each county.

Values range from 0 to 21.33, with a mean of 5.60 and a median of 5.36. The standard deviation is 1.93, and the skewness of 1.25 shows a moderate positive skew, with some counties having significantly higher birthrates. The Shapiro-Wilk test confirms non-normality. There are 104 outliers, including 17 severe ones.

```
variable<-df$birthrate  
table(variable)
```

##	variable					
##		0	0.284090909	0.41322314	0.476758045	0.56980057
##		3	1	1	1	1
##	0.855920114	1.084812623	1.111111111	1.17820324	1.309904153	1.330671989
##	1	1	1	1	1	1
##	1.38121547	1.418067227	1.424501425	1.470588235	1.574468085	1.604584527
##	1	1	1	1	1	1
##	1.638311279	1.660795169	1.750547046	1.754385965	1.811062163	1.879036994
##	1	1	1	1	1	1
##	1.880766501	1.946661476	1.972242513	1.998889506	2.009646302	2.079107505
##	1	1	1	1	1	1
##	2.089552239	2.100668394	2.105263158	2.110199297	2.11038961	2.134146341
##	1	1	1	1	1	1
##	2.147577093	2.164502165	2.181467181	2.190865206	2.195121951	2.198107958

##	1	1	1	1	1	1
##	2.20341093	2.224694105	2.248441674	2.258064516	2.28320067	2.292860865
##	1	1	1	1	1	1
##	2.292993631	2.32	2.328431373	2.353331124	2.363989637	2.387267905
##	1	1	1	1	1	1
##	2.411347518	2.412868633	2.429149798	2.454308094	2.46365105	2.465166131
##	1	1	1	1	1	1
##	2.469135802	2.504393673	2.52476505	2.533172497	2.534226624	2.534634761
##	1	1	1	1	1	1
##	2.5472273	2.550091075	2.553975777	2.555885972	2.584269663	2.628490965
##	1	1	1	1	1	1
##	2.64006449	2.640934485	2.702702703	2.708877285	2.713797035	2.718894009
##	1	1	1	1	1	1
##	2.755102041	2.773207991	2.773556231	2.790178571	2.796024928	2.808716707
##	1	1	1	1	1	1
##	2.826379542	2.843394576	2.851375221	2.872784433	2.874946891	2.876060418
##	1	1	1	1	1	1
##	2.903225806	2.90368272	2.923473775	2.944539737	2.949061662	2.955003358
##	1	1	1	1	1	1
##	2.960526316	2.963473467	2.966381015	2.98792117	3.008098727	3.008595989
##	1	1	1	1	1	1
##	3.014449427	3.045785027	3.0621435	3.072371416	3.084671073	3.087986464
##	1	1	1	1	1	1
##	3.095975232	3.098591549	3.101503759	3.120443278	3.122648608	3.126436782
##	1	1	1	1	1	1
##	3.135414054	3.141875307	3.172641121	3.174132305	3.178368121	3.204859373
##	1	1	1	1	1	1
##	3.205128205	3.216430924	3.227665706	3.236583838	3.237555645	3.238819733
##	1	1	1	1	1	1
##	3.240589198	3.244249435	3.253269113	3.255738239	3.257226644	3.257989088
##	1	1	1	1	1	1
##	3.263086336	3.263424336	3.270009343	3.276955603	3.291139241	3.29153605
##	1	1	1	1	1	1
##	3.293509848	3.298774741	3.301127214	3.304159034	3.309741882	3.314818133
##	1	1	1	1	1	1
##	3.328451604	3.333333333	3.333416627	3.366021733	3.367670365	3.387978142
##	1	1	1	1	1	1
##	3.388278388	3.38904364	3.389715373	3.410283316	3.421727973	3.422926717
##	1	1	1	1	1	1
##	3.43803056	3.440273038	3.450671607	3.453598302	3.454048815	3.458303859
##	1	1	1	1	1	1
##	3.462050599	3.467781604	3.499365035	3.502840141	3.505447655	3.50678733
##	1	1	1	1	1	1
##	3.510758777	3.515151515	3.515263645	3.516921035	3.524229075	3.528435035
##	1	1	1	1	1	1
##	3.536585366	3.53909465	3.545136747	3.545493379	3.546703589	3.558718861
##	1	1	1	1	1	1
##	3.565762635	3.58004827	3.581020591	3.583061889	3.589466089	3.613892365
##	1	1	1	1	1	1
##	3.617945007	3.627311522	3.631082062	3.635690867	3.635817308	3.636363636
##	1	1	1	1	1	1
##	3.636938646	3.640481715	3.641536273	3.646563815	3.646833013	3.648198607
##	1	1	1	1	1	1
##	3.651138835	3.664122137	3.668042182	3.66841353	3.66910141	3.673864003

##	1	1	1	1	1	1
##	3.67487328	3.681800261	3.688606085	3.697661773	3.698558778	3.7027833
##	1	1	1	1	1	1
##	3.704047746	3.706395349	3.708958249	3.712871287	3.726061615	3.728510828
##	1	1	1	1	1	1
##	3.731942215	3.7384301	3.739774055	3.753424658	3.767349636	3.772833827
##	1	1	1	1	1	1
##	3.774056486	3.774792067	3.777923784	3.782242355	3.783783784	3.78540305
##	1	1	1	1	1	1
##	3.788224555	3.788634098	3.794871795	3.797949107	3.807095815	3.812301967
##	1	1	1	1	1	1
##	3.812668868	3.813691779	3.819022136	3.83125269	3.860213737	3.860242322
##	1	1	1	1	1	1
##	3.861652116	3.865213082	3.868613139	3.877752218	3.880659295	3.885024281
##	1	1	1	1	1	1
##	3.887024609	3.888098625	3.896713615	3.897685749	3.899721448	3.900156006
##	1	1	1	1	1	1
##	3.90120498	3.901815696	3.902327704	3.906472769	3.907074974	3.907462265
##	1	1	1	1	1	1
##	3.907580057	3.908475144	3.909798832	3.911735206	3.911889349	3.913308575
##	1	1	1	1	1	1
##	3.917098446	3.919463087	3.920755108	3.923080412	3.923660503	3.929500144
##	1	1	1	1	1	1
##	3.932688732	3.935222672	3.938223938	3.939346569	3.944716384	3.94922426
##	1	1	1	1	1	1
##	3.956148713	3.957556639	3.962223187	3.969171484	3.969883641	3.971422568
##	1	1	1	1	1	1
##	3.991528185	3.995098039	3.996826341	4	4.006754682	4.011231448
##	1	1	1	1	1	1
##	4.012841091	4.017424976	4.01891253	4.019073569	4.025191676	4.038301415
##	1	1	1	1	1	1
##	4.041212898	4.045610747	4.047619048	4.050285512	4.06278855	4.063068526
##	1	1	1	1	1	1
##	4.063558218	4.063986165	4.06824147	4.074920055	4.076692166	4.078719282
##	1	1	1	1	1	1
##	4.079089924	4.079365079	4.079871327	4.090035685	4.095334005	4.101603165
##	1	1	1	1	1	1
##	4.102564103	4.107398367	4.109357003	4.111531191	4.111600587	4.112005292
##	1	1	1	1	1	1
##	4.11444786	4.115421003	4.117165096	4.117879551	4.124513619	4.12979351
##	1	1	1	1	1	1
##	4.133114215	4.135338346	4.138710893	4.138851802	4.141104294	4.14207772
##	1	1	1	1	1	1
##	4.146396651	4.146877226	4.149417541	4.162724693	4.17432129	4.175372554
##	1	1	1	1	1	1
##	4.178438278	4.179907113	4.180500351	4.18906927	4.19412822	4.194566305
##	1	1	1	1	1	1
##	4.195088677	4.200664853	4.202401372	4.202589635	4.202796852	4.203166439
##	1	1	1	1	1	1
##	4.203539823	4.203655352	4.210526316	4.212348529	4.215304799	4.217134885
##	1	1	1	1	1	1
##	4.221954162	4.22223833	4.225482912	4.229447528	4.231027535	4.240824844
##	1	1	1	1	1	1
##	4.243248035	4.250386399	4.262413265	4.269052202	4.270952928	4.272316777

##	1	1	1	1	1	1
##	4.274611399	4.274857505	4.275633043	4.275724276	4.286441109	4.287972051
##	1	1	1	1	1	1
##	4.291640077	4.295098535	4.2987606	4.303337477	4.308202154	4.315304948
##	1	1	1	1	1	1
##	4.31864301	4.318836947	4.322689366	4.326923077	4.330985915	4.337050805
##	1	1	1	1	1	1
##	4.341957255	4.34410443	4.344746163	4.346733668	4.348232654	4.350132626
##	1	1	1	1	1	1
##	4.350706714	4.353189989	4.363776791	4.367123465	4.36746988	4.371921182
##	1	1	1	1	1	1
##	4.372227998	4.373757455	4.376511986	4.377431907	4.378496716	4.379909545
##	1	1	1	1	1	1
##	4.382650748	4.384535624	4.387164057	4.390138408	4.391052196	4.397120317
##	1	1	1	1	1	1
##	4.398148148	4.399845619	4.404237591	4.405016825	4.40907905	4.420790925
##	1	1	1	1	1	1
##	4.422498927	4.425228891	4.426449927	4.426612689	4.428405952	4.428705174
##	1	1	1	1	1	1
##	4.429860541	4.430147059	4.431818182	4.433836917	4.447587616	4.448655477
##	1	1	1	1	1	1
##	4.449778611	4.451140766	4.458527742	4.465658687	4.465958776	4.4729858
##	1	1	1	1	1	1
##	4.474235159	4.474464442	4.480398258	4.483306836	4.490661015	4.493889063
##	1	1	1	1	1	1
##	4.495990836	4.49704142	4.49736915	4.499226458	4.499229584	4.500642949
##	1	1	1	1	1	1
##	4.502177282	4.502231194	4.504381161	4.504504505	4.507693163	4.508674734
##	1	1	1	1	1	1
##	4.517034138	4.518903132	4.519119351	4.519774011	4.523580366	4.527363184
##	1	1	1	1	1	1
##	4.528650647	4.528860796	4.528947447	4.529616725	4.533178353	4.535082715
##	1	1	1	1	1	1
##	4.539151472	4.540620142	4.540884719	4.543403039	4.543828265	4.545454545
##	1	1	1	1	1	1
##	4.546508931	4.550418639	4.550819672	4.553734062	4.557287265	4.557619009
##	1	1	1	1	1	1
##	4.558176729	4.56119093	4.568106312	4.571872443	4.573604784	4.578351666
##	1	1	1	1	1	1
##	4.581280788	4.583083383	4.585798817	4.586129754	4.598628479	4.602473498
##	1	1	1	1	1	1
##	4.603274095	4.603914259	4.609250399	4.610130905	4.61118069	4.612961296
##	1	1	1	1	1	1
##	4.618133504	4.6247512	4.62962963	4.630064868	4.630829016	4.631760644
##	1	1	1	1	1	1
##	4.633003644	4.636200208	4.636217581	4.640520349	4.642829838	4.642857143
##	1	1	1	1	1	1
##	4.643274854	4.645550528	4.646092865	4.646266143	4.646681703	4.650635209
##	1	1	1	1	1	1
##	4.652301665	4.654466449	4.655824075	4.659172628	4.660949114	4.662004662
##	1	1	1	1	1	1
##	4.662698413	4.6635126	4.663623735	4.663810807	4.666419111	4.671312865
##	1	1	1	1	1	1
##	4.673182651	4.685714286	4.686489981	4.689366786	4.689714916	4.691483937

##	1	2	1	1	1	1
##	4.692082111	4.69334869	4.694264339	4.696595168	4.69661296	4.696647392
##	1	1	1	1	1	1
##	4.697986577	4.703476483	4.704009263	4.70528099	4.705882353	4.706601467
##	1	1	1	1	1	1
##	4.706770702	4.707995158	4.708366736	4.708383961	4.710632571	4.712136228
##	1	1	1	1	1	1
##	4.717612918	4.71844142	4.726955479	4.727253413	4.727439304	4.728190104
##	1	1	1	1	1	1
##	4.729251068	4.73206677	4.732327714	4.732510288	4.732975417	4.735242361
##	1	1	1	1	1	1
##	4.737575476	4.73813501	4.739182301	4.740349554	4.740817921	4.741223308
##	1	1	1	1	1	1
##	4.744289032	4.746227709	4.749478079	4.75120604	4.754260311	4.754928419
##	1	1	1	1	1	1
##	4.757548033	4.761498421	4.763673015	4.764332654	4.76817356	4.772020566
##	1	1	1	1	1	1
##	4.774692859	4.775349865	4.779878415	4.780114723	4.783515756	4.785376874
##	1	1	1	1	1	1
##	4.789430223	4.794040316	4.798813776	4.799040192	4.799131379	4.801324503
##	1	1	1	1	1	1
##	4.803398323	4.80349345	4.804003336	4.805591962	4.806902219	4.80729036
##	1	1	1	1	1	1
##	4.807405196	4.807972069	4.808108498	4.812157028	4.815862717	4.817078346
##	1	1	1	1	1	1
##	4.818560381	4.819607038	4.82052462	4.821037253	4.823151125	4.823685961
##	1	1	1	1	1	1
##	4.824422651	4.825363357	4.827807522	4.830100247	4.830467049	4.834403227
##	1	1	1	1	1	1
##	4.835115865	4.83615222	4.836428773	4.839230894	4.839289566	4.839663945
##	1	1	1	1	1	1
##	4.839989421	4.842990108	4.843160444	4.844731809	4.84491139	4.844986927
##	1	1	1	1	1	1
##	4.84544324	4.84804631	4.851190476	4.854021136	4.855936413	4.858604091
##	1	1	1	1	1	1
##	4.860486049	4.863950374	4.863978687	4.865649964	4.867486575	4.871361927
##	1	1	1	1	1	1
##	4.872468117	4.874120242	4.879604208	4.880531448	4.881386861	4.882440161
##	1	1	1	1	1	1
##	4.883318928	4.884815874	4.886769964	4.887361588	4.887587132	4.891062695
##	1	1	1	1	1	1
##	4.891661195	4.891733624	4.893475366	4.894050891	4.894094994	4.895140929
##	1	1	1	1	1	1
##	4.896421846	4.897660819	4.899005125	4.900650039	4.903978052	4.907216495
##	1	1	1	1	1	1
##	4.907650931	4.908292431	4.909090909	4.909253198	4.910307084	4.910516481
##	1	1	1	1	1	1
##	4.912663755	4.912890034	4.9190879	4.919274098	4.919773478	4.92078633
##	1	1	1	1	1	1
##	4.925163106	4.927099045	4.929577465	4.930928626	4.934210526	4.937906564
##	1	1	1	1	1	1
##	4.942780618	4.943288381	4.947730667	4.95049505	4.951690821	4.954367666
##	1	1	1	1	1	1
##	4.955913652	4.956760177	4.96054115	4.961767205	4.962066844	4.962779156

##	1	1	1	1	1	1
##	4.964180034	4.964476021	4.964771639	4.965357968	4.965920156	4.966521485
##	1	1	1	1	1	1
##	4.967320261	4.967904974	4.967948718	4.968421053	4.970178926	4.970419505
##	1	1	1	1	1	1
##	4.970818505	4.974134501	4.977235099	4.979333506	4.983090176	4.987212276
##	1	1	1	1	1	1
##	4.988123515	4.988383217	4.989577905	4.99194847	4.992021883	4.992503748
##	1	1	1	1	1	1
##	4.994346023	4.994880839	4.995038042	4.995519713	5	5.000553159
##	1	1	1	1	2	1
##	5.001246158	5.002432301	5.005740528	5.007153076	5.009078287	5.009079903
##	1	1	1	1	1	1
##	5.011933174	5.015546843	5.015576324	5.015887426	5.016549209	5.024311183
##	1	1	1	1	1	1
##	5.026285421	5.027932961	5.028735632	5.029215359	5.029990529	5.030073987
##	1	1	1	1	1	1
##	5.031446541	5.031948882	5.03246323	5.037284569	5.038143675	5.039267016
##	1	1	1	1	1	1
##	5.041322314	5.041436464	5.042979943	5.043113678	5.051131082	5.0513461
##	1	1	1	1	1	1
##	5.051532411	5.052412881	5.054084297	5.054495913	5.055523086	5.055576216
##	1	1	1	1	1	1
##	5.056955471	5.057135911	5.060304916	5.060383303	5.062186341	5.065070552
##	1	1	1	1	1	1
##	5.066533644	5.066666667	5.072603135	5.073719664	5.076142132	5.078225186
##	1	1	1	1	1	1
##	5.078382982	5.078698846	5.083188478	5.083533771	5.089635277	5.089647195
##	1	1	1	1	1	1
##	5.090116855	5.090180361	5.091311566	5.091597145	5.094111629	5.095541401
##	1	1	1	1	1	1
##	5.096660808	5.099793071	5.103569855	5.104572847	5.104712042	5.106970324
##	1	1	1	1	1	1
##	5.10901206	5.10978957	5.111191026	5.113459745	5.114550334	5.115003933
##	1	1	1	1	1	1
##	5.115327381	5.116068139	5.116648446	5.116736739	5.119076848	5.121798876
##	1	1	1	1	1	1
##	5.124835742	5.125073246	5.130927105	5.1314986	5.134211811	5.134424767
##	1	1	1	1	1	1
##	5.135520685	5.139186296	5.141916907	5.14390692	5.145852764	5.148741419
##	1	1	1	1	1	1
##	5.155718398	5.156824131	5.158324821	5.158672656	5.163242125	5.163792425
##	1	1	1	1	1	1
##	5.163886194	5.165714506	5.168210629	5.169018707	5.169121658	5.170657157
##	1	1	1	1	1	1
##	5.172199041	5.174548155	5.175688509	5.182481752	5.183355631	5.183501765
##	1	1	1	1	1	1
##	5.183987725	5.186500888	5.188559858	5.189317106	5.193753615	5.194197473
##	1	1	1	1	1	1
##	5.19665272	5.197966517	5.200247219	5.205017787	5.205479452	5.208228801
##	1	1	1	1	1	1
##	5.208333333	5.209698558	5.214497041	5.223193741	5.223686116	5.225019069
##	1	1	1	1	1	1
##	5.225693273	5.228870221	5.230870955	5.231143552	5.23255814	5.233611011

##	1	1	1	1	1	1
##	5.234779851	5.235294118	5.235413921	5.236239649	5.246691229	5.248472648
##	1	1	1	1	1	1
##	5.249679898	5.253623188	5.25474016	5.259595862	5.263857504	5.263956311
##	1	1	1	1	1	1
##	5.264732276	5.264795797	5.271431566	5.272822665	5.273833671	5.275186191
##	1	1	1	1	1	1
##	5.275498242	5.278450363	5.280764636	5.281994667	5.282054823	5.282605624
##	1	1	1	1	1	1
##	5.28264626	5.283078379	5.283776996	5.284618101	5.284901395	5.291115354
##	1	1	1	1	1	1
##	5.291214215	5.293976097	5.299539171	5.300353357	5.305574211	5.305999397
##	1	1	1	1	1	1
##	5.31040591	5.312538077	5.314458111	5.315614618	5.317306713	5.317412593
##	1	1	1	1	1	1
##	5.318186956	5.321944809	5.325701036	5.329918283	5.330646143	5.332359386
##	1	1	1	1	1	1
##	5.33869116	5.340223945	5.340297394	5.340540541	5.340817963	5.34181797
##	1	1	1	1	1	1
##	5.344658861	5.349606951	5.34985643	5.350191695	5.354752343	5.354923743
##	1	1	1	1	1	1
##	5.356186395	5.357614146	5.359859567	5.36432284	5.364583333	5.366242744
##	1	1	1	1	1	1
##	5.37037037	5.371555803	5.372694672	5.373276272	5.373953975	5.37428023
##	1	1	1	1	1	1
##	5.375139978	5.379259802	5.380501321	5.382482466	5.383139601	5.384470827
##	1	1	1	1	1	1
##	5.384615385	5.389797883	5.389980977	5.392191187	5.394016013	5.395305384
##	1	1	1	1	1	1
##	5.395462906	5.395731735	5.396074172	5.399151562	5.4001554	5.401029725
##	1	1	1	1	1	1
##	5.404491669	5.408652416	5.410122164	5.410413193	5.414364641	5.416377859
##	1	1	1	1	1	1
##	5.419453329	5.429553265	5.430602081	5.432887856	5.433100881	5.43325019
##	1	1	1	1	1	1
##	5.434122647	5.441478439	5.445266147	5.449538255	5.452638937	5.453791069
##	1	1	1	1	1	1
##	5.455508475	5.457360828	5.46031746	5.461302311	5.461320085	5.465692312
##	1	1	1	1	1	1
##	5.466762526	5.466863951	5.469020422	5.470136014	5.470459519	5.472636816
##	1	1	1	1	1	1
##	5.473379087	5.479735318	5.4866708	5.48894218	5.4900366	5.494453173
##	1	1	1	1	1	1
##	5.499704317	5.499843309	5.501407841	5.501541209	5.501557741	5.503910285
##	1	1	1	1	1	1
##	5.511627907	5.516819572	5.517112399	5.517241379	5.517423442	5.521048999
##	1	1	1	1	1	1
##	5.522288756	5.524669383	5.525568182	5.529393013	5.533430211	5.535521096
##	1	1	1	1	1	1
##	5.539070228	5.539429124	5.540166205	5.540748334	5.541346974	5.541784514
##	1	1	1	1	1	1
##	5.542168675	5.543167508	5.543652919	5.543922993	5.545147996	5.546022599
##	1	1	1	1	1	1
##	5.546875	5.547054323	5.548037889	5.551134489	5.553303608	5.558035714

##	1	1	1	1	1	1
##	5.562130178	5.567338282	5.568862275	5.573419078	5.573940788	5.574778931
##	1	1	1	1	1	1
##	5.57582074	5.577235772	5.586592179	5.587610082	5.591612581	5.593582421
##	1	1	1	1	1	1
##	5.594450511	5.596635235	5.596850638	5.598682663	5.602094241	5.602240896
##	1	1	1	1	1	1
##	5.605176173	5.613126079	5.615292712	5.615628604	5.617495712	5.617591479
##	1	1	1	1	1	1
##	5.618408437	5.620219488	5.621301775	5.621925509	5.625	5.625701056
##	1	1	1	1	1	1
##	5.627462015	5.627676205	5.630397237	5.631220534	5.643104513	5.646027941
##	1	1	1	1	1	1
##	5.647068254	5.647184414	5.64778925	5.65466173	5.65626049	5.662000797
##	1	1	1	1	1	1
##	5.663430421	5.664585192	5.666906226	5.667215815	5.668411867	5.670985089
##	1	1	1	1	1	1
##	5.671766053	5.675616678	5.676241281	5.677852349	5.678233438	5.682006978
##	1	1	1	1	1	1
##	5.682216766	5.682622023	5.683802134	5.684931507	5.687203791	5.68799466
##	1	1	1	1	1	1
##	5.693029274	5.693717277	5.697445973	5.697730565	5.69874932	5.698860228
##	1	1	1	1	1	1
##	5.701401192	5.702647658	5.708508322	5.712144998	5.712260808	5.713596914
##	1	2	1	1	1	1
##	5.713833492	5.713875529	5.713998795	5.717229536	5.718663568	5.719705952
##	1	1	1	1	1	1
##	5.725611099	5.726980992	5.728614138	5.730193853	5.732087227	5.73630137
##	1	1	1	1	1	1
##	5.736951023	5.737356566	5.739464467	5.739599384	5.74048913	5.741042345
##	1	1	1	1	1	1
##	5.74156506	5.742215991	5.74439157	5.744748568	5.746046657	5.746896424
##	1	1	1	1	1	1
##	5.74981682	5.751533742	5.751734773	5.753453598	5.756462494	5.756929638
##	1	1	1	1	1	1
##	5.75990637	5.76001128	5.760869565	5.763286469	5.765351597	5.768189204
##	1	1	1	1	1	1
##	5.769230769	5.771170399	5.772383445	5.775206257	5.778802962	5.782071504
##	1	1	1	1	1	1
##	5.786534285	5.789245447	5.790913336	5.797101449	5.799293999	5.800146592
##	1	1	1	1	1	1
##	5.802028117	5.803003309	5.80474934	5.807872282	5.812641084	5.814864465
##	1	1	1	1	1	1
##	5.818153266	5.822489804	5.823165636	5.828779599	5.83090379	5.831775701
##	1	1	1	1	1	1
##	5.831783426	5.831840933	5.833404093	5.835105126	5.841796248	5.841924399
##	1	1	1	1	1	1
##	5.843469064	5.84433073	5.846508564	5.847110572	5.851979346	5.852417303
##	1	1	1	1	1	1
##	5.856832972	5.858531317	5.860805861	5.866611774	5.87592919	5.876216968
##	1	1	1	1	1	1
##	5.876865672	5.880465757	5.881342381	5.884312202	5.886450714	5.889178996
##	1	1	1	1	1	1
##	5.893845653	5.896980462	5.896994065	5.909161131	5.915444937	5.921976696

##	1	1	1	1	1	1
##	5.925432756	5.925782154	5.925925926	5.928954766	5.929944203	5.930470348
##	1	1	1	1	1	1
##	5.93936457	5.949245101	5.952316445	5.95355245	5.956137583	5.960854093
##	1	1	1	1	1	1
##	5.961538462	5.962622367	5.963302752	5.963398511	5.964154695	5.965766068
##	1	1	1	1	1	1
##	5.968296134	5.969289827	5.971635486	5.975049245	5.975420257	5.98109071
##	1	1	1	1	1	1
##	5.987850738	5.988023952	5.989257845	5.991129949	5.992308937	5.994293642
##	1	1	1	1	1	1
##	6.004383506	6.00624025	6.010723356	6.012145749	6.012884753	6.016466118
##	1	1	1	1	2	1
##	6.018822499	6.021555764	6.021594684	6.021818182	6.027155315	6.028644456
##	1	1	1	1	1	1
##	6.03588907	6.036139946	6.037875061	6.039551042	6.041591968	6.042169307
##	1	1	1	1	1	1
##	6.042296073	6.043704855	6.044769217	6.057735176	6.062213529	6.064227035
##	1	1	1	1	1	1
##	6.066465257	6.068779501	6.072477963	6.07651913	6.077529566	6.078625702
##	1	1	1	1	1	1
##	6.083916084	6.088280061	6.089508437	6.091370558	6.092124814	6.092582124
##	1	1	1	1	1	1
##	6.093113066	6.094022055	6.094222054	6.095143707	6.10380117	6.10521758
##	1	1	1	1	1	1
##	6.110201855	6.11124004	6.112105642	6.125337107	6.12741125	6.128904705
##	1	1	1	1	1	1
##	6.13243835	6.132732926	6.135046611	6.13624545	6.140350877	6.143722718
##	1	1	1	1	2	1
##	6.143948532	6.153360682	6.153846154	6.155507559	6.156408818	6.156510786
##	1	1	1	1	1	1
##	6.157946416	6.161137441	6.165919283	6.17570233	6.176009129	6.177606178
##	1	1	1	1	1	1
##	6.180279523	6.184256155	6.188734455	6.190278355	6.191392604	6.192236599
##	1	1	1	1	1	1
##	6.198748382	6.199261993	6.199460916	6.201122458	6.208251473	6.208291203
##	1	1	1	1	1	1
##	6.210826211	6.211096076	6.213794701	6.215613383	6.216255654	6.218730618
##	1	1	1	1	1	1
##	6.220984215	6.225449895	6.226276831	6.230133503	6.233421751	6.233766234
##	1	1	1	1	1	1
##	6.235912847	6.23870268	6.243007384	6.247570929	6.248990796	6.253360451
##	1	1	1	1	1	1
##	6.25359402	6.255585344	6.260702788	6.266977105	6.270627063	6.274509804
##	1	1	1	1	1	1
##	6.279473815	6.281957633	6.282490875	6.284065406	6.284513403	6.285355123
##	1	1	1	1	1	1
##	6.28798504	6.288032454	6.288258208	6.291165115	6.292364753	6.292374849
##	1	1	1	1	1	1
##	6.296560212	6.298743004	6.301190346	6.307692308	6.308457711	6.309693152
##	1	1	1	1	1	1
##	6.314687286	6.319887594	6.324110672	6.329660834	6.330413359	6.330669945
##	1	1	1	1	1	1
##	6.331168831	6.336972811	6.338028169	6.338625621	6.342209979	6.343283582

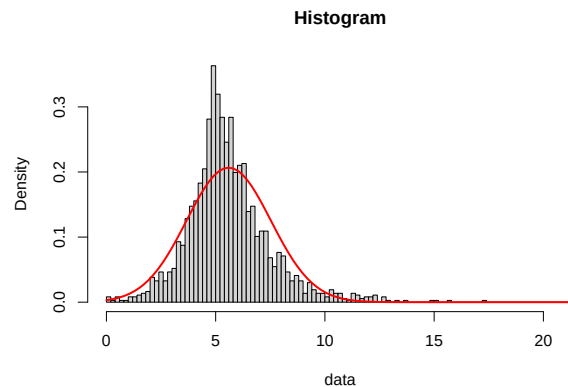
##	1	1	1	1	1	1
##	6.34729556	6.348580442	6.349206349	6.35451505	6.365870002	6.366608963
##	1	1	1	1	1	1
##	6.368497812	6.376628075	6.377617291	6.381381381	6.386292835	6.386701662
##	1	1	1	1	1	1
##	6.387886081	6.389732266	6.390328152	6.393153226	6.393691516	6.394230769
##	1	1	1	1	1	1
##	6.394269037	6.395035078	6.399574544	6.404437721	6.40518172	6.405574323
##	1	1	1	1	1	1
##	6.407674997	6.408629442	6.411140218	6.414172266	6.414219475	6.414368185
##	1	1	1	1	1	1
##	6.41509434	6.416861827	6.417359187	6.420871739	6.422798809	6.423357664
##	1	1	1	1	1	1
##	6.428197261	6.428359096	6.430547019	6.440382942	6.444608704	6.445533358
##	1	1	1	1	1	1
##	6.447299298	6.462372316	6.46314994	6.463878327	6.469002695	6.469648562
##	1	1	1	1	1	1
##	6.476146578	6.483677849	6.487488415	6.491885144	6.495468278	6.505717916
##	1	1	1	1	1	1
##	6.526994359	6.529752501	6.531059017	6.536926148	6.541519012	6.547122602
##	1	1	1	1	1	1
##	6.549831829	6.550868486	6.553245117	6.558627264	6.561704535	6.564217382
##	1	1	1	1	1	1
##	6.575682382	6.579419147	6.593406593	6.593633733	6.59515296	6.598098195
##	1	1	1	1	1	1
##	6.602564103	6.605504587	6.625097428	6.625137665	6.627706717	6.627918833
##	1	1	1	1	1	1
##	6.629834254	6.631716907	6.633380884	6.633574007	6.633919132	6.635885322
##	1	1	1	1	1	1
##	6.637168142	6.638495093	6.643259725	6.645642202	6.653151493	6.659012629
##	1	1	1	1	1	1
##	6.664803689	6.672369547	6.672760512	6.67996012	6.681034483	6.683134349
##	1	1	1	1	1	1
##	6.686006354	6.690267073	6.694613026	6.700733537	6.710020692	6.717676047
##	1	1	1	1	1	1
##	6.720977597	6.724330357	6.724977636	6.74359899	6.744328633	6.744604317
##	1	1	1	1	1	1
##	6.744769874	6.748466258	6.75	6.751054852	6.7516967	6.755829904
##	1	1	1	1	1	1
##	6.761177754	6.763787721	6.768558952	6.768874747	6.77146312	6.774774775
##	1	1	1	1	1	1
##	6.778509815	6.780928638	6.786941581	6.796116505	6.796657382	6.796906797
##	1	1	1	1	1	1
##	6.807405735	6.807727691	6.80980854	6.812933025	6.821562655	6.823375775
##	1	1	1	1	1	1
##	6.825657895	6.82588583	6.831489915	6.838709677	6.844366335	6.850471234
##	1	1	1	1	1	1
##	6.857756326	6.866002215	6.869058571	6.869220608	6.872294372	6.872775801
##	1	1	1	1	1	1
##	6.876456876	6.885245902	6.889676562	6.903026781	6.916426513	6.921416298
##	1	1	1	1	1	1
##	6.924516532	6.93416809	6.938635585	6.943122405	6.944	6.956521739
##	1	1	1	1	1	1
##	6.959459459	6.95970696	6.960408685	6.961613533	6.971510409	6.992367626

##	1	1	1	1	1	1
##	6.993486459	7.013024188	7.016060862	7.01754386	7.0192783	7.020547945
##	1	1	1	1	1	1
##	7.022539698	7.023060797	7.024390244	7.024889509	7.026920032	7.034382214
##	1	1	1	1	1	1
##	7.049608355	7.055749129	7.061503417	7.061933535	7.071713147	7.080078125
##	1	1	1	1	1	1
##	7.08200213	7.088989442	7.092543114	7.098765432	7.104526365	7.1197411
##	1	1	1	1	1	1
##	7.127429806	7.128810226	7.140456989	7.140705032	7.142857143	7.146666667
##	1	1	1	1	3	1
##	7.146724418	7.155147691	7.159837221	7.169811321	7.17472119	7.177836855
##	1	1	1	1	1	1
##	7.18503937	7.18973747	7.197033541	7.205728778	7.207014542	7.209673739
##	1	1	1	1	1	1
##	7.218949743	7.219802888	7.220420251	7.223841918	7.229203118	7.239057239
##	1	1	1	1	1	1
##	7.244318182	7.245912151	7.253229221	7.254261879	7.254779066	7.258064516
##	1	1	1	1	1	1
##	7.258753202	7.265243902	7.27081138	7.272727273	7.273181705	7.28
##	1	1	1	1	1	1
##	7.28003573	7.282913165	7.298720843	7.304204452	7.310838446	7.331771008
##	1	1	1	1	1	1
##	7.333333333	7.342657343	7.352941176	7.357357357	7.366934285	7.368995633
##	1	1	2	1	1	1
##	7.372578522	7.372914241	7.381867511	7.382550336	7.382577604	7.391231312
##	1	1	1	1	1	1
##	7.403380449	7.404550713	7.406593407	7.418397626	7.427277062	7.435158501
##	1	1	1	1	1	1
##	7.437661221	7.439480417	7.441634241	7.446377985	7.467204844	7.48049564
##	1	1	1	1	1	1
##	7.483794932	7.489134069	7.499096494	7.510402923	7.511606314	7.519053578
##	1	1	1	1	1	1
##	7.523638459	7.549668874	7.552083333	7.556742323	7.559938128	7.575757576
##	1	1	1	1	1	1
##	7.598197038	7.603801901	7.608695652	7.615257958	7.638737758	7.641509434
##	1	1	1	1	1	1
##	7.665301346	7.666385847	7.681053402	7.692307692	7.694148618	7.700101317
##	1	1	1	1	1	1
##	7.70252324	7.704581719	7.708225556	7.714016933	7.724034496	7.765151515
##	1	1	1	1	1	1
##	7.78998779	7.791479821	7.799915218	7.800672834	7.80919168	7.810107198
##	1	1	1	1	1	1
##	7.816390603	7.827059262	7.829010567	7.84067086	7.851851852	7.863131656
##	1	1	1	1	1	1
##	7.86717183	7.88863109	7.89058058	7.912885662	7.914410931	7.919234299
##	1	1	1	1	1	1
##	7.936507937	7.938902508	7.939133311	7.947176973	7.948482061	7.955390335
##	1	1	1	1	1	1
##	7.95730228	7.958199357	7.974481659	7.98203179	7.986111111	7.992565056
##	1	1	1	1	1	1
##	7.999080565	8.008741483	8.012820513	8.01384083	8.014184397	8.016032064
##	1	1	1	1	1	1
##	8.019109367	8.022922636	8.043694141	8.043956044	8.048993876	8.05433327

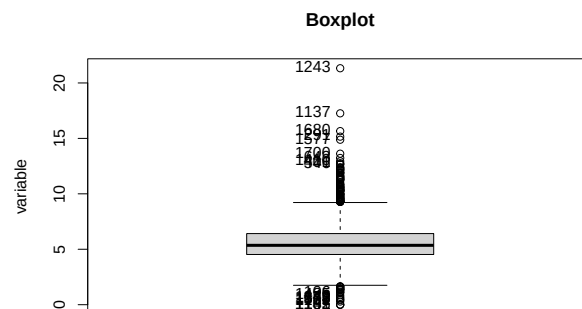
##	1	1	1	1	1	1
##	8.058608059	8.070065687	8.095893103	8.096280088	8.097815193	8.121019108
##	1	1	1	1	1	1
##	8.134642356	8.146487294	8.147459727	8.150028201	8.165019338	8.165882856
##	1	1	1	1	1	1
##	8.186469585	8.189655172	8.198614319	8.205128205	8.208955224	8.209538702
##	1	1	1	1	1	1
##	8.221797323	8.227848101	8.259717314	8.276059565	8.291032149	8.296346876
##	1	1	1	1	1	1
##	8.309915697	8.315256688	8.330673476	8.348623853	8.353221957	8.374384236
##	1	1	1	1	1	1
##	8.382476832	8.388253538	8.42514582	8.433257358	8.451001054	8.463949843
##	1	1	1	1	1	1
##	8.467741935	8.495681864	8.514774495	8.546059933	8.551754776	8.562113933
##	1	1	1	1	1	1
##	8.568548387	8.571428571	8.605012198	8.614426925	8.61618799	8.618283322
##	1	1	1	1	1	1
##	8.620689655	8.62924023	8.634992459	8.672957238	8.686210641	8.704627559
##	1	1	1	1	1	1
##	8.708822416	8.753056235	8.771929825	8.779499824	8.792924037	8.854568855
##	1	1	1	1	1	1
##	8.863294942	8.874172185	8.884297521	8.913784705	8.914100486	8.91959799
##	1	1	1	1	1	1
##	8.929775551	8.958855626	8.967468175	8.977719528	8.986232791	9.004952724
##	1	1	1	1	1	1
##	9.035846724	9.043326548	9.115971329	9.134847753	9.215177413	9.278553175
##	1	1	1	1	1	1
##	9.290030211	9.291187739	9.293394777	9.297604547	9.298780488	9.343715239
##	1	1	1	1	1	1
##	9.370529328	9.375	9.392150771	9.422492401	9.492753623	9.518477044
##	1	1	1	1	1	1
##	9.527108955	9.56284153	9.572250479	9.589041096	9.638554217	9.708737864
##	1	1	1	1	1	1
##	9.725158562	9.777777778	9.781931464	9.86429301	9.93485342	9.935897436
##	1	1	1	1	1	1
##	9.9672438	9.971346705	10.01338091	10.03039514	10.1488498	10.27070064
##	1	1	1	1	1	1
##	10.27131783	10.28708134	10.28806584	10.31227306	10.31353135	10.3626943
##	1	1	1	1	1	1
##	10.43307087	10.43646945	10.47120419	10.5345912	10.58379177	10.60674157
##	1	1	1	1	1	1
##	10.62992126	10.6400665	10.75837743	10.79136691	10.85271318	10.87248322
##	1	1	1	1	1	1
##	11.04133961	11.23809524	11.27029608	11.29568106	11.31436314	11.35038134
##	1	1	1	1	1	1
##	11.43878463	11.46987952	11.51832461	11.55660377	11.63179916	11.7201426
##	1	1	1	1	1	1
##	11.87558907	11.89138577	11.91053624	12.01143947	12.14642263	12.15989251
##	1	1	1	1	1	1
##	12.21973094	12.28389445	12.36017897	12.39291833	12.64845606	12.66766021
##	1	1	1	1	1	1
##	12.76119403	12.93684856	13.23328786	13.62007168	14.87889273	15.15711645
##	1	1	1	1	1	1
##	15.65656566	17.26057906	21.32616487			

```
##           1           1           1
res.var <- summary_and_normality_check(variable)
```

```
## [1] "Summary: "
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##  0.000  4.528   5.355   5.597   6.414   21.326
## [1] "Mean"
## [1] 5.597439
## [1] "Standard deviation"
## [1] 1.930891
```



```
## [1] "Skewness"
## [1] 1.246533
```



```
## [1] "Num Outliers:"
## [1] 104
## [1] "Num. severe outliers:"
## [1] 17
## [1] "Shapiro test: "
##
## Shapiro-Wilk normality test
##
## data: data
## W = 0.93107, p-value < 2.2e-16
##
## [1] "Most repeated value ad times it occurs: "
```

```
## [1] "0"
## [1] 3
outliers_df[res.var,"birthrate"] <- 1
```

Definition of factors

In this section, we convert selected numerical variables into categorical variables (factors). This process enhances the dataset by grouping continuous data into meaningful categories, allowing the modeling process to capture patterns more effectively and potentially improving model fit. By creating factors, we enable the model to identify relationships and trends that might be obscured in the original numerical form.

Note that new factor variables are created both in the training and test sets.

medincome factor

We have created a factor for medincome by dividing it into quartiles, categorizing counties into “Very Low”, “Low”, “Moderate” and “High” income levels. With this factor we’ll try to simplify the analysis and also allows us to capture non-linear relationships between income and the target variable.

```
quantiles <- quantile(df$medincome, probs = c(0, 0.25, 0.5, 0.75, 1))
df$f.income <- cut(
  df$medincome,
  breaks = quantiles,
  labels = c("Very Low", "Low", "Moderate", "High"),
  include.lowest = TRUE
)

quantiles <- quantile(test$medincome, probs = c(0, 0.25, 0.5, 0.75, 1))
test$f.income <- cut(
  test$medincome,
  breaks = quantiles,
  labels = c("Very Low", "Low", "Moderate", "High"),
  include.lowest = TRUE
)
```

popest2015 factor

By defining this factor, we seek to divide the counties in four groups regarding its population. This way we can relate variables with the magnitude of each county.

```
quantiles <- quantile(df$popest2015, probs = c(0, 0.25, 0.5, 0.75, 1))
df$f.population <- cut(
  df$popest2015,
  breaks = quantiles,
  labels = c("Very Low", "Low", "Moderate", "High"),
  include.lowest = TRUE
)

quantiles <- quantile(test$popest2015, probs = c(0, 0.25, 0.5, 0.75, 1))
test$f.population <- cut(
  test$popest2015,
  breaks = quantiles,
  labels = c("Very Low", "Low", "Moderate", "High"),
  include.lowest = TRUE
)
```


geography factor

To potentially enhance the model, we propose extracting the state names from geography and creating a new factor representing states. This new factor could capture regional effects and improve the model's ability to identify patterns related to state-level characteristics.

```
df$f.state <- sapply(strsplit(df$geography, ", "), function(x) x[2])
table(df$f.state)
```

```
##
##      Alabama      Alaska      Arizona      Arkansas      California
##      35          10          8          41          32
##      Colorado    Connecticut    Delaware      Florida      Georgia
##      34          7          1          38          100
##      Hawaii      Idaho      Illinois      Indiana      Iowa
##      2          25          56          56          59
##      Kansas      Kentucky      Louisiana      Maine      Maryland
##      61          75          40          10          14
##      Massachusetts    Michigan    Minnesota    Mississippi    Missouri
##      8          51          51          59          66
##      Montana      Nebraska      Nevada    New Hampshire    New Jersey
##      22          52          14          6          11
##      New Mexico      New York    North Carolina    North Dakota      Ohio
##      20          41          62          32          49
##      Oklahoma      Oregon      Pennsylvania    Rhode Island    South Carolina
##      45          19          42          3          31
##      South Dakota      Tennessee      Texas          Utah          Vermont
##      39          60          136          18          7
##      Virginia      Washington    West Virginia      Wisconsin      Wyoming
##      74          22          33          41          13
```

```
df$f.state <- as.factor(df$f.state)
```

```
test$f.state <- sapply(strsplit(test$geography, ", "), function(x) x[2])
table(test$f.state)
```

```
##
##      Alabama      Alaska      Arizona
##      28          8          7
##      Arkansas      California      Colorado
##      34          25          26
##      Connecticut    Delaware    District of Columbia
##      1          2          1
##      Florida      Georgia      Hawaii
##      28          55          2
##      Idaho      Illinois      Indiana
##      17          46          36
##      Iowa      Kansas      Kentucky
##      40          41          45
##      Louisiana      Maine      Maryland
##      24          6          10
##      Massachusetts    Michigan    Minnesota
##      6          32          36
##      Mississippi      Missouri      Montana
##      23          49          26
##      Nebraska      Nevada      New Hampshire
```

```
##           28           3           4
##       New Jersey       New Mexico       New York
##           10           12           21
##       North Carolina       North Dakota       Ohio
##           37           19           38
##           Oklahoma           Oregon       Pennsylvania
##           32           17           25
##       Rhode Island       South Carolina       South Dakota
##           2           15           20
##       Tennessee           Texas           Utah
##           35           97           9
##           Vermont           Virginia       Washington
##           7           51           17
##       West Virginia       Wisconsin       Wyoming
##           22           31           10
```

```
test$f.state <- as.factor(test$f.state)

combined_levels <- union(levels(df$f.state), levels(test$f.state))
df$f.state <- factor(df$f.state, levels = combined_levels)
test$f.state <- factor(test$f.state, levels = combined_levels)
```

education factor 18_24 factor

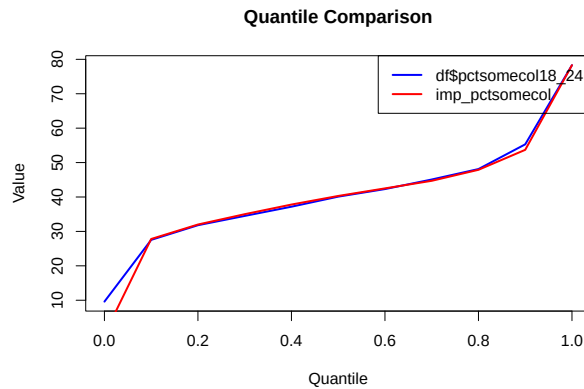
To create the education level factor we first imputed missing values for the `pctsomecol18_24` using its complementary relationship with the other education variables between 18 and 24 years old, ensuring data completeness.

```
miss_ids <- which(is.na(df$pctsomecol18_24))
imp_values <- 100 -
  df$pctnohs18_24[miss_ids] - df$pcths18_24[miss_ids] -
  df$pctbachdeg18_24[miss_ids]
variable <- df$pctsomecol18_24
variable[miss_ids] <- imp_values

quantiles_df <- quantile(df$pctsomecol18_24, probs = seq(0, 1, 0.1), na.rm = TRUE)
quantiles_2 <- quantile(variable, probs = seq(0,1,0.1), na.rm = FALSE)
summary(variable)
```

```
##      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      0.00  33.45   40.30   40.50   46.05   78.30
```

```
plot(seq(0, 1, 0.1), quantiles_df, type = "l", col = "blue", lwd = 2,
     xlab = "Quantile", ylab = "Value", main = "Quantile Comparison")
lines(seq(0, 1, 0.1), quantiles_2, col = "red", lwd = 2);
legend("topright", legend = c("df$pctsomecol18_24", "imp_pctsomecol"),
     col = c("blue", "red"), lwd = 2)
```



```
# add imputed values
df$pctsomecol18_24 <- variable

# Repeat the process for the test set
miss_ids <- which(is.na(test$pctsomecol18_24))
imp_values <- 100 - test$pctnohs18_24[miss_ids] - test$pcths18_24[miss_ids] - test$pctbachdeg18_24[miss_ids]
variable <- test$pctsomecol18_24
variable[miss_ids] <- imp_values

quantiles_df <- quantile(test$pctsomecol18_24, probs = seq(0, 1, 0.1), na.rm = TRUE)
quantiles_2 <- quantile(variable, probs = seq(0,1,0.1), na.rm = FALSE)
summary(variable)

##      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      7.10  33.60   40.50   40.79   46.90   79.00

# add imputed values
test$pctsomecol18_24 <- variable
```

Then we assigned weights to each education level (No HS = 1, HS = 2, Some College = 3, Bachelor = 4) and calculated a weighted average to represent the overall educational attainment for each county. The results were categorized into “Low Education”, “Moderate Education” and “High Education” levels. This factor will help us capturing the impact of educational attainment on the target variable.

```
weights <- c(1,2,3,4)
education_level <- rowSums(df[,c("pctnohs18_24", "pcths18_24",
                                "pctsomecol18_24", "pctbachdeg18_24")] * weights) / 100
df$f.education18_24 <- cut(
  education_level,
  breaks = c(0, 1.5, 2.5, 4),
  labels = c("Low Education", "Moderate Education", "High Education")
)
table(df$f.education18_24)

##
##      Low Education Moderate Education    High Education
##           5           758           1068

education_level <- rowSums(test[,c("pctnohs18_24", "pcths18_24",
                                    "pctsomecol18_24", "pctbachdeg18_24")] * weights) / 100
test$f.education18_24 <- cut(
```

```

education_level,
breaks = c(-Inf, 1.5, 2.5, Inf),
labels = c("Low Education", "Moderate Education", "High Education")
)

```

education factor over-25 factor

The education factor over-25 was created to summarize the educational attainment of county residents aged 25 and over. It is based on the ratio of the percentage of individuals with a bachelor's degree (*pctbachdeg25_over*) to those with only a high school diploma (*pcths25_over*). The factor categorizes counties into three levels: "Mostly High School" for a low prevalence of bachelor's degrees, "Balanced Education" for a similar proportion of both levels, and "Mostly Bachelor's" for counties where bachelor's degree attainment is more common.

```

education_ratio <- df$pctbachdeg25_over / df$pcths25_over

# Categorize based on the ratio
df$f.education25_over <- cut(
  education_ratio,
  breaks = c(-Inf, 0.5, 1.0, Inf),
  labels = c("Mostly High School", "Balanced Eduaction", "Mostly Bachelor's")
)

table(df$f.education25_over)

```

```

##
## Mostly High School Balanced Eduaction Mostly Bachelor's
##           1354           377           100

```

```

#Same for test set
education_ratio <- test$pctbachdeg25_over / test$pcths25_over

# Categorize based on the ratio
test$f.education25_over <- cut(
  education_ratio,
  breaks = c(-Inf, 0.5, 1.0, Inf),
  labels = c("Mostly High School", "Balanced Eduaction", "Mostly Bachelor's")
)

```

unemployment

The unemployment factor categorizes counties into three levels based on the percentage of unemployed residents aged 16 and over. Counties are classified as "Low Unemployment" (<5%), "Moderate Unemployment" (5-10%), or "High Unemployment" (>10%).

```

df$f.unemployment <- cut(
  df$pctunemployed16_over,
  breaks = c(-Inf, 5, 10, Inf),
  labels = c("Low Unemployment", "Moderate Unemployment", "High Unemployment"),
  right = FALSE
)

table(df$f.unemployment)

```

```

##
## Low Unemployment Moderate Unemployment High Unemployment
##           346           1054           431

```

```
#Same for test set:
test$f.unemployment <- cut(
  test$pctunemployed16_over,
  breaks = c(-Inf, 5, 10, Inf),
  labels = c("Low Unemployment", "Moderate Unemployment", "High Unemployment"),
  right = FALSE
)
```

Data Quality Report

This section provides a comprehensive overview of the dataset's quality, complementing the prior identification of univariate outliers and missing values with a visual summary to highlight key patterns and anomalies. The data quality report is structured into two parts: an analysis regarding the variables and a review of individual records (stating the number of outliers per individual and identifying multivariate outliers).

Variables

This first part of the data quality report assesses the quality of the data regarding the variables.

Missing Values and Errors. The dataset originally contained missing values in three variables: *pctemployed16_over* (82 missing values), *pctprivatecoveragealone* (356 missing values), and *pctsomecol18_24* (1376 missing values). To address this, imputation methods will be applied to fill these gaps. Specifically, *pctsomecol18_24* values were calculated based on its numerical relationship with complementary variables *pctnohs18_24*, *pcths18_24*, and *pctbachdeg18_24*, ensuring the integrity of the data prior to creating the *education_level* factor.

For *pctemployed16_over* and *pctprivatecoveragealone*, missing values will be imputed using the *imputePCA* method to preserve variable distributions.

Additionally, we found 126 erroneous values in *avganncount* and *incidencerate* across the same observations. These values were replaced with NA to avoid introducing bias, ensuring a more reliable analysis.

All missing and erroneous values will be imputed in a manner that maintains the original characteristics of the dataset, supporting robust and unbiased modeling.

Outliers. Outliers were thoroughly examined to identify variables with the highest counts of mild and severe cases, as these can distort data distribution and impact model performance. As stated in the statement, all outliers are retained.

The table below highlights the top 5 variables ranked by the number of outliers in each category. This analysis aids in determining variables that may require transformation or additional handling to mitigate their influence.

Rank	Mild Outliers Variable	Count	Severe Outliers Variable	Count
1	popest2015	252	popest2015	162
2	avgdeathsperyear	225	avgdeathsperyear	138
3	pctblack	224	avganncount	131
4	avganncount	207	pctasian	107
5	pctasian	198	pctotherrace	102

This ranking reveals that *pctprivatecoveragealone* stands out as having the most outliers in both mild and severe categories. Variables such as *popest2015* and *avgdeathsperyear*, *pctblack*, *avganncount*, and *pctasian* also exhibit significant outlier counts.

Individuals

This second part of the data quality report assesses the number of outliers and missing values per individual and identifies which individuals are multivariate outliers.

Number of missing values across individuals Out of the total observations in the dataset, 1306 have no missing values across any variable. Additionally, 371 observations contain a missing value in one variable, 128 observations have missing values in two variables, 23 observations have missing values in three variables, and 3 individuals have missing values in 4 variables(“Kiowa County, Kansas”, “Meade County, Kansas”, and “Dakota County, Minnesota”).

```
#Number of missing values
na_counts <- rowSums(is.na(df))
table(na_counts)
```

```
## na_counts
##      0      1      2      3      4
## 1306   371   128    23      3
```

```
ll<- which(rowSums(is.na(df))==4); length(ll)
```

```
## [1] 3
```

```
df$geography[ll]
```

```
## [1] "Kiowa County, Kansas"      "Meade County, Kansas"
## [3] "Dakota County, Minnesota"
```

Number of outliers across individuals Out of all observations, the dataset contains a distribution of outliers across various categories. Specifically, there are 982 observations without any outliers, 343 observations with one outlier, 146 observations with two outliers, 130 observations with three outliers, 93 observations with four outliers, 68 observations with five outliers, 25 observations with six outliers, 21 observations with seven outliers, 15 observations with eight outliers, and 4 observations with nine and ten outliers. The individuals with more outliers are “Loudoun County, Virginia”, “Williamsburg city, Virginia”, “Santa Clara County, California”, and “Hamilton County, Indiana”

```
#Number of outliers per individual
outliers_counts <- rowSums(outliers_df)
table(outliers_counts)
```

```
## outliers_counts
##      0      1      2      3      4      5      6      7      8      9     10
## 982 343 146 130  93  68  25  21  15   4   4
```

```
ll<- which(rowSums(outliers_df)==10); length(ll)
```

```
## [1] 4
```

```
df$geography[ll]
```

```
## [1] "Loudoun County, Virginia"      "Williamsburg city, Virginia"
## [3] "Santa Clara County, California" "Hamilton County, Indiana"
```

Multivariate outliers The *find_valid_moutlier* function identifies the largest combination of numeric variables without missing values that successfully executes the Moutlier function. By iterating through combinations of variables, starting with the largest possible subsets, this function ensures that the process for detecting multivariate outliers incorporates the maximum amount of information available.

```

find_valid_moutlier <- function(df, quantile = 0.95) {
  # Select numeric columns without NA
  valid_numeric_columns <- which(sapply(df, is.numeric)
                                & sapply(df, function(col) all(!is.na(col))))

  # Track total number of variables
  num_vars <- length(valid_numeric_columns)
  cat("Total numeric variables without NAs:", num_vars, "\n")

  # Start testing combinations from the largest size down to 1
  for (k in num_vars:1) {
    cat("Testing combinations of size", k, "\n")

    # Generate all combinations of size k
    combo_indexes <- combn(valid_numeric_columns, k, simplify = FALSE)
    total_combos <- length(combo_indexes)
    cat("Total combinations of size", k, ":", total_combos, "\n")

    for (i in seq_along(combo_indexes)) {
      combo <- combo_indexes[[i]]
      combo_names <- names(df)[combo]

      # Attempt Moutlier function
      result <- try({
        res_mout <- Moutlier(df[, combo], quantile = quantile)

        # If successful, return the result immediately
        cat("Valid combination found! Indexes:",
            paste(combo, collapse = ", "), "\n")
        cat("Variable names:", paste(combo_names, collapse = ", "), "\n")

        multoutliers <- which((res_mout$md > res_mout$cutoff)
                              & (res_mout$rd > res_mout$cutoff))
        cat("Number of multivariate outliers:", length(multoutliers), "\n")

        return(list(res_mout = res_mout, variables = combo_names,
                    multoutliers = multoutliers))
      }, silent = TRUE)

      # Check for errors and move to the next combination
      if (inherits(result, "try-error")) {
        cat("Combination failed.\n")
      } else {
        return(result) # Stop at the first valid combination
      }
    }
  }

  # If no valid combination is found
  cat("No valid combination found.\n")
  return(NULL)
}

```

Since the Moutlier function depends on some stochastic processes, one can get different results when executing

it several times. Thus, we comment the results obtained in one single execution, which has been selected carefully to represent the average values.

We found 144 multivariate outliers, with the following IDs: 12, 18, 62, 70, 71, 73, 76, 94, 98, 101, 103, 104, 106, 110, 113, 123, 128, 162, 168, 218, 226, 229, 237, 254, 279, 284, 291, 294, 301, 364, 368, 372, 411, 434, 466, 474, 506, 527, 529, 540, 558, 562, 574, 588, 600, 603, 608, 613, 615, 623, 632, 634, 636, 652, 654, 719, 720, 734, 753, 784, 786, 793, 815, 817, 844, 847, 864, 883, 890, 892, 899, 902, 904, 934, 960, 971, 1016, 1046, 1091, 1093, 1094, 1095, 1096, 1112, 1113, 1120, 1122, 1124, 1135, 1137, 1139, 1144, 1154, 1157, 1160, 1168, 1171, 1180, 1183, 1190, 1223, 1227, 1238, 1243, 1246, 1247, 1388, 1389, 1395, 1398, 1411, 1413, 1420, 1468, 1496, 1522, 1525, 1547, 1559, 1577, 1599, 1607, 1609, 1615, 1617, 1623, 1633, 1636, 1637, 1639, 1641, 1643, 1648, 1669, 1670, 1675, 1679, 1680, 1681, 1689, 1692, 1700, 1736, 1794.

```
result <- find_valid_moutlier(df, quantile = 0.95)
```

```
## Total numeric variables without NAs: 25
## Testing combinations of size 25
## Total combinations of size 25 : 1

## Warning in covMcd(X): The covariance matrix has become singular during
## the iterations of the MCD algorithm.
## There are 1700 observations (in the entire dataset of 1831 obs.) lying
## on the hyperplane with equation  $a_1(x_{i1} - m_1) + \dots + a_p(x_{ip} -$ 
##  $m_p) = 0$  with  $(m_1, \dots, m_p)$  the mean of these observations and
## coefficients  $a_i$  from the vector  $a \leftarrow c(0, 0, 0, 0, 0, 0, 0, 0, -0.5,$ 
##  $-0.5, -0.5, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0)$ 

## Combination failed.
## Testing combinations of size 24
## Total combinations of size 24 : 25

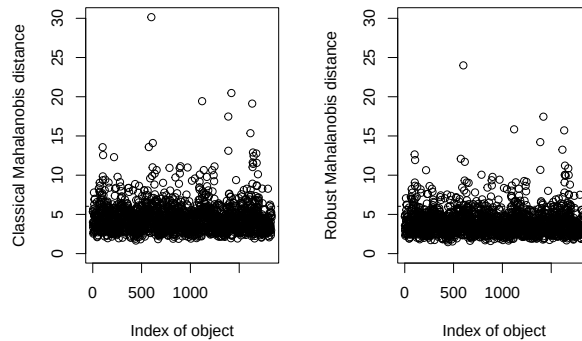
## Warning in covMcd(X): The covariance matrix has become singular during
## the iterations of the MCD algorithm.
## There are 1700 observations (in the entire dataset of 1831 obs.) lying
## on the hyperplane with equation  $a_1(x_{i1} - m_1) + \dots + a_p(x_{ip} -$ 
##  $m_p) = 0$  with  $(m_1, \dots, m_p)$  the mean of these observations and
## coefficients  $a_i$  from the vector  $a \leftarrow c(0, 0, 0, 0, 0, 0, 0, 0, -0.5,$ 
##  $-0.5, -0.5, -0.5, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0)$ 

## Combination failed.

## Warning in covMcd(X): The covariance matrix has become singular during
## the iterations of the MCD algorithm.
## There are 1700 observations (in the entire dataset of 1831 obs.) lying
## on the hyperplane with equation  $a_1(x_{i1} - m_1) + \dots + a_p(x_{ip} -$ 
##  $m_p) = 0$  with  $(m_1, \dots, m_p)$  the mean of these observations and
## coefficients  $a_i$  from the vector  $a \leftarrow c(0, 0, 0, 0, 0, 0, 0, 0, 0.5,$ 
##  $0.5, 0.5, 0.5, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0)$ 

## Combination failed.

## Warning in covMcd(X): The covariance matrix has become singular during
## the iterations of the MCD algorithm.
## There are 1463 observations (in the entire dataset of 1831 obs.) lying
## on the hyperplane with equation  $a_1(x_{i1} - m_1) + \dots + a_p(x_{ip} -$ 
##  $m_p) = 0$  with  $(m_1, \dots, m_p)$  the mean of these observations and
## coefficients  $a_i$  from the vector  $a \leftarrow c(0, 0, 0, 0, 0, 0, 1e-07, -1e-07,$ 
##  $0, -0.5, -0.5, -0.5, -0.4999999, 0, 0, -1e-07, -1e-07, 0, 0, -1e-07, 0,$ 
##  $0, 1e-07, 0, 0)$ 
```

```
## Valid combination found! Indexes: 2, 3, 5, 6, 7, 10, 11, 13, 14, 15, 16, 17, 18, 19, 21, 22, 24, 25,
## Variable names: avgdeathsperyear, target_deathrate, medincome, popest2015, povertypercent, medianage
## Number of multivariate outliers: 147
```

```
res.out <- result$res_mout;
mult.outliers <- result$multoutliers; length(mult.outliers)
```

```
## [1] 147
```

```
# Plot Multivariate Outliers
dev.off()
```

```
## null device
##      1
```

```
plot(res.out$md, res.out$rd)
abline(h=res.out$cutoff, col="red", lty=2)
abline(v=res.out$cutoff, col="red", lty=2)

str(df)
```

```
## 'data.frame':  1831 obs. of  38 variables:
## $ avganncount      : num  57 428 146 4025 113 ...
## $ avgdeathsperyear : num  26 152 71 1380 36 ...
## $ target_deathrate  : num  144 176 184 178 121 ...
## $ incidencerate     : num  350 505 404 511 413 ...
## $ medincome         : num  49955 52313 40189 60397 54721 ...
## $ popest2015        : num  10321 61023 20848 843954 16252 ...
## $ povertypercent    : num  12.5 15.6 17.8 13.1 12.7 15.7 12.6 9.9 19.3 15.7 ...
## $ binnedinc         : chr  "(48021.6, 51046.4]" "(51046.4, 54545.6]" "(37413.8, 40362.7]" "(54
## $ medianage         : num  48.3 45.4 51.7 35.8 54.4 41 45.2 37.6 46.2 36.9 ...
## $ medianagemale     : num  47.8 43.5 50.8 34.7 54 40 44.9 36.6 45.6 35.5 ...
## $ medianagefemale   : num  48.9 48 52.5 37 54.6 42.2 45.5 38.7 46.8 39.1 ...
## $ geography         : chr  "Lincoln County, Washington" "Mason County, Washington" "Pacific Co
## $ percentmarried    : num  57.8 50.4 52.7 50 56.8 53.6 54.4 52.1 57.2 46.5 ...
## $ pctnohs18_24      : num  14.9 29.9 27.3 15.6 17.7 25.5 20 15.4 25.1 13.7 ...
## $ pcths18_24        : num  43 35.1 33.9 36.3 32.4 33.8 43.8 33.3 35.3 29.2 ...
## $ pctsomecol18_24   : num  40 30.5 36.5 41 44.7 37.6 33.8 43 35.7 51.5 ...
## $ pctbachdeg18_24   : num  2 4.5 2.2 7.1 5.2 3.1 2.4 8.3 3.9 5.6 ...
## $ pcths25_over      : num  33.4 30.4 31.6 28.8 17.2 26.7 29.2 24.3 34.4 22.6 ...
## $ pctbachdeg25_over : num  15 11.9 11.3 16.2 26.2 15.9 14.2 20.9 10.7 16.4 ...
## $ pctemployed16_over : num  48.2 44.1 40.9 56.6 54.6 NA 51.5 62.1 46.3 54.5 ...
```

```
## $ pctunemployed16_over : num 4.8 12.9 8.9 9.2 5.9 8.2 8.3 7.5 9.4 6.4 ...
## $ pctprivatecoverage : num 61.6 60 55.8 69.9 67.2 64.4 64.4 73.3 58.3 65.7 ...
## $ pctprivatecoveragealone: num 43.9 38.8 33.1 NA NA NA 49.7 61.6 39.3 47.6 ...
## $ pctempprivcoverage : num 35.1 32.6 25.9 44.4 27.9 38 42.6 54.3 33.9 40 ...
## $ pctpubliccoverage : num 44 43.2 50.9 31.4 41.6 38.1 36.1 25.9 45.8 38 ...
## $ pctpubliccoveragealone : num 22.7 20.2 24.1 16.5 18.3 20.2 20.5 14.1 24.1 18.8 ...
## $ pctwhite : num 94.1 84.9 89.4 74.7 92.6 ...
## $ pctblack : num 0.27 1.653 0.305 6.711 0.652 ...
## $ pctasian : num 0.666 1.538 1.889 6.041 1.429 ...
## $ pctotherrace : num 0.492 3.315 2.286 2.699 2.237 ...
## $ pctmarriedhouseholds : num 54 51.2 49 50.1 50 ...
## $ birthrate : num 6.8 4.96 5.89 5.53 4.59 ...
## $ f.income : Factor w/ 4 levels "Very Low","Low",...: 3 3 2 4 4 3 4 4 2 3 ...
## $ f.population : Factor w/ 4 levels "Very Low","Low",...: 1 3 2 4 2 4 1 4 3 3 ...
## $ f.state : Factor w/ 51 levels "Alabama","Alaska",...: 47 47 47 47 47 47 47 47 47 47 ...
## $ f.education18_24 : Factor w/ 3 levels "Low Education",...: 3 2 2 3 3 2 2 3 3 3 ...
## $ f.education25_over : Factor w/ 3 levels "Mostly High School",...: 1 1 1 2 3 2 1 2 1 2 ...
## $ f.unemployment : Factor w/ 3 levels "Low Unemployment",...: 1 3 2 2 2 2 2 2 2 2 ...
```

Finally, we present the multivariate outlier profiling:

The chi-square test results reveal significant relationships between the multivariate outliers and the categorical variables *f.state*, *f.population*, *f.education25_over*, *f.unemployment*, *f.income*.

Multivariate outliers are more frequently associated with very low population sizes, higher proportions of bachelor's degrees among adults, and states like Texas, Alaska, New Mexico, and South Dakota. These outliers are also associated with extremes in unemployment levels and income categories, highlighting their distinct demographic and socioeconomic characteristics.

The analysis of quantitative variables underscores the importance of several key factors in distinguishing outliers. The Eta-squared values reveal strong associations between outlier status and variables. As expected, multivariate outliers are highly associated with the number of univariate outliers (*n_uout*). Moreover, they are also globally associated with racial composition variables like *pctotherrace*, *pctasian*, *pctwhite* and population estimates (*popest2015*). Other significant variables include average annual case counts (*avganncount*), poverty percentage, and high school education levels. These variables exhibit notable differences between outlier and non-outlier groups, underscoring their explanatory power.

A closer look at the profiles of quantitative variables by cluster reveals distinct patterns. Multivariate outliers are characterized by higher percentages of Asian and Other(not white, black, asian) race populations, lower percentages of White residents, higher poverty levels, larger average annual counts, and younger median ages.

Statistically significant differences between clusters confirm the importance of the identified variables in distinguishing multivariate outliers. Variables related to population demographics, education levels, and racial composition show the most pronounced separations between clusters.

```
bi_mvout_df <- df
#Binary factor if its a multivariate outlier
bi_mvout_df$mout <- 0
bi_mvout_df[mult.outliers, "mout"] <- 1
bi_mvout_df$mout <- factor(bi_mvout_df$mout, labels=c("Not M.O", "Yes M.O"))
table(bi_mvout_df$mout)
```

```
##
## Not M.O Yes M.O
##      1684      147
```

```
#Number of univariate outlier variables for a given observation
bi_mvout_df$n_uout <- rowSums(outliers_df)
```

```
#Number of missing values in variables for a given observation
bi_mvout_df$n_miss <- rowSums(is.na(df))
```

```
summary(bi_mvout_df[, c("mout", "n_uout", "n_miss")]);
```

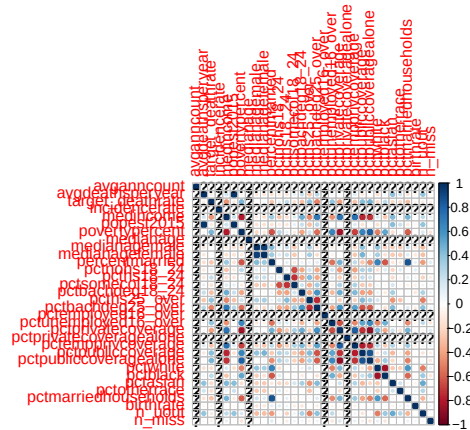
```
##           mout           n_uout           n_miss
## Not M.O:1684   Min.    : 0.000   Min.    :0.0000
## Yes M.O: 147   1st Qu.: 0.000   1st Qu.:0.0000
##               Median : 0.000   Median :0.0000
##               Mean    : 1.218   Mean    :0.3867
##               3rd Qu.: 2.000   3rd Qu.:1.0000
##               Max.    :10.000   Max.    :4.0000
```

```
sd(bi_mvout_df$n_uout); sd(bi_mvout_df$n_miss)
```

```
## [1] 1.83798
```

```
## [1] 0.6872114
```

```
num_vars <- bi_mvout_df[apply(bi_mvout_df, is.numeric)]
corrplot(cor(num_vars))
```



```
catdes(bi_mvout_df, num.var=39, proba=0.05)
```

```
##
## Link between the cluster variable and the categorical variables (chi-square test)
## =====
##           p.value df
## f.state      5.755707e-22 49
## f.population  7.521119e-17  3
## f.education25_over 1.743769e-11  2
## f.unemployment  3.899944e-05  2
## f.income      4.440652e-03  3
## binnedinc     7.322310e-03  9
##
## Description of each cluster by the categories
## =====
## $`Not M.O`
##           Cla/Mod   Mod/Cla   Global
## f.population=Moderate 97.59300 26.4845606 24.9590388
## f.education25_over=Mostly High School 93.94387 75.5344418 73.9486619
## f.unemployment=Moderate Unemployment 94.30740 59.0261283 57.5641726
```

## f.population=Low	95.63319	26.0095012	25.0136537
## f.income=Moderate	95.40481	25.8907363	24.9590388
## f.state=Indiana	100.00000	3.3254157	3.0584380
## binnedinc=(48021.6, 51046.4]	96.41026	11.1638955	10.6499181
## f.state=Oklahoma	100.00000	2.6722090	2.4576734
## f.state=Wisconsin	100.00000	2.4346793	2.2392135
## f.state=Arkansas	100.00000	2.4346793	2.2392135
## f.state=North Carolina	98.38710	3.6223278	3.3861278
## f.education18_24=High Education	93.07116	59.0261283	58.3287821
## f.state=Iowa	98.30508	3.4441805	3.2222829
## f.state=California	81.25000	1.5439430	1.7476789
## binnedinc=[22640, 34218.1]	88.14433	10.1543943	10.5953031
## f.education25_over=Balanced Eduaction	89.38992	20.0118765	20.5898416
## f.state=Nevada	71.42857	0.5938242	0.7646095
## f.income=High	89.30131	24.2874109	25.0136537
## f.state=North Dakota	78.12500	1.4845606	1.7476789
## f.state=Montana	72.72727	0.9501188	1.2015292
## f.state=Hawaii	0.00000	0.0000000	0.1092299
## binnedinc=(61494.5, 125635]	85.48387	9.4418052	10.1583834
## f.state=South Dakota	74.35897	1.7220903	2.1299836
## f.unemployment=Low Unemployment	87.28324	17.9334917	18.8967777
## f.state=New Mexico	65.00000	0.7719715	1.0922993
## f.state=Texas	82.35294	6.6508314	7.4276352
## f.state=Alaska	40.00000	0.2375297	0.5461496
## f.education25_over=Mostly Bachelor's	75.00000	4.4536817	5.4614965
## f.population=Very Low	82.96943	22.5653207	25.0136537
##	p.value	v.test	
## f.population=Moderate	1.719040e-08	5.638133	
## f.education25_over=Mostly High School	6.776195e-07	4.967671	
## f.unemployment=Moderate Unemployment	2.228686e-05	4.240655	
## f.population=Low	4.718878e-04	3.496227	
## f.income=Moderate	1.123965e-03	3.257504	
## f.state=Indiana	8.551364e-03	2.629488	
## binnedinc=(48021.6, 51046.4]	9.758351e-03	2.584277	
## f.state=Oklahoma	2.205806e-02	2.289367	
## f.state=Wisconsin	3.108563e-02	2.155975	
## f.state=Arkansas	3.108563e-02	2.155975	
## f.state=North Carolina	3.873820e-02	2.066957	
## f.education18_24=High Education	4.222144e-02	2.031331	
## f.state=Iowa	4.838102e-02	1.974006	
## f.state=California	4.952258e-02	-1.964065	
## binnedinc=[22640, 34218.1]	4.809015e-02	-1.976571	
## f.education25_over=Balanced Eduaction	4.431931e-02	-2.011058	
## f.state=Nevada	2.457263e-02	-2.248056	
## f.income=High	1.821907e-02	-2.361136	
## f.state=North Dakota	1.417598e-02	-2.452773	
## f.state=Montana	7.264423e-03	-2.684473	
## f.state=Hawaii	6.405171e-03	-2.726285	
## binnedinc=(61494.5, 125635]	1.583543e-03	-3.158921	
## f.state=South Dakota	8.275736e-04	-3.343404	
## f.unemployment=Low Unemployment	7.432482e-04	-3.373108	
## f.state=New Mexico	6.715601e-04	-3.400934	
## f.state=Texas	1.428366e-04	-3.803204	
## f.state=Alaska	4.072251e-05	-4.103342	

```

## f.education25_over=Mostly Bachelor's 9.858148e-08 -5.329319
## f.population=Very Low 2.774055e-14 -7.608449
##
## $`Yes M.O`
##
## Cla/Mod Mod/Cla Global
## f.population=Very Low 17.030568 53.0612245 25.0136537
## f.education25_over=Mostly Bachelor's 25.000000 17.0068027 5.4614965
## f.state=Alaska 60.000000 4.0816327 0.5461496
## f.state=Texas 17.647059 16.3265306 7.4276352
## f.state=New Mexico 35.000000 4.7619048 1.0922993
## f.unemployment=Low Unemployment 12.716763 29.9319728 18.8967777
## f.state=South Dakota 25.641026 6.8027211 2.1299836
## binnedinc=(61494.5, 125635] 14.516129 18.3673469 10.1583834
## f.state=Hawaii 100.000000 1.3605442 0.1092299
## f.state=Montana 27.272727 4.0816327 1.2015292
## f.state=North Dakota 21.875000 4.7619048 1.7476789
## f.income=High 10.698690 33.3333333 25.0136537
## f.state=Nevada 28.571429 2.7210884 0.7646095
## f.education25_over=Balanced Eduaction 10.610080 27.2108844 20.5898416
## binnedinc=[22640, 34218.1] 11.855670 15.6462585 10.5953031
## f.state=California 18.750000 4.0816327 1.7476789
## f.state=Iowa 1.694915 0.6802721 3.2222829
## f.education18_24=High Education 6.928839 50.3401361 58.3287821
## f.state=North Carolina 1.612903 0.6802721 3.3861278
## f.state=Wisconsin 0.000000 0.0000000 2.2392135
## f.state=Arkansas 0.000000 0.0000000 2.2392135
## f.state=Oklahoma 0.000000 0.0000000 2.4576734
## binnedinc=(48021.6, 51046.4] 3.589744 4.7619048 10.6499181
## f.state=Indiana 0.000000 0.0000000 3.0584380
## f.income=Moderate 4.595186 14.2857143 24.9590388
## f.population=Low 4.366812 13.6054422 25.0136537
## f.unemployment=Moderate Unemployment 5.692600 40.8163265 57.5641726
## f.education25_over=Mostly High School 6.056130 55.7823129 73.9486619
## f.population=Moderate 2.407002 7.4829932 24.9590388
##
## p.value v.test
## f.population=Very Low 2.774055e-14 7.608449
## f.education25_over=Mostly Bachelor's 9.858148e-08 5.329319
## f.state=Alaska 4.072251e-05 4.103342
## f.state=Texas 1.428366e-04 3.803204
## f.state=New Mexico 6.715601e-04 3.400934
## f.unemployment=Low Unemployment 7.432482e-04 3.373108
## f.state=South Dakota 8.275736e-04 3.343404
## binnedinc=(61494.5, 125635] 1.583543e-03 3.158921
## f.state=Hawaii 6.405171e-03 2.726285
## f.state=Montana 7.264423e-03 2.684473
## f.state=North Dakota 1.417598e-02 2.452773
## f.income=High 1.821907e-02 2.361136
## f.state=Nevada 2.457263e-02 2.248056
## f.education25_over=Balanced Eduaction 4.431931e-02 2.011058
## binnedinc=[22640, 34218.1] 4.809015e-02 1.976571
## f.state=California 4.952258e-02 1.964065
## f.state=Iowa 4.838102e-02 -1.974006
## f.education18_24=High Education 4.222144e-02 -2.031331
## f.state=North Carolina 3.873820e-02 -2.066957

```

```
## f.state=Wisconsin          3.108563e-02 -2.155975
## f.state=Arkansas          3.108563e-02 -2.155975
## f.state=Oklahoma          2.205806e-02 -2.289367
## binnedinc=(48021.6, 51046.4] 9.758351e-03 -2.584277
## f.state=Indiana            8.551364e-03 -2.629488
## f.income=Moderate          1.123965e-03 -3.257504
## f.population=Low           4.718878e-04 -3.496227
## f.unemployment=Moderate Unemployment 2.228686e-05 -4.240655
## f.education25_over=Mostly High School 6.776195e-07 -4.967671
## f.population=Moderate      1.719040e-08 -5.638133
```

```
##
##
```

```
## Link between the cluster variable and the quantitative variables
```

```
## =====
```

```
##              Eta2      P-value
```

```
## n_uout          0.153313874 3.791904e-68
## pctasian        0.073091457 4.910367e-32
## pctotherrace    0.048007822 2.430867e-21
## avganncount     0.048449762 3.744281e-20
## popest2015      0.043365210 2.185208e-19
## pcths25_over    0.041615192 1.186055e-18
## avgdeathsperyear 0.040268135 4.354212e-18
## pctwhite        0.036332179 1.932492e-16
## percentmarried  0.017320818 1.583045e-08
## pctbachdeg18_24 0.015964274 5.808697e-08
## pctbachdeg25_over 0.015299502 1.098940e-07
## pctempprivcoverage 0.012194498 2.174085e-06
## birthrate       0.011658703 3.644683e-06
## medianagefemale 0.010330444 1.315800e-05
## pctnohs18_24    0.009645324 2.556414e-05
## medianage       0.009207136 4.271565e-05
## pctsomocol18_24 0.008409824 8.505864e-05
## target_deathrate 0.007461924 2.149929e-04
## medianagemale   0.007364931 2.364607e-04
## pctprivatecoverage 0.007259613 2.622232e-04
## incidencerate   0.007376256 3.847938e-04
## povertypercent  0.006709523 4.506085e-04
## medincome       0.006317458 6.637458e-04
## pctmarriedhouseholds 0.005573139 1.390020e-03
## pctpubliccoverage 0.003849613 7.915018e-03
```

```
##
```

```
## Description of each cluster by quantitative variables
```

```
## =====
```

```
## $`Not M.O`
```

```
##              v.test Mean in category Overall mean sd in category
## pcths25_over    8.726729          35.158314 3.473255e+01 6.754396e+00
## pctwhite        8.154011          84.749893 8.384699e+01 1.498047e+01
## percentmarried  5.630017          52.162886 5.190087e+01 6.274106e+00
## pctempprivcoverage 4.723974          41.604691 4.129388e+01 9.092963e+00
## medianagefemale 4.347955          42.325297 4.216931e+01 4.844565e+00
## medianage       4.111581          40.996461 4.085063e+01 4.805576e+00
## pctsomocol18_24 3.923006          40.786520 4.049574e+01 1.033856e+01
## target_deathrate 3.695311          179.532304 1.788197e+02 2.592243e+01
## incidencerate   3.675726          450.123342 4.486551e+02 5.025101e+01
```

```

## medianagemale      3.671215      39.717162 3.958564e+01 4.836604e+00
## pctprivatecoverage  3.644872      64.742280 6.447460e+01 9.964230e+00
## pctmarriedhouseholds 3.193563      51.544564 5.140232e+01 6.074045e+00
## pctpubliccoverage   2.654203      36.296675 3.615309e+01 7.510317e+00
## medincome          -3.400140      46988.199525 4.727794e+04 1.130869e+04
## povertypercent     -3.504059      16.636639 1.679175e+01 6.061278e+00
## pctnohs18_24       -4.201302      18.058017 1.828777e+01 7.184115e+00
## birthrate          -4.619029      5.535857 5.597439e+00 1.728539e+00
## pctbachdeg25_over  -5.291322      13.097684 1.329754e+01 5.119765e+00
## pctbachdeg18_24    -5.405055      6.047625 6.216221e+00 3.883363e+00
## avgdeathsperyear   -8.584328      157.727435 1.916237e+02 2.893479e+02
## popest2015         -8.908329      83683.343230 1.068412e+05 1.751841e+05
## pctotherrace       -9.373063      1.765437 2.003134e+00 2.958972e+00
## avganncount        -9.420436      420.053571 5.242393e+02 7.860380e+02
## pctasian          -11.565352      1.067569 1.274301e+00 1.541322e+00
## n_uout             -16.750056      1.005344 1.217914e+00 1.588063e+00
##
## Overall sd      p.value
## pcths25_over    7.064060e+00 2.621438e-18
## pctwhite        1.603270e+01 3.520502e-16
## percentmarried  6.738289e+00 1.801914e-08
## pctempprivcoverage 9.526266e+00 2.312798e-06
## medianagefemale  5.194588e+00 1.374129e-05
## medianage       5.135293e+00 3.929593e-05
## pctsomecol18_24  1.073207e+01 8.745087e-05
## target_deathrate 2.792279e+01 2.196181e-04
## incidencerate    5.783627e+01 2.371737e-04
## medianagemale    5.187250e+00 2.413997e-04
## pctprivatecoverage 1.063324e+01 2.675249e-04
## pctmarriedhouseholds 6.448935e+00 1.405287e-03
## pctpubliccoverage 7.832935e+00 7.949604e-03
## medincome        1.233832e+04 6.735147e-04
## povertypercent    6.409410e+00 4.582243e-04
## pctnohs18_24     7.917877e+00 2.653837e-05
## birthrate        1.930363e+00 3.855404e-06
## pctbachdeg25_over 5.468847e+00 1.214353e-07
## pctbachdeg18_24  4.516322e+00 6.478842e-08
## avgdeathsperyear 5.717202e+02 9.136892e-18
## popest2015       3.763912e+05 5.180725e-19
## pctotherrace     3.671806e+00 7.045736e-21
## avganncount      1.601309e+03 4.492193e-21
## pctasian         2.588138e+00 6.173759e-31
## n_uout           1.837478e+00 5.657262e-63
##
## $`Yes M.O`
##
## v.test Mean in category Overall mean sd in category
## n_uout          16.750056      3.653061e+00 1.217914e+00 2.591596e+00
## pctasian        11.565352      3.642581e+00 1.274301e+00 7.079617e+00
## avganncount      9.411779      1.716672e+03 5.242393e+02 4.826423e+03
## pctotherrace     9.373063      4.726137e+00 2.003134e+00 7.718002e+00
## popest2015       8.908329      3.721322e+05 1.068412e+05 1.156080e+06
## avgdeathsperyear 8.584328      5.799320e+02 1.916237e+02 1.717061e+03
## pctbachdeg18_24  5.405055      8.147619e+00 6.216221e+00 8.789065e+00
## pctbachdeg25_over 5.291322      1.558707e+01 1.329754e+01 8.157983e+00
## birthrate        4.619029      6.302906e+00 5.597439e+00 3.412441e+00

```

## pctnohs18_24	4.201302	2.091973e+01	1.828777e+01	1.349465e+01
## povertypercent	3.504059	1.856871e+01	1.679175e+01	9.347835e+00
## medincome	3.400140	5.059719e+04	4.727794e+04	2.047381e+04
## pctpubliccoverage	-2.654203	3.450816e+01	3.615309e+01	1.072936e+01
## pctmarriedhouseholds	-3.193563	4.977284e+01	5.140232e+01	9.616843e+00
## pctprivatecoverage	-3.644872	6.140816e+01	6.447460e+01	1.614623e+01
## medianagemale	-3.671215	3.807891e+01	3.958564e+01	8.043895e+00
## incidencerate	-3.672349	4.318504e+02	4.486551e+02	1.114521e+02
## target_deathrate	-3.695311	1.706558e+02	1.788197e+02	4.405811e+01
## pctsomecol18_24	-3.923006	3.716463e+01	4.049574e+01	1.407479e+01
## medianage	-4.097952	3.918562e+01	4.085063e+01	7.796153e+00
## medianagefemale	-4.347955	4.038231e+01	4.216931e+01	7.985392e+00
## pctempprivcoverage	-4.723974	3.773333e+01	4.129388e+01	1.301492e+01
## percentmarried	-5.630017	4.889932e+01	5.190087e+01	1.023734e+01
## pctwhite	-8.154011	7.350356e+01	8.384699e+01	2.268382e+01
## pcths25_over	-8.726729	2.985510e+01	3.473255e+01	8.547221e+00
##	Overall sd	p.value		
## n_uout	1.837478e+00	5.657262e-63		
## pctasian	2.588138e+00	6.173759e-31		
## avganncount	1.601309e+03	4.878079e-21		
## pctotherrace	3.671806e+00	7.045736e-21		
## popest2015	3.763912e+05	5.180725e-19		
## avgdeathsperyear	5.717202e+02	9.136892e-18		
## pctbachdeg18_24	4.516322e+00	6.478842e-08		
## pctbachdeg25_over	5.468847e+00	1.214353e-07		
## birthrate	1.930363e+00	3.855404e-06		
## pctnohs18_24	7.917877e+00	2.653837e-05		
## povertypercent	6.409410e+00	4.582243e-04		
## medincome	1.233832e+04	6.735147e-04		
## pctpubliccoverage	7.832935e+00	7.949604e-03		
## pctmarriedhouseholds	6.448935e+00	1.405287e-03		
## pctprivatecoverage	1.063324e+01	2.675249e-04		
## medianagemale	5.187250e+00	2.413997e-04		
## incidencerate	5.783627e+01	2.403315e-04		
## target_deathrate	2.792279e+01	2.196181e-04		
## pctsomecol18_24	1.073207e+01	8.745087e-05		
## medianage	5.135293e+00	4.168225e-05		
## medianagefemale	5.194588e+00	1.374129e-05		
## pctempprivcoverage	9.526266e+00	2.312798e-06		
## percentmarried	6.738289e+00	1.801914e-08		
## pctwhite	1.603270e+01	3.520502e-16		
## pcths25_over	7.064060e+00	2.621438e-18		

Imputation

In this section, all the missing values previously identified are addressed through an imputation process. To avoid repetitive code, a function is first defined to perform the imputation consistently across variables. The imputation is then carried out for each variable with missing values, and a brief commentary is provided for each to discuss the imputation approach and its impact on the data.

Note that values are imputed both in the training and test sets, as stated in the statement of the project.

```
imputation_function <- function(variable) {
  num_vars <- df[sapply(df, is.numeric)]
```



```

miss_ids <- which(is.na(df[[variable]]))

# Perform PCA-based imputation
res_imp <- imputePCA(num_vars)
imp_values <- res_imp$completeObs[, variable]

# Calculate quantiles for validation
quantiles_df <- quantile(df[[variable]], probs = seq(0, 1, 0.1), na.rm = TRUE)
quantiles_imp <- quantile(imp_values, probs = seq(0, 1, 0.1), na.rm = FALSE)

# Plot the quantile comparison
plot(seq(0, 1, 0.1), quantiles_df, type = "l", col = "blue", lwd = 2,
      xlab = "Quantile", ylab = "Value",
      main = paste("Quantile Comparison for", variable))
lines(seq(0, 1, 0.1), quantiles_imp, col = "red", lwd = 2)
legend("topright", legend = c(variable, paste("imp_", variable, sep = "")),
      col = c("blue", "red"), lwd = 2)

return(list(
  na_indexes = miss_ids,
  imp_values = imp_values[miss_ids]
))
}

imputation_function_test <- function(variable) {

  num_vars <- test[sapply(test, is.numeric)]
  miss_ids <- which(is.na(test[[variable]]))

  # Perform PCA-based imputation
  res_imp <- imputePCA(num_vars)
  imp_values <- res_imp$completeObs[, variable]

  # Calculate quantiles for validation
  quantiles_df <- quantile(test[[variable]],
                           probs = seq(0, 1, 0.1), na.rm = TRUE)
  quantiles_imp <- quantile(imp_values, probs = seq(0, 1, 0.1),
                           na.rm = FALSE)

  # Plot the quantile comparison
  plot(seq(0, 1, 0.1), quantiles_df, type = "l", col = "blue", lwd = 2,
        xlab = "Quantile", ylab = "Value",
        main = paste("Quantile Comparison for", variable))
  lines(seq(0, 1, 0.1), quantiles_imp, col = "red", lwd = 2)
  legend("topright", legend = c(variable, paste("imp_", variable, sep = "")),
        col = c("blue", "red"), lwd = 2)

  return(list(
    na_indexes = miss_ids,
    imp_values = imp_values[miss_ids]
  ))
}

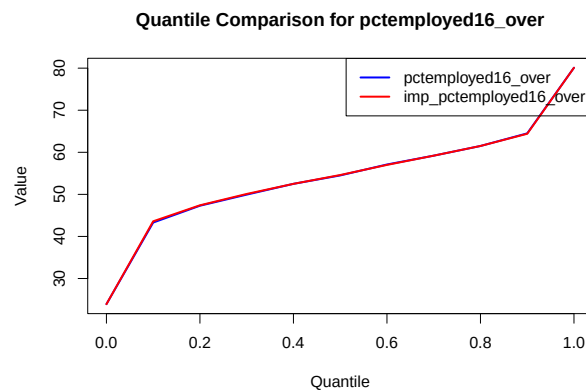
```

```
}
```

pctemployed16_over

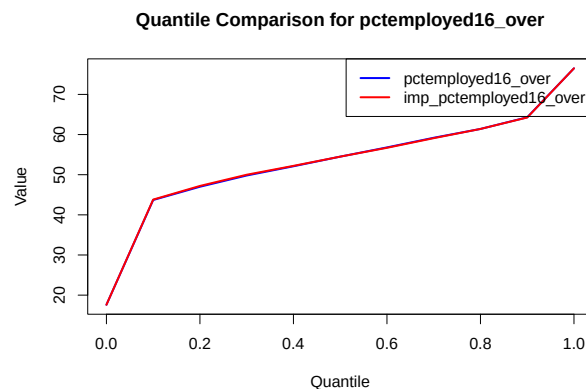
The variable *pctemployed16_over* presented 82 missing values, a relatively small number, making it suitable for imputation to preserve the dataset's completeness. Using the `imputePCA` function, the imputation process successfully replicated the original distribution of the variable. This outcome was validated through a quantile comparison plot, where the imputed values (represented by a red line) closely matched the original distribution (blue line). The strong alignment between the two confirms that the imputation method effectively preserved the integrity of the variable's distribution.

```
imp.res <- imputation_function("pctemployed16_over")
```



```
df[imp.res$na_indexes, "pctemployed16_over"] = imp.res$imp_values
```

```
imp.res <- imputation_function_test("pctemployed16_over")
```

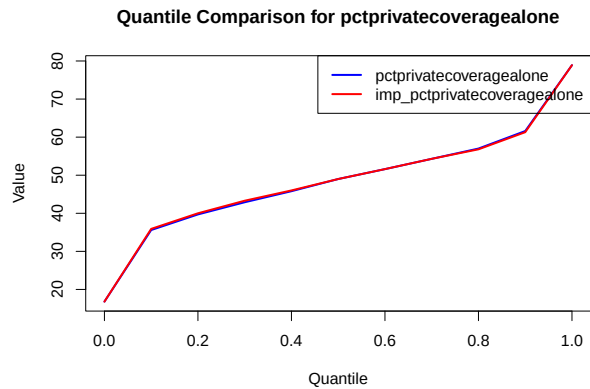


```
test[imp.res$na_indexes, "pctemployed16_over"] = imp.res$imp_values
```

pctprivatecoveragealone

The variable *pctprivatecoveragealone*, with a relatively high number of missing values (356), has shown substantial improvement after imputation. By employing the `imputePCA` function, the original distribution has been effectively preserved, as evidenced by the close alignment of the imputed and observed data.

```
imp.res <- imputation_function("pctprivatecoveragealone")
```



```
df[imp.res$na_indexes,"pctprivatecoveragealone"] = imp.res$imp_values
summary(df)
```

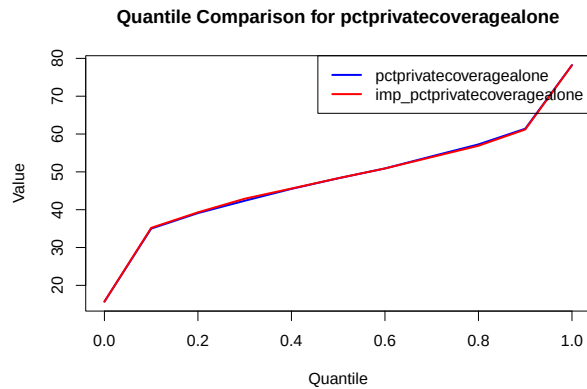
```
##      avganncount      avgdeathspereyear  target_deathrate  incidencerate
## Min.   : 7.0      Min.   : 3.0      Min.   : 59.7      Min.   : 201.3
## 1st Qu.: 74.0     1st Qu.: 29.0     1st Qu.:161.3     1st Qu.: 417.5
## Median : 155.0    Median : 62.0     Median :178.3     Median : 451.8
## Mean   : 524.2    Mean   : 191.6    Mean   :178.8     Mean   : 448.7
## 3rd Qu.: 383.0    3rd Qu.: 140.5    3rd Qu.:195.3     3rd Qu.: 483.8
## Max.   :38150.0   Max.   :14010.0   Max.   :362.8     Max.   :1206.9
## NA's   :126
##      medincome      popest2015      povertypercent  binnedinc
## Min.   : 22640     Min.   : 829      Min.   : 3.70     Length:1831
## 1st Qu.: 39031     1st Qu.: 12191    1st Qu.:12.15     Class :character
## Median : 45454     Median : 27158    Median :15.70     Mode  :character
## Mean   : 47278     Mean   : 106841    Mean   :16.79
## 3rd Qu.: 52612     3rd Qu.: 66880    3rd Qu.:20.40
## Max.   :122641     Max.   :10170292   Max.   :44.00
##
##      medianage  medianagemale  medianagefemale  geography
## Min.   :23.30   Min.   :23.00     Min.   :23.60     Length:1831
## 1st Qu.:37.90   1st Qu.:36.40     1st Qu.:39.20     Class :character
## Median :40.90   Median :39.50     Median :42.40     Mode  :character
## Mean   :40.85   Mean   :39.59     Mean   :42.17
## 3rd Qu.:43.80   3rd Qu.:42.60     3rd Qu.:45.30
## Max.   :59.00   Max.   :60.20     Max.   :58.20
## NA's   :18
##      percentmarried  pctnohs18_24  pcths18_24  pctsomecol18_24  pctbachdeg18_24
## Min.   :23.1      Min.   : 0.50   Min.   : 0.0   Min.   : 0.00     Min.   : 0.000
## 1st Qu.:47.8      1st Qu.:12.90   1st Qu.:29.2   1st Qu.:33.45     1st Qu.: 3.200
## Median :52.5      Median :17.20   Median :34.7   Median :40.30     Median : 5.400
## Mean   :51.9      Mean   :18.29   Mean   :35.0   Mean   :40.50     Mean   : 6.216
## 3rd Qu.:56.4      3rd Qu.:22.70   3rd Qu.:40.5   3rd Qu.:46.05     3rd Qu.: 8.200
## Max.   :68.0      Max.   :59.10   Max.   :72.5   Max.   :78.30     Max.   :51.800
##
##      pcths25_over  pctbachdeg25_over  pctemployed16_over  pctunemployed16_over
## Min.   : 8.30     Min.   : 2.5       Min.   :23.90       Min.   : 0.700
## 1st Qu.:30.35     1st Qu.: 9.3       1st Qu.:48.65       1st Qu.: 5.500
## Median :35.30     Median :12.3       Median :54.60       Median : 7.500
## Mean   :34.73     Mean   :13.3       Mean   :54.25       Mean   : 7.861
```

```

## 3rd Qu.:39.65 3rd Qu.:16.0 3rd Qu.:60.20 3rd Qu.: 9.750
## Max. :52.70 Max. :42.2 Max. :80.10 Max. :29.400
##
## pctprivatecoverage pctprivatecoveragealone pctempprivcoverage
## Min. :23.40 Min. :16.80 Min. :14.30
## 1st Qu.:57.50 1st Qu.:41.80 1st Qu.:34.60
## Median :65.20 Median :49.00 Median :41.10
## Mean :64.47 Mean :48.67 Mean :41.29
## 3rd Qu.:72.10 3rd Qu.:55.50 3rd Qu.:47.70
## Max. :89.60 Max. :78.90 Max. :70.20
##
## pctpubliccoverage pctpubliccoveragealone pctwhite pctblack
## Min. :11.20 Min. : 2.60 Min. :12.27 Min. : 0.000
## 1st Qu.:30.90 1st Qu.:14.90 1st Qu.:77.31 1st Qu.: 0.648
## Median :36.30 Median :18.70 Median :89.90 Median : 2.323
## Mean :36.15 Mean :19.15 Mean :83.85 Mean : 9.082
## 3rd Qu.:41.40 3rd Qu.:23.00 3rd Qu.:95.57 3rd Qu.:10.867
## Max. :62.70 Max. :46.60 Max. :99.69 Max. :85.948
##
## pctasian pctotherrace pctmarriedhouseholds birthrate
## Min. : 0.0000 Min. : 0.0000 Min. :22.99 Min. : 0.000
## 1st Qu.: 0.2582 1st Qu.: 0.2867 1st Qu.:47.85 1st Qu.: 4.528
## Median : 0.5495 Median : 0.7826 Median :51.73 Median : 5.355
## Mean : 1.2743 Mean : 2.0031 Mean :51.40 Mean : 5.597
## 3rd Qu.: 1.2515 3rd Qu.: 2.1066 3rd Qu.:55.48 3rd Qu.: 6.414
## Max. :37.1569 Max. :41.9303 Max. :71.40 Max. :21.326
##
## f.income f.population f.state f.education18_24
## Very Low:458 Very Low:458 Texas : 136 Low Education : 5
## Low :458 Low :458 Georgia : 100 Moderate Education: 758
## Moderate:457 Moderate:457 Kentucky : 75 High Education :1068
## High :458 High :458 Virginia : 74
## Missouri : 66
## North Carolina: 62
## (Other) :1318
## f.education25_over f.unemployment
## Mostly High School:1354 Low Unemployment : 346
## Balanced Eduaction: 377 Moderate Unemployment:1054
## Mostly Bachelor's : 100 High Unemployment : 431
##
##
##
##

```

```
imp.res <- imputation_function_test("pctprivatecoveragealone")
```



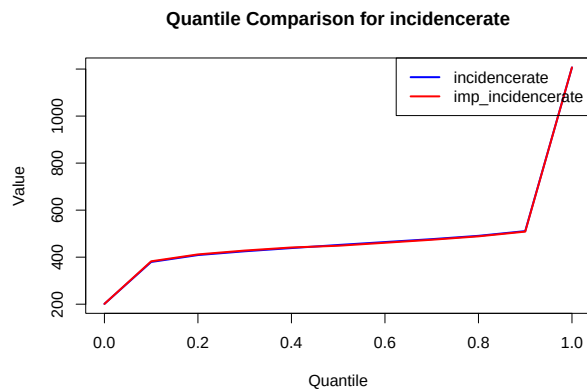
```
test[imp.res$na_indexes,"pctprivatecoveragealone"] = imp.res$imp_values
```

incidencerate, avganncount and medianage

Finally, the imputation process was completed for the two variables where 126 erroneous values were replaced with NA and the *medianage* variable, where 18 erroneous values were similarly addressed. After applying imputation, the resulting values were verified to align closely with the expected distributions, ensuring their reliability. These updated values were subsequently incorporated into the main dataframe. To conclude, a final check using the `summary()` function confirmed that all variables in the dataset are now free of missing values, ensuring data completeness for subsequent analyses.

This is only done in the training set since we can't explore the test dataset, and therefore we can't assume we discovered wrong values in some individuals.

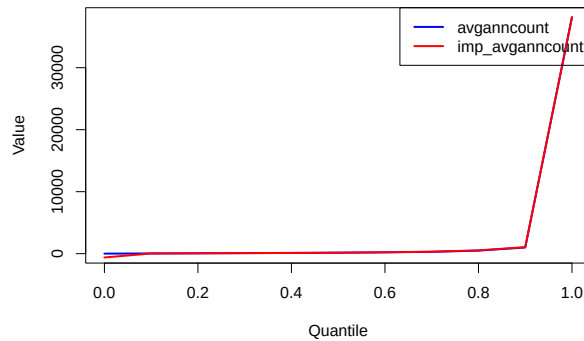
```
imp.res <- imputation_function("incidencerate")
```



```
df[imp.res$na_indexes,"incidencerate"] = imp.res$imp_values
```

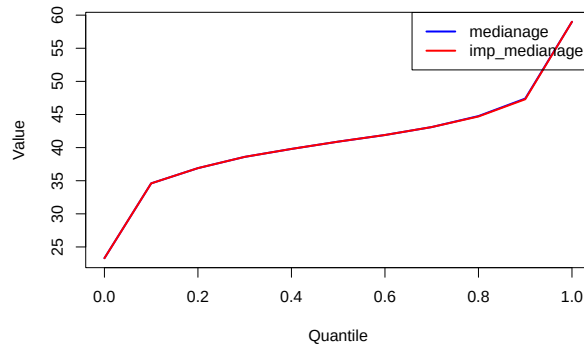
```
imp.res <- imputation_function("avganncount")
```

Quantile Comparison for avganncount



```
df[imp.res$na_indexes,"avganncount"] = imp.res$imp_values
imp.res <- imputation_function("medianage")
```

Quantile Comparison for medianage



```
df[imp.res$na_indexes,"medianage"] = imp.res$imp_values
summary(df)
```

```
##   avganncount   avgdeathspereyear   target_deathrate   incidencerate
##   Min.   : -615.0   Min.   :    3.0   Min.   : 59.7   Min.   : 201.3
##   1st Qu.:   73.0   1st Qu.:   29.0   1st Qu.:161.3   1st Qu.: 421.4
##   Median :  159.0   Median :   62.0   Median :178.3   Median : 448.5
##   Mean   :  524.7   Mean   :  191.6   Mean   :178.8   Mean   : 448.4
##   3rd Qu.:  408.5   3rd Qu.:  140.5   3rd Qu.:195.3   3rd Qu.: 481.3
##   Max.   :38150.0   Max.   :14010.0   Max.   :362.8   Max.   :1206.9
##
##   medincome      popest2015      povertyperecent   binnedinc
##   Min.   : 22640   Min.   :    829   Min.   : 3.70   Length:1831
##   1st Qu.:39031   1st Qu.:  12191   1st Qu.:12.15   Class :character
##   Median :45454   Median :   27158   Median :15.70   Mode  :character
##   Mean   :47278   Mean   : 106841   Mean   :16.79
##   3rd Qu.:52612   3rd Qu.:  66880   3rd Qu.:20.40
##   Max.   :122641   Max.   :10170292   Max.   :44.00
##
##   medianage   medianagemale   medianagefemale   geography
##   Min.   :23.30   Min.   :23.00   Min.   :23.60   Length:1831
```

```

## 1st Qu.:37.90 1st Qu.:36.40 1st Qu.:39.20 Class :character
## Median :40.90 Median :39.50 Median :42.40 Mode :character
## Mean :40.85 Mean :39.59 Mean :42.17
## 3rd Qu.:43.80 3rd Qu.:42.60 3rd Qu.:45.30
## Max. :59.00 Max. :60.20 Max. :58.20
##
## percentmarried pctnohs18_24 pcths18_24 pctsomecol18_24 pctbachdeg18_24
## Min. :23.1 Min. : 0.50 Min. : 0.0 Min. : 0.00 Min. : 0.000
## 1st Qu.:47.8 1st Qu.:12.90 1st Qu.:29.2 1st Qu.:33.45 1st Qu.: 3.200
## Median :52.5 Median :17.20 Median :34.7 Median :40.30 Median : 5.400
## Mean :51.9 Mean :18.29 Mean :35.0 Mean :40.50 Mean : 6.216
## 3rd Qu.:56.4 3rd Qu.:22.70 3rd Qu.:40.5 3rd Qu.:46.05 3rd Qu.: 8.200
## Max. :68.0 Max. :59.10 Max. :72.5 Max. :78.30 Max. :51.800
##
## pcths25_over pctbachdeg25_over pctemployed16_over pctunemployed16_over
## Min. : 8.30 Min. : 2.5 Min. :23.90 Min. : 0.700
## 1st Qu.:30.35 1st Qu.: 9.3 1st Qu.:48.65 1st Qu.: 5.500
## Median :35.30 Median :12.3 Median :54.60 Median : 7.500
## Mean :34.73 Mean :13.3 Mean :54.25 Mean : 7.861
## 3rd Qu.:39.65 3rd Qu.:16.0 3rd Qu.:60.20 3rd Qu.: 9.750
## Max. :52.70 Max. :42.2 Max. :80.10 Max. :29.400
##
## pctprivatecoverage pctprivatecoveragealone pctempprivcoverage
## Min. :23.40 Min. :16.80 Min. :14.30
## 1st Qu.:57.50 1st Qu.:41.80 1st Qu.:34.60
## Median :65.20 Median :49.00 Median :41.10
## Mean :64.47 Mean :48.67 Mean :41.29
## 3rd Qu.:72.10 3rd Qu.:55.50 3rd Qu.:47.70
## Max. :89.60 Max. :78.90 Max. :70.20
##
## pctpubliccoverage pctpubliccoveragealone pctwhite pctblack
## Min. :11.20 Min. : 2.60 Min. :12.27 Min. : 0.000
## 1st Qu.:30.90 1st Qu.:14.90 1st Qu.:77.31 1st Qu.: 0.648
## Median :36.30 Median :18.70 Median :89.90 Median : 2.323
## Mean :36.15 Mean :19.15 Mean :83.85 Mean : 9.082
## 3rd Qu.:41.40 3rd Qu.:23.00 3rd Qu.:95.57 3rd Qu.:10.867
## Max. :62.70 Max. :46.60 Max. :99.69 Max. :85.948
##
## pctasian pctotherrace pctmarriedhouseholds birthrate
## Min. : 0.0000 Min. : 0.0000 Min. :22.99 Min. : 0.000
## 1st Qu.: 0.2582 1st Qu.: 0.2867 1st Qu.:47.85 1st Qu.: 4.528
## Median : 0.5495 Median : 0.7826 Median :51.73 Median : 5.355
## Mean : 1.2743 Mean : 2.0031 Mean :51.40 Mean : 5.597
## 3rd Qu.: 1.2515 3rd Qu.: 2.1066 3rd Qu.:55.48 3rd Qu.: 6.414
## Max. :37.1569 Max. :41.9303 Max. :71.40 Max. :21.326
##
## f.income f.population f.state f.education18_24
## Very Low:458 Very Low:458 Texas : 136 Low Education : 5
## Low :458 Low :458 Georgia : 100 Moderate Education: 758
## Moderate:457 Moderate:457 Kentucky : 75 High Education :1068
## High :458 High :458 Virginia : 74
## Missouri : 66
## North Carolina: 62
## (Other) :1318

```

```
##          f.education25_over          f.unemployment
## Mostly High School:1354      Low Unemployment      : 346
## Balanced Eduaction: 377      Moderate Unemployment:1054
## Mostly Bachelor's : 100      High Unemployment     : 431
##
##
##
##
```

```
summary(test)
```

```
##      avganncount      avgdeathsperyear target_deathrate incidencerate
## Min.   :    6.00   Min.   :    3.0   Min.   : 87.6   Min.   :214.8
## 1st Qu.:   70.75   1st Qu.:   26.0   1st Qu.:161.1   1st Qu.:419.6
## Median :  168.00   Median :   60.0   Median :177.8   Median :453.5
## Mean   :   580.91   Mean   :  177.4   Mean   :178.4   Mean   :447.2
## 3rd Qu.:  534.50   3rd Qu.:  156.0   3rd Qu.:195.0   3rd Qu.:480.0
## Max.   :13294.00   Max.   :4895.0   Max.   :293.9   Max.   :718.9
##
##      medincome      popest2015      povertypercent      binnedinc
## Min.   : 23047   Min.   :    827   Min.   : 3.20   Length:1216
## 1st Qu.: 38492   1st Qu.:  11170   1st Qu.:12.18   Class :character
## Median : 44784   Median :  25962   Median :16.20   Mode  :character
## Mean   : 46740   Mean   :  96307   Mean   :17.01
## 3rd Qu.: 52387   3rd Qu.:  71517   3rd Qu.:20.40
## Max.   :125635   Max.   :3299521   Max.   :47.40
##
##      medianage      medianagemale      medianagefemale      geography
## Min.   : 22.30   Min.   :22.40   Min.   :22.30   Length:1216
## 1st Qu.: 37.60   1st Qu.:36.30   1st Qu.:38.77   Class :character
## Median : 41.00   Median :39.70   Median :42.30   Mode  :character
## Mean   : 45.31   Mean   :39.55   Mean   :42.11
## 3rd Qu.: 43.92   3rd Qu.:42.50   3rd Qu.:45.40
## Max.   :619.20   Max.   :64.70   Max.   :65.70
##
##      percentmarried      pctnohs18_24      pcths18_24      pctsomacol18_24
## Min.   :25.30   Min.   : 0.00   Min.   : 7.10   Min.   : 7.10
## 1st Qu.:47.67   1st Qu.:12.50   1st Qu.:29.00   1st Qu.:33.60
## Median :52.30   Median :17.10   Median :34.85   Median :40.50
## Mean   :51.58   Mean   :18.13   Mean   :35.01   Mean   :40.79
## 3rd Qu.:56.30   3rd Qu.:22.62   3rd Qu.:40.90   3rd Qu.:46.90
## Max.   :72.50   Max.   :64.10   Max.   :66.20   Max.   :79.00
##
##      pctbachdeg18_24      pcths25_over      pctbachdeg25_over      pctemployed16_over
## Min.   : 0.000   Min.   : 7.50   Min.   : 3.20   Min.   :17.60
## 1st Qu.: 2.900   1st Qu.:30.50   1st Qu.: 9.40   1st Qu.:48.60
## Median : 5.300   Median :35.30   Median :12.30   Median :54.50
## Mean   : 6.071   Mean   :34.91   Mean   :13.26   Mean   :54.08
## 3rd Qu.: 8.300   3rd Qu.:39.62   3rd Qu.:16.20   3rd Qu.:60.23
## Max.   :43.400   Max.   :54.80   Max.   :40.40   Max.   :76.50
##
##      pctunemployed16_over      pctprivatecoverage      pctprivatecoveragealone
## Min.   : 0.40   Min.   :22.30   Min.   :15.70
## 1st Qu.: 5.60   1st Qu.:56.80   1st Qu.:41.13
## Median : 7.60   Median :65.00   Median :48.32
```



```

## Mean      : 7.84          Mean      :64.17          Mean      :48.25
## 3rd Qu.: 9.70          3rd Qu.:72.00          3rd Qu.:55.49
## Max.      :27.00        Max.      :92.30          Max.      :78.20
##
## pctempprivcoverage pctpubliccoverage pctpubliccoveragealone pctwhite
## Min.      :13.50        Min.      :13.50        Min.      : 2.70        Min.      : 10.20
## 1st Qu.:34.20          1st Qu.:30.88          1st Qu.:14.78          1st Qu.: 77.31
## Median :41.15          Median :36.50          Median :19.00          Median : 90.12
## Mean      :41.05          Mean      :36.40          Mean      :19.38          Mean      : 83.34
## 3rd Qu.:47.52          3rd Qu.:41.80          3rd Qu.:23.30          3rd Qu.: 95.13
## Max.      :70.70        Max.      :65.10          Max.      :43.30          Max.      :100.00
##
##      pctblack          pctasian          pctotherrace          pctmarriedhouseholds
## Min.      : 0.0000        Min.      : 0.0000        Min.      : 0.0000        Min.      :23.89
## 1st Qu.: 0.6108          1st Qu.: 0.2489          1st Qu.: 0.3074          1st Qu.:47.56
## Median : 2.1584          Median : 0.5503          Median : 0.8908          Median :51.55
## Mean      : 9.1478          Mean      : 1.2233          Mean      : 1.9540          Mean      :51.01
## 3rd Qu.:10.0581          3rd Qu.: 1.1626          3rd Qu.: 2.2318          3rd Qu.:55.10
## Max.      :81.2819        Max.      :42.6194          Max.      :38.7437          Max.      :78.08
##
##      birthrate          f.income          f.population          f.state
## Min.      : 0.000          Very Low:304          Very Low:304          Texas      : 97
## 1st Qu.: 4.515          Low      :304          Low      :304          Georgia   : 55
## Median : 5.451          Moderate:304          Moderate:304          Virginia  : 51
## Mean      : 5.705          High     :304          High     :304          Missouri  : 49
## 3rd Qu.: 6.602                                     Illinois: 46
## Max.      :18.557                                     Kentucky: 45
##                                                         (Other) :873
##      f.education18_24          f.education25_over
## Low Education      :304          Mostly High School:888
## Moderate Education:304          Balanced Eduaction:271
## High Education     :608          Mostly Bachelor's : 57
##
##
##
##
##      f.unemployment
## Low Unemployment      :235
## Moderate Unemployment:710
## High Unemployment     :271
##
##
##
##

```

Target variable: Profiling and correlation analysis.

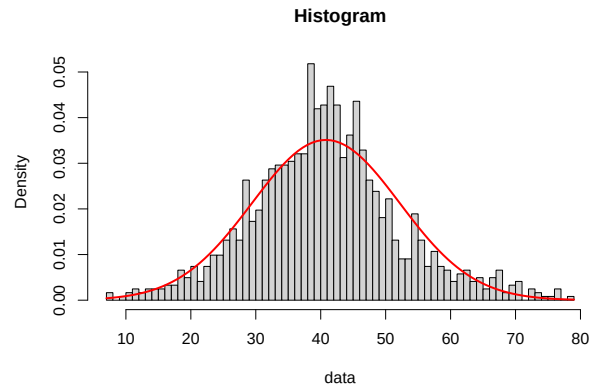
The target variable is complete in the training sample, with no missing values. The median (178.3) and mean (178.8) are nearly identical, suggesting a symmetric distribution. This symmetry is further supported by visual inspections of the boxplot and histogram, both of which confirm the balanced nature of the data around its center.

To evaluate normality, we performed the Shapiro-Wilk test. The resulting small p-value rejected the null hypothesis, indicating that the distribution deviates from normality despite its apparent symmetry. Thus, the target variable is not normally distributed.

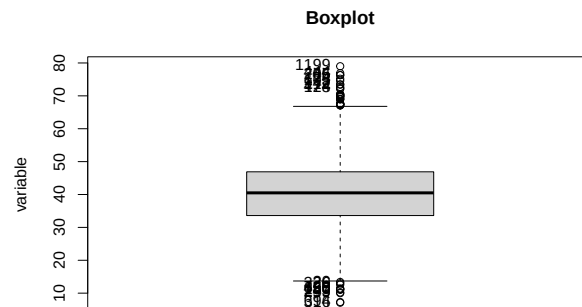
In terms of outliers, the variable contains 35 observations classified as mild outliers, and one severe outlier.

```
summary_and_normality_check(variable)
```

```
## [1] "Summary: "  
##      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.  
##      7.10  33.60   40.50   40.79  46.90   79.00  
## [1] "Mean"  
## [1] 40.79424  
## [1] "Standard deviation"  
## [1] 11.36139
```



```
## [1] "Skewness"  
## [1] 0.2593399
```



```
## [1] "Num Outliers:"  
## [1] 41  
## [1] "Num. severe outliers:"  
## [1] 0  
## [1] "Shapiro test: "  
##  
## Shapiro-Wilk normality test  
##  
## data: data  
## W = 0.98978, p-value = 1.741e-07  
##  
## [1] "Most repeated value ad times it occurs: "
```

```
## [1] "39"
## [1] 14

## [1] 20 46 72 89 105 112 128 174 180 181 205 243 259 260 262
## [16] 277 316 321 326 345 376 444 457 461 474 530 547 655 694 720
## [31] 922 941 942 950 954 1008 1075 1076 1142 1199 1213
```

We use the `condes()` function of the `FactoMiner` package in order to profile the numeric target variable.

Using the *quanti* output, we see how the most positively correlated variables with the target are *pctpubliccoveragealone*(0.466), *incidencerate*(0.456) and *povertypercent*(0.450). Moreover, the more negatively correlated are *pctbachdeg25_over* (-0.497), *medincome*(-0.443) and *pctemployed16_over*(-0.440). Moreover, we see that the following numeric variables are not significantly correlated with the target variable, and thus they will be not included in the model: *medianage*, *medianagemale*, *medianagefemale*.

The *quali* output expresses that all factors defined but *feducation18_24* are significantly globally related with the target variable. Thus, this factor won't be included in the model. The most related one is *f.state* ($R^2 = 0.33$), followed by *binnedinc* ($R^2=0.22$) and *f.income* ($R^2=0.18$).

The *category* output shows how some individuals like those who have *f.income*=Very Low, *binnedinc*=[22640, 34218.1], *f.state*=Mississippi, *f.state*=Kentucky or *f.state*=Arkansas have estimates over the overall mean of the target variable. On the other hand, those who have *f.state*=Utah, *f.state*=Arizona, *f.state*=Colorado, *binnedinc*=(61494.5, 125635], among others, have estimates below the overall mean.

```
cor(df[,which(sapply(df, is.numeric))])
```

```
##               avganncount avgdeathsperyear target_deathrate
## avganncount           1.00000000          0.98711825      -0.114405324
## avgdeathsperyear       0.98711825          1.00000000      -0.091676499
## target_deathrate      -0.11440532         -0.09167650       1.000000000
## incidencerate          0.05630361          0.05310090       0.456497402
## medincome              0.23673999          0.20363073      -0.443682800
## popest2015             0.97230017          0.97888010      -0.119022317
## povertypercent         -0.07893414         -0.05964856       0.450084648
## medianage              -0.16165607         -0.13973241       0.007555468
## medianagemale          -0.16204071         -0.14119835      -0.007769463
## medianagefemale        -0.16142147         -0.13906639       0.022987894
## percentmarried         -0.16317529         -0.17437292      -0.272537329
## pctnohs18_24           -0.13226433         -0.12246684       0.114800600
## pcths18_24             -0.16388275         -0.14379048       0.242434869
## pctsomecol18_24        0.12875184          0.11092383      -0.160219915
## pctbachdeg18_24        0.25162649          0.23693550      -0.301622549
## pcths25_over           -0.29978574         -0.27718024       0.406901721
## pctbachdeg25_over       0.29423292          0.27147247      -0.497771564
## pctemployed16_over      0.15418660          0.12441906      -0.440771339
## pctunemployed16_over    0.04689505          0.06433474       0.397806807
## pctprivatecoverage      0.06513708          0.04267950      -0.403541486
## pctprivatecoveragealone 0.16221551          0.13242918      -0.417525623
## pctemprprivcoverage     0.17141832          0.14251411      -0.288776902
## pctpubliccoverage       -0.15624615         -0.12674534       0.424367508
## pctpubliccoveragealone -0.04613594         -0.02337244       0.466295777
## pctwhite               -0.17533285         -0.17900832      -0.181525529
## pctblack                0.06655371          0.07879809       0.268754523
## pctasian                0.42898109          0.40972382      -0.194364123
## pctotherrace            0.21157938          0.19780430      -0.181535932
## pctmarriedhouseholds   -0.13181224         -0.15163821      -0.294406426
## birthrate              -0.05661714         -0.06141470      -0.067560432
```

##	incidencerate	medincome	popest2015	povertypercent
## avganncount	0.05630361	0.236739994	0.97230017	-0.078934141
## avgdeathsperyear	0.05310090	0.203630735	0.97888010	-0.059648559
## target_deathrate	0.45649740	-0.443682800	-0.11902232	0.450084648
## incidencerate	1.00000000	-0.025497297	0.01839017	0.047516253
## medincome	-0.02549730	1.000000000	0.21580024	-0.791819961
## popest2015	0.01839017	0.215800239	1.00000000	-0.060414429
## povertypercent	0.04751625	-0.791819961	-0.06041443	1.000000000
## medianage	-0.02463730	-0.132487479	-0.16654299	-0.175817342
## medianagemale	-0.02754286	-0.104903540	-0.16447734	-0.200184425
## medianagefemale	-0.03688939	-0.171240301	-0.16795632	-0.124690843
## percentmarried	-0.14655371	0.364990463	-0.15138827	-0.639717469
## pctnohs18_24	-0.12148090	-0.280052313	-0.11389451	0.279449824
## pcths18_24	0.02477989	-0.179227457	-0.14371858	0.092589093
## pctsomcol18_24	0.06231654	0.148875748	0.10902629	-0.116021487
## pctbachdeg18_24	0.01599607	0.492805548	0.22634370	-0.397817087
## pcths25_over	0.12076097	-0.481985765	-0.29201482	0.207014657
## pctbachdeg25_over	-0.05563210	0.718600560	0.27572670	-0.552806078
## pctemployed16_over	-0.03901051	0.705211277	0.13893576	-0.749689018
## pctunemployed16_over	0.14872225	-0.459947782	0.04378062	0.655931387
## pctprivatecoverage	0.05843820	0.732681982	0.04153334	-0.835328641
## pctprivatecoveragealone	0.04087288	0.809019246	0.14030593	-0.788303184
## pctempprivcoverage	0.11418704	0.750214397	0.14134873	-0.688234514
## pctpubliccoverage	0.07258142	-0.766863497	-0.15317821	0.669484990
## pctpubliccoveragealone	0.08505107	-0.728676472	-0.03925597	0.810204969
## pctwhite	-0.03322807	0.161696052	-0.17638420	-0.505022841
## pctblack	0.12260341	-0.277433862	0.06384545	0.526670945
## pctasian	-0.02909845	0.443466066	0.42768474	-0.172152191
## pctotherrace	-0.20523574	0.078731913	0.21911471	0.043272071
## pctmarriedhouseholds	-0.16480537	0.464441967	-0.11753223	-0.609389106
## birthrate	-0.09625752	-0.009384043	-0.04733889	0.009298835
##	medianage	medianagemale	medianagefemale	
## avganncount	-0.161656074	-0.162040706	-0.16142147	
## avgdeathsperyear	-0.139732408	-0.141198346	-0.13906639	
## target_deathrate	0.007555468	-0.007769463	0.02298789	
## incidencerate	-0.024637298	-0.027542864	-0.03688939	
## medincome	-0.132487479	-0.104903540	-0.17124030	
## popest2015	-0.166542991	-0.164477337	-0.16795632	
## povertypercent	-0.175817342	-0.200184425	-0.12469084	
## medianage	1.000000000	0.982531073	0.97625489	
## medianagemale	0.982531073	1.000000000	0.93400344	
## medianagefemale	0.976254888	0.934003441	1.00000000	
## percentmarried	0.416605315	0.436394582	0.36333803	
## pctnohs18_24	0.119622023	0.115204381	0.14240183	
## pcths18_24	0.228405707	0.223391729	0.22175208	
## pctsomcol18_24	-0.262555103	-0.261764145	-0.26732838	
## pctbachdeg18_24	-0.039210339	-0.023364837	-0.05463045	
## pcths25_over	0.326569252	0.314091713	0.33708802	
## pctbachdeg25_over	-0.149580317	-0.131097624	-0.18016197	
## pctemployed16_over	-0.197483209	-0.168282115	-0.25009014	
## pctunemployed16_over	-0.104240861	-0.119146302	-0.08443701	
## pctprivatecoverage	0.054614899	0.073560707	0.02449116	
## pctprivatecoveragealone	-0.204830480	-0.179862699	-0.23882692	
## pctempprivcoverage	-0.243144802	-0.217785889	-0.27163409	

## pctpubliccoverage	0.433106337	0.403445667	0.46651287
## pctpubliccoveragealone	0.035028524	0.012220186	0.06935994
## pctwhite	0.370118054	0.391725742	0.33565953
## pctblack	-0.208054389	-0.243114632	-0.15997254
## pctasian	-0.257176988	-0.244245818	-0.26906985
## pctotherrace	-0.267327145	-0.255078490	-0.25771924
## pctmarriedhouseholds	0.177102285	0.200305510	0.13789701
## birthrate	-0.133848775	-0.140545008	-0.12532033
##	percentmarried	pctnohs18_24	pcths18_24
## avganncount	-0.16317529	-0.13226433	-0.163882754
## avgdeathsperyear	-0.17437292	-0.12246684	-0.143790476
## target_deathrate	-0.27253733	0.11480060	0.242434869
## incidencerate	-0.14655371	-0.12148090	0.024779890
## medincome	0.36499046	-0.28005231	-0.179227457
## popest2015	-0.15138827	-0.11389451	-0.143718577
## povertypercent	-0.63971747	0.27944982	0.092589093
## medianage	0.41660532	0.11962202	0.228405707
## medianagemale	0.43639458	0.11520438	0.223391729
## medianagefemale	0.36333803	0.14240183	0.221752077
## percentmarried	1.00000000	-0.02837631	0.116394365
## pctnohs18_24	-0.02837631	1.00000000	0.085403804
## pcths18_24	0.11639437	0.08540380	1.000000000
## pctsomocol18_24	-0.11190880	-0.64823156	-0.726438352
## pctbachdeg18_24	0.08451843	-0.38202691	-0.407827431
## pcths25_over	0.08759357	0.20964016	0.422898355
## pctbachdeg25_over	0.12901056	-0.37730208	-0.385207793
## pctemployed16_over	0.41656568	-0.33565230	-0.258517321
## pctunemployed16_over	-0.54778416	0.18043771	0.131679696
## pctprivatecoverage	0.47370513	-0.43355521	-0.245830863
## pctprivatecoveragealone	0.37278787	-0.43411902	-0.299106605
## pctempprivcoverage	0.25546010	-0.40827459	-0.237948437
## pctpubliccoverage	-0.26470829	0.30268199	0.246094720
## pctpubliccoveragealone	-0.47493780	0.30900087	0.209420968
## pctwhite	0.67740060	-0.16456470	0.029611195
## pctblack	-0.62498590	0.13162769	-0.006807061
## pctasian	-0.13802055	-0.21403255	-0.196357011
## pctotherrace	-0.11599758	0.13195884	-0.063182864
## pctmarriedhouseholds	0.86879678	-0.02431230	0.107757739
## birthrate	0.10220607	0.12373310	0.073011697
##	pctsomocol18_24	pctbachdeg18_24	pcths25_over
## avganncount	0.12875184	0.251626493	-0.29978574
## avgdeathsperyear	0.11092383	0.236935495	-0.27718024
## target_deathrate	-0.16021991	-0.301622549	0.40690172
## incidencerate	0.06231654	0.015996073	0.12076097
## medincome	0.14887575	0.492805548	-0.48198576
## popest2015	0.10902629	0.226343703	-0.29201482
## povertypercent	-0.11602149	-0.397817087	0.20701466
## medianage	-0.26255510	-0.039210339	0.32656925
## medianagemale	-0.26176414	-0.023364837	0.31409171
## medianagefemale	-0.26732838	-0.054630450	0.33708802
## percentmarried	-0.11190880	0.084518427	0.08759357
## pctnohs18_24	-0.64823156	-0.382026905	0.20964016
## pcths18_24	-0.72643835	-0.407827431	0.42289835
## pctsomocol18_24	1.00000000	0.201547013	-0.35001632

##	pctbachdeg18_24	0.20154701	1.000000000	-0.37520644
##	pcths25_over	-0.35001632	-0.375206438	1.00000000
##	pctbachdeg25_over	0.34936513	0.595713539	-0.74196527
##	pctemployed16_over	0.26897318	0.462312582	-0.34839655
##	pctunemployed16_over	-0.10559799	-0.326673806	0.09209607
##	pctprivatecoverage	0.31738385	0.493429321	-0.22935764
##	pctprivatecoveragealone	0.35296235	0.515681319	-0.35355048
##	pctempprivcoverage	0.31276944	0.444586898	-0.22761779
##	pctpubliccoverage	-0.25751820	-0.407027460	0.43001413
##	pctpubliccoveragealone	-0.23015742	-0.410190905	0.30507667
##	pctwhite	0.06056572	0.085795817	0.18862302
##	pctblack	-0.04163911	-0.118432286	-0.02806826
##	pctasian	0.18170277	0.333184467	-0.44397049
##	pctotherrace	-0.04584295	0.002997127	-0.28369269
##	pctmarriedhouseholds	-0.08901564	0.040075285	0.03522358
##	birthrate	-0.09711464	-0.130925164	0.01899900
##		pctbachdeg25_over	pctemployed16_over	
##	avganncount	0.29423292	0.15418660	
##	avgdeathsperyear	0.27147247	0.12441906	
##	target_deathrate	-0.49777156	-0.44077134	
##	incidencerate	-0.05563210	-0.03901051	
##	medincome	0.71860056	0.70521128	
##	popest2015	0.27572670	0.13893576	
##	povertypercent	-0.55280608	-0.74968902	
##	medianage	-0.14958032	-0.19748321	
##	medianagemale	-0.13109762	-0.16828211	
##	medianagefemale	-0.18016197	-0.25009014	
##	percentmarried	0.12901056	0.41656568	
##	pctnohs18_24	-0.37730208	-0.33565230	
##	pcths18_24	-0.38520779	-0.25851732	
##	pctsomecol18_24	0.34936513	0.26897318	
##	pctbachdeg18_24	0.59571354	0.46231258	
##	pcths25_over	-0.74196527	-0.34839655	
##	pctbachdeg25_over	1.00000000	0.62487168	
##	pctemployed16_over	0.62487168	1.00000000	
##	pctunemployed16_over	-0.38146021	-0.66417622	
##	pctprivatecoverage	0.60826263	0.71882426	
##	pctprivatecoveragealone	0.67517821	0.80876628	
##	pctempprivcoverage	0.53952110	0.71960688	
##	pctpubliccoverage	-0.63873929	-0.77806167	
##	pctpubliccoveragealone	-0.60769670	-0.72241909	
##	pctwhite	0.05844204	0.27737483	
##	pctblack	-0.15811531	-0.34542981	
##	pctasian	0.43380139	0.26248653	
##	pctotherrace	0.03726825	0.02410787	
##	pctmarriedhouseholds	0.13566096	0.34404532	
##	birthrate	-0.09303469	0.06931305	
##		pctunemployed16_over	pctprivatecoverage	
##	avganncount	0.04689505	0.06513708	
##	avgdeathsperyear	0.06433474	0.04267950	
##	target_deathrate	0.39780681	-0.40354149	
##	incidencerate	0.14872225	0.05843820	
##	medincome	-0.45994778	0.73268198	
##	popest2015	0.04378062	0.04153334	

## povertypercent	0.65593139	-0.83532864
## medianage	-0.10424086	0.05461490
## medianagemale	-0.11914630	0.07356071
## medianagefemale	-0.08443701	0.02449116
## percentmarried	-0.54778416	0.47370513
## pctnohs18_24	0.18043771	-0.43355521
## pcths18_24	0.13167970	-0.24583086
## pctsomecol18_24	-0.10559799	0.31738385
## pctbachdeg18_24	-0.32667381	0.49342932
## pcths25_over	0.09209607	-0.22935764
## pctbachdeg25_over	-0.38146021	0.60826263
## pctemployed16_over	-0.66417622	0.71882426
## pctunemployed16_over	1.00000000	-0.64258652
## pctprivatecoverage	-0.64258652	1.00000000
## pctprivatecoveragealone	-0.60971181	0.91839887
## pctempprivcoverage	-0.48457134	0.82529356
## pctpubliccoverage	0.52920643	-0.72786193
## pctpubliccoveragealone	0.64343815	-0.88825549
## pctwhite	-0.49821515	0.43519629
## pctblack	0.48505061	-0.37917991
## pctasian	-0.03539896	0.20197040
## pctotherrace	0.01701224	-0.16822293
## pctmarriedhouseholds	-0.47209889	0.46508301
## birthrate	-0.06139033	-0.04300982
##	pctprivatecoveragealone	pctempprivcoverage
## avganncount	0.16221551	0.17141832
## avgdeathsperyear	0.13242918	0.14251411
## target_deathrate	-0.41752562	-0.28877690
## incidencerate	0.04087288	0.11418704
## medincome	0.80901925	0.75021440
## popest2015	0.14030593	0.14134873
## povertypercent	-0.78830318	-0.68823451
## medianage	-0.20483048	-0.24314480
## medianagemale	-0.17986270	-0.21778589
## medianagefemale	-0.23882692	-0.27163409
## percentmarried	0.37278787	0.25546010
## pctnohs18_24	-0.43411902	-0.40827459
## pcths18_24	-0.29910660	-0.23794844
## pctsomecol18_24	0.35296235	0.31276944
## pctbachdeg18_24	0.51568132	0.44458690
## pcths25_over	-0.35355048	-0.22761779
## pctbachdeg25_over	0.67517821	0.53952110
## pctemployed16_over	0.80876628	0.71960688
## pctunemployed16_over	-0.60971181	-0.48457134
## pctprivatecoverage	0.91839887	0.82529356
## pctprivatecoveragealone	1.00000000	0.90937937
## pctempprivcoverage	0.90937937	1.00000000
## pctpubliccoverage	-0.86667514	-0.78222871
## pctpubliccoveragealone	-0.85642598	-0.72724885
## pctwhite	0.32053709	0.26755301
## pctblack	-0.32538463	-0.25634710
## pctasian	0.31374424	0.29392456
## pctotherrace	-0.06131700	-0.06081810
## pctmarriedhouseholds	0.43641146	0.35174733

## birthrate	-0.03307549	-0.08503768	
##	pctpubliccoverage	pctpubliccoveragealone	pctwhite
## avganncount	-0.15624615	-0.046135939	-0.17533285
## avgdeathsperyear	-0.12674534	-0.023372444	-0.17900832
## target_deathrate	0.42436751	0.466295777	-0.18152553
## incidencerate	0.07258142	0.085051069	-0.03322807
## medincome	-0.76686350	-0.728676472	0.16169605
## popest2015	-0.15317821	-0.039255965	-0.17638420
## povertypercent	0.66948499	0.810204969	-0.50502284
## medianage	0.43310634	0.035028524	0.37011805
## medianagemale	0.40344567	0.012220186	0.39172574
## medianagefemale	0.46651287	0.069359936	0.33565953
## percentmarried	-0.26470829	-0.474937799	0.67740060
## pctnohs18_24	0.30268199	0.309000866	-0.16456470
## pcths18_24	0.24609472	0.209420968	0.02961119
## pctsomecol18_24	-0.25751820	-0.230157422	0.06056572
## pctbachdeg18_24	-0.40702746	-0.410190905	0.08579582
## pcths25_over	0.43001413	0.305076668	0.18862302
## pctbachdeg25_over	-0.63873929	-0.607696697	0.05844204
## pctemployed16_over	-0.77806167	-0.722419093	0.27737483
## pctunemployed16_over	0.52920643	0.643438154	-0.49821515
## pctprivatecoverage	-0.72786193	-0.888255485	0.43519629
## pctprivatecoveragealone	-0.86667514	-0.856425979	0.32053709
## pctempprivcoverage	-0.78222871	-0.727248852	0.26755301
## pctpubliccoverage	1.00000000	0.867526258	-0.13459590
## pctpubliccoveragealone	0.86752626	1.00000000	-0.36131431
## pctwhite	-0.13459590	-0.361314305	1.00000000
## pctblack	0.21577954	0.357786626	-0.84450412
## pctasian	-0.32908476	-0.202136524	-0.26143601
## pctotherrace	-0.07545917	0.075449375	-0.23648580
## pctmarriedhouseholds	-0.39724774	-0.501541964	0.58514155
## birthrate	-0.02754595	0.007083385	-0.02068183
##	pctblack	pctasian	pctotherrace
## avganncount	0.066553708	0.428981090	0.211579378
## avgdeathsperyear	0.078798094	0.409723825	0.197804300
## target_deathrate	0.268754523	-0.194364123	-0.181535932
## incidencerate	0.122603408	-0.029098454	-0.205235741
## medincome	-0.277433862	0.443466066	0.078731913
## popest2015	0.063845452	0.427684743	0.219114713
## povertypercent	0.526670945	-0.172152191	0.043272071
## medianage	-0.208054389	-0.257176988	-0.267327145
## medianagemale	-0.243114632	-0.244245818	-0.255078490
## medianagefemale	-0.159972542	-0.269069855	-0.257719235
## percentmarried	-0.624985896	-0.138020551	-0.115997582
## pctnohs18_24	0.131627687	-0.214032553	0.131958838
## pcths18_24	-0.006807061	-0.196357011	-0.063182864
## pctsomecol18_24	-0.041639111	0.181702770	-0.045842945
## pctbachdeg18_24	-0.118432286	0.333184467	0.002997127
## pcths25_over	-0.028068257	-0.443970494	-0.283692690
## pctbachdeg25_over	-0.158115314	0.433801389	0.037268252
## pctemployed16_over	-0.345429814	0.262486526	0.024107873
## pctunemployed16_over	0.485050612	-0.035398959	0.017012237
## pctprivatecoverage	-0.379179910	0.201970399	-0.168222927
## pctprivatecoveragealone	-0.325384630	0.313744239	-0.061317000


```

## pctempprivcoverage      -0.256347105  0.293924565 -0.060818099
## pctpubliccoverage       0.215779535 -0.329084756 -0.075459174
## pctpubliccoveragealone  0.357786626 -0.202136524  0.075449375
## pctwhite                -0.844504119 -0.261436005 -0.236485799
## pctblack                1.000000000  0.008514247 -0.038731635
## pctasian                0.008514247  1.000000000  0.199572125
## pctotherrace            -0.038731635  0.199572125  1.000000000
## pctmarriedhouseholds    -0.572910903 -0.059156286 -0.028673662
## birthrate               -0.038333247 -0.063211239  0.062895435
##
## pctmarriedhouseholds    birthrate
## avganncount             -0.13181224 -0.056617135
## avgdeathsperyear        -0.15163821 -0.061414705
## target_deathrate        -0.29440643 -0.067560432
## incidencerate           -0.16480537 -0.096257517
## medincome               0.46444197 -0.009384043
## popest2015              -0.11753223 -0.047338890
## povertypercent          -0.60938911  0.009298835
## medianage               0.17710229 -0.133848775
## medianagemale           0.20030551 -0.140545008
## medianagefemale         0.13789701 -0.125320332
## percentmarried          0.86879678  0.102206067
## pctnohs18_24            -0.02431230  0.123733097
## pcths18_24              0.10775774  0.073011697
## pctsomecol18_24         -0.08901564 -0.097114638
## pctbachdeg18_24         0.04007528 -0.130925164
## pcths25_over            0.03522358  0.018998998
## pctbachdeg25_over       0.13566096 -0.093034694
## pctemployed16_over      0.34404532  0.069313052
## pctunemployed16_over    -0.47209889 -0.061390334
## pctprivatecoverage      0.46508301 -0.043009823
## pctprivatecoveragealone 0.43641146 -0.033075491
## pctempprivcoverage      0.35174733 -0.085037683
## pctpubliccoverage       -0.39724774 -0.027545953
## pctpubliccoveragealone  -0.50154196  0.007083385
## pctwhite                0.58514155 -0.020681833
## pctblack                -0.57291090 -0.038333247
## pctasian                -0.05915629 -0.063211239
## pctotherrace            -0.02867366  0.062895435
## pctmarriedhouseholds    1.00000000  0.070842057
## birthrate               0.07084206  1.000000000

```

```

res.con <- condes(df[, -which(names(df) == "geography")],
                  which(names(df) == "target_deathrate") )
res.con$quanti

```

```

## correlation p.value
## pctpubliccoveragealone 0.46629578 1.687857e-99
## incidencerate          0.45649740 6.295910e-95
## povertypercent          0.45008465 5.144264e-92
## pctpubliccoverage       0.42436751 6.056303e-81
## pcths25_over            0.40690172 5.910723e-74
## pctunemployed16_over    0.39780681 1.777035e-70
## pctblack                0.26875452 1.156196e-31
## pcths18_24              0.24243487 6.687515e-26
## pctnohs18_24           0.11480060 8.424149e-07

```

```
## birthrate -0.06756043 3.824884e-03
## avgdeathsperyear -0.09167650 8.549496e-05
## avganncount -0.11440532 9.191383e-07
## popest2015 -0.11902232 3.261480e-07
## pctsomecol18_24 -0.16021991 5.358126e-12
## pctwhite -0.18152553 4.983753e-15
## pctotherrace -0.18153593 4.965708e-15
## pctasian -0.19436412 4.806194e-17
## percentmarried -0.27253733 1.511951e-32
## pctempprivcoverage -0.28877690 1.677032e-36
## pctmarriedhouseholds -0.29440643 6.182702e-38
## pctbachdeg18_24 -0.30162255 8.054861e-40
## pctprivatecoverage -0.40354149 1.173008e-72
## pctprivatecoveragealone -0.41752562 3.712408e-78
## pctemployed16_over -0.44077134 6.780957e-88
## medincome -0.44368280 3.606607e-89
## pctbachdeg25_over -0.49777156 3.106333e-115
```

```
non_significant_vars <- setdiff(names(df), c(rownames(res.con$quanti),
"target_deathrate")); print(non_significant_vars)
```

```
## [1] "binnedinc" "medianage" "medianagemale"
## [4] "medianagefemale" "geography" "f.income"
## [7] "f.population" "f.state" "f.education18_24"
## [10] "f.education25_over" "f.unemployment"
```

```
res.con$quali
```

```
## R2 p.value
## f.state 0.33921765 1.277399e-125
## binnedinc 0.22173238 7.669702e-93
## f.income 0.18739419 7.009760e-82
## f.education25_over 0.14159778 2.474110e-61
## f.unemployment 0.12652650 2.005651e-54
## f.population 0.03968164 5.946091e-16
```

```
res.con$category
```

```
## Estimate p.value
## f.income=Very Low 17.0903924 5.300619e-55
## f.education25_over=Mostly High School 18.4696182 1.778808e-53
## binnedinc=[22640, 34218.1] 25.7637115 9.853130e-45
## f.unemployment=High Unemployment 15.2240361 3.899820e-36
## f.state=Kentucky 39.8572364 4.823986e-30
## f.education25_over=Balanced Eduaction 0.2768434 1.038144e-22
## f.state=Mississippi 29.8648974 9.615495e-13
## binnedinc=(34218.1, 37413.8] 13.5119973 1.196419e-11
## f.state=Tennessee 26.6435697 3.520425e-10
## f.population=Low 6.3268430 1.884523e-08
## f.state=Arkansas 28.1949925 3.518656e-08
## f.state=Louisiana 25.8152364 9.927510e-07
## binnedinc=(37413.8, 40362.7] 9.4545526 3.683926e-06
## f.population=Moderate 4.4015671 9.714319e-05
## f.income=Low 4.1299121 2.554111e-04
## f.state=Missouri 16.1852364 5.309726e-04
## f.state=Oklahoma 18.5985697 6.003716e-04
```

```

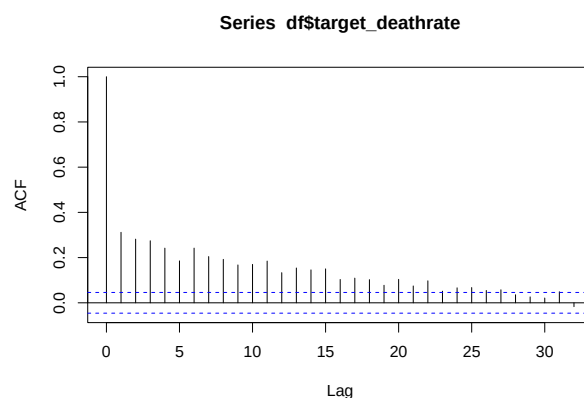
## f.state=West Virginia          20.4064485 9.532054e-04
## f.state=Alabama                16.2309507 1.210845e-02
## f.state=Indiana                13.0066650 2.065389e-02
## binnedinc=(40362.7, 42724.4]   3.9910670 4.088504e-02
## f.state=South Carolina         14.3852364 4.695201e-02
## f.state=Connecticut           -18.6719065 2.779123e-02
## f.state=Wyoming               -13.3070713 2.098262e-02
## f.state=Iowa                  -4.6571365 1.039203e-02
## f.state=Kansas                 -5.5934521 4.051307e-03
## f.state=Texas                 -2.2875577 3.178113e-03
## f.population=Very Low         -3.5231570 1.819819e-03
## f.state=South Dakota          -10.6352764 6.126807e-04
## f.state=North Dakota          -12.4428886 5.259784e-04
## f.state=New Mexico            -17.2597636 4.505662e-04
## f.state=Montana               -17.7829454 1.623894e-04
## f.state=Arizona               -33.8897636 9.494885e-05
## binnedinc=(48021.6, 51046.4]  -7.9528865 3.574891e-05
## f.state=Nebraska              -11.9013021 1.659603e-05
## f.state=California            -17.9210136 4.357341e-06
## binnedinc=(51046.4, 54545.6]  -9.4609546 3.130385e-06
## f.state=Idaho                 -21.8427636 1.917078e-06
## f.income=Moderate             -5.6155004 6.388572e-07
## f.state=Minnesota             -14.7696656 5.345467e-07
## f.state=Utah                  -36.8147636 2.302930e-10
## f.population=High             -7.2052531 1.488645e-10
## binnedinc=(54545.6, 61494.5] -13.0934128 1.157068e-11
## f.state=Colorado              -32.5000577 3.813385e-15
## binnedinc=(61494.5, 125635]   -20.4589080 2.459391e-26
## f.education25_over=Mostly Bachelor's -18.7464616 4.281285e-32
## f.unemployment=Low Unemployment -15.3742144 4.355141e-34
## f.income=High                 -15.6048041 1.112935e-45

```

Serial correlations

The ACF plot for *target_deathrate* shows significant autocorrelation for several lags, with a gradual decline over time. The initial lags exceed the confidence intervals, indicating strong serial correlation. In order to assess the serial correlation, the dataframe is randomized and reset. The new dataframe does not include autocorrelation anymore.

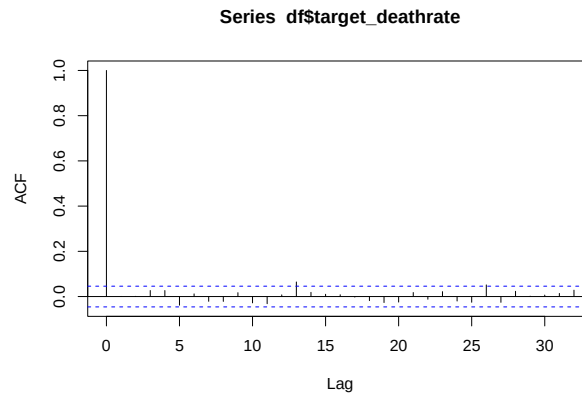
```
acf(df$target_deathrate)
```



```

l1<- sample(1:nrow(df), nrow(df))
df<-df[l1,]
acf(df$target_deathrate)

```



Modeling

Model Exploration

Base model: Based on the results of the `condes` function executed earlier, the most relevant variables have been selected for inclusion in the model. These variables are as follows:

- Numerical: *pctpubliccoveragealone*, *incidencerate*, *povertypercent*, *pctbachdeg25_over*, *medincome*, *pctemployed16_over*
- Factors: *f.state*, *binnedinc*, *f.income*, *f.education25_over*, *f.unemployment*

As observed in the Residuals vs Fitted plot, the data does not appear to be homoscedastic, indicating non-constant variance in the residuals.

The Q-Q Residuals plot reveals deviations from normality, particularly at the extremes of the X-axis, suggesting the presence of outliers.

In the Residuals vs Leverage plot, there are very few high-leverage points (only one notable case), which is favorable for the model. However, these high-leverage observations may be removed if deemed influential.

To ensure the assumptions of linear modeling are met, appropriate transformations will be applied to the data.

```

m1 <- lm(target_deathrate ~ pctpubliccoveragealone + incidencerate +
          povertypercent+pctbachdeg25_over+medincome+pctemployed16_over+
          f.state + binnedinc + f.income + f.education25_over + f.unemployment,
          data = df)
summary(m1)

```

```

##
## Call:
## lm(formula = target_deathrate ~ pctpubliccoveragealone + incidencerate +
##     povertypercent + pctbachdeg25_over + medincome + pctemployed16_over +
##     f.state + binnedinc + f.income + f.education25_over + f.unemployment,
##     data = df)
##
## Residuals:
##      Min       1Q   Median       3Q      Max

```

```

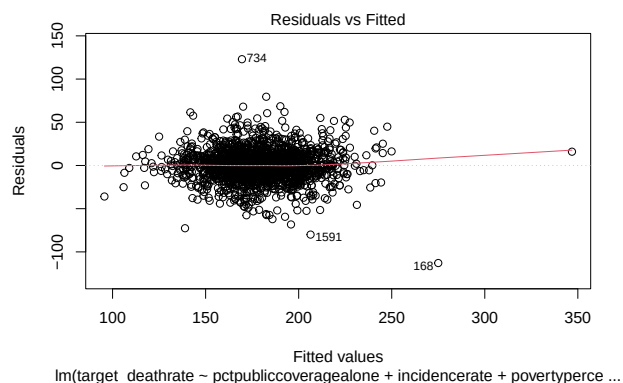
## -112.898   -9.939   -0.042    9.888  122.987
##
## Coefficients:
##
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)    1.021e+02  1.034e+01   9.876 < 2e-16 ***
## pctpubliccoveragealone  5.233e-01  1.639e-01   3.193 0.001435 **
## incidencerate    1.944e-01  9.244e-03  21.034 < 2e-16 ***
## povertypercent    1.872e-01  1.821e-01   1.028 0.303899
## pctbachdeg25_over -1.118e+00  2.005e-01  -5.577 2.83e-08 ***
## medincome       4.736e-06  1.286e-04   0.037 0.970638
## pctemployed16_over -1.429e-01  1.058e-01  -1.351 0.176926
## f.stateAlaska    1.937e+01  6.842e+00   2.831 0.004694 **
## f.stateArizona   -2.656e+01  7.390e+00  -3.594 0.000334 ***
## f.stateArkansas   1.122e+01  4.324e+00   2.596 0.009512 **
## f.stateCalifornia -2.005e+01  4.782e+00  -4.194 2.88e-05 ***
## f.stateColorado  -1.770e+01  4.695e+00  -3.770 0.000169 ***
## f.stateConnecticut -2.054e+01  7.857e+00  -2.614 0.009027 **
## f.stateDelaware  -4.066e+00  1.883e+01  -0.216 0.829087
## f.stateFlorida   -1.463e+00  4.396e+00  -0.333 0.739346
## f.stateGeorgia   -8.264e+00  3.647e+00  -2.266 0.023570 *
## f.stateHawaii    -1.961e+01  1.350e+01  -1.453 0.146509
## f.stateIdaho     -2.146e+01  4.902e+00  -4.378 1.27e-05 ***
## f.stateIllinois  -1.719e+00  4.096e+00  -0.420 0.674769
## f.stateIndiana    7.481e+00  4.058e+00   1.843 0.065453 .
## f.stateIowa      -8.120e+00  4.153e+00  -1.955 0.050749 .
## f.stateKansas    -3.459e+00  4.073e+00  -0.849 0.395824
## f.stateKentucky   9.076e+00  3.862e+00   2.350 0.018869 *
## f.stateLouisiana  3.835e+00  4.302e+00   0.891 0.372821
## f.stateMaine      1.114e+00  6.730e+00   0.166 0.868565
## f.stateMaryland  -6.409e-01  6.099e+00  -0.105 0.916322
## f.stateMassachusetts -9.308e+00  7.476e+00  -1.245 0.213274
## f.stateMichigan  -1.457e+00  4.108e+00  -0.355 0.722908
## f.stateMinnesota -1.143e+01  4.275e+00  -2.673 0.007578 **
## f.stateMississippi  7.234e+00  3.997e+00   1.810 0.070514 .
## f.stateMissouri   9.606e+00  3.894e+00   2.467 0.013713 *
## f.stateMontana   -1.620e+01  5.146e+00  -3.148 0.001670 **
## f.stateNebraska  -5.794e+00  4.291e+00  -1.350 0.177160
## f.stateNevada    -4.187e-01  5.971e+00  -0.070 0.944103
## f.stateNew Hampshire -3.048e+00  8.257e+00  -0.369 0.712024
## f.stateNew Jersey -7.867e+00  6.628e+00  -1.187 0.235417
## f.stateNew Mexico -2.319e+01  5.341e+00  -4.342 1.49e-05 ***
## f.stateNew York  -1.626e+01  4.380e+00  -3.713 0.000212 ***
## f.stateNorth Carolina -1.025e+01  3.931e+00  -2.607 0.009222 **
## f.stateNorth Dakota -2.489e+00  4.792e+00  -0.519 0.603601
## f.stateOhio       3.185e+00  4.147e+00   0.768 0.442545
## f.stateOklahoma   1.327e+01  4.224e+00   3.142 0.001704 **
## f.stateOregon     -9.778e+00  5.315e+00  -1.840 0.065997 .
## f.statePennsylvania -9.793e+00  4.285e+00  -2.286 0.022400 *
## f.stateRhode Island -5.163e+00  1.128e+01  -0.458 0.647151
## f.stateSouth Carolina 1.320e+00  4.591e+00   0.287 0.773809
## f.stateSouth Dakota -7.078e+00  4.537e+00  -1.560 0.118928
## f.stateTennessee   8.670e+00  3.941e+00   2.200 0.027951 *
## f.stateTexas      -5.362e-01  3.597e+00  -0.149 0.881505
## f.stateUtah       -2.378e+01  5.460e+00  -4.355 1.41e-05 ***

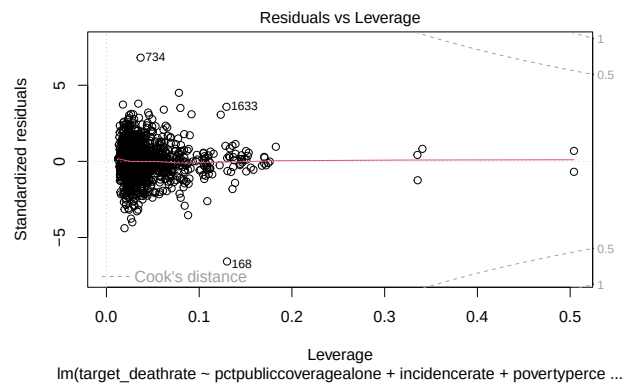
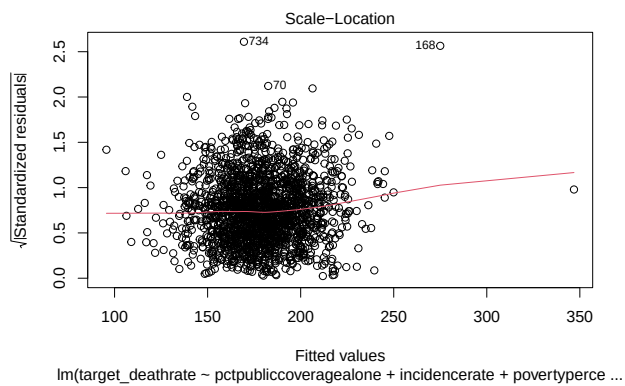
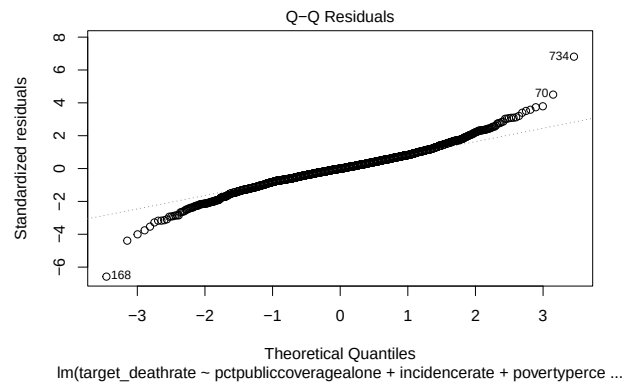
```

```
## f.stateVermont          4.056e+00  7.970e+00   0.509 0.610860
## f.stateVirginia        4.722e+00  3.862e+00   1.223 0.221634
## f.stateWashington      -1.063e+01  5.151e+00  -2.064 0.039148 *
## f.stateWest Virginia   2.970e+00  4.513e+00   0.658 0.510622
## f.stateWisconsin       -3.208e+00  4.374e+00  -0.733 0.463472
## f.stateWyoming         -3.402e+00  6.117e+00  -0.556 0.578129
## binnedinc(37413.8, 40362.7] -2.611e+00  2.457e+00  -1.062 0.288159
## binnedinc(40362.7, 42724.4] -8.190e+00  3.610e+00  -2.269 0.023403 *
## binnedinc(42724.4, 45201]  -6.988e+00  3.691e+00  -1.893 0.058478 .
## binnedinc(45201, 48021.6]  -9.048e+00  5.416e+00  -1.671 0.094991 .
## binnedinc(48021.6, 51046.4] -1.026e+01  5.866e+00  -1.750 0.080341 .
## binnedinc(51046.4, 54545.6] -1.168e+01  6.076e+00  -1.923 0.054674 .
## binnedinc(54545.6, 61494.5] -1.288e+01  6.694e+00  -1.924 0.054487 .
## binnedinc(61494.5, 125635]  -1.124e+01  7.432e+00  -1.513 0.130504
## binnedinc[22640, 34218.1]   4.164e+00  2.143e+00   1.943 0.052181 .
## f.incomeLow            5.596e+00  2.959e+00   1.891 0.058753 .
## f.incomeModerate       7.085e+00  5.320e+00   1.332 0.183143
## f.incomeHigh           9.932e+00  6.039e+00   1.645 0.100238
## f.education25_overBalanced Eduaction  5.283e-01  1.703e+00   0.310 0.756426
## f.education25_overMostly Bachelor's -1.164e+00  3.577e+00  -0.325 0.744891
## f.unemploymentModerate Unemployment  2.090e+00  1.510e+00   1.384 0.166392
## f.unemploymentHigh Unemployment  4.562e+00  2.034e+00   2.243 0.025017 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 18.4 on 1759 degrees of freedom
## Multiple R-squared:  0.5828, Adjusted R-squared:  0.566
## F-statistic: 34.61 on 71 and 1759 DF, p-value: < 2.2e-16
```

```
plot(m1)
```

```
## Warning: not plotting observations with leverage one:
## 1481
```



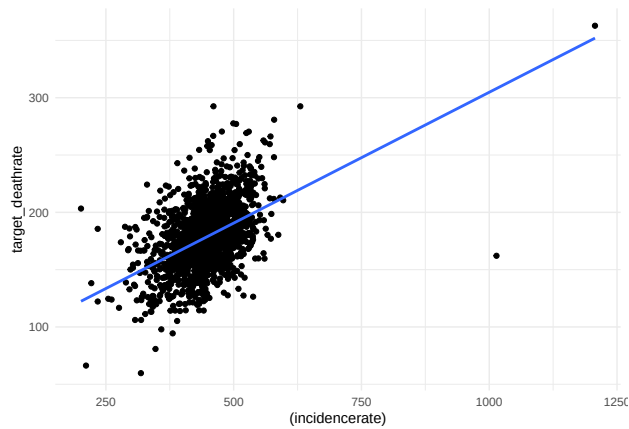


Following, we present several plots of the relationship between the target variable and the numeric variables, in order to know if they should be transformed.

By the interpretation of the scatterplots of the target variable vs. the numeric variables *incidencerate*, *povertypercent*, *medincome*, and *pctemployed16_over* variables seem to require a transformation. In the next model, we transform them to avoid

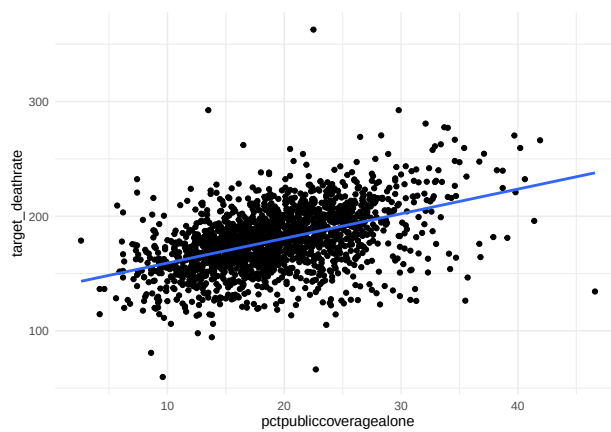
```
#Incidencerate
ggplot(df, aes(x = (incidencerate), y = target_deathrate)) +
  geom_point() +
  geom_smooth(method = "lm", se = FALSE) +
  theme_minimal()
```

```
## `geom_smooth()` using formula = 'y ~ x'
```



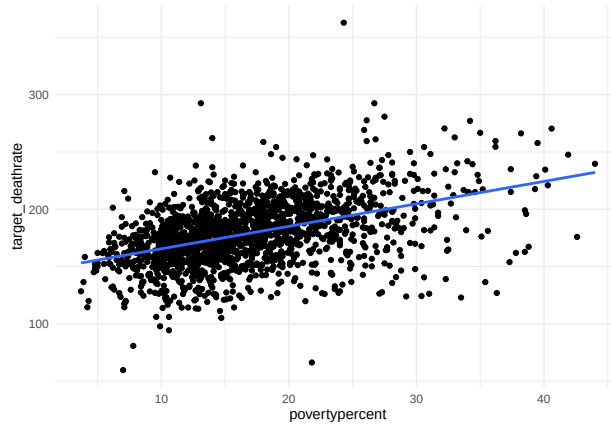
```
#pctpubliccoveragealone  
ggplot(df, aes(x = pctpubliccoveragealone, y = target_deathrate)) +  
  geom_point() +  
  geom_smooth(method = "lm", se = FALSE) +  
  theme_minimal()
```

```
## `geom_smooth()` using formula = 'y ~ x'
```



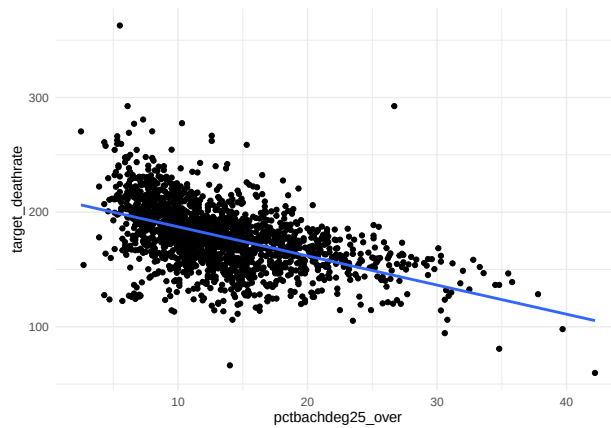
```
#povertypercent  
ggplot(df, aes(x = povertypercent, y = target_deathrate)) +  
  geom_point() +  
  geom_smooth(method = "lm", se = FALSE) +  
  theme_minimal()
```

```
## `geom_smooth()` using formula = 'y ~ x'
```

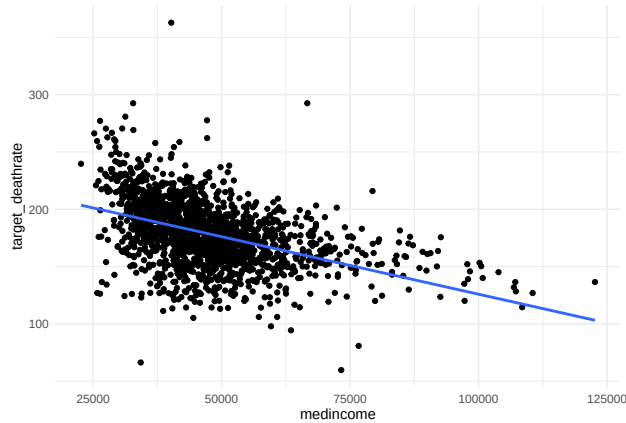
```
#pctbachdeg25_over
ggplot(df, aes(x = pctbachdeg25_over, y = target_deathrate)) +
  geom_point() +
  geom_smooth(method = "lm", se = FALSE) +
  theme_minimal()
```

```
## `geom_smooth()` using formula = 'y ~ x'
```



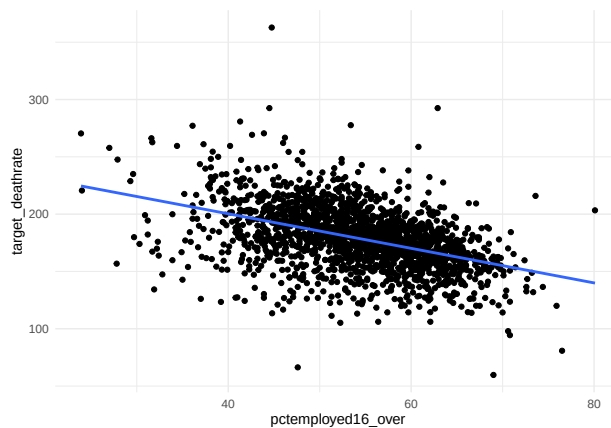
```
#medincome
ggplot(df, aes(x = medincome, y = target_deathrate)) +
  geom_point() +
  geom_smooth(method = "lm", se = FALSE) +
  theme_minimal()
```

```
## `geom_smooth()` using formula = 'y ~ x'
```



```
#pctemployed16_over
ggplot(df, aes(x = pctemployed16_over, y = target_deathrate)) +
  geom_point() +
  geom_smooth(method = "lm", se = FALSE) +
  theme_minimal()
```

```
## `geom_smooth()` using formula = 'y ~ x'
```



Apply transformations to variables which shown non-linear relations due to the high skewness of the distribution:

```
m2 <- lm(target_deathrate ~ (pctpubliccoveragealone) + sqrt(incidencerate) +
  log(povertypercent) + pctbachdeg25_over + log(medincome) +
  log(pctemployed16_over) + f.state + binnedinc + f.income +
  f.education25_over + f.unemployment, data = df)
summary(m2)
```

```
##
## Call:
## lm(formula = target_deathrate ~ (pctpubliccoveragealone) + sqrt(incidencerate) +
##   log(povertypercent) + pctbachdeg25_over + log(medincome) +
##   log(pctemployed16_over) + f.state + binnedinc + f.income +
##   f.education25_over + f.unemployment, data = df)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -95.132  -10.269   -0.134    9.800  122.296
```

```

##
## Coefficients:
##
## Estimate Std. Error t value Pr(>|t|)
## (Intercept) -21.0018 106.0807 -0.198 0.843084
## pctpubliccoveragealone 0.5027 0.1615 3.112 0.001888 **
## sqrt(incidencerate) 8.4755 0.3984 21.275 < 2e-16 ***
## log(povertypercent) 4.5870 3.3270 1.379 0.168163
## pctbachdeg25_over -1.1160 0.1990 -5.609 2.36e-08 ***
## log(medincome) 4.9985 9.6318 0.519 0.603856
## log(pctemployed16_over) -9.6450 5.0352 -1.915 0.055591 .
## f.stateAlaska 20.9049 6.8448 3.054 0.002291 **
## f.stateArizona -24.9218 7.3889 -3.373 0.000760 ***
## f.stateArkansas 11.4387 4.3114 2.653 0.008047 **
## f.stateCalifornia -20.1496 4.7824 -4.213 2.64e-05 ***
## f.stateColorado -16.3163 4.6890 -3.480 0.000514 ***
## f.stateConnecticut -20.5459 7.8425 -2.620 0.008873 **
## f.stateDelaware -3.8828 18.7864 -0.207 0.836282
## f.stateFlorida -0.4637 4.3896 -0.106 0.915875
## f.stateGeorgia -8.1295 3.6312 -2.239 0.025293 *
## f.stateHawaii -19.2343 13.4724 -1.428 0.153562
## f.stateIdaho -20.8284 4.8882 -4.261 2.14e-05 ***
## f.stateIllinois -1.4717 4.0884 -0.360 0.718911
## f.stateIndiana 7.9216 4.0530 1.954 0.050801 .
## f.stateIowa -7.8199 4.1363 -1.891 0.058851 .
## f.stateKansas -3.2013 4.0552 -0.789 0.429965
## f.stateKentucky 9.3305 3.8498 2.424 0.015467 *
## f.stateLouisiana 3.6709 4.3007 0.854 0.393471
## f.stateMaine 1.4042 6.7111 0.209 0.834284
## f.stateMaryland -0.5257 6.0789 -0.086 0.931099
## f.stateMassachusetts -9.2023 7.4586 -1.234 0.217452
## f.stateMichigan -0.9036 4.0988 -0.220 0.825542
## f.stateMinnesota -10.9978 4.2630 -2.580 0.009967 **
## f.stateMississippi 7.4620 3.9829 1.874 0.061161 .
## f.stateMissouri 10.1636 3.8833 2.617 0.008941 **
## f.stateMontana -15.8269 5.1310 -3.085 0.002070 **
## f.stateNebraska -4.9685 4.2754 -1.162 0.245352
## f.stateNevada -0.5051 5.9576 -0.085 0.932448
## f.stateNew Hampshire -2.7265 8.2349 -0.331 0.740612
## f.stateNew Jersey -7.6167 6.6006 -1.154 0.248682
## f.stateNew Mexico -22.2153 5.3300 -4.168 3.22e-05 ***
## f.stateNew York -16.2355 4.3735 -3.712 0.000212 ***
## f.stateNorth Carolina -10.2100 3.9182 -2.606 0.009244 **
## f.stateNorth Dakota -1.6144 4.7845 -0.337 0.735844
## f.stateOhio 3.4626 4.1418 0.836 0.403272
## f.stateOklahoma 13.4081 4.2135 3.182 0.001487 **
## f.stateOregon -9.6791 5.3033 -1.825 0.068153 .
## f.statePennsylvania -9.6464 4.2764 -2.256 0.024210 *
## f.stateRhode Island -4.6961 11.2339 -0.418 0.675978
## f.stateSouth Carolina 1.3632 4.5805 0.298 0.766040
## f.stateSouth Dakota -6.6170 4.5122 -1.466 0.142695
## f.stateTennessee 8.7273 3.9294 2.221 0.026477 *
## f.stateTexas 0.2627 3.5982 0.073 0.941799
## f.stateUtah -22.7932 5.4469 -4.185 3.00e-05 ***
## f.stateVermont 4.5627 7.9369 0.575 0.565453

```

```
## f.stateVirginia          5.6487      3.8559      1.465 0.143112
## f.stateWashington       -10.5433     5.1383     -2.052 0.040327 *
## f.stateWest Virginia    3.2376      4.4953      0.720 0.471486
## f.stateWisconsin        -2.4955      4.3735     -0.571 0.568342
## f.stateWyoming          -2.9547      6.0973     -0.485 0.628029
## binnedinc(37413.8, 40362.7] -2.8432      2.4827     -1.145 0.252283
## binnedinc(40362.7, 42724.4] -9.1064      3.6659     -2.484 0.013081 *
## binnedinc(42724.4, 45201]  -8.0235      3.8114     -2.105 0.035421 *
## binnedinc(45201, 48021.6] -10.2911      5.5328     -1.860 0.063048 .
## binnedinc(48021.6, 51046.4] -11.7063      6.0387     -1.939 0.052719 .
## binnedinc(51046.4, 54545.6] -13.2825      6.3125     -2.104 0.035505 *
## binnedinc(54545.6, 61494.5] -14.4211      7.0065     -2.058 0.039714 *
## binnedinc(61494.5, 125635] -13.1667      7.9536     -1.655 0.098012 .
## binnedinc[22640, 34218.1]   4.8166      2.2838      2.109 0.035085 *
## f.incomeLow             5.9477      2.9547      2.013 0.044271 *
## f.incomeModerate        7.4687      5.3155      1.405 0.160174
## f.incomeHigh           10.3300      6.0430      1.709 0.087551 .
## f.education25_overBalanced Eduaction 0.4166      1.6997      0.245 0.806426
## f.education25_overMostly Bachelor's -1.3055      3.5626     -0.366 0.714073
## f.unemploymentModerate Unemployment 1.6437      1.5172      1.083 0.278795
## f.unemploymentHigh Unemployment 3.9391      2.0355      1.935 0.053121 .
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 18.35 on 1759 degrees of freedom
## Multiple R-squared:  0.585, Adjusted R-squared:  0.5682
## F-statistic: 34.92 on 71 and 1759 DF, p-value: < 2.2e-16
```

```
vif(m2)
```

```
##              GVIF Df GVIF^(1/(2*Df))
## pctpubliccoveragealone  5.200285  1      2.280413
## sqrt(incidencerate)    1.475880  1      1.214858
## log(povertypercent)     8.909008  1      2.984796
## pctbachdeg25_over      6.436157  1      2.536958
## log(medincome)         29.423643  1      5.424356
## log(pctemployed16_over)  3.506749  1      1.872632
## f.state                23.138846 49      1.032576
## binnedinc              3910.128127 9      1.583311
## f.income                843.773831 3      3.074003
## f.education25_over      5.268775  2      1.515052
## f.unemployment         3.086683  2      1.325479
```

The VIF function allows us to see that there is a lot of multicollinearity introduced by some variables. The variables that are introducing more multicollinearity to the model are: *binnedinc* (GVIF = 3910.12), *f.income* (GVIF=843.77). We remove them from the model

```
m3 <- lm(target_deathrate ~ ((pctpubliccoveragealone) + sqrt(incidencerate)
+ log(povertypercent)+ pctbachdeg25_over +
log(medincome)+
log(pctemployed16_over))*
(f.state + f.education25_over
+ f.unemployment), data = df)
summary(m3)
```

```
##
```

```

## Call:
## lm(formula = target_deathrate ~ ((pctpubliccoveragealone) + sqrt(incidencerate) +
##   log(povertypercent) + pctbachdeg25_over + log(medincome) +
##   log(pctemployed16_over)) * (f.state + f.education25_over +
##   f.unemployment), data = df)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -88.758  -8.610   0.000   8.694  83.997
##
## Coefficients: (16 not defined because of singularities)
##
##                                     Estimate
## (Intercept)                        1.179e+02
## pctpubliccoveragealone             -2.968e-03
## sqrt(incidencerate)                7.622e+00
## log(povertypercent)                1.346e+00
## pctbachdeg25_over                 -1.284e+00
## log(medincome)                    -2.890e+01
## log(pctemployed16_over)            5.726e+01
## f.stateAlaska                     -1.253e+03
## f.stateArizona                     -6.247e+02
## f.stateArkansas                    -2.168e+02
## f.stateCalifornia                  2.928e+01
## f.stateColorado                    -2.668e+02
## f.stateConnecticut                 4.266e+03
## f.stateDelaware                    -6.091e+00
## f.stateFlorida                     -3.713e+02
## f.stateGeorgia                     7.900e+02
## f.stateHawaii                      3.509e+02
## f.stateIdaho                       -1.406e+03
## f.stateIllinois                    -7.411e+02
## f.stateIndiana                     -4.544e+02
## f.stateIowa                        -2.016e+02
## f.stateKansas                      -2.111e+03
## f.stateKentucky                    -8.964e+01
## f.stateLouisiana                   1.130e+02
## f.stateMaine                       -9.370e+02
## f.stateMaryland                    -1.686e+03
## f.stateMassachusetts                -1.371e+02
## f.stateMichigan                     -1.962e+02
## f.stateMinnesota                   -2.580e+03
## f.stateMississippi                  -1.092e+03
## f.stateMissouri                     -3.611e+02
## f.stateMontana                      6.037e+00
## f.stateNebraska                     4.986e+02
## f.stateNevada                       1.722e+03
## f.stateNew Hampshire                -2.843e+03
## f.stateNew Jersey                   7.800e+02
## f.stateNew Mexico                   8.517e+02
## f.stateNew York                     2.820e+02
## f.stateNorth Carolina               -8.271e+02
## f.stateNorth Dakota                 -4.020e+02
## f.stateOhio                         -5.239e+02
## f.stateOklahoma                    -5.396e+02

```

## f.stateOregon	-4.234e+02
## f.statePennsylvania	-3.653e+02
## f.stateRhode Island	8.165e+03
## f.stateSouth Carolina	-5.243e+02
## f.stateSouth Dakota	5.158e+02
## f.stateTennessee	-6.934e+02
## f.stateTexas	4.633e+02
## f.stateUtah	2.459e+02
## f.stateVermont	1.228e+04
## f.stateVirginia	-5.071e+02
## f.stateWashington	3.683e+02
## f.stateWest Virginia	3.523e+02
## f.stateWisconsin	3.552e+02
## f.stateWyoming	8.699e+00
## f.education25_overBalanced Eduaction	-1.364e+02
## f.education25_overMostly Bachelor's	-2.271e+02
## f.unemploymentModerate Unemployment	8.226e+01
## f.unemploymentHigh Unemployment	1.864e+01
## pctpubliccoveragealone:f.stateAlaska	3.452e+00
## pctpubliccoveragealone:f.stateArizona	3.909e+00
## pctpubliccoveragealone:f.stateArkansas	1.690e+00
## pctpubliccoveragealone:f.stateCalifornia	6.832e-01
## pctpubliccoveragealone:f.stateColorado	2.719e+00
## pctpubliccoveragealone:f.stateConnecticut	-1.206e+00
## pctpubliccoveragealone:f.stateDelaware	NA
## pctpubliccoveragealone:f.stateFlorida	-3.749e+00
## pctpubliccoveragealone:f.stateGeorgia	-1.593e+00
## pctpubliccoveragealone:f.stateHawaii	-2.079e+01
## pctpubliccoveragealone:f.stateIdaho	-3.482e+00
## pctpubliccoveragealone:f.stateIllinois	-4.749e-01
## pctpubliccoveragealone:f.stateIndiana	5.940e-01
## pctpubliccoveragealone:f.stateIowa	1.591e+00
## pctpubliccoveragealone:f.stateKansas	-1.772e+00
## pctpubliccoveragealone:f.stateKentucky	-4.974e-01
## pctpubliccoveragealone:f.stateLouisiana	9.006e-01
## pctpubliccoveragealone:f.stateMaine	-1.583e+00
## pctpubliccoveragealone:f.stateMaryland	3.398e+00
## pctpubliccoveragealone:f.stateMassachusetts	-5.697e+00
## pctpubliccoveragealone:f.stateMichigan	6.659e-01
## pctpubliccoveragealone:f.stateMinnesota	7.982e-01
## pctpubliccoveragealone:f.stateMississippi	1.990e+00
## pctpubliccoveragealone:f.stateMissouri	1.754e+00
## pctpubliccoveragealone:f.stateMontana	1.302e+00
## pctpubliccoveragealone:f.stateNebraska	1.977e-01
## pctpubliccoveragealone:f.stateNevada	-7.376e+00
## pctpubliccoveragealone:f.stateNew Hampshire	-6.766e+00
## pctpubliccoveragealone:f.stateNew Jersey	-6.238e+00
## pctpubliccoveragealone:f.stateNew Mexico	-1.153e+00
## pctpubliccoveragealone:f.stateNew York	-1.511e+00
## pctpubliccoveragealone:f.stateNorth Carolina	-3.610e-01
## pctpubliccoveragealone:f.stateNorth Dakota	-8.641e-01
## pctpubliccoveragealone:f.stateOhio	3.518e-01
## pctpubliccoveragealone:f.stateOklahoma	3.974e+00
## pctpubliccoveragealone:f.stateOregon	1.258e+00

## pctpubliccoveragealone:f.statePennsylvania	2.290e+00
## pctpubliccoveragealone:f.stateRhode Island	1.445e+02
## pctpubliccoveragealone:f.stateSouth Carolina	6.771e-01
## pctpubliccoveragealone:f.stateSouth Dakota	1.570e+00
## pctpubliccoveragealone:f.stateTennessee	1.389e+00
## pctpubliccoveragealone:f.stateTexas	-1.469e+00
## pctpubliccoveragealone:f.stateUtah	-1.776e+00
## pctpubliccoveragealone:f.stateVermont	9.733e+00
## pctpubliccoveragealone:f.stateVirginia	1.969e+00
## pctpubliccoveragealone:f.stateWashington	-1.551e+00
## pctpubliccoveragealone:f.stateWest Virginia	1.729e+00
## pctpubliccoveragealone:f.stateWisconsin	4.221e-01
## pctpubliccoveragealone:f.stateWyoming	-7.060e-01
## pctpubliccoveragealone:f.education25_overBalanced Eduaction	-1.370e-01
## pctpubliccoveragealone:f.education25_overMostly Bachelor's	1.998e+00
## pctpubliccoveragealone:f.unemploymentModerate Unemployment	2.572e-01
## pctpubliccoveragealone:f.unemploymentHigh Unemployment	-1.777e-02
## sqrt(incidencerate):f.stateAlaska	4.240e+00
## sqrt(incidencerate):f.stateArizona	-1.662e+01
## sqrt(incidencerate):f.stateArkansas	-3.285e+00
## sqrt(incidencerate):f.stateCalifornia	2.035e+00
## sqrt(incidencerate):f.stateColorado	1.178e+00
## sqrt(incidencerate):f.stateConnecticut	-1.406e+00
## sqrt(incidencerate):f.stateDelaware	NA
## sqrt(incidencerate):f.stateFlorida	2.202e+00
## sqrt(incidencerate):f.stateGeorgia	3.591e+00
## sqrt(incidencerate):f.stateHawaii	NA
## sqrt(incidencerate):f.stateIdaho	6.020e+00
## sqrt(incidencerate):f.stateIllinois	2.845e+00
## sqrt(incidencerate):f.stateIndiana	2.344e+00
## sqrt(incidencerate):f.stateIowa	1.782e+00
## sqrt(incidencerate):f.stateKansas	1.417e+02
## sqrt(incidencerate):f.stateKentucky	5.138e+00
## sqrt(incidencerate):f.stateLouisiana	-2.187e+00
## sqrt(incidencerate):f.stateMaine	-1.212e+01
## sqrt(incidencerate):f.stateMaryland	1.685e+01
## sqrt(incidencerate):f.stateMassachusetts	1.397e+01
## sqrt(incidencerate):f.stateMichigan	-5.101e+00
## sqrt(incidencerate):f.stateMinnesota	1.472e+02
## sqrt(incidencerate):f.stateMississippi	-3.740e+00
## sqrt(incidencerate):f.stateMissouri	-1.651e+00
## sqrt(incidencerate):f.stateMontana	-5.944e-01
## sqrt(incidencerate):f.stateNebraska	-2.649e+00
## sqrt(incidencerate):f.stateNevada	8.036e+01
## sqrt(incidencerate):f.stateNew Hampshire	1.490e+01
## sqrt(incidencerate):f.stateNew Jersey	-1.746e+01
## sqrt(incidencerate):f.stateNew Mexico	5.078e+00
## sqrt(incidencerate):f.stateNew York	1.356e+00
## sqrt(incidencerate):f.stateNorth Carolina	-1.019e-01
## sqrt(incidencerate):f.stateNorth Dakota	4.029e+00
## sqrt(incidencerate):f.stateOhio	3.312e+00
## sqrt(incidencerate):f.stateOklahoma	-6.116e+00
## sqrt(incidencerate):f.stateOregon	-4.288e+00
## sqrt(incidencerate):f.statePennsylvania	3.855e-01

## sqrt(incidencerate):f.stateRhode Island	-4.506e+02
## sqrt(incidencerate):f.stateSouth Carolina	2.435e+00
## sqrt(incidencerate):f.stateSouth Dakota	-2.142e+00
## sqrt(incidencerate):f.stateTennessee	2.175e+00
## sqrt(incidencerate):f.stateTexas	3.127e+00
## sqrt(incidencerate):f.stateUtah	-1.042e+00
## sqrt(incidencerate):f.stateVermont	-6.350e+01
## sqrt(incidencerate):f.stateVirginia	-2.375e+00
## sqrt(incidencerate):f.stateWashington	-5.221e+00
## sqrt(incidencerate):f.stateWest Virginia	-7.913e-01
## sqrt(incidencerate):f.stateWisconsin	-5.012e+00
## sqrt(incidencerate):f.stateWyoming	1.394e+01
## sqrt(incidencerate):f.education25_overBalanced Eduaction	1.860e+00
## sqrt(incidencerate):f.education25_overMostly Bachelor's	-1.845e+00
## sqrt(incidencerate):f.unemploymentModerate Unemployment	1.072e+00
## sqrt(incidencerate):f.unemploymentHigh Unemployment	4.931e-01
## log(povertypercent):f.stateAlaska	-9.713e+00
## log(povertypercent):f.stateArizona	-6.373e+01
## log(povertypercent):f.stateArkansas	1.817e+01
## log(povertypercent):f.stateCalifornia	1.739e+01
## log(povertypercent):f.stateColorado	-2.790e+01
## log(povertypercent):f.stateConnecticut	-7.948e+01
## log(povertypercent):f.stateDelaware	NA
## log(povertypercent):f.stateFlorida	4.535e+01
## log(povertypercent):f.stateGeorgia	-2.099e+01
## log(povertypercent):f.stateHawaii	NA
## log(povertypercent):f.stateIdaho	9.253e+01
## log(povertypercent):f.stateIllinois	3.829e+01
## log(povertypercent):f.stateIndiana	2.180e+01
## log(povertypercent):f.stateIowa	1.828e+01
## log(povertypercent):f.stateKansas	-2.747e+01
## log(povertypercent):f.stateKentucky	2.324e+01
## log(povertypercent):f.stateLouisiana	1.831e+01
## log(povertypercent):f.stateMaine	4.377e+01
## log(povertypercent):f.stateMaryland	2.233e+01
## log(povertypercent):f.stateMassachusetts	8.874e+01
## log(povertypercent):f.stateMichigan	1.652e+01
## log(povertypercent):f.stateMinnesota	-4.648e+01
## log(povertypercent):f.stateMississippi	5.095e+01
## log(povertypercent):f.stateMissouri	2.218e+01
## log(povertypercent):f.stateMontana	-1.568e+01
## log(povertypercent):f.stateNebraska	-2.129e+01
## log(povertypercent):f.stateNevada	-2.718e+01
## log(povertypercent):f.stateNew Hampshire	6.268e+01
## log(povertypercent):f.stateNew Jersey	5.759e+01
## log(povertypercent):f.stateNew Mexico	-4.798e+01
## log(povertypercent):f.stateNew York	1.212e+01
## log(povertypercent):f.stateNorth Carolina	5.478e+01
## log(povertypercent):f.stateNorth Dakota	6.918e+00
## log(povertypercent):f.stateOhio	2.227e+01
## log(povertypercent):f.stateOklahoma	3.352e+01
## log(povertypercent):f.stateOregon	9.862e+00
## log(povertypercent):f.statePennsylvania	9.128e+00
## log(povertypercent):f.stateRhode Island	NA

## log(povertypercent):f.stateSouth Carolina	5.536e+01
## log(povertypercent):f.stateSouth Dakota	-8.258e+00
## log(povertypercent):f.stateTennessee	3.796e+01
## log(povertypercent):f.stateTexas	-8.512e+00
## log(povertypercent):f.stateUtah	2.534e+01
## log(povertypercent):f.stateVermont	-7.979e+02
## log(povertypercent):f.stateVirginia	1.633e+01
## log(povertypercent):f.stateWashington	-4.090e-01
## log(povertypercent):f.stateWest Virginia	-3.533e+01
## log(povertypercent):f.stateWisconsin	1.206e+00
## log(povertypercent):f.stateWyoming	-5.020e+01
## log(povertypercent):f.education25_overBalanced Eduaction	-3.944e+00
## log(povertypercent):f.education25_overMostly Bachelor's	-7.566e+00
## log(povertypercent):f.unemploymentModerate Unemployment	-6.283e+00
## log(povertypercent):f.unemploymentHigh Unemployment	-6.687e+00
## pctbachdeg25_over:f.stateAlaska	-4.574e+00
## pctbachdeg25_over:f.stateArizona	5.692e+00
## pctbachdeg25_over:f.stateArkansas	-6.521e-01
## pctbachdeg25_over:f.stateCalifornia	1.660e+00
## pctbachdeg25_over:f.stateColorado	-9.980e-01
## pctbachdeg25_over:f.stateConnecticut	6.017e+00
## pctbachdeg25_over:f.stateDelaware	NA
## pctbachdeg25_over:f.stateFlorida	-2.475e+00
## pctbachdeg25_over:f.stateGeorgia	-1.889e-02
## pctbachdeg25_over:f.stateHawaii	NA
## pctbachdeg25_over:f.stateIdaho	-1.334e+00
## pctbachdeg25_over:f.stateIllinois	-1.251e+00
## pctbachdeg25_over:f.stateIndiana	3.642e-01
## pctbachdeg25_over:f.stateIowa	1.599e+00
## pctbachdeg25_over:f.stateKansas	4.540e-01
## pctbachdeg25_over:f.stateKentucky	-1.030e-03
## pctbachdeg25_over:f.stateLouisiana	6.209e-01
## pctbachdeg25_over:f.stateMaine	-3.206e+00
## pctbachdeg25_over:f.stateMaryland	-1.488e+00
## pctbachdeg25_over:f.stateMassachusetts	-4.333e-02
## pctbachdeg25_over:f.stateMichigan	-9.030e-01
## pctbachdeg25_over:f.stateMinnesota	8.469e-01
## pctbachdeg25_over:f.stateMississippi	1.611e+00
## pctbachdeg25_over:f.stateMissouri	-3.110e-01
## pctbachdeg25_over:f.stateMontana	1.729e+00
## pctbachdeg25_over:f.stateNebraska	1.160e+00
## pctbachdeg25_over:f.stateNevada	-5.769e+00
## pctbachdeg25_over:f.stateNew Hampshire	-1.601e+01
## pctbachdeg25_over:f.stateNew Jersey	-2.127e+00
## pctbachdeg25_over:f.stateNew Mexico	-2.874e+00
## pctbachdeg25_over:f.stateNew York	1.071e-01
## pctbachdeg25_over:f.stateNorth Carolina	-2.479e-01
## pctbachdeg25_over:f.stateNorth Dakota	1.840e-01
## pctbachdeg25_over:f.stateOhio	4.597e-02
## pctbachdeg25_over:f.stateOklahoma	1.532e+00
## pctbachdeg25_over:f.stateOregon	7.845e-01
## pctbachdeg25_over:f.statePennsylvania	5.232e-01
## pctbachdeg25_over:f.stateRhode Island	NA
## pctbachdeg25_over:f.stateSouth Carolina	2.191e-01

## pctbachdeg25_over:f.stateSouth Dakota	1.854e+00
## pctbachdeg25_over:f.stateTennessee	3.810e-01
## pctbachdeg25_over:f.stateTexas	3.580e-01
## pctbachdeg25_over:f.stateUtah	2.190e-01
## pctbachdeg25_over:f.stateVermont	1.811e+01
## pctbachdeg25_over:f.stateVirginia	-5.857e-01
## pctbachdeg25_over:f.stateWashington	-1.891e+00
## pctbachdeg25_over:f.stateWest Virginia	3.735e+00
## pctbachdeg25_over:f.stateWisconsin	1.003e+00
## pctbachdeg25_over:f.stateWyoming	2.973e+00
## pctbachdeg25_over:f.education25_overBalanced Eduaction	-1.472e-01
## pctbachdeg25_over:f.education25_overMostly Bachelor's	-2.232e-01
## pctbachdeg25_over:f.unemploymentModerate Unemployment	2.945e-01
## pctbachdeg25_over:f.unemploymentHigh Unemployment	-1.898e-01
## log(medincome):f.stateAlaska	7.719e+00
## log(medincome):f.stateArizona	1.336e+02
## log(medincome):f.stateArkansas	6.952e+00
## log(medincome):f.stateCalifornia	7.920e-01
## log(medincome):f.stateColorado	1.826e+01
## log(medincome):f.stateConnecticut	-2.378e+02
## log(medincome):f.stateDelaware	NA
## log(medincome):f.stateFlorida	3.763e+01
## log(medincome):f.stateGeorgia	-7.294e+01
## log(medincome):f.stateHawaii	NA
## log(medincome):f.stateIdaho	1.234e+02
## log(medincome):f.stateIllinois	6.055e+01
## log(medincome):f.stateIndiana	5.670e+01
## log(medincome):f.stateIowa	4.893e+00
## log(medincome):f.stateKansas	-1.338e+01
## log(medincome):f.stateKentucky	2.367e+01
## log(medincome):f.stateLouisiana	6.204e+00
## log(medincome):f.stateMaine	7.692e+01
## log(medincome):f.stateMaryland	1.075e+02
## log(medincome):f.stateMassachusetts	1.919e+01
## log(medincome):f.stateMichigan	2.079e+01
## log(medincome):f.stateMinnesota	-4.626e+00
## log(medincome):f.stateMississippi	1.009e+02
## log(medincome):f.stateMissouri	1.399e+01
## log(medincome):f.stateMontana	7.661e+00
## log(medincome):f.stateNebraska	2.232e+01
## log(medincome):f.stateNevada	-3.279e+02
## log(medincome):f.stateNew Hampshire	2.519e+02
## log(medincome):f.stateNew Jersey	8.524e+01
## log(medincome):f.stateNew Mexico	-1.025e+02
## log(medincome):f.stateNew York	-1.726e+01
## log(medincome):f.stateNorth Carolina	7.009e+01
## log(medincome):f.stateNorth Dakota	3.375e+01
## log(medincome):f.stateOhio	4.396e+01
## log(medincome):f.stateOklahoma	1.974e+01
## log(medincome):f.stateOregon	6.467e+01
## log(medincome):f.statePennsylvania	4.331e+01
## log(medincome):f.stateRhode Island	NA
## log(medincome):f.stateSouth Carolina	5.462e+00
## log(medincome):f.stateSouth Dakota	-3.114e+01

## log(medincome):f.stateTennessee	6.903e+01
## log(medincome):f.stateTexas	-2.519e+01
## log(medincome):f.stateUtah	-3.398e+01
## log(medincome):f.stateVermont	-4.051e+02
## log(medincome):f.stateVirginia	5.570e+01
## log(medincome):f.stateWashington	-5.211e+00
## log(medincome):f.stateWest Virginia	4.737e+01
## log(medincome):f.stateWisconsin	1.864e+01
## log(medincome):f.stateWyoming	1.380e+01
## log(medincome):f.education25_overBalanced Eduaction	2.434e+01
## log(medincome):f.education25_overMostly Bachelor's	1.941e+01
## log(medincome):f.unemploymentModerate Unemployment	5.006e-02
## log(medincome):f.unemploymentHigh Unemployment	1.480e+01
## log(pctemployed16_over):f.stateAlaska	2.760e+02
## log(pctemployed16_over):f.stateArizona	-1.271e+02
## log(pctemployed16_over):f.stateArkansas	3.403e+01
## log(pctemployed16_over):f.stateCalifornia	-4.888e+01
## log(pctemployed16_over):f.stateColorado	1.445e+01
## log(pctemployed16_over):f.stateConnecticut	-3.717e+02
## log(pctemployed16_over):f.stateDelaware	NA
## log(pctemployed16_over):f.stateFlorida	-2.503e+01
## log(pctemployed16_over):f.stateGeorgia	-4.646e-01
## log(pctemployed16_over):f.stateHawaii	NA
## log(pctemployed16_over):f.stateIdaho	-5.984e+01
## log(pctemployed16_over):f.stateIllinois	-1.423e+01
## log(pctemployed16_over):f.stateIndiana	-6.872e+01
## log(pctemployed16_over):f.stateIowa	-1.163e+00
## log(pctemployed16_over):f.stateKansas	-1.599e+02
## log(pctemployed16_over):f.stateKentucky	-8.343e+01
## log(pctemployed16_over):f.stateLouisiana	-5.306e+01
## log(pctemployed16_over):f.stateMaine	8.401e+01
## log(pctemployed16_over):f.stateMaryland	1.138e+01
## log(pctemployed16_over):f.stateMassachusetts	-1.220e+02
## log(pctemployed16_over):f.stateMichigan	7.480e+00
## log(pctemployed16_over):f.stateMinnesota	-9.906e+01
## log(pctemployed16_over):f.stateMississippi	-2.659e+01
## log(pctemployed16_over):f.stateMissouri	4.008e+01
## log(pctemployed16_over):f.stateMontana	-2.675e+01
## log(pctemployed16_over):f.stateNebraska	-1.629e+02
## log(pctemployed16_over):f.stateNevada	9.935e+01
## log(pctemployed16_over):f.stateNew Hampshire	NA
## log(pctemployed16_over):f.stateNew Jersey	-3.306e+02
## log(pctemployed16_over):f.stateNew Mexico	8.556e+01
## log(pctemployed16_over):f.stateNew York	-3.781e+01
## log(pctemployed16_over):f.stateNorth Carolina	-1.950e+01
## log(pctemployed16_over):f.stateNorth Dakota	-1.854e+01
## log(pctemployed16_over):f.stateOhio	-2.213e+01
## log(pctemployed16_over):f.stateOklahoma	6.887e+01
## log(pctemployed16_over):f.stateOregon	-6.488e+01
## log(pctemployed16_over):f.statePennsylvania	-4.860e+01
## log(pctemployed16_over):f.stateRhode Island	NA
## log(pctemployed16_over):f.stateSouth Carolina	5.948e+01
## log(pctemployed16_over):f.stateSouth Dakota	-4.578e+01
## log(pctemployed16_over):f.stateTennessee	-5.722e+01

## log(pctemployed16_over):f.stateTexas	-5.307e+01	
## log(pctemployed16_over):f.stateUtah	1.692e+01	
## log(pctemployed16_over):f.stateVermont	-1.246e+03	
## log(pctemployed16_over):f.stateVirginia	-2.766e+01	
## log(pctemployed16_over):f.stateWashington	-3.831e+01	
## log(pctemployed16_over):f.stateWest Virginia	-2.090e+02	
## log(pctemployed16_over):f.stateWisconsin	-1.197e+02	
## log(pctemployed16_over):f.stateWyoming	-8.984e+01	
## log(pctemployed16_over):f.education25_overBalanced Eduaction	-3.726e+01	
## log(pctemployed16_over):f.education25_overMostly Bachelor's	1.259e+01	
## log(pctemployed16_over):f.unemploymentModerate Unemployment	-2.420e+01	
## log(pctemployed16_over):f.unemploymentHigh Unemployment	-4.095e+01	
##	Std. Error	t value
## (Intercept)	9.297e+02	0.127
## pctpubliccoveragealone	2.238e+00	-0.001
## sqrt(incidencerate)	4.843e+00	1.574
## log(povertypercent)	3.811e+01	0.035
## pctbachdeg25_over	1.399e+00	-0.918
## log(medincome)	6.953e+01	-0.416
## log(pctemployed16_over)	5.141e+01	1.114
## f.stateAlaska	1.261e+03	-0.994
## f.stateArizona	3.821e+03	-0.163
## f.stateArkansas	1.036e+03	-0.209
## f.stateCalifornia	1.001e+03	0.029
## f.stateColorado	1.033e+03	-0.258
## f.stateConnecticut	2.375e+04	0.180
## f.stateDelaware	2.435e+01	-0.250
## f.stateFlorida	1.132e+03	-0.328
## f.stateGeorgia	9.523e+02	0.830
## f.stateHawaii	6.071e+02	0.578
## f.stateIdaho	1.165e+03	-1.207
## f.stateIllinois	1.010e+03	-0.734
## f.stateIndiana	1.048e+03	-0.433
## f.stateIowa	1.096e+03	-0.184
## f.stateKansas	1.209e+03	-1.747
## f.stateKentucky	1.019e+03	-0.088
## f.stateLouisiana	1.035e+03	0.109
## f.stateMaine	1.685e+03	-0.556
## f.stateMaryland	1.313e+03	-1.285
## f.stateMassachusetts	2.007e+03	-0.068
## f.stateMichigan	9.974e+02	-0.197
## f.stateMinnesota	1.452e+03	-1.776
## f.stateMississippi	1.020e+03	-1.071
## f.stateMissouri	1.015e+03	-0.356
## f.stateMontana	1.093e+03	0.006
## f.stateNebraska	1.012e+03	0.493
## f.stateNevada	2.183e+03	0.789
## f.stateNew Hampshire	2.596e+04	-0.110
## f.stateNew Jersey	2.535e+03	0.308
## f.stateNew Mexico	1.006e+03	0.847
## f.stateNew York	1.010e+03	0.279
## f.stateNorth Carolina	9.956e+02	-0.831
## f.stateNorth Dakota	1.052e+03	-0.382
## f.stateOhio	1.047e+03	-0.500

## f.stateOklahoma	1.084e+03	-0.498
## f.stateOregon	1.220e+03	-0.347
## f.statePennsylvania	1.064e+03	-0.343
## f.stateRhode Island	6.994e+03	1.167
## f.stateSouth Carolina	1.289e+03	-0.407
## f.stateSouth Dakota	1.040e+03	0.496
## f.stateTennessee	1.012e+03	-0.685
## f.stateTexas	9.294e+02	0.498
## f.stateUtah	1.121e+03	0.219
## f.stateVermont	2.502e+04	0.491
## f.stateVirginia	9.557e+02	-0.531
## f.stateWashington	1.194e+03	0.309
## f.stateWest Virginia	1.120e+03	0.314
## f.stateWisconsin	1.127e+03	0.315
## f.stateWyoming	1.933e+03	0.004
## f.education25_overBalanced Eduaction	1.710e+02	-0.797
## f.education25_overMostly Bachelor's	3.255e+02	-0.698
## f.unemploymentModerate Unemployment	2.156e+02	0.381
## f.unemploymentHigh Unemployment	2.765e+02	0.067
## pctpubliccoveragealone:f.stateAlaska	2.898e+00	1.191
## pctpubliccoveragealone:f.stateArizona	6.882e+00	0.568
## pctpubliccoveragealone:f.stateArkansas	2.587e+00	0.653
## pctpubliccoveragealone:f.stateCalifornia	2.414e+00	0.283
## pctpubliccoveragealone:f.stateColorado	2.623e+00	1.037
## pctpubliccoveragealone:f.stateConnecticut	1.044e+01	-0.116
## pctpubliccoveragealone:f.stateDelaware	NA	NA
## pctpubliccoveragealone:f.stateFlorida	2.477e+00	-1.513
## pctpubliccoveragealone:f.stateGeorgia	2.249e+00	-0.708
## pctpubliccoveragealone:f.stateHawaii	3.397e+01	-0.612
## pctpubliccoveragealone:f.stateIdaho	2.591e+00	-1.344
## pctpubliccoveragealone:f.stateIllinois	2.387e+00	-0.199
## pctpubliccoveragealone:f.stateIndiana	2.430e+00	0.244
## pctpubliccoveragealone:f.stateIowa	2.522e+00	0.631
## pctpubliccoveragealone:f.stateKansas	2.440e+00	-0.726
## pctpubliccoveragealone:f.stateKentucky	2.345e+00	-0.212
## pctpubliccoveragealone:f.stateLouisiana	2.444e+00	0.368
## pctpubliccoveragealone:f.stateMaine	6.610e+00	-0.239
## pctpubliccoveragealone:f.stateMaryland	4.511e+00	0.753
## pctpubliccoveragealone:f.stateMassachusetts	9.922e+00	-0.574
## pctpubliccoveragealone:f.stateMichigan	2.483e+00	0.268
## pctpubliccoveragealone:f.stateMinnesota	2.565e+00	0.311
## pctpubliccoveragealone:f.stateMississippi	2.306e+00	0.863
## pctpubliccoveragealone:f.stateMissouri	2.380e+00	0.737
## pctpubliccoveragealone:f.stateMontana	2.744e+00	0.474
## pctpubliccoveragealone:f.stateNebraska	2.418e+00	0.082
## pctpubliccoveragealone:f.stateNevada	3.540e+00	-2.084
## pctpubliccoveragealone:f.stateNew Hampshire	9.762e+01	-0.069
## pctpubliccoveragealone:f.stateNew Jersey	1.139e+01	-0.548
## pctpubliccoveragealone:f.stateNew Mexico	2.323e+00	-0.496
## pctpubliccoveragealone:f.stateNew York	2.487e+00	-0.607
## pctpubliccoveragealone:f.stateNorth Carolina	2.410e+00	-0.150
## pctpubliccoveragealone:f.stateNorth Dakota	2.665e+00	-0.324
## pctpubliccoveragealone:f.stateOhio	2.464e+00	0.143
## pctpubliccoveragealone:f.stateOklahoma	2.577e+00	1.542

## pctpubliccoveragealone:f.stateOregon	2.500e+00	0.503
## pctpubliccoveragealone:f.statePennsylvania	2.471e+00	0.927
## pctpubliccoveragealone:f.stateRhode Island	1.178e+02	1.227
## pctpubliccoveragealone:f.stateSouth Carolina	3.084e+00	0.220
## pctpubliccoveragealone:f.stateSouth Dakota	2.323e+00	0.676
## pctpubliccoveragealone:f.stateTennessee	2.434e+00	0.571
## pctpubliccoveragealone:f.stateTexas	2.244e+00	-0.655
## pctpubliccoveragealone:f.stateUtah	3.369e+00	-0.527
## pctpubliccoveragealone:f.stateVermont	1.719e+01	0.566
## pctpubliccoveragealone:f.stateVirginia	2.326e+00	0.847
## pctpubliccoveragealone:f.stateWashington	2.668e+00	-0.581
## pctpubliccoveragealone:f.stateWest Virginia	2.356e+00	0.734
## pctpubliccoveragealone:f.stateWisconsin	2.851e+00	0.148
## pctpubliccoveragealone:f.stateWyoming	6.015e+00	-0.117
## pctpubliccoveragealone:f.education25_overBalanced Eduaction	4.108e-01	-0.334
## pctpubliccoveragealone:f.education25_overMostly Bachelor's	9.934e-01	2.012
## pctpubliccoveragealone:f.unemploymentModerate Unemployment	5.522e-01	0.466
## pctpubliccoveragealone:f.unemploymentHigh Unemployment	6.316e-01	-0.028
## sqrt(incidencerate):f.stateAlaska	7.017e+00	0.604
## sqrt(incidencerate):f.stateArizona	1.623e+01	-1.024
## sqrt(incidencerate):f.stateArkansas	5.841e+00	-0.562
## sqrt(incidencerate):f.stateCalifornia	7.212e+00	0.282
## sqrt(incidencerate):f.stateColorado	5.515e+00	0.214
## sqrt(incidencerate):f.stateConnecticut	5.705e+01	-0.025
## sqrt(incidencerate):f.stateDelaware	NA	NA
## sqrt(incidencerate):f.stateFlorida	4.870e+00	0.452
## sqrt(incidencerate):f.stateGeorgia	4.985e+00	0.720
## sqrt(incidencerate):f.stateHawaii	NA	NA
## sqrt(incidencerate):f.stateIdaho	5.722e+00	1.052
## sqrt(incidencerate):f.stateIllinois	5.717e+00	0.498
## sqrt(incidencerate):f.stateIndiana	6.004e+00	0.390
## sqrt(incidencerate):f.stateIowa	5.816e+00	0.306
## sqrt(incidencerate):f.stateKansas	4.927e+01	2.876
## sqrt(incidencerate):f.stateKentucky	5.180e+00	0.992
## sqrt(incidencerate):f.stateLouisiana	6.571e+00	-0.333
## sqrt(incidencerate):f.stateMaine	2.314e+01	-0.524
## sqrt(incidencerate):f.stateMaryland	9.863e+00	1.708
## sqrt(incidencerate):f.stateMassachusetts	3.284e+01	0.425
## sqrt(incidencerate):f.stateMichigan	5.038e+00	-1.013
## sqrt(incidencerate):f.stateMinnesota	6.788e+01	2.168
## sqrt(incidencerate):f.stateMississippi	5.416e+00	-0.690
## sqrt(incidencerate):f.stateMissouri	4.930e+00	-0.335
## sqrt(incidencerate):f.stateMontana	5.801e+00	-0.102
## sqrt(incidencerate):f.stateNebraska	5.178e+00	-0.512
## sqrt(incidencerate):f.stateNevada	9.322e+01	0.862
## sqrt(incidencerate):f.stateNew Hampshire	1.044e+02	0.143
## sqrt(incidencerate):f.stateNew Jersey	3.541e+01	-0.493
## sqrt(incidencerate):f.stateNew Mexico	5.851e+00	0.868
## sqrt(incidencerate):f.stateNew York	7.340e+00	0.185
## sqrt(incidencerate):f.stateNorth Carolina	5.789e+00	-0.018
## sqrt(incidencerate):f.stateNorth Dakota	5.317e+00	0.758
## sqrt(incidencerate):f.stateOhio	6.215e+00	0.533
## sqrt(incidencerate):f.stateOklahoma	5.846e+00	-1.046
## sqrt(incidencerate):f.stateOregon	6.692e+00	-0.641

## sqrt(incidencerate):f.statePennsylvania	6.282e+00	0.061
## sqrt(incidencerate):f.stateRhode Island	3.818e+02	-1.180
## sqrt(incidencerate):f.stateSouth Carolina	6.653e+00	0.366
## sqrt(incidencerate):f.stateSouth Dakota	5.418e+00	-0.395
## sqrt(incidencerate):f.stateTennessee	5.604e+00	0.388
## sqrt(incidencerate):f.stateTexas	4.844e+00	0.645
## sqrt(incidencerate):f.stateUtah	7.294e+00	-0.143
## sqrt(incidencerate):f.stateVermont	1.197e+02	-0.530
## sqrt(incidencerate):f.stateVirginia	4.904e+00	-0.484
## sqrt(incidencerate):f.stateWashington	6.356e+00	-0.821
## sqrt(incidencerate):f.stateWest Virginia	5.113e+00	-0.155
## sqrt(incidencerate):f.stateWisconsin	5.479e+00	-0.915
## sqrt(incidencerate):f.stateWyoming	1.279e+01	1.090
## sqrt(incidencerate):f.education25_overBalanced Eduaction	1.174e+00	1.585
## sqrt(incidencerate):f.education25_overMostly Bachelor's	2.108e+00	-0.875
## sqrt(incidencerate):f.unemploymentModerate Unemployment	1.286e+00	0.833
## sqrt(incidencerate):f.unemploymentHigh Unemployment	1.486e+00	0.332
## log(povertypercent):f.stateAlaska	6.256e+01	-0.155
## log(povertypercent):f.stateArizona	1.196e+02	-0.533
## log(povertypercent):f.stateArkansas	4.566e+01	0.398
## log(povertypercent):f.stateCalifornia	4.229e+01	0.411
## log(povertypercent):f.stateColorado	4.398e+01	-0.634
## log(povertypercent):f.stateConnecticut	7.049e+02	-0.113
## log(povertypercent):f.stateDelaware	NA	NA
## log(povertypercent):f.stateFlorida	4.643e+01	0.977
## log(povertypercent):f.stateGeorgia	3.890e+01	-0.539
## log(povertypercent):f.stateHawaii	NA	NA
## log(povertypercent):f.stateIdaho	5.723e+01	1.617
## log(povertypercent):f.stateIllinois	4.344e+01	0.881
## log(povertypercent):f.stateIndiana	4.108e+01	0.531
## log(povertypercent):f.stateIowa	4.232e+01	0.432
## log(povertypercent):f.stateKansas	4.551e+01	-0.604
## log(povertypercent):f.stateKentucky	4.235e+01	0.549
## log(povertypercent):f.stateLouisiana	4.767e+01	0.384
## log(povertypercent):f.stateMaine	1.110e+02	0.394
## log(povertypercent):f.stateMaryland	6.797e+01	0.329
## log(povertypercent):f.stateMassachusetts	8.889e+01	0.998
## log(povertypercent):f.stateMichigan	4.214e+01	0.392
## log(povertypercent):f.stateMinnesota	4.622e+01	-1.006
## log(povertypercent):f.stateMississippi	4.458e+01	1.143
## log(povertypercent):f.stateMissouri	4.140e+01	0.536
## log(povertypercent):f.stateMontana	5.267e+01	-0.298
## log(povertypercent):f.stateNebraska	4.095e+01	-0.520
## log(povertypercent):f.stateNevada	5.746e+01	-0.473
## log(povertypercent):f.stateNew Hampshire	2.543e+02	0.247
## log(povertypercent):f.stateNew Jersey	1.182e+02	0.487
## log(povertypercent):f.stateNew Mexico	4.800e+01	-1.000
## log(povertypercent):f.stateNew York	4.513e+01	0.268
## log(povertypercent):f.stateNorth Carolina	4.216e+01	1.299
## log(povertypercent):f.stateNorth Dakota	5.028e+01	0.138
## log(povertypercent):f.stateOhio	4.120e+01	0.541
## log(povertypercent):f.stateOklahoma	4.442e+01	0.755
## log(povertypercent):f.stateOregon	5.622e+01	0.175
## log(povertypercent):f.statePennsylvania	4.391e+01	0.208

## log(povertypercent):f.stateRhode Island	NA	NA
## log(povertypercent):f.stateSouth Carolina	6.192e+01	0.894
## log(povertypercent):f.stateSouth Dakota	4.271e+01	-0.193
## log(povertypercent):f.stateTennessee	4.385e+01	0.866
## log(povertypercent):f.stateTexas	3.857e+01	-0.221
## log(povertypercent):f.stateUtah	4.743e+01	0.534
## log(povertypercent):f.stateVermont	1.496e+03	-0.534
## log(povertypercent):f.stateVirginia	3.962e+01	0.412
## log(povertypercent):f.stateWashington	5.022e+01	-0.008
## log(povertypercent):f.stateWest Virginia	5.166e+01	-0.684
## log(povertypercent):f.stateWisconsin	4.123e+01	0.029
## log(povertypercent):f.stateWyoming	6.313e+01	-0.795
## log(povertypercent):f.education25_overBalanced Eduaction	8.343e+00	-0.473
## log(povertypercent):f.education25_overMostly Bachelor's	1.394e+01	-0.543
## log(povertypercent):f.unemploymentModerate Unemployment	1.030e+01	-0.610
## log(povertypercent):f.unemploymentHigh Unemployment	1.377e+01	-0.486
## pctbachdeg25_over:f.stateAlaska	2.566e+00	-1.783
## pctbachdeg25_over:f.stateArizona	3.383e+00	1.683
## pctbachdeg25_over:f.stateArkansas	2.031e+00	-0.321
## pctbachdeg25_over:f.stateCalifornia	1.906e+00	0.871
## pctbachdeg25_over:f.stateColorado	1.519e+00	-0.657
## pctbachdeg25_over:f.stateConnecticut	3.591e+01	0.168
## pctbachdeg25_over:f.stateDelaware	NA	NA
## pctbachdeg25_over:f.stateFlorida	1.709e+00	-1.448
## pctbachdeg25_over:f.stateGeorgia	1.441e+00	-0.013
## pctbachdeg25_over:f.stateHawaii	NA	NA
## pctbachdeg25_over:f.stateIdaho	1.644e+00	-0.812
## pctbachdeg25_over:f.stateIllinois	1.588e+00	-0.788
## pctbachdeg25_over:f.stateIndiana	1.523e+00	0.239
## pctbachdeg25_over:f.stateIowa	1.566e+00	1.021
## pctbachdeg25_over:f.stateKansas	1.514e+00	0.300
## pctbachdeg25_over:f.stateKentucky	1.491e+00	-0.001
## pctbachdeg25_over:f.stateLouisiana	1.724e+00	0.360
## pctbachdeg25_over:f.stateMaine	4.115e+00	-0.779
## pctbachdeg25_over:f.stateMaryland	2.403e+00	-0.619
## pctbachdeg25_over:f.stateMassachusetts	6.522e+00	-0.007
## pctbachdeg25_over:f.stateMichigan	1.628e+00	-0.554
## pctbachdeg25_over:f.stateMinnesota	1.555e+00	0.545
## pctbachdeg25_over:f.stateMississippi	1.587e+00	1.015
## pctbachdeg25_over:f.stateMissouri	1.605e+00	-0.194
## pctbachdeg25_over:f.stateMontana	1.639e+00	1.055
## pctbachdeg25_over:f.stateNebraska	1.587e+00	0.731
## pctbachdeg25_over:f.stateNevada	2.253e+00	-2.561
## pctbachdeg25_over:f.stateNew Hampshire	1.991e+02	-0.080
## pctbachdeg25_over:f.stateNew Jersey	4.609e+00	-0.461
## pctbachdeg25_over:f.stateNew Mexico	2.395e+00	-1.200
## pctbachdeg25_over:f.stateNew York	1.821e+00	0.059
## pctbachdeg25_over:f.stateNorth Carolina	1.572e+00	-0.158
## pctbachdeg25_over:f.stateNorth Dakota	2.243e+00	0.082
## pctbachdeg25_over:f.stateOhio	1.602e+00	0.029
## pctbachdeg25_over:f.stateOklahoma	1.922e+00	0.797
## pctbachdeg25_over:f.stateOregon	1.867e+00	0.420
## pctbachdeg25_over:f.statePennsylvania	1.764e+00	0.297
## pctbachdeg25_over:f.stateRhode Island	NA	NA

## pctbachdeg25_over:f.stateSouth Carolina	1.788e+00	0.122
## pctbachdeg25_over:f.stateSouth Dakota	1.678e+00	1.105
## pctbachdeg25_over:f.stateTennessee	1.594e+00	0.239
## pctbachdeg25_over:f.stateTexas	1.422e+00	0.252
## pctbachdeg25_over:f.stateUtah	1.820e+00	0.120
## pctbachdeg25_over:f.stateVermont	3.457e+01	0.524
## pctbachdeg25_over:f.stateVirginia	1.462e+00	-0.401
## pctbachdeg25_over:f.stateWashington	1.815e+00	-1.042
## pctbachdeg25_over:f.stateWest Virginia	1.977e+00	1.889
## pctbachdeg25_over:f.stateWisconsin	1.582e+00	0.634
## pctbachdeg25_over:f.stateWyoming	6.743e+00	0.441
## pctbachdeg25_over:f.education25_overBalanced Eduaction	5.545e-01	-0.265
## pctbachdeg25_over:f.education25_overMostly Bachelor's	8.769e-01	-0.255
## pctbachdeg25_over:f.unemploymentModerate Unemployment	4.048e-01	0.728
## pctbachdeg25_over:f.unemploymentHigh Unemployment	5.848e-01	-0.325
## log(medincome):f.stateAlaska	8.223e+01	0.094
## log(medincome):f.stateArizona	3.708e+02	0.360
## log(medincome):f.stateArkansas	8.013e+01	0.087
## log(medincome):f.stateCalifornia	7.630e+01	0.010
## log(medincome):f.stateColorado	8.048e+01	0.227
## log(medincome):f.stateConnecticut	1.120e+03	-0.212
## log(medincome):f.stateDelaware	NA	NA
## log(medincome):f.stateFlorida	9.230e+01	0.408
## log(medincome):f.stateGeorgia	7.230e+01	-1.009
## log(medincome):f.stateHawaii	NA	NA
## log(medincome):f.stateIdaho	9.609e+01	1.284
## log(medincome):f.stateIllinois	7.660e+01	0.791
## log(medincome):f.stateIndiana	8.160e+01	0.695
## log(medincome):f.stateIowa	8.817e+01	0.056
## log(medincome):f.stateKansas	7.545e+01	-0.177
## log(medincome):f.stateKentucky	7.602e+01	0.311
## log(medincome):f.stateLouisiana	8.077e+01	0.077
## log(medincome):f.stateMaine	1.378e+02	0.558
## log(medincome):f.stateMaryland	1.095e+02	0.982
## log(medincome):f.stateMassachusetts	2.182e+02	0.088
## log(medincome):f.stateMichigan	7.962e+01	0.261
## log(medincome):f.stateMinnesota	8.225e+01	-0.056
## log(medincome):f.stateMississippi	7.497e+01	1.345
## log(medincome):f.stateMissouri	7.786e+01	0.180
## log(medincome):f.stateMontana	8.640e+01	0.089
## log(medincome):f.stateNebraska	7.589e+01	0.294
## log(medincome):f.stateNevada	1.042e+02	-3.147
## log(medincome):f.stateNew Hampshire	2.603e+03	0.097
## log(medincome):f.stateNew Jersey	1.777e+02	0.480
## log(medincome):f.stateNew Mexico	8.198e+01	-1.250
## log(medincome):f.stateNew York	7.546e+01	-0.229
## log(medincome):f.stateNorth Carolina	7.720e+01	0.908
## log(medincome):f.stateNorth Dakota	7.670e+01	0.440
## log(medincome):f.stateOhio	7.956e+01	0.552
## log(medincome):f.stateOklahoma	8.092e+01	0.244
## log(medincome):f.stateOregon	1.057e+02	0.612
## log(medincome):f.statePennsylvania	8.034e+01	0.539
## log(medincome):f.stateRhode Island	NA	NA
## log(medincome):f.stateSouth Carolina	9.822e+01	0.056

## log(medincome):f.stateSouth Dakota	8.111e+01	-0.384
## log(medincome):f.stateTennessee	7.703e+01	0.896
## log(medincome):f.stateTexas	6.977e+01	-0.361
## log(medincome):f.stateUtah	8.334e+01	-0.408
## log(medincome):f.stateVermont	9.485e+02	-0.427
## log(medincome):f.stateVirginia	7.177e+01	0.776
## log(medincome):f.stateWashington	1.041e+02	-0.050
## log(medincome):f.stateWest Virginia	8.429e+01	0.562
## log(medincome):f.stateWisconsin	9.208e+01	0.202
## log(medincome):f.stateWyoming	2.225e+02	0.062
## log(medincome):f.education25_overBalanced Eduaction	1.458e+01	1.669
## log(medincome):f.education25_overMostly Bachelor's	2.806e+01	0.692
## log(medincome):f.unemploymentModerate Unemployment	1.697e+01	0.003
## log(medincome):f.unemploymentHigh Unemployment	2.277e+01	0.650
## log(pctemployed16_over):f.stateAlaska	1.157e+02	2.386
## log(pctemployed16_over):f.stateArizona	1.458e+02	-0.872
## log(pctemployed16_over):f.stateArkansas	5.728e+01	0.594
## log(pctemployed16_over):f.stateCalifornia	5.791e+01	-0.844
## log(pctemployed16_over):f.stateColorado	5.553e+01	0.260
## log(pctemployed16_over):f.stateConnecticut	2.586e+03	-0.144
## log(pctemployed16_over):f.stateDelaware	NA	NA
## log(pctemployed16_over):f.stateFlorida	6.266e+01	-0.399
## log(pctemployed16_over):f.stateGeorgia	5.122e+01	-0.009
## log(pctemployed16_over):f.stateHawaii	NA	NA
## log(pctemployed16_over):f.stateIdaho	7.006e+01	-0.854
## log(pctemployed16_over):f.stateIllinois	5.521e+01	-0.258
## log(pctemployed16_over):f.stateIndiana	6.815e+01	-1.008
## log(pctemployed16_over):f.stateIowa	7.779e+01	-0.015
## log(pctemployed16_over):f.stateKansas	6.690e+01	-2.390
## log(pctemployed16_over):f.stateKentucky	5.551e+01	-1.503
## log(pctemployed16_over):f.stateLouisiana	6.271e+01	-0.846
## log(pctemployed16_over):f.stateMaine	2.233e+02	0.376
## log(pctemployed16_over):f.stateMaryland	1.960e+02	0.058
## log(pctemployed16_over):f.stateMassachusetts	8.139e+02	-0.150
## log(pctemployed16_over):f.stateMichigan	5.698e+01	0.131
## log(pctemployed16_over):f.stateMinnesota	7.602e+01	-1.303
## log(pctemployed16_over):f.stateMississippi	5.686e+01	-0.468
## log(pctemployed16_over):f.stateMissouri	5.360e+01	0.748
## log(pctemployed16_over):f.stateMontana	7.112e+01	-0.376
## log(pctemployed16_over):f.stateNebraska	6.502e+01	-2.505
## log(pctemployed16_over):f.stateNevada	7.374e+01	1.347
## log(pctemployed16_over):f.stateNew Hampshire	NA	NA
## log(pctemployed16_over):f.stateNew Jersey	5.213e+02	-0.634
## log(pctemployed16_over):f.stateNew Mexico	7.540e+01	1.135
## log(pctemployed16_over):f.stateNew York	7.292e+01	-0.519
## log(pctemployed16_over):f.stateNorth Carolina	5.786e+01	-0.337
## log(pctemployed16_over):f.stateNorth Dakota	1.145e+02	-0.162
## log(pctemployed16_over):f.stateOhio	7.619e+01	-0.290
## log(pctemployed16_over):f.stateOklahoma	6.381e+01	1.079
## log(pctemployed16_over):f.stateOregon	7.687e+01	-0.844
## log(pctemployed16_over):f.statePennsylvania	6.819e+01	-0.713
## log(pctemployed16_over):f.stateRhode Island	NA	NA
## log(pctemployed16_over):f.stateSouth Carolina	6.344e+01	0.938
## log(pctemployed16_over):f.stateSouth Dakota	6.221e+01	-0.736

## log(pctemployed16_over):f.stateTennessee	5.660e+01	-1.011
## log(pctemployed16_over):f.stateTexas	5.057e+01	-1.049
## log(pctemployed16_over):f.stateUtah	8.589e+01	0.197
## log(pctemployed16_over):f.stateVermont	2.435e+03	-0.512
## log(pctemployed16_over):f.stateVirginia	5.572e+01	-0.496
## log(pctemployed16_over):f.stateWashington	7.141e+01	-0.536
## log(pctemployed16_over):f.stateWest Virginia	6.947e+01	-3.009
## log(pctemployed16_over):f.stateWisconsin	7.620e+01	-1.570
## log(pctemployed16_over):f.stateWyoming	2.764e+02	-0.325
## log(pctemployed16_over):f.education25_overBalanced Eduaction	1.332e+01	-2.798
## log(pctemployed16_over):f.education25_overMostly Bachelor's	3.101e+01	0.406
## log(pctemployed16_over):f.unemploymentModerate Unemployment	1.622e+01	-1.492
## log(pctemployed16_over):f.unemploymentHigh Unemployment	1.839e+01	-2.228
##	Pr(> t)	
## (Intercept)	0.89908	
## pctpubliccoveragealone	0.99894	
## sqrt(incidencerate)	0.11571	
## log(povertypercent)	0.97182	
## pctbachdeg25_over	0.35866	
## log(medincome)	0.67768	
## log(pctemployed16_over)	0.26560	
## f.stateAlaska	0.32048	
## f.stateArizona	0.87015	
## f.stateArkansas	0.83428	
## f.stateCalifornia	0.97666	
## f.stateColorado	0.79615	
## f.stateConnecticut	0.85751	
## f.stateDelaware	0.80252	
## f.stateFlorida	0.74290	
## f.stateGeorgia	0.40692	
## f.stateHawaii	0.56329	
## f.stateIdaho	0.22752	
## f.stateIllinois	0.46309	
## f.stateIndiana	0.66479	
## f.stateIowa	0.85408	
## f.stateKansas	0.08089	
## f.stateKentucky	0.92994	
## f.stateLouisiana	0.91307	
## f.stateMaine	0.57812	
## f.stateMaryland	0.19910	
## f.stateMassachusetts	0.94555	
## f.stateMichigan	0.84408	
## f.stateMinnesota	0.07590	
## f.stateMississippi	0.28451	
## f.stateMissouri	0.72216	
## f.stateMontana	0.99559	
## f.stateNebraska	0.62219	
## f.stateNevada	0.43040	
## f.stateNew Hampshire	0.91281	
## f.stateNew Jersey	0.75838	
## f.stateNew Mexico	0.39714	
## f.stateNew York	0.78025	
## f.stateNorth Carolina	0.40624	
## f.stateNorth Dakota	0.70241	

## f.stateOhio	0.61703
## f.stateOklahoma	0.61885
## f.stateOregon	0.72861
## f.statePennsylvania	0.73139
## f.stateRhode Island	0.24325
## f.stateSouth Carolina	0.68429
## f.stateSouth Dakota	0.61994
## f.stateTennessee	0.49333
## f.stateTexas	0.61823
## f.stateUtah	0.82646
## f.stateVermont	0.62362
## f.stateVirginia	0.59574
## f.stateWashington	0.75772
## f.stateWest Virginia	0.75319
## f.stateWisconsin	0.75272
## f.stateWyoming	0.99641
## f.education25_overBalanced Eduaction	0.42537
## f.education25_overMostly Bachelor's	0.48539
## f.unemploymentModerate Unemployment	0.70289
## f.unemploymentHigh Unemployment	0.94628
## pctpubliccoveragealone:f.stateAlaska	0.23382
## pctpubliccoveragealone:f.stateArizona	0.57014
## pctpubliccoveragealone:f.stateArkansas	0.51364
## pctpubliccoveragealone:f.stateCalifornia	0.77718
## pctpubliccoveragealone:f.stateColorado	0.30006
## pctpubliccoveragealone:f.stateConnecticut	0.90805
## pctpubliccoveragealone:f.stateDelaware	NA
## pctpubliccoveragealone:f.stateFlorida	0.13048
## pctpubliccoveragealone:f.stateGeorgia	0.47903
## pctpubliccoveragealone:f.stateHawaii	0.54067
## pctpubliccoveragealone:f.stateIdaho	0.17916
## pctpubliccoveragealone:f.stateIllinois	0.84230
## pctpubliccoveragealone:f.stateIndiana	0.80692
## pctpubliccoveragealone:f.stateIowa	0.52840
## pctpubliccoveragealone:f.stateKansas	0.46791
## pctpubliccoveragealone:f.stateKentucky	0.83204
## pctpubliccoveragealone:f.stateLouisiana	0.71260
## pctpubliccoveragealone:f.stateMaine	0.81081
## pctpubliccoveragealone:f.stateMaryland	0.45141
## pctpubliccoveragealone:f.stateMassachusetts	0.56594
## pctpubliccoveragealone:f.stateMichigan	0.78860
## pctpubliccoveragealone:f.stateMinnesota	0.75567
## pctpubliccoveragealone:f.stateMississippi	0.38822
## pctpubliccoveragealone:f.stateMissouri	0.46106
## pctpubliccoveragealone:f.stateMontana	0.63530
## pctpubliccoveragealone:f.stateNebraska	0.93483
## pctpubliccoveragealone:f.stateNevada	0.03736 *
## pctpubliccoveragealone:f.stateNew Hampshire	0.94475
## pctpubliccoveragealone:f.stateNew Jersey	0.58410
## pctpubliccoveragealone:f.stateNew Mexico	0.61978
## pctpubliccoveragealone:f.stateNew York	0.54363
## pctpubliccoveragealone:f.stateNorth Carolina	0.88098
## pctpubliccoveragealone:f.stateNorth Dakota	0.74578
## pctpubliccoveragealone:f.stateOhio	0.88648

## pctpubliccoveragealone:f.stateOklahoma	0.12328
## pctpubliccoveragealone:f.stateOregon	0.61501
## pctpubliccoveragealone:f.statePennsylvania	0.35432
## pctpubliccoveragealone:f.stateRhode Island	0.22021
## pctpubliccoveragealone:f.stateSouth Carolina	0.82627
## pctpubliccoveragealone:f.stateSouth Dakota	0.49936
## pctpubliccoveragealone:f.stateTennessee	0.56833
## pctpubliccoveragealone:f.stateTexas	0.51268
## pctpubliccoveragealone:f.stateUtah	0.59823
## pctpubliccoveragealone:f.stateVermont	0.57130
## pctpubliccoveragealone:f.stateVirginia	0.39733
## pctpubliccoveragealone:f.stateWashington	0.56125
## pctpubliccoveragealone:f.stateWest Virginia	0.46311
## pctpubliccoveragealone:f.stateWisconsin	0.88231
## pctpubliccoveragealone:f.stateWyoming	0.90657
## pctpubliccoveragealone:f.education25_overBalanced Eduaction	0.73874
## pctpubliccoveragealone:f.education25_overMostly Bachelor's	0.04443 *
## pctpubliccoveragealone:f.unemploymentModerate Unemployment	0.64143
## pctpubliccoveragealone:f.unemploymentHigh Unemployment	0.97756
## sqrt(incidencerate):f.stateAlaska	0.54581
## sqrt(incidencerate):f.stateArizona	0.30598
## sqrt(incidencerate):f.stateArkansas	0.57401
## sqrt(incidencerate):f.stateCalifornia	0.77787
## sqrt(incidencerate):f.stateColorado	0.83089
## sqrt(incidencerate):f.stateConnecticut	0.98035
## sqrt(incidencerate):f.stateDelaware	NA
## sqrt(incidencerate):f.stateFlorida	0.65126
## sqrt(incidencerate):f.stateGeorgia	0.47141
## sqrt(incidencerate):f.stateHawaii	NA
## sqrt(incidencerate):f.stateIdaho	0.29289
## sqrt(incidencerate):f.stateIllinois	0.61883
## sqrt(incidencerate):f.stateIndiana	0.69630
## sqrt(incidencerate):f.stateIowa	0.75940
## sqrt(incidencerate):f.stateKansas	0.00408 **
## sqrt(incidencerate):f.stateKentucky	0.32141
## sqrt(incidencerate):f.stateLouisiana	0.73934
## sqrt(incidencerate):f.stateMaine	0.60056
## sqrt(incidencerate):f.stateMaryland	0.08781 .
## sqrt(incidencerate):f.stateMassachusetts	0.67061
## sqrt(incidencerate):f.stateMichigan	0.31140
## sqrt(incidencerate):f.stateMinnesota	0.03031 *
## sqrt(incidencerate):f.stateMississippi	0.49000
## sqrt(incidencerate):f.stateMissouri	0.73771
## sqrt(incidencerate):f.stateMontana	0.91840
## sqrt(incidencerate):f.stateNebraska	0.60906
## sqrt(incidencerate):f.stateNevada	0.38881
## sqrt(incidencerate):f.stateNew Hampshire	0.88653
## sqrt(incidencerate):f.stateNew Jersey	0.62211
## sqrt(incidencerate):f.stateNew Mexico	0.38564
## sqrt(incidencerate):f.stateNew York	0.85346
## sqrt(incidencerate):f.stateNorth Carolina	0.98596
## sqrt(incidencerate):f.stateNorth Dakota	0.44878
## sqrt(incidencerate):f.stateOhio	0.59417
## sqrt(incidencerate):f.stateOklahoma	0.29566

## sqrt(incidencerate):f.stateOregon	0.52180
## sqrt(incidencerate):f.statePennsylvania	0.95108
## sqrt(incidencerate):f.stateRhode Island	0.23810
## sqrt(incidencerate):f.stateSouth Carolina	0.71445
## sqrt(incidencerate):f.stateSouth Dakota	0.69267
## sqrt(incidencerate):f.stateTennessee	0.69799
## sqrt(incidencerate):f.stateTexas	0.51874
## sqrt(incidencerate):f.stateUtah	0.88638
## sqrt(incidencerate):f.stateVermont	0.59594
## sqrt(incidencerate):f.stateVirginia	0.62824
## sqrt(incidencerate):f.stateWashington	0.41151
## sqrt(incidencerate):f.stateWest Virginia	0.87704
## sqrt(incidencerate):f.stateWisconsin	0.36051
## sqrt(incidencerate):f.stateWyoming	0.27569
## sqrt(incidencerate):f.education25_overBalanced Eduaction	0.11323
## sqrt(incidencerate):f.education25_overMostly Bachelor's	0.38161
## sqrt(incidencerate):f.unemploymentModerate Unemployment	0.40482
## sqrt(incidencerate):f.unemploymentHigh Unemployment	0.74013
## log(povertypercent):f.stateAlaska	0.87664
## log(povertypercent):f.stateArizona	0.59428
## log(povertypercent):f.stateArkansas	0.69073
## log(povertypercent):f.stateCalifornia	0.68102
## log(povertypercent):f.stateColorado	0.52597
## log(povertypercent):f.stateConnecticut	0.91023
## log(povertypercent):f.stateDelaware	NA
## log(povertypercent):f.stateFlorida	0.32888
## log(povertypercent):f.stateGeorgia	0.58969
## log(povertypercent):f.stateHawaii	NA
## log(povertypercent):f.stateIdaho	0.10615
## log(povertypercent):f.stateIllinois	0.37823
## log(povertypercent):f.stateIndiana	0.59561
## log(povertypercent):f.stateIowa	0.66587
## log(povertypercent):f.stateKansas	0.54621
## log(povertypercent):f.stateKentucky	0.58320
## log(povertypercent):f.stateLouisiana	0.70096
## log(povertypercent):f.stateMaine	0.69337
## log(povertypercent):f.stateMaryland	0.74253
## log(povertypercent):f.stateMassachusetts	0.31828
## log(povertypercent):f.stateMichigan	0.69518
## log(povertypercent):f.stateMinnesota	0.31473
## log(povertypercent):f.stateMississippi	0.25328
## log(povertypercent):f.stateMissouri	0.59214
## log(povertypercent):f.stateMontana	0.76591
## log(povertypercent):f.stateNebraska	0.60321
## log(povertypercent):f.stateNevada	0.63624
## log(povertypercent):f.stateNew Hampshire	0.80530
## log(povertypercent):f.stateNew Jersey	0.62625
## log(povertypercent):f.stateNew Mexico	0.31768
## log(povertypercent):f.stateNew York	0.78836
## log(povertypercent):f.stateNorth Carolina	0.19402
## log(povertypercent):f.stateNorth Dakota	0.89058
## log(povertypercent):f.stateOhio	0.58892
## log(povertypercent):f.stateOklahoma	0.45055
## log(povertypercent):f.stateOregon	0.86076

## log(povertypercent):f.statePennsylvania	0.83536
## log(povertypercent):f.stateRhode Island	NA
## log(povertypercent):f.stateSouth Carolina	0.37143
## log(povertypercent):f.stateSouth Dakota	0.84670
## log(povertypercent):f.stateTennessee	0.38670
## log(povertypercent):f.stateTexas	0.82537
## log(povertypercent):f.stateUtah	0.59322
## log(povertypercent):f.stateVermont	0.59376
## log(povertypercent):f.stateVirginia	0.68032
## log(povertypercent):f.stateWashington	0.99350
## log(povertypercent):f.stateWest Virginia	0.49407
## log(povertypercent):f.stateWisconsin	0.97667
## log(povertypercent):f.stateWyoming	0.42658
## log(povertypercent):f.education25_overBalanced Eduaction	0.63647
## log(povertypercent):f.education25_overMostly Bachelor's	0.58727
## log(povertypercent):f.unemploymentModerate Unemployment	0.54203
## log(povertypercent):f.unemploymentHigh Unemployment	0.62737
## pctbachdeg25_over:f.stateAlaska	0.07484
## pctbachdeg25_over:f.stateArizona	0.09266
## pctbachdeg25_over:f.stateArkansas	0.74821
## pctbachdeg25_over:f.stateCalifornia	0.38400
## pctbachdeg25_over:f.stateColorado	0.51139
## pctbachdeg25_over:f.stateConnecticut	0.86695
## pctbachdeg25_over:f.stateDelaware	NA
## pctbachdeg25_over:f.stateFlorida	0.14781
## pctbachdeg25_over:f.stateGeorgia	0.98954
## pctbachdeg25_over:f.stateHawaii	NA
## pctbachdeg25_over:f.stateIdaho	0.41697
## pctbachdeg25_over:f.stateIllinois	0.43105
## pctbachdeg25_over:f.stateIndiana	0.81107
## pctbachdeg25_over:f.stateIowa	0.30746
## pctbachdeg25_over:f.stateKansas	0.76427
## pctbachdeg25_over:f.stateKentucky	0.99945
## pctbachdeg25_over:f.stateLouisiana	0.71881
## pctbachdeg25_over:f.stateMaine	0.43602
## pctbachdeg25_over:f.stateMaryland	0.53574
## pctbachdeg25_over:f.stateMassachusetts	0.99470
## pctbachdeg25_over:f.stateMichigan	0.57933
## pctbachdeg25_over:f.stateMinnesota	0.58598
## pctbachdeg25_over:f.stateMississippi	0.31020
## pctbachdeg25_over:f.stateMissouri	0.84633
## pctbachdeg25_over:f.stateMontana	0.29156
## pctbachdeg25_over:f.stateNebraska	0.46503
## pctbachdeg25_over:f.stateNevada	0.01053 *
## pctbachdeg25_over:f.stateNew Hampshire	0.93594
## pctbachdeg25_over:f.stateNew Jersey	0.64452
## pctbachdeg25_over:f.stateNew Mexico	0.23044
## pctbachdeg25_over:f.stateNew York	0.95313
## pctbachdeg25_over:f.stateNorth Carolina	0.87473
## pctbachdeg25_over:f.stateNorth Dakota	0.93462
## pctbachdeg25_over:f.stateOhio	0.97711
## pctbachdeg25_over:f.stateOklahoma	0.42543
## pctbachdeg25_over:f.stateOregon	0.67440
## pctbachdeg25_over:f.statePennsylvania	0.76680

## pctbachdeg25_over:f.stateRhode Island	NA
## pctbachdeg25_over:f.stateSouth Carolina	0.90253
## pctbachdeg25_over:f.stateSouth Dakota	0.26948
## pctbachdeg25_over:f.stateTennessee	0.81118
## pctbachdeg25_over:f.stateTexas	0.80118
## pctbachdeg25_over:f.stateUtah	0.90422
## pctbachdeg25_over:f.stateVermont	0.60045
## pctbachdeg25_over:f.stateVirginia	0.68870
## pctbachdeg25_over:f.stateWashington	0.29775
## pctbachdeg25_over:f.stateWest Virginia	0.05905
## pctbachdeg25_over:f.stateWisconsin	0.52624
## pctbachdeg25_over:f.stateWyoming	0.65940
## pctbachdeg25_over:f.education25_overBalanced Eduaction	0.79072
## pctbachdeg25_over:f.education25_overMostly Bachelor's	0.79914
## pctbachdeg25_over:f.unemploymentModerate Unemployment	0.46701
## pctbachdeg25_over:f.unemploymentHigh Unemployment	0.74553
## log(medincome):f.stateAlaska	0.92522
## log(medincome):f.stateArizona	0.71863
## log(medincome):f.stateArkansas	0.93087
## log(medincome):f.stateCalifornia	0.99172
## log(medincome):f.stateColorado	0.82054
## log(medincome):f.stateConnecticut	0.83189
## log(medincome):f.stateDelaware	NA
## log(medincome):f.stateFlorida	0.68354
## log(medincome):f.stateGeorgia	0.31319
## log(medincome):f.stateHawaii	NA
## log(medincome):f.stateIdaho	0.19940
## log(medincome):f.stateIllinois	0.42935
## log(medincome):f.stateIndiana	0.48726
## log(medincome):f.stateIowa	0.95575
## log(medincome):f.stateKansas	0.85927
## log(medincome):f.stateKentucky	0.75562
## log(medincome):f.stateLouisiana	0.93879
## log(medincome):f.stateMaine	0.57680
## log(medincome):f.stateMaryland	0.32629
## log(medincome):f.stateMassachusetts	0.92993
## log(medincome):f.stateMichigan	0.79403
## log(medincome):f.stateMinnesota	0.95515
## log(medincome):f.stateMississippi	0.17875
## log(medincome):f.stateMissouri	0.85740
## log(medincome):f.stateMontana	0.92935
## log(medincome):f.stateNebraska	0.76875
## log(medincome):f.stateNevada	0.00168 **
## log(medincome):f.stateNew Hampshire	0.92292
## log(medincome):f.stateNew Jersey	0.63155
## log(medincome):f.stateNew Mexico	0.21141
## log(medincome):f.stateNew York	0.81913
## log(medincome):f.stateNorth Carolina	0.36408
## log(medincome):f.stateNorth Dakota	0.66000
## log(medincome):f.stateOhio	0.58070
## log(medincome):f.stateOklahoma	0.80728
## log(medincome):f.stateOregon	0.54067
## log(medincome):f.statePennsylvania	0.58992
## log(medincome):f.stateRhode Island	NA

## log(medincome):f.stateSouth Carolina	0.95566
## log(medincome):f.stateSouth Dakota	0.70103
## log(medincome):f.stateTennessee	0.37035
## log(medincome):f.stateTexas	0.71809
## log(medincome):f.stateUtah	0.68353
## log(medincome):f.stateVermont	0.66934
## log(medincome):f.stateVirginia	0.43788
## log(medincome):f.stateWashington	0.96007
## log(medincome):f.stateWest Virginia	0.57420
## log(medincome):f.stateWisconsin	0.83957
## log(medincome):f.stateWyoming	0.95056
## log(medincome):f.education25_overBalanced Eduaction	0.09528 .
## log(medincome):f.education25_overMostly Bachelor's	0.48916
## log(medincome):f.unemploymentModerate Unemployment	0.99765
## log(medincome):f.unemploymentHigh Unemployment	0.51577
## log(pctemployed16_over):f.stateAlaska	0.01716 *
## log(pctemployed16_over):f.stateArizona	0.38338
## log(pctemployed16_over):f.stateArkansas	0.55253
## log(pctemployed16_over):f.stateCalifornia	0.39879
## log(pctemployed16_over):f.stateColorado	0.79468
## log(pctemployed16_over):f.stateConnecticut	0.88572
## log(pctemployed16_over):f.stateDelaware	NA
## log(pctemployed16_over):f.stateFlorida	0.68959
## log(pctemployed16_over):f.stateGeorgia	0.99276
## log(pctemployed16_over):f.stateHawaii	NA
## log(pctemployed16_over):f.stateIdaho	0.39317
## log(pctemployed16_over):f.stateIllinois	0.79661
## log(pctemployed16_over):f.stateIndiana	0.31339
## log(pctemployed16_over):f.stateIowa	0.98808
## log(pctemployed16_over):f.stateKansas	0.01696 *
## log(pctemployed16_over):f.stateKentucky	0.13306
## log(pctemployed16_over):f.stateLouisiana	0.39763
## log(pctemployed16_over):f.stateMaine	0.70684
## log(pctemployed16_over):f.stateMaryland	0.95371
## log(pctemployed16_over):f.stateMassachusetts	0.88091
## log(pctemployed16_over):f.stateMichigan	0.89557
## log(pctemployed16_over):f.stateMinnesota	0.19273
## log(pctemployed16_over):f.stateMississippi	0.64004
## log(pctemployed16_over):f.stateMissouri	0.45471
## log(pctemployed16_over):f.stateMontana	0.70686
## log(pctemployed16_over):f.stateNebraska	0.01235 *
## log(pctemployed16_over):f.stateNevada	0.17811
## log(pctemployed16_over):f.stateNew Hampshire	NA
## log(pctemployed16_over):f.stateNew Jersey	0.52605
## log(pctemployed16_over):f.stateNew Mexico	0.25667
## log(pctemployed16_over):f.stateNew York	0.60414
## log(pctemployed16_over):f.stateNorth Carolina	0.73621
## log(pctemployed16_over):f.stateNorth Dakota	0.87138
## log(pctemployed16_over):f.stateOhio	0.77154
## log(pctemployed16_over):f.stateOklahoma	0.28060
## log(pctemployed16_over):f.stateOregon	0.39881
## log(pctemployed16_over):f.statePennsylvania	0.47615
## log(pctemployed16_over):f.stateRhode Island	NA
## log(pctemployed16_over):f.stateSouth Carolina	0.34858

```
## log(pctemployed16_over):f.stateSouth Dakota          0.46197
## log(pctemployed16_over):f.stateTennessee             0.31219
## log(pctemployed16_over):f.stateTexas                 0.29415
## log(pctemployed16_over):f.stateUtah                  0.84386
## log(pctemployed16_over):f.stateVermont               0.60898
## log(pctemployed16_over):f.stateVirginia              0.61968
## log(pctemployed16_over):f.stateWashington            0.59171
## log(pctemployed16_over):f.stateWest Virginia         0.00267 **
## log(pctemployed16_over):f.stateWisconsin             0.11655
## log(pctemployed16_over):f.stateWyoming               0.74517
## log(pctemployed16_over):f.education25_overBalanced Eduaction 0.00522 **
## log(pctemployed16_over):f.education25_overMostly Bachelor's 0.68485
## log(pctemployed16_over):f.unemploymentModerate Unemployment 0.13586
## log(pctemployed16_over):f.unemploymentHigh Unemployment 0.02606 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 17.26 on 1469 degrees of freedom
## Multiple R-squared:  0.6934, Adjusted R-squared:  0.618
## F-statistic: 9.203 on 361 and 1469 DF,  p-value: < 2.2e-16
```

We use the `step()` function in order to see if there is any variable which is not significantly apporting to the model. The obtained model is the following:

```
lm(target_deathrate ~ pctpubliccoveragealone + sqrt(incidencerate) + pctbachdeg25_over + log(medincome)
+ log(pctemployed16_over) + f.state + f.education25_over + f.unemployment + pctpubliccover-
agealone:f.state + pctpubliccoveragealone:f.education25_over + sqrt(incidencerate):f.state + sqrt(incidencerate):f.education25_
+ pctbachdeg25_over:f.state + log(medincome):f.state + log(medincome):f.education25_over +
log(medincome):f.unemployment + log(pctemployed16_over):f.state + log(pctemployed16_over):f.education25_over
+ log(pctemployed16_over):f.unemployment, data = df)
```

In order to simplify the model further, we applied the `Anova` function from the `car` package. After performing the `Anova` function multiple times, we found that the variables `f.education25_over`, `log(pctemployed16_over)`, `log(medincome):f.unemployment`, `log(pctemployed16_over):f.unemployment`, and `sqrt(incidencerate):f.education25_over` do not significantly contribute to the model. As a result, we decided to remove them to simplify the model.

```
m4 <- step(m3)
```

```
## Start:  AIC=10751.83
## target_deathrate ~ ((pctpubliccoveragealone) + sqrt(incidencerate) +
##   log(povertypercent) + pctbachdeg25_over + log(medincome) +
##   log(pctemployed16_over)) * (f.state + f.education25_over +
##   f.unemployment)
##
##                                     Df Sum of Sq   RSS   AIC
## - log(povertypercent):f.state      45  13887.6 451597 10719
## - pctbachdeg25_over:f.state         45  19242.7 456952 10741
## - log(medincome):f.state           45  20421.8 458131 10745
## - pctbachdeg25_over:f.education25_over    2    32.4 437742 10748
## - log(povertypercent):f.unemployment    2    112.5 437822 10748
## - log(povertypercent):f.education25_over  2    113.7 437823 10748
## - pctpubliccoveragealone:f.unemployment  2    179.3 437889 10749
## - log(medincome):f.unemployment        2    242.6 437952 10749
## - sqrt(incidencerate):f.unemployment    2    262.8 437972 10749
## - pctbachdeg25_over:f.unemployment      2    539.6 438249 10750
```

```

## - log(medincome):f.education25_over      2      830.6 438540 10751
## <none>                                     437710 10752
## - sqrt(incidencerate):f.education25_over  2      1223.2 438933 10753
## - pctpubliccoveragealone:f.education25_over  2      1376.8 439086 10754
## - log(pctemployed16_over):f.unemployment   2      1487.8 439197 10754
## - sqrt(incidencerate):f.state             45     23524.2 461234 10758
## - log(pctemployed16_over):f.education25_over  2      2641.5 440351 10759
## - pctpubliccoveragealone:f.state          45     28970.6 466680 10779
## - log(pctemployed16_over):f.state          45     29359.3 467069 10781
##
## Step: AIC=10719.02
## target_deathrate ~ pctpubliccoveragealone + sqrt(incidencerate) +
##   log(povertypercent) + pctbachdeg25_over + log(medincome) +
##   log(pctemployed16_over) + f.state + f.education25_over +
##   f.unemployment + pctpubliccoveragealone:f.state + pctpubliccoveragealone:f.education25_over +
##   pctpubliccoveragealone:f.unemployment + sqrt(incidencerate):f.state +
##   sqrt(incidencerate):f.education25_over + sqrt(incidencerate):f.unemployment +
##   log(povertypercent):f.education25_over + log(povertypercent):f.unemployment +
##   pctbachdeg25_over:f.state + pctbachdeg25_over:f.education25_over +
##   pctbachdeg25_over:f.unemployment + log(medincome):f.state +
##   log(medincome):f.education25_over + log(medincome):f.unemployment +
##   log(pctemployed16_over):f.state + log(pctemployed16_over):f.education25_over +
##   log(pctemployed16_over):f.unemployment
##
##                                     Df Sum of Sq    RSS    AIC
## - pctbachdeg25_over:f.education25_over      2         12 451609 10715
## - log(povertypercent):f.education25_over      2         16 451613 10715
## - log(povertypercent):f.unemployment          2        171 451768 10716
## - pctpubliccoveragealone:f.unemployment        2        240 451837 10716
## - sqrt(incidencerate):f.unemployment          2        307 451904 10716
## - pctbachdeg25_over:f.unemployment            2        392 451989 10717
## - pctbachdeg25_over:f.state                   46     22896 474493 10718
## - log(medincome):f.unemployment               2         639 452236 10718
## - pctpubliccoveragealone:f.education25_over    2         821 452418 10718
## <none>                                         451597 10719
## - sqrt(incidencerate):f.state                 46     23289 474887 10719
## - sqrt(incidencerate):f.education25_over       2        1035 452632 10719
## - log(medincome):f.state                     46     23330 474927 10719
## - log(medincome):f.education25_over           2        1114 452711 10720
## - log(pctemployed16_over):f.unemployment       2        1731 453328 10722
## - log(pctemployed16_over):f.education25_over    2        2189 453786 10724
## - log(pctemployed16_over):f.state              46     28224 479822 10738
## - pctpubliccoveragealone:f.state              46     34830 486427 10763
##
## Step: AIC=10715.07
## target_deathrate ~ pctpubliccoveragealone + sqrt(incidencerate) +
##   log(povertypercent) + pctbachdeg25_over + log(medincome) +
##   log(pctemployed16_over) + f.state + f.education25_over +
##   f.unemployment + pctpubliccoveragealone:f.state + pctpubliccoveragealone:f.education25_over +
##   pctpubliccoveragealone:f.unemployment + sqrt(incidencerate):f.state +
##   sqrt(incidencerate):f.education25_over + sqrt(incidencerate):f.unemployment +
##   log(povertypercent):f.education25_over + log(povertypercent):f.unemployment +
##   pctbachdeg25_over:f.state + pctbachdeg25_over:f.unemployment +
##   log(medincome):f.state + log(medincome):f.education25_over +

```

```

##      log(medincome):f.unemployment + log(pctemployed16_over):f.state +
##      log(pctemployed16_over):f.education25_over + log(pctemployed16_over):f.unemployment
##
##                                     Df Sum of Sq    RSS    AIC
## - log(povertypercent):f.education25_over      2      18 451627 10711
## - log(povertypercent):f.unemployment          2      173 451782 10712
## - pctpubliccoveragealone:f.unemployment        2      240 451849 10712
## - sqrt(incidencerate):f.unemployment           2      310 451919 10712
## - pctbachdeg25_over:f.unemployment             2      394 452003 10713
## - log(medincome):f.unemployment               2      636 452245 10714
## - pctpubliccoveragealone:f.education25_over    2      894 452503 10715
## <none>                                         451609 10715
## - pctbachdeg25_over:f.state                   46     23291 474900 10715
## - sqrt(incidencerate):f.state                 46     23327 474936 10715
## - sqrt(incidencerate):f.education25_over       2      1043 452652 10715
## - log(medincome):f.state                     46     23350 474959 10715
## - log(medincome):f.education25_over           2      1117 452726 10716
## - log(pctemployed16_over):f.unemployment        2      1727 453336 10718
## - log(pctemployed16_over):f.education25_over    2      2205 453814 10720
## - log(pctemployed16_over):f.state              46     28216 479825 10734
## - pctpubliccoveragealone:f.state              46     34952 486561 10760
##
## Step:  AIC=10711.14
## target_deathrate ~ pctpubliccoveragealone + sqrt(incidencerate) +
##      log(povertypercent) + pctbachdeg25_over + log(medincome) +
##      log(pctemployed16_over) + f.state + f.education25_over +
##      f.unemployment + pctpubliccoveragealone:f.state + pctpubliccoveragealone:f.education25_over +
##      pctpubliccoveragealone:f.unemployment + sqrt(incidencerate):f.state +
##      sqrt(incidencerate):f.education25_over + sqrt(incidencerate):f.unemployment +
##      log(povertypercent):f.unemployment + pctbachdeg25_over:f.state +
##      pctbachdeg25_over:f.unemployment + log(medincome):f.state +
##      log(medincome):f.education25_over + log(medincome):f.unemployment +
##      log(pctemployed16_over):f.state + log(pctemployed16_over):f.education25_over +
##      log(pctemployed16_over):f.unemployment
##
##                                     Df Sum of Sq    RSS    AIC
## - log(povertypercent):f.unemployment          2      177 451804 10708
## - pctpubliccoveragealone:f.unemployment        2      248 451875 10708
## - sqrt(incidencerate):f.unemployment           2      310 451937 10708
## - pctbachdeg25_over:f.unemployment             2      388 452015 10709
## - log(medincome):f.unemployment               2      641 452268 10710
## <none>                                         451627 10711
## - sqrt(incidencerate):f.education25_over       2      1029 452656 10711
## - sqrt(incidencerate):f.state                 46     23330 474957 10711
## - log(medincome):f.state                     46     23335 474961 10711
## - pctpubliccoveragealone:f.education25_over    2      1136 452763 10712
## - pctbachdeg25_over:f.state                   46     23461 475088 10712
## - log(pctemployed16_over):f.unemployment        2      1726 453353 10714
## - log(pctemployed16_over):f.education25_over    2      2193 453820 10716
## - log(medincome):f.education25_over           2      2464 454091 10717
## - log(pctemployed16_over):f.state              46     28331 479958 10730
## - pctpubliccoveragealone:f.state              46     35014 486641 10756
##
## Step:  AIC=10707.85

```

```

## target_deathrate ~ pctpubliccoveragealone + sqrt(incidencerate) +
##   log(povertypercent) + pctbachdeg25_over + log(medincome) +
##   log(pctemployed16_over) + f.state + f.education25_over +
##   f.unemployment + pctpubliccoveragealone:f.state + pctpubliccoveragealone:f.education25_over +
##   pctpubliccoveragealone:f.unemployment + sqrt(incidencerate):f.state +
##   sqrt(incidencerate):f.education25_over + sqrt(incidencerate):f.unemployment +
##   pctbachdeg25_over:f.state + pctbachdeg25_over:f.unemployment +
##   log(medincome):f.state + log(medincome):f.education25_over +
##   log(medincome):f.unemployment + log(pctemployed16_over):f.state +
##   log(pctemployed16_over):f.education25_over + log(pctemployed16_over):f.unemployment
##
##
##           Df Sum of Sq   RSS   AIC
## - pctpubliccoveragealone:f.unemployment      2      202 452006 10705
## - sqrt(incidencerate):f.unemployment          2      296 452100 10705
## - pctbachdeg25_over:f.unemployment            2      404 452208 10706
## - log(medincome):f.unemployment              2      560 452364 10706
## - log(povertypercent)                        1       95 451899 10706
## - sqrt(incidencerate):f.state                46     23259 475063 10708
## - log(medincome):f.state                    46     23262 475066 10708
## <none>                                         451804 10708
## - sqrt(incidencerate):f.education25_over      2      1032 452836 10708
## - pctpubliccoveragealone:f.education25_over  2      1143 452947 10708
## - pctbachdeg25_over:f.state                  46     23591 475394 10709
## - log(pctemployed16_over):f.unemployment      2      1924 453728 10712
## - log(pctemployed16_over):f.education25_over  2      2257 454061 10713
## - log(medincome):f.education25_over          2      2557 454361 10714
## - log(pctemployed16_over):f.state            46     28489 480293 10728
## - pctpubliccoveragealone:f.state            46     35584 487388 10755
##
## Step: AIC=10704.67
## target_deathrate ~ pctpubliccoveragealone + sqrt(incidencerate) +
##   log(povertypercent) + pctbachdeg25_over + log(medincome) +
##   log(pctemployed16_over) + f.state + f.education25_over +
##   f.unemployment + pctpubliccoveragealone:f.state + pctpubliccoveragealone:f.education25_over +
##   sqrt(incidencerate):f.state + sqrt(incidencerate):f.education25_over +
##   sqrt(incidencerate):f.unemployment + pctbachdeg25_over:f.state +
##   pctbachdeg25_over:f.unemployment + log(medincome):f.state +
##   log(medincome):f.education25_over + log(medincome):f.unemployment +
##   log(pctemployed16_over):f.state + log(pctemployed16_over):f.education25_over +
##   log(pctemployed16_over):f.unemployment
##
##
##           Df Sum of Sq   RSS   AIC
## - sqrt(incidencerate):f.unemployment          2      310 452315 10702
## - pctbachdeg25_over:f.unemployment            2      372 452377 10702
## - log(povertypercent)                        1       93 452099 10703
## - sqrt(incidencerate):f.state                46     23284 475290 10705
## <none>                                         452006 10705
## - log(medincome):f.state                    46     23336 475341 10705
## - sqrt(incidencerate):f.education25_over      2      1045 453050 10705
## - pctpubliccoveragealone:f.education25_over  2      1253 453258 10706
## - pctbachdeg25_over:f.state                  46     23711 475716 10706
## - log(medincome):f.unemployment              2      1530 453536 10707
## - log(pctemployed16_over):f.unemployment      2      1901 453907 10708
## - log(pctemployed16_over):f.education25_over  2      2321 454326 10710

```

```

## - log(medincome):f.education25_over          2      2710 454716 10712
## - log(pctemployed16_over):f.state             46      28335 480341 10724
## - pctpubliccoveragealone:f.state             46      35403 487409 10751
##
## Step: AIC=10701.93
## target_deathrate ~ pctpubliccoveragealone + sqrt(incidencerate) +
##   log(povertypercent) + pctbachdeg25_over + log(medincome) +
##   log(pctemployed16_over) + f.state + f.education25_over +
##   f.unemployment + pctpubliccoveragealone:f.state + pctpubliccoveragealone:f.education25_over +
##   sqrt(incidencerate):f.state + sqrt(incidencerate):f.education25_over +
##   pctbachdeg25_over:f.state + pctbachdeg25_over:f.unemployment +
##   log(medincome):f.state + log(medincome):f.education25_over +
##   log(medincome):f.unemployment + log(pctemployed16_over):f.state +
##   log(pctemployed16_over):f.education25_over + log(pctemployed16_over):f.unemployment
##
##
##              Df Sum of Sq    RSS    AIC
## - pctbachdeg25_over:f.unemployment      2      302 452617 10699
## - log(povertypercent)                   1      101 452416 10700
## <none>                                   452315 10702
## - sqrt(incidencerate):f.state           46     23338 475653 10702
## - sqrt(incidencerate):f.education25_over  2      1035 453351 10702
## - log(medincome):f.state                46     23465 475780 10702
## - pctpubliccoveragealone:f.education25_over  2      1268 453583 10703
## - pctbachdeg25_over:f.state             46     23602 475917 10703
## - log(medincome):f.unemployment          2      1474 453790 10704
## - log(pctemployed16_over):f.unemployment  2      1866 454182 10706
## - log(pctemployed16_over):f.education25_over  2      2314 454629 10707
## - log(medincome):f.education25_over      2      2649 454964 10709
## - log(pctemployed16_over):f.state        46     28437 480752 10722
## - pctpubliccoveragealone:f.state        46     35697 488012 10749
##
## Step: AIC=10699.15
## target_deathrate ~ pctpubliccoveragealone + sqrt(incidencerate) +
##   log(povertypercent) + pctbachdeg25_over + log(medincome) +
##   log(pctemployed16_over) + f.state + f.education25_over +
##   f.unemployment + pctpubliccoveragealone:f.state + pctpubliccoveragealone:f.education25_over +
##   sqrt(incidencerate):f.state + sqrt(incidencerate):f.education25_over +
##   pctbachdeg25_over:f.state + log(medincome):f.state + log(medincome):f.education25_over +
##   log(medincome):f.unemployment + log(pctemployed16_over):f.state +
##   log(pctemployed16_over):f.education25_over + log(pctemployed16_over):f.unemployment
##
##
##              Df Sum of Sq    RSS    AIC
## - log(povertypercent)                   1       99 452716 10698
## - sqrt(incidencerate):f.state           46     23196 475813 10699
## <none>                                   452617 10699
## - sqrt(incidencerate):f.education25_over  2      1005 453623 10699
## - pctpubliccoveragealone:f.education25_over  2      1327 453945 10700
## - log(medincome):f.state                46     23770 476388 10701
## - pctbachdeg25_over:f.state             46     23942 476559 10702
## - log(medincome):f.unemployment          2      1643 454261 10702
## - log(pctemployed16_over):f.unemployment  2      1954 454571 10703
## - log(pctemployed16_over):f.education25_over  2      2276 454893 10704
## - log(medincome):f.education25_over      2      2606 455223 10706
## - log(pctemployed16_over):f.state        46     28526 481143 10719

```

```
## - pctpubliccoveragealone:f.state          46      35956 488573 10747
##
## Step: AIC=10697.55
## target_deathrate ~ pctpubliccoveragealone + sqrt(incidencerate) +
##   pctbachdeg25_over + log(medincome) + log(pctemployed16_over) +
##   f.state + f.education25_over + f.unemployment + pctpubliccoveragealone:f.state +
##   pctpubliccoveragealone:f.education25_over + sqrt(incidencerate):f.state +
##   sqrt(incidencerate):f.education25_over + pctbachdeg25_over:f.state +
##   log(medincome):f.state + log(medincome):f.education25_over +
##   log(medincome):f.unemployment + log(pctemployed16_over):f.state +
##   log(pctemployed16_over):f.education25_over + log(pctemployed16_over):f.unemployment
##
##              Df Sum of Sq    RSS   AIC
## <none>                        452716 10698
## - sqrt(incidencerate):f.state      46    23341 476057 10698
## - sqrt(incidencerate):f.education25_over    2     1015 453731 10698
## - pctpubliccoveragealone:f.education25_over    2     1322 454038 10699
## - log(medincome):f.state      46    23908 476624 10700
## - log(medincome):f.unemployment    2     1648 454364 10700
## - log(pctemployed16_over):f.unemployment    2     1907 454623 10701
## - pctbachdeg25_over:f.state      46    24305 477021 10701
## - log(pctemployed16_over):f.education25_over    2     2212 454928 10702
## - log(medincome):f.education25_over    2     2515 455231 10704
## - log(pctemployed16_over):f.state      46    28723 481439 10718
## - pctpubliccoveragealone:f.state      46     36170 488886 10746
```

```
summary(m4)
```

```
##
## Call:
## lm(formula = target_deathrate ~ pctpubliccoveragealone + sqrt(incidencerate) +
##   pctbachdeg25_over + log(medincome) + log(pctemployed16_over) +
##   f.state + f.education25_over + f.unemployment + pctpubliccoveragealone:f.state +
##   pctpubliccoveragealone:f.education25_over + sqrt(incidencerate):f.state +
##   sqrt(incidencerate):f.education25_over + pctbachdeg25_over:f.state +
##   log(medincome):f.state + log(medincome):f.education25_over +
##   log(medincome):f.unemployment + log(pctemployed16_over):f.state +
##   log(pctemployed16_over):f.education25_over + log(pctemployed16_over):f.unemployment,
##   data = df)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -88.712  -8.315  -0.199   8.669  83.397
##
## Coefficients: (12 not defined because of singularities)
##
##              Estimate
## (Intercept)      1.601e+02
## pctpubliccoveragealone    -3.268e-02
## sqrt(incidencerate)      8.665e+00
## pctbachdeg25_over     -1.378e+00
## log(medincome)     -3.484e+01
## log(pctemployed16_over)    5.836e+01
## f.stateAlaska     -1.352e+03
## f.stateArizona    -2.606e+03
## f.stateArkansas     2.098e+01
```

## f.stateCalifornia	2.653e+02
## f.stateColorado	-6.049e+02
## f.stateConnecticut	1.496e+03
## f.stateDelaware	-4.740e+00
## f.stateFlorida	4.981e+02
## f.stateGeorgia	3.678e+02
## f.stateHawaii	3.808e+02
## f.stateIdaho	-2.754e+02
## f.stateIllinois	-1.791e+02
## f.stateIndiana	-1.600e+01
## f.stateIowa	2.073e+02
## f.stateKansas	-1.535e+03
## f.stateKentucky	3.333e+02
## f.stateLouisiana	3.837e+02
## f.stateMaine	-7.414e+02
## f.stateMaryland	-1.502e+03
## f.stateMassachusetts	-8.981e+02
## f.stateMichigan	6.307e+01
## f.stateMinnesota	-1.714e+03
## f.stateMississippi	-3.183e+02
## f.stateMissouri	3.608e+01
## f.stateMontana	-2.216e+02
## f.stateNebraska	6.295e+01
## f.stateNevada	2.498e+03
## f.stateNew Hampshire	-2.966e+04
## f.stateNew Jersey	3.783e+02
## f.stateNew Mexico	3.830e+02
## f.stateNew York	3.899e+02
## f.stateNorth Carolina	-3.696e+01
## f.stateNorth Dakota	-3.561e+02
## f.stateOhio	-9.536e+01
## f.stateOklahoma	1.062e+02
## f.stateOregon	-2.930e+02
## f.statePennsylvania	-2.369e+02
## f.stateRhode Island	7.586e+03
## f.stateSouth Carolina	2.391e+02
## f.stateSouth Dakota	3.252e+02
## f.stateTennessee	-1.399e+02
## f.stateTexas	3.594e+02
## f.stateUtah	5.545e+02
## f.stateVermont	-1.600e+03
## f.stateVirginia	-2.735e+02
## f.stateWashington	2.460e+02
## f.stateWest Virginia	-1.983e+02
## f.stateWisconsin	3.681e+02
## f.stateWyoming	-9.540e+02
## f.education25_overBalanced Eduaction	-1.763e+02
## f.education25_overMostly Bachelor's	-3.297e+02
## f.unemploymentModerate Unemployment	-2.607e+01
## f.unemploymentHigh Unemployment	-1.168e+02
## pctpubliccoveragealone:f.stateAlaska	3.075e+00
## pctpubliccoveragealone:f.stateArizona	6.416e+00
## pctpubliccoveragealone:f.stateArkansas	2.220e+00
## pctpubliccoveragealone:f.stateCalifornia	9.213e-01

## pctpubliccoveragealone:f.stateColorado	2.067e+00
## pctpubliccoveragealone:f.stateConnecticut	-1.974e+00
## pctpubliccoveragealone:f.stateDelaware	NA
## pctpubliccoveragealone:f.stateFlorida	-4.047e+00
## pctpubliccoveragealone:f.stateGeorgia	-1.546e+00
## pctpubliccoveragealone:f.stateHawaii	-2.240e+01
## pctpubliccoveragealone:f.stateIdaho	-2.626e+00
## pctpubliccoveragealone:f.stateIllinois	3.780e-01
## pctpubliccoveragealone:f.stateIndiana	9.629e-01
## pctpubliccoveragealone:f.stateIowa	1.898e+00
## pctpubliccoveragealone:f.stateKansas	-1.550e+00
## pctpubliccoveragealone:f.stateKentucky	-3.108e-01
## pctpubliccoveragealone:f.stateLouisiana	1.153e+00
## pctpubliccoveragealone:f.stateMaine	2.754e-01
## pctpubliccoveragealone:f.stateMaryland	4.185e+00
## pctpubliccoveragealone:f.stateMassachusetts	1.207e+00
## pctpubliccoveragealone:f.stateMichigan	1.002e+00
## pctpubliccoveragealone:f.stateMinnesota	1.659e+00
## pctpubliccoveragealone:f.stateMississippi	2.493e+00
## pctpubliccoveragealone:f.stateMissouri	2.012e+00
## pctpubliccoveragealone:f.stateMontana	8.663e-01
## pctpubliccoveragealone:f.stateNebraska	-4.283e-02
## pctpubliccoveragealone:f.stateNevada	-7.206e+00
## pctpubliccoveragealone:f.stateNew Hampshire	-1.316e+02
## pctpubliccoveragealone:f.stateNew Jersey	-1.493e+00
## pctpubliccoveragealone:f.stateNew Mexico	-1.491e+00
## pctpubliccoveragealone:f.stateNew York	-9.993e-01
## pctpubliccoveragealone:f.stateNorth Carolina	3.890e-01
## pctpubliccoveragealone:f.stateNorth Dakota	-5.379e-01
## pctpubliccoveragealone:f.stateOhio	7.931e-01
## pctpubliccoveragealone:f.stateOklahoma	4.285e+00
## pctpubliccoveragealone:f.stateOregon	1.500e+00
## pctpubliccoveragealone:f.statePennsylvania	2.595e+00
## pctpubliccoveragealone:f.stateRhode Island	1.352e+02
## pctpubliccoveragealone:f.stateSouth Carolina	1.648e+00
## pctpubliccoveragealone:f.stateSouth Dakota	1.416e+00
## pctpubliccoveragealone:f.stateTennessee	2.122e+00
## pctpubliccoveragealone:f.stateTexas	-1.573e+00
## pctpubliccoveragealone:f.stateUtah	-6.443e-01
## pctpubliccoveragealone:f.stateVermont	5.521e+00
## pctpubliccoveragealone:f.stateVirginia	2.291e+00
## pctpubliccoveragealone:f.stateWashington	-1.115e+00
## pctpubliccoveragealone:f.stateWest Virginia	1.820e+00
## pctpubliccoveragealone:f.stateWisconsin	6.863e-01
## pctpubliccoveragealone:f.stateWyoming	-3.023e-01
## pctpubliccoveragealone:f.education25_overBalanced Eduaction	-2.532e-01
## pctpubliccoveragealone:f.education25_overMostly Bachelor's	1.474e+00
## sqrt(incidencerate):f.stateAlaska	3.760e+00
## sqrt(incidencerate):f.stateArizona	-2.109e+01
## sqrt(incidencerate):f.stateArkansas	-4.297e+00
## sqrt(incidencerate):f.stateCalifornia	5.032e-01
## sqrt(incidencerate):f.stateColorado	1.198e-01
## sqrt(incidencerate):f.stateConnecticut	-1.925e+00
## sqrt(incidencerate):f.stateDelaware	NA

## sqrt(incidencerate):f.stateFlorida	1.708e+00
## sqrt(incidencerate):f.stateGeorgia	3.069e+00
## sqrt(incidencerate):f.stateHawaii	NA
## sqrt(incidencerate):f.stateIdaho	4.218e+00
## sqrt(incidencerate):f.stateIllinois	3.714e+00
## sqrt(incidencerate):f.stateIndiana	3.295e+00
## sqrt(incidencerate):f.stateIowa	1.632e+00
## sqrt(incidencerate):f.stateKansas	9.433e+01
## sqrt(incidencerate):f.stateKentucky	4.548e+00
## sqrt(incidencerate):f.stateLouisiana	-2.535e+00
## sqrt(incidencerate):f.stateMaine	-4.039e+00
## sqrt(incidencerate):f.stateMaryland	1.740e+01
## sqrt(incidencerate):f.stateMassachusetts	-6.349e+00
## sqrt(incidencerate):f.stateMichigan	-4.894e+00
## sqrt(incidencerate):f.stateMinnesota	6.143e+01
## sqrt(incidencerate):f.stateMississippi	-4.712e+00
## sqrt(incidencerate):f.stateMissouri	-1.550e+00
## sqrt(incidencerate):f.stateMontana	-9.159e-01
## sqrt(incidencerate):f.stateNebraska	-3.788e+00
## sqrt(incidencerate):f.stateNevada	3.491e+01
## sqrt(incidencerate):f.stateNew Hampshire	1.230e+02
## sqrt(incidencerate):f.stateNew Jersey	-5.253e+00
## sqrt(incidencerate):f.stateNew Mexico	2.368e+00
## sqrt(incidencerate):f.stateNew York	1.946e+00
## sqrt(incidencerate):f.stateNorth Carolina	-1.864e+00
## sqrt(incidencerate):f.stateNorth Dakota	3.085e+00
## sqrt(incidencerate):f.stateOhio	3.358e+00
## sqrt(incidencerate):f.stateOklahoma	-7.080e+00
## sqrt(incidencerate):f.stateOregon	-4.261e+00
## sqrt(incidencerate):f.statePennsylvania	-1.471e-01
## sqrt(incidencerate):f.stateRhode Island	-4.191e+02
## sqrt(incidencerate):f.stateSouth Carolina	1.563e+00
## sqrt(incidencerate):f.stateSouth Dakota	-2.980e+00
## sqrt(incidencerate):f.stateTennessee	9.212e-01
## sqrt(incidencerate):f.stateTexas	2.498e+00
## sqrt(incidencerate):f.stateUtah	-1.185e+00
## sqrt(incidencerate):f.stateVermont	-1.398e+00
## sqrt(incidencerate):f.stateVirginia	-2.642e+00
## sqrt(incidencerate):f.stateWashington	-5.667e+00
## sqrt(incidencerate):f.stateWest Virginia	-1.112e+00
## sqrt(incidencerate):f.stateWisconsin	-4.957e+00
## sqrt(incidencerate):f.stateWyoming	1.706e+01
## sqrt(incidencerate):f.education25_overBalanced Eduaction	1.680e+00
## sqrt(incidencerate):f.education25_overMostly Bachelor's	-1.514e+00
## pctbachdeg25_over:f.stateAlaska	-4.495e+00
## pctbachdeg25_over:f.stateArizona	4.752e+00
## pctbachdeg25_over:f.stateArkansas	-9.768e-02
## pctbachdeg25_over:f.stateCalifornia	1.780e+00
## pctbachdeg25_over:f.stateColorado	-8.296e-01
## pctbachdeg25_over:f.stateConnecticut	1.861e+00
## pctbachdeg25_over:f.stateDelaware	NA
## pctbachdeg25_over:f.stateFlorida	-2.905e+00
## pctbachdeg25_over:f.stateGeorgia	-2.500e-01
## pctbachdeg25_over:f.stateHawaii	NA

## pctbachdeg25_over:f.stateIdaho	-1.305e+00
## pctbachdeg25_over:f.stateIllinois	-3.651e-01
## pctbachdeg25_over:f.stateIndiana	8.738e-01
## pctbachdeg25_over:f.stateIowa	2.089e+00
## pctbachdeg25_over:f.stateKansas	4.357e-01
## pctbachdeg25_over:f.stateKentucky	3.037e-01
## pctbachdeg25_over:f.stateLouisiana	7.611e-01
## pctbachdeg25_over:f.stateMaine	-2.776e+00
## pctbachdeg25_over:f.stateMaryland	-1.369e+00
## pctbachdeg25_over:f.stateMassachusetts	-1.272e+00
## pctbachdeg25_over:f.stateMichigan	-6.744e-01
## pctbachdeg25_over:f.stateMinnesota	7.131e-01
## pctbachdeg25_over:f.stateMississippi	2.489e+00
## pctbachdeg25_over:f.stateMissouri	1.459e-01
## pctbachdeg25_over:f.stateMontana	1.644e+00
## pctbachdeg25_over:f.stateNebraska	9.389e-01
## pctbachdeg25_over:f.stateNevada	-5.498e+00
## pctbachdeg25_over:f.stateNew Hampshire	-2.627e+02
## pctbachdeg25_over:f.stateNew Jersey	-1.701e+00
## pctbachdeg25_over:f.stateNew Mexico	-1.351e+00
## pctbachdeg25_over:f.stateNew York	4.221e-01
## pctbachdeg25_over:f.stateNorth Carolina	-9.827e-02
## pctbachdeg25_over:f.stateNorth Dakota	2.846e-01
## pctbachdeg25_over:f.stateOhio	5.463e-01
## pctbachdeg25_over:f.stateOklahoma	2.254e+00
## pctbachdeg25_over:f.stateOregon	1.033e+00
## pctbachdeg25_over:f.statePennsylvania	7.539e-01
## pctbachdeg25_over:f.stateRhode Island	NA
## pctbachdeg25_over:f.stateSouth Carolina	7.821e-01
## pctbachdeg25_over:f.stateSouth Dakota	1.898e+00
## pctbachdeg25_over:f.stateTennessee	7.729e-01
## pctbachdeg25_over:f.stateTexas	5.269e-01
## pctbachdeg25_over:f.stateUtah	4.421e-01
## pctbachdeg25_over:f.stateVermont	1.379e+00
## pctbachdeg25_over:f.stateVirginia	-3.731e-01
## pctbachdeg25_over:f.stateWashington	-1.579e+00
## pctbachdeg25_over:f.stateWest Virginia	3.663e+00
## pctbachdeg25_over:f.stateWisconsin	1.181e+00
## pctbachdeg25_over:f.stateWyoming	2.734e+00
## log(medincome):f.stateAlaska	1.063e+01
## log(medincome):f.stateArizona	3.263e+02
## log(medincome):f.stateArkansas	-8.217e+00
## log(medincome):f.stateCalifornia	-1.427e+01
## log(medincome):f.stateColorado	4.830e+01
## log(medincome):f.stateConnecticut	-1.136e+02
## log(medincome):f.stateDelaware	NA
## log(medincome):f.stateFlorida	-3.817e+01
## log(medincome):f.stateGeorgia	-3.701e+01
## log(medincome):f.stateHawaii	NA
## log(medincome):f.stateIdaho	4.174e+01
## log(medincome):f.stateIllinois	1.871e+01
## log(medincome):f.stateIndiana	2.301e+01
## log(medincome):f.stateIowa	-2.506e+01
## log(medincome):f.stateKansas	1.471e+01

## log(medincome):f.stateKentucky	-8.437e+00
## log(medincome):f.stateLouisiana	-1.468e+01
## log(medincome):f.stateMaine	5.636e+01
## log(medincome):f.stateMaryland	1.090e+02
## log(medincome):f.stateMassachusetts	6.388e+00
## log(medincome):f.stateMichigan	-1.339e+00
## log(medincome):f.stateMinnesota	6.433e+01
## log(medincome):f.stateMississippi	5.453e+01
## log(medincome):f.stateMissouri	-1.659e+01
## log(medincome):f.stateMontana	2.988e+01
## log(medincome):f.stateNebraska	5.468e+01
## log(medincome):f.stateNevada	-3.166e+02
## log(medincome):f.stateNew Hampshire	2.947e+03
## log(medincome):f.stateNew Jersey	3.507e+01
## log(medincome):f.stateNew Mexico	-4.222e+01
## log(medincome):f.stateNew York	-2.647e+01
## log(medincome):f.stateNorth Carolina	6.656e+00
## log(medincome):f.stateNorth Dakota	3.531e+01
## log(medincome):f.stateOhio	1.150e+01
## log(medincome):f.stateOklahoma	-2.332e+01
## log(medincome):f.stateOregon	5.334e+01
## log(medincome):f.statePennsylvania	3.505e+01
## log(medincome):f.stateRhode Island	NA
## log(medincome):f.stateSouth Carolina	-4.794e+01
## log(medincome):f.stateSouth Dakota	-1.171e+01
## log(medincome):f.stateTennessee	2.898e+01
## log(medincome):f.stateTexas	-1.633e+01
## log(medincome):f.stateUtah	-5.496e+01
## log(medincome):f.stateVermont	1.295e+02
## log(medincome):f.stateVirginia	3.973e+01
## log(medincome):f.stateWashington	9.619e+00
## log(medincome):f.stateWest Virginia	8.304e+01
## log(medincome):f.stateWisconsin	1.803e+01
## log(medincome):f.stateWyoming	1.187e+02
## log(medincome):f.education25_overBalanced Eduaction	2.600e+01
## log(medincome):f.education25_overMostly Bachelor's	2.962e+01
## log(medincome):f.unemploymentModerate Unemployment	1.070e+01
## log(medincome):f.unemploymentHigh Unemployment	2.690e+01
## log(pctemployed16_over):f.stateAlaska	2.902e+02
## log(pctemployed16_over):f.stateArizona	-1.857e+02
## log(pctemployed16_over):f.stateArkansas	2.890e+01
## log(pctemployed16_over):f.stateCalifornia	-4.857e+01
## log(pctemployed16_over):f.stateColorado	8.251e+00
## log(pctemployed16_over):f.stateConnecticut	-5.297e+01
## log(pctemployed16_over):f.stateDelaware	NA
## log(pctemployed16_over):f.stateFlorida	-4.162e-01
## log(pctemployed16_over):f.stateGeorgia	-3.242e+00
## log(pctemployed16_over):f.stateHawaii	NA
## log(pctemployed16_over):f.stateIdaho	-5.575e+01
## log(pctemployed16_over):f.stateIllinois	-2.824e+01
## log(pctemployed16_over):f.stateIndiana	-8.125e+01
## log(pctemployed16_over):f.stateIowa	-1.263e+01
## log(pctemployed16_over):f.stateKansas	-1.485e+02
## log(pctemployed16_over):f.stateKentucky	-8.543e+01

## log(pctemployed16_over):f.stateLouisiana	-5.114e+01	
## log(pctemployed16_over):f.stateMaine	6.513e+01	
## log(pctemployed16_over):f.stateMaryland	-3.104e+01	
## log(pctemployed16_over):f.stateMassachusetts	2.319e+02	
## log(pctemployed16_over):f.stateMichigan	1.012e+01	
## log(pctemployed16_over):f.stateMinnesota	-8.283e+01	
## log(pctemployed16_over):f.stateMississippi	-5.980e+01	
## log(pctemployed16_over):f.stateMissouri	3.483e+01	
## log(pctemployed16_over):f.stateMontana	-3.589e+01	
## log(pctemployed16_over):f.stateNebraska	-1.474e+02	
## log(pctemployed16_over):f.stateNevada	9.582e+01	
## log(pctemployed16_over):f.stateNew Hampshire	4.139e+02	
## log(pctemployed16_over):f.stateNew Jersey	-1.492e+02	
## log(pctemployed16_over):f.stateNew Mexico	1.589e+01	
## log(pctemployed16_over):f.stateNew York	-3.833e+01	
## log(pctemployed16_over):f.stateNorth Carolina	-2.898e+00	
## log(pctemployed16_over):f.stateNorth Dakota	-2.613e+01	
## log(pctemployed16_over):f.stateOhio	-3.072e+01	
## log(pctemployed16_over):f.stateOklahoma	4.723e+01	
## log(pctemployed16_over):f.stateOregon	-6.214e+01	
## log(pctemployed16_over):f.statePennsylvania	-5.151e+01	
## log(pctemployed16_over):f.stateRhode Island	NA	
## log(pctemployed16_over):f.stateSouth Carolina	4.916e+01	
## log(pctemployed16_over):f.stateSouth Dakota	-5.085e+01	
## log(pctemployed16_over):f.stateTennessee	-5.991e+01	
## log(pctemployed16_over):f.stateTexas	-5.361e+01	
## log(pctemployed16_over):f.stateUtah	9.337e+00	
## log(pctemployed16_over):f.stateVermont	1.494e+01	
## log(pctemployed16_over):f.stateVirginia	-3.289e+01	
## log(pctemployed16_over):f.stateWashington	-4.906e+01	
## log(pctemployed16_over):f.stateWest Virginia	-1.891e+02	
## log(pctemployed16_over):f.stateWisconsin	-1.221e+02	
## log(pctemployed16_over):f.stateWyoming	-1.796e+02	
## log(pctemployed16_over):f.education25_overBalanced Eduaction	-3.355e+01	
## log(pctemployed16_over):f.education25_overMostly Bachelor's	4.693e+00	
## log(pctemployed16_over):f.unemploymentModerate Unemployment	-2.224e+01	
## log(pctemployed16_over):f.unemploymentHigh Unemployment	-4.226e+01	
##	Std. Error	t value
## (Intercept)	6.131e+02	0.261
## pctpubliccoveragealone	2.136e+00	-0.015
## sqrt(incidencerate)	4.616e+00	1.877
## pctbachdeg25_over	1.062e+00	-1.297
## log(medincome)	4.905e+01	-0.710
## log(pctemployed16_over)	4.796e+01	1.217
## f.stateAlaska	8.850e+02	-1.528
## f.stateArizona	1.953e+03	-1.334
## f.stateArkansas	7.234e+02	0.029
## f.stateCalifornia	6.893e+02	0.385
## f.stateColorado	7.088e+02	-0.853
## f.stateConnecticut	3.731e+03	0.401
## f.stateDelaware	2.404e+01	-0.197
## f.stateFlorida	6.812e+02	0.731
## f.stateGeorgia	6.285e+02	0.585
## f.stateHawaii	5.734e+02	0.664

## f.stateIdaho	7.858e+02	-0.351
## f.stateIllinois	6.707e+02	-0.267
## f.stateIndiana	6.842e+02	-0.023
## f.stateIowa	7.018e+02	0.295
## f.stateKansas	8.907e+02	-1.724
## f.stateKentucky	6.600e+02	0.505
## f.stateLouisiana	6.587e+02	0.582
## f.stateMaine	1.406e+03	-0.527
## f.stateMaryland	1.088e+03	-1.380
## f.stateMassachusetts	1.677e+03	-0.536
## f.stateMichigan	6.711e+02	0.094
## f.stateMinnesota	1.168e+03	-1.468
## f.stateMississippi	6.600e+02	-0.482
## f.stateMissouri	6.663e+02	0.054
## f.stateMontana	6.920e+02	-0.320
## f.stateNebraska	6.768e+02	0.093
## f.stateNevada	1.910e+03	1.308
## f.stateNew Hampshire	1.398e+05	-0.212
## f.stateNew Jersey	2.317e+03	0.163
## f.stateNew Mexico	6.757e+02	0.567
## f.stateNew York	6.964e+02	0.560
## f.stateNorth Carolina	6.704e+02	-0.055
## f.stateNorth Dakota	7.019e+02	-0.507
## f.stateOhio	6.957e+02	-0.137
## f.stateOklahoma	6.871e+02	0.155
## f.stateOregon	7.141e+02	-0.410
## f.statePennsylvania	6.890e+02	-0.344
## f.stateRhode Island	6.791e+03	1.117
## f.stateSouth Carolina	8.474e+02	0.282
## f.stateSouth Dakota	6.765e+02	0.481
## f.stateTennessee	6.728e+02	-0.208
## f.stateTexas	6.252e+02	0.575
## f.stateUtah	7.920e+02	0.700
## f.stateVermont	3.079e+03	-0.520
## f.stateVirginia	6.376e+02	-0.429
## f.stateWashington	7.076e+02	0.348
## f.stateWest Virginia	6.916e+02	-0.287
## f.stateWisconsin	8.103e+02	0.454
## f.stateWyoming	1.541e+03	-0.619
## f.education25_overBalanced Eduaction	1.032e+02	-1.709
## f.education25_overMostly Bachelor's	1.820e+02	-1.811
## f.unemploymentModerate Unemployment	7.911e+01	-0.330
## f.unemploymentHigh Unemployment	1.061e+02	-1.100
## pctpubliccoveragealone:f.stateAlaska	2.561e+00	1.201
## pctpubliccoveragealone:f.stateArizona	5.854e+00	1.096
## pctpubliccoveragealone:f.stateArkansas	2.422e+00	0.917
## pctpubliccoveragealone:f.stateCalifornia	2.361e+00	0.390
## pctpubliccoveragealone:f.stateColorado	2.518e+00	0.821
## pctpubliccoveragealone:f.stateConnecticut	6.022e+00	-0.328
## pctpubliccoveragealone:f.stateDelaware	NA	NA
## pctpubliccoveragealone:f.stateFlorida	2.422e+00	-1.671
## pctpubliccoveragealone:f.stateGeorgia	2.220e+00	-0.696
## pctpubliccoveragealone:f.stateHawaii	3.206e+01	-0.699
## pctpubliccoveragealone:f.stateIdaho	2.540e+00	-1.034

## pctpubliccoveragealone:f.stateIllinois	2.305e+00	0.164
## pctpubliccoveragealone:f.stateIndiana	2.384e+00	0.404
## pctpubliccoveragealone:f.stateIowa	2.465e+00	0.770
## pctpubliccoveragealone:f.stateKansas	2.393e+00	-0.648
## pctpubliccoveragealone:f.stateKentucky	2.317e+00	-0.134
## pctpubliccoveragealone:f.stateLouisiana	2.398e+00	0.481
## pctpubliccoveragealone:f.stateMaine	5.607e+00	0.049
## pctpubliccoveragealone:f.stateMaryland	3.345e+00	1.251
## pctpubliccoveragealone:f.stateMassachusetts	7.093e+00	0.170
## pctpubliccoveragealone:f.stateMichigan	2.401e+00	0.417
## pctpubliccoveragealone:f.stateMinnesota	2.481e+00	0.669
## pctpubliccoveragealone:f.stateMississippi	2.262e+00	1.102
## pctpubliccoveragealone:f.stateMissouri	2.341e+00	0.859
## pctpubliccoveragealone:f.stateMontana	2.406e+00	0.360
## pctpubliccoveragealone:f.stateNebraska	2.332e+00	-0.018
## pctpubliccoveragealone:f.stateNevada	3.412e+00	-2.112
## pctpubliccoveragealone:f.stateNew Hampshire	6.228e+02	-0.211
## pctpubliccoveragealone:f.stateNew Jersey	6.210e+00	-0.240
## pctpubliccoveragealone:f.stateNew Mexico	2.271e+00	-0.657
## pctpubliccoveragealone:f.stateNew York	2.321e+00	-0.430
## pctpubliccoveragealone:f.stateNorth Carolina	2.368e+00	0.164
## pctpubliccoveragealone:f.stateNorth Dakota	2.308e+00	-0.233
## pctpubliccoveragealone:f.stateOhio	2.419e+00	0.328
## pctpubliccoveragealone:f.stateOklahoma	2.535e+00	1.691
## pctpubliccoveragealone:f.stateOregon	2.434e+00	0.616
## pctpubliccoveragealone:f.statePennsylvania	2.410e+00	1.077
## pctpubliccoveragealone:f.stateRhode Island	1.139e+02	1.187
## pctpubliccoveragealone:f.stateSouth Carolina	2.945e+00	0.560
## pctpubliccoveragealone:f.stateSouth Dakota	2.245e+00	0.631
## pctpubliccoveragealone:f.stateTennessee	2.363e+00	0.898
## pctpubliccoveragealone:f.stateTexas	2.197e+00	-0.716
## pctpubliccoveragealone:f.stateUtah	3.071e+00	-0.210
## pctpubliccoveragealone:f.stateVermont	1.133e+01	0.487
## pctpubliccoveragealone:f.stateVirginia	2.276e+00	1.007
## pctpubliccoveragealone:f.stateWashington	2.616e+00	-0.426
## pctpubliccoveragealone:f.stateWest Virginia	2.320e+00	0.784
## pctpubliccoveragealone:f.stateWisconsin	2.806e+00	0.245
## pctpubliccoveragealone:f.stateWyoming	5.968e+00	-0.051
## pctpubliccoveragealone:f.education25_overBalanced Eduaction	3.589e-01	-0.705
## pctpubliccoveragealone:f.education25_overMostly Bachelor's	8.463e-01	1.742
## sqrt(incidencerate):f.stateAlaska	6.845e+00	0.549
## sqrt(incidencerate):f.stateArizona	1.486e+01	-1.420
## sqrt(incidencerate):f.stateArkansas	5.595e+00	-0.768
## sqrt(incidencerate):f.stateCalifornia	6.988e+00	0.072
## sqrt(incidencerate):f.stateColorado	5.379e+00	0.022
## sqrt(incidencerate):f.stateConnecticut	5.686e+01	-0.034
## sqrt(incidencerate):f.stateDelaware	NA	NA
## sqrt(incidencerate):f.stateFlorida	4.778e+00	0.357
## sqrt(incidencerate):f.stateGeorgia	4.923e+00	0.623
## sqrt(incidencerate):f.stateHawaii	NA	NA
## sqrt(incidencerate):f.stateIdaho	5.597e+00	0.754
## sqrt(incidencerate):f.stateIllinois	5.632e+00	0.659
## sqrt(incidencerate):f.stateIndiana	5.853e+00	0.563
## sqrt(incidencerate):f.stateIowa	5.629e+00	0.290

## sqrt(incidencerate):f.stateKansas	3.073e+01	3.070
## sqrt(incidencerate):f.stateKentucky	5.106e+00	0.891
## sqrt(incidencerate):f.stateLouisiana	6.506e+00	-0.390
## sqrt(incidencerate):f.stateMaine	1.423e+01	-0.284
## sqrt(incidencerate):f.stateMaryland	9.516e+00	1.829
## sqrt(incidencerate):f.stateMassachusetts	2.358e+01	-0.269
## sqrt(incidencerate):f.stateMichigan	4.929e+00	-0.993
## sqrt(incidencerate):f.stateMinnesota	4.699e+01	1.307
## sqrt(incidencerate):f.stateMississippi	5.332e+00	-0.884
## sqrt(incidencerate):f.stateMissouri	4.861e+00	-0.319
## sqrt(incidencerate):f.stateMontana	5.519e+00	-0.166
## sqrt(incidencerate):f.stateNebraska	4.984e+00	-0.760
## sqrt(incidencerate):f.stateNevada	7.210e+01	0.484
## sqrt(incidencerate):f.stateNew Hampshire	5.518e+02	0.223
## sqrt(incidencerate):f.stateNew Jersey	2.285e+01	-0.230
## sqrt(incidencerate):f.stateNew Mexico	5.640e+00	0.420
## sqrt(incidencerate):f.stateNew York	7.192e+00	0.271
## sqrt(incidencerate):f.stateNorth Carolina	5.695e+00	-0.327
## sqrt(incidencerate):f.stateNorth Dakota	5.096e+00	0.605
## sqrt(incidencerate):f.stateOhio	6.157e+00	0.545
## sqrt(incidencerate):f.stateOklahoma	5.743e+00	-1.233
## sqrt(incidencerate):f.stateOregon	6.613e+00	-0.644
## sqrt(incidencerate):f.statePennsylvania	6.173e+00	-0.024
## sqrt(incidencerate):f.stateRhode Island	3.703e+02	-1.132
## sqrt(incidencerate):f.stateSouth Carolina	6.565e+00	0.238
## sqrt(incidencerate):f.stateSouth Dakota	5.296e+00	-0.563
## sqrt(incidencerate):f.stateTennessee	5.512e+00	0.167
## sqrt(incidencerate):f.stateTexas	4.763e+00	0.524
## sqrt(incidencerate):f.stateUtah	7.214e+00	-0.164
## sqrt(incidencerate):f.stateVermont	1.919e+01	-0.073
## sqrt(incidencerate):f.stateVirginia	4.824e+00	-0.548
## sqrt(incidencerate):f.stateWashington	6.054e+00	-0.936
## sqrt(incidencerate):f.stateWest Virginia	5.045e+00	-0.220
## sqrt(incidencerate):f.stateWisconsin	5.394e+00	-0.919
## sqrt(incidencerate):f.stateWyoming	1.245e+01	1.370
## sqrt(incidencerate):f.education25_overBalanced Eduaction	1.123e+00	1.496
## sqrt(incidencerate):f.education25_overMostly Bachelor's	1.998e+00	-0.758
## pctbachdeg25_over:f.stateAlaska	2.315e+00	-1.942
## pctbachdeg25_over:f.stateArizona	2.821e+00	1.684
## pctbachdeg25_over:f.stateArkansas	1.695e+00	-0.058
## pctbachdeg25_over:f.stateCalifornia	1.700e+00	1.047
## pctbachdeg25_over:f.stateColorado	1.214e+00	-0.683
## pctbachdeg25_over:f.stateConnecticut	7.630e+00	0.244
## pctbachdeg25_over:f.stateDelaware	NA	NA
## pctbachdeg25_over:f.stateFlorida	1.457e+00	-1.993
## pctbachdeg25_over:f.stateGeorgia	1.186e+00	-0.211
## pctbachdeg25_over:f.stateHawaii	NA	NA
## pctbachdeg25_over:f.stateIdaho	1.414e+00	-0.923
## pctbachdeg25_over:f.stateIllinois	1.274e+00	-0.287
## pctbachdeg25_over:f.stateIndiana	1.227e+00	0.712
## pctbachdeg25_over:f.stateIowa	1.227e+00	1.702
## pctbachdeg25_over:f.stateKansas	1.254e+00	0.347
## pctbachdeg25_over:f.stateKentucky	1.243e+00	0.244
## pctbachdeg25_over:f.stateLouisiana	1.514e+00	0.503

## pctbachdeg25_over:f.stateMaine	4.003e+00	-0.693
## pctbachdeg25_over:f.stateMaryland	2.174e+00	-0.629
## pctbachdeg25_over:f.stateMassachusetts	6.322e+00	-0.201
## pctbachdeg25_over:f.stateMichigan	1.403e+00	-0.481
## pctbachdeg25_over:f.stateMinnesota	1.302e+00	0.548
## pctbachdeg25_over:f.stateMississippi	1.258e+00	1.978
## pctbachdeg25_over:f.stateMissouri	1.371e+00	0.106
## pctbachdeg25_over:f.stateMontana	1.355e+00	1.213
## pctbachdeg25_over:f.stateNebraska	1.316e+00	0.713
## pctbachdeg25_over:f.stateNevada	2.007e+00	-2.740
## pctbachdeg25_over:f.stateNew Hampshire	1.238e+03	-0.212
## pctbachdeg25_over:f.stateNew Jersey	4.124e+00	-0.413
## pctbachdeg25_over:f.stateNew Mexico	2.122e+00	-0.637
## pctbachdeg25_over:f.stateNew York	1.572e+00	0.268
## pctbachdeg25_over:f.stateNorth Carolina	1.342e+00	-0.073
## pctbachdeg25_over:f.stateNorth Dakota	1.944e+00	0.146
## pctbachdeg25_over:f.stateOhio	1.334e+00	0.409
## pctbachdeg25_over:f.stateOklahoma	1.686e+00	1.338
## pctbachdeg25_over:f.stateOregon	1.503e+00	0.687
## pctbachdeg25_over:f.statePennsylvania	1.453e+00	0.519
## pctbachdeg25_over:f.stateRhode Island	NA	NA
## pctbachdeg25_over:f.stateSouth Carolina	1.535e+00	0.509
## pctbachdeg25_over:f.stateSouth Dakota	1.416e+00	1.340
## pctbachdeg25_over:f.stateTennessee	1.345e+00	0.575
## pctbachdeg25_over:f.stateTexas	1.165e+00	0.452
## pctbachdeg25_over:f.stateUtah	1.587e+00	0.278
## pctbachdeg25_over:f.stateVermont	6.007e+00	0.230
## pctbachdeg25_over:f.stateVirginia	1.205e+00	-0.310
## pctbachdeg25_over:f.stateWashington	1.601e+00	-0.986
## pctbachdeg25_over:f.stateWest Virginia	1.760e+00	2.081
## pctbachdeg25_over:f.stateWisconsin	1.337e+00	0.883
## pctbachdeg25_over:f.stateWyoming	6.649e+00	0.411
## log(medincome):f.stateAlaska	5.966e+01	0.178
## log(medincome):f.stateArizona	2.057e+02	1.586
## log(medincome):f.stateArkansas	6.199e+01	-0.133
## log(medincome):f.stateCalifornia	5.656e+01	-0.252
## log(medincome):f.stateColorado	5.856e+01	0.825
## log(medincome):f.stateConnecticut	2.454e+02	-0.463
## log(medincome):f.stateDelaware	NA	NA
## log(medincome):f.stateFlorida	5.652e+01	-0.675
## log(medincome):f.stateGeorgia	5.088e+01	-0.727
## log(medincome):f.stateHawaii	NA	NA
## log(medincome):f.stateIdaho	7.387e+01	0.565
## log(medincome):f.stateIllinois	5.434e+01	0.344
## log(medincome):f.stateIndiana	5.636e+01	0.408
## log(medincome):f.stateIowa	6.124e+01	-0.409
## log(medincome):f.stateKansas	5.327e+01	0.276
## log(medincome):f.stateKentucky	5.227e+01	-0.161
## log(medincome):f.stateLouisiana	5.487e+01	-0.268
## log(medincome):f.stateMaine	1.163e+02	0.485
## log(medincome):f.stateMaryland	9.797e+01	1.112
## log(medincome):f.stateMassachusetts	2.091e+02	0.031
## log(medincome):f.stateMichigan	5.704e+01	-0.023
## log(medincome):f.stateMinnesota	5.469e+01	1.176

## log(medincome):f.stateMississippi	5.340e+01	1.021
## log(medincome):f.stateMissouri	5.437e+01	-0.305
## log(medincome):f.stateMontana	5.947e+01	0.502
## log(medincome):f.stateNebraska	5.488e+01	0.996
## log(medincome):f.stateNevada	8.607e+01	-3.678
## log(medincome):f.stateNew Hampshire	1.398e+04	0.211
## log(medincome):f.stateNew Jersey	1.119e+02	0.313
## log(medincome):f.stateNew Mexico	5.595e+01	-0.755
## log(medincome):f.stateNew York	5.497e+01	-0.482
## log(medincome):f.stateNorth Carolina	5.585e+01	0.119
## log(medincome):f.stateNorth Dakota	5.583e+01	0.632
## log(medincome):f.stateOhio	5.613e+01	0.205
## log(medincome):f.stateOklahoma	5.591e+01	-0.417
## log(medincome):f.stateOregon	6.816e+01	0.783
## log(medincome):f.statePennsylvania	5.694e+01	0.616
## log(medincome):f.stateRhode Island	NA	NA
## log(medincome):f.stateSouth Carolina	7.064e+01	-0.679
## log(medincome):f.stateSouth Dakota	5.582e+01	-0.210
## log(medincome):f.stateTennessee	5.523e+01	0.525
## log(medincome):f.stateTexas	5.026e+01	-0.325
## log(medincome):f.stateUtah	6.272e+01	-0.876
## log(medincome):f.stateVermont	3.415e+02	0.379
## log(medincome):f.stateVirginia	5.103e+01	0.779
## log(medincome):f.stateWashington	6.391e+01	0.150
## log(medincome):f.stateWest Virginia	5.890e+01	1.410
## log(medincome):f.stateWisconsin	6.986e+01	0.258
## log(medincome):f.stateWyoming	1.898e+02	0.625
## log(medincome):f.education25_overBalanced Eduaction	9.510e+00	2.735
## log(medincome):f.education25_overMostly Bachelor's	1.462e+01	2.027
## log(medincome):f.unemploymentModerate Unemployment	8.917e+00	1.200
## log(medincome):f.unemploymentHigh Unemployment	1.173e+01	2.294
## log(pctemployed16_over):f.stateAlaska	1.092e+02	2.657
## log(pctemployed16_over):f.stateArizona	1.182e+02	-1.571
## log(pctemployed16_over):f.stateArkansas	5.384e+01	0.537
## log(pctemployed16_over):f.stateCalifornia	5.493e+01	-0.884
## log(pctemployed16_over):f.stateColorado	5.231e+01	0.158
## log(pctemployed16_over):f.stateConnecticut	4.263e+02	-0.124
## log(pctemployed16_over):f.stateDelaware	NA	NA
## log(pctemployed16_over):f.stateFlorida	5.616e+01	-0.007
## log(pctemployed16_over):f.stateGeorgia	4.822e+01	-0.067
## log(pctemployed16_over):f.stateHawaii	NA	NA
## log(pctemployed16_over):f.stateIdaho	6.752e+01	-0.826
## log(pctemployed16_over):f.stateIllinois	5.180e+01	-0.545
## log(pctemployed16_over):f.stateIndiana	6.514e+01	-1.247
## log(pctemployed16_over):f.stateIowa	7.522e+01	-0.168
## log(pctemployed16_over):f.stateKansas	6.275e+01	-2.367
## log(pctemployed16_over):f.stateKentucky	5.244e+01	-1.629
## log(pctemployed16_over):f.stateLouisiana	5.942e+01	-0.861
## log(pctemployed16_over):f.stateMaine	2.161e+02	0.301
## log(pctemployed16_over):f.stateMaryland	1.817e+02	-0.171
## log(pctemployed16_over):f.stateMassachusetts	7.428e+02	0.312
## log(pctemployed16_over):f.stateMichigan	5.345e+01	0.189
## log(pctemployed16_over):f.stateMinnesota	7.210e+01	-1.149
## log(pctemployed16_over):f.stateMississippi	5.072e+01	-1.179

## log(pctemployed16_over):f.stateMissouri	5.063e+01	0.688
## log(pctemployed16_over):f.stateMontana	6.829e+01	-0.526
## log(pctemployed16_over):f.stateNebraska	6.082e+01	-2.423
## log(pctemployed16_over):f.stateNevada	7.154e+01	1.340
## log(pctemployed16_over):f.stateNew Hampshire	1.881e+03	0.220
## log(pctemployed16_over):f.stateNew Jersey	3.338e+02	-0.447
## log(pctemployed16_over):f.stateNew Mexico	6.409e+01	0.248
## log(pctemployed16_over):f.stateNew York	6.995e+01	-0.548
## log(pctemployed16_over):f.stateNorth Carolina	5.466e+01	-0.053
## log(pctemployed16_over):f.stateNorth Dakota	1.097e+02	-0.238
## log(pctemployed16_over):f.stateOhio	7.366e+01	-0.417
## log(pctemployed16_over):f.stateOklahoma	5.927e+01	0.797
## log(pctemployed16_over):f.stateOregon	7.252e+01	-0.857
## log(pctemployed16_over):f.statePennsylvania	6.445e+01	-0.799
## log(pctemployed16_over):f.stateRhode Island	NA	NA
## log(pctemployed16_over):f.stateSouth Carolina	6.021e+01	0.816
## log(pctemployed16_over):f.stateSouth Dakota	5.911e+01	-0.860
## log(pctemployed16_over):f.stateTennessee	5.382e+01	-1.113
## log(pctemployed16_over):f.stateTexas	4.745e+01	-1.130
## log(pctemployed16_over):f.stateUtah	8.163e+01	0.114
## log(pctemployed16_over):f.stateVermont	2.343e+02	0.064
## log(pctemployed16_over):f.stateVirginia	5.269e+01	-0.624
## log(pctemployed16_over):f.stateWashington	6.391e+01	-0.768
## log(pctemployed16_over):f.stateWest Virginia	6.400e+01	-2.955
## log(pctemployed16_over):f.stateWisconsin	7.294e+01	-1.675
## log(pctemployed16_over):f.stateWyoming	2.585e+02	-0.695
## log(pctemployed16_over):f.education25_overBalanced Eduaction	1.271e+01	-2.640
## log(pctemployed16_over):f.education25_overMostly Bachelor's	2.788e+01	0.168
## log(pctemployed16_over):f.unemploymentModerate Unemployment	1.449e+01	-1.535
## log(pctemployed16_over):f.unemploymentHigh Unemployment	1.680e+01	-2.516
##	Pr(> t)	
## (Intercept)	0.794026	
## pctpubliccoveragealone	0.987791	
## sqrt(incidencerate)	0.060676	.
## pctbachdeg25_over	0.194791	
## log(medincome)	0.477691	
## log(pctemployed16_over)	0.223863	
## f.stateAlaska	0.126830	
## f.stateArizona	0.182277	
## f.stateArkansas	0.976862	
## f.stateCalifornia	0.700343	
## f.stateColorado	0.393519	
## f.stateConnecticut	0.688556	
## f.stateDelaware	0.843739	
## f.stateFlorida	0.464759	
## f.stateGeorgia	0.558511	
## f.stateHawaii	0.506702	
## f.stateIdaho	0.726010	
## f.stateIllinois	0.789544	
## f.stateIndiana	0.981345	
## f.stateIowa	0.767748	
## f.stateKansas	0.084944	.
## f.stateKentucky	0.613598	
## f.stateLouisiana	0.560354	

## f.stateMaine	0.598172
## f.stateMaryland	0.167753
## f.stateMassachusetts	0.592315
## f.stateMichigan	0.925140
## f.stateMinnesota	0.142308
## f.stateMississippi	0.629681
## f.stateMissouri	0.956817
## f.stateMontana	0.748859
## f.stateNebraska	0.925904
## f.stateNevada	0.191076
## f.stateNew Hampshire	0.831933
## f.stateNew Jersey	0.870335
## f.stateNew Mexico	0.570961
## f.stateNew York	0.575646
## f.stateNorth Carolina	0.956044
## f.stateNorth Dakota	0.612029
## f.stateOhio	0.890995
## f.stateOklahoma	0.877223
## f.stateOregon	0.681654
## f.statePennsylvania	0.730987
## f.stateRhode Island	0.264167
## f.stateSouth Carolina	0.777858
## f.stateSouth Dakota	0.630816
## f.stateTennessee	0.835329
## f.stateTexas	0.565486
## f.stateUtah	0.483946
## f.stateVermont	0.603440
## f.stateVirginia	0.668078
## f.stateWashington	0.728104
## f.stateWest Virginia	0.774336
## f.stateWisconsin	0.649677
## f.stateWyoming	0.535876
## f.education25_overBalanced Eduaction	0.087735 .
## f.education25_overMostly Bachelor's	0.070266 .
## f.unemploymentModerate Unemployment	0.741823
## f.unemploymentHigh Unemployment	0.271375
## pctpubliccoveragealone:f.stateAlaska	0.230006
## pctpubliccoveragealone:f.stateArizona	0.273277
## pctpubliccoveragealone:f.stateArkansas	0.359332
## pctpubliccoveragealone:f.stateCalifornia	0.696373
## pctpubliccoveragealone:f.stateColorado	0.411829
## pctpubliccoveragealone:f.stateConnecticut	0.743170
## pctpubliccoveragealone:f.stateDelaware	NA
## pctpubliccoveragealone:f.stateFlorida	0.094939 .
## pctpubliccoveragealone:f.stateGeorgia	0.486296
## pctpubliccoveragealone:f.stateHawaii	0.484967
## pctpubliccoveragealone:f.stateIdaho	0.301345
## pctpubliccoveragealone:f.stateIllinois	0.869748
## pctpubliccoveragealone:f.stateIndiana	0.686306
## pctpubliccoveragealone:f.stateIowa	0.441591
## pctpubliccoveragealone:f.stateKansas	0.517072
## pctpubliccoveragealone:f.stateKentucky	0.893303
## pctpubliccoveragealone:f.stateLouisiana	0.630552
## pctpubliccoveragealone:f.stateMaine	0.960831

## pctpubliccoveragealone:f.stateMaryland	0.211079
## pctpubliccoveragealone:f.stateMassachusetts	0.864899
## pctpubliccoveragealone:f.stateMichigan	0.676384
## pctpubliccoveragealone:f.stateMinnesota	0.503740
## pctpubliccoveragealone:f.stateMississippi	0.270660
## pctpubliccoveragealone:f.stateMissouri	0.390238
## pctpubliccoveragealone:f.stateMontana	0.718841
## pctpubliccoveragealone:f.stateNebraska	0.985350
## pctpubliccoveragealone:f.stateNevada	0.034861 *
## pctpubliccoveragealone:f.stateNew Hampshire	0.832724
## pctpubliccoveragealone:f.stateNew Jersey	0.809990
## pctpubliccoveragealone:f.stateNew Mexico	0.511544
## pctpubliccoveragealone:f.stateNew York	0.666916
## pctpubliccoveragealone:f.stateNorth Carolina	0.869570
## pctpubliccoveragealone:f.stateNorth Dakota	0.815758
## pctpubliccoveragealone:f.stateOhio	0.743031
## pctpubliccoveragealone:f.stateOklahoma	0.091135 .
## pctpubliccoveragealone:f.stateOregon	0.537728
## pctpubliccoveragealone:f.statePennsylvania	0.281607
## pctpubliccoveragealone:f.stateRhode Island	0.235365
## pctpubliccoveragealone:f.stateSouth Carolina	0.575712
## pctpubliccoveragealone:f.stateSouth Dakota	0.528382
## pctpubliccoveragealone:f.stateTennessee	0.369268
## pctpubliccoveragealone:f.stateTexas	0.474101
## pctpubliccoveragealone:f.stateUtah	0.833859
## pctpubliccoveragealone:f.stateVermont	0.626156
## pctpubliccoveragealone:f.stateVirginia	0.314267
## pctpubliccoveragealone:f.stateWashington	0.670072
## pctpubliccoveragealone:f.stateWest Virginia	0.432992
## pctpubliccoveragealone:f.stateWisconsin	0.806829
## pctpubliccoveragealone:f.stateWyoming	0.959610
## pctpubliccoveragealone:f.education25_overBalanced Eduaction	0.480650
## pctpubliccoveragealone:f.education25_overMostly Bachelor's	0.081714 .
## sqrt(incidencerate):f.stateAlaska	0.582908
## sqrt(incidencerate):f.stateArizona	0.155955
## sqrt(incidencerate):f.stateArkansas	0.442650
## sqrt(incidencerate):f.stateCalifornia	0.942607
## sqrt(incidencerate):f.stateColorado	0.982227
## sqrt(incidencerate):f.stateConnecticut	0.973001
## sqrt(incidencerate):f.stateDelaware	NA
## sqrt(incidencerate):f.stateFlorida	0.720859
## sqrt(incidencerate):f.stateGeorgia	0.533074
## sqrt(incidencerate):f.stateHawaii	NA
## sqrt(incidencerate):f.stateIdaho	0.451239
## sqrt(incidencerate):f.stateIllinois	0.509750
## sqrt(incidencerate):f.stateIndiana	0.573531
## sqrt(incidencerate):f.stateIowa	0.771906
## sqrt(incidencerate):f.stateKansas	0.002178 **
## sqrt(incidencerate):f.stateKentucky	0.373265
## sqrt(incidencerate):f.stateLouisiana	0.696827
## sqrt(incidencerate):f.stateMaine	0.776659
## sqrt(incidencerate):f.stateMaryland	0.067588 .
## sqrt(incidencerate):f.stateMassachusetts	0.787775
## sqrt(incidencerate):f.stateMichigan	0.320912

## sqrt(incidencerate):f.stateMinnesota	0.191299
## sqrt(incidencerate):f.stateMississippi	0.376983
## sqrt(incidencerate):f.stateMissouri	0.749951
## sqrt(incidencerate):f.stateMontana	0.868210
## sqrt(incidencerate):f.stateNebraska	0.447288
## sqrt(incidencerate):f.stateNevada	0.628334
## sqrt(incidencerate):f.stateNew Hampshire	0.823678
## sqrt(incidencerate):f.stateNew Jersey	0.818228
## sqrt(incidencerate):f.stateNew Mexico	0.674622
## sqrt(incidencerate):f.stateNew York	0.786789
## sqrt(incidencerate):f.stateNorth Carolina	0.743497
## sqrt(incidencerate):f.stateNorth Dakota	0.545045
## sqrt(incidencerate):f.stateOhio	0.585542
## sqrt(incidencerate):f.stateOklahoma	0.217790
## sqrt(incidencerate):f.stateOregon	0.519446
## sqrt(incidencerate):f.statePennsylvania	0.980996
## sqrt(incidencerate):f.stateRhode Island	0.257947
## sqrt(incidencerate):f.stateSouth Carolina	0.811897
## sqrt(incidencerate):f.stateSouth Dakota	0.573786
## sqrt(incidencerate):f.stateTennessee	0.867285
## sqrt(incidencerate):f.stateTexas	0.600024
## sqrt(incidencerate):f.stateUtah	0.869602
## sqrt(incidencerate):f.stateVermont	0.941964
## sqrt(incidencerate):f.stateVirginia	0.584043
## sqrt(incidencerate):f.stateWashington	0.349415
## sqrt(incidencerate):f.stateWest Virginia	0.825610
## sqrt(incidencerate):f.stateWisconsin	0.358164
## sqrt(incidencerate):f.stateWyoming	0.170803
## sqrt(incidencerate):f.education25_overBalanced Eduaction	0.134865
## sqrt(incidencerate):f.education25_overMostly Bachelor's	0.448750
## pctbachdeg25_over:f.stateAlaska	0.052330 .
## pctbachdeg25_over:f.stateArizona	0.092320 .
## pctbachdeg25_over:f.stateArkansas	0.954063
## pctbachdeg25_over:f.stateCalifornia	0.295325
## pctbachdeg25_over:f.stateColorado	0.494399
## pctbachdeg25_over:f.stateConnecticut	0.807290
## pctbachdeg25_over:f.stateDelaware	NA
## pctbachdeg25_over:f.stateFlorida	0.046394 *
## pctbachdeg25_over:f.stateGeorgia	0.833116
## pctbachdeg25_over:f.stateHawaii	NA
## pctbachdeg25_over:f.stateIdaho	0.356233
## pctbachdeg25_over:f.stateIllinois	0.774463
## pctbachdeg25_over:f.stateIndiana	0.476319
## pctbachdeg25_over:f.stateIowa	0.089014 .
## pctbachdeg25_over:f.stateKansas	0.728264
## pctbachdeg25_over:f.stateKentucky	0.806922
## pctbachdeg25_over:f.stateLouisiana	0.615305
## pctbachdeg25_over:f.stateMaine	0.488121
## pctbachdeg25_over:f.stateMaryland	0.529187
## pctbachdeg25_over:f.stateMassachusetts	0.840548
## pctbachdeg25_over:f.stateMichigan	0.630855
## pctbachdeg25_over:f.stateMinnesota	0.584068
## pctbachdeg25_over:f.stateMississippi	0.048130 *
## pctbachdeg25_over:f.stateMissouri	0.915242

## pctbachdeg25_over:f.stateMontana	0.225266
## pctbachdeg25_over:f.stateNebraska	0.475658
## pctbachdeg25_over:f.stateNevada	0.006214 **
## pctbachdeg25_over:f.stateNew Hampshire	0.831918
## pctbachdeg25_over:f.stateNew Jersey	0.679999
## pctbachdeg25_over:f.stateNew Mexico	0.524494
## pctbachdeg25_over:f.stateNew York	0.788361
## pctbachdeg25_over:f.stateNorth Carolina	0.941626
## pctbachdeg25_over:f.stateNorth Dakota	0.883604
## pctbachdeg25_over:f.stateOhio	0.682317
## pctbachdeg25_over:f.stateOklahoma	0.181254
## pctbachdeg25_over:f.stateOregon	0.492052
## pctbachdeg25_over:f.statePennsylvania	0.603972
## pctbachdeg25_over:f.stateRhode Island	NA
## pctbachdeg25_over:f.stateSouth Carolina	0.610564
## pctbachdeg25_over:f.stateSouth Dakota	0.180311
## pctbachdeg25_over:f.stateTennessee	0.565669
## pctbachdeg25_over:f.stateTexas	0.651037
## pctbachdeg25_over:f.stateUtah	0.780671
## pctbachdeg25_over:f.stateVermont	0.818442
## pctbachdeg25_over:f.stateVirginia	0.756949
## pctbachdeg25_over:f.stateWashington	0.324090
## pctbachdeg25_over:f.stateWest Virginia	0.037604 *
## pctbachdeg25_over:f.stateWisconsin	0.377166
## pctbachdeg25_over:f.stateWyoming	0.680949
## log(medincome):f.stateAlaska	0.858561
## log(medincome):f.stateArizona	0.112912
## log(medincome):f.stateArkansas	0.894557
## log(medincome):f.stateCalifornia	0.800818
## log(medincome):f.stateColorado	0.409598
## log(medincome):f.stateConnecticut	0.643460
## log(medincome):f.stateDelaware	NA
## log(medincome):f.stateFlorida	0.499505
## log(medincome):f.stateGeorgia	0.467049
## log(medincome):f.stateHawaii	NA
## log(medincome):f.stateIdaho	0.572101
## log(medincome):f.stateIllinois	0.730629
## log(medincome):f.stateIndiana	0.683153
## log(medincome):f.stateIowa	0.682466
## log(medincome):f.stateKansas	0.782528
## log(medincome):f.stateKentucky	0.871789
## log(medincome):f.stateLouisiana	0.789091
## log(medincome):f.stateMaine	0.627925
## log(medincome):f.stateMaryland	0.266220
## log(medincome):f.stateMassachusetts	0.975637
## log(medincome):f.stateMichigan	0.981275
## log(medincome):f.stateMinnesota	0.239628
## log(medincome):f.stateMississippi	0.307384
## log(medincome):f.stateMissouri	0.760285
## log(medincome):f.stateMontana	0.615440
## log(medincome):f.stateNebraska	0.319220
## log(medincome):f.stateNevada	0.000243 ***
## log(medincome):f.stateNew Hampshire	0.833074
## log(medincome):f.stateNew Jersey	0.753955

## log(medincome):f.stateNew Mexico	0.450571
## log(medincome):f.stateNew York	0.630193
## log(medincome):f.stateNorth Carolina	0.905162
## log(medincome):f.stateNorth Dakota	0.527202
## log(medincome):f.stateOhio	0.837719
## log(medincome):f.stateOklahoma	0.676649
## log(medincome):f.stateOregon	0.433961
## log(medincome):f.statePennsylvania	0.538310
## log(medincome):f.stateRhode Island	NA
## log(medincome):f.stateSouth Carolina	0.497431
## log(medincome):f.stateSouth Dakota	0.833879
## log(medincome):f.stateTennessee	0.599873
## log(medincome):f.stateTexas	0.745347
## log(medincome):f.stateUtah	0.380987
## log(medincome):f.stateVermont	0.704664
## log(medincome):f.stateVirginia	0.436363
## log(medincome):f.stateWashington	0.880393
## log(medincome):f.stateWest Virginia	0.158804
## log(medincome):f.stateWisconsin	0.796430
## log(medincome):f.stateWyoming	0.531990
## log(medincome):f.education25_overBalanced Eduaction	0.006319 **
## log(medincome):f.education25_overMostly Bachelor's	0.042877 *
## log(medincome):f.unemploymentModerate Unemployment	0.230503
## log(medincome):f.unemploymentHigh Unemployment	0.021898 *
## log(pctemployed16_over):f.stateAlaska	0.007962 **
## log(pctemployed16_over):f.stateArizona	0.116345
## log(pctemployed16_over):f.stateArkansas	0.591461
## log(pctemployed16_over):f.stateCalifornia	0.376646
## log(pctemployed16_over):f.stateColorado	0.874676
## log(pctemployed16_over):f.stateConnecticut	0.901129
## log(pctemployed16_over):f.stateDelaware	NA
## log(pctemployed16_over):f.stateFlorida	0.994088
## log(pctemployed16_over):f.stateGeorgia	0.946401
## log(pctemployed16_over):f.stateHawaii	NA
## log(pctemployed16_over):f.stateIdaho	0.409090
## log(pctemployed16_over):f.stateIllinois	0.585639
## log(pctemployed16_over):f.stateIndiana	0.212492
## log(pctemployed16_over):f.stateIowa	0.866645
## log(pctemployed16_over):f.stateKansas	0.018064 *
## log(pctemployed16_over):f.stateKentucky	0.103518
## log(pctemployed16_over):f.stateLouisiana	0.389496
## log(pctemployed16_over):f.stateMaine	0.763154
## log(pctemployed16_over):f.stateMaryland	0.864408
## log(pctemployed16_over):f.stateMassachusetts	0.754906
## log(pctemployed16_over):f.stateMichigan	0.849812
## log(pctemployed16_over):f.stateMinnesota	0.250814
## log(pctemployed16_over):f.stateMississippi	0.238592
## log(pctemployed16_over):f.stateMissouri	0.491539
## log(pctemployed16_over):f.stateMontana	0.599248
## log(pctemployed16_over):f.stateNebraska	0.015488 *
## log(pctemployed16_over):f.stateNevada	0.180607
## log(pctemployed16_over):f.stateNew Hampshire	0.825869
## log(pctemployed16_over):f.stateNew Jersey	0.654975
## log(pctemployed16_over):f.stateNew Mexico	0.804266


```

## log(pctemployed16_over):f.stateNew York 0.583781
## log(pctemployed16_over):f.stateNorth Carolina 0.957724
## log(pctemployed16_over):f.stateNorth Dakota 0.811793
## log(pctemployed16_over):f.stateOhio 0.676740
## log(pctemployed16_over):f.stateOklahoma 0.425665
## log(pctemployed16_over):f.stateOregon 0.391634
## log(pctemployed16_over):f.statePennsylvania 0.424263
## log(pctemployed16_over):f.stateRhode Island NA
## log(pctemployed16_over):f.stateSouth Carolina 0.414361
## log(pctemployed16_over):f.stateSouth Dakota 0.389819
## log(pctemployed16_over):f.stateTennessee 0.265737
## log(pctemployed16_over):f.stateTexas 0.258719
## log(pctemployed16_over):f.stateUtah 0.908945
## log(pctemployed16_over):f.stateVermont 0.949166
## log(pctemployed16_over):f.stateVirginia 0.532560
## log(pctemployed16_over):f.stateWashington 0.442803
## log(pctemployed16_over):f.stateWest Virginia 0.003177 **
## log(pctemployed16_over):f.stateWisconsin 0.094236 .
## log(pctemployed16_over):f.stateWyoming 0.487170
## log(pctemployed16_over):f.education25_overBalanced Eduaction 0.008364 **
## log(pctemployed16_over):f.education25_overMostly Bachelor's 0.866345
## log(pctemployed16_over):f.unemploymentModerate Unemployment 0.124888
## log(pctemployed16_over):f.unemploymentHigh Unemployment 0.011981 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 17.22 on 1527 degrees of freedom
## Multiple R-squared:  0.6829, Adjusted R-squared:  0.62
## F-statistic: 10.85 on 303 and 1527 DF,  p-value: < 2.2e-16

m4<- lm(target_deathrate ~ pctpubliccoveragealone + sqrt(incidencerate) +
  pctbachdeg25_over + log(medincome) + f.state + f.unemployment +
  pctpubliccoveragealone:f.state +
  pctpubliccoveragealone:f.education25_over + sqrt(incidencerate):f.state +
  pctbachdeg25_over:f.state + log(medincome):f.state +
  + log(medincome):f.education25_over +
  + log(pctemployed16_over):f.state +
  log(pctemployed16_over):f.education25_over , data = df)

summary(m4)

##
## Call:
## lm(formula = target_deathrate ~ pctpubliccoveragealone + sqrt(incidencerate) +
##     pctbachdeg25_over + log(medincome) + f.state + f.unemployment +
##     pctpubliccoveragealone:f.state + pctpubliccoveragealone:f.education25_over +
##     sqrt(incidencerate):f.state + pctbachdeg25_over:f.state +
##     log(medincome):f.state + +log(medincome):f.education25_over +
##     +log(pctemployed16_over):f.state + log(pctemployed16_over):f.education25_over,
##     data = df)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -90.639  -8.501  -0.011   8.875  85.827
##

```

```

## Coefficients: (12 not defined because of singularities)
##
## (Intercept)
## pctpubliccoveragealone
## sqrt(incidencerate)
## pctbachdeg25_over
## log(medincome)
## f.stateAlaska
## f.stateArizona
## f.stateArkansas
## f.stateCalifornia
## f.stateColorado
## f.stateConnecticut
## f.stateDelaware
## f.stateFlorida
## f.stateGeorgia
## f.stateHawaii
## f.stateIdaho
## f.stateIllinois
## f.stateIndiana
## f.stateIowa
## f.stateKansas
## f.stateKentucky
## f.stateLouisiana
## f.stateMaine
## f.stateMaryland
## f.stateMassachusetts
## f.stateMichigan
## f.stateMinnesota
## f.stateMississippi
## f.stateMissouri
## f.stateMontana
## f.stateNebraska
## f.stateNevada
## f.stateNew Hampshire
## f.stateNew Jersey
## f.stateNew Mexico
## f.stateNew York
## f.stateNorth Carolina
## f.stateNorth Dakota
## f.stateOhio
## f.stateOklahoma
## f.stateOregon
## f.statePennsylvania
## f.stateRhode Island
## f.stateSouth Carolina
## f.stateSouth Dakota
## f.stateTennessee
## f.stateTexas
## f.stateUtah
## f.stateVermont
## f.stateVirginia
## f.stateWashington
## f.stateWest Virginia

```

	Estimate
(Intercept)	1.089e+02
pctpubliccoveragealone	-8.983e-02
sqrt(incidencerate)	8.076e+00
pctbachdeg25_over	-1.431e+00
log(medincome)	-1.631e+01
f.stateAlaska	-1.501e+03
f.stateArizona	-2.585e+03
f.stateArkansas	-3.332e+01
f.stateCalifornia	3.215e+01
f.stateColorado	-6.840e+02
f.stateConnecticut	1.137e+03
f.stateDelaware	9.727e+01
f.stateFlorida	4.687e+02
f.stateGeorgia	3.311e+02
f.stateHawaii	4.037e+02
f.stateIdaho	-3.319e+02
f.stateIllinois	-2.392e+02
f.stateIndiana	-5.990e+01
f.stateIowa	7.499e+01
f.stateKansas	-1.321e+03
f.stateKentucky	3.631e+02
f.stateLouisiana	3.747e+02
f.stateMaine	-7.860e+02
f.stateMaryland	-1.574e+03
f.stateMassachusetts	-1.076e+03
f.stateMichigan	4.304e+01
f.stateMinnesota	-1.742e+03
f.stateMississippi	-5.347e+02
f.stateMissouri	5.920e+01
f.stateMontana	-2.675e+02
f.stateNebraska	3.547e+01
f.stateNevada	2.253e+03
f.stateNew Hampshire	-3.049e+04
f.stateNew Jersey	7.100e+02
f.stateNew Mexico	3.074e+02
f.stateNew York	1.960e+02
f.stateNorth Carolina	-9.241e+01
f.stateNorth Dakota	-3.775e+02
f.stateOhio	-1.246e+02
f.stateOklahoma	7.230e+01
f.stateOregon	-2.669e+02
f.statePennsylvania	-3.070e+02
f.stateRhode Island	7.507e+03
f.stateSouth Carolina	2.014e+02
f.stateSouth Dakota	2.780e+02
f.stateTennessee	-1.656e+02
f.stateTexas	3.625e+02
f.stateUtah	4.942e+02
f.stateVermont	-1.757e+03
f.stateVirginia	-4.271e+02
f.stateWashington	7.137e+01
f.stateWest Virginia	-1.984e+02

## f.stateWisconsin	3.804e+02
## f.stateWyoming	-8.931e+02
## f.unemploymentModerate Unemployment	-1.030e+00
## f.unemploymentHigh Unemployment	2.428e+00
## pctpubliccoveragealone:f.stateAlaska	3.307e+00
## pctpubliccoveragealone:f.stateArizona	6.207e+00
## pctpubliccoveragealone:f.stateArkansas	2.361e+00
## pctpubliccoveragealone:f.stateCalifornia	1.331e+00
## pctpubliccoveragealone:f.stateColorado	2.358e+00
## pctpubliccoveragealone:f.stateConnecticut	-1.283e+00
## pctpubliccoveragealone:f.stateDelaware	NA
## pctpubliccoveragealone:f.stateFlorida	-3.842e+00
## pctpubliccoveragealone:f.stateGeorgia	-1.479e+00
## pctpubliccoveragealone:f.stateHawaii	-1.785e+01
## pctpubliccoveragealone:f.stateIdaho	-2.356e+00
## pctpubliccoveragealone:f.stateIllinois	6.651e-01
## pctpubliccoveragealone:f.stateIndiana	1.098e+00
## pctpubliccoveragealone:f.stateIowa	2.387e+00
## pctpubliccoveragealone:f.stateKansas	-1.113e+00
## pctpubliccoveragealone:f.stateKentucky	-3.669e-01
## pctpubliccoveragealone:f.stateLouisiana	1.276e+00
## pctpubliccoveragealone:f.stateMaine	8.511e-01
## pctpubliccoveragealone:f.stateMaryland	4.835e+00
## pctpubliccoveragealone:f.stateMassachusetts	1.414e+00
## pctpubliccoveragealone:f.stateMichigan	1.060e+00
## pctpubliccoveragealone:f.stateMinnesota	1.794e+00
## pctpubliccoveragealone:f.stateMississippi	2.935e+00
## pctpubliccoveragealone:f.stateMissouri	1.959e+00
## pctpubliccoveragealone:f.stateMontana	1.082e+00
## pctpubliccoveragealone:f.stateNebraska	1.931e-01
## pctpubliccoveragealone:f.stateNevada	-7.016e+00
## pctpubliccoveragealone:f.stateNew Hampshire	-1.342e+02
## pctpubliccoveragealone:f.stateNew Jersey	-2.147e+00
## pctpubliccoveragealone:f.stateNew Mexico	-1.371e+00
## pctpubliccoveragealone:f.stateNew York	-6.066e-01
## pctpubliccoveragealone:f.stateNorth Carolina	5.100e-01
## pctpubliccoveragealone:f.stateNorth Dakota	-4.021e-01
## pctpubliccoveragealone:f.stateOhio	1.037e+00
## pctpubliccoveragealone:f.stateOklahoma	4.505e+00
## pctpubliccoveragealone:f.stateOregon	1.617e+00
## pctpubliccoveragealone:f.statePennsylvania	2.766e+00
## pctpubliccoveragealone:f.stateRhode Island	1.358e+02
## pctpubliccoveragealone:f.stateSouth Carolina	1.542e+00
## pctpubliccoveragealone:f.stateSouth Dakota	1.525e+00
## pctpubliccoveragealone:f.stateTennessee	2.221e+00
## pctpubliccoveragealone:f.stateTexas	-1.542e+00
## pctpubliccoveragealone:f.stateUtah	-7.433e-01
## pctpubliccoveragealone:f.stateVermont	6.085e+00
## pctpubliccoveragealone:f.stateVirginia	2.892e+00
## pctpubliccoveragealone:f.stateWashington	-7.378e-01
## pctpubliccoveragealone:f.stateWest Virginia	1.886e+00
## pctpubliccoveragealone:f.stateWisconsin	6.687e-01
## pctpubliccoveragealone:f.stateWyoming	-2.757e-01
## pctpubliccoveragealone:f.education25_overBalanced Eduaction	-5.707e-01

## pctpubliccoveragealone:f.education25_overMostly Bachelor's	5.349e-01
## sqrt(incidencerate):f.stateAlaska	4.663e+00
## sqrt(incidencerate):f.stateArizona	-1.603e+01
## sqrt(incidencerate):f.stateArkansas	-4.358e+00
## sqrt(incidencerate):f.stateCalifornia	1.425e+00
## sqrt(incidencerate):f.stateColorado	7.399e-01
## sqrt(incidencerate):f.stateConnecticut	2.607e+00
## sqrt(incidencerate):f.stateDelaware	NA
## sqrt(incidencerate):f.stateFlorida	2.409e+00
## sqrt(incidencerate):f.stateGeorgia	3.717e+00
## sqrt(incidencerate):f.stateHawaii	NA
## sqrt(incidencerate):f.stateIdaho	4.862e+00
## sqrt(incidencerate):f.stateIllinois	3.859e+00
## sqrt(incidencerate):f.stateIndiana	3.828e+00
## sqrt(incidencerate):f.stateIowa	2.144e+00
## sqrt(incidencerate):f.stateKansas	8.225e+01
## sqrt(incidencerate):f.stateKentucky	5.169e+00
## sqrt(incidencerate):f.stateLouisiana	-1.973e+00
## sqrt(incidencerate):f.stateMaine	-2.911e+00
## sqrt(incidencerate):f.stateMaryland	1.681e+01
## sqrt(incidencerate):f.stateMassachusetts	-5.629e+00
## sqrt(incidencerate):f.stateMichigan	-3.879e+00
## sqrt(incidencerate):f.stateMinnesota	6.168e+01
## sqrt(incidencerate):f.stateMississippi	-3.673e+00
## sqrt(incidencerate):f.stateMissouri	-1.025e+00
## sqrt(incidencerate):f.stateMontana	4.029e-01
## sqrt(incidencerate):f.stateNebraska	-3.210e+00
## sqrt(incidencerate):f.stateNevada	4.353e+01
## sqrt(incidencerate):f.stateNew Hampshire	1.281e+02
## sqrt(incidencerate):f.stateNew Jersey	-5.359e+00
## sqrt(incidencerate):f.stateNew Mexico	3.465e+00
## sqrt(incidencerate):f.stateNew York	3.792e+00
## sqrt(incidencerate):f.stateNorth Carolina	-1.034e+00
## sqrt(incidencerate):f.stateNorth Dakota	4.768e+00
## sqrt(incidencerate):f.stateOhio	3.805e+00
## sqrt(incidencerate):f.stateOklahoma	-6.161e+00
## sqrt(incidencerate):f.stateOregon	-3.266e+00
## sqrt(incidencerate):f.statePennsylvania	6.847e-01
## sqrt(incidencerate):f.stateRhode Island	-4.110e+02
## sqrt(incidencerate):f.stateSouth Carolina	2.600e+00
## sqrt(incidencerate):f.stateSouth Dakota	-1.590e+00
## sqrt(incidencerate):f.stateTennessee	1.425e+00
## sqrt(incidencerate):f.stateTexas	3.917e+00
## sqrt(incidencerate):f.stateUtah	-1.128e+00
## sqrt(incidencerate):f.stateVermont	1.508e+00
## sqrt(incidencerate):f.stateVirginia	-3.177e+00
## sqrt(incidencerate):f.stateWashington	-3.049e+00
## sqrt(incidencerate):f.stateWest Virginia	-4.545e-01
## sqrt(incidencerate):f.stateWisconsin	-3.804e+00
## sqrt(incidencerate):f.stateWyoming	1.717e+01
## pctbachdeg25_over:f.stateAlaska	-4.760e+00
## pctbachdeg25_over:f.stateArizona	3.700e+00
## pctbachdeg25_over:f.stateArkansas	6.518e-02
## pctbachdeg25_over:f.stateCalifornia	1.825e+00

## pctbachdeg25_over:f.stateColorado	-5.035e-01
## pctbachdeg25_over:f.stateConnecticut	1.890e+00
## pctbachdeg25_over:f.stateDelaware	NA
## pctbachdeg25_over:f.stateFlorida	-2.885e+00
## pctbachdeg25_over:f.stateGeorgia	-2.039e-01
## pctbachdeg25_over:f.stateHawaii	NA
## pctbachdeg25_over:f.stateIdaho	-1.118e+00
## pctbachdeg25_over:f.stateIllinois	-6.136e-02
## pctbachdeg25_over:f.stateIndiana	9.254e-01
## pctbachdeg25_over:f.stateIowa	2.125e+00
## pctbachdeg25_over:f.stateKansas	4.315e-01
## pctbachdeg25_over:f.stateKentucky	3.678e-01
## pctbachdeg25_over:f.stateLouisiana	1.079e+00
## pctbachdeg25_over:f.stateMaine	-2.331e+00
## pctbachdeg25_over:f.stateMaryland	-5.213e-01
## pctbachdeg25_over:f.stateMassachusetts	-1.138e+00
## pctbachdeg25_over:f.stateMichigan	-6.426e-01
## pctbachdeg25_over:f.stateMinnesota	9.730e-01
## pctbachdeg25_over:f.stateMississippi	2.190e+00
## pctbachdeg25_over:f.stateMissouri	2.168e-01
## pctbachdeg25_over:f.stateMontana	1.573e+00
## pctbachdeg25_over:f.stateNebraska	7.566e-01
## pctbachdeg25_over:f.stateNevada	-5.552e+00
## pctbachdeg25_over:f.stateNew Hampshire	-2.686e+02
## pctbachdeg25_over:f.stateNew Jersey	-6.181e-01
## pctbachdeg25_over:f.stateNew Mexico	-1.546e+00
## pctbachdeg25_over:f.stateNew York	4.750e-01
## pctbachdeg25_over:f.stateNorth Carolina	1.643e-02
## pctbachdeg25_over:f.stateNorth Dakota	7.215e-02
## pctbachdeg25_over:f.stateOhio	7.154e-01
## pctbachdeg25_over:f.stateOklahoma	2.091e+00
## pctbachdeg25_over:f.stateOregon	1.165e+00
## pctbachdeg25_over:f.statePennsylvania	8.415e-01
## pctbachdeg25_over:f.stateRhode Island	NA
## pctbachdeg25_over:f.stateSouth Carolina	7.281e-01
## pctbachdeg25_over:f.stateSouth Dakota	1.994e+00
## pctbachdeg25_over:f.stateTennessee	8.358e-01
## pctbachdeg25_over:f.stateTexas	4.626e-01
## pctbachdeg25_over:f.stateUtah	4.884e-01
## pctbachdeg25_over:f.stateVermont	1.906e+00
## pctbachdeg25_over:f.stateVirginia	-4.550e-01
## pctbachdeg25_over:f.stateWashington	-1.466e+00
## pctbachdeg25_over:f.stateWest Virginia	3.754e+00
## pctbachdeg25_over:f.stateWisconsin	1.275e+00
## pctbachdeg25_over:f.stateWyoming	2.100e+00
## log(medincome):f.stateAlaska	1.567e+01
## log(medincome):f.stateArizona	3.061e+02
## log(medincome):f.stateArkansas	-2.649e+00
## log(medincome):f.stateCalifornia	7.797e+00
## log(medincome):f.stateColorado	4.876e+01
## log(medincome):f.stateConnecticut	-9.036e+01
## log(medincome):f.stateDelaware	NA
## log(medincome):f.stateFlorida	-3.786e+01
## log(medincome):f.stateGeorgia	-3.443e+01

## log(medincome):f.stateHawaii	NA
## log(medincome):f.stateIdaho	4.255e+01
## log(medincome):f.stateIllinois	2.040e+01
## log(medincome):f.stateIndiana	2.268e+01
## log(medincome):f.stateIowa	-2.599e+01
## log(medincome):f.stateKansas	7.566e+00
## log(medincome):f.stateKentucky	-1.327e+01
## log(medincome):f.stateLouisiana	-1.783e+01
## log(medincome):f.stateMaine	5.451e+01
## log(medincome):f.stateMaryland	1.081e+02
## log(medincome):f.stateMassachusetts	8.591e+00
## log(medincome):f.stateMichigan	-1.118e-01
## log(medincome):f.stateMinnesota	5.807e+01
## log(medincome):f.stateMississippi	7.252e+01
## log(medincome):f.stateMissouri	-2.153e+01
## log(medincome):f.stateMontana	2.536e+01
## log(medincome):f.stateNebraska	4.765e+01
## log(medincome):f.stateNevada	-3.055e+02
## log(medincome):f.stateNew Hampshire	3.017e+03
## log(medincome):f.stateNew Jersey	-1.855e+00
## log(medincome):f.stateNew Mexico	-3.944e+01
## log(medincome):f.stateNew York	-1.806e+01
## log(medincome):f.stateNorth Carolina	1.047e+01
## log(medincome):f.stateNorth Dakota	2.262e+01
## log(medincome):f.stateOhio	7.997e+00
## log(medincome):f.stateOklahoma	-3.200e+01
## log(medincome):f.stateOregon	4.267e+01
## log(medincome):f.statePennsylvania	3.673e+01
## log(medincome):f.stateRhode Island	NA
## log(medincome):f.stateSouth Carolina	-4.450e+01
## log(medincome):f.stateSouth Dakota	-1.795e+01
## log(medincome):f.stateTennessee	2.942e+01
## log(medincome):f.stateTexas	-2.493e+01
## log(medincome):f.stateUtah	-5.123e+01
## log(medincome):f.stateVermont	1.301e+02
## log(medincome):f.stateVirginia	4.730e+01
## log(medincome):f.stateWashington	1.911e+01
## log(medincome):f.stateWest Virginia	7.856e+01
## log(medincome):f.stateWisconsin	9.723e+00
## log(medincome):f.stateWyoming	1.020e+02
## log(medincome):f.education25_overBalanced Eduaction	1.485e+01
## log(medincome):f.education25_overMostly Bachelor's	2.690e+00
## f.stateAlabama:log(pctemployed16_over)	2.518e+01
## f.stateAlaska:log(pctemployed16_over)	3.337e+02
## f.stateArizona:log(pctemployed16_over)	-1.306e+02
## f.stateArkansas:log(pctemployed16_over)	5.207e+01
## f.stateCalifornia:log(pctemployed16_over)	-3.178e+01
## f.stateColorado:log(pctemployed16_over)	4.613e+01
## f.stateConnecticut:log(pctemployed16_over)	-2.967e+01
## f.stateDelaware:log(pctemployed16_over)	NA
## f.stateFlorida:log(pctemployed16_over)	2.642e+01
## f.stateGeorgia:log(pctemployed16_over)	2.030e+01
## f.stateHawaii:log(pctemployed16_over)	NA
## f.stateIdaho:log(pctemployed16_over)	-2.371e+01

## f.stateIllinois:log(pctemployed16_over)	4.379e+00	
## f.stateIndiana:log(pctemployed16_over)	-4.780e+01	
## f.stateIowa:log(pctemployed16_over)	4.280e+01	
## f.stateKansas:log(pctemployed16_over)	-9.600e+01	
## f.stateKentucky:log(pctemployed16_over)	-5.786e+01	
## f.stateLouisiana:log(pctemployed16_over)	-1.965e+01	
## f.stateMaine:log(pctemployed16_over)	9.593e+01	
## f.stateMaryland:log(pctemployed16_over)	1.109e+01	
## f.stateMassachusetts:log(pctemployed16_over)	2.903e+02	
## f.stateMichigan:log(pctemployed16_over)	3.129e+01	
## f.stateMinnesota:log(pctemployed16_over)	-3.675e+01	
## f.stateMississippi:log(pctemployed16_over)	-3.527e+01	
## f.stateMissouri:log(pctemployed16_over)	6.482e+01	
## f.stateMontana:log(pctemployed16_over)	5.256e+00	
## f.stateNebraska:log(pctemployed16_over)	-9.993e+01	
## f.stateNevada:log(pctemployed16_over)	1.061e+02	
## f.stateNew Hampshire:log(pctemployed16_over)	4.612e+02	
## f.stateNew Jersey:log(pctemployed16_over)	-1.060e+02	
## f.stateNew Mexico:log(pctemployed16_over)	4.683e+01	
## f.stateNew York:log(pctemployed16_over)	5.528e-01	
## f.stateNorth Carolina:log(pctemployed16_over)	2.054e+01	
## f.stateNorth Dakota:log(pctemployed16_over)	2.962e+01	
## f.stateOhio:log(pctemployed16_over)	7.182e+00	
## f.stateOklahoma:log(pctemployed16_over)	9.885e+01	
## f.stateOregon:log(pctemployed16_over)	-2.091e+01	
## f.statePennsylvania:log(pctemployed16_over)	-1.876e+01	
## f.stateRhode Island:log(pctemployed16_over)		NA
## f.stateSouth Carolina:log(pctemployed16_over)	6.977e+01	
## f.stateSouth Dakota:log(pctemployed16_over)	-5.244e+00	
## f.stateTennessee:log(pctemployed16_over)	-3.277e+01	
## f.stateTexas:log(pctemployed16_over)	-1.341e+01	
## f.stateUtah:log(pctemployed16_over)	3.886e+01	
## f.stateVermont:log(pctemployed16_over)	5.665e+01	
## f.stateVirginia:log(pctemployed16_over)	1.069e+01	
## f.stateWashington:log(pctemployed16_over)	-2.205e+01	
## f.stateWest Virginia:log(pctemployed16_over)	-1.558e+02	
## f.stateWisconsin:log(pctemployed16_over)	-8.395e+01	
## f.stateWyoming:log(pctemployed16_over)	-1.236e+02	
## f.education25_overBalanced Eduaction:log(pctemployed16_over)	-3.727e+01	
## f.education25_overMostly Bachelor's:log(pctemployed16_over)	-8.441e+00	
##	Std. Error	t value
## (Intercept)	6.091e+02	0.179
## pctpubliccoveragealone	2.139e+00	-0.042
## sqrt(incidencerate)	4.614e+00	1.750
## pctbachdeg25_over	1.062e+00	-1.347
## log(medincome)	4.825e+01	-0.338
## f.stateAlaska	8.839e+02	-1.698
## f.stateArizona	1.955e+03	-1.322
## f.stateArkansas	7.247e+02	-0.046
## f.stateCalifornia	6.843e+02	0.047
## f.stateColorado	7.035e+02	-0.972
## f.stateConnecticut	3.730e+03	0.305
## f.stateDelaware	1.781e+02	0.546
## f.stateFlorida	6.809e+02	0.688

## f.stateGeorgia	6.287e+02	0.527
## f.stateHawaii	6.136e+02	0.658
## f.stateIdaho	7.871e+02	-0.422
## f.stateIllinois	6.710e+02	-0.356
## f.stateIndiana	6.844e+02	-0.088
## f.stateIowa	7.000e+02	0.107
## f.stateKansas	8.832e+02	-1.496
## f.stateKentucky	6.614e+02	0.549
## f.stateLouisiana	6.602e+02	0.568
## f.stateMaine	1.408e+03	-0.558
## f.stateMaryland	1.084e+03	-1.452
## f.stateMassachusetts	1.668e+03	-0.645
## f.stateMichigan	6.721e+02	0.064
## f.stateMinnesota	1.165e+03	-1.495
## f.stateMississippi	6.550e+02	-0.816
## f.stateMissouri	6.675e+02	0.089
## f.stateMontana	6.902e+02	-0.388
## f.stateNebraska	6.716e+02	0.053
## f.stateNevada	1.909e+03	1.180
## f.stateNew Hampshire	1.401e+05	-0.218
## f.stateNew Jersey	2.313e+03	0.307
## f.stateNew Mexico	6.751e+02	0.455
## f.stateNew York	6.915e+02	0.283
## f.stateNorth Carolina	6.713e+02	-0.138
## f.stateNorth Dakota	6.978e+02	-0.541
## f.stateOhio	6.968e+02	-0.179
## f.stateOklahoma	6.870e+02	0.105
## f.stateOregon	7.157e+02	-0.373
## f.statePennsylvania	6.899e+02	-0.445
## f.stateRhode Island	6.799e+03	1.104
## f.stateSouth Carolina	8.491e+02	0.237
## f.stateSouth Dakota	6.733e+02	0.413
## f.stateTennessee	6.740e+02	-0.246
## f.stateTexas	6.258e+02	0.579
## f.stateUtah	7.920e+02	0.624
## f.stateVermont	3.082e+03	-0.570
## f.stateVirginia	6.328e+02	-0.675
## f.stateWashington	7.021e+02	0.102
## f.stateWest Virginia	6.927e+02	-0.286
## f.stateWisconsin	8.099e+02	0.470
## f.stateWyoming	1.544e+03	-0.579
## f.unemploymentModerate Unemployment	1.637e+00	-0.629
## f.unemploymentHigh Unemployment	2.112e+00	1.150
## pctpubliccoveragealone:f.stateAlaska	2.556e+00	1.294
## pctpubliccoveragealone:f.stateArizona	5.860e+00	1.059
## pctpubliccoveragealone:f.stateArkansas	2.426e+00	0.973
## pctpubliccoveragealone:f.stateCalifornia	2.355e+00	0.565
## pctpubliccoveragealone:f.stateColorado	2.507e+00	0.940
## pctpubliccoveragealone:f.stateConnecticut	6.022e+00	-0.213
## pctpubliccoveragealone:f.stateDelaware	NA	NA
## pctpubliccoveragealone:f.stateFlorida	2.424e+00	-1.585
## pctpubliccoveragealone:f.stateGeorgia	2.219e+00	-0.667
## pctpubliccoveragealone:f.stateHawaii	3.181e+01	-0.561
## pctpubliccoveragealone:f.stateIdaho	2.543e+00	-0.926

## pctpubliccoveragealone:f.stateIllinois	2.307e+00	0.288
## pctpubliccoveragealone:f.stateIndiana	2.383e+00	0.461
## pctpubliccoveragealone:f.stateIowa	2.462e+00	0.970
## pctpubliccoveragealone:f.stateKansas	2.392e+00	-0.465
## pctpubliccoveragealone:f.stateKentucky	2.320e+00	-0.158
## pctpubliccoveragealone:f.stateLouisiana	2.401e+00	0.531
## pctpubliccoveragealone:f.stateMaine	5.613e+00	0.152
## pctpubliccoveragealone:f.stateMaryland	3.341e+00	1.447
## pctpubliccoveragealone:f.stateMassachusetts	7.075e+00	0.200
## pctpubliccoveragealone:f.stateMichigan	2.402e+00	0.441
## pctpubliccoveragealone:f.stateMinnesota	2.484e+00	0.722
## pctpubliccoveragealone:f.stateMississippi	2.256e+00	1.301
## pctpubliccoveragealone:f.stateMissouri	2.345e+00	0.836
## pctpubliccoveragealone:f.stateMontana	2.406e+00	0.450
## pctpubliccoveragealone:f.stateNebraska	2.334e+00	0.083
## pctpubliccoveragealone:f.stateNevada	3.402e+00	-2.062
## pctpubliccoveragealone:f.stateNew Hampshire	6.244e+02	-0.215
## pctpubliccoveragealone:f.stateNew Jersey	6.212e+00	-0.346
## pctpubliccoveragealone:f.stateNew Mexico	2.271e+00	-0.604
## pctpubliccoveragealone:f.stateNew York	2.320e+00	-0.262
## pctpubliccoveragealone:f.stateNorth Carolina	2.372e+00	0.215
## pctpubliccoveragealone:f.stateNorth Dakota	2.310e+00	-0.174
## pctpubliccoveragealone:f.stateOhio	2.421e+00	0.428
## pctpubliccoveragealone:f.stateOklahoma	2.537e+00	1.776
## pctpubliccoveragealone:f.stateOregon	2.439e+00	0.663
## pctpubliccoveragealone:f.statePennsylvania	2.411e+00	1.147
## pctpubliccoveragealone:f.stateRhode Island	1.141e+02	1.191
## pctpubliccoveragealone:f.stateSouth Carolina	2.949e+00	0.523
## pctpubliccoveragealone:f.stateSouth Dakota	2.246e+00	0.679
## pctpubliccoveragealone:f.stateTennessee	2.368e+00	0.938
## pctpubliccoveragealone:f.stateTexas	2.201e+00	-0.701
## pctpubliccoveragealone:f.stateUtah	3.074e+00	-0.242
## pctpubliccoveragealone:f.stateVermont	1.134e+01	0.537
## pctpubliccoveragealone:f.stateVirginia	2.260e+00	1.280
## pctpubliccoveragealone:f.stateWashington	2.614e+00	-0.282
## pctpubliccoveragealone:f.stateWest Virginia	2.325e+00	0.811
## pctpubliccoveragealone:f.stateWisconsin	2.811e+00	0.238
## pctpubliccoveragealone:f.stateWyoming	5.981e+00	-0.046
## pctpubliccoveragealone:f.education25_overBalanced Eduaction	2.580e-01	-2.212
## pctpubliccoveragealone:f.education25_overMostly Bachelor's	6.245e-01	0.857
## sqrt(incidencerate):f.stateAlaska	6.816e+00	0.684
## sqrt(incidencerate):f.stateArizona	1.473e+01	-1.089
## sqrt(incidencerate):f.stateArkansas	5.606e+00	-0.777
## sqrt(incidencerate):f.stateCalifornia	6.939e+00	0.205
## sqrt(incidencerate):f.stateColorado	5.339e+00	0.139
## sqrt(incidencerate):f.stateConnecticut	5.691e+01	0.046
## sqrt(incidencerate):f.stateDelaware	NA	NA
## sqrt(incidencerate):f.stateFlorida	4.775e+00	0.504
## sqrt(incidencerate):f.stateGeorgia	4.920e+00	0.755
## sqrt(incidencerate):f.stateHawaii	NA	NA
## sqrt(incidencerate):f.stateIdaho	5.596e+00	0.869
## sqrt(incidencerate):f.stateIllinois	5.637e+00	0.685
## sqrt(incidencerate):f.stateIndiana	5.856e+00	0.654
## sqrt(incidencerate):f.stateIowa	5.629e+00	0.381

## sqrt(incidencerate):f.stateKansas	3.033e+01	2.711
## sqrt(incidencerate):f.stateKentucky	5.100e+00	1.014
## sqrt(incidencerate):f.stateLouisiana	6.515e+00	-0.303
## sqrt(incidencerate):f.stateMaine	1.426e+01	-0.204
## sqrt(incidencerate):f.stateMaryland	9.394e+00	1.790
## sqrt(incidencerate):f.stateMassachusetts	2.359e+01	-0.239
## sqrt(incidencerate):f.stateMichigan	4.920e+00	-0.789
## sqrt(incidencerate):f.stateMinnesota	4.688e+01	1.316
## sqrt(incidencerate):f.stateMississippi	5.332e+00	-0.689
## sqrt(incidencerate):f.stateMissouri	4.860e+00	-0.211
## sqrt(incidencerate):f.stateMontana	5.515e+00	0.073
## sqrt(incidencerate):f.stateNebraska	4.981e+00	-0.644
## sqrt(incidencerate):f.stateNevada	7.212e+01	0.604
## sqrt(incidencerate):f.stateNew Hampshire	5.532e+02	0.232
## sqrt(incidencerate):f.stateNew Jersey	2.282e+01	-0.235
## sqrt(incidencerate):f.stateNew Mexico	5.628e+00	0.616
## sqrt(incidencerate):f.stateNew York	7.174e+00	0.529
## sqrt(incidencerate):f.stateNorth Carolina	5.694e+00	-0.182
## sqrt(incidencerate):f.stateNorth Dakota	5.055e+00	0.943
## sqrt(incidencerate):f.stateOhio	6.166e+00	0.617
## sqrt(incidencerate):f.stateOklahoma	5.741e+00	-1.073
## sqrt(incidencerate):f.stateOregon	6.619e+00	-0.493
## sqrt(incidencerate):f.statePennsylvania	6.179e+00	0.111
## sqrt(incidencerate):f.stateRhode Island	3.703e+02	-1.110
## sqrt(incidencerate):f.stateSouth Carolina	6.565e+00	0.396
## sqrt(incidencerate):f.stateSouth Dakota	5.286e+00	-0.301
## sqrt(incidencerate):f.stateTennessee	5.518e+00	0.258
## sqrt(incidencerate):f.stateTexas	4.753e+00	0.824
## sqrt(incidencerate):f.stateUtah	7.220e+00	-0.156
## sqrt(incidencerate):f.stateVermont	1.919e+01	0.079
## sqrt(incidencerate):f.stateVirginia	4.741e+00	-0.670
## sqrt(incidencerate):f.stateWashington	6.022e+00	-0.506
## sqrt(incidencerate):f.stateWest Virginia	5.042e+00	-0.090
## sqrt(incidencerate):f.stateWisconsin	5.386e+00	-0.706
## sqrt(incidencerate):f.stateWyoming	1.246e+01	1.378
## pctbachdeg25_over:f.stateAlaska	2.279e+00	-2.089
## pctbachdeg25_over:f.stateArizona	2.780e+00	1.331
## pctbachdeg25_over:f.stateArkansas	1.698e+00	0.038
## pctbachdeg25_over:f.stateCalifornia	1.678e+00	1.088
## pctbachdeg25_over:f.stateColorado	1.202e+00	-0.419
## pctbachdeg25_over:f.stateConnecticut	7.648e+00	0.247
## pctbachdeg25_over:f.stateDelaware	NA	NA
## pctbachdeg25_over:f.stateFlorida	1.458e+00	-1.978
## pctbachdeg25_over:f.stateGeorgia	1.185e+00	-0.172
## pctbachdeg25_over:f.stateHawaii	NA	NA
## pctbachdeg25_over:f.stateIdaho	1.409e+00	-0.794
## pctbachdeg25_over:f.stateIllinois	1.264e+00	-0.049
## pctbachdeg25_over:f.stateIndiana	1.222e+00	0.757
## pctbachdeg25_over:f.stateIowa	1.226e+00	1.734
## pctbachdeg25_over:f.stateKansas	1.254e+00	0.344
## pctbachdeg25_over:f.stateKentucky	1.239e+00	0.297
## pctbachdeg25_over:f.stateLouisiana	1.513e+00	0.713
## pctbachdeg25_over:f.stateMaine	4.006e+00	-0.582
## pctbachdeg25_over:f.stateMaryland	2.141e+00	-0.243

## pctbachdeg25_over:f.stateMassachusetts	6.291e+00	-0.181
## pctbachdeg25_over:f.stateMichigan	1.404e+00	-0.458
## pctbachdeg25_over:f.stateMinnesota	1.295e+00	0.751
## pctbachdeg25_over:f.stateMississippi	1.249e+00	1.753
## pctbachdeg25_over:f.stateMissouri	1.370e+00	0.158
## pctbachdeg25_over:f.stateMontana	1.353e+00	1.163
## pctbachdeg25_over:f.stateNebraska	1.311e+00	0.577
## pctbachdeg25_over:f.stateNevada	2.007e+00	-2.766
## pctbachdeg25_over:f.stateNew Hampshire	1.241e+03	-0.216
## pctbachdeg25_over:f.stateNew Jersey	4.109e+00	-0.150
## pctbachdeg25_over:f.stateNew Mexico	2.104e+00	-0.735
## pctbachdeg25_over:f.stateNew York	1.571e+00	0.302
## pctbachdeg25_over:f.stateNorth Carolina	1.342e+00	0.012
## pctbachdeg25_over:f.stateNorth Dakota	1.942e+00	0.037
## pctbachdeg25_over:f.stateOhio	1.331e+00	0.537
## pctbachdeg25_over:f.stateOklahoma	1.684e+00	1.242
## pctbachdeg25_over:f.stateOregon	1.504e+00	0.775
## pctbachdeg25_over:f.statePennsylvania	1.452e+00	0.580
## pctbachdeg25_over:f.stateRhode Island	NA	NA
## pctbachdeg25_over:f.stateSouth Carolina	1.538e+00	0.473
## pctbachdeg25_over:f.stateSouth Dakota	1.416e+00	1.408
## pctbachdeg25_over:f.stateTennessee	1.343e+00	0.622
## pctbachdeg25_over:f.stateTexas	1.157e+00	0.400
## pctbachdeg25_over:f.stateUtah	1.584e+00	0.308
## pctbachdeg25_over:f.stateVermont	5.998e+00	0.318
## pctbachdeg25_over:f.stateVirginia	1.203e+00	-0.378
## pctbachdeg25_over:f.stateWashington	1.598e+00	-0.917
## pctbachdeg25_over:f.stateWest Virginia	1.762e+00	2.130
## pctbachdeg25_over:f.stateWisconsin	1.335e+00	0.955
## pctbachdeg25_over:f.stateWyoming	6.656e+00	0.316
## log(medincome):f.stateAlaska	5.970e+01	0.262
## log(medincome):f.stateArizona	2.057e+02	1.488
## log(medincome):f.stateArkansas	6.208e+01	-0.043
## log(medincome):f.stateCalifornia	5.599e+01	0.139
## log(medincome):f.stateColorado	5.812e+01	0.839
## log(medincome):f.stateConnecticut	2.445e+02	-0.370
## log(medincome):f.stateDelaware	NA	NA
## log(medincome):f.stateFlorida	5.637e+01	-0.672
## log(medincome):f.stateGeorgia	5.089e+01	-0.677
## log(medincome):f.stateHawaii	NA	NA
## log(medincome):f.stateIdaho	7.396e+01	0.575
## log(medincome):f.stateIllinois	5.423e+01	0.376
## log(medincome):f.stateIndiana	5.627e+01	0.403
## log(medincome):f.stateIowa	6.092e+01	-0.427
## log(medincome):f.stateKansas	5.295e+01	0.143
## log(medincome):f.stateKentucky	5.232e+01	-0.254
## log(medincome):f.stateLouisiana	5.491e+01	-0.325
## log(medincome):f.stateMaine	1.164e+02	0.468
## log(medincome):f.stateMaryland	9.790e+01	1.104
## log(medincome):f.stateMassachusetts	2.088e+02	0.041
## log(medincome):f.stateMichigan	5.709e+01	-0.002
## log(medincome):f.stateMinnesota	5.450e+01	1.065
## log(medincome):f.stateMississippi	5.302e+01	1.368
## log(medincome):f.stateMissouri	5.441e+01	-0.396

## log(medincome):f.stateMontana	5.918e+01	0.429
## log(medincome):f.stateNebraska	5.414e+01	0.880
## log(medincome):f.stateNevada	8.592e+01	-3.556
## log(medincome):f.stateNew Hampshire	1.401e+04	0.215
## log(medincome):f.stateNew Jersey	1.111e+02	-0.017
## log(medincome):f.stateNew Mexico	5.592e+01	-0.705
## log(medincome):f.stateNew York	5.454e+01	-0.331
## log(medincome):f.stateNorth Carolina	5.591e+01	0.187
## log(medincome):f.stateNorth Dakota	5.505e+01	0.411
## log(medincome):f.stateOhio	5.617e+01	0.142
## log(medincome):f.stateOklahoma	5.568e+01	-0.575
## log(medincome):f.stateOregon	6.820e+01	0.626
## log(medincome):f.statePennsylvania	5.697e+01	0.645
## log(medincome):f.stateRhode Island	NA	NA
## log(medincome):f.stateSouth Carolina	7.078e+01	-0.629
## log(medincome):f.stateSouth Dakota	5.527e+01	-0.325
## log(medincome):f.stateTennessee	5.530e+01	0.532
## log(medincome):f.stateTexas	5.014e+01	-0.497
## log(medincome):f.stateUtah	6.252e+01	-0.819
## log(medincome):f.stateVermont	3.416e+02	0.381
## log(medincome):f.stateVirginia	5.070e+01	0.933
## log(medincome):f.stateWashington	6.326e+01	0.302
## log(medincome):f.stateWest Virginia	5.888e+01	1.334
## log(medincome):f.stateWisconsin	6.964e+01	0.140
## log(medincome):f.stateWyoming	1.901e+02	0.537
## log(medincome):f.education25_overBalanced Eduaction	4.727e+00	3.141
## log(medincome):f.education25_overMostly Bachelor's	9.179e+00	0.293
## f.stateAlabama:log(pctemployed16_over)	4.579e+01	0.550
## f.stateAlaska:log(pctemployed16_over)	9.899e+01	3.371
## f.stateArizona:log(pctemployed16_over)	1.083e+02	-1.206
## f.stateArkansas:log(pctemployed16_over)	2.894e+01	1.799
## f.stateCalifornia:log(pctemployed16_over)	3.045e+01	-1.044
## f.stateColorado:log(pctemployed16_over)	2.506e+01	1.841
## f.stateConnecticut:log(pctemployed16_over)	4.241e+02	-0.070
## f.stateDelaware:log(pctemployed16_over)	NA	NA
## f.stateFlorida:log(pctemployed16_over)	3.283e+01	0.805
## f.stateGeorgia:log(pctemployed16_over)	1.599e+01	1.270
## f.stateHawaii:log(pctemployed16_over)	NA	NA
## f.stateIdaho:log(pctemployed16_over)	4.988e+01	-0.475
## f.stateIllinois:log(pctemployed16_over)	2.343e+01	0.187
## f.stateIndiana:log(pctemployed16_over)	4.631e+01	-1.032
## f.stateIowa:log(pctemployed16_over)	5.867e+01	0.729
## f.stateKansas:log(pctemployed16_over)	4.129e+01	-2.325
## f.stateKentucky:log(pctemployed16_over)	2.606e+01	-2.220
## f.stateLouisiana:log(pctemployed16_over)	3.747e+01	-0.525
## f.stateMaine:log(pctemployed16_over)	2.113e+02	0.454
## f.stateMaryland:log(pctemployed16_over)	1.756e+02	0.063
## f.stateMassachusetts:log(pctemployed16_over)	7.428e+02	0.391
## f.stateMichigan:log(pctemployed16_over)	2.829e+01	1.106
## f.stateMinnesota:log(pctemployed16_over)	5.477e+01	-0.671
## f.stateMississippi:log(pctemployed16_over)	2.212e+01	-1.595
## f.stateMissouri:log(pctemployed16_over)	2.221e+01	2.918
## f.stateMontana:log(pctemployed16_over)	5.052e+01	0.104
## f.stateNebraska:log(pctemployed16_over)	3.939e+01	-2.537

## f.stateNevada:log(pctemployed16_over)	5.477e+01	1.937
## f.stateNew Hampshire:log(pctemployed16_over)	1.885e+03	0.245
## f.stateNew Jersey:log(pctemployed16_over)	3.300e+02	-0.321
## f.stateNew Mexico:log(pctemployed16_over)	4.394e+01	1.066
## f.stateNew York:log(pctemployed16_over)	5.277e+01	0.010
## f.stateNorth Carolina:log(pctemployed16_over)	3.014e+01	0.681
## f.stateNorth Dakota:log(pctemployed16_over)	9.877e+01	0.300
## f.stateOhio:log(pctemployed16_over)	5.766e+01	0.125
## f.stateOklahoma:log(pctemployed16_over)	3.629e+01	2.724
## f.stateOregon:log(pctemployed16_over)	5.628e+01	-0.371
## f.statePennsylvania:log(pctemployed16_over)	4.502e+01	-0.417
## f.stateRhode Island:log(pctemployed16_over)	NA	NA
## f.stateSouth Carolina:log(pctemployed16_over)	3.945e+01	1.768
## f.stateSouth Dakota:log(pctemployed16_over)	3.665e+01	-0.143
## f.stateTennessee:log(pctemployed16_over)	2.861e+01	-1.146
## f.stateTexas:log(pctemployed16_over)	1.142e+01	-1.175
## f.stateUtah:log(pctemployed16_over)	6.722e+01	0.578
## f.stateVermont:log(pctemployed16_over)	2.291e+02	0.247
## f.stateVirginia:log(pctemployed16_over)	2.478e+01	0.431
## f.stateWashington:log(pctemployed16_over)	4.433e+01	-0.497
## f.stateWest Virginia:log(pctemployed16_over)	4.468e+01	-3.487
## f.stateWisconsin:log(pctemployed16_over)	5.632e+01	-1.491
## f.stateWyoming:log(pctemployed16_over)	2.543e+02	-0.486
## f.education25_overBalanced Eduaction:log(pctemployed16_over)	1.245e+01	-2.993
## f.education25_overMostly Bachelor's:log(pctemployed16_over)	2.457e+01	-0.344
##	Pr(> t)	
## (Intercept)	0.858144	
## pctpubliccoveragealone	0.966505	
## sqrt(incidencerate)	0.080283	.
## pctbachdeg25_over	0.178028	
## log(medincome)	0.735312	
## f.stateAlaska	0.089668	.
## f.stateArizona	0.186329	
## f.stateArkansas	0.963337	
## f.stateCalifornia	0.962531	
## f.stateColorado	0.331032	
## f.stateConnecticut	0.760492	
## f.stateDelaware	0.584987	
## f.stateFlorida	0.491294	
## f.stateGeorgia	0.598448	
## f.stateHawaii	0.510684	
## f.stateIdaho	0.673355	
## f.stateIllinois	0.721562	
## f.stateIndiana	0.930273	
## f.stateIowa	0.914704	
## f.stateKansas	0.134932	
## f.stateKentucky	0.583066	
## f.stateLouisiana	0.570410	
## f.stateMaine	0.576802	
## f.stateMaryland	0.146629	
## f.stateMassachusetts	0.519216	
## f.stateMichigan	0.948946	
## f.stateMinnesota	0.135160	
## f.stateMississippi	0.414416	

## f.stateMissouri	0.929342
## f.stateMontana	0.698352
## f.stateNebraska	0.957891
## f.stateNevada	0.238223
## f.stateNew Hampshire	0.827755
## f.stateNew Jersey	0.758917
## f.stateNew Mexico	0.648959
## f.stateNew York	0.776891
## f.stateNorth Carolina	0.890521
## f.stateNorth Dakota	0.588571
## f.stateOhio	0.858146
## f.stateOklahoma	0.916197
## f.stateOregon	0.709217
## f.statePennsylvania	0.656438
## f.stateRhode Island	0.269710
## f.stateSouth Carolina	0.812578
## f.stateSouth Dakota	0.679705
## f.stateTennessee	0.805889
## f.stateTexas	0.562507
## f.stateUtah	0.532727
## f.stateVermont	0.568775
## f.stateVirginia	0.499874
## f.stateWashington	0.919044
## f.stateWest Virginia	0.774568
## f.stateWisconsin	0.638694
## f.stateWyoming	0.562925
## f.unemploymentModerate Unemployment	0.529372
## f.unemploymentHigh Unemployment	0.250496
## pctpubliccoveragealone:f.stateAlaska	0.195950
## pctpubliccoveragealone:f.stateArizona	0.289667
## pctpubliccoveragealone:f.stateArkansas	0.330696
## pctpubliccoveragealone:f.stateCalifornia	0.572070
## pctpubliccoveragealone:f.stateColorado	0.347155
## pctpubliccoveragealone:f.stateConnecticut	0.831288
## pctpubliccoveragealone:f.stateDelaware	NA
## pctpubliccoveragealone:f.stateFlorida	0.113111
## pctpubliccoveragealone:f.stateGeorgia	0.505147
## pctpubliccoveragealone:f.stateHawaii	0.574648
## pctpubliccoveragealone:f.stateIdaho	0.354375
## pctpubliccoveragealone:f.stateIllinois	0.773124
## pctpubliccoveragealone:f.stateIndiana	0.644995
## pctpubliccoveragealone:f.stateIowa	0.332435
## pctpubliccoveragealone:f.stateKansas	0.641788
## pctpubliccoveragealone:f.stateKentucky	0.874347
## pctpubliccoveragealone:f.stateLouisiana	0.595245
## pctpubliccoveragealone:f.stateMaine	0.879481
## pctpubliccoveragealone:f.stateMaryland	0.148013
## pctpubliccoveragealone:f.stateMassachusetts	0.841600
## pctpubliccoveragealone:f.stateMichigan	0.659160
## pctpubliccoveragealone:f.stateMinnesota	0.470382
## pctpubliccoveragealone:f.stateMississippi	0.193480
## pctpubliccoveragealone:f.stateMissouri	0.403514
## pctpubliccoveragealone:f.stateMontana	0.652848
## pctpubliccoveragealone:f.stateNebraska	0.934072

## pctpubliccoveragealone:f.stateNevada	0.039381 *
## pctpubliccoveragealone:f.stateNew Hampshire	0.829863
## pctpubliccoveragealone:f.stateNew Jersey	0.729740
## pctpubliccoveragealone:f.stateNew Mexico	0.546207
## pctpubliccoveragealone:f.stateNew York	0.793737
## pctpubliccoveragealone:f.stateNorth Carolina	0.829752
## pctpubliccoveragealone:f.stateNorth Dakota	0.861860
## pctpubliccoveragealone:f.stateOhio	0.668590
## pctpubliccoveragealone:f.stateOklahoma	0.075974 .
## pctpubliccoveragealone:f.stateOregon	0.507443
## pctpubliccoveragealone:f.statePennsylvania	0.251580
## pctpubliccoveragealone:f.stateRhode Island	0.233882
## pctpubliccoveragealone:f.stateSouth Carolina	0.600997
## pctpubliccoveragealone:f.stateSouth Dakota	0.497411
## pctpubliccoveragealone:f.stateTennessee	0.348401
## pctpubliccoveragealone:f.stateTexas	0.483558
## pctpubliccoveragealone:f.stateUtah	0.808974
## pctpubliccoveragealone:f.stateVermont	0.591589
## pctpubliccoveragealone:f.stateVirginia	0.200876
## pctpubliccoveragealone:f.stateWashington	0.777809
## pctpubliccoveragealone:f.stateWest Virginia	0.417362
## pctpubliccoveragealone:f.stateWisconsin	0.811959
## pctpubliccoveragealone:f.stateWyoming	0.963232
## pctpubliccoveragealone:f.education25_overBalanced Eduaction	0.027079 *
## pctpubliccoveragealone:f.education25_overMostly Bachelor's	0.391826
## sqrt(incidencerate):f.stateAlaska	0.493986
## sqrt(incidencerate):f.stateArizona	0.276475
## sqrt(incidencerate):f.stateArkansas	0.437108
## sqrt(incidencerate):f.stateCalifornia	0.837381
## sqrt(incidencerate):f.stateColorado	0.889790
## sqrt(incidencerate):f.stateConnecticut	0.963470
## sqrt(incidencerate):f.stateDelaware	NA
## sqrt(incidencerate):f.stateFlorida	0.614016
## sqrt(incidencerate):f.stateGeorgia	0.450067
## sqrt(incidencerate):f.stateHawaii	NA
## sqrt(incidencerate):f.stateIdaho	0.385121
## sqrt(incidencerate):f.stateIllinois	0.493736
## sqrt(incidencerate):f.stateIndiana	0.513422
## sqrt(incidencerate):f.stateIowa	0.703323
## sqrt(incidencerate):f.stateKansas	0.006775 **
## sqrt(incidencerate):f.stateKentucky	0.310938
## sqrt(incidencerate):f.stateLouisiana	0.762002
## sqrt(incidencerate):f.stateMaine	0.838238
## sqrt(incidencerate):f.stateMaryland	0.073688 .
## sqrt(incidencerate):f.stateMassachusetts	0.811440
## sqrt(incidencerate):f.stateMichigan	0.430510
## sqrt(incidencerate):f.stateMinnesota	0.188534
## sqrt(incidencerate):f.stateMississippi	0.491013
## sqrt(incidencerate):f.stateMissouri	0.832920
## sqrt(incidencerate):f.stateMontana	0.941765
## sqrt(incidencerate):f.stateNebraska	0.519350
## sqrt(incidencerate):f.stateNevada	0.546263
## sqrt(incidencerate):f.stateNew Hampshire	0.816890
## sqrt(incidencerate):f.stateNew Jersey	0.814380

## sqrt(incidencerate):f.stateNew Mexico	0.538182
## sqrt(incidencerate):f.stateNew York	0.597209
## sqrt(incidencerate):f.stateNorth Carolina	0.855938
## sqrt(incidencerate):f.stateNorth Dakota	0.345727
## sqrt(incidencerate):f.stateOhio	0.537227
## sqrt(incidencerate):f.stateOklahoma	0.283441
## sqrt(incidencerate):f.stateOregon	0.621779
## sqrt(incidencerate):f.statePennsylvania	0.911785
## sqrt(incidencerate):f.stateRhode Island	0.267220
## sqrt(incidencerate):f.stateSouth Carolina	0.692174
## sqrt(incidencerate):f.stateSouth Dakota	0.763599
## sqrt(incidencerate):f.stateTennessee	0.796190
## sqrt(incidencerate):f.stateTexas	0.410014
## sqrt(incidencerate):f.stateUtah	0.875873
## sqrt(incidencerate):f.stateVermont	0.937384
## sqrt(incidencerate):f.stateVirginia	0.502807
## sqrt(incidencerate):f.stateWashington	0.612656
## sqrt(incidencerate):f.stateWest Virginia	0.928186
## sqrt(incidencerate):f.stateWisconsin	0.480117
## sqrt(incidencerate):f.stateWyoming	0.168439
## pctbachdeg25_over:f.stateAlaska	0.036909 *
## pctbachdeg25_over:f.stateArizona	0.183311
## pctbachdeg25_over:f.stateArkansas	0.969387
## pctbachdeg25_over:f.stateCalifornia	0.276919
## pctbachdeg25_over:f.stateColorado	0.675401
## pctbachdeg25_over:f.stateConnecticut	0.804864
## pctbachdeg25_over:f.stateDelaware	NA
## pctbachdeg25_over:f.stateFlorida	0.048057 *
## pctbachdeg25_over:f.stateGeorgia	0.863383
## pctbachdeg25_over:f.stateHawaii	NA
## pctbachdeg25_over:f.stateIdaho	0.427443
## pctbachdeg25_over:f.stateIllinois	0.961293
## pctbachdeg25_over:f.stateIndiana	0.448890
## pctbachdeg25_over:f.stateIowa	0.083191 .
## pctbachdeg25_over:f.stateKansas	0.730689
## pctbachdeg25_over:f.stateKentucky	0.766674
## pctbachdeg25_over:f.stateLouisiana	0.476046
## pctbachdeg25_over:f.stateMaine	0.560790
## pctbachdeg25_over:f.stateMaryland	0.807658
## pctbachdeg25_over:f.stateMassachusetts	0.856513
## pctbachdeg25_over:f.stateMichigan	0.647282
## pctbachdeg25_over:f.stateMinnesota	0.452685
## pctbachdeg25_over:f.stateMississippi	0.079774 .
## pctbachdeg25_over:f.stateMissouri	0.874243
## pctbachdeg25_over:f.stateMontana	0.245128
## pctbachdeg25_over:f.stateNebraska	0.564022
## pctbachdeg25_over:f.stateNevada	0.005747 **
## pctbachdeg25_over:f.stateNew Hampshire	0.828662
## pctbachdeg25_over:f.stateNew Jersey	0.880461
## pctbachdeg25_over:f.stateNew Mexico	0.462637
## pctbachdeg25_over:f.stateNew York	0.762362
## pctbachdeg25_over:f.stateNorth Carolina	0.990240
## pctbachdeg25_over:f.stateNorth Dakota	0.970369
## pctbachdeg25_over:f.stateOhio	0.591088

## pctbachdeg25_over:f.stateOklahoma	0.214545
## pctbachdeg25_over:f.stateOregon	0.438578
## pctbachdeg25_over:f.statePennsylvania	0.562326
## pctbachdeg25_over:f.stateRhode Island	NA
## pctbachdeg25_over:f.stateSouth Carolina	0.635985
## pctbachdeg25_over:f.stateSouth Dakota	0.159243
## pctbachdeg25_over:f.stateTennessee	0.533933
## pctbachdeg25_over:f.stateTexas	0.689344
## pctbachdeg25_over:f.stateUtah	0.757812
## pctbachdeg25_over:f.stateVermont	0.750742
## pctbachdeg25_over:f.stateVirginia	0.705217
## pctbachdeg25_over:f.stateWashington	0.359105
## pctbachdeg25_over:f.stateWest Virginia	0.033327 *
## pctbachdeg25_over:f.stateWisconsin	0.339730
## pctbachdeg25_over:f.stateWyoming	0.752401
## log(medincome):f.stateAlaska	0.793006
## log(medincome):f.stateArizona	0.136918
## log(medincome):f.stateArkansas	0.965974
## log(medincome):f.stateCalifornia	0.889264
## log(medincome):f.stateColorado	0.401630
## log(medincome):f.stateConnecticut	0.711757
## log(medincome):f.stateDelaware	NA
## log(medincome):f.stateFlorida	0.501997
## log(medincome):f.stateGeorgia	0.498693
## log(medincome):f.stateHawaii	NA
## log(medincome):f.stateIdaho	0.565135
## log(medincome):f.stateIllinois	0.706881
## log(medincome):f.stateIndiana	0.686959
## log(medincome):f.stateIowa	0.669690
## log(medincome):f.stateKansas	0.886391
## log(medincome):f.stateKentucky	0.799793
## log(medincome):f.stateLouisiana	0.745496
## log(medincome):f.stateMaine	0.639689
## log(medincome):f.stateMaryland	0.269671
## log(medincome):f.stateMassachusetts	0.967184
## log(medincome):f.stateMichigan	0.998438
## log(medincome):f.stateMinnesota	0.286830
## log(medincome):f.stateMississippi	0.171567
## log(medincome):f.stateMissouri	0.692400
## log(medincome):f.stateMontana	0.668261
## log(medincome):f.stateNebraska	0.378928
## log(medincome):f.stateNevada	0.000388 ***
## log(medincome):f.stateNew Hampshire	0.829543
## log(medincome):f.stateNew Jersey	0.986686
## log(medincome):f.stateNew Mexico	0.480744
## log(medincome):f.stateNew York	0.740620
## log(medincome):f.stateNorth Carolina	0.851536
## log(medincome):f.stateNorth Dakota	0.681172
## log(medincome):f.stateOhio	0.886802
## log(medincome):f.stateOklahoma	0.565539
## log(medincome):f.stateOregon	0.531663
## log(medincome):f.statePennsylvania	0.519159
## log(medincome):f.stateRhode Island	NA
## log(medincome):f.stateSouth Carolina	0.529616

## log(medincome):f.stateSouth Dakota	0.745438
## log(medincome):f.stateTennessee	0.594766
## log(medincome):f.stateTexas	0.619079
## log(medincome):f.stateUtah	0.412734
## log(medincome):f.stateVermont	0.703444
## log(medincome):f.stateVirginia	0.350923
## log(medincome):f.stateWashington	0.762599
## log(medincome):f.stateWest Virginia	0.182365
## log(medincome):f.stateWisconsin	0.888989
## log(medincome):f.stateWyoming	0.591561
## log(medincome):f.education25_overBalanced Eduaction	0.001716 **
## log(medincome):f.education25_overMostly Bachelor's	0.769501
## f.stateAlabama:log(pctemployed16_over)	0.582404
## f.stateAlaska:log(pctemployed16_over)	0.000768 ***
## f.stateArizona:log(pctemployed16_over)	0.227980
## f.stateArkansas:log(pctemployed16_over)	0.072208 .
## f.stateCalifornia:log(pctemployed16_over)	0.296777
## f.stateColorado:log(pctemployed16_over)	0.065842 .
## f.stateConnecticut:log(pctemployed16_over)	0.944234
## f.stateDelaware:log(pctemployed16_over)	NA
## f.stateFlorida:log(pctemployed16_over)	0.420937
## f.stateGeorgia:log(pctemployed16_over)	0.204355
## f.stateHawaii:log(pctemployed16_over)	NA
## f.stateIdaho:log(pctemployed16_over)	0.634597
## f.stateIllinois:log(pctemployed16_over)	0.851772
## f.stateIndiana:log(pctemployed16_over)	0.302171
## f.stateIowa:log(pctemployed16_over)	0.465862
## f.stateKansas:log(pctemployed16_over)	0.020182 *
## f.stateKentucky:log(pctemployed16_over)	0.026547 *
## f.stateLouisiana:log(pctemployed16_over)	0.599981
## f.stateMaine:log(pctemployed16_over)	0.649918
## f.stateMaryland:log(pctemployed16_over)	0.949654
## f.stateMassachusetts:log(pctemployed16_over)	0.695999
## f.stateMichigan:log(pctemployed16_over)	0.268815
## f.stateMinnesota:log(pctemployed16_over)	0.502400
## f.stateMississippi:log(pctemployed16_over)	0.110997
## f.stateMissouri:log(pctemployed16_over)	0.003575 **
## f.stateMontana:log(pctemployed16_over)	0.917146
## f.stateNebraska:log(pctemployed16_over)	0.011274 *
## f.stateNevada:log(pctemployed16_over)	0.052946 .
## f.stateNew Hampshire:log(pctemployed16_over)	0.806748
## f.stateNew Jersey:log(pctemployed16_over)	0.748160
## f.stateNew Mexico:log(pctemployed16_over)	0.286699
## f.stateNew York:log(pctemployed16_over)	0.991644
## f.stateNorth Carolina:log(pctemployed16_over)	0.495730
## f.stateNorth Dakota:log(pctemployed16_over)	0.764310
## f.stateOhio:log(pctemployed16_over)	0.900888
## f.stateOklahoma:log(pctemployed16_over)	0.006524 **
## f.stateOregon:log(pctemployed16_over)	0.710335
## f.statePennsylvania:log(pctemployed16_over)	0.676992
## f.stateRhode Island:log(pctemployed16_over)	NA
## f.stateSouth Carolina:log(pctemployed16_over)	0.077195 .
## f.stateSouth Dakota:log(pctemployed16_over)	0.886242
## f.stateTennessee:log(pctemployed16_over)	0.252129

```
## f.stateTexas:log(pctemployed16_over) 0.240314
## f.stateUtah:log(pctemployed16_over) 0.563299
## f.stateVermont:log(pctemployed16_over) 0.804783
## f.stateVirginia:log(pctemployed16_over) 0.666282
## f.stateWashington:log(pctemployed16_over) 0.618967
## f.stateWest Virginia:log(pctemployed16_over) 0.000502 ***
## f.stateWisconsin:log(pctemployed16_over) 0.136282
## f.stateWyoming:log(pctemployed16_over) 0.627021
## f.education25_overBalanced Eduaction:log(pctemployed16_over) 0.002807 **
## f.education25_overMostly Bachelor's:log(pctemployed16_over) 0.731202
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 17.26 on 1535 degrees of freedom
## Multiple R-squared:  0.6797, Adjusted R-squared:  0.6181
## F-statistic: 11.04 on 295 and 1535 DF, p-value: < 2.2e-16
```

```
Anova(m4)
```

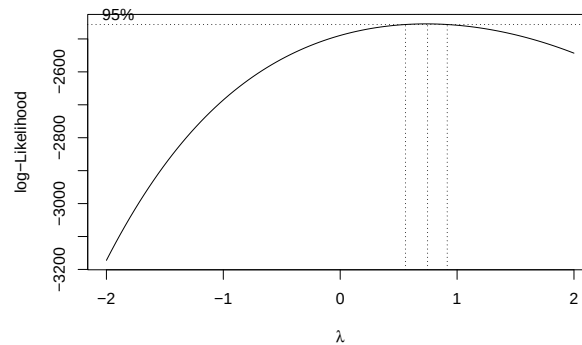
```
## Note: model has aliased coefficients
##      sums of squares computed by model comparison

## Anova Table (Type II tests)
##
## Response: target_deathrate
##
##              Sum Sq   Df F value    Pr(>F)
## pctpubliccoveragealone      1736     1   5.8267 0.0159015 *
## sqrt(incidencerate)    136401     1 457.8308 < 2.2e-16 ***
## pctbachdeg25_over        7073     1  23.7413 1.216e-06 ***
## log(medincome)           732      1   2.4574 0.1171789
## f.state                 138090    49   9.4592 < 2.2e-16 ***
## f.unemployment           1973      2   3.3116 0.0367179 *
## pctpubliccoveragealone:f.state    39366    46   2.8725 9.741e-10 ***
## pctpubliccoveragealone:f.education25_over    2041      2   3.4254 0.0327838 *
## sqrt(incidencerate):f.state    25174    46   1.8369 0.0006095 ***
## pctbachdeg25_over:f.state    23017    46   1.6795 0.0031911 **
## log(medincome):f.state    25452    46   1.8572 0.0004874 ***
## log(medincome):f.education25_over    3008      2   5.0490 0.0065226 **
## f.state:log(pctemployed16_over)    27078    47   1.9338 0.0001805 ***
## f.education25_over:log(pctemployed16_over)    2721      2   4.5664 0.0105374 *
## Residuals                457319 1535
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

The residual standard error seems a bit too high. We will now assess a transformation to the target variable in order to try to reduce it while maintaining the R^2 values. First we try a boxcox transformation of the target model. Since the lambda is around 0.8, we apply the boxcox transformation with this value. Thanks to the boxcox transformation, we reduced the residual standard error to 4.633 (which previously was 17.22), while the Adjusted R-squared keeps approximately the same(0.62), while the homoskedasticity has been greatly improved looking at the plots.

```
res_box <- boxcox(target_deathrate ~ pctpubliccoveragealone + sqrt(incidencerate)
+ pctbachdeg25_over + log(medincome) + f.state + f.unemployment
+ pctpubliccoveragealone:f.state + pctpubliccoveragealone:f.education25_over
+ sqrt(incidencerate):f.state + pctbachdeg25_over:f.state +
log(medincome):f.state + log(medincome):f.education25_over
```

```
+ log(pctemployed16_over):f.state + log(pctemployed16_over):f.education25_over,
data=df)
```



```
index <- which.max(res_box$y)
lambda <- res_box$x[index];lambda
```

```
## [1] 0.7474747
```

```
df$target_transformed <- (df$target_deathrate^lambda - 1) / lambda
```

```
m5 <- lm(formula = target_transformed ~ pctpubliccoveragealone
+ sqrt(incidencerate) + pctbachdeg25_over + log(medincome) + f.state
+ f.unemployment + pctpubliccoveragealone:f.state +
  pctpubliccoveragealone:f.education25_over + sqrt(incidencerate):f.state
+ pctbachdeg25_over:f.state + log(medincome):f.state
+ log(medincome):f.education25_over + + log(pctemployed16_over):f.state + log(pctemployed16_o

summary(m5)
```

```
##
```

```
## Call:
```

```
## lm(formula = target_transformed ~ pctpubliccoveragealone + sqrt(incidencerate) +
##   pctbachdeg25_over + log(medincome) + f.state + f.unemployment +
##   pctpubliccoveragealone:f.state + pctpubliccoveragealone:f.education25_over +
##   sqrt(incidencerate):f.state + pctbachdeg25_over:f.state +
##   log(medincome):f.state + log(medincome):f.education25_over +
##   +log(pctemployed16_over):f.state + log(pctemployed16_over):f.education25_over,
##   data = df)
```

```
##
```

```
## Residuals:
```

```
##      Min       1Q   Median       3Q      Max
## -24.979  -2.266   0.000   2.449  21.994
```

```
##
```

```
## Coefficients: (12 not defined because of singularities)
```

```
##
```

	Estimate
## (Intercept)	4.393e+01
## pctpubliccoveragealone	-2.120e-02
## sqrt(incidencerate)	2.128e+00
## pctbachdeg25_over	-3.784e-01
## log(medincome)	-4.274e+00
## f.stateAlaska	-3.939e+02

## f.stateArizona	-7.500e+02
## f.stateArkansas	-5.289e+00
## f.stateCalifornia	6.778e+00
## f.stateColorado	-2.094e+02
## f.stateConnecticut	3.171e+02
## f.stateDelaware	2.622e+01
## f.stateFlorida	1.317e+02
## f.stateGeorgia	8.770e+01
## f.stateHawaii	1.110e+02
## f.stateIdaho	-8.098e+01
## f.stateIllinois	-6.113e+01
## f.stateIndiana	-1.466e+01
## f.stateIowa	2.156e+01
## f.stateKansas	-3.668e+02
## f.stateKentucky	9.369e+01
## f.stateLouisiana	1.015e+02
## f.stateMaine	-2.100e+02
## f.stateMaryland	-4.229e+02
## f.stateMassachusetts	-2.930e+02
## f.stateMichigan	1.181e+01
## f.stateMinnesota	-4.749e+02
## f.stateMississippi	-1.367e+02
## f.stateMissouri	1.482e+01
## f.stateMontana	-7.648e+01
## f.stateNebraska	8.913e+00
## f.stateNevada	5.719e+02
## f.stateNew Hampshire	-8.293e+03
## f.stateNew Jersey	1.904e+02
## f.stateNew Mexico	8.458e+01
## f.stateNew York	5.493e+01
## f.stateNorth Carolina	-2.340e+01
## f.stateNorth Dakota	-1.068e+02
## f.stateOhio	-3.453e+01
## f.stateOklahoma	1.696e+01
## f.stateOregon	-7.405e+01
## f.statePennsylvania	-8.721e+01
## f.stateRhode Island	2.085e+03
## f.stateSouth Carolina	5.209e+01
## f.stateSouth Dakota	7.699e+01
## f.stateTennessee	-4.162e+01
## f.stateTexas	9.501e+01
## f.stateUtah	1.353e+02
## f.stateVermont	-4.644e+02
## f.stateVirginia	-1.173e+02
## f.stateWashington	1.970e+01
## f.stateWest Virginia	-5.241e+01
## f.stateWisconsin	1.040e+02
## f.stateWyoming	-2.324e+02
## f.unemploymentModerate Unemployment	-2.401e-01
## f.unemploymentHigh Unemployment	6.961e-01
## pctpubliccoveragealone:f.stateAlaska	8.142e-01
## pctpubliccoveragealone:f.stateArizona	1.803e+00
## pctpubliccoveragealone:f.stateArkansas	6.039e-01
## pctpubliccoveragealone:f.stateCalifornia	3.635e-01

## pctpubliccoveragealone:f.stateColorado	6.835e-01
## pctpubliccoveragealone:f.stateConnecticut	-3.628e-01
## pctpubliccoveragealone:f.stateDelaware	NA
## pctpubliccoveragealone:f.stateFlorida	-1.048e+00
## pctpubliccoveragealone:f.stateGeorgia	-3.936e-01
## pctpubliccoveragealone:f.stateHawaii	-4.957e+00
## pctpubliccoveragealone:f.stateIdaho	-6.930e-01
## pctpubliccoveragealone:f.stateIllinois	1.711e-01
## pctpubliccoveragealone:f.stateIndiana	2.875e-01
## pctpubliccoveragealone:f.stateIowa	6.378e-01
## pctpubliccoveragealone:f.stateKansas	-3.148e-01
## pctpubliccoveragealone:f.stateKentucky	-9.091e-02
## pctpubliccoveragealone:f.stateLouisiana	3.497e-01
## pctpubliccoveragealone:f.stateMaine	2.220e-01
## pctpubliccoveragealone:f.stateMaryland	1.265e+00
## pctpubliccoveragealone:f.stateMassachusetts	3.969e-01
## pctpubliccoveragealone:f.stateMichigan	2.779e-01
## pctpubliccoveragealone:f.stateMinnesota	4.976e-01
## pctpubliccoveragealone:f.stateMississippi	7.560e-01
## pctpubliccoveragealone:f.stateMissouri	5.154e-01
## pctpubliccoveragealone:f.stateMontana	3.027e-01
## pctpubliccoveragealone:f.stateNebraska	3.988e-02
## pctpubliccoveragealone:f.stateNevada	-1.903e+00
## pctpubliccoveragealone:f.stateNew Hampshire	-3.651e+01
## pctpubliccoveragealone:f.stateNew Jersey	-5.784e-01
## pctpubliccoveragealone:f.stateNew Mexico	-3.766e-01
## pctpubliccoveragealone:f.stateNew York	-1.745e-01
## pctpubliccoveragealone:f.stateNorth Carolina	1.354e-01
## pctpubliccoveragealone:f.stateNorth Dakota	-1.137e-01
## pctpubliccoveragealone:f.stateOhio	2.722e-01
## pctpubliccoveragealone:f.stateOklahoma	1.209e+00
## pctpubliccoveragealone:f.stateOregon	4.387e-01
## pctpubliccoveragealone:f.statePennsylvania	7.478e-01
## pctpubliccoveragealone:f.stateRhode Island	3.770e+01
## pctpubliccoveragealone:f.stateSouth Carolina	4.220e-01
## pctpubliccoveragealone:f.stateSouth Dakota	4.003e-01
## pctpubliccoveragealone:f.stateTennessee	5.869e-01
## pctpubliccoveragealone:f.stateTexas	-4.195e-01
## pctpubliccoveragealone:f.stateUtah	-2.008e-01
## pctpubliccoveragealone:f.stateVermont	1.625e+00
## pctpubliccoveragealone:f.stateVirginia	7.841e-01
## pctpubliccoveragealone:f.stateWashington	-2.218e-01
## pctpubliccoveragealone:f.stateWest Virginia	4.920e-01
## pctpubliccoveragealone:f.stateWisconsin	1.692e-01
## pctpubliccoveragealone:f.stateWyoming	-1.011e-01
## pctpubliccoveragealone:f.education25_overBalanced Eduaction	-1.575e-01
## pctpubliccoveragealone:f.education25_overMostly Bachelor's	1.658e-01
## sqrt(incidencerate):f.stateAlaska	1.245e+00
## sqrt(incidencerate):f.stateArizona	-4.445e+00
## sqrt(incidencerate):f.stateArkansas	-1.156e+00
## sqrt(incidencerate):f.stateCalifornia	5.411e-01
## sqrt(incidencerate):f.stateColorado	4.663e-01
## sqrt(incidencerate):f.stateConnecticut	8.322e-01
## sqrt(incidencerate):f.stateDelaware	NA

## sqrt(incidencerate):f.stateFlorida	4.385e-01
## sqrt(incidencerate):f.stateGeorgia	1.059e+00
## sqrt(incidencerate):f.stateHawaii	NA
## sqrt(incidencerate):f.stateIdaho	1.498e+00
## sqrt(incidencerate):f.stateIllinois	1.045e+00
## sqrt(incidencerate):f.stateIndiana	1.049e+00
## sqrt(incidencerate):f.stateIowa	6.977e-01
## sqrt(incidencerate):f.stateKansas	2.273e+01
## sqrt(incidencerate):f.stateKentucky	1.251e+00
## sqrt(incidencerate):f.stateLouisiana	-5.880e-01
## sqrt(incidencerate):f.stateMaine	-7.703e-01
## sqrt(incidencerate):f.stateMaryland	4.703e+00
## sqrt(incidencerate):f.stateMassachusetts	-1.546e+00
## sqrt(incidencerate):f.stateMichigan	-9.651e-01
## sqrt(incidencerate):f.stateMinnesota	1.673e+01
## sqrt(incidencerate):f.stateMississippi	-9.464e-01
## sqrt(incidencerate):f.stateMissouri	-2.549e-01
## sqrt(incidencerate):f.stateMontana	1.998e-01
## sqrt(incidencerate):f.stateNebraska	-7.569e-01
## sqrt(incidencerate):f.stateNevada	1.300e+01
## sqrt(incidencerate):f.stateNew Hampshire	3.495e+01
## sqrt(incidencerate):f.stateNew Jersey	-1.359e+00
## sqrt(incidencerate):f.stateNew Mexico	1.016e+00
## sqrt(incidencerate):f.stateNew York	1.116e+00
## sqrt(incidencerate):f.stateNorth Carolina	-2.046e-01
## sqrt(incidencerate):f.stateNorth Dakota	1.422e+00
## sqrt(incidencerate):f.stateOhio	1.146e+00
## sqrt(incidencerate):f.stateOklahoma	-1.656e+00
## sqrt(incidencerate):f.stateOregon	-7.919e-01
## sqrt(incidencerate):f.statePennsylvania	2.632e-01
## sqrt(incidencerate):f.stateRhode Island	-1.142e+02
## sqrt(incidencerate):f.stateSouth Carolina	6.981e-01
## sqrt(incidencerate):f.stateSouth Dakota	-3.638e-01
## sqrt(incidencerate):f.stateTennessee	3.773e-01
## sqrt(incidencerate):f.stateTexas	1.227e+00
## sqrt(incidencerate):f.stateUtah	-1.010e-01
## sqrt(incidencerate):f.stateVermont	4.249e-01
## sqrt(incidencerate):f.stateVirginia	-7.913e-01
## sqrt(incidencerate):f.stateWashington	-7.292e-01
## sqrt(incidencerate):f.stateWest Virginia	-9.073e-02
## sqrt(incidencerate):f.stateWisconsin	-9.724e-01
## sqrt(incidencerate):f.stateWyoming	4.775e+00
## pctbachdeg25_over:f.stateAlaska	-1.323e+00
## pctbachdeg25_over:f.stateArizona	1.046e+00
## pctbachdeg25_over:f.stateArkansas	2.061e-02
## pctbachdeg25_over:f.stateCalifornia	4.849e-01
## pctbachdeg25_over:f.stateColorado	-2.033e-01
## pctbachdeg25_over:f.stateConnecticut	5.079e-01
## pctbachdeg25_over:f.stateDelaware	NA
## pctbachdeg25_over:f.stateFlorida	-7.776e-01
## pctbachdeg25_over:f.stateGeorgia	-5.454e-02
## pctbachdeg25_over:f.stateHawaii	NA
## pctbachdeg25_over:f.stateIdaho	-3.379e-01
## pctbachdeg25_over:f.stateIllinois	-1.140e-02

## pctbachdeg25_over:f.stateIndiana	2.477e-01
## pctbachdeg25_over:f.stateIowa	5.708e-01
## pctbachdeg25_over:f.stateKansas	1.134e-01
## pctbachdeg25_over:f.stateKentucky	9.971e-02
## pctbachdeg25_over:f.stateLouisiana	2.996e-01
## pctbachdeg25_over:f.stateMaine	-6.260e-01
## pctbachdeg25_over:f.stateMaryland	-1.751e-01
## pctbachdeg25_over:f.stateMassachusetts	-3.073e-01
## pctbachdeg25_over:f.stateMichigan	-1.858e-01
## pctbachdeg25_over:f.stateMinnesota	2.635e-01
## pctbachdeg25_over:f.stateMississippi	5.506e-01
## pctbachdeg25_over:f.stateMissouri	5.463e-02
## pctbachdeg25_over:f.stateMontana	4.191e-01
## pctbachdeg25_over:f.stateNebraska	2.058e-01
## pctbachdeg25_over:f.stateNevada	-1.467e+00
## pctbachdeg25_over:f.stateNew Hampshire	-7.307e+01
## pctbachdeg25_over:f.stateNew Jersey	-1.605e-01
## pctbachdeg25_over:f.stateNew Mexico	-4.455e-01
## pctbachdeg25_over:f.stateNew York	1.206e-01
## pctbachdeg25_over:f.stateNorth Carolina	6.853e-03
## pctbachdeg25_over:f.stateNorth Dakota	-1.668e-05
## pctbachdeg25_over:f.stateOhio	1.929e-01
## pctbachdeg25_over:f.stateOklahoma	5.719e-01
## pctbachdeg25_over:f.stateOregon	3.134e-01
## pctbachdeg25_over:f.statePennsylvania	2.131e-01
## pctbachdeg25_over:f.stateRhode Island	NA
## pctbachdeg25_over:f.stateSouth Carolina	2.013e-01
## pctbachdeg25_over:f.stateSouth Dakota	5.382e-01
## pctbachdeg25_over:f.stateTennessee	2.262e-01
## pctbachdeg25_over:f.stateTexas	1.107e-01
## pctbachdeg25_over:f.stateUtah	1.184e-01
## pctbachdeg25_over:f.stateVermont	5.027e-01
## pctbachdeg25_over:f.stateVirginia	-1.261e-01
## pctbachdeg25_over:f.stateWashington	-4.313e-01
## pctbachdeg25_over:f.stateWest Virginia	1.027e+00
## pctbachdeg25_over:f.stateWisconsin	3.446e-01
## pctbachdeg25_over:f.stateWyoming	5.189e-01
## log(medincome):f.stateAlaska	4.859e+00
## log(medincome):f.stateArizona	8.841e+01
## log(medincome):f.stateArkansas	-8.550e-01
## log(medincome):f.stateCalifornia	2.036e+00
## log(medincome):f.stateColorado	1.497e+01
## log(medincome):f.stateConnecticut	-2.536e+01
## log(medincome):f.stateDelaware	NA
## log(medincome):f.stateFlorida	-1.025e+01
## log(medincome):f.stateGeorgia	-9.410e+00
## log(medincome):f.stateHawaii	NA
## log(medincome):f.stateIdaho	1.015e+01
## log(medincome):f.stateIllinois	5.202e+00
## log(medincome):f.stateIndiana	5.743e+00
## log(medincome):f.stateIowa	-7.520e+00
## log(medincome):f.stateKansas	1.931e+00
## log(medincome):f.stateKentucky	-3.446e+00
## log(medincome):f.stateLouisiana	-4.813e+00

## log(medincome):f.stateMaine	1.441e+01
## log(medincome):f.stateMaryland	2.881e+01
## log(medincome):f.stateMassachusetts	2.748e+00
## log(medincome):f.stateMichigan	-2.961e-01
## log(medincome):f.stateMinnesota	1.589e+01
## log(medincome):f.stateMississippi	1.857e+01
## log(medincome):f.stateMissouri	-5.872e+00
## log(medincome):f.stateMontana	6.935e+00
## log(medincome):f.stateNebraska	1.278e+01
## log(medincome):f.stateNevada	-8.175e+01
## log(medincome):f.stateNew Hampshire	8.206e+02
## log(medincome):f.stateNew Jersey	-7.173e-01
## log(medincome):f.stateNew Mexico	-1.131e+01
## log(medincome):f.stateNew York	-5.295e+00
## log(medincome):f.stateNorth Carolina	2.453e+00
## log(medincome):f.stateNorth Dakota	6.074e+00
## log(medincome):f.stateOhio	1.717e+00
## log(medincome):f.stateOklahoma	-8.151e+00
## log(medincome):f.stateOregon	1.145e+01
## log(medincome):f.statePennsylvania	1.009e+01
## log(medincome):f.stateRhode Island	NA
## log(medincome):f.stateSouth Carolina	-1.198e+01
## log(medincome):f.stateSouth Dakota	-5.246e+00
## log(medincome):f.stateTennessee	7.410e+00
## log(medincome):f.stateTexas	-6.730e+00
## log(medincome):f.stateUtah	-1.447e+01
## log(medincome):f.stateVermont	3.427e+01
## log(medincome):f.stateVirginia	1.278e+01
## log(medincome):f.stateWashington	4.992e+00
## log(medincome):f.stateWest Virginia	2.106e+01
## log(medincome):f.stateWisconsin	2.548e+00
## log(medincome):f.stateWyoming	2.624e+01
## log(medincome):f.education25_overBalanced Eduaction	4.082e+00
## log(medincome):f.education25_overMostly Bachelor's	7.532e-01
## f.stateAlabama:log(pctemployed16_over)	6.778e+00
## f.stateAlaska:log(pctemployed16_over)	8.605e+01
## f.stateArizona:log(pctemployed16_over)	-3.820e+01
## f.stateArkansas:log(pctemployed16_over)	1.353e+01
## f.stateCalifornia:log(pctemployed16_over)	-8.790e+00
## f.stateColorado:log(pctemployed16_over)	1.236e+01
## f.stateConnecticut:log(pctemployed16_over)	-8.500e+00
## f.stateDelaware:log(pctemployed16_over)	NA
## f.stateFlorida:log(pctemployed16_over)	7.027e+00
## f.stateGeorgia:log(pctemployed16_over)	5.848e+00
## f.stateHawaii:log(pctemployed16_over)	NA
## f.stateIdaho:log(pctemployed16_over)	-5.643e+00
## f.stateIllinois:log(pctemployed16_over)	1.131e+00
## f.stateIndiana:log(pctemployed16_over)	-1.234e+01
## f.stateIowa:log(pctemployed16_over)	1.194e+01
## f.stateKansas:log(pctemployed16_over)	-2.580e+01
## f.stateKentucky:log(pctemployed16_over)	-1.416e+01
## f.stateLouisiana:log(pctemployed16_over)	-5.154e+00
## f.stateMaine:log(pctemployed16_over)	2.608e+01
## f.stateMaryland:log(pctemployed16_over)	2.941e+00

## f.stateMassachusetts:log(pctemployed16_over)	7.788e+01	
## f.stateMichigan:log(pctemployed16_over)	8.735e+00	
## f.stateMinnesota:log(pctemployed16_over)	-9.904e+00	
## f.stateMississippi:log(pctemployed16_over)	-8.713e+00	
## f.stateMissouri:log(pctemployed16_over)	1.787e+01	
## f.stateMontana:log(pctemployed16_over)	1.683e+00	
## f.stateNebraska:log(pctemployed16_over)	-2.717e+01	
## f.stateNevada:log(pctemployed16_over)	2.897e+01	
## f.stateNew Hampshire:log(pctemployed16_over)	1.254e+02	
## f.stateNew Jersey:log(pctemployed16_over)	-2.824e+01	
## f.stateNew Mexico:log(pctemployed16_over)	1.370e+01	
## f.stateNew York:log(pctemployed16_over)	3.320e-01	
## f.stateNorth Carolina:log(pctemployed16_over)	5.734e+00	
## f.stateNorth Dakota:log(pctemployed16_over)	8.599e+00	
## f.stateOhio:log(pctemployed16_over)	2.727e+00	
## f.stateOklahoma:log(pctemployed16_over)	2.594e+01	
## f.stateOregon:log(pctemployed16_over)	-5.489e+00	
## f.statePennsylvania:log(pctemployed16_over)	-4.856e+00	
## f.stateRhode Island:log(pctemployed16_over)	NA	
## f.stateSouth Carolina:log(pctemployed16_over)	1.926e+01	
## f.stateSouth Dakota:log(pctemployed16_over)	-1.180e+00	
## f.stateTennessee:log(pctemployed16_over)	-8.129e+00	
## f.stateTexas:log(pctemployed16_over)	-3.763e+00	
## f.stateUtah:log(pctemployed16_over)	1.067e+01	
## f.stateVermont:log(pctemployed16_over)	1.518e+01	
## f.stateVirginia:log(pctemployed16_over)	2.972e+00	
## f.stateWashington:log(pctemployed16_over)	-5.893e+00	
## f.stateWest Virginia:log(pctemployed16_over)	-4.212e+01	
## f.stateWisconsin:log(pctemployed16_over)	-2.305e+01	
## f.stateWyoming:log(pctemployed16_over)	-3.252e+01	
## f.education25_overBalanced Eduaction:log(pctemployed16_over)	-1.026e+01	
## f.education25_overMostly Bachelor's:log(pctemployed16_over)	-2.456e+00	
##	Std. Error	t value
## (Intercept)	1.645e+02	0.267
## pctpubliccoveragealone	5.778e-01	-0.037
## sqrt(incidencerate)	1.247e+00	1.707
## pctbachdeg25_over	2.868e-01	-1.319
## log(medincome)	1.303e+01	-0.328
## f.stateAlaska	2.388e+02	-1.650
## f.stateArizona	5.281e+02	-1.420
## f.stateArkansas	1.958e+02	-0.027
## f.stateCalifornia	1.849e+02	0.037
## f.stateColorado	1.901e+02	-1.102
## f.stateConnecticut	1.008e+03	0.315
## f.stateDelaware	4.811e+01	0.545
## f.stateFlorida	1.839e+02	0.716
## f.stateGeorgia	1.698e+02	0.516
## f.stateHawaii	1.658e+02	0.670
## f.stateIdaho	2.126e+02	-0.381
## f.stateIllinois	1.813e+02	-0.337
## f.stateIndiana	1.849e+02	-0.079
## f.stateIowa	1.891e+02	0.114
## f.stateKansas	2.386e+02	-1.537
## f.stateKentucky	1.787e+02	0.524

## f.stateLouisiana	1.784e+02	0.569
## f.stateMaine	3.804e+02	-0.552
## f.stateMaryland	2.928e+02	-1.445
## f.stateMassachusetts	4.507e+02	-0.650
## f.stateMichigan	1.816e+02	0.065
## f.stateMinnesota	3.148e+02	-1.508
## f.stateMississippi	1.770e+02	-0.773
## f.stateMissouri	1.803e+02	0.082
## f.stateMontana	1.865e+02	-0.410
## f.stateNebraska	1.814e+02	0.049
## f.stateNevada	5.158e+02	1.109
## f.stateNew Hampshire	3.785e+04	-0.219
## f.stateNew Jersey	6.249e+02	0.305
## f.stateNew Mexico	1.824e+02	0.464
## f.stateNew York	1.868e+02	0.294
## f.stateNorth Carolina	1.813e+02	-0.129
## f.stateNorth Dakota	1.885e+02	-0.566
## f.stateOhio	1.882e+02	-0.183
## f.stateOklahoma	1.856e+02	0.091
## f.stateOregon	1.933e+02	-0.383
## f.statePennsylvania	1.864e+02	-0.468
## f.stateRhode Island	1.837e+03	1.135
## f.stateSouth Carolina	2.294e+02	0.227
## f.stateSouth Dakota	1.819e+02	0.423
## f.stateTennessee	1.821e+02	-0.229
## f.stateTexas	1.691e+02	0.562
## f.stateUtah	2.140e+02	0.632
## f.stateVermont	8.325e+02	-0.558
## f.stateVirginia	1.710e+02	-0.686
## f.stateWashington	1.897e+02	0.104
## f.stateWest Virginia	1.871e+02	-0.280
## f.stateWisconsin	2.188e+02	0.475
## f.stateWyoming	4.170e+02	-0.557
## f.unemploymentModerate Unemployment	4.422e-01	-0.543
## f.unemploymentHigh Unemployment	5.705e-01	1.220
## pctpubliccoveragealone:f.stateAlaska	6.905e-01	1.179
## pctpubliccoveragealone:f.stateArizona	1.583e+00	1.139
## pctpubliccoveragealone:f.stateArkansas	6.555e-01	0.921
## pctpubliccoveragealone:f.stateCalifornia	6.361e-01	0.571
## pctpubliccoveragealone:f.stateColorado	6.773e-01	1.009
## pctpubliccoveragealone:f.stateConnecticut	1.627e+00	-0.223
## pctpubliccoveragealone:f.stateDelaware	NA	NA
## pctpubliccoveragealone:f.stateFlorida	6.547e-01	-1.601
## pctpubliccoveragealone:f.stateGeorgia	5.995e-01	-0.657
## pctpubliccoveragealone:f.stateHawaii	8.593e+00	-0.577
## pctpubliccoveragealone:f.stateIdaho	6.869e-01	-1.009
## pctpubliccoveragealone:f.stateIllinois	6.231e-01	0.275
## pctpubliccoveragealone:f.stateIndiana	6.438e-01	0.447
## pctpubliccoveragealone:f.stateIowa	6.652e-01	0.959
## pctpubliccoveragealone:f.stateKansas	6.462e-01	-0.487
## pctpubliccoveragealone:f.stateKentucky	6.267e-01	-0.145
## pctpubliccoveragealone:f.stateLouisiana	6.486e-01	0.539
## pctpubliccoveragealone:f.stateMaine	1.516e+00	0.146
## pctpubliccoveragealone:f.stateMaryland	9.025e-01	1.401

## pctpubliccoveragealone:f.stateMassachusetts	1.911e+00	0.208
## pctpubliccoveragealone:f.stateMichigan	6.488e-01	0.428
## pctpubliccoveragealone:f.stateMinnesota	6.711e-01	0.741
## pctpubliccoveragealone:f.stateMississippi	6.096e-01	1.240
## pctpubliccoveragealone:f.stateMissouri	6.334e-01	0.814
## pctpubliccoveragealone:f.stateMontana	6.499e-01	0.466
## pctpubliccoveragealone:f.stateNebraska	6.306e-01	0.063
## pctpubliccoveragealone:f.stateNevada	9.192e-01	-2.070
## pctpubliccoveragealone:f.stateNew Hampshire	1.687e+02	-0.216
## pctpubliccoveragealone:f.stateNew Jersey	1.678e+00	-0.345
## pctpubliccoveragealone:f.stateNew Mexico	6.136e-01	-0.614
## pctpubliccoveragealone:f.stateNew York	6.267e-01	-0.279
## pctpubliccoveragealone:f.stateNorth Carolina	6.407e-01	0.211
## pctpubliccoveragealone:f.stateNorth Dakota	6.242e-01	-0.182
## pctpubliccoveragealone:f.stateOhio	6.540e-01	0.416
## pctpubliccoveragealone:f.stateOklahoma	6.854e-01	1.763
## pctpubliccoveragealone:f.stateOregon	6.588e-01	0.666
## pctpubliccoveragealone:f.statePennsylvania	6.514e-01	1.148
## pctpubliccoveragealone:f.stateRhode Island	3.082e+01	1.223
## pctpubliccoveragealone:f.stateSouth Carolina	7.965e-01	0.530
## pctpubliccoveragealone:f.stateSouth Dakota	6.069e-01	0.660
## pctpubliccoveragealone:f.stateTennessee	6.396e-01	0.918
## pctpubliccoveragealone:f.stateTexas	5.946e-01	-0.705
## pctpubliccoveragealone:f.stateUtah	8.305e-01	-0.242
## pctpubliccoveragealone:f.stateVermont	3.063e+00	0.530
## pctpubliccoveragealone:f.stateVirginia	6.105e-01	1.284
## pctpubliccoveragealone:f.stateWashington	7.062e-01	-0.314
## pctpubliccoveragealone:f.stateWest Virginia	6.282e-01	0.783
## pctpubliccoveragealone:f.stateWisconsin	7.593e-01	0.223
## pctpubliccoveragealone:f.stateWyoming	1.616e+00	-0.063
## pctpubliccoveragealone:f.education25_overBalanced Eduaction	6.968e-02	-2.261
## pctpubliccoveragealone:f.education25_overMostly Bachelor's	1.687e-01	0.983
## sqrt(incidencerate):f.stateAlaska	1.841e+00	0.676
## sqrt(incidencerate):f.stateArizona	3.978e+00	-1.117
## sqrt(incidencerate):f.stateArkansas	1.515e+00	-0.764
## sqrt(incidencerate):f.stateCalifornia	1.875e+00	0.289
## sqrt(incidencerate):f.stateColorado	1.442e+00	0.323
## sqrt(incidencerate):f.stateConnecticut	1.537e+01	0.054
## sqrt(incidencerate):f.stateDelaware	NA	NA
## sqrt(incidencerate):f.stateFlorida	1.290e+00	0.340
## sqrt(incidencerate):f.stateGeorgia	1.329e+00	0.797
## sqrt(incidencerate):f.stateHawaii	NA	NA
## sqrt(incidencerate):f.stateIdaho	1.512e+00	0.991
## sqrt(incidencerate):f.stateIllinois	1.523e+00	0.686
## sqrt(incidencerate):f.stateIndiana	1.582e+00	0.663
## sqrt(incidencerate):f.stateIowa	1.521e+00	0.459
## sqrt(incidencerate):f.stateKansas	8.194e+00	2.773
## sqrt(incidencerate):f.stateKentucky	1.378e+00	0.908
## sqrt(incidencerate):f.stateLouisiana	1.760e+00	-0.334
## sqrt(incidencerate):f.stateMaine	3.852e+00	-0.200
## sqrt(incidencerate):f.stateMaryland	2.538e+00	1.853
## sqrt(incidencerate):f.stateMassachusetts	6.373e+00	-0.243
## sqrt(incidencerate):f.stateMichigan	1.329e+00	-0.726
## sqrt(incidencerate):f.stateMinnesota	1.267e+01	1.321

## sqrt(incidencerate):f.stateMississippi	1.440e+00	-0.657
## sqrt(incidencerate):f.stateMissouri	1.313e+00	-0.194
## sqrt(incidencerate):f.stateMontana	1.490e+00	0.134
## sqrt(incidencerate):f.stateNebraska	1.346e+00	-0.563
## sqrt(incidencerate):f.stateNevada	1.948e+01	0.667
## sqrt(incidencerate):f.stateNew Hampshire	1.494e+02	0.234
## sqrt(incidencerate):f.stateNew Jersey	6.165e+00	-0.220
## sqrt(incidencerate):f.stateNew Mexico	1.520e+00	0.668
## sqrt(incidencerate):f.stateNew York	1.938e+00	0.576
## sqrt(incidencerate):f.stateNorth Carolina	1.538e+00	-0.133
## sqrt(incidencerate):f.stateNorth Dakota	1.365e+00	1.042
## sqrt(incidencerate):f.stateOhio	1.666e+00	0.688
## sqrt(incidencerate):f.stateOklahoma	1.551e+00	-1.068
## sqrt(incidencerate):f.stateOregon	1.788e+00	-0.443
## sqrt(incidencerate):f.statePennsylvania	1.669e+00	0.158
## sqrt(incidencerate):f.stateRhode Island	1.000e+02	-1.141
## sqrt(incidencerate):f.stateSouth Carolina	1.774e+00	0.394
## sqrt(incidencerate):f.stateSouth Dakota	1.428e+00	-0.255
## sqrt(incidencerate):f.stateTennessee	1.491e+00	0.253
## sqrt(incidencerate):f.stateTexas	1.284e+00	0.956
## sqrt(incidencerate):f.stateUtah	1.950e+00	-0.052
## sqrt(incidencerate):f.stateVermont	5.183e+00	0.082
## sqrt(incidencerate):f.stateVirginia	1.281e+00	-0.618
## sqrt(incidencerate):f.stateWashington	1.627e+00	-0.448
## sqrt(incidencerate):f.stateWest Virginia	1.362e+00	-0.067
## sqrt(incidencerate):f.stateWisconsin	1.455e+00	-0.668
## sqrt(incidencerate):f.stateWyoming	3.366e+00	1.419
## pctbachdeg25_over:f.stateAlaska	6.156e-01	-2.149
## pctbachdeg25_over:f.stateArizona	7.509e-01	1.393
## pctbachdeg25_over:f.stateArkansas	4.587e-01	0.045
## pctbachdeg25_over:f.stateCalifornia	4.533e-01	1.070
## pctbachdeg25_over:f.stateColorado	3.247e-01	-0.626
## pctbachdeg25_over:f.stateConnecticut	2.066e+00	0.246
## pctbachdeg25_over:f.stateDelaware	NA	NA
## pctbachdeg25_over:f.stateFlorida	3.939e-01	-1.974
## pctbachdeg25_over:f.stateGeorgia	3.201e-01	-0.170
## pctbachdeg25_over:f.stateHawaii	NA	NA
## pctbachdeg25_over:f.stateIdaho	3.806e-01	-0.888
## pctbachdeg25_over:f.stateIllinois	3.415e-01	-0.033
## pctbachdeg25_over:f.stateIndiana	3.300e-01	0.751
## pctbachdeg25_over:f.stateIowa	3.311e-01	1.724
## pctbachdeg25_over:f.stateKansas	3.386e-01	0.335
## pctbachdeg25_over:f.stateKentucky	3.348e-01	0.298
## pctbachdeg25_over:f.stateLouisiana	4.087e-01	0.733
## pctbachdeg25_over:f.stateMaine	1.082e+00	-0.578
## pctbachdeg25_over:f.stateMaryland	5.784e-01	-0.303
## pctbachdeg25_over:f.stateMassachusetts	1.700e+00	-0.181
## pctbachdeg25_over:f.stateMichigan	3.794e-01	-0.490
## pctbachdeg25_over:f.stateMinnesota	3.499e-01	0.753
## pctbachdeg25_over:f.stateMississippi	3.375e-01	1.631
## pctbachdeg25_over:f.stateMissouri	3.700e-01	0.148
## pctbachdeg25_over:f.stateMontana	3.655e-01	1.147
## pctbachdeg25_over:f.stateNebraska	3.542e-01	0.581
## pctbachdeg25_over:f.stateNevada	5.423e-01	-2.705

## pctbachdeg25_over:f.stateNew Hampshire	3.352e+02	-0.218
## pctbachdeg25_over:f.stateNew Jersey	1.110e+00	-0.145
## pctbachdeg25_over:f.stateNew Mexico	5.685e-01	-0.784
## pctbachdeg25_over:f.stateNew York	4.243e-01	0.284
## pctbachdeg25_over:f.stateNorth Carolina	3.627e-01	0.019
## pctbachdeg25_over:f.stateNorth Dakota	5.247e-01	0.000
## pctbachdeg25_over:f.stateOhio	3.596e-01	0.536
## pctbachdeg25_over:f.stateOklahoma	4.549e-01	1.257
## pctbachdeg25_over:f.stateOregon	4.064e-01	0.771
## pctbachdeg25_over:f.statePennsylvania	3.923e-01	0.543
## pctbachdeg25_over:f.stateRhode Island	NA	NA
## pctbachdeg25_over:f.stateSouth Carolina	4.155e-01	0.485
## pctbachdeg25_over:f.stateSouth Dakota	3.824e-01	1.407
## pctbachdeg25_over:f.stateTennessee	3.629e-01	0.623
## pctbachdeg25_over:f.stateTexas	3.125e-01	0.354
## pctbachdeg25_over:f.stateUtah	4.278e-01	0.277
## pctbachdeg25_over:f.stateVermont	1.620e+00	0.310
## pctbachdeg25_over:f.stateVirginia	3.249e-01	-0.388
## pctbachdeg25_over:f.stateWashington	4.318e-01	-0.999
## pctbachdeg25_over:f.stateWest Virginia	4.761e-01	2.157
## pctbachdeg25_over:f.stateWisconsin	3.607e-01	0.955
## pctbachdeg25_over:f.stateWyoming	1.798e+00	0.289
## log(medincome):f.stateAlaska	1.613e+01	0.301
## log(medincome):f.stateArizona	5.557e+01	1.591
## log(medincome):f.stateArkansas	1.677e+01	-0.051
## log(medincome):f.stateCalifornia	1.513e+01	0.135
## log(medincome):f.stateColorado	1.570e+01	0.953
## log(medincome):f.stateConnecticut	6.605e+01	-0.384
## log(medincome):f.stateDelaware	NA	NA
## log(medincome):f.stateFlorida	1.523e+01	-0.673
## log(medincome):f.stateGeorgia	1.375e+01	-0.685
## log(medincome):f.stateHawaii	NA	NA
## log(medincome):f.stateIdaho	1.998e+01	0.508
## log(medincome):f.stateIllinois	1.465e+01	0.355
## log(medincome):f.stateIndiana	1.520e+01	0.378
## log(medincome):f.stateIowa	1.646e+01	-0.457
## log(medincome):f.stateKansas	1.430e+01	0.135
## log(medincome):f.stateKentucky	1.414e+01	-0.244
## log(medincome):f.stateLouisiana	1.483e+01	-0.324
## log(medincome):f.stateMaine	3.145e+01	0.458
## log(medincome):f.stateMaryland	2.645e+01	1.089
## log(medincome):f.stateMassachusetts	5.640e+01	0.049
## log(medincome):f.stateMichigan	1.542e+01	-0.019
## log(medincome):f.stateMinnesota	1.472e+01	1.079
## log(medincome):f.stateMississippi	1.432e+01	1.296
## log(medincome):f.stateMissouri	1.470e+01	-0.400
## log(medincome):f.stateMontana	1.599e+01	0.434
## log(medincome):f.stateNebraska	1.463e+01	0.874
## log(medincome):f.stateNevada	2.321e+01	-3.522
## log(medincome):f.stateNew Hampshire	3.786e+03	0.217
## log(medincome):f.stateNew Jersey	3.002e+01	-0.024
## log(medincome):f.stateNew Mexico	1.511e+01	-0.749
## log(medincome):f.stateNew York	1.473e+01	-0.359
## log(medincome):f.stateNorth Carolina	1.510e+01	0.162

## log(medincome):f.stateNorth Dakota	1.487e+01	0.408
## log(medincome):f.stateOhio	1.517e+01	0.113
## log(medincome):f.stateOklahoma	1.504e+01	-0.542
## log(medincome):f.stateOregon	1.842e+01	0.622
## log(medincome):f.statePennsylvania	1.539e+01	0.656
## log(medincome):f.stateRhode Island	NA	NA
## log(medincome):f.stateSouth Carolina	1.912e+01	-0.627
## log(medincome):f.stateSouth Dakota	1.493e+01	-0.351
## log(medincome):f.stateTennessee	1.494e+01	0.496
## log(medincome):f.stateTexas	1.355e+01	-0.497
## log(medincome):f.stateUtah	1.689e+01	-0.857
## log(medincome):f.stateVermont	9.228e+01	0.371
## log(medincome):f.stateVirginia	1.370e+01	0.933
## log(medincome):f.stateWashington	1.709e+01	0.292
## log(medincome):f.stateWest Virginia	1.591e+01	1.324
## log(medincome):f.stateWisconsin	1.881e+01	0.135
## log(medincome):f.stateWyoming	5.135e+01	0.511
## log(medincome):f.education25_overBalanced Eduaction	1.277e+00	3.197
## log(medincome):f.education25_overMostly Bachelor's	2.480e+00	0.304
## f.stateAlabama:log(pctemployed16_over)	1.237e+01	0.548
## f.stateAlaska:log(pctemployed16_over)	2.674e+01	3.218
## f.stateArizona:log(pctemployed16_over)	2.925e+01	-1.306
## f.stateArkansas:log(pctemployed16_over)	7.818e+00	1.731
## f.stateCalifornia:log(pctemployed16_over)	8.226e+00	-1.069
## f.stateColorado:log(pctemployed16_over)	6.770e+00	1.826
## f.stateConnecticut:log(pctemployed16_over)	1.146e+02	-0.074
## f.stateDelaware:log(pctemployed16_over)	NA	NA
## f.stateFlorida:log(pctemployed16_over)	8.868e+00	0.792
## f.stateGeorgia:log(pctemployed16_over)	4.320e+00	1.354
## f.stateHawaii:log(pctemployed16_over)	NA	NA
## f.stateIdaho:log(pctemployed16_over)	1.347e+01	-0.419
## f.stateIllinois:log(pctemployed16_over)	6.330e+00	0.179
## f.stateIndiana:log(pctemployed16_over)	1.251e+01	-0.986
## f.stateIowa:log(pctemployed16_over)	1.585e+01	0.753
## f.stateKansas:log(pctemployed16_over)	1.115e+01	-2.314
## f.stateKentucky:log(pctemployed16_over)	7.040e+00	-2.012
## f.stateLouisiana:log(pctemployed16_over)	1.012e+01	-0.509
## f.stateMaine:log(pctemployed16_over)	5.709e+01	0.457
## f.stateMaryland:log(pctemployed16_over)	4.744e+01	0.062
## f.stateMassachusetts:log(pctemployed16_over)	2.007e+02	0.388
## f.stateMichigan:log(pctemployed16_over)	7.641e+00	1.143
## f.stateMinnesota:log(pctemployed16_over)	1.480e+01	-0.669
## f.stateMississippi:log(pctemployed16_over)	5.975e+00	-1.458
## f.stateMissouri:log(pctemployed16_over)	6.001e+00	2.977
## f.stateMontana:log(pctemployed16_over)	1.365e+01	0.123
## f.stateNebraska:log(pctemployed16_over)	1.064e+01	-2.553
## f.stateNevada:log(pctemployed16_over)	1.479e+01	1.958
## f.stateNew Hampshire:log(pctemployed16_over)	5.093e+02	0.246
## f.stateNew Jersey:log(pctemployed16_over)	8.914e+01	-0.317
## f.stateNew Mexico:log(pctemployed16_over)	1.187e+01	1.154
## f.stateNew York:log(pctemployed16_over)	1.426e+01	0.023
## f.stateNorth Carolina:log(pctemployed16_over)	8.142e+00	0.704
## f.stateNorth Dakota:log(pctemployed16_over)	2.668e+01	0.322
## f.stateOhio:log(pctemployed16_over)	1.558e+01	0.175

## f.stateOklahoma:log(pctemployed16_over)	9.803e+00	2.646
## f.stateOregon:log(pctemployed16_over)	1.520e+01	-0.361
## f.statePennsylvania:log(pctemployed16_over)	1.216e+01	-0.399
## f.stateRhode Island:log(pctemployed16_over)	NA	NA
## f.stateSouth Carolina:log(pctemployed16_over)	1.066e+01	1.807
## f.stateSouth Dakota:log(pctemployed16_over)	9.901e+00	-0.119
## f.stateTennessee:log(pctemployed16_over)	7.728e+00	-1.052
## f.stateTexas:log(pctemployed16_over)	3.084e+00	-1.220
## f.stateUtah:log(pctemployed16_over)	1.816e+01	0.588
## f.stateVermont:log(pctemployed16_over)	6.190e+01	0.245
## f.stateVirginia:log(pctemployed16_over)	6.695e+00	0.444
## f.stateWashington:log(pctemployed16_over)	1.198e+01	-0.492
## f.stateWest Virginia:log(pctemployed16_over)	1.207e+01	-3.490
## f.stateWisconsin:log(pctemployed16_over)	1.522e+01	-1.515
## f.stateWyoming:log(pctemployed16_over)	6.869e+01	-0.473
## f.education25_overBalanced Eduaction:log(pctemployed16_over)	3.364e+00	-3.049
## f.education25_overMostly Bachelor's:log(pctemployed16_over)	6.637e+00	-0.370
##	Pr(> t)	
## (Intercept)	0.789512	
## pctpubliccoveragealone	0.970743	
## sqrt(incidencerate)	0.088001	.
## pctbachdeg25_over	0.187209	
## log(medincome)	0.743052	
## f.stateAlaska	0.099230	.
## f.stateArizona	0.155758	
## f.stateArkansas	0.978449	
## f.stateCalifornia	0.970760	
## f.stateColorado	0.270764	
## f.stateConnecticut	0.753016	
## f.stateDelaware	0.585857	
## f.stateFlorida	0.474209	
## f.stateGeorgia	0.605646	
## f.stateHawaii	0.503034	
## f.stateIdaho	0.703368	
## f.stateIllinois	0.736010	
## f.stateIndiana	0.936829	
## f.stateIowa	0.909241	
## f.stateKansas	0.124421	
## f.stateKentucky	0.600091	
## f.stateLouisiana	0.569563	
## f.stateMaine	0.580990	
## f.stateMaryland	0.148750	
## f.stateMassachusetts	0.515755	
## f.stateMichigan	0.948159	
## f.stateMinnesota	0.131634	
## f.stateMississippi	0.439806	
## f.stateMissouri	0.934534	
## f.stateMontana	0.681752	
## f.stateNebraska	0.960824	
## f.stateNevada	0.267777	
## f.stateNew Hampshire	0.826581	
## f.stateNew Jersey	0.760617	
## f.stateNew Mexico	0.642876	
## f.stateNew York	0.768790	

## f.stateNorth Carolina	0.897329
## f.stateNorth Dakota	0.571258
## f.stateOhio	0.854490
## f.stateOklahoma	0.927202
## f.stateOregon	0.701773
## f.statePennsylvania	0.639921
## f.stateRhode Island	0.256585
## f.stateSouth Carolina	0.820374
## f.stateSouth Dakota	0.672149
## f.stateTennessee	0.819210
## f.stateTexas	0.574180
## f.stateUtah	0.527248
## f.stateVermont	0.577053
## f.stateVirginia	0.492676
## f.stateWashington	0.917286
## f.stateWest Virginia	0.779464
## f.stateWisconsin	0.634597
## f.stateWyoming	0.577387
## f.unemploymentModerate Unemployment	0.587154
## f.unemploymentHigh Unemployment	0.222578
## pctpubliccoveragealone:f.stateAlaska	0.238537
## pctpubliccoveragealone:f.stateArizona	0.254767
## pctpubliccoveragealone:f.stateArkansas	0.357016
## pctpubliccoveragealone:f.stateCalifornia	0.567817
## pctpubliccoveragealone:f.stateColorado	0.313091
## pctpubliccoveragealone:f.stateConnecticut	0.823528
## pctpubliccoveragealone:f.stateDelaware	NA
## pctpubliccoveragealone:f.stateFlorida	0.109509
## pctpubliccoveragealone:f.stateGeorgia	0.511535
## pctpubliccoveragealone:f.stateHawaii	0.564123
## pctpubliccoveragealone:f.stateIdaho	0.313158
## pctpubliccoveragealone:f.stateIllinois	0.783723
## pctpubliccoveragealone:f.stateIndiana	0.655237
## pctpubliccoveragealone:f.stateIowa	0.337838
## pctpubliccoveragealone:f.stateKansas	0.626239
## pctpubliccoveragealone:f.stateKentucky	0.884693
## pctpubliccoveragealone:f.stateLouisiana	0.589845
## pctpubliccoveragealone:f.stateMaine	0.883627
## pctpubliccoveragealone:f.stateMaryland	0.161360
## pctpubliccoveragealone:f.stateMassachusetts	0.835526
## pctpubliccoveragealone:f.stateMichigan	0.668494
## pctpubliccoveragealone:f.stateMinnesota	0.458536
## pctpubliccoveragealone:f.stateMississippi	0.215084
## pctpubliccoveragealone:f.stateMissouri	0.415949
## pctpubliccoveragealone:f.stateMontana	0.641415
## pctpubliccoveragealone:f.stateNebraska	0.949587
## pctpubliccoveragealone:f.stateNevada	0.038605 *
## pctpubliccoveragealone:f.stateNew Hampshire	0.828669
## pctpubliccoveragealone:f.stateNew Jersey	0.730401
## pctpubliccoveragealone:f.stateNew Mexico	0.539466
## pctpubliccoveragealone:f.stateNew York	0.780649
## pctpubliccoveragealone:f.stateNorth Carolina	0.832623
## pctpubliccoveragealone:f.stateNorth Dakota	0.855446
## pctpubliccoveragealone:f.stateOhio	0.677296

## pctpubliccoveragealone:f.stateOklahoma	0.078019 .
## pctpubliccoveragealone:f.stateOregon	0.505626
## pctpubliccoveragealone:f.statePennsylvania	0.251120
## pctpubliccoveragealone:f.stateRhode Island	0.221410
## pctpubliccoveragealone:f.stateSouth Carolina	0.596300
## pctpubliccoveragealone:f.stateSouth Dakota	0.509645
## pctpubliccoveragealone:f.stateTennessee	0.358944
## pctpubliccoveragealone:f.stateTexas	0.480640
## pctpubliccoveragealone:f.stateUtah	0.809010
## pctpubliccoveragealone:f.stateVermont	0.595929
## pctpubliccoveragealone:f.stateVirginia	0.199203
## pctpubliccoveragealone:f.stateWashington	0.753524
## pctpubliccoveragealone:f.stateWest Virginia	0.433675
## pctpubliccoveragealone:f.stateWisconsin	0.823657
## pctpubliccoveragealone:f.stateWyoming	0.950129
## pctpubliccoveragealone:f.education25_overBalanced Eduaction	0.023908 *
## pctpubliccoveragealone:f.education25_overMostly Bachelor's	0.325825
## sqrt(incidencerate):f.stateAlaska	0.499036
## sqrt(incidencerate):f.stateArizona	0.263981
## sqrt(incidencerate):f.stateArkansas	0.445270
## sqrt(incidencerate):f.stateCalifornia	0.772895
## sqrt(incidencerate):f.stateColorado	0.746512
## sqrt(incidencerate):f.stateConnecticut	0.956835
## sqrt(incidencerate):f.stateDelaware	NA
## sqrt(incidencerate):f.stateFlorida	0.733955
## sqrt(incidencerate):f.stateGeorgia	0.425719
## sqrt(incidencerate):f.stateHawaii	NA
## sqrt(incidencerate):f.stateIdaho	0.321738
## sqrt(incidencerate):f.stateIllinois	0.492833
## sqrt(incidencerate):f.stateIndiana	0.507191
## sqrt(incidencerate):f.stateIowa	0.646446
## sqrt(incidencerate):f.stateKansas	0.005617 **
## sqrt(incidencerate):f.stateKentucky	0.363984
## sqrt(incidencerate):f.stateLouisiana	0.738347
## sqrt(incidencerate):f.stateMaine	0.841529
## sqrt(incidencerate):f.stateMaryland	0.064006 .
## sqrt(incidencerate):f.stateMassachusetts	0.808325
## sqrt(incidencerate):f.stateMichigan	0.467867
## sqrt(incidencerate):f.stateMinnesota	0.186622
## sqrt(incidencerate):f.stateMississippi	0.511270
## sqrt(incidencerate):f.stateMissouri	0.846085
## sqrt(incidencerate):f.stateMontana	0.893318
## sqrt(incidencerate):f.stateNebraska	0.573857
## sqrt(incidencerate):f.stateNevada	0.504696
## sqrt(incidencerate):f.stateNew Hampshire	0.815127
## sqrt(incidencerate):f.stateNew Jersey	0.825532
## sqrt(incidencerate):f.stateNew Mexico	0.504192
## sqrt(incidencerate):f.stateNew York	0.564741
## sqrt(incidencerate):f.stateNorth Carolina	0.894210
## sqrt(incidencerate):f.stateNorth Dakota	0.297765
## sqrt(incidencerate):f.stateOhio	0.491510
## sqrt(incidencerate):f.stateOklahoma	0.285702
## sqrt(incidencerate):f.stateOregon	0.657940
## sqrt(incidencerate):f.statePennsylvania	0.874714

## sqrt(incidencerate):f.stateRhode Island	0.254010
## sqrt(incidencerate):f.stateSouth Carolina	0.693920
## sqrt(incidencerate):f.stateSouth Dakota	0.798909
## sqrt(incidencerate):f.stateTennessee	0.800209
## sqrt(incidencerate):f.stateTexas	0.339316
## sqrt(incidencerate):f.stateUtah	0.958711
## sqrt(incidencerate):f.stateVermont	0.934683
## sqrt(incidencerate):f.stateVirginia	0.536738
## sqrt(incidencerate):f.stateWashington	0.654055
## sqrt(incidencerate):f.stateWest Virginia	0.946902
## sqrt(incidencerate):f.stateWisconsin	0.504044
## sqrt(incidencerate):f.stateWyoming	0.156235
## pctbachdeg25_over:f.stateAlaska	0.031777 *
## pctbachdeg25_over:f.stateArizona	0.163905
## pctbachdeg25_over:f.stateArkansas	0.964174
## pctbachdeg25_over:f.stateCalifornia	0.284926
## pctbachdeg25_over:f.stateColorado	0.531307
## pctbachdeg25_over:f.stateConnecticut	0.805872
## pctbachdeg25_over:f.stateDelaware	NA
## pctbachdeg25_over:f.stateFlorida	0.048548 *
## pctbachdeg25_over:f.stateGeorgia	0.864727
## pctbachdeg25_over:f.stateHawaii	NA
## pctbachdeg25_over:f.stateIdaho	0.374804
## pctbachdeg25_over:f.stateIllinois	0.973370
## pctbachdeg25_over:f.stateIndiana	0.452960
## pctbachdeg25_over:f.stateIowa	0.084937 .
## pctbachdeg25_over:f.stateKansas	0.737843
## pctbachdeg25_over:f.stateKentucky	0.765868
## pctbachdeg25_over:f.stateLouisiana	0.463624
## pctbachdeg25_over:f.stateMaine	0.563094
## pctbachdeg25_over:f.stateMaryland	0.762073
## pctbachdeg25_over:f.stateMassachusetts	0.856533
## pctbachdeg25_over:f.stateMichigan	0.624413
## pctbachdeg25_over:f.stateMinnesota	0.451617
## pctbachdeg25_over:f.stateMississippi	0.103034
## pctbachdeg25_over:f.stateMissouri	0.882643
## pctbachdeg25_over:f.stateMontana	0.251714
## pctbachdeg25_over:f.stateNebraska	0.561376
## pctbachdeg25_over:f.stateNevada	0.006912 **
## pctbachdeg25_over:f.stateNew Hampshire	0.827467
## pctbachdeg25_over:f.stateNew Jersey	0.885032
## pctbachdeg25_over:f.stateNew Mexico	0.433379
## pctbachdeg25_over:f.stateNew York	0.776233
## pctbachdeg25_over:f.stateNorth Carolina	0.984928
## pctbachdeg25_over:f.stateNorth Dakota	0.999975
## pctbachdeg25_over:f.stateOhio	0.591855
## pctbachdeg25_over:f.stateOklahoma	0.208900
## pctbachdeg25_over:f.stateOregon	0.440691
## pctbachdeg25_over:f.statePennsylvania	0.586963
## pctbachdeg25_over:f.stateRhode Island	NA
## pctbachdeg25_over:f.stateSouth Carolina	0.628030
## pctbachdeg25_over:f.stateSouth Dakota	0.159574
## pctbachdeg25_over:f.stateTennessee	0.533156
## pctbachdeg25_over:f.stateTexas	0.723285

## pctbachdeg25_over:f.stateUtah	0.782071
## pctbachdeg25_over:f.stateVermont	0.756407
## pctbachdeg25_over:f.stateVirginia	0.698092
## pctbachdeg25_over:f.stateWashington	0.318042
## pctbachdeg25_over:f.stateWest Virginia	0.031159 *
## pctbachdeg25_over:f.stateWisconsin	0.339556
## pctbachdeg25_over:f.stateWyoming	0.772935
## log(medincome):f.stateAlaska	0.763233
## log(medincome):f.stateArizona	0.111803
## log(medincome):f.stateArkansas	0.959346
## log(medincome):f.stateCalifornia	0.892930
## log(medincome):f.stateColorado	0.340495
## log(medincome):f.stateConnecticut	0.701042
## log(medincome):f.stateDelaware	NA
## log(medincome):f.stateFlorida	0.501060
## log(medincome):f.stateGeorgia	0.493743
## log(medincome):f.stateHawaii	NA
## log(medincome):f.stateIdaho	0.611626
## log(medincome):f.stateIllinois	0.722571
## log(medincome):f.stateIndiana	0.705618
## log(medincome):f.stateIowa	0.647779
## log(medincome):f.stateKansas	0.892655
## log(medincome):f.stateKentucky	0.807399
## log(medincome):f.stateLouisiana	0.745631
## log(medincome):f.stateMaine	0.646783
## log(medincome):f.stateMaryland	0.276142
## log(medincome):f.stateMassachusetts	0.961147
## log(medincome):f.stateMichigan	0.984686
## log(medincome):f.stateMinnesota	0.280575
## log(medincome):f.stateMississippi	0.195060
## log(medincome):f.stateMissouri	0.689561
## log(medincome):f.stateMontana	0.664468
## log(medincome):f.stateNebraska	0.382415
## log(medincome):f.stateNevada	0.000441 ***
## log(medincome):f.stateNew Hampshire	0.828419
## log(medincome):f.stateNew Jersey	0.980942
## log(medincome):f.stateNew Mexico	0.453950
## log(medincome):f.stateNew York	0.719336
## log(medincome):f.stateNorth Carolina	0.871015
## log(medincome):f.stateNorth Dakota	0.683022
## log(medincome):f.stateOhio	0.909936
## log(medincome):f.stateOklahoma	0.587968
## log(medincome):f.stateOregon	0.534263
## log(medincome):f.statePennsylvania	0.512096
## log(medincome):f.stateRhode Island	NA
## log(medincome):f.stateSouth Carolina	0.530940
## log(medincome):f.stateSouth Dakota	0.725360
## log(medincome):f.stateTennessee	0.619990
## log(medincome):f.stateTexas	0.619364
## log(medincome):f.stateUtah	0.391730
## log(medincome):f.stateVermont	0.710405
## log(medincome):f.stateVirginia	0.350881
## log(medincome):f.stateWashington	0.770222
## log(medincome):f.stateWest Virginia	0.185676

```

## log(medincome):f.stateWisconsin 0.892284
## log(medincome):f.stateWyoming 0.609442
## log(medincome):f.education25_overBalanced Eduaction 0.001419 **
## log(medincome):f.education25_overMostly Bachelor's 0.761366
## f.stateAlabama:log(pctemployed16_over) 0.583807
## f.stateAlaska:log(pctemployed16_over) 0.001318 **
## f.stateArizona:log(pctemployed16_over) 0.191801
## f.stateArkansas:log(pctemployed16_over) 0.083625 .
## f.stateCalifornia:log(pctemployed16_over) 0.285433
## f.stateColorado:log(pctemployed16_over) 0.068033 .
## f.stateConnecticut:log(pctemployed16_over) 0.940872
## f.stateDelaware:log(pctemployed16_over) NA
## f.stateFlorida:log(pctemployed16_over) 0.428250
## f.stateGeorgia:log(pctemployed16_over) 0.175990
## f.stateHawaii:log(pctemployed16_over) NA
## f.stateIdaho:log(pctemployed16_over) 0.675406
## f.stateIllinois:log(pctemployed16_over) 0.858250
## f.stateIndiana:log(pctemployed16_over) 0.324222
## f.stateIowa:log(pctemployed16_over) 0.451290
## f.stateKansas:log(pctemployed16_over) 0.020821 *
## f.stateKentucky:log(pctemployed16_over) 0.044435 *
## f.stateLouisiana:log(pctemployed16_over) 0.610702
## f.stateMaine:log(pctemployed16_over) 0.647830
## f.stateMaryland:log(pctemployed16_over) 0.950578
## f.stateMassachusetts:log(pctemployed16_over) 0.697963
## f.stateMichigan:log(pctemployed16_over) 0.253173
## f.stateMinnesota:log(pctemployed16_over) 0.503388
## f.stateMississippi:log(pctemployed16_over) 0.145012
## f.stateMissouri:log(pctemployed16_over) 0.002954 **
## f.stateMontana:log(pctemployed16_over) 0.901886
## f.stateNebraska:log(pctemployed16_over) 0.010774 *
## f.stateNevada:log(pctemployed16_over) 0.050405 .
## f.stateNew Hampshire:log(pctemployed16_over) 0.805492
## f.stateNew Jersey:log(pctemployed16_over) 0.751397
## f.stateNew Mexico:log(pctemployed16_over) 0.248780
## f.stateNew York:log(pctemployed16_over) 0.981426
## f.stateNorth Carolina:log(pctemployed16_over) 0.481344
## f.stateNorth Dakota:log(pctemployed16_over) 0.747304
## f.stateOhio:log(pctemployed16_over) 0.861023
## f.stateOklahoma:log(pctemployed16_over) 0.008224 **
## f.stateOregon:log(pctemployed16_over) 0.718136
## f.statePennsylvania:log(pctemployed16_over) 0.689737
## f.stateRhode Island:log(pctemployed16_over) NA
## f.stateSouth Carolina:log(pctemployed16_over) 0.070911 .
## f.stateSouth Dakota:log(pctemployed16_over) 0.905187
## f.stateTennessee:log(pctemployed16_over) 0.293022
## f.stateTexas:log(pctemployed16_over) 0.222589
## f.stateUtah:log(pctemployed16_over) 0.556824
## f.stateVermont:log(pctemployed16_over) 0.806279
## f.stateVirginia:log(pctemployed16_over) 0.657170
## f.stateWashington:log(pctemployed16_over) 0.622742
## f.stateWest Virginia:log(pctemployed16_over) 0.000497 ***
## f.stateWisconsin:log(pctemployed16_over) 0.129965
## f.stateWyoming:log(pctemployed16_over) 0.636008

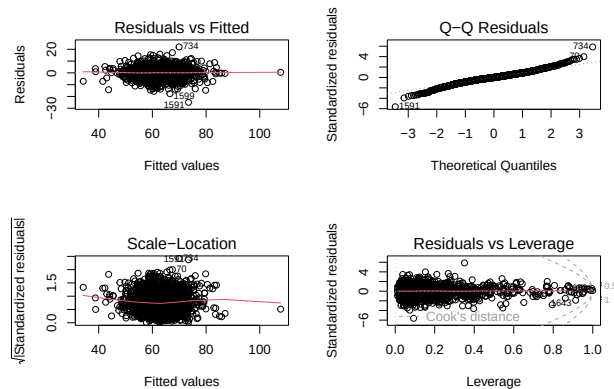
```

```
## f.education25_overBalanced Eduaction:log(pctemployed16_over) 0.002339 **
## f.education25_overMostly Bachelor's:log(pctemployed16_over) 0.711425
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 4.663 on 1535 degrees of freedom
## Multiple R-squared:  0.6791, Adjusted R-squared:  0.6175
## F-statistic: 11.01 on 295 and 1535 DF,  p-value: < 2.2e-16
```

```
par(mfrow=c(2,2))
plot(m5)
```

```
## Warning: not plotting observations with leverage one:
## 237, 429, 503, 711, 760, 1020, 1050, 1168, 1481, 1543, 1600, 1806

## Warning in sqrt(crit * p * (1 - hh)/hh): NaNs produced
## Warning in sqrt(crit * p * (1 - hh)/hh): NaNs produced
```



```

## -0.42447 -0.03448 0.00129 0.04050 0.30078
##
## Coefficients: (12 not defined because of singularities)
##
## (Intercept) 3.817e+00
## pctpubliccoveragealone -1.766e-04
## sqrt(incidencerate) 3.130e-02
## pctbachdeg25_over -5.647e-03
## log(medincome) -6.148e-02
## f.stateAlaska -5.844e+00
## f.stateArizona -1.479e+01
## f.stateArkansas 7.249e-02
## f.stateCalifornia 1.579e-02
## f.stateColorado -4.850e+00
## f.stateConnecticut 5.652e+00
## f.stateDelaware 4.161e-01
## f.stateFlorida 2.351e+00
## f.stateGeorgia 1.320e+00
## f.stateHawaii 1.878e+00
## f.stateIdaho -7.660e-01
## f.stateIllinois -7.976e-01
## f.stateIndiana -1.614e-01
## f.stateIowa 4.229e-01
## f.stateKansas -6.410e+00
## f.stateKentucky 1.307e+00
## f.stateLouisiana 1.636e+00
## f.stateMaine -3.215e+00
## f.stateMaryland -6.620e+00
## f.stateMassachusetts -4.808e+00
## f.stateMichigan 1.986e-01
## f.stateMinnesota -7.818e+00
## f.stateMississippi -1.874e+00
## f.stateMissouri 1.796e-01
## f.stateMontana -1.436e+00
## f.stateNebraska 1.101e-01
## f.stateNevada 7.425e+00
## f.stateNew Hampshire -1.345e+02
## f.stateNew Jersey 2.996e+00
## f.stateNew Mexico 1.432e+00
## f.stateNew York 9.775e-01
## f.stateNorth Carolina -2.850e-01
## f.stateNorth Dakota -1.947e+00
## f.stateOhio -6.035e-01
## f.stateOklahoma 1.381e-01
## f.stateOregon -1.276e+00
## f.statePennsylvania -1.602e+00
## f.stateRhode Island 3.607e+01
## f.stateSouth Carolina 7.454e-01
## f.stateSouth Dakota 1.336e+00
## f.stateTennessee -5.222e-01
## f.stateTexas 1.308e+00
## f.stateUtah 2.243e+00
## f.stateVermont -6.897e+00
## f.stateVirginia -1.987e+00

```

## f.stateWashington	3.382e-01
## f.stateWest Virginia	-7.704e-01
## f.stateWisconsin	1.729e+00
## f.stateWyoming	-3.239e+00
## f.unemploymentModerate Unemployment	-2.342e-03
## f.unemploymentHigh Unemployment	1.275e-02
## pctpubliccoveragealone:f.stateAlaska	9.587e-03
## pctpubliccoveragealone:f.stateArizona	3.576e-02
## pctpubliccoveragealone:f.stateArkansas	8.104e-03
## pctpubliccoveragealone:f.stateCalifornia	5.964e-03
## pctpubliccoveragealone:f.stateColorado	1.359e-02
## pctpubliccoveragealone:f.stateConnecticut	-6.708e-03
## pctpubliccoveragealone:f.stateDelaware	NA
## pctpubliccoveragealone:f.stateFlorida	-1.730e-02
## pctpubliccoveragealone:f.stateGeorgia	-6.027e-03
## pctpubliccoveragealone:f.stateHawaii	-8.605e-02
## pctpubliccoveragealone:f.stateIdaho	-1.422e-02
## pctpubliccoveragealone:f.stateIllinois	2.276e-03
## pctpubliccoveragealone:f.stateIndiana	4.169e-03
## pctpubliccoveragealone:f.stateIowa	9.817e-03
## pctpubliccoveragealone:f.stateKansas	-5.776e-03
## pctpubliccoveragealone:f.stateKentucky	-1.153e-03
## pctpubliccoveragealone:f.stateLouisiana	5.826e-03
## pctpubliccoveragealone:f.stateMaine	3.119e-03
## pctpubliccoveragealone:f.stateMaryland	1.803e-02
## pctpubliccoveragealone:f.stateMassachusetts	7.694e-03
## pctpubliccoveragealone:f.stateMichigan	3.978e-03
## pctpubliccoveragealone:f.stateMinnesota	8.595e-03
## pctpubliccoveragealone:f.stateMississippi	1.050e-02
## pctpubliccoveragealone:f.stateMissouri	7.658e-03
## pctpubliccoveragealone:f.stateMontana	5.415e-03
## pctpubliccoveragealone:f.stateNebraska	4.362e-06
## pctpubliccoveragealone:f.stateNevada	-3.092e-02
## pctpubliccoveragealone:f.stateNew Hampshire	-5.923e-01
## pctpubliccoveragealone:f.stateNew Jersey	-9.125e-03
## pctpubliccoveragealone:f.stateNew Mexico	-6.318e-03
## pctpubliccoveragealone:f.stateNew York	-3.301e-03
## pctpubliccoveragealone:f.stateNorth Carolina	1.999e-03
## pctpubliccoveragealone:f.stateNorth Dakota	-2.064e-03
## pctpubliccoveragealone:f.stateOhio	3.964e-03
## pctpubliccoveragealone:f.stateOklahoma	1.897e-02
## pctpubliccoveragealone:f.stateOregon	7.069e-03
## pctpubliccoveragealone:f.statePennsylvania	1.195e-02
## pctpubliccoveragealone:f.stateRhode Island	6.511e-01
## pctpubliccoveragealone:f.stateSouth Carolina	6.845e-03
## pctpubliccoveragealone:f.stateSouth Dakota	5.820e-03
## pctpubliccoveragealone:f.stateTennessee	8.787e-03
## pctpubliccoveragealone:f.stateTexas	-6.762e-03
## pctpubliccoveragealone:f.stateUtah	-3.070e-03
## pctpubliccoveragealone:f.stateVermont	2.507e-02
## pctpubliccoveragealone:f.stateVirginia	1.275e-02
## pctpubliccoveragealone:f.stateWashington	-4.757e-03
## pctpubliccoveragealone:f.stateWest Virginia	7.045e-03
## pctpubliccoveragealone:f.stateWisconsin	2.111e-03

## pctpubliccoveragealone:f.stateWyoming	-2.965e-03
## pctpubliccoveragealone:f.education25_overBalanced Eduaction	-2.717e-03
## pctpubliccoveragealone:f.education25_overMostly Bachelor's	3.983e-03
## sqrt(incidencerate):f.stateAlaska	1.998e-02
## sqrt(incidencerate):f.stateArizona	-7.745e-02
## sqrt(incidencerate):f.stateArkansas	-1.736e-02
## sqrt(incidencerate):f.stateCalifornia	1.645e-02
## sqrt(incidencerate):f.stateColorado	2.300e-02
## sqrt(incidencerate):f.stateConnecticut	1.823e-02
## sqrt(incidencerate):f.stateDelaware	NA
## sqrt(incidencerate):f.stateFlorida	-3.581e-04
## sqrt(incidencerate):f.stateGeorgia	1.969e-02
## sqrt(incidencerate):f.stateHawaii	NA
## sqrt(incidencerate):f.stateIdaho	3.324e-02
## sqrt(incidencerate):f.stateIllinois	1.677e-02
## sqrt(incidencerate):f.stateIndiana	1.737e-02
## sqrt(incidencerate):f.stateIowa	1.692e-02
## sqrt(incidencerate):f.stateKansas	3.906e-01
## sqrt(incidencerate):f.stateKentucky	1.410e-02
## sqrt(incidencerate):f.stateLouisiana	-1.184e-02
## sqrt(incidencerate):f.stateMaine	-1.152e-02
## sqrt(incidencerate):f.stateMaryland	8.360e-02
## sqrt(incidencerate):f.stateMassachusetts	-2.771e-02
## sqrt(incidencerate):f.stateMichigan	-1.125e-02
## sqrt(incidencerate):f.stateMinnesota	2.714e-01
## sqrt(incidencerate):f.stateMississippi	-1.294e-02
## sqrt(incidencerate):f.stateMissouri	-2.806e-03
## sqrt(incidencerate):f.stateMontana	7.647e-03
## sqrt(incidencerate):f.stateNebraska	-6.482e-03
## sqrt(incidencerate):f.stateNevada	2.689e-01
## sqrt(incidencerate):f.stateNew Hampshire	5.714e-01
## sqrt(incidencerate):f.stateNew Jersey	-1.809e-02
## sqrt(incidencerate):f.stateNew Mexico	2.028e-02
## sqrt(incidencerate):f.stateNew York	2.221e-02
## sqrt(incidencerate):f.stateNorth Carolina	3.664e-04
## sqrt(incidencerate):f.stateNorth Dakota	2.972e-02
## sqrt(incidencerate):f.stateOhio	2.436e-02
## sqrt(incidencerate):f.stateOklahoma	-2.596e-02
## sqrt(incidencerate):f.stateOregon	-8.129e-03
## sqrt(incidencerate):f.statePennsylvania	7.886e-03
## sqrt(incidencerate):f.stateRhode Island	-1.977e+00
## sqrt(incidencerate):f.stateSouth Carolina	1.108e-02
## sqrt(incidencerate):f.stateSouth Dakota	-2.287e-03
## sqrt(incidencerate):f.stateTennessee	5.832e-03
## sqrt(incidencerate):f.stateTexas	2.926e-02
## sqrt(incidencerate):f.stateUtah	9.395e-03
## sqrt(incidencerate):f.stateVermont	7.924e-03
## sqrt(incidencerate):f.stateVirginia	-9.245e-03
## sqrt(incidencerate):f.stateWashington	-6.860e-03
## sqrt(incidencerate):f.stateWest Virginia	2.521e-04
## sqrt(incidencerate):f.stateWisconsin	-1.281e-02
## sqrt(incidencerate):f.stateWyoming	8.242e-02
## pctbachdeg25_over:f.stateAlaska	-2.298e-02
## pctbachdeg25_over:f.stateArizona	1.917e-02

## pctbachdeg25_over:f.stateArkansas	4.879e-04
## pctbachdeg25_over:f.stateCalifornia	7.168e-03
## pctbachdeg25_over:f.stateColorado	-7.723e-03
## pctbachdeg25_over:f.stateConnecticut	7.993e-03
## pctbachdeg25_over:f.stateDelaware	NA
## pctbachdeg25_over:f.stateFlorida	-1.235e-02
## pctbachdeg25_over:f.stateGeorgia	-8.793e-04
## pctbachdeg25_over:f.stateHawaii	NA
## pctbachdeg25_over:f.stateIdaho	-7.502e-03
## pctbachdeg25_over:f.stateIllinois	3.071e-05
## pctbachdeg25_over:f.stateIndiana	3.845e-03
## pctbachdeg25_over:f.stateIowa	8.956e-03
## pctbachdeg25_over:f.stateKansas	1.639e-03
## pctbachdeg25_over:f.stateKentucky	1.610e-03
## pctbachdeg25_over:f.stateLouisiana	5.207e-03
## pctbachdeg25_over:f.stateMaine	-9.806e-03
## pctbachdeg25_over:f.stateMaryland	-4.512e-03
## pctbachdeg25_over:f.stateMassachusetts	-4.917e-03
## pctbachdeg25_over:f.stateMichigan	-3.578e-03
## pctbachdeg25_over:f.stateMinnesota	4.213e-03
## pctbachdeg25_over:f.stateMississippi	7.002e-03
## pctbachdeg25_over:f.stateMissouri	7.614e-04
## pctbachdeg25_over:f.stateMontana	6.386e-03
## pctbachdeg25_over:f.stateNebraska	3.326e-03
## pctbachdeg25_over:f.stateNevada	-2.195e-02
## pctbachdeg25_over:f.stateNew Hampshire	-1.185e+00
## pctbachdeg25_over:f.stateNew Jersey	-2.175e-03
## pctbachdeg25_over:f.stateNew Mexico	-8.614e-03
## pctbachdeg25_over:f.stateNew York	1.511e-03
## pctbachdeg25_over:f.stateNorth Carolina	2.219e-04
## pctbachdeg25_over:f.stateNorth Dakota	-1.035e-03
## pctbachdeg25_over:f.stateOhio	3.065e-03
## pctbachdeg25_over:f.stateOklahoma	9.461e-03
## pctbachdeg25_over:f.stateOregon	4.835e-03
## pctbachdeg25_over:f.statePennsylvania	2.688e-03
## pctbachdeg25_over:f.stateRhode Island	NA
## pctbachdeg25_over:f.stateSouth Carolina	3.357e-03
## pctbachdeg25_over:f.stateSouth Dakota	8.591e-03
## pctbachdeg25_over:f.stateTennessee	3.644e-03
## pctbachdeg25_over:f.stateTexas	8.914e-04
## pctbachdeg25_over:f.stateUtah	1.134e-03
## pctbachdeg25_over:f.stateVermont	7.424e-03
## pctbachdeg25_over:f.stateVirginia	-2.155e-03
## pctbachdeg25_over:f.stateWashington	-8.808e-03
## pctbachdeg25_over:f.stateWest Virginia	1.700e-02
## pctbachdeg25_over:f.stateWisconsin	5.508e-03
## pctbachdeg25_over:f.stateWyoming	5.656e-03
## log(medincome):f.stateAlaska	1.066e-01
## log(medincome):f.stateArizona	1.722e+00
## log(medincome):f.stateArkansas	-1.918e-02
## log(medincome):f.stateCalifornia	2.823e-02
## log(medincome):f.stateColorado	3.533e-01
## log(medincome):f.stateConnecticut	-4.621e-01
## log(medincome):f.stateDelaware	NA

## log(medincome):f.stateFlorida	-1.713e-01
## log(medincome):f.stateGeorgia	-1.563e-01
## log(medincome):f.stateHawaii	NA
## log(medincome):f.stateIdaho	8.674e-02
## log(medincome):f.stateIllinois	6.734e-02
## log(medincome):f.stateIndiana	7.375e-02
## log(medincome):f.stateIowa	-1.459e-01
## log(medincome):f.stateKansas	2.607e-02
## log(medincome):f.stateKentucky	-4.907e-02
## log(medincome):f.stateLouisiana	-7.750e-02
## log(medincome):f.stateMaine	2.144e-01
## log(medincome):f.stateMaryland	4.327e-01
## log(medincome):f.stateMassachusetts	8.482e-02
## log(medincome):f.stateMichigan	-1.812e-02
## log(medincome):f.stateMinnesota	2.647e-01
## log(medincome):f.stateMississippi	2.537e-01
## log(medincome):f.stateMissouri	-9.762e-02
## log(medincome):f.stateMontana	1.150e-01
## log(medincome):f.stateNebraska	1.989e-01
## log(medincome):f.stateNevada	-1.277e+00
## log(medincome):f.stateNew Hampshire	1.330e+01
## log(medincome):f.stateNew Jersey	-2.058e-02
## log(medincome):f.stateNew Mexico	-2.149e-01
## log(medincome):f.stateNew York	-1.055e-01
## log(medincome):f.stateNorth Carolina	1.998e-02
## log(medincome):f.stateNorth Dakota	9.367e-02
## log(medincome):f.stateOhio	5.485e-03
## log(medincome):f.stateOklahoma	-1.067e-01
## log(medincome):f.stateOregon	1.784e-01
## log(medincome):f.statePennsylvania	1.694e-01
## log(medincome):f.stateRhode Island	NA
## log(medincome):f.stateSouth Carolina	-1.908e-01
## log(medincome):f.stateSouth Dakota	-1.047e-01
## log(medincome):f.stateTennessee	9.337e-02
## log(medincome):f.stateTexas	-1.031e-01
## log(medincome):f.stateUtah	-2.630e-01
## log(medincome):f.stateVermont	5.017e-01
## log(medincome):f.stateVirginia	2.053e-01
## log(medincome):f.stateWashington	7.066e-02
## log(medincome):f.stateWest Virginia	3.271e-01
## log(medincome):f.stateWisconsin	3.614e-02
## log(medincome):f.stateWyoming	3.498e-01
## log(medincome):f.education25_overBalanced Eduaction	6.952e-02
## log(medincome):f.education25_overMostly Bachelor's	1.351e-02
## f.stateAlabama:log(pctemployed16_over)	1.071e-01
## f.stateAlaska:log(pctemployed16_over)	1.203e+00
## f.stateArizona:log(pctemployed16_over)	-7.650e-01
## f.stateArkansas:log(pctemployed16_over)	1.919e-01
## f.stateCalifornia:log(pctemployed16_over)	-1.481e-01
## f.stateColorado:log(pctemployed16_over)	1.878e-01
## f.stateConnecticut:log(pctemployed16_over)	-1.483e-01
## f.stateDelaware:log(pctemployed16_over)	NA
## f.stateFlorida:log(pctemployed16_over)	1.086e-01
## f.stateGeorgia:log(pctemployed16_over)	1.118e-01

## f.stateHawaii:log(pctemployed16_over)	NA	
## f.stateIdaho:log(pctemployed16_over)	-5.208e-02	
## f.stateIllinois:log(pctemployed16_over)	1.586e-02	
## f.stateIndiana:log(pctemployed16_over)	-1.697e-01	
## f.stateIowa:log(pctemployed16_over)	2.097e-01	
## f.stateKansas:log(pctemployed16_over)	-4.058e-01	
## f.stateKentucky:log(pctemployed16_over)	-1.650e-01	
## f.stateLouisiana:log(pctemployed16_over)	-7.509e-02	
## f.stateMaine:log(pctemployed16_over)	4.203e-01	
## f.stateMaryland:log(pctemployed16_over)	5.806e-02	
## f.stateMassachusetts:log(pctemployed16_over)	1.173e+00	
## f.stateMichigan:log(pctemployed16_over)	1.539e-01	
## f.stateMinnesota:log(pctemployed16_over)	-1.564e-01	
## f.stateMississippi:log(pctemployed16_over)	-1.029e-01	
## f.stateMissouri:log(pctemployed16_over)	3.039e-01	
## f.stateMontana:log(pctemployed16_over)	4.156e-02	
## f.stateNebraska:log(pctemployed16_over)	-4.415e-01	
## f.stateNevada:log(pctemployed16_over)	4.811e-01	
## f.stateNew Hampshire:log(pctemployed16_over)	2.035e+00	
## f.stateNew Jersey:log(pctemployed16_over)	-4.387e-01	
## f.stateNew Mexico:log(pctemployed16_over)	2.733e-01	
## f.stateNew York:log(pctemployed16_over)	1.573e-02	
## f.stateNorth Carolina:log(pctemployed16_over)	1.005e-01	
## f.stateNorth Dakota:log(pctemployed16_over)	1.707e-01	
## f.stateOhio:log(pctemployed16_over)	8.362e-02	
## f.stateOklahoma:log(pctemployed16_over)	3.798e-01	
## f.stateOregon:log(pctemployed16_over)	-7.753e-02	
## f.statePennsylvania:log(pctemployed16_over)	-6.617e-02	
## f.stateRhode Island:log(pctemployed16_over)	NA	
## f.stateSouth Carolina:log(pctemployed16_over)	3.254e-01	
## f.stateSouth Dakota:log(pctemployed16_over)	-8.391e-03	
## f.stateTennessee:log(pctemployed16_over)	-9.669e-02	
## f.stateTexas:log(pctemployed16_over)	-6.809e-02	
## f.stateUtah:log(pctemployed16_over)	1.778e-01	
## f.stateVermont:log(pctemployed16_over)	2.381e-01	
## f.stateVirginia:log(pctemployed16_over)	5.219e-02	
## f.stateWashington:log(pctemployed16_over)	-8.911e-02	
## f.stateWest Virginia:log(pctemployed16_over)	-6.729e-01	
## f.stateWisconsin:log(pctemployed16_over)	-3.858e-01	
## f.stateWyoming:log(pctemployed16_over)	-4.696e-01	
## f.education25_overBalanced Eduaction:log(pctemployed16_over)	-1.750e-01	
## f.education25_overMostly Bachelor's:log(pctemployed16_over)	-4.893e-02	
##	Std. Error	t value
## (Intercept)	2.681e+00	1.424
## pctpubliccoveragealone	9.416e-03	-0.019
## sqrt(incidencerate)	2.031e-02	1.541
## pctbachdeg25_over	4.673e-03	-1.208
## log(medincome)	2.124e-01	-0.289
## f.stateAlaska	3.891e+00	-1.502
## f.stateArizona	8.606e+00	-1.719
## f.stateArkansas	3.190e+00	0.023
## f.stateCalifornia	3.013e+00	0.005
## f.stateColorado	3.097e+00	-1.566
## f.stateConnecticut	1.642e+01	0.344

## f.stateDelaware	7.839e-01	0.531
## f.stateFlorida	2.997e+00	0.785
## f.stateGeorgia	2.767e+00	0.477
## f.stateHawaii	2.701e+00	0.695
## f.stateIdaho	3.465e+00	-0.221
## f.stateIllinois	2.954e+00	-0.270
## f.stateIndiana	3.013e+00	-0.054
## f.stateIowa	3.082e+00	0.137
## f.stateKansas	3.888e+00	-1.649
## f.stateKentucky	2.911e+00	0.449
## f.stateLouisiana	2.906e+00	0.563
## f.stateMaine	6.199e+00	-0.519
## f.stateMaryland	4.771e+00	-1.388
## f.stateMassachusetts	7.344e+00	-0.655
## f.stateMichigan	2.959e+00	0.067
## f.stateMinnesota	5.130e+00	-1.524
## f.stateMississippi	2.883e+00	-0.650
## f.stateMissouri	2.938e+00	0.061
## f.stateMontana	3.038e+00	-0.473
## f.stateNebraska	2.956e+00	0.037
## f.stateNevada	8.406e+00	0.883
## f.stateNew Hampshire	6.167e+02	-0.218
## f.stateNew Jersey	1.018e+01	0.294
## f.stateNew Mexico	2.972e+00	0.482
## f.stateNew York	3.044e+00	0.321
## f.stateNorth Carolina	2.955e+00	-0.096
## f.stateNorth Dakota	3.072e+00	-0.634
## f.stateOhio	3.067e+00	-0.197
## f.stateOklahoma	3.024e+00	0.046
## f.stateOregon	3.150e+00	-0.405
## f.statePennsylvania	3.037e+00	-0.527
## f.stateRhode Island	2.993e+01	1.205
## f.stateSouth Carolina	3.738e+00	0.199
## f.stateSouth Dakota	2.964e+00	0.451
## f.stateTennessee	2.967e+00	-0.176
## f.stateTexas	2.755e+00	0.475
## f.stateUtah	3.486e+00	0.643
## f.stateVermont	1.357e+01	-0.508
## f.stateVirginia	2.786e+00	-0.713
## f.stateWashington	3.091e+00	0.109
## f.stateWest Virginia	3.049e+00	-0.253
## f.stateWisconsin	3.565e+00	0.485
## f.stateWyoming	6.795e+00	-0.477
## f.unemploymentModerate Unemployment	7.205e-03	-0.325
## f.unemploymentHigh Unemployment	9.296e-03	1.372
## pctpubliccoveragealone:f.stateAlaska	1.125e-02	0.852
## pctpubliccoveragealone:f.stateArizona	2.579e-02	1.386
## pctpubliccoveragealone:f.stateArkansas	1.068e-02	0.759
## pctpubliccoveragealone:f.stateCalifornia	1.037e-02	0.575
## pctpubliccoveragealone:f.stateColorado	1.104e-02	1.231
## pctpubliccoveragealone:f.stateConnecticut	2.651e-02	-0.253
## pctpubliccoveragealone:f.stateDelaware	NA	NA
## pctpubliccoveragealone:f.stateFlorida	1.067e-02	-1.621
## pctpubliccoveragealone:f.stateGeorgia	9.769e-03	-0.617

## pctpubliccoveragealone:f.stateHawaii	1.400e-01	-0.615
## pctpubliccoveragealone:f.stateIdaho	1.119e-02	-1.270
## pctpubliccoveragealone:f.stateIllinois	1.015e-02	0.224
## pctpubliccoveragealone:f.stateIndiana	1.049e-02	0.397
## pctpubliccoveragealone:f.stateIowa	1.084e-02	0.906
## pctpubliccoveragealone:f.stateKansas	1.053e-02	-0.549
## pctpubliccoveragealone:f.stateKentucky	1.021e-02	-0.113
## pctpubliccoveragealone:f.stateLouisiana	1.057e-02	0.551
## pctpubliccoveragealone:f.stateMaine	2.471e-02	0.126
## pctpubliccoveragealone:f.stateMaryland	1.471e-02	1.226
## pctpubliccoveragealone:f.stateMassachusetts	3.114e-02	0.247
## pctpubliccoveragealone:f.stateMichigan	1.057e-02	0.376
## pctpubliccoveragealone:f.stateMinnesota	1.093e-02	0.786
## pctpubliccoveragealone:f.stateMississippi	9.933e-03	1.057
## pctpubliccoveragealone:f.stateMissouri	1.032e-02	0.742
## pctpubliccoveragealone:f.stateMontana	1.059e-02	0.511
## pctpubliccoveragealone:f.stateNebraska	1.028e-02	0.000
## pctpubliccoveragealone:f.stateNevada	1.498e-02	-2.065
## pctpubliccoveragealone:f.stateNew Hampshire	2.748e+00	-0.216
## pctpubliccoveragealone:f.stateNew Jersey	2.735e-02	-0.334
## pctpubliccoveragealone:f.stateNew Mexico	9.999e-03	-0.632
## pctpubliccoveragealone:f.stateNew York	1.021e-02	-0.323
## pctpubliccoveragealone:f.stateNorth Carolina	1.044e-02	0.191
## pctpubliccoveragealone:f.stateNorth Dakota	1.017e-02	-0.203
## pctpubliccoveragealone:f.stateOhio	1.066e-02	0.372
## pctpubliccoveragealone:f.stateOklahoma	1.117e-02	1.699
## pctpubliccoveragealone:f.stateOregon	1.074e-02	0.658
## pctpubliccoveragealone:f.statePennsylvania	1.061e-02	1.126
## pctpubliccoveragealone:f.stateRhode Island	5.021e-01	1.297
## pctpubliccoveragealone:f.stateSouth Carolina	1.298e-02	0.527
## pctpubliccoveragealone:f.stateSouth Dakota	9.889e-03	0.588
## pctpubliccoveragealone:f.stateTennessee	1.042e-02	0.843
## pctpubliccoveragealone:f.stateTexas	9.690e-03	-0.698
## pctpubliccoveragealone:f.stateUtah	1.353e-02	-0.227
## pctpubliccoveragealone:f.stateVermont	4.992e-02	0.502
## pctpubliccoveragealone:f.stateVirginia	9.948e-03	1.282
## pctpubliccoveragealone:f.stateWashington	1.151e-02	-0.413
## pctpubliccoveragealone:f.stateWest Virginia	1.024e-02	0.688
## pctpubliccoveragealone:f.stateWisconsin	1.237e-02	0.171
## pctpubliccoveragealone:f.stateWyoming	2.633e-02	-0.113
## pctpubliccoveragealone:f.education25_overBalanced Eduaction	1.136e-03	-2.392
## pctpubliccoveragealone:f.education25_overMostly Bachelor's	2.749e-03	1.449
## sqrt(incidencerate):f.stateAlaska	3.000e-02	0.666
## sqrt(incidencerate):f.stateArizona	6.482e-02	-1.195
## sqrt(incidencerate):f.stateArkansas	2.468e-02	-0.703
## sqrt(incidencerate):f.stateCalifornia	3.055e-02	0.538
## sqrt(incidencerate):f.stateColorado	2.350e-02	0.978
## sqrt(incidencerate):f.stateConnecticut	2.505e-01	0.073
## sqrt(incidencerate):f.stateDelaware	NA	NA
## sqrt(incidencerate):f.stateFlorida	2.102e-02	-0.017
## sqrt(incidencerate):f.stateGeorgia	2.166e-02	0.909
## sqrt(incidencerate):f.stateHawaii	NA	NA
## sqrt(incidencerate):f.stateIdaho	2.463e-02	1.349
## sqrt(incidencerate):f.stateIllinois	2.481e-02	0.676

## sqrt(incidencerate):f.stateIndiana	2.578e-02	0.674
## sqrt(incidencerate):f.stateIowa	2.478e-02	0.683
## sqrt(incidencerate):f.stateKansas	1.335e-01	2.925
## sqrt(incidencerate):f.stateKentucky	2.245e-02	0.628
## sqrt(incidencerate):f.stateLouisiana	2.868e-02	-0.413
## sqrt(incidencerate):f.stateMaine	6.277e-02	-0.184
## sqrt(incidencerate):f.stateMaryland	4.135e-02	2.022
## sqrt(incidencerate):f.stateMassachusetts	1.039e-01	-0.267
## sqrt(incidencerate):f.stateMichigan	2.166e-02	-0.519
## sqrt(incidencerate):f.stateMinnesota	2.064e-01	1.315
## sqrt(incidencerate):f.stateMississippi	2.347e-02	-0.552
## sqrt(incidencerate):f.stateMissouri	2.139e-02	-0.131
## sqrt(incidencerate):f.stateMontana	2.428e-02	0.315
## sqrt(incidencerate):f.stateNebraska	2.193e-02	-0.296
## sqrt(incidencerate):f.stateNevada	3.175e-01	0.847
## sqrt(incidencerate):f.stateNew Hampshire	2.435e+00	0.235
## sqrt(incidencerate):f.stateNew Jersey	1.005e-01	-0.180
## sqrt(incidencerate):f.stateNew Mexico	2.477e-02	0.818
## sqrt(incidencerate):f.stateNew York	3.158e-02	0.703
## sqrt(incidencerate):f.stateNorth Carolina	2.506e-02	0.015
## sqrt(incidencerate):f.stateNorth Dakota	2.225e-02	1.336
## sqrt(incidencerate):f.stateOhio	2.714e-02	0.897
## sqrt(incidencerate):f.stateOklahoma	2.527e-02	-1.027
## sqrt(incidencerate):f.stateOregon	2.914e-02	-0.279
## sqrt(incidencerate):f.statePennsylvania	2.720e-02	0.290
## sqrt(incidencerate):f.stateRhode Island	1.630e+00	-1.213
## sqrt(incidencerate):f.stateSouth Carolina	2.890e-02	0.383
## sqrt(incidencerate):f.stateSouth Dakota	2.327e-02	-0.098
## sqrt(incidencerate):f.stateTennessee	2.429e-02	0.240
## sqrt(incidencerate):f.stateTexas	2.093e-02	1.398
## sqrt(incidencerate):f.stateUtah	3.178e-02	0.296
## sqrt(incidencerate):f.stateVermont	8.446e-02	0.094
## sqrt(incidencerate):f.stateVirginia	2.087e-02	-0.443
## sqrt(incidencerate):f.stateWashington	2.651e-02	-0.259
## sqrt(incidencerate):f.stateWest Virginia	2.220e-02	0.011
## sqrt(incidencerate):f.stateWisconsin	2.371e-02	-0.540
## sqrt(incidencerate):f.stateWyoming	5.486e-02	1.502
## pctbachdeg25_over:f.stateAlaska	1.003e-02	-2.291
## pctbachdeg25_over:f.stateArizona	1.224e-02	1.567
## pctbachdeg25_over:f.stateArkansas	7.475e-03	0.065
## pctbachdeg25_over:f.stateCalifornia	7.386e-03	0.971
## pctbachdeg25_over:f.stateColorado	5.292e-03	-1.459
## pctbachdeg25_over:f.stateConnecticut	3.367e-02	0.237
## pctbachdeg25_over:f.stateDelaware	NA	NA
## pctbachdeg25_over:f.stateFlorida	6.419e-03	-1.925
## pctbachdeg25_over:f.stateGeorgia	5.216e-03	-0.169
## pctbachdeg25_over:f.stateHawaii	NA	NA
## pctbachdeg25_over:f.stateIdaho	6.202e-03	-1.210
## pctbachdeg25_over:f.stateIllinois	5.565e-03	0.006
## pctbachdeg25_over:f.stateIndiana	5.378e-03	0.715
## pctbachdeg25_over:f.stateIowa	5.395e-03	1.660
## pctbachdeg25_over:f.stateKansas	5.518e-03	0.297
## pctbachdeg25_over:f.stateKentucky	5.455e-03	0.295
## pctbachdeg25_over:f.stateLouisiana	6.661e-03	0.782

## pctbachdeg25_over:f.stateMaine	1.764e-02	-0.556
## pctbachdeg25_over:f.stateMaryland	9.425e-03	-0.479
## pctbachdeg25_over:f.stateMassachusetts	2.769e-02	-0.178
## pctbachdeg25_over:f.stateMichigan	6.182e-03	-0.579
## pctbachdeg25_over:f.stateMinnesota	5.702e-03	0.739
## pctbachdeg25_over:f.stateMississippi	5.500e-03	1.273
## pctbachdeg25_over:f.stateMissouri	6.030e-03	0.126
## pctbachdeg25_over:f.stateMontana	5.957e-03	1.072
## pctbachdeg25_over:f.stateNebraska	5.772e-03	0.576
## pctbachdeg25_over:f.stateNevada	8.837e-03	-2.484
## pctbachdeg25_over:f.stateNew Hampshire	5.462e+00	-0.217
## pctbachdeg25_over:f.stateNew Jersey	1.809e-02	-0.120
## pctbachdeg25_over:f.stateNew Mexico	9.264e-03	-0.930
## pctbachdeg25_over:f.stateNew York	6.914e-03	0.219
## pctbachdeg25_over:f.stateNorth Carolina	5.910e-03	0.038
## pctbachdeg25_over:f.stateNorth Dakota	8.549e-03	-0.121
## pctbachdeg25_over:f.stateOhio	5.860e-03	0.523
## pctbachdeg25_over:f.stateOklahoma	7.413e-03	1.276
## pctbachdeg25_over:f.stateOregon	6.622e-03	0.730
## pctbachdeg25_over:f.statePennsylvania	6.392e-03	0.421
## pctbachdeg25_over:f.stateRhode Island	NA	NA
## pctbachdeg25_over:f.stateSouth Carolina	6.770e-03	0.496
## pctbachdeg25_over:f.stateSouth Dakota	6.232e-03	1.378
## pctbachdeg25_over:f.stateTennessee	5.914e-03	0.616
## pctbachdeg25_over:f.stateTexas	5.093e-03	0.175
## pctbachdeg25_over:f.stateUtah	6.972e-03	0.163
## pctbachdeg25_over:f.stateVermont	2.640e-02	0.281
## pctbachdeg25_over:f.stateVirginia	5.294e-03	-0.407
## pctbachdeg25_over:f.stateWashington	7.036e-03	-1.252
## pctbachdeg25_over:f.stateWest Virginia	7.758e-03	2.191
## pctbachdeg25_over:f.stateWisconsin	5.878e-03	0.937
## pctbachdeg25_over:f.stateWyoming	2.930e-02	0.193
## log(medincome):f.stateAlaska	2.628e-01	0.406
## log(medincome):f.stateArizona	9.055e-01	1.902
## log(medincome):f.stateArkansas	2.733e-01	-0.070
## log(medincome):f.stateCalifornia	2.465e-01	0.115
## log(medincome):f.stateColorado	2.559e-01	1.381
## log(medincome):f.stateConnecticut	1.076e+00	-0.429
## log(medincome):f.stateDelaware	NA	NA
## log(medincome):f.stateFlorida	2.482e-01	-0.690
## log(medincome):f.stateGeorgia	2.240e-01	-0.698
## log(medincome):f.stateHawaii	NA	NA
## log(medincome):f.stateIdaho	3.256e-01	0.266
## log(medincome):f.stateIllinois	2.387e-01	0.282
## log(medincome):f.stateIndiana	2.477e-01	0.298
## log(medincome):f.stateIowa	2.682e-01	-0.544
## log(medincome):f.stateKansas	2.331e-01	0.112
## log(medincome):f.stateKentucky	2.303e-01	-0.213
## log(medincome):f.stateLouisiana	2.417e-01	-0.321
## log(medincome):f.stateMaine	5.125e-01	0.418
## log(medincome):f.stateMaryland	4.310e-01	1.004
## log(medincome):f.stateMassachusetts	9.191e-01	0.092
## log(medincome):f.stateMichigan	2.513e-01	-0.072
## log(medincome):f.stateMinnesota	2.399e-01	1.103

## log(medincome):f.stateMississippi	2.334e-01	1.087
## log(medincome):f.stateMissouri	2.395e-01	-0.408
## log(medincome):f.stateMontana	2.605e-01	0.441
## log(medincome):f.stateNebraska	2.383e-01	0.835
## log(medincome):f.stateNevada	3.782e-01	-3.377
## log(medincome):f.stateNew Hampshire	6.169e+01	0.216
## log(medincome):f.stateNew Jersey	4.892e-01	-0.042
## log(medincome):f.stateNew Mexico	2.461e-01	-0.873
## log(medincome):f.stateNew York	2.401e-01	-0.440
## log(medincome):f.stateNorth Carolina	2.461e-01	0.081
## log(medincome):f.stateNorth Dakota	2.423e-01	0.387
## log(medincome):f.stateOhio	2.473e-01	0.022
## log(medincome):f.stateOklahoma	2.451e-01	-0.435
## log(medincome):f.stateOregon	3.002e-01	0.594
## log(medincome):f.statePennsylvania	2.508e-01	0.676
## log(medincome):f.stateRhode Island	NA	NA
## log(medincome):f.stateSouth Carolina	3.116e-01	-0.612
## log(medincome):f.stateSouth Dakota	2.433e-01	-0.430
## log(medincome):f.stateTennessee	2.434e-01	0.384
## log(medincome):f.stateTexas	2.207e-01	-0.467
## log(medincome):f.stateUtah	2.752e-01	-0.956
## log(medincome):f.stateVermont	1.504e+00	0.334
## log(medincome):f.stateVirginia	2.232e-01	0.920
## log(medincome):f.stateWashington	2.785e-01	0.254
## log(medincome):f.stateWest Virginia	2.592e-01	1.262
## log(medincome):f.stateWisconsin	3.066e-01	0.118
## log(medincome):f.stateWyoming	8.368e-01	0.418
## log(medincome):f.education25_overBalanced Eduaction	2.081e-02	3.341
## log(medincome):f.education25_overMostly Bachelor's	4.041e-02	0.334
## f.stateAlabama:log(pctemployed16_over)	2.016e-01	0.532
## f.stateAlaska:log(pctemployed16_over)	4.357e-01	2.760
## f.stateArizona:log(pctemployed16_over)	4.767e-01	-1.605
## f.stateArkansas:log(pctemployed16_over)	1.274e-01	1.506
## f.stateCalifornia:log(pctemployed16_over)	1.340e-01	-1.105
## f.stateColorado:log(pctemployed16_over)	1.103e-01	1.703
## f.stateConnecticut:log(pctemployed16_over)	1.867e+00	-0.079
## f.stateDelaware:log(pctemployed16_over)	NA	NA
## f.stateFlorida:log(pctemployed16_over)	1.445e-01	0.752
## f.stateGeorgia:log(pctemployed16_over)	7.039e-02	1.589
## f.stateHawaii:log(pctemployed16_over)	NA	NA
## f.stateIdaho:log(pctemployed16_over)	2.196e-01	-0.237
## f.stateIllinois:log(pctemployed16_over)	1.031e-01	0.154
## f.stateIndiana:log(pctemployed16_over)	2.039e-01	-0.833
## f.stateIowa:log(pctemployed16_over)	2.583e-01	0.812
## f.stateKansas:log(pctemployed16_over)	1.817e-01	-2.233
## f.stateKentucky:log(pctemployed16_over)	1.147e-01	-1.438
## f.stateLouisiana:log(pctemployed16_over)	1.649e-01	-0.455
## f.stateMaine:log(pctemployed16_over)	9.302e-01	0.452
## f.stateMaryland:log(pctemployed16_over)	7.730e-01	0.075
## f.stateMassachusetts:log(pctemployed16_over)	3.270e+00	0.359
## f.stateMichigan:log(pctemployed16_over)	1.245e-01	1.236
## f.stateMinnesota:log(pctemployed16_over)	2.411e-01	-0.649
## f.stateMississippi:log(pctemployed16_over)	9.737e-02	-1.057
## f.stateMissouri:log(pctemployed16_over)	9.779e-02	3.108

## f.stateMontana:log(pctemployed16_over)	2.224e-01	0.187
## f.stateNebraska:log(pctemployed16_over)	1.734e-01	-2.546
## f.stateNevada:log(pctemployed16_over)	2.411e-01	1.996
## f.stateNew Hampshire:log(pctemployed16_over)	8.299e+00	0.245
## f.stateNew Jersey:log(pctemployed16_over)	1.452e+00	-0.302
## f.stateNew Mexico:log(pctemployed16_over)	1.934e-01	1.413
## f.stateNew York:log(pctemployed16_over)	2.323e-01	0.068
## f.stateNorth Carolina:log(pctemployed16_over)	1.327e-01	0.757
## f.stateNorth Dakota:log(pctemployed16_over)	4.348e-01	0.393
## f.stateOhio:log(pctemployed16_over)	2.538e-01	0.329
## f.stateOklahoma:log(pctemployed16_over)	1.597e-01	2.377
## f.stateOregon:log(pctemployed16_over)	2.477e-01	-0.313
## f.statePennsylvania:log(pctemployed16_over)	1.982e-01	-0.334
## f.stateRhode Island:log(pctemployed16_over)	NA	NA
## f.stateSouth Carolina:log(pctemployed16_over)	1.737e-01	1.874
## f.stateSouth Dakota:log(pctemployed16_over)	1.613e-01	-0.052
## f.stateTennessee:log(pctemployed16_over)	1.259e-01	-0.768
## f.stateTexas:log(pctemployed16_over)	5.025e-02	-1.355
## f.stateUtah:log(pctemployed16_over)	2.959e-01	0.601
## f.stateVermont:log(pctemployed16_over)	1.009e+00	0.236
## f.stateVirginia:log(pctemployed16_over)	1.091e-01	0.478
## f.stateWashington:log(pctemployed16_over)	1.951e-01	-0.457
## f.stateWest Virginia:log(pctemployed16_over)	1.967e-01	-3.421
## f.stateWisconsin:log(pctemployed16_over)	2.479e-01	-1.556
## f.stateWyoming:log(pctemployed16_over)	1.119e+00	-0.420
## f.education25_overBalanced Eduaction:log(pctemployed16_over)	5.482e-02	-3.193
## f.education25_overMostly Bachelor's:log(pctemployed16_over)	1.081e-01	-0.452
##	Pr(> t)	
## (Intercept)	0.154775	
## pctpubliccoveragealone	0.985040	
## sqrt(incidencerate)	0.123584	
## pctbachdeg25_over	0.227104	
## log(medincome)	0.772254	
## f.stateAlaska	0.133356	
## f.stateArizona	0.085842	.
## f.stateArkansas	0.981876	
## f.stateCalifornia	0.995817	
## f.stateColorado	0.117553	
## f.stateConnecticut	0.730743	
## f.stateDelaware	0.595609	
## f.stateFlorida	0.432864	
## f.stateGeorgia	0.633391	
## f.stateHawaii	0.487070	
## f.stateIdaho	0.825062	
## f.stateIllinois	0.787175	
## f.stateIndiana	0.957291	
## f.stateIowa	0.890869	
## f.stateKansas	0.099433	.
## f.stateKentucky	0.653460	
## f.stateLouisiana	0.573572	
## f.stateMaine	0.604027	
## f.stateMaryland	0.165464	
## f.stateMassachusetts	0.512774	
## f.stateMichigan	0.946501	

## f.stateMinnesota	0.127679
## f.stateMississippi	0.515889
## f.stateMissouri	0.951269
## f.stateMontana	0.636470
## f.stateNebraska	0.970301
## f.stateNevada	0.377194
## f.stateNew Hampshire	0.827418
## f.stateNew Jersey	0.768586
## f.stateNew Mexico	0.630006
## f.stateNew York	0.748170
## f.stateNorth Carolina	0.923168
## f.stateNorth Dakota	0.526338
## f.stateOhio	0.844046
## f.stateOklahoma	0.963593
## f.stateOregon	0.685559
## f.statePennsylvania	0.597934
## f.stateRhode Island	0.228279
## f.stateSouth Carolina	0.841970
## f.stateSouth Dakota	0.652268
## f.stateTennessee	0.860304
## f.stateTexas	0.634883
## f.stateUtah	0.520132
## f.stateVermont	0.611228
## f.stateVirginia	0.475807
## f.stateWashington	0.912885
## f.stateWest Virginia	0.800595
## f.stateWisconsin	0.627848
## f.stateWyoming	0.633641
## f.unemploymentModerate Unemployment	0.745192
## f.unemploymentHigh Unemployment	0.170282
## pctpubliccoveragealone:f.stateAlaska	0.394323
## pctpubliccoveragealone:f.stateArizona	0.165858
## pctpubliccoveragealone:f.stateArkansas	0.448125
## pctpubliccoveragealone:f.stateCalifornia	0.565117
## pctpubliccoveragealone:f.stateColorado	0.218414
## pctpubliccoveragealone:f.stateConnecticut	0.800249
## pctpubliccoveragealone:f.stateDelaware	NA
## pctpubliccoveragealone:f.stateFlorida	0.105181
## pctpubliccoveragealone:f.stateGeorgia	0.537364
## pctpubliccoveragealone:f.stateHawaii	0.538953
## pctpubliccoveragealone:f.stateIdaho	0.204259
## pctpubliccoveragealone:f.stateIllinois	0.822655
## pctpubliccoveragealone:f.stateIndiana	0.691162
## pctpubliccoveragealone:f.stateIowa	0.365261
## pctpubliccoveragealone:f.stateKansas	0.583394
## pctpubliccoveragealone:f.stateKentucky	0.910157
## pctpubliccoveragealone:f.stateLouisiana	0.581573
## pctpubliccoveragealone:f.stateMaine	0.899569
## pctpubliccoveragealone:f.stateMaryland	0.220479
## pctpubliccoveragealone:f.stateMassachusetts	0.804906
## pctpubliccoveragealone:f.stateMichigan	0.706750
## pctpubliccoveragealone:f.stateMinnesota	0.431960
## pctpubliccoveragealone:f.stateMississippi	0.290479
## pctpubliccoveragealone:f.stateMissouri	0.458222

## pctpubliccoveragealone:f.stateMontana	0.609206
## pctpubliccoveragealone:f.stateNebraska	0.999661
## pctpubliccoveragealone:f.stateNevada	0.039118 *
## pctpubliccoveragealone:f.stateNew Hampshire	0.829399
## pctpubliccoveragealone:f.stateNew Jersey	0.738661
## pctpubliccoveragealone:f.stateNew Mexico	0.527580
## pctpubliccoveragealone:f.stateNew York	0.746553
## pctpubliccoveragealone:f.stateNorth Carolina	0.848199
## pctpubliccoveragealone:f.stateNorth Dakota	0.839227
## pctpubliccoveragealone:f.stateOhio	0.709943
## pctpubliccoveragealone:f.stateOklahoma	0.089599 .
## pctpubliccoveragealone:f.stateOregon	0.510365
## pctpubliccoveragealone:f.statePennsylvania	0.260311
## pctpubliccoveragealone:f.stateRhode Island	0.194971
## pctpubliccoveragealone:f.stateSouth Carolina	0.598006
## pctpubliccoveragealone:f.stateSouth Dakota	0.556288
## pctpubliccoveragealone:f.stateTennessee	0.399302
## pctpubliccoveragealone:f.stateTexas	0.485395
## pctpubliccoveragealone:f.stateUtah	0.820552
## pctpubliccoveragealone:f.stateVermont	0.615627
## pctpubliccoveragealone:f.stateVirginia	0.200073
## pctpubliccoveragealone:f.stateWashington	0.679397
## pctpubliccoveragealone:f.stateWest Virginia	0.491442
## pctpubliccoveragealone:f.stateWisconsin	0.864518
## pctpubliccoveragealone:f.stateWyoming	0.910340
## pctpubliccoveragealone:f.education25_overBalanced Eduaction	0.016856 *
## pctpubliccoveragealone:f.education25_overMostly Bachelor's	0.147620
## sqrt(incidencerate):f.stateAlaska	0.505589
## sqrt(incidencerate):f.stateArizona	0.232379
## sqrt(incidencerate):f.stateArkansas	0.481981
## sqrt(incidencerate):f.stateCalifornia	0.590359
## sqrt(incidencerate):f.stateColorado	0.327983
## sqrt(incidencerate):f.stateConnecticut	0.942009
## sqrt(incidencerate):f.stateDelaware	NA
## sqrt(incidencerate):f.stateFlorida	0.986408
## sqrt(incidencerate):f.stateGeorgia	0.363399
## sqrt(incidencerate):f.stateHawaii	NA
## sqrt(incidencerate):f.stateIdaho	0.177389
## sqrt(incidencerate):f.stateIllinois	0.499164
## sqrt(incidencerate):f.stateIndiana	0.500513
## sqrt(incidencerate):f.stateIowa	0.494735
## sqrt(incidencerate):f.stateKansas	0.003496 **
## sqrt(incidencerate):f.stateKentucky	0.530050
## sqrt(incidencerate):f.stateLouisiana	0.679883
## sqrt(incidencerate):f.stateMaine	0.854379
## sqrt(incidencerate):f.stateMaryland	0.043368 *
## sqrt(incidencerate):f.stateMassachusetts	0.789685
## sqrt(incidencerate):f.stateMichigan	0.603645
## sqrt(incidencerate):f.stateMinnesota	0.188635
## sqrt(incidencerate):f.stateMississippi	0.581360
## sqrt(incidencerate):f.stateMissouri	0.895672
## sqrt(incidencerate):f.stateMontana	0.752809
## sqrt(incidencerate):f.stateNebraska	0.767541
## sqrt(incidencerate):f.stateNevada	0.397214

## sqrt(incidencerate):f.stateNew Hampshire	0.814511
## sqrt(incidencerate):f.stateNew Jersey	0.857123
## sqrt(incidencerate):f.stateNew Mexico	0.413246
## sqrt(incidencerate):f.stateNew York	0.482111
## sqrt(incidencerate):f.stateNorth Carolina	0.988338
## sqrt(incidencerate):f.stateNorth Dakota	0.181811
## sqrt(incidencerate):f.stateOhio	0.369640
## sqrt(incidencerate):f.stateOklahoma	0.304441
## sqrt(incidencerate):f.stateOregon	0.780300
## sqrt(incidencerate):f.statePennsylvania	0.771926
## sqrt(incidencerate):f.stateRhode Island	0.225496
## sqrt(incidencerate):f.stateSouth Carolina	0.701544
## sqrt(incidencerate):f.stateSouth Dakota	0.921722
## sqrt(incidencerate):f.stateTennessee	0.810278
## sqrt(incidencerate):f.stateTexas	0.162220
## sqrt(incidencerate):f.stateUtah	0.767567
## sqrt(incidencerate):f.stateVermont	0.925264
## sqrt(incidencerate):f.stateVirginia	0.657830
## sqrt(incidencerate):f.stateWashington	0.795828
## sqrt(incidencerate):f.stateWest Virginia	0.990941
## sqrt(incidencerate):f.stateWisconsin	0.589169
## sqrt(incidencerate):f.stateWyoming	0.133195
## pctbachdeg25_over:f.stateAlaska	0.022091 *
## pctbachdeg25_over:f.stateArizona	0.117415
## pctbachdeg25_over:f.stateArkansas	0.947963
## pctbachdeg25_over:f.stateCalifornia	0.331943
## pctbachdeg25_over:f.stateColorado	0.144633
## pctbachdeg25_over:f.stateConnecticut	0.812383
## pctbachdeg25_over:f.stateDelaware	NA
## pctbachdeg25_over:f.stateFlorida	0.054450 .
## pctbachdeg25_over:f.stateGeorgia	0.866146
## pctbachdeg25_over:f.stateHawaii	NA
## pctbachdeg25_over:f.stateIdaho	0.226640
## pctbachdeg25_over:f.stateIllinois	0.995598
## pctbachdeg25_over:f.stateIndiana	0.474725
## pctbachdeg25_over:f.stateIowa	0.097131 .
## pctbachdeg25_over:f.stateKansas	0.766508
## pctbachdeg25_over:f.stateKentucky	0.767877
## pctbachdeg25_over:f.stateLouisiana	0.434512
## pctbachdeg25_over:f.stateMaine	0.578282
## pctbachdeg25_over:f.stateMaryland	0.632211
## pctbachdeg25_over:f.stateMassachusetts	0.859091
## pctbachdeg25_over:f.stateMichigan	0.562795
## pctbachdeg25_over:f.stateMinnesota	0.460129
## pctbachdeg25_over:f.stateMississippi	0.203158
## pctbachdeg25_over:f.stateMissouri	0.899523
## pctbachdeg25_over:f.stateMontana	0.283860
## pctbachdeg25_over:f.stateNebraska	0.564502
## pctbachdeg25_over:f.stateNevada	0.013093 *
## pctbachdeg25_over:f.stateNew Hampshire	0.828219
## pctbachdeg25_over:f.stateNew Jersey	0.904333
## pctbachdeg25_over:f.stateNew Mexico	0.352623
## pctbachdeg25_over:f.stateNew York	0.826982
## pctbachdeg25_over:f.stateNorth Carolina	0.970051

## pctbachdeg25_over:f.stateNorth Dakota	0.903652
## pctbachdeg25_over:f.stateOhio	0.601012
## pctbachdeg25_over:f.stateOklahoma	0.202061
## pctbachdeg25_over:f.stateOregon	0.465386
## pctbachdeg25_over:f.statePennsylvania	0.674132
## pctbachdeg25_over:f.stateRhode Island	NA
## pctbachdeg25_over:f.stateSouth Carolina	0.620068
## pctbachdeg25_over:f.stateSouth Dakota	0.168251
## pctbachdeg25_over:f.stateTennessee	0.537807
## pctbachdeg25_over:f.stateTexas	0.861077
## pctbachdeg25_over:f.stateUtah	0.870816
## pctbachdeg25_over:f.stateVermont	0.778623
## pctbachdeg25_over:f.stateVirginia	0.684051
## pctbachdeg25_over:f.stateWashington	0.210861
## pctbachdeg25_over:f.stateWest Virginia	0.028627 *
## pctbachdeg25_over:f.stateWisconsin	0.348939
## pctbachdeg25_over:f.stateWyoming	0.846965
## log(medincome):f.stateAlaska	0.685120
## log(medincome):f.stateArizona	0.057405 .
## log(medincome):f.stateArkansas	0.944048
## log(medincome):f.stateCalifornia	0.908842
## log(medincome):f.stateColorado	0.167477
## log(medincome):f.stateConnecticut	0.667737
## log(medincome):f.stateDelaware	NA
## log(medincome):f.stateFlorida	0.490058
## log(medincome):f.stateGeorgia	0.485321
## log(medincome):f.stateHawaii	NA
## log(medincome):f.stateIdaho	0.789963
## log(medincome):f.stateIllinois	0.777933
## log(medincome):f.stateIndiana	0.765948
## log(medincome):f.stateIowa	0.586581
## log(medincome):f.stateKansas	0.910955
## log(medincome):f.stateKentucky	0.831328
## log(medincome):f.stateLouisiana	0.748532
## log(medincome):f.stateMaine	0.675683
## log(medincome):f.stateMaryland	0.315521
## log(medincome):f.stateMassachusetts	0.926484
## log(medincome):f.stateMichigan	0.942514
## log(medincome):f.stateMinnesota	0.270172
## log(medincome):f.stateMississippi	0.277174
## log(medincome):f.stateMissouri	0.683640
## log(medincome):f.stateMontana	0.659005
## log(medincome):f.stateNebraska	0.404053
## log(medincome):f.stateNevada	0.000752 ***
## log(medincome):f.stateNew Hampshire	0.829364
## log(medincome):f.stateNew Jersey	0.966452
## log(medincome):f.stateNew Mexico	0.382729
## log(medincome):f.stateNew York	0.660351
## log(medincome):f.stateNorth Carolina	0.935309
## log(medincome):f.stateNorth Dakota	0.699137
## log(medincome):f.stateOhio	0.982306
## log(medincome):f.stateOklahoma	0.663495
## log(medincome):f.stateOregon	0.552470
## log(medincome):f.statePennsylvania	0.499397

## log(medincome):f.stateRhode Island	NA
## log(medincome):f.stateSouth Carolina	0.540418
## log(medincome):f.stateSouth Dakota	0.666963
## log(medincome):f.stateTennessee	0.701361
## log(medincome):f.stateTexas	0.640554
## log(medincome):f.stateUtah	0.339368
## log(medincome):f.stateVermont	0.738718
## log(medincome):f.stateVirginia	0.357743
## log(medincome):f.stateWashington	0.799715
## log(medincome):f.stateWest Virginia	0.207223
## log(medincome):f.stateWisconsin	0.906184
## log(medincome):f.stateWyoming	0.676007
## log(medincome):f.education25_overBalanced Eduaction	0.000854 ***
## log(medincome):f.education25_overMostly Bachelor's	0.738126
## f.stateAlabama:log(pctemployed16_over)	0.595146
## f.stateAlaska:log(pctemployed16_over)	0.005848 **
## f.stateArizona:log(pctemployed16_over)	0.108730
## f.stateArkansas:log(pctemployed16_over)	0.132231
## f.stateCalifornia:log(pctemployed16_over)	0.269386
## f.stateColorado:log(pctemployed16_over)	0.088827 .
## f.stateConnecticut:log(pctemployed16_over)	0.936716
## f.stateDelaware:log(pctemployed16_over)	NA
## f.stateFlorida:log(pctemployed16_over)	0.452409
## f.stateGeorgia:log(pctemployed16_over)	0.112370
## f.stateHawaii:log(pctemployed16_over)	NA
## f.stateIdaho:log(pctemployed16_over)	0.812536
## f.stateIllinois:log(pctemployed16_over)	0.877849
## f.stateIndiana:log(pctemployed16_over)	0.405176
## f.stateIowa:log(pctemployed16_over)	0.416930
## f.stateKansas:log(pctemployed16_over)	0.025710 *
## f.stateKentucky:log(pctemployed16_over)	0.150597
## f.stateLouisiana:log(pctemployed16_over)	0.648987
## f.stateMaine:log(pctemployed16_over)	0.651469
## f.stateMaryland:log(pctemployed16_over)	0.940131
## f.stateMassachusetts:log(pctemployed16_over)	0.719797
## f.stateMichigan:log(pctemployed16_over)	0.216636
## f.stateMinnesota:log(pctemployed16_over)	0.516712
## f.stateMississippi:log(pctemployed16_over)	0.290592
## f.stateMissouri:log(pctemployed16_over)	0.001917 **
## f.stateMontana:log(pctemployed16_over)	0.851789
## f.stateNebraska:log(pctemployed16_over)	0.010981 *
## f.stateNevada:log(pctemployed16_over)	0.046141 *
## f.stateNew Hampshire:log(pctemployed16_over)	0.806355
## f.stateNew Jersey:log(pctemployed16_over)	0.762671
## f.stateNew Mexico:log(pctemployed16_over)	0.157816
## f.stateNew York:log(pctemployed16_over)	0.946026
## f.stateNorth Carolina:log(pctemployed16_over)	0.448939
## f.stateNorth Dakota:log(pctemployed16_over)	0.694728
## f.stateOhio:log(pctemployed16_over)	0.741857
## f.stateOklahoma:log(pctemployed16_over)	0.017561 *
## f.stateOregon:log(pctemployed16_over)	0.754364
## f.statePennsylvania:log(pctemployed16_over)	0.738532
## f.stateRhode Island:log(pctemployed16_over)	NA
## f.stateSouth Carolina:log(pctemployed16_over)	0.061178 .

```
## f.stateSouth Dakota:log(pctemployed16_over) 0.958525
## f.stateTennessee:log(pctemployed16_over) 0.442694
## f.stateTexas:log(pctemployed16_over) 0.175617
## f.stateUtah:log(pctemployed16_over) 0.547942
## f.stateVermont:log(pctemployed16_over) 0.813392
## f.stateVirginia:log(pctemployed16_over) 0.632473
## f.stateWashington:log(pctemployed16_over) 0.647973
## f.stateWest Virginia:log(pctemployed16_over) 0.000641 ***
## f.stateWisconsin:log(pctemployed16_over) 0.119944
## f.stateWyoming:log(pctemployed16_over) 0.674895
## f.education25_overBalanced Eduaction:log(pctemployed16_over) 0.001436 **
## f.education25_overMostly Bachelor's:log(pctemployed16_over) 0.651022
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.07598 on 1535 degrees of freedom
## Multiple R-squared:  0.6757, Adjusted R-squared:  0.6134
## F-statistic: 10.84 on 295 and 1535 DF,  p-value: < 2.2e-16
```

This model has obtained the following results in the training data:

- Residual standard error: 0.08089 on 1535 degrees of freedom
- Multiple R-squared: 0.6757,
- Adjusted R-squared: 0.6134

Bias-variance tradeoff

These models yield excellent results but are susceptible to overfitting the training data. To address this issue, we aim to balance the bias-variance tradeoff by constructing a simpler model. The steps we follow are:

1. Build an initial model using all numeric variables.
2. Address multicollinearity to eliminate insignificant variables.
3. Introduce factors into the model.
4. Use stepwise selection to refine and finalize the model.
5. Incorporate meaningful interactions between numeric variables and factors, ensuring a limited number to minimize the risk of overfitting.

The model obtains the following results in the training data:

- Residual standard error: 18.1 on 1745 degrees of freedom
- Multiple R-squared: 0.5996
- Adjusted R-squared: 0.5801

```
m8 <- lm( target_transformed ~ avganncount + avgdeathspereyear + sqrt(incidencerate)
+ pctblack + pctwhite+ pctotherrace+ pctasian +pctpubliccoveragealone +
pctbachdeg25_over + pctnohs18_24 + pctsomacol18_24 + pctbachdeg18_24 +
pctemployed16_over + ptempprivcoverage + pctmarriedhouseholds +
f.state + f.unemployment + f.income + f.education25_over
+ sqrt(incidencerate):f.unemployment + sqrt(incidencerate):f.income + sqrt(incidencerate):f.educati
+ pctnohs18_24:f.unemployment + pctwhite:f.income
, data=df)

summary(m8)
```

```
##
## Call:
```



```

## lm(formula = target_transformed ~ avganncount + avgdeathsperyear +
##     sqrt(incidencerate) + pctblack + pctwhite + pctotherrace +
##     pctasian + pctpubliccoveragealone + pctbachdeg25_over + pctnohs18_24 +
##     pctsomocol18_24 + pctbachdeg18_24 + pctemployed16_over +
##     pctempprivcoverage + pctmarriedhouseholds + f.state + f.unemployment +
##     f.income + f.education25_over + sqrt(incidencerate):f.unemployment +
##     sqrt(incidencerate):f.income + sqrt(incidencerate):f.education25_over +
##     avgdeathsperyear:f.unemployment + pctnohs18_24:f.unemployment +
##     pctwhite:f.income, data = df)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -21.3577  -2.6734  -0.0073   2.5161  29.5396
##
## Coefficients:
##                                     Estimate Std. Error
## (Intercept)                      3.405e+01  7.490e+00
## avganncount                      -1.392e-03  5.192e-04
## avgdeathsperyear                  4.499e-03  3.266e-03
## sqrt(incidencerate)               2.129e+00  3.059e-01
## pctblack                         -4.690e-02  2.641e-02
## pctwhite                        -2.573e-02  2.563e-02
## pctotherrace                     -1.527e-01  4.547e-02
## pctasian                        1.803e-03  8.202e-02
## pctpubliccoveragealone            1.248e-01  4.593e-02
## pctbachdeg25_over                -3.592e-01  5.621e-02
## pctnohs18_24                    -3.784e-02  3.431e-02
## pctsomocol18_24                 -2.852e-02  1.621e-02
## pctbachdeg18_24                 -9.462e-02  3.741e-02
## pctemployed16_over              -6.605e-02  3.023e-02
## pctempprivcoverage               4.629e-02  2.701e-02
## pctmarriedhouseholds            -1.154e-01  2.976e-02
## f.stateAlaska                    7.286e-01  2.141e+00
## f.stateArizona                   -5.269e+00  2.074e+00
## f.stateArkansas                   3.580e+00  1.158e+00
## f.stateCalifornia                 -5.019e+00  1.395e+00
## f.stateColorado                  -4.268e+00  1.301e+00
## f.stateConnecticut               -5.233e+00  2.116e+00
## f.stateDelaware                  -2.076e+00  5.032e+00
## f.stateFlorida                    5.841e-02  1.195e+00
## f.stateGeorgia                   -1.800e+00  9.676e-01
## f.stateHawaii                    -1.259e+01  4.323e+00
## f.stateIdaho                     -5.255e+00  1.360e+00
## f.stateIllinois                  -8.865e-02  1.129e+00
## f.stateIndiana                    2.172e+00  1.125e+00
## f.stateIowa                     -1.566e+00  1.164e+00
## f.stateKansas                     3.788e-01  1.147e+00
## f.stateKentucky                  2.556e+00  1.082e+00
## f.stateLouisiana                  5.999e-01  1.152e+00
## f.stateMaine                      7.648e-01  1.825e+00
## f.stateMaryland                  -7.501e-01  1.602e+00
## f.stateMassachusetts              -1.771e+00  2.050e+00
## f.stateMichigan                   -1.293e-01  1.132e+00
## f.stateMinnesota                 -2.519e+00  1.193e+00

```

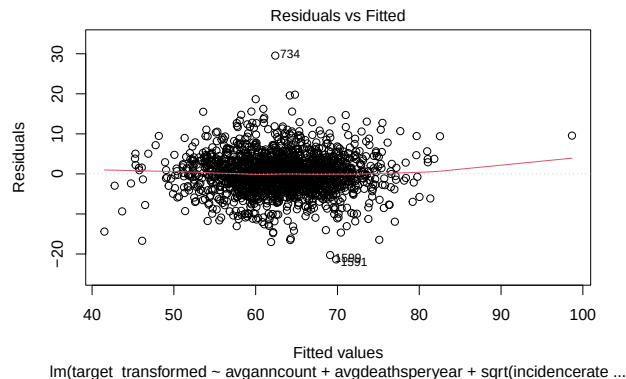
## f.stateMississippi	2.221e+00	1.080e+00
## f.stateMissouri	2.874e+00	1.071e+00
## f.stateMontana	-4.015e+00	1.447e+00
## f.stateNebraska	-1.062e+00	1.190e+00
## f.stateNevada	-1.309e+00	1.653e+00
## f.stateNew Hampshire	-6.232e-01	2.252e+00
## f.stateNew Jersey	-1.691e+00	1.802e+00
## f.stateNew Mexico	-4.598e+00	1.554e+00
## f.stateNew York	-4.159e+00	1.231e+00
## f.stateNorth Carolina	-2.165e+00	1.051e+00
## f.stateNorth Dakota	-5.888e-01	1.348e+00
## f.stateOhio	8.143e-01	1.151e+00
## f.stateOklahoma	3.233e+00	1.213e+00
## f.stateOregon	-2.186e+00	1.460e+00
## f.statePennsylvania	-2.354e+00	1.183e+00
## f.stateRhode Island	-5.920e-01	3.018e+00
## f.stateSouth Carolina	2.735e-01	1.215e+00
## f.stateSouth Dakota	-1.900e+00	1.309e+00
## f.stateTennessee	2.125e+00	1.089e+00
## f.stateTexas	6.859e-01	1.002e+00
## f.stateUtah	-5.944e+00	1.493e+00
## f.stateVermont	1.859e+00	2.163e+00
## f.stateVirginia	1.054e+00	1.037e+00
## f.stateWashington	-2.766e+00	1.433e+00
## f.stateWest Virginia	5.972e-01	1.255e+00
## f.stateWisconsin	-7.342e-01	1.206e+00
## f.stateWyoming	-1.510e+00	1.674e+00
## f.unemploymentModerate Unemployment	-7.372e+00	5.725e+00
## f.unemploymentHigh Unemployment	-1.125e+01	6.765e+00
## f.incomeLow	1.260e+01	5.412e+00
## f.incomeModerate	1.385e+01	6.925e+00
## f.incomeHigh	1.426e+01	7.147e+00
## f.education25_overBalanced Eduaction	-8.981e+00	5.688e+00
## f.education25_overMostly Bachelor's	1.254e+01	8.357e+00
## sqrt(incidencerate):f.unemploymentModerate Unemployment	3.492e-01	2.691e-01
## sqrt(incidencerate):f.unemploymentHigh Unemployment	5.842e-01	3.145e-01
## sqrt(incidencerate):f.incomeLow	-4.897e-01	2.349e-01
## sqrt(incidencerate):f.incomeModerate	-4.811e-01	3.076e-01
## sqrt(incidencerate):f.incomeHigh	-2.978e-01	3.126e-01
## sqrt(incidencerate):f.education25_overBalanced Eduaction	4.403e-01	2.704e-01
## sqrt(incidencerate):f.education25_overMostly Bachelor's	-5.966e-01	3.929e-01
## avgdeathsperyear:f.unemploymentModerate Unemployment	-1.093e-03	2.877e-03
## avgdeathsperyear:f.unemploymentHigh Unemployment	-1.141e-03	2.907e-03
## pctnohs18_24:f.unemploymentModerate Unemployment	1.507e-02	3.856e-02
## pctnohs18_24:f.unemploymentHigh Unemployment	-1.120e-02	4.500e-02
## pctwhite:f.incomeLow	-3.630e-02	2.190e-02
## pctwhite:f.incomeModerate	-5.666e-02	2.566e-02
## pctwhite:f.incomeHigh	-9.948e-02	3.096e-02
##	t value	Pr(> t)
## (Intercept)	4.546	5.84e-06 ***
## avganncount	-2.681	0.007412 **
## avgdeathsperyear	1.377	0.168599
## sqrt(incidencerate)	6.961	4.77e-12 ***
## pctblack	-1.776	0.075937 .

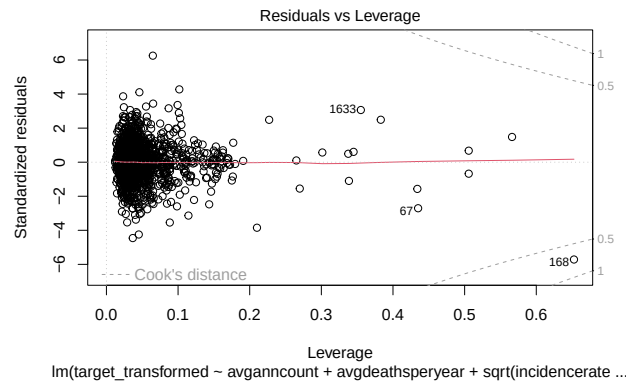
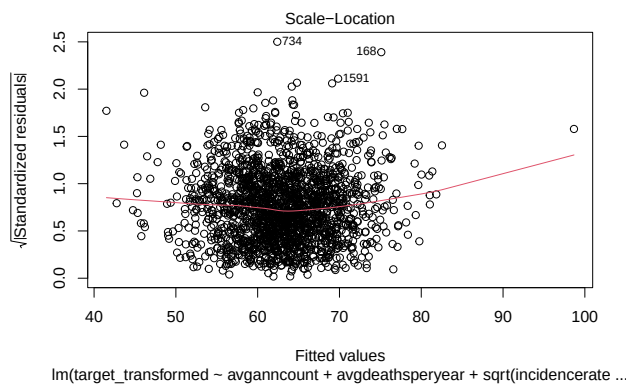
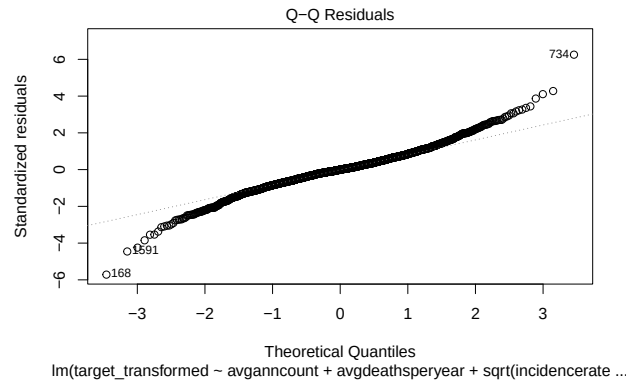
## pctwhite	-1.004	0.315568	
## pctotherrace	-3.359	0.000799	***
## pctasian	0.022	0.982460	
## pctpubliccoveragealone	2.718	0.006638	**
## pctbachdeg25_over	-6.391	2.11e-10	***
## pctnohs18_24	-1.103	0.270196	
## pctsomecol18_24	-1.759	0.078795	.
## pctbachdeg18_24	-2.529	0.011513	*
## pctemployed16_over	-2.185	0.029017	*
## pctempprivcoverage	1.714	0.086665	.
## pctmarriedhouseholds	-3.876	0.000110	***
## f.stateAlaska	0.340	0.733638	
## f.stateArizona	-2.541	0.011144	*
## f.stateArkansas	3.091	0.002023	**
## f.stateCalifornia	-3.597	0.000331	***
## f.stateColorado	-3.281	0.001053	**
## f.stateConnecticut	-2.473	0.013488	*
## f.stateDelaware	-0.412	0.680069	
## f.stateFlorida	0.049	0.961025	
## f.stateGeorgia	-1.860	0.063084	.
## f.stateHawaii	-2.912	0.003639	**
## f.stateIdaho	-3.863	0.000116	***
## f.stateIllinois	-0.079	0.937433	
## f.stateIndiana	1.930	0.053713	.
## f.stateIowa	-1.346	0.178556	
## f.stateKansas	0.330	0.741217	
## f.stateKentucky	2.362	0.018263	*
## f.stateLouisiana	0.521	0.602550	
## f.stateMaine	0.419	0.675227	
## f.stateMaryland	-0.468	0.639573	
## f.stateMassachusetts	-0.864	0.387541	
## f.stateMichigan	-0.114	0.909056	
## f.stateMinnesota	-2.112	0.034852	*
## f.stateMississippi	2.057	0.039856	*
## f.stateMissouri	2.683	0.007355	**
## f.stateMontana	-2.775	0.005584	**
## f.stateNebraska	-0.893	0.372029	
## f.stateNevada	-0.792	0.428571	
## f.stateNew Hampshire	-0.277	0.781980	
## f.stateNew Jersey	-0.939	0.347982	
## f.stateNew Mexico	-2.959	0.003133	**
## f.stateNew York	-3.378	0.000745	***
## f.stateNorth Carolina	-2.060	0.039563	*
## f.stateNorth Dakota	-0.437	0.662287	
## f.stateOhio	0.708	0.479324	
## f.stateOklahoma	2.665	0.007774	**
## f.stateOregon	-1.497	0.134445	
## f.statePennsylvania	-1.991	0.046653	*
## f.stateRhode Island	-0.196	0.844494	
## f.stateSouth Carolina	0.225	0.821975	
## f.stateSouth Dakota	-1.451	0.146881	
## f.stateTennessee	1.952	0.051149	.
## f.stateTexas	0.684	0.493840	
## f.stateUtah	-3.983	7.09e-05	***

```
## f.stateVermont                0.859 0.390327
## f.stateVirginia               1.016 0.309697
## f.stateWashington            -1.931 0.053632 .
## f.stateWest Virginia         0.476 0.634268
## f.stateWisconsin             -0.609 0.542762
## f.stateWyoming               -0.903 0.366884
## f.unemploymentModerate Unemployment -1.288 0.197967
## f.unemploymentHigh Unemployment -1.663 0.096528 .
## f.incomeLow                  2.328 0.020034 *
## f.incomeModerate             2.000 0.045613 *
## f.incomeHigh                 1.996 0.046111 *
## f.education25_overBalanced Eduaction -1.579 0.114545
## f.education25_overMostly Bachelor's 1.501 0.133502
## sqrt(incidencerate):f.unemploymentModerate Unemployment 1.298 0.194584
## sqrt(incidencerate):f.unemploymentHigh Unemployment 1.857 0.063431 .
## sqrt(incidencerate):f.incomeLow -2.084 0.037268 *
## sqrt(incidencerate):f.incomeModerate -1.564 0.118029
## sqrt(incidencerate):f.incomeHigh -0.953 0.340836
## sqrt(incidencerate):f.education25_overBalanced Eduaction 1.628 0.103666
## sqrt(incidencerate):f.education25_overMostly Bachelor's -1.518 0.129140
## avgdeathsperyear:f.unemploymentModerate Unemployment -0.380 0.704073
## avgdeathsperyear:f.unemploymentHigh Unemployment -0.392 0.694839
## pctnohs18_24:f.unemploymentModerate Unemployment 0.391 0.695901
## pctnohs18_24:f.unemploymentHigh Unemployment -0.249 0.803410
## pctwhite:f.incomeLow -1.658 0.097564 .
## pctwhite:f.incomeModerate -2.208 0.027347 *
## pctwhite:f.incomeHigh -3.213 0.001336 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 4.884 on 1745 degrees of freedom
## Multiple R-squared:  0.5999, Adjusted R-squared:  0.5804
## F-statistic: 30.78 on 85 and 1745 DF, p-value: < 2.2e-16
```

```
plot(m8)
```

```
## Warning: not plotting observations with leverage one:
## 1481
```





The interpretation of the plots given by the `lm` function is crucial to assess whether the data respects the assumption of the linear model:

- 1- Residuals vs Fitted plot: Here, the residuals exhibit some spread across the range of fitted values, but there may be mild patterns, suggesting potential non-linearity or heteroscedasticity (unequal variance of residuals). Outliers (734 and 1591) are labeled, indicating unusual points that could influence the model.
- 2- Q-Q Plot: In this case, the residuals follow the diagonal fairly well, except at the extremes, where some points deviate. This indicates potential slight departures from normality in the tails, which may need attention depending on the context.
- 3- Scale-Location Plot: Here, the red trend line shows slight curvature, and there is some non-random spread at higher fitted values, suggesting mild heteroscedasticity.
- 4- Residuals vs Leverage: The plot identifies influential data points in a regression model. Most residuals are close to zero, indicating a good overall fit, but some notable outliers are present. Point 168 has high leverage (~ 0.6) and is close to the Cook's distance threshold (1.0), marking it as highly influential.

Point 1633 shows a large positive residual with moderate leverage, while point 67 has moderate values for both leverage and residuals. These points may significantly affect the model and require further investigation. Most data points, however, lie within acceptable ranges, suggesting the model performs well for the majority of observations.

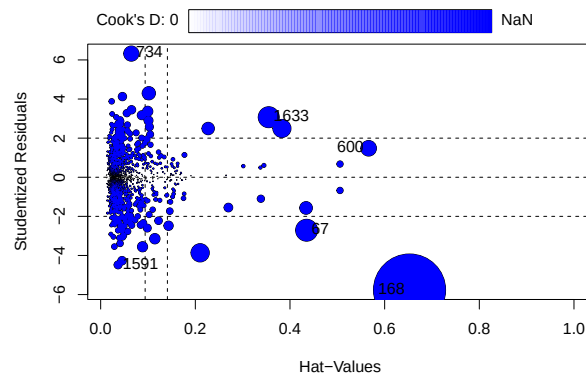
Influential data study

Given that the models are being quite negatively affected by influential data, an study of the aforementioned will be performed to address this issue.

Thanks to the `influencePlot` function, we can see that the following observations (83,1168,1650,80,1389,1591,67,734,168) are potentially influencing strongly in the regression, and thus we want to remove them:

“Queen Anne’s County, Maryland”, “Merrimack County, New Hampshire”, “Tippecanoe County, Indiana”, “Dooley County, Georgia”, “Marion County, West Virginia”, “Grand County, Colorado”, “St. Joseph County, Indiana”, Natrona County, Wyoming”, “Ocean County, New Jersey” .

```
influential <- influencePlot(m8,pch=19,id=list(method="noteworthy",n=3))
```



```
rownames(influential)
```

```
## [1] "1633" "1591" "67" "734" "860" "600" "168"
```

```
df$geography[c(899,70,67,734,860,600,168)]
```

```
## [1] "Rains County, Texas" "Sheridan County, North Dakota"
## [3] "St. Joseph County, Indiana" "Natrona County, Wyoming"
## [5] "Benton County, Indiana" "Nevada County, California"
## [7] "Ocean County, New Jersey"
```

```
df<- df[-c(899,70,67,734,860,600,168), ]
test[(which(test$f.state=="District of Columbia")),]
```

```
##      avganncount avgdeathspereyear target_deathrate incidencerate medincome
## 562      2881      1084      182.3      483.7      69992
##      popest2015 povertypercent      binnedinc medianage medianagemale
## 562      672228      18.4 (61494.5, 125635]      33.7      33.5
##      medianagefemale      geography percentmarried
## 562      33.9 District of Columbia, District of Columbia      27.3
##      pctnohs18_24 pcths18_24 pctsomecol18_24 pctbachdeg18_24 pcths25_over
## 562      10.5      22.6      43.9      23      18
##      pctbachdeg25_over pctemployed16_over pctunemployed16_over
## 562      23.3      61.8      9.6
##      pctprivatecoverage pctprivatecoveragealone pctempprivcoverage
```

```
## 562          69.6          55.6          48.2
##      pctpubliccoverage pctpubliccoveragealone pctwhite pctblack pctasian
## 562          35.1          21.7 40.20563 48.87333 3.679164
##      pctotherrace pctmarriedhouseholds birthrate f.income f.population
## 562    4.194544          23.91565 4.581892    High    High
##          f.state  f.education18_24 f.education25_over
## 562 District of Columbia Moderate Education  Mostly Bachelor's
##          f.unemployment
## 562 Moderate Unemployment
test<-test[-562,]
```

Model evaluation

Once the model has been thoroughly studied, this last part evaluates the model in the test data to assess its forecasting capabilities.

We will test Model6 and Model8, to assess the difference between a model carefully built to handle the bias-variance tradeoff and avoid overfitting (Model6), and a model with better results in the training set which was built to get the best results in the training set.

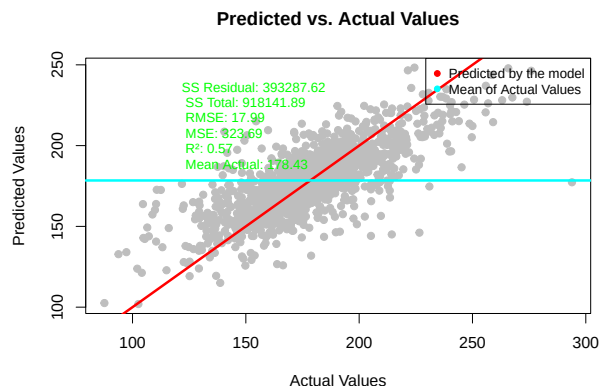
```
test$target_transformed <- (test$target_deathrate^lambda - 1) / lambda
test$target_transformed <- log(test$target_transformed)
test$f.state <- factor(test$f.state, levels = levels(df$f.state))
```

Model 8

Using Model 8, we obtain the following results in the test set: - Mean Squared Error (MSE): 324.9554 - Root Mean Squared Error (RMSE): 18.02652 - SS_{total} is: 918141.9 - SS_{residual} is: 394820.8 - R-squared (R^2): -0.5706384

```
par(mfrow=c(1,1))
evaluate_model(m8, test)
```

```
## Mean Squared Error (MSE): 323.6935
## Root Mean Squared Error (RMSE): 17.99148
## SStotal is: 918141.9
## SSresidual is: 393287.6
## R-squared ( $R^2$ ): 0.5722682
```



Model 6

Using Model 6, we obtain the following results in the test set: - Mean Squared Error (MSE): 0.0583265 - Root Mean Squared Error (RMSE): 0.2415088 - SS_{total} is: 17.62189 - SS_{residual} is: 70.8667 - R-squared (R^2): -3.021516

Note that the negative R-squared shows how the model performs worse than just using a basic model with the average of the test set. The evaluation of this model is very useful to see that obtaining best results in the training data is not always better; the bias-variance trade off must be considered in order to avoid overfitting and end up in a situation in which the model is not able to generalize properly.

Conclusion

In conclusion, this project has shown the importance of several foundational aspects of a successful modeling process:

- *Exploratory Data Analysis (EDA)*: The EDA process is fundamental to any data-driven project. It allows us to thoroughly understand the structure and nature of the data, uncover hidden patterns, and identify potential issues such as errors or missing values. EDA is also a key step in creating new, meaningful variables that can improve model performance. Without this critical step, a robust and reliable modeling project cannot exist. It lays the groundwork for all subsequent stages and ensures the data is well-prepared for analysis.
- *Overfitting Awareness*: While achieving good results on the training set can be satisfying, it is essential to remain aware against overfitting. Overfitting occurs when a model becomes too tailored to the training data, capturing noise or insignificant patterns that do not generalize well to unseen data. This can result in poor forecasting and unreliable predictions. It is crucial to remember that the primary objective of modeling is not to optimize performance on the existing training data but to accurately predict new, unseen values. Additionally, overly small mean squared error (MSE) values might signal overfitting and should not be interpreted as the unique indicator of success. Instead, a comprehensive evaluation should include the R^2 metric, which measures the proportion of variance in the data that the model captures. A high R^2 suggests that the model effectively explains the data's variability, making it a critical measure to consider alongside MSE.